FRM PART II • 2026 • EXAM 25 NOVEMBER 2026

Liquidity and Treasury Risk

Notes + Question Workbook

P1 • P2 • P3, in order of importance

47 learning objectives from 17 readings, banded by the Master LO Map v3. P4 is excluded.

Built from Vardeez's notes, the GARP 2026 Liquidity and Treasury Risk book, the Schweser-Notes 2026 book, the Tier A and B–D Recall Workbooks and the Liquidity and Treasury Risk Question Bank.

How to use this book

- The book runs **P1, then P2, then P3**. Inside each part, LOs follow syllabus order. The syllabus index at the front lists all 47 LOs in GARP order with their band and page.
- Each LO is one block: **focus box** (why it sits in its band, Vardeez's pointer and the reported exam shapes), **notes** limited to that focus, a **formula card** or comparison table, **exam traps**, then **6 to 10 practice questions**. **Solutions** sit at the end of each block, so attempt first.
- LO wording is verbatim from the official GARP 2026 learning objectives, as listed in the Master LO Map.
- Type tags: CALC numerical questions expected; CONCEPT statement questions (only the reported workbook calculations appear); BOTH a mix set by the reported shapes and Vardeez's focus.
- Question sources, in priority order: Tier workbook (August 2026 recall and mapped shapes), GARP-tagged bank questions (practice exams, GARP book, mocks), other bank questions, and *New* questions written in GARP style from the readings where an LO was short. Every number has been recomputed.
- Green-edged boxes mark corrections to the source notes or to a bank key. Figures are drawn for black-and-white printing.

Contents by priority

P1 Most important 6

LTR-1 • Liquidity Risk

LTR-1 a	BOTH	Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR).	7

LTR-2 • Liquidity and Leverage

LTR-2 f	CALC	Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days.	23

LTR-4 • The Investment Function in Financial-Services Management

LTR-4 a	CONCEPT	Compare various money market and capital market instruments and discuss their advantages and disadvantages.	36
LTR-4 b	BOTH	Identify and discuss various factors that affect the choice of investment securities by a bank.	49

LTR-5 • Liquidity and Reserves Management: Strategies and Policies

LTR-5 a	BOTH	Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank.	63
LTR-5 b	CONCEPT	Compare strategies that a bank can use to meet demands for additional liquidity.	74
LTR-5 c	BOTH	Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators).	85

LTR-7 • Monitoring Liquidity

LTR-7 d	BOTH	Interpret the term structure of expected cash flows and cumulative cash flows.	99
LTR-7 e	BOTH	Discuss the impact of available asset transactions on cash flows and liquidity generation capacity.	114

LTR-12 • Managing Non-Deposit Liabilities

LTR-12 a	CONCEPT	Distinguish between the various sources of non-deposit liabilities at a bank.	127
LTR-12 c	CONCEPT	Discuss factors affecting the choice of non-deposit funding sources.	138
LTR-12 d	CALC	Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach.	149

LTR-14 • Liquidity Transfer Pricing: A Guide to Better Practice

LTR-14 a	CONCEPT	Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process.	162
LTR-14 c	BOTH	Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost).	172

LTR-17 • Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques

LTR-17 b	BOTH	Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin.	183
LTR-17 c	BOTH	Describe duration gap management and apply this strategy to protect a bank's net worth.	200

P2 Second priority 216

LTR-3 • Early Warning Indicators

LTR-3 a	CONCEPT	Evaluate the characteristics of sound Early Warning Indicators (EWI) measures.	217
LTR-3 b	CONCEPT	Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve).	227
LTR-3 c	CONCEPT	Discuss the applications of EWIs in the context of the liquidity risk management process.	236

LTR-4 • The Investment Function in Financial-Services Management

LTR-4 c	BOTH	Apply investment maturity strategies and maturity management tools based on the yield curve and duration.	246

LTR-11 • Contingency Funding Planning

LTR-11 a	CONCEPT	Discuss the relationship between contingency funding planning and liquidity stress testing.	258
LTR-11 b	CONCEPT	Describe best practices in the design of a sound contingency funding plan.	268
LTR-11 c	CONCEPT	Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting).	276

LTR-13 • Repurchase Agreements and Financing

LTR-13 e	CONCEPT	Compare the use of general and special collateral in repo transactions.	290
LTR-13 f	CONCEPT	Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle.	301

LTR-15 • The US Dollar Shortage in Global Banking and the International Policy Response

LTR-15 a	CONCEPT	Identify the causes of the U.S. dollar shortage during the Great Financial Crisis.	313
LTR-15 b	CONCEPT	Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities.	324
LTR-15 c	CONCEPT	Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness.	335

LTR-17 • Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques

LTR-17 a	CONCEPT	Discuss how asset-liability management strategies can help a bank hedge against interest rate risk.	344
LTR-17 d	CONCEPT	Discuss the limitations of interest-sensitive gap management and duration gap management.	353

P3 Third priority 361

LTR-1 • Liquidity Risk

LTR-1 b	CONCEPT	Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft.	362
LTR-1 d	CONCEPT	Explain liquidity black holes and identify the causes of positive feedback trading.	371

LTR-6 • Intraday Liquidity Risk Management

LTR-6 a	CONCEPT	Identify and explain the uses and sources of intraday liquidity.	379
LTR-6 c	CONCEPT	Differentiate between methods for tracking intraday flows and monitoring risk levels.	391

LTR-7 • Monitoring Liquidity

LTR-7 a	CONCEPT	Differentiate between deterministic and stochastic cash flows and provide examples of each.	402
LTR-7 b	CONCEPT	Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process.	410

LTR-8 • The Failure Mechanics of Dealer Banks

LTR-8 a	CONCEPT	Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business.	417
LTR-8 b	CONCEPT	Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks.	427
LTR-8 c	CONCEPT	Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.	437

LTR-9 • Liquidity Stress Testing

LTR-9 b	CALC	Estimate contingent liquidity via the liquid asset buffer.	444

LTR-10 • Liquidity Risk Reporting and Stress Testing

LTR-10 b	CONCEPT	Compare and interpret different types of liquidity risk reports.	452

LTR-12 • Managing Non-Deposit Liabilities

LTR-12 b	BOTH	Describe and calculate the available funds gap.	463

LTR-13 • Repurchase Agreements and Financing

LTR-13 a	BOTH	Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction.	472
LTR-13 c	CONCEPT	Explain how counterparty risk and liquidity risk can arise through the use of repo transactions.	482

LTR-16 • Covered Interest Parity Lost: Understanding the Cross-Currency Basis

LTR-16 a	CONCEPT	Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps.	491
LTR-16 b	CONCEPT	Identify key factors that affect the cross-currency swap basis.	499
LTR-16 c	BOTH	Assess the causes of covered interest rate parity violations after the financial crisis of 2008.	509

Syllabus index (GARP order)

Every LO in this book, in the order of the official 2026 learning objectives. P4 LOs are not in this book.

LTR-1 • Liquidity Risk

LTR-1 a P1 BOTH 7
LTR-1 b P3 CONCEPT 362
LTR-1 d P3 CONCEPT 371

LTR-2 • Liquidity and Leverage

LTR-2 f P1 CALC 23

LTR-3 • Early Warning Indicators

LTR-3 a P2 CONCEPT 217
LTR-3 b P2 CONCEPT 227
LTR-3 c P2 CONCEPT 236

LTR-4 • The Investment Function in Financial-Services Management

LTR-4 a P1 CONCEPT 36
LTR-4 b P1 BOTH 49
LTR-4 c P2 BOTH 246

LTR-5 • Liquidity and Reserves Management: Strategies and Policies

LTR-5 a P1 BOTH 63
LTR-5 b P1 CONCEPT 74
LTR-5 c P1 BOTH 85

LTR-6 • Intraday Liquidity Risk Management

LTR-6 a P3 CONCEPT 379
LTR-6 c P3 CONCEPT 391

LTR-7 • Monitoring Liquidity

LTR-7 a P3 CONCEPT 402
LTR-7 b P3 CONCEPT 410
LTR-7 d P1 BOTH 99
LTR-7 e P1 BOTH 114

LTR-8 • The Failure Mechanics of Dealer Banks

LTR-8 a P3 CONCEPT 417
LTR-8 b P3 CONCEPT 427
LTR-8 c P3 CONCEPT 437

LTR-9 • Liquidity Stress Testing

LTR-9 b P3 CALC 444

LTR-10 • Liquidity Risk Reporting and Stress Testing

LTR-10 b P3 CONCEPT 452

LTR-11 • Contingency Funding Planning

LTR-11 a P2 CONCEPT 258
LTR-11 b P2 CONCEPT 268
LTR-11 c P2 CONCEPT 276

LTR-12 • Managing Non-Deposit Liabilities

LTR-12 a P1 CONCEPT 127
LTR-12 b P3 BOTH 463
LTR-12 c P1 CONCEPT 138
LTR-12 d P1 CALC 149

LTR-13 • Repurchase Agreements and Financing

LTR-13 a P3 BOTH 472
LTR-13 c P3 CONCEPT 482
LTR-13 e P2 CONCEPT 290
LTR-13 f P2 CONCEPT 301

LTR-14 • Liquidity Transfer Pricing: A Guide to Better Practice

LTR-14 a P1 CONCEPT 162
LTR-14 c P1 BOTH 172

LTR-15 • The US Dollar Shortage in Global Banking and the International Policy Response

LTR-15 a P2 CONCEPT 313
LTR-15 b P2 CONCEPT 324
LTR-15 c P2 CONCEPT 335

LTR-16 • Covered Interest Parity Lost: Understanding the Cross-Currency Basis

LTR-16 a P3 CONCEPT 491
LTR-16 b P3 CONCEPT 499
LTR-16 c P3 BOTH 509

LTR-17 • Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques

LTR-17 a P2 CONCEPT 344
LTR-17 b P1 BOTH 183
LTR-17 c P1 BOTH 200
LTR-17 d P2 CONCEPT 353

LIQUIDITY AND TREASURY RISK • PART

P1

Most important

Tier A recall topics and the concepts Vardeez flags in Very Important chapters. 16 LOs from 8 readings.

LTR-1 • Liquidity Risk • Vardeez: IMPORTANT

LTR-1 a

P1 BOTH

Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR).

Workbook: A11 Liquidity-Adjusted VaR and Cost of Liquidation • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A11, recalled in the August 2026 sitting ("LVaR CoL" and "liquidity-adjusted VaR theory"). Reported in 6 of 16 recall sources (Map #11): "1 numerical on liquidity adjusted VaR", "liquidity var numerical", and an LVaR question paired with a capital adequacy ratio (CAR) question (CAR itself sits in workbook D3).
Vardeez: LVaR calculation (liquidity trading risk) is the first of his two pointers for this reading; liquidity black holes (LTR-1 d) is the second.

Reported exam shapes

- **S1** LVaR numerical with the normal-market (constant-spread) cost of liquidation.
- **S2** Stressed (exogenous-spread) cost of liquidation $\frac{1}{2}\alpha(\mu + \lambda\sigma)$, with the current quoted spread planted as the sibling-swap input.
- **S3** Cost of liquidation as its own number, the plain form of the August recall.
- **S4** LVaR theory: normal versus stressed spread, perfect spread correlation, liquidation horizon.

What the questions look like: mostly numerical. A table of positions with bid and offer quotes, or with the mean and standard deviation of the spread, sometimes a VaR figure or a price volatility, and the question asks for the cost of liquidation or the LVaR. The stem usually carries inputs for *both* constructions, and only one belongs to the definition asked. Expect one statement question on the theory.

1. Liquidity trading risk

Hull separates two liquidity risks. **Liquidity trading risk** is the risk of not being able to sell (or buy) a position at close to its fair value: the act of trading moves the price against you. **Liquidity funding risk** (LTR-1 b) is the risk of not meeting cash needs as they arise. Malz calls the same pair transactions liquidity risk and funding liquidity risk (LTR-2 e). Liquidity is not solvency: a solvent bank, assets above liabilities, can still fail if it cannot raise cash in time.

The price at which an asset can be sold depends on four things:

1. the **mid-market price** of the asset, or an estimate of its value;
2. **how much** of the asset is to be sold;
3. **how quickly** it is to be sold;
4. the **economic environment** (a "seller's market" or a market where liquidity is tight, as after the 1998 Russian default and in 2007 to 2008).

Spreads, trade size and the market maker

A market maker quotes a bid (where you sell) and an offer (where you buy) that are good up to a certain size. Beyond that size limit the bid falls and the offer rises, because a bigger trade is harder for the market maker to hedge (Figure 1). Where there is no market maker there is still an implicit

spread: the price depends on which side you want. Spreads run from about 0.05% of the mid-market price for the most liquid assets to 5%, or even 10%, for illiquid ones. In retail markets the pattern can reverse: a bigger foreign exchange deal at a bank branch usually gets a *better* quote.

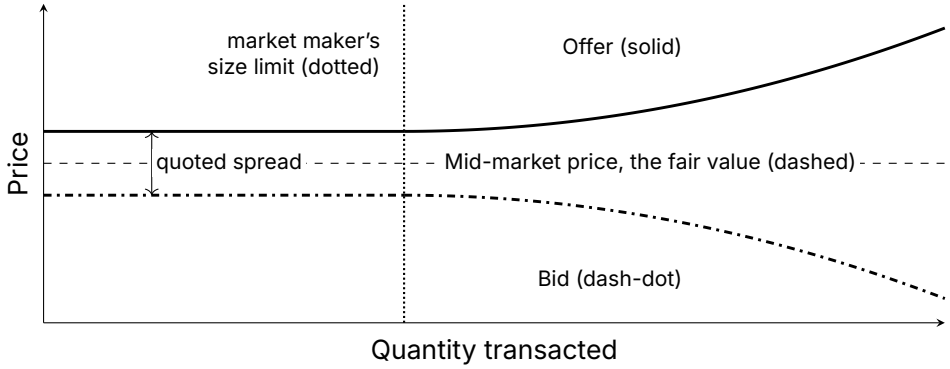


Figure 1. Bid and offer as a function of the quantity traded, after Hull's Figure 1.1. The quotes are flat up to the market maker's size limit and then diverge, so a large position costs more per unit to exit than a small one. This is why position-size limits reduce liquidity trading risk.

What else makes an exit expensive

- **Predatory trading.** If others guess that Company X must unwind a large position, they trade ahead of it (short the stock before X sells), so X exits at worse prices. Firms keep positions and trading plans confidential. Hull cites the London Whale (2012) and LTCM, whose positions were unwound slowly under Federal Reserve supervision.
- **Crowded exits.** When many institutions hold the same position and all need to sell, the liquidity normally in the market evaporates: the liquidity black hole of LTR-1 d.
- **Transparency.** Opaque products do not stay liquid. After August 2007 nobody could value ABS CDO tranches; in July 2008 Merrill Lynch sold USD 30.6 billion of formerly AAA-rated tranches to Lone Star Funds for 22 cents on the dollar. Plain credit default swaps kept trading.

2. Measuring the spread

$$p = \text{Offer} - \text{Bid} \qquad s = \frac{\text{Offer} - \text{Bid}}{\text{Mid-market price}}, \qquad \text{Mid} = \tfrac{1}{2}(\text{Bid} + \text{Offer})$$

p is the **dollar** spread, s the **proportional** spread (a pure number). The mid-market price is the fair price. A sale is done at the bid and a purchase at the offer, each *half* a spread away from mid, so liquidating a position of mid-market value α costs

$$\boxed{\text{cost} = \frac{s\,\alpha}{2}} = \frac{p}{2} \times \text{units held}.$$

The two forms are identical because $\alpha = \text{Mid} \times \text{units}$ and $s = p/\text{Mid}$. Use whichever the stem makes easier: with bid, offer and a share count, half the dollar spread times units is one line. Long and short positions both pay: a short is closed at the offer.

3. Cost of liquidation in a normal market

$$\boxed{\text{Cost of liquidation (normal market)} = \sum_{i=1}^{n} \frac{s_i\,\alpha_i}{2}} \qquad \text{(Hull eq. 1.1)}$$

s_i is an estimate of the proportional spread of instrument i in normal conditions, usually the current quote, and α_i its mid-market value. The workbook and the recall map call this the **constant-spread** approach.

Example (Hull Example 1.1). A firm has bought 10 million shares quoted bid \$89.5, offer \$90.5, and 50 million ounces of a commodity quoted bid \$15.00, offer \$15.10.

- Mid values: shares $90 \times 10 = \$900$m; commodity $15.05 \times 50 = \$752.5$m.
- Proportional spreads: shares $1/90 = 0.01111$; commodity $0.10/15.05 = 0.006645$.
- Cost: $900 \times 0.01111/2 + 752.5 \times 0.006645/2 = 5.00 + 2.50 = \$7.5$ million.

$\Rightarrow$ \$7.5 million is the cost of walking the book out of the door on a quiet day. Prices have not moved at all: it is a cost, not a market loss.

4. Cost of liquidation in a stressed market

Replace the normal spread with a "worst case" spread. If μ_i and σ_i are the mean and standard deviation of the proportional spread,

$$\boxed{\text{Cost of liquidation (stressed market)} = \sum_{i=1}^{n} \frac{(\mu_i + \lambda\sigma_i)\,\alpha_i}{2}} \qquad \text{(Hull eq. 1.2)}$$

- **λ is a confidence parameter for the spread.** For a spread exceeded only 1% of the time, with normally distributed spreads, $\lambda = 2.326$. Spreads are not normal in practice, so λ can be set from the empirical distribution: if the 99th percentile is 3.6 standard deviations above the mean, $\lambda = 3.6$. Schweser writes the same parameter as γ.
- **The bad tail is the right tail.** A *wider* spread hurts, so λ is one-tailed and added to the mean (Figure 2). VaR looks at the left tail of P/L; the stressed spread looks at the right tail of the spread distribution.
- **Moments must be proportional.** If the stem gives the mean and standard deviation of the *dollar* spread, divide each by the mid price first (or, equivalently, work in dollars per unit and multiply by units).
- The workbook and the recall map call this the **exogenous-spread** approach, after Bangia, Diebold, Schuermann and Stroughair (1999), who proposed the stressed LVaR. These labels are not Hull's; read which construction the stem describes.

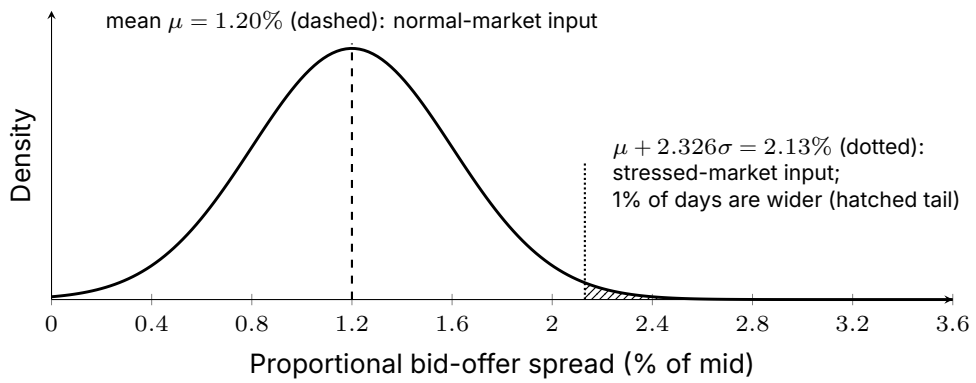


Figure 2. Distribution of a position's proportional spread with mean 1.20% and standard deviation 0.40% (the inputs of the worked LVaR example in Section 5). The stressed cost uses the right-tail value $\mu + \lambda\sigma$, not the mean and not the current quote.

Example (Hull Example 1.2). Same book. The dollar spread of the shares has mean \$1.0 and standard deviation \$2.0; for the commodity both are \$0.1. Spreads are normal; 99% confidence.

- Proportional moments: shares $\mu = 1.0/90 = 0.01111$, $\sigma = 2.0/90 = 0.02222$; commodity $\mu = \sigma = 0.1/15.05 = 0.006645$.
- Stressed spreads: shares $0.01111 + 2.326 \times 0.02222 = 0.06280$; commodity $0.006645 \times (1 + 2.326) = 0.02210$.
- Cost: $0.5 \times 900 \times 0.06280 + 0.5 \times 752.5 \times 0.02210 = 28.26 + 8.32 = \$36.58$ million.

$\Rightarrow$ Almost five times $(4.88\times)$ the normal-market cost, with no change in any price. The whole difference is spread widening, and the shares, whose spread volatility is twice its mean, take most of it (Figure 3).

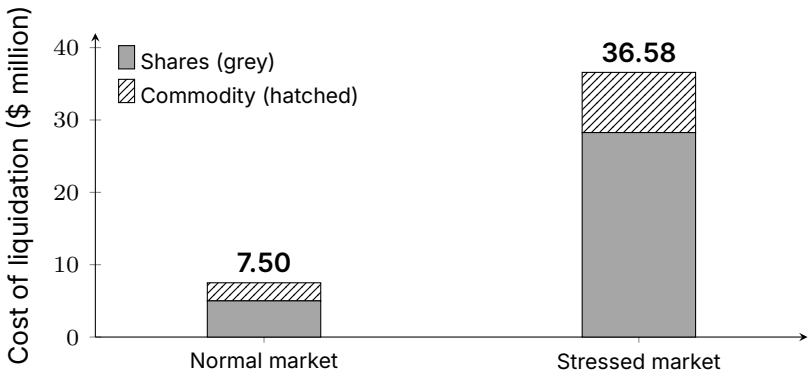


Figure 3. Hull's Examples 1.1 and 1.2: the same positions and the same prices under the two spread assumptions. Stressing the spread alone multiplies the cost by 4.88.

Properties Hull states, and GARP turns into statements

- **Perfect correlation is assumed, and it is not over-conservative.** Equation 1.2 simply adds the worst-case spreads of all instruments. Hull: this "may seem overly conservative, but it is not", because when liquidity is tight spreads widen together.
- **Diversification does not necessarily reduce liquidity trading risk.** It reduces market risk. Since s_i rises with the size of position i, many small positions carry less liquidity risk than a few large ones, and position-size limits reduce it.

- **Both costs fall as the liquidation period lengthens.** Spreads depend on how quickly you sell (Section 6).
- **Monitor both measures regularly**, and base funding analysis on the stressed one: a firm that plans to raise cash by selling its trading book (LTR-1 b) must assume stressed spreads.

5. Liquidity-adjusted VaR

VaR estimates a worst-case change in the *mark-to-market value* of the book. The cost of liquidation estimates what it costs to sell the book *if market prices do not change*. They measure different risks, so some researchers combine them:

$$\text{LVaR} = \text{VaR} + \sum_{i=1}^{n} \frac{s_i\,\alpha_i}{2} \quad \text{(Hull eq. 1.3)} \qquad \text{LVaR} = \text{VaR} + \sum_{i=1}^{n} \frac{(\mu_i + \lambda\sigma_i)\,\alpha_i}{2}$$

The first adds the normal-market cost, the second the stressed-market cost (the Bangia et al. version). The two pieces are **added in full**: no square-root combination, no averaging, no diversification credit between them.

Worked example. A USD 20 million position has daily return volatility 2.00%. Its proportional spread has mean 1.20% and standard deviation 0.40%. Use 99% ($z = \lambda = 2.326$) for both parts.

$$\begin{aligned}
\text{VaR} &= 20{,}000{,}000 \times 2.326 \times 0.02 = 930{,}400 \\
\text{Normal-market cost} &= \tfrac{1}{2} \times 20{,}000{,}000 \times 0.012 = 120{,}000 \\
\text{LVaR (normal market)} &= 930{,}400 + 120{,}000 = 1{,}050{,}400 \\
\text{Stressed cost} &= \tfrac{1}{2} \times 20{,}000{,}000 \times (0.012 + 2.326 \times 0.004) = 213{,}040 \\
\text{LVaR (stressed market)} &= 930{,}400 + 213{,}040 = 1{,}143{,}440
\end{aligned}$$

$\Rightarrow$ Liquidity adds 13% to VaR on normal spreads and 23% on stressed spreads. Combining the two as the square root of the summed squares (954,479) is wrong: there is no correlation term between the pieces.

Why the liquidity add-on matters most at short horizons

VaR grows with $\sqrt{t}$; the cost of liquidation is a one-off charge that does not. Over 10 days the VaR above becomes $930{,}400 \times \sqrt{10} = 2{,}942{,}183$, and the same stressed cost is only a 7% add-on, against 23% at one day. A one-day LVaR with even a modest spread can therefore sit well above VaR.

6. The liquidation horizon and optimal unwinding

- **Fast unwind:** large spreads, small risk that the mid-market price moves against you.
- **Slow unwind:** smaller daily spreads, larger risk of adverse mid-market moves.

Neither extreme is always right. Almgren and Chriss (2001) choose the daily trades q_i that minimise a liquidity-adjusted VaR of the whole unwind: the total half-spread cost, $\sum q_i\,p(q_i)/2$ with the dollar spread $p(q)$ rising in daily volume, plus λ times the standard deviation of the price moves on the unsold position. In this formula λ is the VaR confidence parameter, not the spread one. The answer is to trade **more than** $1/n$ **on the first day**, because the longer any part of the position is held, the greater the risk of adverse moves (Figure 4).

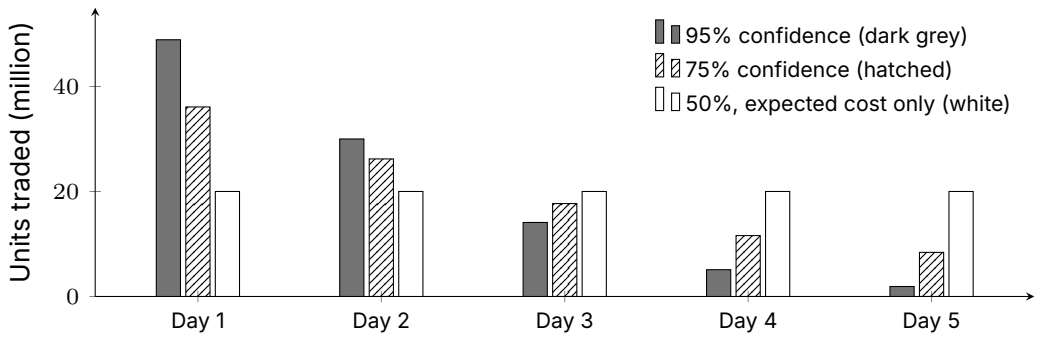


Figure 4. Hull's Example 1.3: unwinding 100 million units over five days with $p(q) = 0.1 + 0.05\,e^{0.03q}$ and a price volatility of \$0.1 a day. At 95% the trader sells 48.9 million on day 1; the more weight on price risk, the more front-loaded the schedule. At 50% only expected cost matters and 20 million a day is optimal.

For a position liquidated in equal parts over T days, Malz's adjustment VaR$\times\sqrt{(1+T)(1+2T)/(6T)}$ is the calculation of LTR-2 f (workbook A11.3 and A11.4 are worked there). Here you only need the direction: the factor is above 1 and below $\sqrt{T}$, because the position shrinks each day.

7. Other measures of market liquidity

- **Trading volume per day.** For a highly illiquid asset it is often zero.
- **Price impact** of a trade of a given size.
- **Amihud (2002) illiquidity measure:** the average over the period of $\dfrac{|\text{daily return}|}{\text{daily dollar volume}}$. Higher means less liquid. Amihud shows expected return rises as liquidity falls: investors are compensated for illiquidity.

Formula card: LTR-1 a

Spreads	$p = \text{Offer} - \text{Bid}, \quad s = p/\text{Mid}, \quad \text{Mid} = \frac{1}{2}(\text{Bid} + \text{Offer})$
Cost of one exit	$s\alpha/2 = \frac{1}{2}\,p \times$ units (long or short)
Normal-market cost	$\sum_i s_i\alpha_i/2$ (current or normal spread)
Stressed-market cost	$\sum_i(\mu_i + \lambda\sigma_i)\alpha_i/2$ (μ_i, σ_i proportional; dollar moments ÷ Mid)
LVaR	VaR+ normal cost, or VaR+ stressed cost (added in full)
Price VaR (when you build it)	$\alpha \times z \times \sigma_{\text{daily}}$, scaled by $\sqrt{t}$; the cost is not scaled
λ	99%: 2.326 95%: 1.645 one-tailed (two-tailed 2.576, 1.96 are traps); empirical value allowed
Amihud	average of $\lvert\text{return}\rvert/\text{dollar volume}$

	Normal-market cost	Stressed-market cost
Hull	eq. 1.1, LVaR eq. 1.3	eq. 1.2, LVaR from Bangia et al.
Map / workbook label	Constant spread	Exogenous spread
Spread input	Current or normal s_i	$\mu_i + \lambda\sigma_i$ of the proportional spread
Inputs you ignore	Spread mean and SD	The current quote (unless no mean is given)
Size	Smaller	Larger (4.88× in Hull's example)
Use	Day-to-day monitoring	Funding plans, stressed conditions

Exam traps

- **Sibling swap.** Using the mean and SD when the definition is normal-market, or the current quote as μ when the mean is given. Read which LVaR definition the stem uses.
- **Full spread instead of half.** The exit is half a spread from mid. Its twin is halving twice: computing the half-spread as "s" and then applying the $\frac{1}{2}$ again.
- λ **on the wrong term.** $\mu + \lambda\sigma$, not $\lambda(\mu + \sigma)$, and not $\lambda\sigma$ alone.
- **Two-tailed** z**.** Only a wider spread hurts: 1.645 at 95%, 2.326 at 99%, not 1.96 or 2.576.
- **Dollar moments on mid values.** Dividing a dollar spread by the bid or the offer, or applying dollar μ, σ to a dollar position value. Divide by the mid.
- **Intermediate result.** The cost of liquidation is not the LVaR, and a VaR alone is not the LVaR. Read what is asked.
- **Leaving out a position.** Each position's cost is added; GARP plants the single-position totals.
- **Polarity flips in theory.** Perfect spread correlation is *not* over-conservative; diversification does *not* necessarily cut liquidity risk; a longer horizon cuts spread cost but *raises* market-move risk; the liquidity add-on *increases* VaR.

Practice questions: LTR-1 a

Question LTR-1 a.1

Bank Q6 • AP Mock 3 Q72 (rebuilt)

QPR Bank holds 25 million shares of a company and 35 million ounces of a commodity. The shares are quoted bid USD 90.8 / offer USD 92.4 and the commodity bid USD 24.0 / offer USD 26.2. What is the cost of liquidation in a normal market?

A. USD 20.0 million

B. USD 38.5 million

C. USD 58.5 million

D. USD 117.0 million

Question LTR-1 a.2

Bank Q4 • AP Mock 2 Q50 (rebuilt, $z = 2.326$)

Samir Khan manages a 10,000-share position in an illiquid stock, XYZ, trading at INR 200 (the midpoint of the bid-ask spread). XYZ's daily return volatility is 2% and the average bid-ask spread is INR 0.80. Assuming normally distributed returns with zero mean, which of the following is closest to the 1-day 99% liquidity-adjusted VaR under the constant-spread approach?

A. INR 4,000

B. INR 65,800

C. INR 69,800

D. INR 97,040

Question LTR-1 a.3

Bank Q26 • AP Mock 1 Q59 (rebuilt)

A financial manager estimates a position's 1-day 95% liquidity-adjusted VaR as VaR plus the cost of liquidation at a constant proportional bid-ask spread. All figures are in return terms: the expected daily return is 0, the daily volatility is 2.21%, and the proportional bid-ask spread is 2.0%. Which of the following statements is correct?

A. The liquidity adjustment raises VaR by about 27.5%.

B. The liquidity adjustment reduces VaR by about 27.5%.

C. The liquidity adjustment raises VaR by about 55%.

D. A spread as small as 2% cannot produce a large liquidity adjustment to VaR.

Question LTR-1 a.4

Workbook A11.1 • S1, S2, S3 • Aug-2026 • Map #11

A market risk analyst at a UK asset manager is reporting liquidity-adjusted VaR, defined by the firm as regular VaR plus the cost of unwinding positions in a normal market. The one-day 99% VaR of the book is GBP 9.4 million. Positions and spread data are:

Position	Mid-market value	Current quote	Mean prop. spread	SD prop. spread
Equity block	GBP 200 million	bid 49.80 / offer 50.20	0.90%	0.60%
Corporate bond	GBP 150 million	bid 98.70 / offer 99.30	0.70%	0.40%

Which of the following is closest to the liquidity-adjusted VaR?

A. GBP 10.03 million

B. GBP 10.65 million

C. GBP 11.91 million

D. GBP 12.92 million

Question LTR-1 a.5

Bank Q22 • GARP PE2 Q29 • Mock 4 Q30

A manager at an asset management firm requests that an analyst calculate the cost of liquidation of one of the fund's stock positions. The position consists of 100,000 shares of company ABC and the stock has a current bid price of USD 53.5 and an offer price of USD 54.5. The mean and standard deviation for the stock's proportional bid-offer spread are 0.0185 and 0.0250 respectively. The analyst calculates the cost of liquidation for this entire stock position under a stressed market scenario based on a 99% confidence level. What is the correct cost of liquidation for this stock position?

A. USD 49,950

B. USD 99,900

C. USD 206,955

D. USD 413,910

Question LTR-1 a.6

Bank Q23 • Mock 2 Q47

A portfolio manager is attempting to determine the cost of liquidation of a specific holding in a stressed market. The manager holds 50,000 shares of Lear, Inc. (Lear), with bid and offer prices of USD 22.20 and USD 23.10, respectively. The standard deviation of the bid-offer spread (assumed normal distribution) is USD 1.70. The cost of liquidation of Lear at a 95% level of confidence (1.645 z-score for one-tail and 1.96 z-score for two-tail) is closest to:

A. USD 69,900

B. USD 92,400

C. USD 105,800

D. USD 184,800

Question LTR-1 a.7

Workbook A11.2 • S2, S3 • Aug-2026 • Map #11

The head of market risk at a Canadian dealer computes liquidity-adjusted VaR as regular VaR plus the cost of unwinding in a stressed market, with spreads assumed normally distributed and a 99% spread confidence level ($\lambda = 2.326$). The one-day 99% VaR is CAD 7.5 million. The book holds:

- Provincial bonds: mid-market value CAD 120 million; proportional spread mean 1.0%, standard deviation 1.2%
- 2 million shares of a mid-cap stock, currently quoted bid CAD 39.60 / offer CAD 40.40; the dollar bid-offer spread has a mean of CAD 0.16 and a standard deviation of CAD 0.20

Which of the following is closest to the stressed liquidity-adjusted VaR?

A. CAD 10.40 million

B. CAD 11.04 million

C. CAD 11.41 million

D. CAD 13.30 million

Question LTR-1 a.8 Bank Q5 • AP Mock 2 Q6 (rebuilt)

Two trading desks hold identical bond portfolios and use the same market-risk VaR. The liquidity-adjusted VaR reported by the Manchester desk is lower than that reported by the London desk. Which is the most likely explanation?

A. Manchester adds a normal-market cost of liquidation (mean spread only), while London adds a stressed-market cost using $\mu + \lambda\sigma$ of the spread.

B. Manchester adds a stressed-market cost using $\mu + \lambda\sigma$ of the spread, while London uses the mean spread only.

C. Manchester uses the full bid-offer spread as the cost of liquidation, while London uses half of it.

D. Manchester uses a higher confidence level for the spread distribution than London.

Question LTR-1 a.9 Workbook A11.5 • S4 • Aug-2026 • Map #11

The head of market risk at a Mumbai bank is reviewing a consultant's note on adding liquidity to the bank's VaR. The bank holds large corporate bond and mid-cap equity positions that would take several days to unwind. Which of the following statements in the note is correct?

A. The stressed cost of liquidation assumes spreads on all instruments are perfectly correlated, so it overstates liquidity risk and should be reduced to reflect diversification across positions.

B. Allowing more days to unwind the book lowers both the bid-offer spread cost and the exposure to adverse mid-market moves, so the longest feasible liquidation horizon should always be assumed.

C. The normal-market cost charges half of each position's current proportional spread on its mid-market value, while the stressed cost replaces that spread with its mean plus a confidence multiple of its standard deviation.

D. If the book is liquidated in equal parts over T days, its liquidity-adjusted VaR equals the one-day VaR scaled by the square root of T, because the whole position stays exposed for all T days.

Solutions • LTR-1 a

Solution LTR-1 a.1

Answer: C

Bank Q6 • AP Mock 3 Q72

Step 1 – Cost per position as half the dollar spread times units

$\frac{1}{2}\,p\times$ units is the same as $s\alpha/2$, and needs no mid values.

Step 2 – Shares

$$\tfrac{1}{2}\times(92.4-90.8)\times 25\text{m}=\tfrac{1}{2}\times 1.6\times 25\text{m}=\text{USD }20.0\text{ million}$$

Step 3 – Commodity

$$\tfrac{1}{2}\times(26.2-24.0)\times 35\text{m}=\tfrac{1}{2}\times 2.2\times 35\text{m}=\text{USD }38.5\text{ million}$$

Step 4 – Total

$$20.0+38.5=\text{USD }58.5\text{ million}$$

$\Rightarrow$ Selling both positions at the bid instead of mid costs USD 58.5 million if prices do not move.

Interpretation

In proportional terms the spreads are $1.6/91.6=1.75\%$ and $2.2/25.1=8.76\%$: the commodity is the illiquid leg and costs almost twice as much to exit, though the share position is worth more (USD 2,290 million against USD 878.5 million).

Why the other options are wrong

A is incorrect. Included only the shares: 20.0.
B is incorrect. Included only the commodity: 38.5.
D is incorrect. Charged the full spread instead of half: $2\times 58.5=117.0$.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). • **Reference:** Hull, Risk Management and Financial Institutions 5e, Ch.24, 1.1.2 Measuring Market Liquidity, eq. (1.1), Example 1.1

Solution LTR-1 a.2

Answer: D

Bank Q4 • AP Mock 2 Q50

Step 1 – Position value

$$\alpha=10{,}000\times 200=\text{INR }2{,}000{,}000$$

Step 2 – Market VaR (zero mean)

$$\text{VaR}=2{,}000{,}000\times 2.326\times 0.02=\text{INR }93{,}040$$

Step 3 – Constant-spread cost of liquidation

$$s=\frac{0.80}{200}=0.004,\qquad \tfrac{1}{2}\,s\,\alpha=\tfrac{1}{2}\times 0.004\times 2{,}000{,}000=\text{INR }4{,}000$$

Step 4 – LVaR

$$\text{LVaR}=93{,}040+4{,}000=\text{INR }97{,}040$$

$\Rightarrow$ A 1% chance of losing more than about INR 97,000 in a day once the cost of exiting is included.

Interpretation

The original mock used $z = 2.33$, giving INR 97,200; the book uses 2.326 throughout. With a 0.4% spread the liquidity charge adds only 4% to VaR. The constant-spread approach treats the average spread as known; it gives no credit for, and no charge on, spread volatility.

Why the other options are wrong

A is incorrect. The cost of liquidation alone, without the market VaR.

B is incorrect. The 95% VaR ($z = 1.645$) with no liquidity cost: $2{,}000{,}000 \times 1.645 \times 0.02 = 65{,}800$.

C is incorrect. Used the 95% z: $65{,}800 + 4{,}000 = 69{,}800$.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). • **Reference:** Hull, Ch.24, 1.1.3 Liquidity-Adjusted VaR, eq. (1.3)

Solution LTR-1 a.3

Answer: A

Bank Q26 • AP Mock 1 Q59

Step 1 – VaR in return terms

$$\text{VaR} = 1.645 \times 0.0221 = 0.036355 = 3.64\%$$

Step 2 – Constant-spread liquidity cost

$$\tfrac{1}{2} \times 0.020 = 0.010 = 1.00\%$$

Step 3 – Relative effect

$$\frac{0.010}{0.036355} = 27.5\% \qquad \text{LVaR} = 3.64\% + 1.00\% = 4.64\%$$

$\Rightarrow$ The liquidity charge raises the 1-day VaR by about 27.5%.

Interpretation

The adjustment is large because the VaR is a one-day figure: VaR scales with $\sqrt{t}$, the cost of liquidation does not, so at short horizons a 2% spread is material (Section 5).

Why the other options are wrong

B is incorrect. Wrong direction: the cost of liquidation is an additional loss, so it always raises VaR.

C is incorrect. Used the full spread: $0.020/0.036355 = 55\%$.

D is incorrect. Absolute claim, and false here: the effect depends on the spread relative to VaR, not on the spread alone.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). • **Reference:** Hull, Ch.24, 1.1.3 Liquidity-Adjusted VaR, eq. (1.3)

Solution LTR-1 a.4

Answer: B

Workbook A11.1 • S1, S2, S3 • Aug-2026

Step 1 – Proportional spreads from current quotes

$$s_1 = \frac{50.20 - 49.80}{50.00} = 0.008000, \qquad s_2 = \frac{99.30 - 98.70}{99.00} = 0.006061$$

$\Rightarrow$ A round trip in the equity costs 0.80% of mid value; in the bond 0.61%.

Step 2 – Normal-market cost of liquidation

$$\text{Cost}_{\text{normal}} = \tfrac{1}{2}(200)(0.008000) + \tfrac{1}{2}(150)(0.006061) = 0.8000 + 0.4545 = \text{GBP } 1.2545 \text{ million}$$

$\Rightarrow$ Selling everything at the bid rather than mid would cost about GBP 1.25 million if prices did not move.

Step 3 – Liquidity-adjusted VaR

$$\text{LVaR} = 9.4 + 1.2545 = \text{GBP } 10.65 \text{ million}$$

Interpretation

Liquidity adds about 13% to the market VaR in normal conditions. The spread mean and SD columns are red herrings under the firm's definition; they would only enter a stressed-market LVaR (at 95%, $\lambda = 1.645$, that would be GBP 12.31 million).

Why the other options are wrong

A is incorrect. Halved twice: took the half-spread $(\text{offer} - \text{bid})/(2\times\text{mid})$ as s and then applied the $\frac{1}{2}$ of eq. 1.1 again, $9.4 + 1.2545/2 = 10.03$.
C is incorrect. Charged the full spread instead of half: $9.4 + 2(1.2545) = 11.91$.
D is incorrect. Sibling swap: used the stressed (exogenous-spread) cost at 99% when the firm's definition is normal-market, $9.4 + \frac{1}{2}(200)(0.009 + 2.326 \times 0.006) + \frac{1}{2}(150)(0.007 + 2.326 \times 0.004) = 9.4 + 3.518 = 12.92$.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). • **Reference:** Hull, Ch.24, 1.1.2 Measuring Market Liquidity, eq. (1.1); 1.1.3 Liquidity-Adjusted VaR, eq. (1.3)

Solution LTR-1 a.5 **Answer: C**

Bank Q22 • GARP PE2 Q29

Step 1 – Mid-market value of the position

$$\text{Mid} = \tfrac{1}{2}(53.5 + 54.5) = 54.0, \qquad \alpha = 100{,}000 \times 54.0 = \text{USD } 5{,}400{,}000$$

Step 2 – 99% worst-case proportional spread

$$\mu + \lambda\sigma = 0.0185 + 2.326 \times 0.0250 = 0.07665$$

$\Rightarrow$ On the worst 1% of days the spread is 7.7% of mid, four times its mean.

Step 3 – Stressed cost of liquidation

$$\tfrac{1}{2} \times 5{,}400{,}000 \times 0.07665 = \text{USD } 206{,}955$$

Interpretation

The current quote (USD 1.0, or 1.85% of mid) happens to equal the mean spread here; under a stressed scenario it is only the starting point. The answer is the cost alone, with no VaR added.

Why the other options are wrong

A is incorrect. Normal-market cost from the mean spread: $\frac{1}{2} \times 5{,}400{,}000 \times 0.0185 = 49{,}950$.
B is incorrect. Mean spread without the half: $5{,}400{,}000 \times 0.0185 = 99{,}900$.
D is incorrect. Stressed spread without the half: $5{,}400{,}000 \times 0.07665 = 413{,}910$.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). • **Reference:** Hull, Ch.24, 1.1.2 Measuring Market Liquidity, eq. (1.2)

Solution LTR-1 a.6 — Answer: B

Bank Q23 • Mock 2 Q47

Step 1 – Spread inputs in dollars per share

The stem gives no mean spread, so the current quote stands in for it: $\mu = 23.10 - 22.20 =$ USD 0.90; $\sigma =$ USD 1.70.

Step 2 – Worst-case spread at 95%, one tail

$$\mu + 1.645\,\sigma = 0.90 + 1.645 \times 1.70 = \text{USD } 3.6965$$

$\Rightarrow$ Only a wider spread hurts, so the one-tailed 1.645 applies.

Step 3 – Stressed cost

$$\tfrac{1}{2} \times 50{,}000 \times 3.6965 = \text{USD } 92{,}412 \approx 92{,}400$$

Interpretation

Working in dollars per share avoids the mid-price conversion: $\frac{1}{2}\alpha(\mu_s + \lambda\sigma_s)$ with proportional moments gives the same USD 92,412, since $\alpha = 22.65 \times 50{,}000$ and both moments are divided by 22.65. When a mean spread *is* given, the current quote must not replace it (Question LTR-1 a.7).

Why the other options are wrong

A is incorrect. Dropped the mean spread: $\frac{1}{2} \times 50{,}000 \times 1.645 \times 1.70 = 69{,}912$.
C is incorrect. Two-tailed z: $\frac{1}{2} \times 50{,}000 \times (0.90 + 1.96 \times 1.70) = 105{,}800$.
D is incorrect. Forgot the half: $50{,}000 \times 3.6965 = 184{,}825$.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). • **Reference:** Hull, Ch.24, 1.1.2 Measuring Market Liquidity, eq. (1.2)

Solution LTR-1 a.7 — Answer: A

Workbook A11.2 • S2, S3 • Aug-2026

Step 1 – Convert the stock's dollar spread moments to proportional

$$\text{Mid} = \tfrac{1}{2}(39.60 + 40.40) = 40.00, \qquad \alpha_2 = 2 \times 40.00 = \text{CAD } 80 \text{ million}$$
$$\mu_2 = \frac{0.16}{40.00} = 0.0040, \qquad \sigma_2 = \frac{0.20}{40.00} = 0.0050$$

$\Rightarrow$ Equation 1.2 needs proportional spreads, so dollar moments are divided by the mid price, as in Hull's Example 1.2.

Step 2 – Stressed spread for each position

$$\mu_1 + \lambda\sigma_1 = 0.010 + 2.326(0.012) = 0.037912, \qquad \mu_2 + \lambda\sigma_2 = 0.004 + 2.326(0.005) = 0.015630$$

$\Rightarrow$ A 1-in-100 spread for the bonds is 3.79% of mid, almost four times its average.

Step 3 – Stressed cost of liquidation

$$\text{Cost}_{\text{stressed}} = \tfrac{1}{2}(120)(0.037912) + \tfrac{1}{2}(80)(0.015630) = 2.2747 + 0.6252 = \text{CAD } 2.8999 \text{ million}$$

Step 4 – Stressed LVaR

$$\text{LVaR} = 7.5 + 2.8999 = \text{CAD } 10.40 \text{ million}$$

Interpretation

The stressed liquidation cost is about 3.8 times the normal-market cost built from the mean spreads (CAD 0.76 million). Equation 1.2 assumes spreads move together, which Hull argues is realistic when liquidity is tight, so no diversification credit is taken. The current quote (2.0% of mid) is the red herring.

Why the other options are wrong

B is incorrect. Sibling swap: used the current quoted spread $0.80/40.00 = 2.0\%$ as μ_2, so $7.5 + 2.2747 + \frac{1}{2}(80)(0.020 + 2.326 \times 0.005) = 11.04$.
C is incorrect. Applied λ to the sum of mean and SD, $\lambda(\mu + \sigma)$: $7.5 + \frac{1}{2}(120)(2.326)(0.022) + \frac{1}{2}(80)(2.326)(0.009) = 11.41$.
D is incorrect. Charged the full stressed spread instead of half: $7.5 + 2(2.8999) = 13.30$.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). • **Reference:** Hull, Ch.24, 1.1.2 Measuring Market Liquidity, eq. (1.2), Example 1.2; 1.1.3 Liquidity-Adjusted VaR

Solution LTR-1 a.8

Answer: A

Bank Q5 • AP Mock 2 Q6

Step 1 – Write both LVaRs

$$\text{LVaR}_{\text{normal}} = \text{VaR} + \tfrac{1}{2}\sum \alpha_i \mu_i, \qquad \text{LVaR}_{\text{stressed}} = \text{VaR} + \tfrac{1}{2}\sum \alpha_i (\mu_i + \lambda \sigma_i)$$

Step 2 – Compare

VaR and holdings are identical, so the stressed figure is larger by $\frac{1}{2}\sum \alpha_i \lambda \sigma_i > 0$.
$\Rightarrow$ The desk with the lower LVaR is the one that ignores spread volatility: Manchester uses the normal-market measure.

Interpretation

Any change that widens the spread input (stressing it, raising λ, charging the full spread) raises LVaR. Only dropping the stress term lowers it.

Why the other options are wrong

B is incorrect. Reversed: a stressed cost would make Manchester's LVaR higher, not lower.
C is incorrect. Charging the full spread doubles the cost and makes Manchester higher.
D is incorrect. A higher confidence level raises λ and so the stressed cost: Manchester would be higher.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). • **Reference:** Hull, Ch.24, 1.1.2, eqs. (1.1) and (1.2); 1.1.3 Liquidity-Adjusted VaR

Solution LTR-1 a.9

Answer: C

Workbook A11.5 • S4 • Aug-2026

Step 1 – Why C is correct

Hull's eq. 1.1 charges $s_i \alpha_i / 2$ using the proportional spread in normal conditions; eq. 1.2 replaces s_i with $\mu_i + \lambda \sigma_i$, where "the parameter λ gives the required confidence level for the spread" (2.326 at 99% if normal, or an empirical multiple such as 3.6). Either cost is added to regular VaR to give an LVaR.

Interpretation

Keeping the two constructions apart is the reported sibling swap: the current quote belongs in the normal-market measure, the mean and volatility of the spread in the stressed one. For a book that takes days to exit, the bank should also price the horizon trade-off and use Malz's declining-position adjustment (LTR-2 f).

Why the other options are wrong

A is incorrect. Polarity flip: Hull says the perfect-correlation assumption "may seem overly conservative, but it is not", because spreads widen together when liquidity is tight; diversification does not necessarily reduce liquidity trading risk.

B is incorrect. Absolute claim: a longer horizon lowers the spread cost but raises the risk of adverse mid-market moves, which is why Almgren and Chriss optimise the schedule (trading more than $1/n$ on day 1) rather than always stretching it.

D is incorrect. Scope error: dropped the condition that the position shrinks each day. Holding $1, \frac{T-1}{T}, \ldots, \frac{1}{T}$ of the position gives the factor $\sqrt{(1+T)(1+2T)/(6T)}$, so $\sqrt{T}$ overstates the LVaR.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). • **Reference:** Hull, Ch.24, 1.1.2 to 1.1.4; Malz, Financial Risk Management, Ch.12, Measuring the Risk of Adverse Price Impact

LTR-2 • Liquidity and Leverage • Vardeez: NOT SO IMP

LTR-2 f

P1 CALC

Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days.

Workbook: A11 Liquidity-Adjusted VaR and Cost of Liquidation • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A11, recalled in the August 2026 sitting and reported in 6 of 16 recall sources (Map #11) as "liquidity adjusted VaR" numericals. The recalls do not say whose construction was asked: A11 covers Hull's cost of liquidation (LTR-1 a) and Malz's two measures (this LO). The two workbook questions on Malz (A11.3, A11.4) are self-generated, not recalled. **Vardeez:** rates the reading NOT SO IMP, and his one pointer for it is exactly this calculation: the expected transaction cost. The other LTR-2 LOs are P4 and are not built, so this block carries its own background.

Reported exam shapes

- **S5** Malz expected transactions cost and the 99% spread risk factor.
- **S6** Malz multi-day liquidation adjustment, with the time to escape worked out from trading volume.

What the questions look like: all numerical and short. Either a bid, an ask (or a midprice and a dollar spread) and the standard deviation of the proportional spread, with a request for the expected cost, the 99% cost or the spread risk factor, per share or for the position; or a one-day VaR and a number of days (or a position and a daily volume), with a request for the adjusted VaR or its percentage increase. The distractors are the missing $\frac{1}{2}$, the expected cost offered as the 99% cost, $\sqrt{T}$ in place of Malz's factor, and the factor without its square root.

1. Malz's view of transactions liquidity risk

Malz splits liquidity risk into **transactions liquidity risk** (assets and markets: moving the price against yourself when you trade) and **funding liquidity risk** (liabilities and creditors: lenders withdraw credit or change its terms). Hull's names for the same pair are liquidity trading risk and liquidity funding risk (LTR-1 a, Section 1). Malz lists three ways in which a lack of liquidity shows up, all observable and all hard to measure:

- **Bid-ask spread.** A constant spread would be a predictable cost of trading. The spread fluctuates, so it is a *risk*: this is **transaction cost liquidity risk** (Section 2).
- **Adverse price impact.** The effect of the trader's own activity on the equilibrium price. It is measured with a liquidity-adjusted VaR over the days needed to sell (Section 3).
- **Slippage.** The deterioration in the market price caused by the time it takes to get a trade done; in a trending market it hurts even a small order. It is a definition only (Schweser's Module Quiz 67.3 asks for the name); there is no formula for it in this LO.

Malz adds that quantitative transactions-liquidity measures are less widely used than market and credit risk models, partly because they are not built into the regulatory framework.

2. Expected transactions cost and the spread risk factor

The distributional assumption. Daily *changes* in the relative (proportional) bid-ask spread are normal with a mean of zero and a constant variance σ_s^2, estimated by the sample variance. A zero mean is "unobjectionable", since spreads can neither rise indefinitely nor shrink to zero. Because the changes have zero mean, tomorrow's expected spread is today's: $\bar{s}$ is set to the most recent observed relative spread, and the next-day midprice $\bar{P}$ to the most recent midprice.

$$\bar{s} = \frac{\text{ask} - \text{bid}}{\text{midprice}} = 2 \times \frac{\text{ask} - \text{bid}}{\text{ask} + \text{bid}}, \qquad \text{midprice} = \tfrac{1}{2}(\text{ask} + \text{bid})$$

$$\boxed{\text{Expected transactions cost} = \bar{P} \times \frac{\bar{s}}{2}}$$

$$\boxed{\text{99\% confidence interval on the transactions cost} = \pm\, \bar{P} \times \tfrac{1}{2}\,(\bar{s} + 2.33\,\sigma_s)}$$

- **The spread risk factor** is the bracket with its half: $\frac{1}{2}(\bar{s} + 2.33\sigma_s)$. It is a pure number, a fraction of the price. Times $\bar{P}$ it is the 99% transactions cost per unit; times the position value it is the 99% cost of the position. Malz calls it "the worst case, at a 99 percent confidence level, of the bid-ask spread cost of changing a position." In practice it can also be read off history: the 99th percentile of actual daily changes in the half-spread over, say, the past two years.
- **The expected cost is the half-spread**, mid to bid for a sale (mid to ask for a purchase). The $\pm$ in Malz's interval is the side of the trade: the cost is paid below the midprice on a sale and above it on a purchase, the same size either way.
- **The $\frac{1}{2}$ halves both terms.** Halving $\bar{s}$ but adding the full $2.33\sigma_s$ is the commonest slip after forgetting the half altogether.
- **2.33 is Malz's printed multiplier** for 99%, one-tailed (a wider spread is the bad side). This book uses 2.326 everywhere else; inside this formula keep 2.33.
- **Units.** Formulas are per unit of the asset; Vardeez writes P as the value of the position, which gives the position cost directly. Both are right if you are consistent. σ_s is a proportion: 20 basis points is 0.0020, not 0.20.

Example 1 (Schweser's Brieton). Ask USD 100, bid USD 99; the standard deviation of the proportional spread is 0.0002.

- Midprice 99.50; $\bar{s} = 1/99.50 = 0.010050$.
- Expected transactions cost $= 99.50 \times 0.010050/2 =$ USD 0.50 a share: half the dollar spread, as it must be.
- Spread risk factor $= \frac{1}{2}(0.010050 + 2.33 \times 0.0002) = \frac{1}{2}(0.010516) = 0.005258$.
- 99% transactions cost $= 99.50 \times 0.005258 =$ USD 0.5232 a share.

$\Rightarrow$ The spread risk adds only 2.3 cents a share, because this spread hardly moves (σ_s is 2% of $\bar{s}$). On 400,000 shares the costs are USD 200,000 expected and USD 209,273 at 99%.

Example 2 (position level). A desk holds 250,000 shares quoted bid USD 39.84, ask USD 40.16; $\sigma_s = 0.20\%$.

$$\bar{P} = 40.00, \qquad \bar{s} = \frac{0.32}{40.00} = 0.0080$$

$$\text{Position value} = 250{,}000 \times 40.00 = \text{USD } 10{,}000{,}000$$

$$\text{Expected cost} = \tfrac{1}{2} \times 0.0080 \times 10{,}000{,}000 = \text{USD } 40{,}000$$

$$\text{Spread risk factor} = \tfrac{1}{2}(0.0080 + 2.33 \times 0.0020) = \tfrac{1}{2}(0.01266) = 0.00633$$

$$99\%\text{ transactions cost} = 0.00633 \times 10{,}000{,}000 = \text{USD } 63{,}300$$

$\Rightarrow$ Spread volatility adds USD 23,300 to the cost of exiting. The planted wrong answers are USD 126,600 (no half), USD 86,600 (half on $\bar{s}$ only) and USD 56,450 (1.645 for 99%).

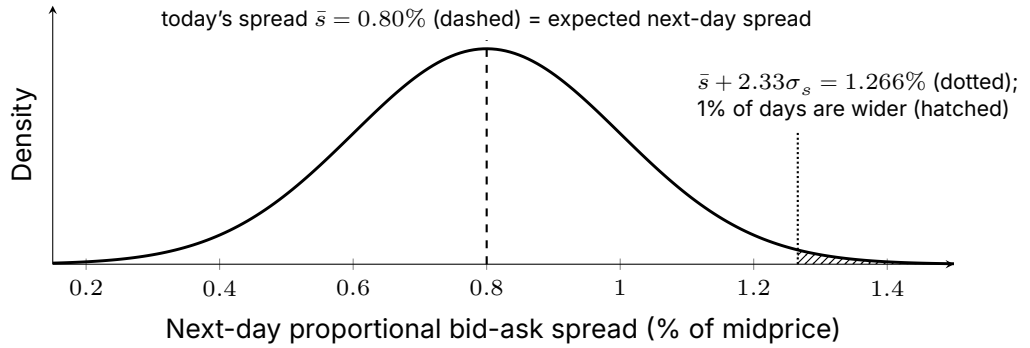


Figure 1. Malz's hypothesis for Example 2: daily changes in the spread are normal with zero mean, so the distribution is centred on today's spread, with $\sigma_s = 0.20\%$. Half of each marked spread is a cost per unit of price: 0.40% expected, 0.633% (the spread risk factor) at 99%.

Malz and Hull on the spread: same shape, different reading

Malz's 99% cost $\frac{1}{2}\bar{P}(\bar{s} + 2.33\sigma_s)$ looks like Hull's stressed cost $\frac{1}{2}\alpha(\mu + \lambda\sigma)$ (LTR-1 a, Section 4), and with μ equal to today's spread the two give almost the same number (USD 63,260 with $\lambda = 2.326$ in Example 2). The differences are in what you feed in and what you do next:

- Hull's μ and σ are the mean and standard deviation of the spread *level*, and the current quote is a different input (the normal-market cost). Malz has no separate mean: zero-mean *changes* make today's spread the expected spread.
- Hull adds the cost of liquidation to VaR (LVaR = VaR + cost). Malz's liquidity-adjusted VaR is a separate, *multiplicative* adjustment for the days needed to sell (Section 3), and Malz notes that it is large compared with the spread measures.

3. Adverse price impact: VaR for a position sold over T days

Step 1: the time to escape. Estimate T, the number of trading days needed for an *orderly* liquidation, one that does not move the price. Malz: estimates of the "time to escape" are usually based on a comparison of the position size with the daily transactions volume:

$$T = \frac{\text{position}}{\text{units the firm will sell per day}} = \frac{\text{position}}{\text{participation rate} \times \text{daily volume}}$$

Use the volume the firm will actually trade. A fund that caps itself at 25% of daily volume needs four times as long as the raw volume suggests (Question LTR-2 f.8).

Step 2: the holding pattern. Sold in equal parts at the end of each day, the position is held for one day in full, a second day on a fraction $\frac{T-1}{T}$, a third day on $\frac{T-2}{T}$, and so on down to $\frac{1}{T}$ (Figure 2).

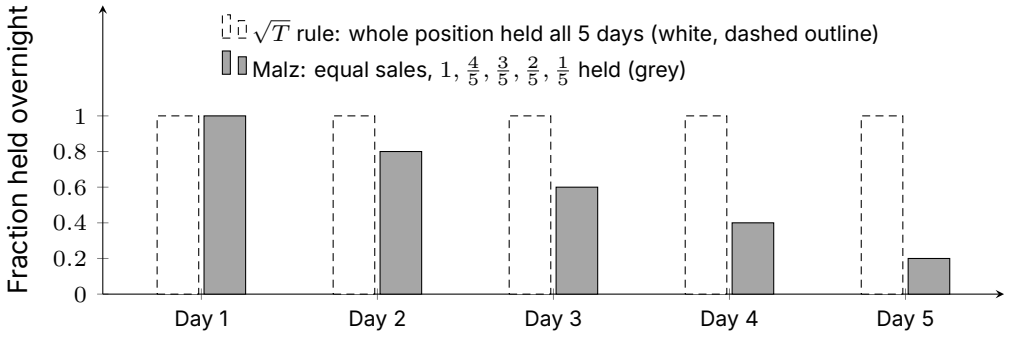


Figure 2. Liquidating over $T = 5$ days. With independent daily returns the variances add, each in proportion to the square of the amount held: $1 + 0.64 + 0.36 + 0.16 + 0.04 = 2.2$ for Malz against 5 for the $\sqrt{T}$ rule. The VaR multipliers are $\sqrt{2.2} = 1.483$ and $\sqrt{5} = 2.236$.

Step 3: the adjustment. With i.i.d. daily returns and a one-day VaR on the full position X, the variance of the total loss is $\sigma^2 X^2 \sum_{k=1}^{T} (k/T)^2 = \sigma^2 X^2 \, (1+T)(1+2T)/(6T)$, so

$$\boxed{\text{LVaR}_T = \text{VaR}_{\text{1-day}} \times \sqrt{\frac{(1+T)(1+2T)}{6T}}} \qquad 1 < \sqrt{\frac{(1+T)(1+2T)}{6T}} < \sqrt{T} \quad (T > 1).$$

The square-root-of-time rule, $\text{VaR}_{\text{1-day}} \times \sqrt{T}$, assumes the whole position is held for all T days and **overstates** the VaR. Malz: the answer "has to be greater than the one-day position VaR, but less than the one-day position VaR $\times\sqrt{T}$".

T (days)	1	2	4	5	6	8	10	20
Malz factor	1.000	1.118	1.369	1.483	1.590	1.785	1.962	2.679
$\sqrt{T}$	1.000	1.414	2.000	2.236	2.449	2.828	3.162	4.472

Malz's own numbers: $T = 5$ raises the overnight VaR by 48%; $T = 10$ roughly doubles it. Schweser's: $T = 4$ gives 1.3693, a 37% increase. Run the check $T = 1 \Rightarrow 1.000$ before trusting a factor. For large T the factor approaches $\sqrt{T/3} \approx 0.577\sqrt{T}$ (Figure 3).

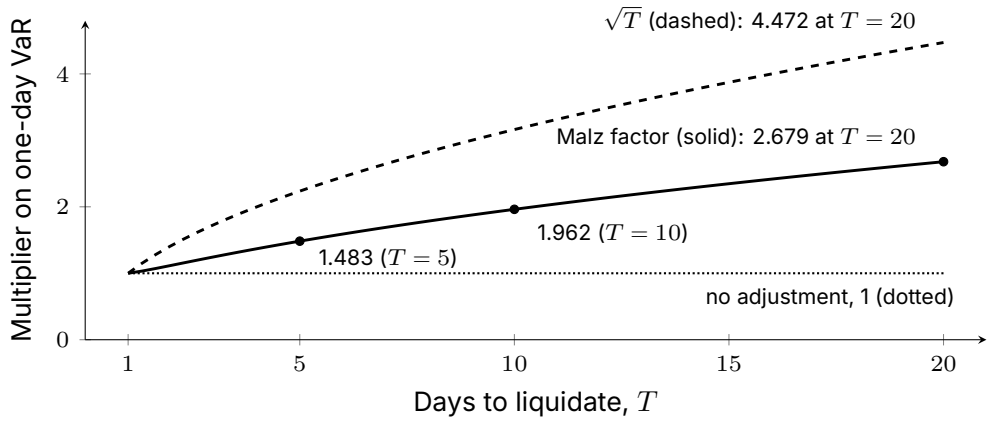


Figure 3. Malz's multiplier $\sqrt{(1+T)(1+2T)/(6T)}$ (solid) against the $\sqrt{T}$ rule (dashed). The two coincide at $T = 1$; afterwards the gap widens because the shrinking position is exposed for fewer days on average. For large T the solid curve approaches $\sqrt{T/3}$, about 58% of $\sqrt{T}$.

Example 3 (time to escape). A fund holds 1.5 million shares; the stock trades 0.75 million shares a day and the fund will sell no more than 40% of daily volume. The one-day 99% VaR of the position is USD 2.0 million.

$$T = \frac{1.5}{0.40 \times 0.75} = \frac{1.5}{0.30} = 5 \text{ days}, \qquad \sqrt{\frac{(6)(11)}{30}} = \sqrt{2.2} = 1.4832$$

$$\text{LVaR} = 2.0 \times 1.4832 = \text{USD } 2.966 \text{ million} \qquad (\text{increment USD } 0.966 \text{ million}, \ +48\%)$$

$\Rightarrow$ Ignoring the participation cap gives $T = 2$ and USD 2.236 million; the $\sqrt{T}$ rule gives USD 4.472 million. Both are offered in GARP-style questions.

Correction to the source notes. Vardeez says the multiplier "grows roughly as $\sqrt{T}/3$ for large T". The limit is $\sqrt{T/3}$: for large T, $(1+T)(1+2T)/(6T) \to 2T^2/(6T) = T/3$. At $T = 20$, $\sqrt{T/3} = 2.58$ (the factor is 2.679), while $\sqrt{T}/3 = 1.49$ is below even the $T = 5$ factor. His table values ($T = 5$: 1.483; $T = 10$: 1.962; $T = 20$: 2.679) are correct.

Formula card: LTR-2 f

Relative spread	$\bar{s} = (\text{ask} - \text{bid})/\text{mid}$, $\text{mid} = \frac{1}{2}(\text{ask} + \text{bid})$; today's spread is the expected spread
Expected transactions cost	$\frac{1}{2}\,\bar{s}\,\bar{P}$ per unit (half the dollar spread); $\times$ units for the position
Spread risk factor	$\frac{1}{2}(\bar{s} + 2.33\,\sigma_s)$, a fraction of price; the $\frac{1}{2}$ covers both terms
99% transactions cost	$\bar{P} \times \frac{1}{2}(\bar{s} + 2.33\,\sigma_s)$ per unit, $\pm$ for the side of the trade
Time to escape	$T = \text{position}/(\text{participation} \times \text{daily volume})$
Multi-day LVaR	$\text{VaR}_{\text{1-day}} \times \sqrt{(1+T)(1+2T)/(6T)}$, a factor between 1 and $\sqrt{T}$
Percentage increase	factor -1 (48% at $T = 5$; about 96% at $T = 10$)
Building the one-day VaR	$X \times z \times \sigma_{\text{annual}}/\sqrt{250}$, $z = 2.326$ at 99%

	Hull (LTR-1 a)	Malz (LTR-2 f)
Spread inputs	Current s (normal market) or $\mu + \lambda\sigma$ of the spread level (stressed)	Today's $\bar{s}$ plus $2.33\sigma_s$ of daily changes
Multiplier	$\lambda = 2.326$ at 99% if normal, or empirical	2.33 as printed
Output	Cost of liquidation of the book	Expected and 99% transactions cost, per unit
Link to VaR	LVaR = VaR + cost (additive)	LVaR = VaR $\times$ factor for T days (multiplicative)
Horizon	Exit now, prices unchanged	Equal sales over T days

Exam traps

- **The missing half.** $\bar{P}(\bar{s} + 2.33\sigma_s)$ is a round trip. Its twin is halving only $\bar{s}$: $\bar{P}(\frac{1}{2}\bar{s} + 2.33\sigma_s)$.

- **Intermediate result.** The expected cost $\frac{1}{2}\bar{s}\bar{P}$ is not the 99% cost; the spread risk factor is a fraction, not dollars; the increment is not the adjusted VaR, and a percentage increase is factor -1, not the factor.
- $\sqrt{T}$ **or no square root.** The position shrinks each day, so $\sqrt{T}$ overstates. Dropping the root on $(1+T)(1+2T)/(6T)$ overstates by even more.
- **Wrong** T**.** Divide by the volume the firm will actually trade (participation cap), not the market's whole daily volume.
- **Units and multipliers.** 20 bp is 0.0020; divide the dollar spread by the mid, not the bid or the ask; 1.645 is the 95% multiplier, not 99%.
- **Hull/Malz cross-wiring.** Adding a transactions cost to VaR in a multi-day Malz question, or scaling Hull's one-off cost of liquidation by $\sqrt{T}$.

The statement version of this LO, with the $\sqrt{T}$ claim as the distractor, is workbook A11.5, worked as Question LTR-1 a.9.

Practice questions: LTR-2 f

Question LTR-2 f.1

Bank Q107 • BT P2.T8.24.3.2 (rebuilt)

A trader notes an ask price of USD 10.50 and a bid price of USD 7.50 for a micro-cap stock. The standard deviation of the proportional spread is 8% (0.08), and daily changes in the spread are assumed normally distributed with zero mean. Using Malz's approach, what are the expected transactions cost and the 99% upper bound on the transactions cost per share?

A. Expected USD 1.50; upper bound USD 2.34

B. Expected USD 1.50; upper bound USD 3.18

C. Expected USD 3.00; upper bound USD 3.84

D. Expected USD 3.00; upper bound USD 4.68

Question LTR-2 f.2

Bank Q110 • AP Mock 1 Q49 • AP Mock 3 Q73 (rebuilt)

A risk analyst is assessing transaction cost liquidity risk on shares of Bell Aviation, quoted bid USD 32.45 and ask USD 32.90. The sample standard deviation of the proportional bid-ask spread is 0.004. Using Malz's 99% multiplier of 2.33, which of the following is closest to the 99% transactions cost per share and the 99% spread risk factor?

A. Cost USD 0.2250; spread risk factor 0.00689

B. Cost USD 0.3773; spread risk factor 0.01155

C. Cost USD 0.5295; spread risk factor 0.01621

D. Cost USD 0.7545; spread risk factor 0.02309

Question LTR-2 f.3 Bank Q108 • AP Mock 2 Q19 (rebuilt)

Delight Inc. recently traded at an ask of USD 200.00 and a bid of USD 199.00. The sample standard deviation of the proportional spread is 0.0004. Using $z = 2.33$, what is the 99% upper bound on the per-share transactions cost?

A. USD 0.5000

B. USD 0.5009

C. USD 0.5930

D. USD 1.1859

Question LTR-2 f.4 Workbook A11.3 • S5 • self-generated

A risk analyst at a Singapore hedge fund measures transactions cost liquidity risk following Malz. The fund holds 2 million shares of a stock quoted bid SGD 25.10 / ask SGD 25.30. Daily changes in the relative spread are assumed normal with zero mean and a volatility of 0.45%. The one-day 99% market VaR of the position is SGD 2.6 million. Which of the following is closest to the 99% transactions cost of liquidating the position?

A. SGD 0.200 million

B. SGD 0.387 million

C. SGD 0.464 million

D. SGD 0.728 million

Question LTR-2 f.5 Bank Q109 • BT P2.T8.24.3.1 (rebuilt)

A fund holds a thinly traded position with a one-day 99% VaR of USD 5 million. The position can be liquidated in equal parts at the end of each of $T = 10$ trading days. Using Malz's approach, what is the liquidity-adjusted VaR?

A. USD 5.00 million

B. USD 9.81 million

C. USD 15.81 million

D. USD 19.25 million

Question LTR-2 f.6 Bank Q114 • AnalystPrep Q.3883 (rebuilt, $T = 6$)

A trader estimates that a position can be liquidated in equal parts over $T = 6$ trading days. Using Malz's adjustment for a position liquidated over a number of trading days, by approximately what percentage should the one-day VaR be increased?

A. 59%

B. 145%

C. 153%

D. 159%

Question LTR-2 f.7 Bank Q115 • BT P2.T7.511.3 (rebuilt, $z = 2.326$)

An equity position has a value of USD 1,000,000 and an annual volatility of 34.0%. Assume 250 trading days per year, i.i.d. normal returns with zero mean, and a 99% one-tailed quantile of 2.326. The position requires five trading days to liquidate, in equal parts at the end of each day. Which of the following is nearest to the 99% liquidity-adjusted VaR?

A. USD 50,020

B. USD 74,190

C. USD 110,040

D. USD 111,840

Question LTR-2 f.8 Workbook A11.4 • S6 • self-generated

A portfolio risk manager at an Australian superannuation fund holds 2.4 million shares of a mid-cap stock. The position's one-day 99% VaR is AUD 4.0 million. The stock trades about 1.2 million shares a day, and the fund's execution policy caps its own selling at 25% of daily volume. The position would be liquidated in equal parts at the end of each trading day. Using Malz's approach, which of the following is closest to the liquidity-adjusted VaR of the position?

A. AUD 4.47 million

B. AUD 7.14 million

C. AUD 11.31 million

D. AUD 12.75 million

Solutions • LTR-2 f

Solution LTR-2 f.1

Answer: A

Bank Q107 • BT P2.T8.24.3.2

Step 1 – Midprice and relative spread

$$\bar{P} = \tfrac{1}{2}(10.50 + 7.50) = 9.00, \qquad \bar{s} = \frac{3.00}{9.00} = 0.3333$$

Step 2 – Expected transactions cost

$$\tfrac{1}{2} \times 0.3333 \times 9.00 = \text{USD } 1.50$$

$\Rightarrow$ Half the USD 3.00 dollar spread: a sale is done at the bid, USD 1.50 below mid.

Step 3 – 99% upper bound

$$9.00 \times \tfrac{1}{2}(0.3333 + 2.33 \times 0.08) = 9.00 \times \tfrac{1}{2}(0.5197) = \text{USD } 2.34$$

Interpretation

Spread risk adds USD 0.84 a share to the expected USD 1.50. A 33% spread with an 8% standard deviation is an extreme micro-cap case; the normal hypothesis is rough here, but the arithmetic is the exam's.

Why the other options are wrong

B is incorrect. Dropped the $\frac{1}{2}$ on the risk term: $1.50 + 9.00 \times 2.33 \times 0.08 = 3.18$.
C is incorrect. Took the full spread as the expected cost (3.00) and then added the halved risk term (0.84): 3.84.
D is incorrect. Dropped the $\frac{1}{2}$ everywhere: $9.00 \times (0.3333 + 0.1864) = 4.68$, a round-trip cost.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. • **Reference:** Malz, Financial Risk Management, Ch.12, 2.5 Liquidity Risk Measurement, Transaction Cost Liquidity Risk

Solution LTR-2 f.2

Answer: B

Bank Q110 • AP Mock 1 Q49 • AP Mock 3 Q73

Step 1 – Midprice and relative spread

$$\bar{P} = \tfrac{1}{2}(32.45 + 32.90) = 32.675, \qquad \bar{s} = \frac{0.45}{32.675} = 0.013772$$

Step 2 – Spread risk factor

$$\tfrac{1}{2}(\bar{s} + 2.33\sigma_s) = \tfrac{1}{2}(0.013772 + 2.33 \times 0.004) = \tfrac{1}{2}(0.023092) = 0.011546$$

$\Rightarrow$ On the worst 1% of days, selling costs 1.15% of the midprice.

Step 3 – 99% transactions cost per share

$$32.675 \times 0.011546 = \text{USD } 0.3773$$

Interpretation

The expected cost is USD 0.2250 (half the USD 0.45 spread); spread volatility adds about 68% to it. The factor is a fraction of price, the cost is in dollars: the question asks for both, and each wrong option pairs a wrong factor with the cost it implies.

Why the other options are wrong

A is incorrect. The expected half-spread only: factor $0.5 \times 0.013772 = 0.00689$, cost 0.2250.
C is incorrect. Halved $\bar{s}$ but not the volatility term: $0.006886 + 2.33 \times 0.004 = 0.01621$, cost 0.5295.
D is incorrect. No $\frac{1}{2}$ at all: factor 0.02309, cost 0.7545.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. • **Reference:** Malz, Ch.12, 2.5 Liquidity Risk Measurement, Transaction Cost Liquidity Risk

Solution LTR-2 f.3

Answer: C

Bank Q108 • AP Mock 2 Q19

Step 1 – Midprice and relative spread

$$\bar{P} = 199.50, \qquad \bar{s} = \frac{1.00}{199.50} = 0.0050125$$

Step 2 – Spread plus 2.33 standard deviations

$$\bar{s} + 2.33\sigma_s = 0.0050125 + 2.33 \times 0.0004 = 0.0059445$$

Step 3 – 99% cost per share

$$199.50 \times \tfrac{1}{2} \times 0.0059445 = \text{USD } 0.5930$$

Interpretation

The spread risk factor is $\frac{1}{2} \times 0.0059445 = 0.00297$ (the same Delight data appear as bank Q113, which asks for it). Spread volatility adds about 9 cents a share to the 50-cent half-spread.

Why the other options are wrong

A is incorrect. The expected half-spread only, $199.50 \times 0.0050125/2 = 0.5000$.
B is incorrect. Added the proportional term $2.33 \times 0.0004 = 0.000932$ to the dollar half-spread without scaling by price or halving: 0.5009.
D is incorrect. Forgot the $\frac{1}{2}$: $199.50 \times 0.0059445 = 1.1859$.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. • **Reference:** Malz, Ch.12, 2.5 Liquidity Risk Measurement, Transaction Cost Liquidity Risk

Solution LTR-2 f.4

Answer: C

Workbook A11.3 • S5 • self-generated

Step 1 – Midprice, position value and relative spread

$$\bar{P} = \tfrac{1}{2}(25.10 + 25.30) = 25.20, \qquad \text{Value} = 2{,}000{,}000 \times 25.20 = \text{SGD } 50.4 \text{ million}$$

$$\bar{s} = \frac{25.30 - 25.10}{25.20} = 0.0079365$$

Step 2 – Expected transactions cost (intermediate)

$$\tfrac{1}{2}\,\bar{s} \times \text{Value} = \tfrac{1}{2}(0.0079365)(50.4) = \text{SGD } 0.200 \text{ million}$$

$\Rightarrow$ This is the typical mid-to-bid cost of selling out on an ordinary day.

Step 3 – 99% spread risk factor

$$\tfrac{1}{2}(\bar{s} + 2.33\sigma_s) = \tfrac{1}{2}(0.0079365 + 2.33 \times 0.0045) = \tfrac{1}{2}(0.0184215) = 0.0092108$$

Step 4 – 99% transactions cost

$$0.0092108 \times 50.4 = \text{SGD } 0.464 \text{ million}$$

Interpretation

At 99% confidence the bid-ask cost of changing the position is SGD 0.46 million, more than twice the expected cost. The market VaR (SGD 2.6 million) is a red herring for this measure; it would matter only for the multi-day adverse-price-impact adjustment.

Why the other options are wrong

A is incorrect. Intermediate result: stopped at the expected transactions cost $\frac{1}{2}\bar{s}\bar{P}$, 0.200.
B is incorrect. Wrong-input twin: used the 95% multiplier 1.645 instead of 2.33, $50.4 \times \frac{1}{2}(0.0079365 + 1.645 \times 0.0045) = 0.387$.
D is incorrect. Halved only $\bar{s}$ and not the volatility term: $50.4 \times (0.0039683 + 2.33 \times 0.0045) = 0.728$.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. • **Reference:** Malz, Financial Risk Management, Ch.12, 2.5 Liquidity Risk Measurement, Transaction Cost Liquidity Risk

Solution LTR-2 f.5 **Answer: B**

Bank Q109 • BT P2.T8.24.3.1

Step 1 – Adjustment factor for $T = 10$

$$\sqrt{\frac{(1+10)(1+20)}{60}} = \sqrt{\frac{231}{60}} = \sqrt{3.85} = 1.9621$$

Step 2 – Liquidity-adjusted VaR

$$5 \times 1.9621 = \text{USD } 9.81 \text{ million}$$

$\Rightarrow$ A 96% increase: Malz's "doubles the overnight VaR" at $T = 10$.

Interpretation

Because the position is sold down gradually, each extra day exposes only part of it, so the factor lies above 1 and well below $\sqrt{10} = 3.162$.

Why the other options are wrong

A is incorrect. No liquidity adjustment: treats the one-day VaR as enough for a ten-day exit.
C is incorrect. Square-root-of-time rule, $5\sqrt{10} = 15.81$: as if the whole position were held all ten days.
D is incorrect. Forgot the square root: $5 \times 3.85 = 19.25$.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. • **Reference:** Malz, Ch.12, 2.5 Liquidity Risk Measurement, Measuring the Risk of Adverse Price Impact

Solution LTR-2 f.6

Answer: A

Bank Q114 • AnalystPrep Q.3883 (rebuilt, $T = 6$)

Step 1 – Adjustment factor for $T = 6$

$$\sqrt{\frac{(1+6)(1+12)}{36}} = \sqrt{\frac{91}{36}} = \sqrt{2.5278} = 1.5899$$

Step 2 – Percentage increase

$$1.5899 - 1 = 0.590 \approx 59\%$$

$\Rightarrow$ The one-day VaR should be raised by about 59%.

Interpretation

The original bank question used $T = 8$ (79%); it is rebuilt at $T = 6$ because $T = 8$ is the setting of workbook A11.4 (Question LTR-2 f.8). Read whether the stem asks for the adjusted VaR (factor), the increment, or the percentage increase (factor -1).

Why the other options are wrong

B is incorrect. Used the square-root-of-time rule: $\sqrt{6} - 1 = 1.449$, 145%.
C is incorrect. Forgot the square root: $2.5278 - 1 = 1.528$, 153%.
D is incorrect. Reported the factor 1.59 itself as the increase, 159%.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. • **Reference:** Malz, Ch.12, 2.5 Liquidity Risk Measurement, Measuring the Risk of Adverse Price Impact

Solution LTR-2 f.7

Answer: B

Bank Q115 • BT P2.T7.511.3 (rebuilt)

Step 1 – Daily volatility

$$\sigma_{\text{daily}} = \frac{0.34}{\sqrt{250}} = 0.021503$$

Step 2 – One-day 99% VaR

$$\text{VaR}_{\text{1-day}} = 1{,}000{,}000 \times 2.326 \times 0.021503 = \text{USD } 50{,}017$$

Step 3 – Factor for $T = 5$

$$\sqrt{\frac{(6)(11)}{30}} = \sqrt{2.2} = 1.48324$$

Step 4 – Liquidity-adjusted VaR

$$50{,}017 \times 1.48324 = \text{USD } 74{,}187 \approx 74{,}190$$

Interpretation

The adjustment adds about USD 24,000 (48%). The source question used 2.33, giving USD 74,315; this book uses 2.326 for a 99% VaR and keeps 2.33 only inside Malz's spread risk factor.

Why the other options are wrong

A is incorrect. The unadjusted one-day VaR, 50,017.

C is incorrect. Applied $(1+T)(1+2T)/(6T) = 2.2$ without the square root: $2.2 \times 50{,}017 = 110{,}038$.
D is incorrect. Square-root-of-time rule: $\sqrt{5} \times 50{,}017 = 111{,}842$, as if the whole position were held five days.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. • **Reference:** Malz, Ch.12, 2.5 Liquidity Risk Measurement, Measuring the Risk of Adverse Price Impact

Solution LTR-2 f.8

Answer: B

Workbook A11.4 • S6 • self-generated

Step 1 – Time to escape

$$T = \frac{\text{position}}{\text{shares sold per day}} = \frac{2.4}{0.25 \times 1.2} = \frac{2.4}{0.30} = 8 \text{ days}$$

$\Rightarrow$ Malz: the time to liquidate is usually based on comparing position size with daily transactions volume; here the volume the fund will actually trade is 0.30 million shares a day.

Step 2 – Liquidity adjustment factor

$$\sqrt{\frac{(1+T)(1+2T)}{6T}} = \sqrt{\frac{(9)(17)}{48}} = \sqrt{3.1875} = 1.7854$$

$\Rightarrow$ Holding a declining fraction $1, \frac{7}{8}, \ldots, \frac{1}{8}$ raises VaR by 78.5%, far less than $\sqrt{8} = 2.83$.

Step 3 – Liquidity-adjusted VaR

$$\text{LVaR} = 4.0 \times 1.7854 = \text{AUD } 7.14 \text{ million}$$

Interpretation

The fund should hold risk capital or set limits on the basis of AUD 7.14 million. The figure lies between the one-day VaR (4.0) and the square-root-of-time VaR (11.31), as Malz requires; the increment alone is AUD 3.14 million.

Why the other options are wrong

A is incorrect. Ignored the 25% participation cap: $T = 2.4/1.2 = 2$, factor $\sqrt{3 \times 5/12} = 1.1180$, $4.0 \times 1.1180 = 4.47$.
C is incorrect. Scope error: scaled by $\sqrt{T}$ as if the whole position were held for 8 days, $4.0 \times \sqrt{8} = 11.31$.
D is incorrect. Dropped the square root on the adjustment factor: $4.0 \times 3.1875 = 12.75$.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. LTR-2 e: Describe and compare methods to measure and manage funding liquidity risk and transactions liquidity risk. • **Reference:** Malz, Ch.12, 2.5 Liquidity Risk Measurement, Measuring the Risk of Adverse Price Impact

LTR-4 • The Investment Function in Financial-Services Management • Vardeez: VERY IMPORTANT

LTR-4 a

P1 CONCEPT

Compare various money market and capital market instruments and discuss their advantages and disadvantages.

Workbook: A30 Asset vs Borrowed Liquidity Management and the Investment Function • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A30. The section was recalled in the August 2026 sitting for its liquidity-strategy half ("asset liquidity management and borrowed liquidity management", LTR-5 b). Map #63 (1 of 16 sources) lists "investment instruments available to banks, the factors affecting the choice of securities, tax-exposure considerations, and reserve-management strategy", rated high by the provider but not yet confirmed by candidate reports.
Vardeez: VERY IMPORTANT reading; his pointer for it is "Money market investments".

Reported exam shapes

- **A30 S3** Investment instruments available to banks and their advantages and disadvantages (workbook A30.2, which also carries the S4 tax and risk factors of LTR-4 b).

What the questions look like: nothing to compute. Either a "which statement is correct" list in which each wrong option pins an advantage or a disadvantage on the wrong instrument, or a short investor profile (horizon, safety, tax rate, counterparty limits, yield appetite) followed by a choice of four instruments. The discriminators are backing or insurance, tax status, resale market and maturity. The bank's Treasury bill discount-yield calculations test a skill this compare-and-discuss LO does not name, so they are left out.

1. Why banks hold securities at all

A bank's first job is lending, but not every dollar can go into loans. Loans are illiquid, they carry the highest default rates of any asset, they are concentrated in the bank's own market area, and their income is fully taxable. Investment securities fill those gaps, and they typically make up between a fifth and a third of a bank's assets. Rose and Hudgins list what the portfolio does:

- **stabilises income** over the business cycle (securities income can rise when loan income falls);
- **offsets credit risk** in the loan book with high-quality securities, and gives **geographic diversification**;
- is a **backup source of liquidity**: securities can be sold, or used as collateral to borrow;
- **reduces tax exposure**, mainly through tax-exempt municipals that offset taxable loan revenue;
- serves as **pledged collateral** behind federal, state and local government deposits;
- helps **hedge interest rate risk**, adds **flexibility** (securities trade quickly, loans do not), and **dresses up the balance sheet**, because high-quality securities make the bank look stronger.

Regulators expect a written investment policy that sets the acceptable default risk, the desired maturity range and marketability, the goals of the portfolio and the degree of diversification.

The crossroads account. Investments sit between cash, loans and deposits (Figure 1). When cash is high, the surplus goes into securities; when cash is low, securities are sold. When loan demand is strong, securities are sold to fund loans; when it is weak, securities rise to keep assets earning. When

deposits grow slowly, securities serve as collateral for nondeposit borrowing. No other account sits at such a busy intersection.

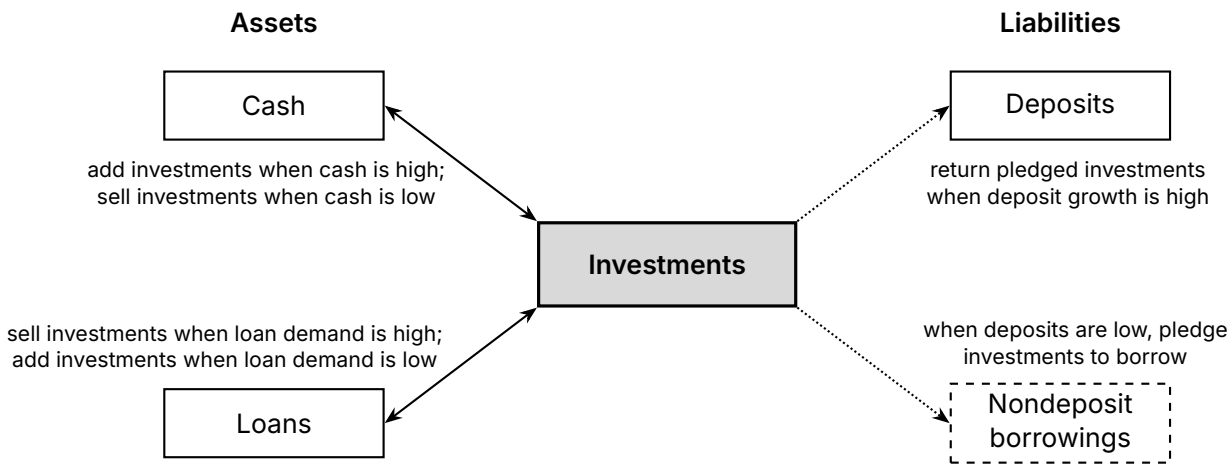


Figure 1. Investments as the crossroads account, after Rose and Hudgins' Exhibit 4.1. Solid two-way arrows: funds move into securities when cash is plentiful or loan demand is weak, and out of them when cash is short or loan demand is strong. Dotted arrows: securities pledged as collateral for nondeposit borrowing (dashed box) when deposits fall short, and released when deposits grow.

2. The dividing line: money market versus capital market

- **Money market instruments** reach maturity within one year. They are noted for **low risk and ready marketability**, and their yields are low.
- **Capital market instruments** have *remaining* maturities beyond one year. They are noted for a **higher expected rate of return and capital gains potential**, and carry more price risk.

The test is remaining maturity, not original maturity. A 10-year Treasury note with eight months to run is a money market instrument. Schweser lists each group in roughly ascending order of yield and risk.

3. Money market instruments

- **Treasury bills.** Direct obligations of the US government that must mature within one year (original maturities of 3, 6 and 12 months), sold at weekly and monthly auctions at a discount and redeemed at par; the return is pure price appreciation, quoted by the bank discount method. Backed by the federal taxing power, stable in price, readily marketable, good collateral and pledgeable behind government deposits. *Against:* the lowest yields of the group, and taxable income. Government backing is not deposit insurance.
- **Short-term Treasury notes and bonds** (within one year of maturity). Same safety and resale market as bills, usually a higher yield and some capital gains potential. *Against:* more price risk than bills; taxable gains and income.
- **Federal agency securities** (Fannie Mae, the Farm Credit System, the Federal Land Banks, Freddie Mac). Mostly *not* formally guaranteed; the support is **implied** (Congress is expected to rescue an agency in trouble), which keeps yields close to, but above, Treasury yields and keeps them liquid. *Against:* less marketable than Treasuries; interest federally taxable and usually subject to state and local tax too.

- **Certificates of deposit.** Interest-bearing receipts for a fixed-term deposit, with a penalty for early withdrawal; consumer CDs from $500 to $100,000, business "jumbo" or negotiable CDs above $100,000. Rose's table says "insured to at least $100,000"; the text notes that only the first $250,000, the current limit, is federally insured. Yields above T-bills; dealers make a secondary market in $100,000-plus CDs maturing within six months. *Against:* limited resale market for longer CDs; taxable income.
- **International Eurocurrency deposits.** Fixed-maturity time deposits (mostly 30, 60 or 90 days) issued in million-dollar units by the world's largest banks, centred on London. Low risk but **not insured**; because of their perceived higher credit risk they yield **slightly more** than domestic time deposits of comparable-size US banks. *Against:* volatile interest rates; taxable income.
- **Bankers' acceptances.** A bank's promise to pay a set amount on a set date, created when it guarantees a customer's import, export, storage or currency purchase; the bank becomes the primary obligor for a fee. Among the **safest** money market instruments (multiple credit guarantees), sold at a discount, with a resale market, and yields close to those on Eurocurrency deposits and T-bills. Eligible acceptances (in US dollars, normally six months or less, arising from trade or the storage of marketable commodities) can be discounted at the Federal Reserve. *Against:* limited availability at specific maturities, odd denominations, taxable income.
- **Commercial paper.** Short-term unsecured IOUs of major corporations with the highest credit ratings; the bulk matures in 90 days or less in the US, mostly sold at a discount. Low risk, safer than many loans. *Against:* a volatile market and a poor resale market in the US (Europaper has longer maturities, higher rates and a more active resale market); taxable income.
- **Short-term municipal obligations.** Tax-anticipation notes (TANs, repaid from future taxes) and revenue-anticipation notes (RANs, repaid from project revenues such as a toll bridge). **Interest is exempt from federal income tax**, so they suit high-bracket investors. *Against:* limited resale market, taxable capital gains, and credit quality weakened by the fiscal problems of some states and cities after 2007 to 2009.

4. Capital market instruments

- **Treasury notes and bonds.** Notes have original maturities of 1 to 10 years, bonds more than 10 years (denominations $1,000 to $1 million). Safe, good resale market, good collateral, pledgeable behind government deposits, higher expected returns than bills. *Against:* low yields relative to long-term private securities, more price and liquidity risk than bills (T-bonds trade in a thinner market with wider swings), taxable gains and income, limited supply of the longest issues.
- **Municipal notes and bonds. General obligation (GO)** bonds are backed by the full faith and credit of the issuer and can be paid from any revenue, including new taxes; **revenue bonds** are payable only from stipulated sources, such as a toll road. Interest is tax-exempt if the issue funds public, not private, projects; credit quality is generally high. *Against:* **capital gains are fully taxable** (except the gain from a discounted purchase price up to par, which counts as tax-exempt interest), the market is volatile and often illiquid (few issues trade on any given day). Banks have long underwritten GO bonds and gained the power to underwrite revenue bonds directly in 1999 (Gramm-Leach-Bliley).
- **Corporate notes and bonds.** Notes mature within five years, bonds later; secured by mortgages or unsecured debentures. Higher pretax yields than government securities, and a way to lock in high long-term rates. *Against:* limited resale market, inflexible terms, taxable gains and income; the spread over governments widens when investors worry about credit and recessions. They are more attractive to **insurance companies and pension funds** than to banks.

- **Asset-backed securities.** Higher pretax yields than Treasuries and usable as collateral for borrowing. *Against:* less marketable and less stable in price than Treasuries, possible substantial default risk (as 2007 to 2009 showed), prepayment risk, taxable gains and income.

5. Newer instruments

- **Structured notes** (1990s): dealer packages of federal agency securities whose coupon resets with a reference rate (a Treasury bond rate), sometimes with a floor and a cap, step-up coupons or formula coupons. Their complexity caused substantial losses when rates moved.
- **Securitized assets**, backed by pools of loans of uniform type and quality. **Pass-throughs** move the mortgages off the lender's balance sheet to a trustee and pass principal and interest through to investors (Ginnie Mae guarantees for a small fee; Fannie Mae securitises conventional and insured loans). **CMOs** (Freddie Mac, 1983) split a pass-through into tranches with different coupons and risk; **REMICs** also split cash flows into maturity classes to cut cash-flow uncertainty. **Mortgage-backed bonds** are the exception: the mortgages *stay* on the issuer's balance sheet, segregated and pledged, watched by a trustee. Their appeal: guarantees, higher yields than governments, the shortage of other good-quality assets, and better liquidity than the loans themselves.
- **Stripped securities**: dealers separate a Treasury or mortgage-backed security into **PO** (principal-only) and **IO** (interest-only) claims. Both are zero-coupon, sold at a discount, with *zero reinvestment risk*. **POs are more price-sensitive** to interest rates than the underlying bond; **IOs are less**. They offer interest-rate hedging possibilities.

What banks actually hold. Mostly US Treasury and agency-backed debt, with mortgage-backed securities more than half of bank investment holdings; then municipals and private asset-backed securities. Smaller banks hold a larger share of assets in securities and favour low-risk government paper to balance their riskier loans; the largest banks accept more risk (foreign securities, private debt and equity) for yield.

Instrument	Backing, insurance, safety	Tax status	Resale market and yield
T-bills	Full faith and credit; not insured	Taxable	Most active; lowest yield
Short T-notes, bonds	Full faith and credit	Taxable	Good; above bills, more price risk
Agency securities	Implied support only	Taxable (federal and usually local)	Good to average; above Treasuries
CDs	Deposit-insured up to the limit	Taxable	Limited for longer CDs; above bills
Eurocurrency deposits	Not insured; largest banks	Taxable	Slightly above domestic CDs; volatile rates
Bankers' acceptances	Bank is primary obligor; among the safest	Taxable	Resale market; odd sizes, limited maturities
Commercial paper	Top-rated corporate, unsecured	Taxable	Poor US resale market
Short municipals	Issuer credit	Interest exempt; gains taxed	Limited resale
Municipal bonds	GO: full faith and credit; revenue: project	Interest exempt; gains taxed	Often illiquid
Corporate notes, bonds	Issuer credit	Taxable	Limited resale; higher pretax yield
Asset-backed	Agency or private guarantees	Taxable	Less marketable than Treasuries; prepayment risk

Exam traps

- **Municipals are half-exempt.** Interest is tax-exempt; capital gains are taxable. "Fully tax-exempt" is the standard distractor.
- **Backing is not insurance.** T-bills are backed by the taxing power but are not insured; CDs are insured; Eurocurrency deposits are neither insured nor government-backed, and so yield *more* than domestic CDs.
- **Agencies: implied, not explicit.** Higher yields than Treasuries and less marketable; both directions get flipped. A T-note's guarantee is explicit.
- **Bankers' acceptances.** The disadvantage is limited availability at specific maturities and odd denominations, not default risk (the bank is the primary obligor) and not the lack of a resale market (there is one).
- **Commercial paper.** Poor resale market and 90 days or less: it cannot fit a six-month hold-to-maturity need, and it is not exposure to banks.
- **Remaining maturity decides the class**; corporate notes are within five years, bonds beyond; corporates suit insurers and pension funds more than banks.
- **Strips.** POs more rate-sensitive, IOs less; mortgage-backed bonds stay on the balance sheet, pass-throughs do not.

Practice questions: LTR-4 a

Question LTR-4 a.1

Bank Q161 • Schweser 69.1-1

An investments officer is listing the merits of Treasury bills for the bank's asset-liability committee. Which of the following characteristics is *not* an advantage of Treasury bills?

A. They have stable market prices.

B. They can serve as collateral for borrowers.

C. They have higher yields than agency securities.

D. They are backed by the taxing power of the federal government.

Question LTR-4 a.2

Bank Q157 • AnalystPrep Q.5425

A member of a US bank's credit committee asks which of the bank's short-term holdings rely on an *implied* rather than an explicit promise of support from the US government. Which of the following securities is considered to carry an implicit guarantee of the US government?

A. Commercial paper

B. Federal agency securities issued by government-sponsored enterprises

C. International Eurocurrency deposits

D. Short-term Treasury notes

Question LTR-4 a.3

Bank Q156 • AnalystPrep Q.5407, Q.5424 (key as corrected in the bank)

Joshua is a portfolio manager considering bankers' acceptances for his firm's short-term portfolio. He knows they are short-term trade-finance instruments in which a bank guarantees a customer's payment, that they offer yields close to those on T-bills and Eurocurrency deposits, and that eligible acceptances can be discounted at the Federal Reserve. Which of the following is the most valid disadvantage of bankers' acceptances?

A. The risk that the bank will default on its guarantee of the acceptance.

B. The limited availability of bankers' acceptances at certain maturities.

C. The possibility that bankers' acceptances could be subject to negative interest rates.

D. The inability to trade bankers' acceptances in the secondary market.

Question LTR-4 a.4

New • built from GARP Ch.4, Sections 4.3 and 4.4

The investments officer of a US savings bank in the top federal tax bracket is preparing a training note on the bank's municipal and corporate holdings. Which of the following statements in the note is correct?

A. Interest on municipal bonds issued for public projects is exempt from federal income tax, but capital gains on municipals are taxable, apart from the accretion to par on bonds bought at a discount.

B. Municipal bonds are fully tax-exempt, on interest and on capital gains alike, which is why high-bracket banks prefer them to Treasury bonds.

C. Revenue bonds are backed by the full faith and credit of the issuing government and may be repaid from any source of revenue, including new taxes.

D. Corporate notes and bonds appeal more to banks than to insurance companies and pension funds, because their resale market is deeper than that of government securities.

Question LTR-4 a.5

New • built from GARP Ch.4, Section 4.5

A risk analyst reviewing a bank's securities book finds structured notes, collateralized mortgage obligations, mortgage-backed bonds and Treasury principal-only and interest-only strips. Which of the following statements about these instruments is correct?

A. Principal-only strips are more price-sensitive to interest-rate changes than the underlying bonds, while interest-only strips are less price-sensitive.

B. Mortgage-backed bonds remove the pledged mortgages from the issuer's balance sheet, in the same way as pass-through securities.

C. Both principal-only and interest-only strips pay periodic coupons, so they carry the same reinvestment risk as the underlying Treasury bond.

D. Structured notes are plain fixed-rate agency notes, which is why investing institutions suffered few losses on them when rates moved in the 1990s.

Question LTR-4 a.6

Bank Q147 • Mock 2 Q10 (rebuilt)

An investor wants to place USD 500,000 for nine months. He requires absolute safety for the entire amount, does not want the lowest possible yield, and wants the potential to earn moderate capital gains over his horizon. Based solely on these objectives, which investment is most suitable?

A. Certificate of deposit

B. Treasury bill

C. Commercial paper

D. Treasury note

Question LTR-4 a.7

Bank Q160 • Mock 1 Q65 (rebuilt)

An institutional investor has USD 50 million of cash to invest until a known USD payment date six months away and plans to hold to maturity. It wants the highest return available but places top priority on safety of principal, and it wants a single instrument maturing on the six-month date. Which of the following instruments is most suitable?

A. Bankers' acceptance

B. Commercial paper

C. International Eurocurrency deposit

D. Revenue anticipation note

Question LTR-4 a.8

Bank Q159 • GARP PE2 Q25

The treasurer at a US-based bank wants to invest some excess cash on the bank's balance sheet and is comparing different types of money market instruments for the potential investment. The investment must meet the following criteria:

- The investment horizon is 3 months and the bank plans to hold the investment to maturity.
- The bank faces a very low incremental tax rate, so it is not looking for tax-exempt interest.
- The bank does not want to increase its counterparty exposure to other banks but is willing to accept exposure to government entities or industrial firms with a high credit rating.

If the bank wants to earn the highest yield while meeting the above criteria, which money market instrument should it invest in?

A. Short-term municipal obligations

B. Eurocurrency deposits

C. US Treasury bills

D. Commercial paper

Question LTR-4 a.9

Workbook A30.2 • S3, S4 • Map #63

The investments officer of a regional US bank in the 35% federal tax bracket is drafting a memo for the asset-liability committee on securities purchases for the coming year. Loan demand in the bank's home market is concentrated in agriculture. Which of the following statements in the memo is correct?

A. A tax-exempt municipal note yielding 4.55% is worth the same to the bank as a fully taxable security yielding 2.96%, so taxable Treasuries above that yield are preferable.

B. Eurocurrency deposits are insured in the same way as domestic CDs, so they usually yield slightly less than time deposits of comparable-size US banks.

C. Adding a large volume of liquid, readily marketable securities raises the average yield on earning assets, because it lowers the bank's exposure to liquidity risk.

D. Buying investment-grade municipals issued outside the bank's home region helps offset business risk in the local loan book, and banks are generally limited to Baa/BBB or better.

Solutions • LTR-4 a

Solution LTR-4 a.1 — Answer: C

Bank Q161 • Schweser 69.1-1

Step 1 – List what Rose credits T-bills with

Safety (the federal taxing power), relatively stable market prices, ready marketability, use as collateral and as pledged assets behind government deposits.

Step 2 – Test C

Agency securities carry only implied support, so they must pay *more* than Treasuries; T-bills, the safest paper, carry the lowest yields of the group.

$\Rightarrow$ "Higher yields than agency securities" is the reverse of the truth and is not an advantage.

Interpretation

Low yield is the price of safety and liquidity; Rose lists "low yields relative to other financial instruments" as the T-bill's main disadvantage.

Why the other options are wrong

A is incorrect. A genuine advantage: very short maturities keep prices stable.
B is incorrect. A genuine advantage: bills are standard collateral for borrowing.
D is incorrect. A genuine advantage: bills are direct obligations backed by the taxing power.

LO: LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.3 Popular Money Market Investment Instruments, Table 4.2

Solution LTR-4 a.2 — Answer: B

Bank Q157 • AnalystPrep Q.5425

Step 1 – Sort the four by kind of backing

Treasury notes: explicit full faith and credit. Agency securities (Fannie Mae, Freddie Mac, Farm Credit System): "not formally guaranteed by the federal government, though many believe Congress would rescue an agency in trouble". Commercial paper and Eurocurrency deposits: no government support.

$\Rightarrow$ Only the agency securities rely on an implied guarantee.

Interpretation

The implied support is why agency yields stay close to, but above, Treasury yields and why agency paper stays liquid. The 2008 conservatorship of Fannie Mae and Freddie Mac showed both sides: the support was real, and the risk had been real too.

Why the other options are wrong

A is incorrect. Unsecured corporate paper with no government support.
C is incorrect. Uninsured time deposits at large banks; no guarantee of any kind.
D is incorrect. The guarantee on a Treasury note is explicit, not implicit.

LO: LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.3, Federal Agency Securities

Solution LTR-4 a.3 — Answer: B

Bank Q156 • AnalystPrep Q.5407, Q.5424

Step 1 – Recall Rose's disadvantages for acceptances

Table 4.2: "limited availability at specific maturities; issued in odd denominations; taxable income".

Step 2 – Check each option against the instrument

Only B is on that list. A acceptance arises from a particular trade transaction, so its amount and maturity follow the transaction, not the investor's needs.

$\Rightarrow$ The binding drawback is availability, not safety or resale.

Interpretation

The question bank marks this key as corrected from its source; B matches Rose's table and Schweser ("lack of availability for certain maturities").

Why the other options are wrong

A is incorrect. The accepting bank is the primary obligor and the customer is also liable; Rose calls acceptances "among the safest of all money market instruments".

C is incorrect. Acceptances are pure discount instruments maturing at par; negative rates are not a feature Rose discusses.

D is incorrect. Acceptances do have a resale market; selling one transfers interest rate risk to another investor.

LO: LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.3, Bankers' Acceptances, Table 4.2

Solution LTR-4 a.4 — Answer: A

New • built from GARP Ch.4

Step 1 – Why A is correct

Rose: interest on most municipals "is exempt from U.S. federal income tax provided they are issued to fund public, not private, projects. Capital gains on municipals are fully taxable, however, except for bonds sold at a discounted price, where the gain from purchase price to par value is considered a portion of tax-exempt interest earnings."

Interpretation

This half-exemption is Vardeez's discriminator for the whole objective: "taxable income" appears on almost every line of Table 4.2, and the two municipal lines split the tax treatment between interest and gains.

Why the other options are wrong

B is incorrect. Absolute claim: only the interest is exempt; capital gains are taxed.

C is incorrect. Sibling swap: full faith and credit and repayment from any revenue describe general obligation bonds; revenue bonds are payable only from stipulated sources.

D is incorrect. Polarity flip: corporates are more attractive to insurers and pension funds than to banks, because of their higher credit risk and more limited resale market.

LO: LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.4 Popular Capital Market Investment Instruments

Solution LTR-4 a.5 — Answer: A

New • built from GARP Ch.4

Step 1 – Why A is correct

Rose: "POs tend to be more price sensitive to interest-rate changes than regular bonds, whereas IOs tend to be less price sensitive." A PO is a single deep-discount payment far in the future; an IO's value falls when rates rise but its cash flows also shift, damping the price response.

Interpretation

Strips, CMOs and structured notes all re-cut the cash flows of familiar securities; the exam asks which way the re-cut changes risk.

Why the other options are wrong

B is incorrect. Sibling swap: pass-throughs and CMOs take the mortgages off the balance sheet; mortgage-backed bonds keep them on it, segregated and pledged.
C is incorrect. Both strips are zero-coupon securities, so they carry *zero* reinvestment risk.
D is incorrect. Structured notes carry reset, cap, floor or step-up coupons; their complexity "has resulted in substantial losses for some investing institutions".

LO: LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.5 Investment Instruments Developed More Recently

Solution LTR-4 a.6 — Answer: D

Bank Q147 • Mock 2 Q10

Step 1 – Absolute safety on USD 500,000

Only US government debt qualifies. A CD is insured only up to the deposit insurance limit (USD 250,000), and commercial paper carries issuer credit risk.

Step 2 – Not the lowest yield

T-bills are the lowest-yielding government instruments.

Step 3 – Moderate capital gains potential

A coupon-bearing Treasury note, priced in the market, can gain if rates fall and usually yields more than bills.
$\Rightarrow$ Only the Treasury note meets all three objectives.

Interpretation

With nine months to maturity the note is a money market instrument ("short-term Treasury notes and bonds" in Table 4.2), whose advantage over bills is a higher yield and some capital gains potential, at the cost of more price risk.

Why the other options are wrong

A is incorrect. Fails safety: half of the USD 500,000 would be uninsured bank credit exposure.
B is incorrect. Safe, but the lowest-yielding choice, with return fixed by the discount.
C is incorrect. Uninsured corporate credit, so it cannot give absolute safety.

LO: LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.3, Table 4.2

Solution LTR-4 a.7 — Answer: A

Bank Q160 • Mock 1 Q65

Step 1 – Maturity fit

US commercial paper mostly matures in 90 days or less; Eurocurrency deposits run mostly 30,

60 or 90 days. Neither matches six months for a single instrument.

Step 2 – Safety and yield

An acceptance can run to six months, the accepting bank is the primary obligor (among the safest money market instruments), and its yield sits close to Eurocurrency yields, above T-bills.
⇒ The bankers' acceptance meets maturity, safety and yield at once.

Interpretation

The revenue anticipation note is the planted tax feature: tax exemption is worthless to an investor who did not ask for it, and its repayment depends on project revenues.

Why the other options are wrong

B is incorrect. Typically 90 days or less and unsecured; misses the maturity and the safety priority.
C is incorrect. Uninsured large-bank deposit with standard 30, 60 or 90-day terms.
D is incorrect. Repaid only from the revenues of a municipal project; credit quality of such notes has weakened.

LO: LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.3, Bankers' Acceptances and Commercial Paper

Solution LTR-4 a.8 **Answer: D**

Bank Q159 • GARP PE2 Q25

Step 1 – Strike out on tax

At a very low tax rate, municipal tax exemption has little value and municipals' lower pretax yield is a cost: A is out.

Step 2 – Strike out on counterparty

Eurocurrency deposits are time deposits placed with large banks: B adds bank exposure and is out.

Step 3 – Highest yield among what is left

Both T-bills (government) and commercial paper (highly rated industrial issuers, 90 days or less) fit the horizon. Commercial paper yields more than T-bills.
⇒ Commercial paper is the highest-yielding instrument that meets every criterion.

Interpretation

GARP builds these questions by listing constraints that each eliminate one option; the last criterion (highest yield) then chooses between the two survivors.

Why the other options are wrong

A is incorrect. Tax-exempt yield is valuable only to high-tax investors; here it is a lower pretax return.
B is incorrect. Yields well, but creates the bank counterparty exposure the treasurer wants to avoid.
C is incorrect. Meets the criteria but offers the lowest yield.

LO: LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.3, Table 4.2

Solution LTR-4 a.9

Answer: D

Workbook A30.2 • S3, S4

Step 1 – Why D is correct

Business risk, the risk that the economy of the market area turns down, "can be reflected quickly in the loan portfolio", so many institutions buy securities from issuers outside their principal loan market ("bank examiners often encourage out-of-market security purchases"). Credit risk is controlled by regulation: US banks "generally are allowed to buy only investment-grade securities, rated at least Baa or BBB".

Interpretation

The choice of securities weighs ten factors at once (LTR-4 b). For an agricultural lender, out-of-market investment-grade municipals address business risk and credit risk while also offering tax-exempt income.

Why the other options are wrong

A is incorrect. Polarity flip on the tax adjustment: multiplied by $(1 - t)$ instead of dividing. TEY $= 4.55\%/(1 - 0.35) = 7.00\%$, so a taxable security must yield 7.00% before tax to match the note; $4.55\% \times 0.65 = 2.96\%$ is not the equivalent yield.

B is incorrect. Sibling swap: deposit insurance belongs to domestic CDs. Eurocurrency deposits "are not insured, and due to their perceived higher credit risk, they often carry slightly higher market yields than domestic time deposits".

C is incorrect. Polarity flip: "the purchase of a large volume of liquid, readily marketable securities tends to lower the average yield" and reduce profitability; management faces a trade-off between profitability and liquidity.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. • **Reference:** Rose and Hudgins, Ch.10, 4.3 Popular Money Market Investment Instruments; 4.7 Factors Affecting Choice of Investment Securities (Equations 4.3, 4.4)

LTR-4 • The Investment Function in Financial-Services Management • Vardeez: VERY IMPORTANT

LTR-4 b

P1 BOTH

Identify and discuss various factors that affect the choice of investment securities by a bank.

Workbook: A30 Asset vs Borrowed Liquidity Management and the Investment Function • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A30; Map #63 names "the factors affecting the choice of securities, tax-exposure considerations". The LO says "identify and discuss", but GARP's practice material calculates here (tax-equivalent yield, net after-tax return on bank-qualified municipals, holding period yield), so the user tagged it BOTH and those questions are kept.
Vardeez: VERY IMPORTANT reading; no separate pointer, but he works the bank-qualified municipal calculation and flags its sign.

Reported exam shapes

- **A30 S4** Factors affecting the choice of securities, including tax exposure. Workbook A30.2 carries this shape and is worked as Question LTR-4 a.9.

What the questions look like: about half numerical: a tax-equivalent yield, the net after-tax return on a bank-qualified municipal (the tax term is *added*), a qualified versus nonqualified comparison, or a holding period yield on a bond sold early. The other half are statements and scenarios: which risk a feature addresses (call risk on falling rates, business risk offset by out-of-market securities, the BBB boundary), and which tax tool a profitable bank should use, with a rate-environment detail planted to rule out yield-curve strategies.

1. Ten factors

The securities themselves were compared in LTR-4 a. Here the investments officer decides which to buy, sell or hold. Rose and Hudgins name ten factors:

1. Expected rate of return
2. Tax exposure
3. Interest rate risk
4. Credit or default risk
5. Business risk
6. Liquidity risk
7. Call risk
8. Prepayment risk
9. Inflation risk
10. Pledging requirements

Bank size shapes how they are weighed: small banks hold more government paper to balance risky loans; large banks take more credit risk for yield and are in the top tax bracket, so tax management matters more to them.

2. Expected rate of return: YTM or HPY

For most securities (not equities) held to maturity, the expected return is the **yield to maturity**: the discount rate that equates the price with the promised coupons and principal. If the security will be sold before maturity, or has no maturity, use the **holding period yield**: the same equation with the expected sale price as the terminal cash flow.

$$P = \sum_{t=1}^{n} \frac{C}{(1+\text{YTM})^t} + \frac{\text{Par}}{(1+\text{YTM})^n} \quad \text{(eq. 4.1)} \qquad P = \sum_{t=1}^{h} \frac{C}{(1+\text{HPY})^t} + \frac{P_{\text{sale}}}{(1+\text{HPY})^h} \quad \text{(eq. 4.2)}$$

Example (GARP). A \$1,000 five-year Treasury note with an 8% annual coupon costs \$900. Solving eq. 4.1 gives YTM $= 10.68\%$. If instead the note is sold after two years for \$950, $900 = 80/(1+h) + (80+950)/(1+h)^2$ gives HPY $= 11.51\%$.

$\Rightarrow$ The HPY is higher because the \$50 discount is earned in two years, not five. The calculated return is then compared with other securities and loans.

> **Correction to the source notes.** GARP's text (Rose and Hudgins, eq. 4.1 example) states the note's YTM as 10.74%. At 10.74% the five cash flows are worth \$898.07, not \$900; the yield that prices the note at \$900 with annual coupons is **10.68%** (with semiannual coupons it would be 10.63%). The HPY of 11.51% is correct.

3. Tax exposure

Interest and capital gains are taxed as ordinary income, and banks pay a high marginal rate, so they compare securities **after tax**. Tax-exempt institutions (credit unions, mutual funds) care about the pretax yield.

$$\text{After-tax gross yield} = \text{Before-tax gross yield} \times (1 - t) \qquad \text{(eq. 4.3)}$$

$$\text{TEY} = \frac{\text{After-tax return on a tax-exempt investment}}{1 - t} \qquad \text{(eq. 4.4)}$$

Example (GARP). A bank in the 35% bracket compares an Aaa corporate bond at 7%, a prime-rated loan at 6% and an Aaa municipal at 5.5%.

- After tax: corporate $7.00 \times 0.65 = 4.55\%$; loan $6.00 \times 0.65 = 3.90\%$; municipal 5.50% (no tax).
- TEY of the municipal: $5.50/0.65 = 8.46\%$. A taxable bond or loan must yield 8.46% before tax to match it.

$\Rightarrow$ The municipal wins on yield alone; Rose adds that loans bring customer relationships, and that tax law and municipal credit quality change.

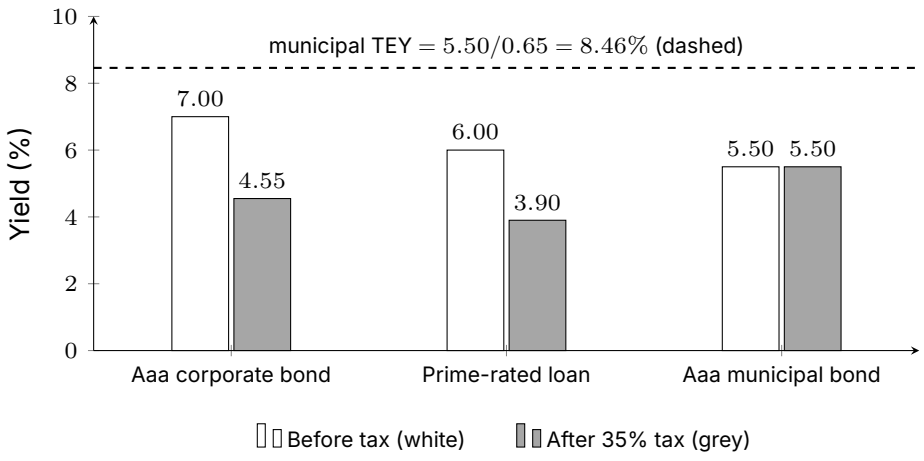


Figure 1. GARP's tax example. The municipal yields least before tax but most after tax; the dashed line is the pretax yield a taxable asset would need to match it.

Bank-qualified municipals

Before the 1986 Tax Reform Act a bank could deduct all the interest it paid on funds borrowed to buy municipals. Since then it can deduct **80%** of that interest, and only on **bank-qualified bonds**:

those issued by small local governments issuing no more than $10 million of public securities a year. Nonqualified bonds get no deduction.

$$\text{Net after-tax return} = (\text{Nominal} - \text{Interest expense}) + \underbrace{t \times 0.80 \times \text{Interest expense}}_{\text{tax advantage of a qualified bond}} \quad \text{(eqs. 4.5, 4.6)}$$

Example (GARP eq. 4.7). A bank-qualified bond yields 7.0%; the bank funds it at 6.5% and is in the 35% bracket.

$$(7.0 - 6.5) + (0.35 \times 0.80 \times 6.5) = 0.50 + 1.82 = 2.32\%$$

$\Rightarrow$ The tax shield on the funding is worth more than three times the raw carry. Had the bond been nonqualified, the net return would be the 0.50% carry alone. Vardeez's version (6% bond, 5% funding, 35%) gives $1.0 + 1.4 = 2.4\%$.

The sign. Municipal interest is already exempt, so no tax is taken off the nominal return. The only tax term is a deduction against the funding cost, so it is *added*. Equivalently, the after-tax funding cost is Interest $\times$ $(1 - 0.80t)$.

Why banks hold fewer municipals today. Declining tax advantages after 1986 (the 80% rule), lower corporate tax rates (the top bracket fell from 46% to 35%, or 34% for smaller firms), and fewer qualified tax-exempt issues (an issue is a fully taxable "private activity" bond if 10% or more of the proceeds benefit a private party; ceilings on industrial development bonds). Municipal interest also enters the alternative minimum tax.

Tax swapping and portfolio shifting

- **Tax swap.** In a year of high loan revenue, sell lower-yielding securities *at a loss* to cut current taxable income, and at the same time buy higher-yielding securities to raise future income. Most useful to large lenders in the top bracket. The limit: no bank can use unlimited tax-exempt income; some taxable income is needed, for example to absorb the allowable deduction for loan losses.
- **Portfolio shifting.** Selling selected securities at a loss to offset large loan income, or simply replacing old below-market yields with new higher yields: short-run losses for long-run profits.

Example (GARP). First National sells $10 million of 10-year New York City 7% bonds (book $10 million, market $9.5 million) and buys $10 million of 10-year Los Angeles County 9% bonds at par. Pretax loss $500,000; after 35% tax only $500{,}000 \times 0.65 = \$325{,}000$; in exchange, $200,000 a year more tax-exempt income for ten years. A bond sold at a *gain* does the opposite: it adds to this year's taxable income.

4. The risk factors

- **Interest rate risk.** Rising rates lower the value of existing bonds, the longest most. They often coincide with surging loan demand, so securities must be sold at a loss just when prices fall; when rates and loan demand fall, officers buy at high prices. Hedges: futures, options, swaps, gap management, duration.
- **Credit or default risk.** Regulation bars speculative securities: US banks generally may buy only **investment grade**, rated at least **Baa (Moody's) or BBB** (S&P, Fitch); unrated securities only if the bank can show they are of investment-grade quality. Moody's adds modifiers 1 (top of a grade), 2 and 3 (bottom). Credit options and swaps protect expected yields.
- **Business risk.** The local economy turns down and loans go delinquent. The offset is to buy securities from issuers **outside** the principal loan market (a Dallas bank buying Los Angeles or New York municipals); examiners encourage out-of-market purchases.

- **Liquidity risk.** The bank may have to sell before maturity. Liquid securities have a ready market, a stable price and a high probability of recovering principal. Treasuries are the most liquid, then agencies, municipals and mortgage-backed securities. Buying many liquid securities **lowers** the average yield: a daily trade-off between profitability and liquidity.
- **Call risk.** Issuers call when rates have **fallen**, so the bank must reinvest at lower rates. Defences: callable bonds with longer call deferments, or no callable bonds. A callable bond trading at a premium is priced to its call date (yield to call), one at a discount to maturity.
- **Prepayment risk**, specific to asset-backed securities: refinancings accelerate when rates fall, and asset turnover (moves, defaults) ends loans early. Officers estimate prepayments with the **PSA model** (Figure 2). Faster prepayment is not always bad: cash returns sooner and lower rates raise the present value of remaining flows; the security loses value when lost interest and lower reinvestment rates outweigh those benefits.
- **Inflation risk.** Erosion of the purchasing power of interest and principal, and of the bank's net worth. Short-term and variable-rate securities give some protection; **TIPS** (5, 10 and 30-year, since 1997) adjust principal and coupon payments to the CPI, and the spread between ordinary Treasuries and TIPS measures expected inflation. Banks have used them little: inflation has been moderate and TIPS are less liquid.
- **Pledging requirements.** Banks cannot accept federal, state or local government deposits without posting collateral. At least the first $100,000 is covered by deposit insurance; the rest must be backed by Treasury and agency securities **valued at par**, or by municipals rated at least A valued at a **discount** (often 80% to 90% of face). Discount window borrowing and repurchase agreements also require pledged securities. Just over half of bank-held securities are pledged.

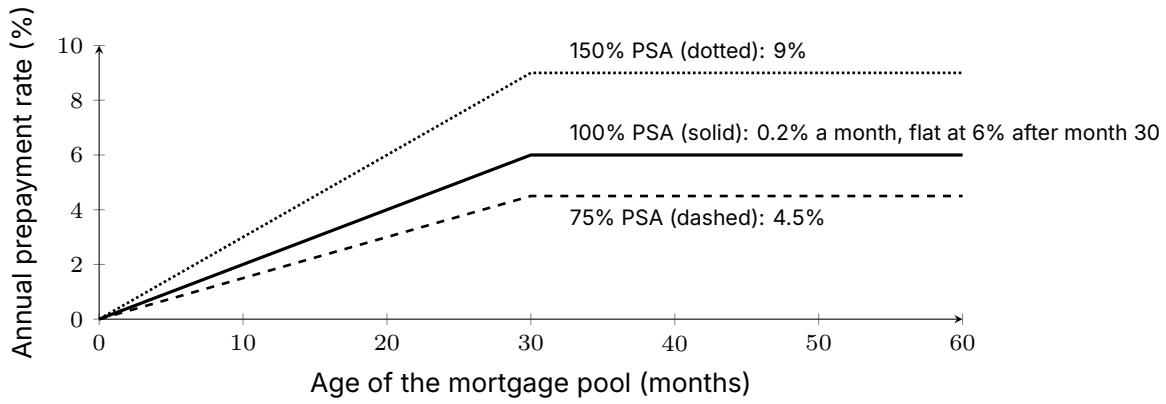


Figure 2. The PSA prepayment benchmark as Rose describes it: 0.2% (annual rate) in the first month, rising 0.2% a month for 30 months, then level at 6%. An officer who expects slower or faster prepayment than the standard scales it (75% PSA dashed, 150% PSA dotted).

Formula card: LTR-4 b

YTM (held to maturity)	$P = \sum_{t=1}^{n} C/(1+y)^t + \text{Par}/(1+y)^n$; for most securities except equities
HPY (sold before maturity)	same equation with the sale price P_{sale} at the sale date
After-tax gross yield	before-tax yield $\times(1-t)$
Tax-equivalent yield	tax-exempt yield $\div(1-t)$ (divide, never multiply)
Qualified municipal, net after tax	$(\text{nominal} - \text{interest expense}) + t \times 0.80 \times \text{interest expense}$
Nonqualified municipal	nominal $-$ interest expense (no deduction)
Bank-qualified	issuer sells $\leq$ \$10 million of public securities a year; 80% of funding interest deductible
Tax swap result	after-tax cost of a realised loss $=$ loss $\times\ (1-t)$; a realised gain adds gain $\times\ t$ of tax
Credit boundary	investment grade: Baa/BBB and above; Ba/BB and below is speculative
PSA 100%	$0.2\%\times$month for 30 months, then 6% a year

Risk	What triggers it	The usual defence or remedy
Interest rate	Rates rise; often with loan demand	Shorter maturities, hedges, duration
Credit	Issuer defaults or is downgraded	Investment grade only (Baa/BBB+); credit options, swaps
Business	Local economy weakens	Out-of-market securities
Liquidity	Forced sale before maturity	Treasuries, agencies; accept lower yield
Call	Rates *fall*, issuer refinances	Longer call deferment; avoid callables
Prepayment	Rates *fall* (refinancing), turnover	PSA-based estimates; price the option
Inflation	Prices rise	Short or variable-rate securities, TIPS
Pledging	Public deposits, discount window, RPs	Hold Treasuries, agencies, A-rated municipals

Exam traps

- **TEY direction.** Divide by $(1-t)$. Multiplying by $(1-t)$ gives a net-of-tax figure, by $(1+t)$ a wrong gross-up, dividing by t a nonsense number.
- **Qualified-municipal sign.** The tax term is *added*: $6-5+1.4=2.4\%$, not $6-5-1.4$. The 1% raw carry and the 1.4% shield on their own are the intermediate-result distractors.
- **Which part is deductible.** 80% of the interest expense, times t. Distractors use the 20% non-deductible part, 100%, or $(1-t)$ in place of t.
- **Nonqualified bonds** get no deduction at all; a higher nominal yield usually signals more credit or business risk.
- **Tax swaps need losses.** Selling at a *gain* raises this year's taxable income; swaps suit high-

profit years, and need no yield-curve slope.

- **Falling-rate risks.** Call and prepayment risk bite when rates fall. Business risk is cut by *out-of-market* buying.
- **YTM vs HPY.** HPY when sold early; the coupon rate and the current yield are both intermediate results.

Practice questions: LTR-4 b

Question LTR-4 b.1

Bank Q183 • Schweser 69.2-2

A bank's investment policy prohibits it from holding speculative-grade securities. The bank would have to sell a security when its rating changes from:

A. AAA to AA.

B. BBB to BB.

C. BB to B.

D. B to CCC.

Question LTR-4 b.2

Bank Q174 • Schweser Module Quiz 12.3 Q3 (rebuilt)

An investments officer is reviewing the call features of bonds the bank might buy. Which of the following statements about call risk on callable bonds is *false*?

A. When interest rates fall, the issuer may call the bond and the investor must reinvest the proceeds at lower rates.

B. An investor can reduce call risk by buying callable bonds with a longer call deferment period.

C. The price appreciation of a callable bond is limited when rates fall, because the issuer can repurchase it at a predetermined price.

D. Call risk is greatest when interest rates rise, because issuers then repurchase bonds at the call price.

Question LTR-4 b.3

New • built from GARP Ch.4, Section 4.7

The investments officer of a US community bank is briefing the asset-liability committee on the risks that should shape next year's securities purchases. Which of the following statements in the briefing is correct?

A. Prepayments on mortgage-backed securities accelerate when market interest rates rise, because borrowers refinance into new loans.

B. Public deposits above the insured amount must be backed by pledged Treasury and agency securities valued at par, while eligible municipals are usually valued at a discount from par.

C. TIPS protect against inflation by adjusting only the coupon to the CPI, while the principal stays fixed at par.

D. Business risk is best offset by concentrating security purchases on issuers in the bank's own market area, whose finances its lending officers know well.

Question LTR-4 b.4 Bank Q166 • GARP book Ch.18 end-of-chapter Q2 (rebuilt)

A savings institution buys a government bond for USD 1,195. The bond pays USD 75 of interest at the end of each year, has 14 years to maturity and a redemption value of USD 1,000. The institution sells the bond after five years for USD 940, just after receiving the fifth coupon. Which of the following is closest to the annual yield it earned over the five years?

A. 2.01%

B. 2.19%

C. 5.47%

D. 6.28%

Question LTR-4 b.5 Bank Q180 • BT Mock Q24 (rebuilt)

A bank investments officer compares an A-rated municipal bond with a 4.20% gross yield to maturity against taxable alternatives. The bank's marginal tax rate is 30%. What is the municipal bond's tax-equivalent yield?

A. 2.94%

B. 5.46%

C. 6.00%

D. 14.00%

Question LTR-4 b.6 Bank Q181 • Mock 1 Q68 (rebuilt)

A bank purchases a municipal bond with a 5% gross nominal rate of return. The bank's pretax borrowing cost is 3.7%, and it is subject to an income tax rate of 40%. The bond is a bank-qualified bond, but only 75% of the interest expense is tax deductible. The net after-tax return on the municipal bond is closest to:

A. 1.67%

B. 2.41%

C. 2.78%

D. 2.97%

Question LTR-4 b.7

Bank Q170, Q171 • GARP book Ch.4 end-of-chapter Q5, Q6 (rebuilt)

Forever Savings Bank is in the 35% corporate tax bracket and funds its security purchases at an average borrowing cost of 4.25%. It is choosing between two 15-year general obligation municipal issues:

Issue	Issuer's annual issuance	Nominal return	Tax status for the bank
York school district	USD 8 million (its only issue this year)	6.75%	Bank-qualified
City of Richmond	Large regular issuer	9.75%	Nonqualified

By how much does the Richmond issue's net after-tax return exceed the York issue's?

A. 1.51 percentage points

B. 1.81 percentage points

C. 3.00 percentage points

D. 4.19 percentage points

Question LTR-4 b.8

Bank Q172 • GARP book Ch.4 end-of-chapter Q7 (rebuilt)

Lakeway Thrift Savings and Trust is in the 35% tax bracket, and its loan revenue is up 16% this year. The investments officer proposes one of two moves:

(a) sell USD 4 million par of 12-year City of Dallas bonds (7.5% coupon, book value par, market value USD 3,750,000) and buy USD 4 million of 12-year Bexar County bonds at par (8% coupon);

(b) sell USD 4 million par of 12-year US Treasury bonds (12% coupon, book value par, market value USD 4,330,000).

Which recommendation is most consistent with Rose and Hudgins?

A. Choose (a): it realises a USD 250,000 loss that cuts this year's tax by USD 87,500 and adds USD 20,000 a year of coupon income.

B. Choose (b): it realises a USD 330,000 gain, which offsets the high loan income and so lowers this year's tax bill.

C. Choose (a): its USD 250,000 loss has an after-tax cost of USD 337,500, which the higher coupon recovers within the 12 years.

D. Choose (b): the Treasuries' 12% coupon is below current market yields, so selling them and reinvesting raises future income.

Question LTR-4 b.9 Bank Q184 • GARP PE1 Q11 • Mock 3 Q69

A bank treasurer is choosing the most appropriate investment strategy given the bank's balance sheet and economic conditions. The treasurer gathers the following information:

- The bank faces a high interest rate environment with a flat yield curve.
- The bank's asset mix includes a high proportion of US Treasury bonds purchased years ago when interest rates were significantly lower.
- The bank has a high proportion of revenues from loans and anticipates a record level of profitability this year.

Which strategy is most appropriate to maintain the bank's current level of total income?

A. Barbell strategy

B. Carry trade

C. Riding the yield curve

D. Tax swap

Solutions • LTR-4 b

Solution LTR-4 b.1 — Answer: B

Bank Q183 • Schweser 69.2-2

Step 1 – Locate the investment-grade boundary

Investment grade runs from AAA down to BBB (Baa at Moody's); BB (Ba) and below is speculative.

Step 2 – Find the move that crosses it

Only BBB to BB takes an eligible security into the prohibited zone.
$\Rightarrow$ A downgrade across the BBB/BB line forces the sale.

Interpretation

This is the credit-risk factor: regulators bar US banks from speculative securities "in order to protect depositors against excessive risk". Unrated securities are allowed only if the bank can show investment-grade quality.

Why the other options are wrong

A is incorrect. AAA to AA stays inside investment grade.
C is incorrect. A BB security was already ineligible; BB to B is a move within speculative grade.
D is incorrect. Same: the bond should not have been held at B.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.7, Credit or Default Risk, Table 4.4

Solution LTR-4 b.2 — Answer: D

Bank Q174 • Schweser Module Quiz 12.3 Q3

Step 1 – When do issuers call?

Calls "usually take place when market interest rates have declined (and the borrower can get lower interest costs)". An issuer has no reason to repay cheap debt when rates have risen.
$\Rightarrow$ D reverses the rate direction, so it is the false statement.

Interpretation

Call risk and prepayment risk are the same mechanism on different instruments, and both are falling-rate events. The call price caps the bond's gain exactly when rates fall, which is also why a premium callable bond is priced to its call date.

Why the other options are wrong

A is incorrect. True: this is the definition of call risk and its reinvestment cost.
B is incorrect. True: a longer deferment delays the first call date; Rose names it as the defence.
C is incorrect. True: the call price acts as a ceiling on price appreciation.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.7, Call Risk

Solution LTR-4 b.3 — Answer: B

New • built from GARP Ch.4

Step 1 – Why B is correct

Rose: "At least the first $100,000 of these public deposits is covered by federal deposit insurance; the rest must be backed up by holdings of Treasury and federal agency securities valued at par. Some municipal bonds (provided they are at least A-rated) can also be used … but these securities

must be valued at a discount from par (often 80 to 90 percent of their face value)."

Interpretation

Pledging is a reason to hold Treasuries, agencies and good municipals that has nothing to do with their yield; just over half of bank-held securities are pledged.

Why the other options are wrong

A is incorrect. Polarity flip: refinancing accelerates when rates *fall*.
C is incorrect. TIPS adjust the principal (face) value to the CPI, and the coupon payments move with it; Rose says both the coupon and the principal are adjusted.
D is incorrect. Polarity flip: business risk is offset by buying securities from issuers *outside* the bank's loan market; examiners encourage out-of-market purchases.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.7, Business Risk, Prepayment Risk, Inflation Risk, Pledging Requirements

Solution LTR-4 b.4 — Answer: B

Bank Q166 • GARP book Ch.18 Q2

Step 1 – Choose the measure

The bond is sold before maturity, so the return is the holding period yield, with the sale price as the terminal cash flow.

Step 2 – Set up and solve

$$1{,}195 = \sum_{t=1}^{5} \frac{75}{(1+h)^t} + \frac{940}{(1+h)^5} \quad \Rightarrow \quad h = 2.19\%$$

$\Rightarrow$ The USD 255 capital loss drags the return far below the 7.5% coupon rate.

Interpretation

Check: at 2.19% the five coupons are worth USD 351.6 and the sale price USD 843.4, total USD 1,195. The 14-year maturity and the USD 1,000 redemption value are red herrings for an early sale; they only matter for the YTM.

Why the other options are wrong

A is incorrect. Simple, uncompounded average: $(5 \times 75 - 255)/5/1{,}195 = 2.01\%$.
C is incorrect. The YTM to the 14-year maturity, the sibling measure: $1{,}195 = \sum_{t=1}^{14} 75/(1+y)^t + 1{,}000/(1+y)^{14}$ gives 5.47%.
D is incorrect. The current yield, $75/1{,}195 = 6.28\%$, which ignores the capital loss.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.7, Expected Rate of Return, eq. (4.2)

Solution LTR-4 b.5 — Answer: C

Bank Q180 • BT Mock Q24

Step 1 – Gross up the tax-exempt yield

$$\text{TEY} = \frac{4.20\%}{1 - 0.30} = \frac{4.20\%}{0.70} = 6.00\%$$

Step 2 – Check

$6.00\% \times (1 - 0.30) = 4.20\%$: a taxable bond yielding 6.00% leaves the bank with the municipal's 4.20% after tax.

$\Rightarrow$ Any taxable alternative must yield more than 6.00% before tax to beat the municipal.

Interpretation

The municipal's tax exemption is worth 1.80 percentage points of pretax yield to a 30% taxpayer; to a tax-exempt investor it is worth nothing (Question LTR-4 a.8).

Why the other options are wrong

A is incorrect. Multiplied by $(1-t)$: $4.20 \times 0.70 = 2.94\%$, the after-tax yield of a 4.20% *taxable* bond.
B is incorrect. Multiplied by $(1+t)$: $4.20 \times 1.30 = 5.46\%$.
D is incorrect. Divided by the tax rate: $4.20/0.30 = 14.00\%$.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.7, Tax Exposure, eq. (4.4)

Solution LTR-4 b.6 — Answer: B

Bank Q181 • Mock 1 Q68

Step 1 – Pretax carry

Municipal interest is tax-exempt, so the gross spread is $5.00\% - 3.70\% = 1.30\%$.

Step 2 – Tax shield on the deductible interest

$$0.40 \times 0.75 \times 3.70\% = 1.11\%$$

Step 3 – Net after-tax return

$$1.30\% + 1.11\% = 2.41\%$$

$\Rightarrow$ The shield nearly doubles the carry; it is added because it is a deduction on an expense.

Interpretation

The stem changes Rose's 80% deductible share to 75%: use the share given. With 80% the answer would be 2.48%.

Why the other options are wrong

A is incorrect. Used the non-deductible 25%: $1.30 + 0.40 \times 0.25 \times 3.70 = 1.67\%$.
C is incorrect. Assumed all interest is deductible: $1.30 + 0.40 \times 3.70 = 2.78\%$.
D is incorrect. Applied $(1-t) = 0.60$ instead of t: $1.30 + 0.60 \times 0.75 \times 3.70 = 2.97\%$.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.7, Bank Qualified Bonds, eqs. (4.5) to (4.7)

Solution LTR-4 b.7 — Answer: B

Bank Q170, Q171 • GARP book Ch.4 Q5, Q6

Step 1 – Which issue qualifies

York issues USD 8 million a year, under the USD 10 million limit, so 80% of the funding interest is deductible. Richmond is nonqualified: no deduction.

Step 2 – York (qualified)

$$(6.75 - 4.25) + 0.35 \times 0.80 \times 4.25 = 2.50 + 1.19 = 3.69\%$$

Step 3 – Richmond (nonqualified)

$$9.75 - 4.25 = 5.50\%$$

Step 4 – Difference

$$5.50 - 3.69 = 1.81 \text{ percentage points}$$

$\Rightarrow$ Richmond still earns more after tax, but the 3-point nominal gap shrinks to 1.81 points once York's tax shield is counted.

Interpretation

Rose's discussion question asks whether the extra 1.81 points pays for the extra risk: a much higher municipal yield usually signals more credit or business risk. The 15-year maturity is not an input.

Why the other options are wrong

A is incorrect. Gave York a 100% deduction: $2.50 + 0.35 \times 4.25 = 3.99\%$, so $5.50 - 3.99 = 1.51$.
C is incorrect. Gave Richmond the 80% deduction as well $(6.69 - 3.69)$, which is the same as comparing nominal returns, $9.75 - 6.75 = 3.00$.
D is incorrect. Sign flip on York's shield: $2.50 - 1.19 = 1.31\%$, so $5.50 - 1.31 = 4.19$.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.7, Bank Qualified Bonds, eqs. (4.5) to (4.7)

Solution LTR-4 b.8 Answer: A

Bank Q172 • GARP book Ch.4 Q7

Step 1 – Option (a): a realised loss

$$\text{Loss} = 4{,}000{,}000 - 3{,}750{,}000 = 250{,}000; \qquad \text{tax saved} = 0.35 \times 250{,}000 = 87{,}500$$

$$\text{Coupon pickup} = (8.0\% - 7.5\%) \times 4{,}000{,}000 = 20{,}000 \text{ a year}$$

Step 2 – Option (b): a realised gain

$$\text{Gain} = 4{,}330{,}000 - 4{,}000{,}000 = 330{,}000; \qquad \text{extra tax} = 0.35 \times 330{,}000 = 115{,}500$$

$\Rightarrow$ In a year of high loan income the bank wants losses, not gains: (a) is a textbook tax swap.

Interpretation

The Treasuries' market value above par says their 12% coupon is *above* current market yields: they are worth keeping, and they also give liquidity and pledgeable collateral. Rose adds the information still needed: the credit quality and liquidity of the two counties, how much tax-exempt income the thrift can use, the rate outlook and transaction costs.

Why the other options are wrong

B is incorrect. A gain adds USD 115,500 to this year's tax; only losses offset loan income.
C is incorrect. Wrong after-tax arithmetic: the after-tax cost of the loss is $250{,}000 \times (1 - 0.35) = 162{,}500$, not $250{,}000 \times 1.35 = 337{,}500$.
D is incorrect. Polarity flip: a price above par means the coupon is above market yields; selling gives up income.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.7, The Tax Swapping Tool; The Portfolio Shifting Tool

Solution LTR-4 b.9 **Answer: D**

Bank Q184 • GARP PE1 Q11

Step 1 – Read the old Treasuries

Bought when rates were much lower and now facing high rates, they trade below cost: unrealised losses.

Step 2 – Read the profit line

Record profitability means large taxable income that realised losses can shelter.

Step 3 – Match the tool

A tax swap sells the low-yielding bonds at a loss to cut this year's tax and reinvests at today's higher yields, which supports total income.

$\Rightarrow$ The tax swap uses all three facts.

Interpretation

The flat curve is planted to kill the yield-curve strategies of LTR-4 c: with no term spread there is nothing to carry and nothing to ride.

Why the other options are wrong

A is incorrect. A barbell mixes short and long maturities for liquidity and income; no liquidity or return-mix problem is described.

B is incorrect. A carry trade borrows short to invest long; on a flat curve the long yield does not exceed the borrowing cost.

C is incorrect. Riding the yield curve needs a steep positive slope so securities roll down to lower yields and higher prices.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.7, The Tax Swapping Tool; 4.9 Maturity Management Tools

LTR-5 • Liquidity and Reserves Management: Strategies and Policies • Vardeez: MODERATE

LTR-5 a

P1 BOTH

Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank.

Workbook: A30 Asset vs Borrowed Liquidity Management and the Investment Function • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A30, recalled in the August 2026 sitting ("asset liquidity management and borrowed liquidity management"). Map #63 (1 of 16 sources) includes "reserve-management strategy", and the workbook treats the net liquidity position as its starting point. **Vardeez:** MODERATE reading, but his reading note names exactly this LO and LTR-5 b ("LO seventy A and LO seventy B"). His trap: acceptable loan requests are a demand, loan repayments a supply.

Reported exam shapes

- **A30 S2** Reserve-management strategy: the net liquidity position as the starting point (workbook A30.1, which then asks for the strategy of LTR-5 b).

What the questions look like: a list of up to ten projected cash flows to sort into supplies and demands. The planted errors are money market borrowings (a supply) against repayments of borrowings (a demand), loan repayments (a supply) against acceptable loan requests (a demand), and dropping dividends or operating expenses. Variants back-solve one missing flow from a stated deficit or surplus, or attach the net position to a recommendation. One or two statement questions on why liquidity demand and supply move (interest rates, timing, public confidence).

1. What it means to be liquid

A financial firm is **liquid** if it has ready access to immediately spendable funds **at reasonable cost at precisely the time those funds are needed**: either the right amount of spendable funds on hand, or the ability to raise them in time by borrowing or selling assets. Liquidity is not solvency. A firm can be closed for lack of liquidity while technically solvent: in the 1990s the Federal Reserve forced the closure of the $10 billion Southeast Bank of Miami because it could not raise the liquidity to repay its Fed loans. A shortage of liquidity is often one of the first signs of trouble: a bank losing deposits sells its safest, most liquid assets, and other lenders demand security or a higher rate, which cuts earnings further. This background serves the whole reading; LTR-5 b and LTR-5 c point back here.

2. Demands for and supplies of liquidity

Demands (cash out)	Supplies (cash in)
Customers withdrawing money from their accounts	Incoming customer deposits (the most important source)
Credit requests from customers the firm wishes to keep: new loans and drawings on existing lines	Customers repaying their loans
Repaying previous borrowings (from other financial firms or the central bank)	Sales of assets, especially marketable securities
Other operating expenses	Revenues (fees) from selling nondeposit services
Income taxes and cash dividends to stockholders	Borrowings in the money market

The two most pressing demands are withdrawals and credit requests. Together they give the firm's **net liquidity position** at time t:

$$\begin{aligned} L_t = \quad & (\text{Deposit inflows} + \text{Nondeposit service revenues} + \text{Loan repayments} \\ & + \text{Asset sales} + \text{Money market borrowings}) \\ & - (\text{Deposit withdrawals} + \text{Acceptable loan requests} + \text{Repayments of borrowings} \\ & + \text{Other operating expenses} + \text{Dividends}) \end{aligned} \qquad \text{(eq. 5.1)}$$

- $L_t < 0$: a **liquidity deficit**. Management must decide *when and where* to raise additional funds.
- $L_t > 0$: a **liquidity surplus**. Management must decide when and where to *invest* the surplus profitably until it is needed.

Worked example (Rose's Concept Check 5.3). A bank projects for the coming week (USD million): withdrawals 33, loan repayments 108, operating expenses 51, acceptable new loan requests 294, asset sales 18, new deposits 670, money market borrowings 43, nondeposit service fees 27, repayment of previous borrowings 23, dividend payment 140.

$$\begin{aligned} \text{Supplies} &= 670 + 108 + 18 + 27 + 43 = 866 \\ \text{Demands} &= 33 + 294 + 23 + 51 + 140 = 541 \\ L &= 866 - 541 = +\text{USD } 325 \text{ million} \end{aligned}$$

$\Rightarrow$ A surplus of USD 325 million to invest until it is needed (Figure 1). The week is dominated by two items: new deposits on the supply side and loan requests on the demand side.

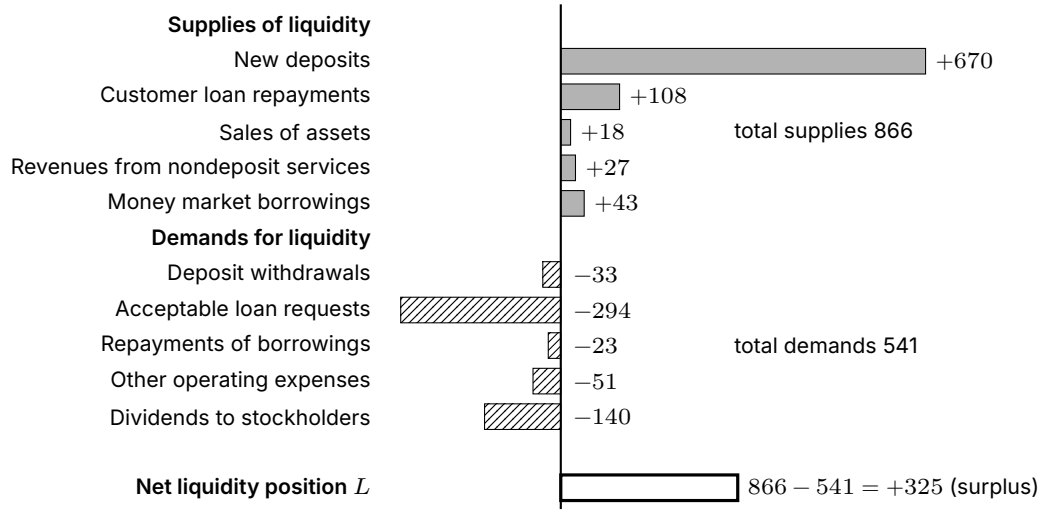


Figure 1. Rose's Concept Check 5.3 (USD million). Grey bars to the right of the axis are supplies of liquidity; hatched bars to the left are demands; the white bar is the net position. Money market borrowings sit with the supplies; repayments of borrowings sit with the demands.

Same item, two directions

Item	Supply when...	Demand when...
Loans	customers *repay* them	acceptable requests are *granted*, lines *drawn*
Borrowings	the bank *borrows* in the money market	the bank *repays* earlier borrowings
Deposits	customers *deposit*	customers *withdraw*
Assets	the bank *sells* them	(buying assets uses cash, but is not in eq. 5.1)

Vardeez calls the loan line "the single most reliable polarity trap in the reading": a loan the bank wants to make is cash going out.

3. Timing: the critical dimension

- **Immediate needs.** Several large CDs mature tomorrow and the holders say they will withdraw rather than roll over. Only sources that can be tapped at once will do.
- **Longer-term needs** arise from **seasonal, cyclical and trend** factors: liquid funds are in greater demand in the fall and summer (school, holidays, travel). With warning, management can use a wider range of sources: sell accumulated liquid assets, advertise new services, negotiate long-term borrowings.
- **Within the month**, deposit inflows are heavy at the start of each month as payrolls are paid, with a secondary peak mid-month as bills are paid; cash demands peak near the end of a week, the first of the month and certain seasons.
- Not every demand needs a sale or a loan: deposits or loan repayments may arrive just when funds are needed.

4. The essence of liquidity management

1. **Rarely are demands for liquidity equal to the supply** at any moment. The firm is always in either a deficit or a surplus.
2. **There is a trade-off between liquidity and profitability.** The more resources tied up in readiness to meet demands, the lower the expected profitability, other factors held constant.

Resolving liquidity problems costs money: the **interest cost** of borrowed funds, the **transactions cost** of time and money in finding them, and the **opportunity cost** of future earnings forgone when earning assets are sold. Excess liquidity has a cost too: surplus funds must be invested at once to avoid idle balances. Liquidity management is exposed to **interest rate risk** (if rates rise, assets to be sold fall in value and some must be sold at a loss, while borrowing costs more) and **availability risk** (liquid funds may not be available in the volume needed; lenders who see more risk charge more or refuse).

5. Why depository institutions face significant liquidity problems

- **Maturity mismatch.** They borrow short from individuals, businesses and other lenders and lend long; incoming cash from assets rarely matches outgoing cash on liabilities.
- **A high share of liabilities payable immediately**, especially demand (checkable) deposits and money market borrowings.
- **Sensitivity to market rates.** When rates *rise*, some depositors withdraw in search of higher returns, many loan customers postpone new requests or *speed up* drawings on lines that carry lower rates, the market value of assets to be sold falls, and borrowing costs rise. Rates move both deposit demand and loan demand.
- **Public confidence.** Failing to meet demands, for example closing teller windows and ATMs for lack of cash (as happened to a Montana bank), can severely damage confidence. A liquidity manager must stay in close contact with the largest fund suppliers and the holders of large unused credit lines.

Most liquidity problems arise from outside the firm, from customers: their liquidity problems gravitate to their liquidity suppliers. After the October 1987 crash, investors who had bought stock on

margin flooded their lenders with loan requests, turning a liquidity crisis in the capital market into one for lenders.

Formula card: LTR-5 a

Net liquidity position	$L_t =$ supplies $-$ demands (eq. 5.1)
Supplies (5)	deposit inflows, nondeposit service revenues, loan repayments, asset sales, money market borrowings
Demands (5)	deposit withdrawals, acceptable loan requests, repayments of borrowings, other operating expenses, dividends
Sign	$L < 0$ deficit (raise funds); $L > 0$ surplus (invest it)
Back-solve	missing item $=$ the amount that makes supplies $-$ demands equal the stated L
Costs; risks	interest, transactions, opportunity costs; interest rate risk, availability risk

Exam traps

- **Loan polarity.** Acceptable loan requests are a demand; loan repayments are a supply.
- **Borrowing polarity.** New money market borrowings are a supply; repaying earlier borrowings is a demand. Leaving borrowings out of supply, or counting them as a demand, are the two commonest builds.
- **Dropped items.** Dividends and operating expenses are demands; fee income from nondeposit services is a supply.
- **Sign.** Report supplies minus demands; a negative number is a deficit, not a surplus.
- **Intermediate results.** Total supplies or total demands alone is never the answer.
- **Rising rates** increase withdrawals and drawings on cheap credit lines, lower the value of assets to be sold and raise borrowing costs. They do not slow withdrawals.
- **More liquid assets $\neq$ liquid.** Liquidity also depends on the demands against the firm (LTR-5 b).

Practice questions: LTR-5 a

Question LTR-5 a.1

Bank Q205 • Schweser Module Quiz 70.1 Q1 (rebuilt)

Genny Richards is the liquidity manager for Legend Bank. In evaluating the bank's net liquidity position, Richards anticipates the following amounts:

Asset sales	USD 1,325,000	Deposit withdrawals	USD 1,015,000
Incoming deposits	USD 2,500,000	Dividend payments	USD 470,000
Nondeposit services revenue	USD 950,000	Loan requests	USD 845,000

Legend Bank's net liquidity position is closest to:

A. USD 2,330,000

B. USD 2,445,000

C. USD 2,915,000

D. USD 3,385,000

Question LTR-5 a.2

Bank Q203 • Mock 3 Q60

A treasurer at a small regional bank is assessing the bank's liquidity position. The treasurer estimates that the following cash inflows and outflows will occur in the next week:

Cash flow	USD million	Cash flow	USD million
Deposit withdrawals	30	Borrowings from money market	60
Deposit inflows	70	Operating expenses	40
Scheduled loan repayments	80	Stockholder dividend payments	20
Acceptable loan requests	50	Repayment of bank borrowings	30

Which of the following is the correct amount (in USD million), at the week's end, for the bank's net liquidity position?

A. -80

B. -20

C. 40

D. 100

Question LTR-5 a.3 Bank Q196 • GARP book Ch.5 end-of-chapter problem (Ocean View State Bank, rebuilt)

Ocean View State Bank estimates that over the next 24 hours the following cash inflows and outflows will occur (USD million):

Deposit withdrawals	100	Sales of bank assets	40
Deposit inflows	95	Stockholder dividend payments	150
Scheduled loan repayments	90	Revenues from sale of nondeposit services	95
Acceptable loan requests	60	Repayments of bank borrowings	60
Borrowings from the money market	80	Operating expenses	50

Which of the following best describes the bank's projected net liquidity position and how it can respond?

A. A deficit of USD 180 million, to be covered by selling liquid assets, borrowing, or a mix of the two.

B. A deficit of USD 100 million, to be covered by selling liquid assets, borrowing, or a mix of the two.

C. A deficit of USD 20 million, to be covered by selling liquid assets, borrowing, or a mix of the two.

D. A surplus of USD 130 million, to be invested in earning assets until it is needed.

Question LTR-5 a.4 Bank Q197 • GARP book Ch.5 end-of-chapter problem (Mountain Top Savings, rebuilt)

Mountain Top Savings projects a net liquidity deficit of USD 10 million next week. Expected demands for liquidity are quality loan demand of USD 32 million, repayments of previous borrowings of USD 15 million, stockholder dividends of USD 10 million and other operating expenses of USD 15 million, in addition to deposit withdrawals. Expected supplies are deposit inflows of USD 26 million, revenues from nondeposit services of USD 18 million, scheduled loan repayments of USD 23 million, asset sales of USD 10 million and money market borrowings of USD 15 million. What must Mountain Top's expected deposit withdrawals be?

A. USD 10 million

B. USD 12 million

C. USD 20 million

D. USD 30 million

Question LTR-5 a.5 New • built from GARP Ch.5, Sections 5.2 and 5.3

The liquidity manager of a mid-sized US bank expects market interest rates to rise sharply next quarter. Which of the following effects on the bank's liquidity position should she plan for, according to Rose and Hudgins?

A. Deposit withdrawals slow, because higher market rates make the bank's existing deposits more attractive to customers.

B. Customers speed up drawings on credit lines that carry lower interest rates, while assets the bank may need to sell fall in value and borrowed liquidity costs more.

C. Loan customers bring forward new loan requests, while the market value of the bank's securities available for sale rises.

D. Liquidity needs are unchanged, because interest rate movements affect only the bank's profitability, not its cash flows.

Question LTR-5 a.6 New • built from GARP Ch.5, Sections 5.1 to 5.3

The treasurer of a Kenyan commercial bank presents a liquidity policy paper to the board. Which statement in the paper is consistent with Rose and Hudgins?

A. The bank should aim to keep supplies of liquidity exactly equal to demands at every moment, so that it is never in surplus or deficit.

B. Because most liquidity problems originate inside the bank, tight internal cost control is the main defence against them.

C. The more resources the bank ties up in readiness to meet liquidity demands, the lower its expected profitability, other factors held constant.

D. A liquidity surplus carries no cost, so excess funds can wait until the next scheduled investment date.

Question LTR-5 a.7 Workbook A30.1 • S1, S2 • Aug-2026 • Map #63

The treasurer of a CAD 900 million community savings bank in Manitoba is preparing next week's liquidity plan for the board. Projected flows (CAD million) are: deposit inflows 120, loan repayments 45, revenues from nondeposit services 8; deposit withdrawals 140, acceptable new loan requests 95, repayment of previous borrowings 10, other operating expenses 12, dividend payment 6. The bank holds CAD 150 million of marketable Treasury and agency securities. Which of the following recommendations to the board is consistent with Rose and Hudgins?

A. Cover the CAD 90 million deficit by borrowing, since liability management is the less risky strategy and the one smaller institutions mainly use.

B. Cover the CAD 90 million deficit by selling Treasury bills, since this leaves the size and composition of the bank's asset portfolio unchanged.

C. Treat the CAD 90 million deficit as no concern, since securities holdings above the deficit guarantee the bank is liquid whatever demands it faces.

D. Cover the CAD 90 million deficit with a balanced strategy: part from marketable securities, part from prearranged credit lines, unexpected needs from near-term borrowing.

Solutions • LTR-5 a

Solution LTR-5 a.1

Answer: B

Bank Q205 • Schweser Module Quiz 70.1 Q1

Step 1 – Supplies

$$1{,}325{,}000 + 2{,}500{,}000 + 950{,}000 = 4{,}775{,}000$$

Step 2 – Demands

$$1{,}015{,}000 + 470{,}000 + 845{,}000 = 2{,}330{,}000$$

Step 3 – Net position

$$L = 4{,}775{,}000 - 2{,}330{,}000 = \text{USD } 2{,}445{,}000$$

$\Rightarrow$ A surplus of about USD 2.4 million to invest until needed.

Interpretation

Only six of the ten flows in eq. 5.1 appear; the missing ones are simply zero. Asset sales, new deposits and fee income supply liquidity; withdrawals, dividends and loan requests demand it.

Why the other options are wrong

A is incorrect. Total demands alone, an intermediate result: 2,330,000.
C is incorrect. Left dividends out of demands: $4{,}775{,}000 - 1{,}860{,}000 = 2{,}915{,}000$.
D is incorrect. Counted dividends as a supply: $2{,}445{,}000 + 2 \times 470{,}000 = 3{,}385{,}000$.

LO: LTR-5 a: Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.2 The Demand for and Supply of Liquidity, eq. (5.1)

Solution LTR-5 a.2

Answer: C

Bank Q203 • Mock 3 Q60

Step 1 – Supplies

Deposit inflows 70 + loan repayments 80 + money market borrowings 60 = 210.

Step 2 – Demands

Withdrawals 30 + loan requests 50 + operating expenses 40 + dividends 20 + repayment of borrowings 30 = 170.

Step 3 – Net position

$$L = 210 - 170 = +\text{USD } 40 \text{ million}$$

$\Rightarrow$ A surplus: new borrowing raised this week outweighs the borrowing repaid.

Interpretation

The stem lists borrowings twice in opposite directions; money raised is a supply and money repaid is a demand.

Why the other options are wrong

A is incorrect. Treated money market borrowings as a demand: $150 - 230 = -80$.
B is incorrect. Left money market borrowings out of supply: $150 - 170 = -20$.
D is incorrect. Treated withdrawals as a supply: $240 - 140 = 100$.

LO: LTR-5 a: Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.2, eq. (5.1)

Solution LTR-5 a.3 — Answer: C

Bank Q196 • GARP book Ch.5 (Ocean View State Bank)

Step 1 – Supplies

Deposit inflows 95 + loan repayments 90 + money market borrowings 80 + asset sales 40 + nondeposit revenues 95 = 400.

Step 2 – Demands

Withdrawals 100 + loan requests 60 + dividends 150 + repayments of borrowings 60 + operating expenses 50 = 420.

Step 3 – Net position

$$L = 400 - 420 = -\text{USD 20 million}$$

$\Rightarrow$ A deficit: the bank must decide when and where to raise USD 20 million.

Interpretation

Rose's answer to "from what sources" is the menu of LTR-5 b: convert assets (sell liquid securities), borrow (Fed funds, repos, CDs, the discount window), or combine the two. The USD 150 million dividend is the largest single demand; the bank could also reconsider its size or timing.

Why the other options are wrong

A is incorrect. Counted money market borrowings as a demand: $320 - 500 = -180$.
B is incorrect. Left money market borrowings out of supply: $320 - 420 = -100$.
D is incorrect. Left the USD 150 million dividend out of demands: $400 - 270 = +130$.

LO: LTR-5 a: Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.2, eq. (5.1); 5.4 Strategies for Liquidity Managers

Solution LTR-5 a.4 — Answer: D

Bank Q197 • GARP book Ch.5 (Mountain Top Savings)

Step 1 – Supplies

$$26 + 18 + 23 + 10 + 15 = 92$$

Step 2 – Known demands

$$32 + 15 + 10 + 15 = 72$$

Step 3 – Solve for withdrawals W

$$92 - (72 + W) = -10 \quad \Rightarrow \quad W = 30$$

$\Rightarrow$ Expected withdrawals of USD 30 million produce the projected USD 10 million deficit.

Interpretation

Check: demands $72+30 = 102$ against supplies 92 gives -10. A back-solve is the same equation read the other way; keep the sign of the stated position.

Why the other options are wrong

A is incorrect. Treated the deficit as a surplus: $20 - W = +10$ gives 10.

B is incorrect. Left nondeposit revenues out of supply: $74 - 72 - W = -10$ gives 12.
C is incorrect. Ignored the deficit: set withdrawals equal to $92 - 72 = 20$.

LO: LTR-5 a: Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.2, eq. (5.1)

Solution LTR-5 a.5 — Answer: B

New • built from GARP Ch.5

Step 1 – Why B is correct

Rose: "When interest rates rise ... some customers will withdraw their funds in search of higher returns elsewhere. Many loan customers may postpone new loan requests or speed up their drawings on those credit lines that carry lower interest rates." Rising rates also lower the market value of assets the firm may need to sell, and "raising liquid funds by borrowing will cost more".

Interpretation

These are the two risks of liquidity management: interest rate risk (asset values and borrowing costs) and availability risk (funds may not be there in the volume needed).

Why the other options are wrong

A is incorrect. Polarity flip: higher market rates draw deposits *away* to higher returns elsewhere.
C is incorrect. Polarity flip twice: customers postpone new loan requests, and security values fall when rates rise.
D is incorrect. Rates change both deposit and loan demand, and so the bank's cash flows, not only its profit.

LO: LTR-5 a: Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.2; 5.3 Why Financial Firms Often Face Significant Liquidity Problems

Solution LTR-5 a.6 — Answer: C

New • built from GARP Ch.5

Step 1 – Why C is correct

It is the second of Rose's two statements of the essence of liquidity management: "There is a trade-off between liquidity and profitability. The more resources tied up in readiness to meet demands for liquidity, the lower is that financial firm's expected profitability (other factors held constant)."

Interpretation

The first statement is the reason option A cannot be a policy: demands and supplies are rarely equal, so the firm is always managing a deficit or a surplus.

Why the other options are wrong

A is incorrect. "Rarely are demands for liquidity equal to the supply of liquidity at any particular moment"; the goal is managing the gap, not abolishing it.
B is incorrect. Polarity flip: "most liquidity problems arise from outside the financial firm as a result of the activities of customers".
D is incorrect. Excess liquidity has an opportunity cost; Rose says surplus funds must be invested immediately.

LO: LTR-5 a: Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.1 Introduction; 5.2; 5.3

Solution LTR-5 a.7 **Answer: D**

Workbook A30.1 • S1, S2 • Aug-2026

Step 1 – Net liquidity position

$$L_t = (120 + 45 + 8) - (140 + 95 + 10 + 12 + 6) = 173 - 263 = -\text{CAD } 90 \text{ million}$$

$\Rightarrow$ Demands exceed supplies, so the bank faces a liquidity deficit and must decide when and where to raise funds.

Step 2 – Why D is correct

"Most financial firms compromise by using both asset and liability management": under a balanced strategy some expected demands are stored in assets (principally marketable securities), other anticipated needs are backstopped by advance lines of credit, and unexpected cash needs are met from near-term borrowings. This avoids both the storage cost of pure asset conversion and the risk of pure borrowing.

Interpretation

For a small bank the securities portfolio is the natural first line, but selling CAD 90 million of it would forgo future earnings, incur transaction costs, risk selling into a falling market and shrink the balance sheet; borrowing the whole amount would expose earnings to rate volatility and availability risk. Mixing the two, and selling the securities with the least profit potential first, is what the book recommends.

Why the other options are wrong

A is incorrect. Polarity flip: borrowing liquidity is "the most risky approach" (with the highest expected return) and is used most by the largest banks; asset conversion is the less risky approach used mainly by smaller institutions.

B is incorrect. Sibling swap: leaving the asset portfolio's volume and composition unchanged is an advantage of borrowed liquidity. Selling assets to meet liability-driven demands shrinks the size of the firm.

C is incorrect. Absolute claim: a firm holding more liquid assets "will not necessarily be a liquid institution", because its position also depends on the demands made against it; liquidity means access at reasonable cost at precisely the time funds are needed.

LO: LTR-5 a: Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank. • LTR-5 b: Compare strategies that a bank can use to meet demands for additional liquidity. • **Reference:** Rose and Hudgins, Ch.11, 5.2 The Demand for and Supply of Liquidity (Equation 5.1); 5.4 Strategies for Liquidity Managers

LTR-5 • Liquidity and Reserves Management: Strategies and Policies • Vardeez: MODERATE

LTR-5 b

P1 CONCEPT

Compare strategies that a bank can use to meet demands for additional liquidity.

Workbook: A30 Asset vs Borrowed Liquidity Management and the Investment Function • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. This is the content recalled from A30 in the August 2026 sitting: "asset liquidity management and borrowed liquidity management". Map #63 lists reserve-management strategy. **Vardeez:** named in his reading note ("LO seventy B"); his discriminator is institution size paired with risk appetite.

Reported exam shapes

- **A30 S1** Asset liquidity (asset conversion) versus borrowed liquidity (liability management) versus balanced strategy: pros, cons and who uses which, as seen in August. Workbook A30.1 carries it with the net position and is worked as Question LTR-5 a.7.

What the questions look like: statements to judge (which strategy is riskier, which is used by small and which by large banks, which leaves the asset portfolio unchanged), the classification of a source (selling securities under repurchase agreement is borrowing; buying them is an asset), and the balance-sheet size after a deposit drain or a loan request under each strategy. GARP's own end-of-chapter problems ask for that size effect; the arithmetic is one line, so the question is kept as a comparison of the strategies.

1. Three strategies

A firm is liquid only if it has access, at reasonable cost, to liquid funds in exactly the amounts required at precisely the time they are needed (LTR-5 a, Section 1). Liquidity managers have three ways to meet a deficit: **store liquidity in assets** (asset liquidity management, or asset conversion), **borrow it** (borrowed liquidity, purchased liquidity, or liability management), or **combine the two** (balanced liquidity management).

2. Asset liquidity management (asset conversion)

The oldest approach. In its purest form, liquidity is stored in assets, mainly cash and marketable securities, and selected assets are converted into cash until all demands for cash are met. A **liquid asset** has three characteristics:

1. a **ready market**, so it can be converted into cash without delay;
2. a **reasonably stable price**: the market is deep enough to absorb even a quick or large sale without a significant price decline;
3. **reversibility**: the seller can recover the original investment with little risk of loss.

Principal options: Treasury bills, federal funds loans to other institutions, purchases of securities under repurchase agreements, correspondent deposits, municipal bonds and notes, federal agency securities, negotiable CDs, Eurocurrency loans.

Who uses it: mainly **smaller institutions**, which find it a **less risky** approach than relying on borrowings.

Its costs (it is "not a costless approach"):

- the **opportunity cost** of future earnings the sold assets would have generated, so the assets with the *least profit potential* should be sold first;
- **transactions costs** (brokers' commissions), and the risk of selling into a market with declining prices;
- a **weaker-looking balance sheet**, because the assets sold are often low-risk securities that signal strength;
- liquid assets carry **the lowest rates of return** of all assets, so storing liquidity means forgoing higher returns.

And it is not a guarantee: holding more liquid assets strengthens the liquidity position, but a firm "will not necessarily be a liquid institution" because its position also depends on the demands made against it.

3. Borrowed liquidity (liability) management

Raise liquid funds by borrowing in the money market; in its purest form, borrow to cover *all* anticipated demands. Principal sources: federal funds borrowings, selling liquid low-risk securities under **repurchase agreements**, issuing **jumbo ($100,000+) negotiable CDs**, issuing Eurocurrency deposits, advances from the **Federal Home Loan Banks**, and borrowing at the central bank's **discount window**.

Who uses it: most extensively the **largest banks**, which often borrow close to 100% of their liquidity needs.

Advantages

- Borrow **only when funds are actually needed**, instead of holding a low-yielding storehouse of liquid assets at all times.
- The **volume and composition of the asset portfolio stay unchanged**. Selling assets to meet liability-driven demands such as withdrawals shrinks the firm.
- Its own **control lever, the offer rate**: raise it until the needed funds flow in, lower it when fewer are needed.

Disadvantages. It is **the most risky approach** to liquidity problems, though it also carries **the highest expected return**, because of interest rate volatility and the speed with which credit availability can change. Borrowing costs are always uncertain, adding uncertainty to earnings. Firms often must buy liquidity when it is hardest to get, and a firm in trouble needs borrowed liquidity most precisely when others are least willing to lend to it.

4. What each strategy does to the balance sheet

Take Loud Bank (USD million): cash 48, securities 940, loans 3,932, total assets 4,920; deposits 3,948, other liabilities 440, equity 532.

- **Depositors withdraw 200.** Asset conversion: sell 200 of securities and pay out; securities fall to 740, deposits to 3,748, and the bank **shrinks** to 4,720. Liability management: borrow 200 and pay out; deposits fall to 3,748, borrowings rise to 200, and total assets **stay** at 4,920 (Figure 1).
- **The loan officer wants to make 400 of new loans.** Asset conversion: sell 400 of securities; the **size is unchanged** at 4,920 but the mix shifts from securities to loans. Liability management: borrow 400; loans and liabilities both rise and the bank **grows** to 5,320.

⇒ Selling assets shrinks or reshuffles the balance sheet; borrowing replaces a liability or enlarges both sides.

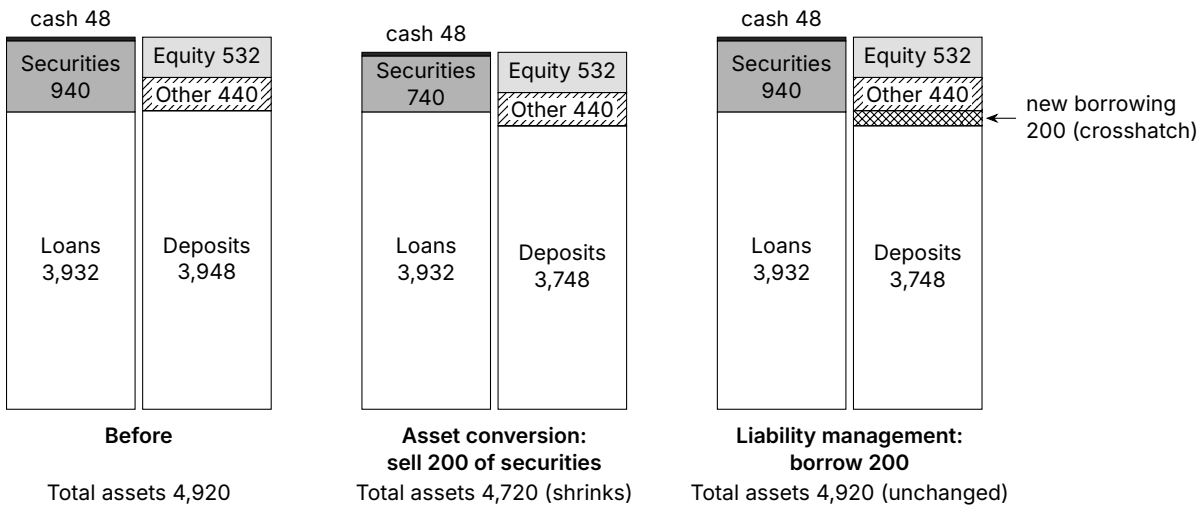


Figure 1. Loud Bank meets a USD 200 million deposit drain two ways. In each panel the left column is assets (loans white, securities grey, cash black) and the right column is liabilities and equity (deposits white, new borrowing crosshatched, other liabilities hatched, equity light grey). Asset conversion shrinks the bank; liability management swaps one liability for another and leaves the asset portfolio untouched.

5. Balanced liquidity management

Because borrowing is risky and storing liquidity in assets is costly, **most financial firms compromise** and use both:

- some **expected** demands are stored in assets, principally marketable securities;
- other **anticipated** needs are backstopped by **advance arrangements for lines of credit**;
- **unexpected** cash needs are met from **near-term borrowings**;
- **longer-term** needs are planned for, and the funds parked in short- and medium-term assets that provide cash as those needs arise.

6. Guidelines for liquidity managers

1. Keep track of the activities of **all departments** using or supplying funds (a new credit line means possible drawings; large expected deposits should be passed on).
2. Know **in advance** when the biggest credit or funds-supplying customers plan to withdraw or add funds.
3. Keep the firm's **priorities clear**. In the past liquidity got top priority, on the view that a bank controls its uses of funds but not its sources; today liquidity management usually plays a **supporting role** to the first priority, making loans and supplying fee-generating services to all qualified customers.
4. Analyse liquidity needs **continuously**: excess liquidity not reinvested the day it appears loses income, and deficits must be dealt with quickly to avoid hurried borrowing or loss sales.

	Asset conversion	Borrowed liquidity	Balanced
How	Store liquidity in cash and marketable securities; sell as needed	Buy liquidity in the money market when needed	Expected needs in securities, anticipated needs on credit lines, unexpected needs borrowed
Main users	Smaller institutions	Largest banks (close to 100% of needs)	Most financial firms
Risk and return	Less risky; liquid assets earn the lowest returns	Most risky; highest expected return	In between
Balance sheet	Shrinks after a drain; composition changes	Assets unchanged after a drain; grows with new loans	Mixed
Cost	Opportunity, transactions, falling-market and window-dressing costs	Uncertain rate; availability can vanish	Planning and time management
Control lever	Which assets to sell (least profitable first)	The offer rate	Both

Exam traps

- **Size and risk swap.** Small banks: asset conversion, the less risky route. Largest banks: borrowed liquidity, the most risky route with the highest expected return. Keep both halves together.
- **Asset portfolio unchanged** and **the offer rate as a lever** belong to borrowing, not to asset conversion.
- **Repurchase agreements cut both ways.** Selling securities under an RP is borrowing (liability management); buying securities under an RP is a liquid asset (asset conversion).
- **Balance-sheet size.** Deposit drain: asset conversion shrinks, borrowing keeps size. New loans: asset conversion keeps size, borrowing grows.
- **Absolute claims.** More liquid assets do not guarantee liquidity; borrowing only when needed does not make borrowing safe.
- **Priorities.** Liquidity management is now a supporting function to lending, not the top priority.

Practice questions: LTR-5 b

Question LTR-5 b.1

Bank Q214 • Schweser Module Quiz 70.1 Q2

A bank uses the liability management strategy to meet its liquidity needs. Which of the following sources of liquidity is it most likely to use?

A. US Treasury securities

B. Federal agency securities

C. Repurchase agreement sales

D. Municipal bond investments

Question LTR-5 b.2

New • built from GARP Ch.5, Section 5.4

A USD 600 million community bank in Iowa relies on asset conversion to meet its liquidity needs. Its CFO lists the drawbacks of the strategy for the board. Which of the following is *not* a cost or drawback of asset conversion according to Rose and Hudgins?

A. The future earnings the sold assets would have generated are lost.

B. Commissions are paid to security brokers, and the assets may have to be sold into a falling market.

C. Selling low-risk securities weakens the appearance of the balance sheet.

D. The cost of the funds raised is always uncertain, which adds uncertainty to earnings.

Question LTR-5 b.3

Bank Q213 • BT answers P2.T8.20.6.2

Experienced liquidity managers tend to employ one of three strategies: asset liquidity management (asset conversion), borrowed liquidity (purchased liquidity or liability management), or balanced liquidity management. About these strategies, each of the following statements is true *except* which is false?

A. A liquid asset has a ready market, a reasonably stable price and is reversible.

B. If a bank's primary goal is to avoid shrinking its balance sheet (and weakening its appearance), then a borrowed liquidity strategy is better than an asset liquidity strategy.

C. If a bank's primary goal is to minimise risk, then a borrowed liquidity strategy is better than an asset liquidity strategy.

D. A balanced liquidity strategy balances the opportunity cost of storing liquidity in assets against the risks of interest rate volatility and credit availability associated with borrowing liquidity.

Question LTR-5 b.4 Bank Q206, Q207 • GARP book Ch.5 end-of-chapter Q6, Q7 (rebuilt)

The Bank of Your Dreams has the following balance sheet (USD million):

Cash	100	Deposits	4,000
Securities	1,000	Other liabilities	500
Loans	4,000	Equity	600
Total assets	5,100	Total liabilities and equity	5,100

The liquidity manager models two events separately: depositors withdraw USD 500 million, and the loan officer wants to make USD 600 million of new loans. Which of the following statements is correct?

A. If the deposit drain is met by selling securities, total assets fall to USD 4,600 million; if it is met by borrowing, total assets stay at USD 5,100 million.

B. If the deposit drain is met by borrowing, total assets fall to USD 4,600 million; if it is met by selling securities, total assets stay at USD 5,100 million.

C. If the new loans are funded by selling securities, total assets rise to USD 5,700 million, because loans rise by USD 600 million.

D. If the new loans are funded by borrowing, total assets stay at USD 5,100 million, because the new liability offsets the new loans.

Question LTR-5 b.5 New • built from GARP Ch.5, Section 5.4

An analyst compares the liquidity strategies of a USD 2 billion community bank and a USD 900 billion money-center bank. Which statement is most consistent with Rose and Hudgins?

A. The community bank is more likely to rely on borrowed liquidity, because smaller banks find it the less risky approach.

B. The money-center bank may borrow close to 100% of its liquidity needs, accepting the most risky approach in exchange for the highest expected return.

C. Borrowed liquidity is the safest approach, because the bank borrows only when funds are actually needed.

D. Asset conversion carries the highest expected return, because the bank keeps control of its offer rate.

Question LTR-5 b.6 New • built from GARP Ch.5, Section 5.4

The treasurer of a regional bank is setting up a balanced liquidity management strategy. Which assignment of funding sources to liquidity needs matches the strategy as Rose and Hudgins describe it?

A. Expected demands from marketable securities, other anticipated needs from prearranged lines of credit, and unexpected needs from near-term borrowings.

B. Expected demands from near-term borrowings, and unexpected needs from a storehouse of marketable securities.

C. All needs from money market borrowing, with the asset portfolio left unchanged.

D. All needs from cash and Treasury bills, sold until every demand is met.

Question LTR-5 b.7 New • built from GARP Ch.5, Guidelines for Liquidity Managers

A newly appointed liquidity manager drafts operating guidelines. Which of the following guidelines is consistent with Rose and Hudgins?

A. Liquidity should take top priority over lending when funds are allocated, since the bank controls its uses of funds but not its sources.

B. The liquidity manager should track the activities of all departments that use or supply funds, such as new credit lines granted by the loan department.

C. Liquidity needs should be reviewed at the end of each quarter, since more frequent analysis adds cost without benefit.

D. The plans of the largest funds-supplying customers are confidential to relationship managers and need not be known in advance by the liquidity manager.

Solutions • LTR-5 b

Solution LTR-5 b.1

Answer: C

Bank Q214 • Schweser Module Quiz 70.1 Q2

Step 1 – What liability management uses

Borrowed funds: federal funds, jumbo CDs, Eurocurrency deposits, FHLB advances, the discount window, and selling securities under repurchase agreements.

Step 2 – Classify each option

A repo sale brings in cash now against a promise to buy the securities back: a collateralised borrowing, a liability. Treasuries, agencies and municipals held by the bank are assets.

$\Rightarrow$ Only the repurchase agreement sale is a borrowed source.

Interpretation

The direction matters: buying securities under a repurchase agreement is on Rose's list of liquid assets for asset conversion.

Why the other options are wrong

A is incorrect. Treasury securities are the classic liquid asset for asset conversion.

B is incorrect. Agency securities are also stored liquidity.

D is incorrect. Municipal bonds are assets held for asset conversion (and tax), not borrowings.

LO: LTR-5 b: Compare strategies that a bank can use to meet demands for additional liquidity. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.4, Borrowing Liquidity: The Principal Options

Solution LTR-5 b.2

Answer: D

New • built from GARP Ch.5

Step 1 – List the costs of asset conversion

Rose: lost future earnings (opportunity cost), transactions costs (commissions), possible sale in a market with declining prices, a weaker-looking balance sheet, and the lowest returns of all assets.

Step 2 – Test D

"Borrowing cost is always uncertain, which adds greater uncertainty to earnings" is Rose's description of *borrowed* liquidity.

$\Rightarrow$ D is a drawback of the other strategy.

Interpretation

Asset conversion trades certainty about the cost of liquidity (the bank knows what it gives up) for a permanent drag on returns; borrowing trades a lighter balance sheet for uncertain cost and availability.

Why the other options are wrong

A is incorrect. The opportunity cost, which is why assets with the least profit potential should be sold first.

B is incorrect. Transactions costs and the risk of a falling market are both on Rose's list.

C is incorrect. The securities sold are often low-risk ones that give the impression the firm is strong.

LO: LTR-5 b: Compare strategies that a bank can use to meet demands for additional liquidity. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.4, Asset Liquidity Management (or Asset Conversion) Strategies

Solution LTR-5 b.3 — Answer: C

Bank Q213 • BT answers P2.T8.20.6.2

Step 1 – Test C

Borrowing liquidity is "the most risky approach to solving liquidity problems" because of interest rate volatility and the speed with which credit availability can change. A risk-minimising bank prefers asset conversion, which smaller institutions use because they find it less risky.
⇒ C is false, so it is the answer.

Interpretation

Each true statement names a real advantage: B is borrowing's "leave the asset portfolio unchanged", D is the reason most firms choose the balanced middle.

Why the other options are wrong

A is incorrect. True: Rose's three characteristics of a liquid asset.
B is incorrect. True: selling assets to meet withdrawals shrinks the firm; borrowing does not.
D is incorrect. True: the trade-off a balanced strategy manages.

LO: LTR-5 b: Compare strategies that a bank can use to meet demands for additional liquidity. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.4 Strategies for Liquidity Managers

Solution LTR-5 b.4 — Answer: A

Bank Q206, Q207 • GARP book Ch.5 Q6, Q7

Step 1 – Deposit drain of 500

Asset conversion: securities fall from 1,000 to 500 and deposits from 4,000 to 3,500; total assets $5{,}100 - 500 = 4{,}600$. Liability management: deposits fall 500, borrowings rise 500, assets untouched at 5,100.

Step 2 – New loans of 600

Asset conversion: securities 1,000 to 400, loans 4,000 to 4,600; total assets still 5,100. Liability management: loans rise 600 and borrowings rise 600; total assets 5,700.
⇒ A matches the deposit-drain results.

Interpretation

Rose's point: "selling assets to provide liquidity for liability-derived demands, such as deposit withdrawals, shrinks the size of a financial firm", while borrowing leaves the volume and composition of the asset portfolio unchanged.

Why the other options are wrong

B is incorrect. Swaps the two strategies.
C is incorrect. Selling securities to fund loans reshuffles assets; the total stays at 5,100 (5,700 is the borrowing outcome).
D is incorrect. Borrowing adds a liability *and* an asset, so the bank grows to 5,700.

LO: LTR-5 b: Compare strategies that a bank can use to meet demands for additional liquidity. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.4, Borrowed Liquidity (Liability) Management Strategies

Solution LTR-5 b.5 — Answer: B

New • built from GARP Ch.5

Step 1 – Why B is correct

Rose: liability management techniques "are used most extensively by the largest banks, which often borrow close to 100 percent of their liquidity needs"; borrowing liquidity "is the most risky approach ... (but also carries the highest expected return)".

Interpretation

Vardeez's discriminator: size goes with risk appetite. A distractor that keeps one half and flips the other ("the largest banks borrow because it is safest") passes a careless read.

Why the other options are wrong

A is incorrect. Polarity flip: smaller institutions use *asset conversion*, which they find less risky than borrowing.

C is incorrect. Uses a true advantage (borrow only when needed) to reach a false conclusion: borrowing is the riskiest approach.

D is incorrect. Sibling swap: the offer rate is the control lever of liability management, and liquid assets carry the lowest returns.

LO: LTR-5 b: Compare strategies that a bank can use to meet demands for additional liquidity. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.4 Strategies for Liquidity Managers

Solution LTR-5 b.6 Answer: A

New • built from GARP Ch.5

Step 1 – Why A is correct

Rose: under a balanced strategy "some of the expected demands for liquidity are stored in assets (principally holdings of marketable securities), while other anticipated liquidity needs are backstopped by advance arrangements for lines of credit ... Unexpected cash needs are typically met from near-term borrowings."

Interpretation

Longer-term needs complete the picture: they are planned for and parked in short- and medium-term assets that turn into cash as the needs arise.

Why the other options are wrong

B is incorrect. Reverses the roles: expected demands are stored in securities, and it is the *unexpected* needs that are borrowed.

C is incorrect. Pure liability management, not a balance.

D is incorrect. Pure asset conversion, not a balance.

LO: LTR-5 b: Compare strategies that a bank can use to meet demands for additional liquidity. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.4, Balanced Liquidity Management Strategies

Solution LTR-5 b.7 Answer: B

New • built from GARP Ch.5

Step 1 – Why B is correct

Rose's first guideline: "the liquidity manager must keep track of the activities of all departments using and/or supplying funds. Whenever the loan department grants a new credit line to a customer, for example, the liquidity manager must prepare for possible drawings against that line."

Interpretation

The four guidelines (track all departments, know the big customers' plans, keep priorities clear, analyse continuously) are what make any of the three strategies work.

Why the other options are wrong

A is incorrect. The old view: today liquidity management "has generally been relegated to a supporting role" behind making loans and supplying fee-generating services.

C is incorrect. Needs must be analysed on a continuing basis; excess liquidity not reinvested the

day it occurs loses income.

D is incorrect. The second guideline: know in advance when the biggest credit and funds-supplying customers plan to withdraw or add funds.

LO: LTR-5 b: Compare strategies that a bank can use to meet demands for additional liquidity. • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.4, Guidelines for Liquidity Managers

LTR-5 • Liquidity and Reserves Management: Strategies and Policies • Vardeez: MODERATE

LTR-5 c

P1 BOTH

Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators).

Workbook: A26 Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A26 (recalled in August 2026 for its TSECF and TSAA numericals, LTR-7). Map #64 (2 of 16 sources) reports a "liquidity risk trend, seasonal component question": decomposing a liquidity requirement into trend, seasonal and cyclical components, which is this LO's sources and uses of funds method.
Vardeez: MODERATE reading; no separate pointer, but his notes work all three methods and list the traps (factors descend 0.95, 0.30, 0.15; loan gap at 1.00; legal reserves inside each bracket; the hot money ratio versus hot money liabilities).

Reported exam shapes

- **A26 S2** Liquidity requirement built from trend, seasonal and cyclical components, then the liquidity deficit or surplus (workbook A26.4).

What the questions look like: numerical tables. A forecast of deposits and loans by month or week (find the deficit months, or build a week's totals from trend, seasonal and cyclical elements); a structure-of-funds reserve with legal reserves netted out and a loan gap; a probability-weighted expected requirement; one liquidity ratio from a balance sheet. Statement questions ask which way an indicator points.

1. Four ways to estimate liquidity needs

Rose and Hudgins describe the **sources and uses of funds**, **structure of funds** and **liquidity indicator** approaches, the three this LO names, and a fourth, the **market signals (discipline)** approach. Each yields only an approximation, so estimates are fine-tuned as information arrives. Liquidity reserves have a **planned component** (what the latest forecast calls for) and a **protective component** (an extra margin, large or small depending on management's attitude to the risk of a "cash-out").

2. Method 1: sources and uses of funds

Two simple facts: liquidity **rises** as deposits increase and loans decrease; liquidity **declines** as deposits decrease and loans increase. When sources (deposit growth, loan run-off) exceed uses (deposit run-off, loan growth), there is a **positive liquidity gap (surplus)** to invest; when uses exceed sources, a **negative gap (deficit)** to fund from the cheapest and most timely sources.

Steps. (1) Forecast loans and deposits for the planning period; (2) calculate the estimated change in each; (3) compare them:

$$\boxed{\text{Liquidity deficit } (-) \text{ or surplus } (+) = \text{Estimated } \Delta\text{Deposits} - \text{Estimated } \Delta\text{Loans}} \quad \text{(eq. 5.4)}$$

The forecasts come from statistical models plus judgement. Loans are a function of projected economic growth (GDP, business sales), corporate earnings, money supply growth, the prime rate less the commercial paper or CD rate, and inflation (eq. 5.2). Deposits are a function of growth in personal income, retail sales, money supply growth, money market yields on deposits and inflation (eq. 5.3).

Correction to the source notes. Schweser writes that the estimated surplus or deficit "is therefore equal to the estimated change in deposits subtracted from the estimated change in loans", which is $\Delta\text{Loans} - \Delta\text{Deposits}$ and gives every gap the wrong sign. Rose's eq. 5.4 and Table 5.2 use $\Delta\text{Deposits} - \Delta\text{Loans}$ (for example $-100 - 50 = -150$ in week 2).

Trend, seasonal and cyclical components

A simpler route splits each forecast of deposits (or other sources) and loans (or other uses) into three elements and adds them:

1. **Trend**: a constant-growth line through year-end, quarterly or monthly totals over at least the last 10 years.
2. **Seasonal**: how deposits and loans are expected to behave in that week or month for seasonal reasons, compared with the most recent year-end level. Rose compares the 10-year average for each week with the average for the final week of December and assumes the same seasonal pattern applies this year.
3. **Cyclical**: positive or negative deviations from the trend-plus-seasonal path, depending on the strength of the economy. Rose measures it as last year's **actual** level for that week minus last year's **trend-plus-seasonal** estimate, and assumes the same cyclical pressure recurs this year.

$$\text{Estimated total} = \text{Trend} + \text{Seasonal} + \text{Cyclical}, \qquad \text{Cyclical} = \text{Actual}_{\text{last year}} - (\text{Trend} + \text{Seasonal})_{\text{last year}}$$

Example (Rose, Tables 5.1 and 5.2). A bank's deposits stood at \$1,200 million and loans at \$800 million at year-end; trend growth has been 10% a year for deposits and 8% for loans. The deposit forecast for the first six weeks:

Week	Trend	Seasonal	Cyclical	Total deposits
January 1	1,210	−4	−6	1,200
January 2	1,212	−54	−58	1,100
January 3	1,214	−121	−93	1,000
January 4	1,216	−165	−101	950
February 1	1,218	+70	−38	1,250
February 2	1,220	+32	−52	1,200

The loans are built the same way (for example January week 3: $801 + 174 - 25 = 950$; February week 1: $803 + 27 - 80 = 750$). Then eq. 5.4 is applied week by week:

Week	Deposits	Loans	ΔDeposits	ΔLoans	Deficit (−) / surplus (+)
January 1	1,200	800			
January 2	1,100	850	−100	+50	−150
January 3	1,000	950	−100	+100	−200
January 4	950	1,000	−50	+50	−100
February 1	1,250	750	+300	−250	+550
February 2	1,200	900	−50	+150	−200

⇒ Raise funds in weeks 2, 3, 4 and 6 from the cheapest and most reliable sources, and invest the week-5 surplus. Management first checks which liquid assets are available, then whether credit lines with its principal correspondent are large enough.

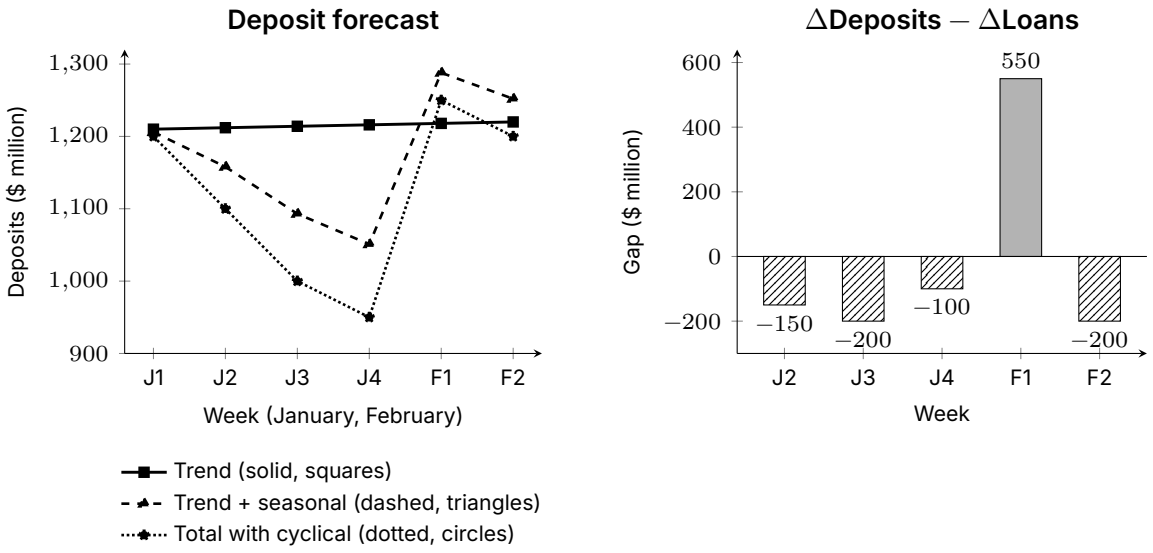


Figure 1. Rose's Tables 5.1 and 5.2. Left: the deposit forecast built up from the trend (solid), plus the seasonal element (dashed), plus the cyclical element (dotted); the January seasonal and cyclical elements both pull deposits down. Right: the weekly liquidity gap; hatched bars are deficits, the grey bar the week-5 surplus.

3. Method 2: structure of funds

Step 1. Divide deposits and other funds sources by their estimated probability of being withdrawn:

- **"Hot money" liabilities** (volatile liabilities): deposits and borrowed funds (such as federal funds borrowings) that are very interest-sensitive or that management is sure will be withdrawn this period;
- **vulnerable funds**: customer deposits of which a substantial portion, perhaps 25% to 30%, will probably be withdrawn this period;
- **stable funds** (core deposits or core liabilities): unlikely to be removed, except for a minor percentage.

Step 2. Set aside liquid funds (liquid assets or borrowing capacity) by an operating rule for each, **net of the required legal reserves** already held behind it, for example 95%, 30% and 15%:

$$\text{Liability reserve} = 0.95(\text{Hot} - \text{LR}) + 0.30(\text{Vuln.} - \text{LR}) + 0.15(\text{Stable} - \text{LR}) \quad \text{(eq. 5.5)}$$

Step 3. Loans. The bank must be ready at all times to make every **good loan** (customers who meet its quality standards): loan customers bring deposits and fee income, and relationship banking sells them other services. So it holds **100%** of the gap between the maximum potential loans and the actual loans outstanding:

$$\boxed{\text{Total liquidity requirement} = \text{eq. } 5.5 + 1.00 \times (\text{Potential loans} - \text{Actual loans})} \quad \text{(eq. 5.6)}$$

The percentages are management's judgement and differ by bank: read them, and the reserve rule, from the stem (Rose's worked example uses 3% on every category; his Queen Savings problem applies 3% to checkable deposits only).

Example (Rose, Table 5.3). First National Bank has hot money $25 million, vulnerable funds $24 million and stable funds $100 million; legal reserves are 3%; reserve rules 95%, 30%, 15%. Loans are

\$135 million, recently as high as \$140 million, with 10% trend growth, so potential loans are $140 \times 1.10 = 154$.

$$0.95(25 - 0.75) + 0.30(24 - 0.72) + 0.15(100 - 3) + (154 - 135)$$
$$= 23.04 + 6.98 + 14.55 + 19.00 = \$63.57 \text{ million}$$

$\Rightarrow$ About \$64 million held in liquid assets and additional borrowing capacity. The loan term is the largest single piece.

Probabilities: best, most likely and worst cases

Management can define the **worst** possible liquidity position (deposits at the bottom of their historical track, loan demand at its top), the **best** (maximum deposits, minimum loans) and the most likely in between, assign probabilities, and compute

$$\text{Expected liquidity requirement} = \sum_i \text{Prob}(i) \times \text{Surplus or deficit in outcome } i \quad \text{(eq. 5.7)}$$

Example (Rose). Best: deposits \$170m, loans \$110m, surplus $+60$, probability 15%. Most likely: \$150m, \$140m, $+10$, 60%. Worst: \$130m, \$150m, -20, 25%.

$$0.15(+60) + 0.60(+10) + 0.25(-20) = 9 + 6 - 5 = +\$10 \text{ million}$$

$\Rightarrow$ Plan to invest an expected \$10 million surplus next week, with a contingency plan for the \$20 million worst case.

4. Method 3: liquidity indicators

Banks also judge needs from experience and industry averages, using balance-sheet ratios. What matters is the **direction** of each ratio and its **change**, judged against peers of similar size in similar markets.

Indicator	Definition	A rise means
Cash position indicator	(Cash + deposits due from depository institutions) / total assets	More liquid
Liquid securities indicator	US government securities / total assets	More liquid
Net federal funds and repo position	(Fed funds sold + reverse repos − Fed funds purchased − repos) / total assets	More liquid
Hot money ratio	Money market assets / volatile liabilities	More liquid
Core deposit ratio	Core deposits (total deposits less those over \$100,000) / total assets	More liquid
Capacity ratio	Net loans and leases / total assets	Less liquid
Pledged securities ratio	Pledged securities / total security holdings	Less liquid
Deposit brokerage index	Brokered deposits / total deposits	Less liquid
Deposit composition ratio	Demand deposits / time deposits	Less stable funding
Loan commitments ratio	Unused loan commitments / total assets	Greater future needs

- The **hot money ratio** has *money market assets* (cash and due from banks, short-term securities, Fed funds loans, reverse repos) on top and *volatile liabilities* (large CDs, Eurocurrency deposits, Fed funds borrowings, repos) below: it shows whether volatile funding is matched by assets that could be sold quickly. Do not confuse it with the hot money *liabilities* of Method 2.

- About half the indicators measure **stored liquidity** (liquid assets); they fall in booms of rising loan demand and rise in sluggish periods. The others measure **purchased liquidity** and commitments; they rise quickly when credit demand is heavy.

Example (Rose's Concept Check 5.13). Net loans and leases \$3,502m; cash and deposits at other banks \$633m; Fed funds sold \$48m; US government securities \$185m; Fed funds purchased \$62m; demand deposits \$988m; time deposits \$2,627m; total assets \$4,446m. Cash position $633/4{,}446 = 14.24\%$; liquid securities $185/4{,}446 = 4.16\%$; net Fed funds $(48 - 62)/4{,}446 = -0.31\%$; capacity ratio $3{,}502/4{,}446 = 78.77\%$; deposit composition $988/2{,}627 = 37.61\%$.

5. The market signals (discipline) approach

Many analysts see the market's verdict as the ultimate standard: no firm knows it has enough liquidity until it has passed the market's test. Watch for: (1) loss of **public confidence**; (2) a falling **stock price**; (3) **risk premiums** on CDs and other borrowings above those of similar institutions; (4) **loss sales of assets** to meet liquidity demands; (5) inability to meet **commitments to credit customers**; (6) larger and more frequent **central bank borrowings**. A yes to any of them calls for a review of liquidity policy.

Formula card: LTR-5 c

Sources and uses	Deficit $(-)$ or surplus $(+) = \Delta$Deposits $-\Delta$Loans, period by period from levels
Components	Total = Trend + Seasonal + Cyclical; Cyclical = last year's actual − last year's (trend + seasonal)
Structure of funds	a(Hot − LR) + b(Vulnerable − LR) + c(Stable − LR) + 1.00(Potential − Actual loans)
Typical factors	$a = 0.95$, $b = 0.30$ (25% to 30%), $c = 0.15$ or less; use the stem's own
Potential loans	recent peak $\times(1 +$ growth$)$ in Rose's example
Expected requirement	$\sum_i p_i \times \text{gap}_i$ (best, most likely, worst)
Indicators	cash, liquid securities, net Fed funds, hot money, core deposits: higher is more liquid; capacity, pledged, brokerage, deposit composition, commitments: higher is less liquid

Exam traps

- **Gap sign.** ΔDeposits $-\Delta$Loans; loan growth is a use, deposit growth a source. Reversing it turns every deficit into a surplus.
- **Changes, not levels.** Deposits below loans in every month does not mean a deficit every month.
- **Cyclical element.** Actual minus estimate (last year), not estimate minus actual; and do not drop it.
- **Legal reserves inside each bracket**, before the factor. The loan term is at 1.00 and is excluded when only the liability (deposit) reserve is asked.
- **Factors descend with stability:** hot money highest, stable funds lowest.

- **Expected requirement** is probability-weighted over all outcomes, surpluses included: not the simple average, not the deficits only, not the worst case.
- **Hot money ratio** is an asset ratio (higher is more liquid); **capacity, pledged, brokerage, deposit composition and commitments** ratios point the other way. A *fall* in one of those is good news.

Practice questions: LTR-5 c

Question LTR-5 c.1

Bank Q216 • GARP book Ch.5 end-of-chapter Q3 (rebuilt)

First National Bank of Belle Mead has forecast its deposits and loans for the next eight months (USD million):

Month	Checkable deposits	Time and savings deposits	Commercial loans	Consumer loans
January	120	550	650	160
February	115	500	650	230
March	100	500	700	210
April	90	485	700	175
May	105	465	710	160
June	80	490	700	200
July	90	525	700	175
August	100	515	675	150

Using the sources and uses of funds approach, in which months is the bank likely to face a liquidity deficit?

A. February, March and June

B. February, March, April and May

C. April, July and August

D. Every month from January to August

Question LTR-5 c.2

Bank Q230 • Mock 1 Q34 (rebuilt; key as corrected in the bank)

A bank has the following deposit and nondeposit liabilities, broken down into three categories:

Category	Amount (USD million)	Desired reserve
Hot money	50	80%
Vulnerable funds	55	30%
Core funds	100	20%

Legal reserves are 3% of each category, and the desired reserve percentages are applied after deducting legal reserves. Actual loans are USD 165 million, with the potential to reach USD 225 million, and the bank wishes to be able to meet all good loans. The bank's liability liquidity reserve (excluding the loan requirement) is closest to:

A. USD 60 million

B. USD 74 million

C. USD 77 million

D. USD 134 million

Question LTR-5 c.3 New • built from GARP Ch.5, Concept Check 5.11

A thrift's liquidity division estimates that it holds USD 19 million in hot money deposits and other IOUs, against which it will hold an 80% liquidity reserve; USD 54 million in vulnerable funds, against which it plans a 25% reserve; and USD 112 million in stable (core) funds, against which it will hold a 5% reserve. Legal reserve requirements are 3% on all of these liabilities, and the liquidity percentages apply net of them. Loans total USD 117 million today but recently reached USD 132 million, and loans are expected to grow 8% a year. The thrift wants to be ready to meet all good loans. Using the structure of funds approach, the thrift's total liquidity requirement is closest to:

A. USD 33.27 million

B. USD 42.63 million

C. USD 58.83 million

D. USD 75.08 million

Question LTR-5 c.4 Bank Q220 • GARP book Ch.5 end-of-chapter Q8 (rebuilt)

Abigail Savings Bank's liquidity manager estimates the bank's liquidity position next month as follows:

Outcome	Probability
USD 375 million deficit	15%
USD 200 million deficit	35%
USD 100 million surplus	35%
USD 250 million surplus	15%

Which of the following best states the bank's expected liquidity requirement and the appropriate response?

A. A deficit of USD 126.25 million; arrange funding of that amount.

B. A deficit of USD 56.25 million; arrange funding of that amount.

C. A deficit of USD 53.75 million; arrange funding of about that amount and keep a contingency plan for the worst case.

D. A surplus of USD 53.75 million; plan to invest it in earning assets.

Question LTR-5 c.5

Bank Q223 • AnalystPrep Q.3894

The following financial information pertains to Daylight Bank (USD million):

Cash and due from depository inst.	546	Demand deposits	741
US Treasury securities	382	Savings deposits	976
Other securities	550	Time deposits	1,351
Pledged securities	503	Total deposits	2,686
Fed funds sold and reverse repos	396	Core deposits	1,111
Net loans and leases	2,374	Brokered deposits	214
Total assets	3,431	Fed funds purchased and repos	488
		Other money market borrowings	186

What is Daylight Bank's net federal funds and repurchase agreements position?

A. -11.54%

B. -2.68%

C. 2.68%

D. 11.54%

Question LTR-5 c.6

Bank Q229 • BT answers P2.T8.20.7.3 (rebuilt)

Consider five of the indicators used in the liquidity indicator approach:

- Capacity ratio = net loans and leases / total assets
- Pledged securities ratio = pledged securities / total securities holdings
- Hot money ratio = money market assets / volatile liabilities
- Core deposit ratio = core deposits / total assets
- Deposit composition ratio = demand deposits / time deposits

A positive indicator is one where an increase signals greater liquidity, so an unexpected decrease is a warning sign; a negative indicator is the reverse. The core deposit ratio is a positive indicator. For which other indicator above would an unexpected decrease be a cause for concern?

A. Capacity ratio

B. Hot money ratio

C. Pledged securities ratio

D. Deposit composition ratio

Question LTR-5 c.7

Bank Q226 • AnalystPrep Q.5382 • Mock 3 Q45

A credit analyst is evaluating the liquidity of a small regional bank for a credit committee. Using quarterly financial statements, the analyst calculates liquidity indicators over the past three years. Which of the following trends should the analyst be most concerned about?

A. The bank's average net federal funds and repurchase agreements position has been increasing.

B. The bank's capacity ratio has been increasing.

C. The bank's pledged securities ratio has been decreasing.

D. The bank's loan commitments ratio has been decreasing.

Question LTR-5 c.8 Workbook A26.4 • S2 • Map #64

The liquidity manager of a Brazilian commercial bank forecasts deposits and loans with the sources and uses of funds approach, splitting each forecast into trend, seasonal and cyclical elements (BRL million). The cyclical element for a week is last year's actual level for that week minus last year's trend-plus-seasonal estimate, assumed to recur this year.

	Deposits	Loans
Week 2: estimated total (all three elements)	2,350	1,610
Week 3: trend element	2,416	1,605
Week 3: seasonal element	−96	+95
Week 3 last year: actual level	2,170	1,560
Week 3 last year: trend-plus-seasonal estimate	2,210	1,540

Which of the following is closest to the estimated liquidity position for week 3?

A. Deficit of BRL 180 million

B. Deficit of BRL 120 million

C. Deficit of BRL 60 million

D. Surplus of BRL 180 million

Solutions • LTR-5 c

Solution LTR-5 c.1 — Answer: A

Bank Q216 • GARP book Ch.5 Q3

Step 1 – Totals by month

Deposits (checkable + time and savings): 670, 615, 600, 575, 570, 570, 615, 615. Loans (commercial + consumer): 810, 880, 910, 875, 870, 900, 875, 825.

Step 2 – Monthly changes

ΔDeposits: $-55, -15, -25, -5, 0, +45, 0$. ΔLoans: $+70, +30, -35, -5, +30, -25, -50$ (February to August).

Step 3 – Gap $=\Delta$Deposits $-\Delta$Loans

February -125, March -45, April $+10$, May 0, June -30, July $+70$, August $+50$.
$\Rightarrow$ Deficits in February, March and June.

Interpretation

Rose's discussion: line up funding in advance for the deficit months (February is the largest), invest the April, July and August surpluses in short-term liquid assets or repay borrowings, and simply monitor May, which is balanced.

Why the other options are wrong

B is incorrect. Looked at deposits only: deposits fall in February to May, but loans fall faster in April and by the same amount in May.
C is incorrect. Sign reversed (ΔLoans $-\Delta$Deposits): those are the surplus months.
D is incorrect. Compared levels instead of changes: deposits are below loans every month, but that is the balance sheet, not the liquidity gap.

LO: LTR-5 c: Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators). • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.5, The Sources and Uses of Funds Approach, eq. (5.4)

Solution LTR-5 c.2 — Answer: B

Bank Q230 • Mock 1 Q34

Step 1 – Net each category of its 3% legal reserves

Hot money $50 \times 0.97 = 48.50$; vulnerable $55 \times 0.97 = 53.35$; core $100 \times 0.97 = 97.00$.

Step 2 – Apply the desired reserves

$$0.80(48.50) + 0.30(53.35) + 0.20(97.00) = 38.80 + 16.005 + 19.40 = \text{USD } 74.2 \text{ million}$$

$\Rightarrow$ About USD 74 million of liquid assets or borrowing capacity behind the deposits and other liabilities.

Interpretation

The loan requirement, $225 - 165 = 60$, would be added only for the *total* requirement, 134.2. The question bank marks this key as corrected from the source mock; the stem's "after deducting legal reserves" fixes the order of operations.

Why the other options are wrong

A is incorrect. The loan requirement alone, $225 - 165 = 60$.
C is incorrect. Applied the percentages to gross funds and forgot the legal reserves: 76.5.
D is incorrect. Added the loan requirement that the question excludes: $74.2 + 60 = 134.2$.

LO: LTR-5 c: Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators). • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.5, The Structure of Funds Approach, eq. (5.5)

Solution LTR-5 c.3

Answer: C

New • built from GARP Ch.5

Step 1 – Liability reserve, net of 3% legal reserves

$$0.80(19 \times 0.97) + 0.25(54 \times 0.97) + 0.05(112 \times 0.97) = 14.744 + 13.095 + 5.432 = 33.271$$

Step 2 – Loan requirement at 100%

Potential loans $= 132 \times 1.08 = 142.56$, as in Rose's example (recent peak grown at the trend rate).

$$142.56 - 117 = 25.56$$

Step 3 – Total

$$33.27 + 25.56 = \text{USD } 58.83 \text{ million}$$

$\Rightarrow$ The loan gap is nearly as large as the whole deposit-side reserve.

Interpretation

The recent peak of 132 is the signal that loan demand can return to that level; growing it at 8% gives the maximum the thrift must be able to fund.

Why the other options are wrong

A is incorrect. The liability reserve only, without the loan requirement: 33.27.
B is incorrect. Applied the growth to current loans: $33.27 + 117 \times 0.08 = 42.63$.
D is incorrect. Used the textbook factors 0.95, 0.30, 0.15 instead of the thrift's own 80%, 25%, 5%: $49.52 + 25.56 = 75.08$.

LO: LTR-5 c: Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators). • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.5, The Structure of Funds Approach, eq. (5.6), Table 5.3; Concept Check 5.11

Solution LTR-5 c.4

Answer: C

Bank Q220 • GARP book Ch.5 Q8

Step 1 – Weight each outcome

$$0.15(-375) + 0.35(-200) + 0.35(+100) + 0.15(+250) = -56.25 - 70.00 + 35.00 + 37.50$$

Step 2 – Sum

$$= -\text{USD } 53.75 \text{ million}$$

$\Rightarrow$ On average a deficit of about USD 54 million: arrange that much funding (asset sales or money market borrowing) and keep a contingency plan for the USD 375 million worst case.

Interpretation

The expected requirement is a planning figure, not a ceiling: a 15% chance of a deficit seven times larger is why Rose adds the contingency plan.

Why the other options are wrong

A is incorrect. Weighted the two deficit outcomes and ignored the surpluses: $-56.25 - 70 = -126.25$.
B is incorrect. Simple average of the four outcomes, ignoring the probabilities: $(-375 - 200 + 100 + 250)/4 = -56.25$ (by coincidence equal to the first weighted term).
D is incorrect. Right size, wrong sign: the weighted outcome is a deficit.

LO: LTR-5 c: Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators). • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.5, The Structure of Funds Approach, eq. (5.7)

Solution LTR-5 c.5

Answer: B

Bank Q223 • AnalystPrep Q.3894

Step 1 – Definition

Net federal funds and repo position = (Fed funds sold and reverse repos − Fed funds purchased and repos) / total assets.

Step 2 – Calculate

$$\frac{396 - 488}{3{,}431} = \frac{-92}{3{,}431} = -2.68\%$$

$\Rightarrow$ The bank borrows more overnight than it lends: a net overnight borrower, a weaker liquidity reading.

Interpretation

Most of the table is there for other ratios: cash position $546/3{,}431 = 15.91\%$, liquid securities $382/3{,}431 = 11.13\%$, core deposit ratio $1{,}111/3{,}431 = 32.38\%$, deposit brokerage index $214/2{,}686 = 7.97\%$. Other money market borrowings are not part of this ratio.

Why the other options are wrong

A is incorrect. Used Fed funds sold alone and attached a minus sign: $-396/3{,}431$.
C is incorrect. Reversed the subtraction (purchased minus sold).
D is incorrect. Used Fed funds sold alone: $396/3{,}431 = 11.54\%$.

LO: LTR-5 c: Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators). • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.5, Liquidity Indicator Approach

Solution LTR-5 c.6

Answer: B

Bank Q229 • BT answers P2.T8.20.7.3

Step 1 – Classify the four candidates

Hot money ratio: money market assets over volatile liabilities, so a higher value means volatile funding is better covered by quickly saleable assets: positive. Capacity, pledged securities and deposit composition ratios: negative.

Step 2 – Apply the definition

For a positive indicator, an unexpected decrease is the warning.
$\Rightarrow$ The hot money ratio.

Interpretation

This is Vardeez's confusable pair: hot money *liabilities* (Method 2) are the most volatile funds, while the hot money *ratio* (Method 3) puts liquid assets on top, so a higher ratio is good.

Why the other options are wrong

A is incorrect. The capacity ratio is negative (loans are the least liquid assets): its *rise* is the concern.
C is incorrect. The pledged securities ratio is negative: a fall frees securities for sale.
D is incorrect. The deposit composition ratio is negative: a decline "suggests greater deposit stability and a lesser need for liquidity".

LO: LTR-5 c: Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators). • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.5, Liquidity Indicator Approach

Solution LTR-5 c.7 — Answer: B

Bank Q226 • AnalystPrep Q.5382 • Mock 3 Q45

Step 1 – Classify

Higher means more liquid: net Fed funds and repo position. Higher means less liquid: capacity, pledged securities and loan commitments ratios.

Step 2 – Read each trend

A rising capacity ratio means illiquid loans are taking a larger share of assets. The other three trends are favourable: the bank is a growing net overnight lender, more securities are free to sell, and fewer commitments may be drawn.
$\Rightarrow$ Only the rising capacity ratio is adverse.

Interpretation

Rose: the capacity ratio "is really a negative liquidity indicator because loans and leases are often the most illiquid of assets". Analysts read the change, not the level, and compare with peers.

Why the other options are wrong

A is incorrect. A rising net Fed funds position improves liquidity ("liquidity tends to increase when this ratio rises").
C is incorrect. A falling pledged securities ratio leaves more securities available to sell when liquidity is needed.
D is incorrect. A falling loan commitments ratio means fewer future takedowns, so lower future liquidity needs.

LO: LTR-5 c: Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators). • **Reference:** Rose and Hudgins, Ch.11 (GARP Ch.5), 5.5, Liquidity Indicator Approach

Solution LTR-5 c.8 — Answer: A

Workbook A26.4 • S2

Step 1 – Cyclical elements for week 3

$$\text{Deposits: } 2{,}170 - 2{,}210 = -40, \qquad \text{Loans: } 1{,}560 - 1{,}540 = +20$$

$\Rightarrow$ Last year deposits ran 40 below their trend-plus-seasonal path and loans 20 above it; the same cyclical pressure is assumed to apply this year.

Step 2 – Estimated totals for week 3

$$\text{Deposits} = 2{,}416 - 96 - 40 = 2{,}280, \qquad \text{Loans} = 1{,}605 + 95 + 20 = 1{,}720$$

Step 3 – Changes from week 2

$$\Delta\text{deposits} = 2{,}280 - 2{,}350 = -70, \qquad \Delta\text{loans} = 1{,}720 - 1{,}610 = +110$$

$\Rightarrow$ Sources shrink by 70 while uses grow by 110.

Step 4 – Liquidity deficit or surplus

$$\Delta\text{deposits} - \Delta\text{loans} = -70 - 110 = -\text{BRL } 180 \text{ million}$$

Interpretation

With deposits falling and loans rising, the bank faces a BRL 180 million liquidity deficit in week 3. The manager must raise funds from the cheapest and most reliable sources, first checking which liquid assets are available and whether credit lines with correspondents are large enough.

Why the other options are wrong

B is incorrect. Dropped the cyclical element in week 3: deposits 2,320 (-30), loans 1,700 $(+90)$, gap $-30 - 90 = -120$.

C is incorrect. Sign error on the cyclical element (estimate minus actual): deposits 2,360 $(+10)$, loans 1,680 $(+70)$, gap $10 - 70 = -60$.

D is incorrect. Polarity flip: treated loan growth as a source and the deposit change as a use, $110 - (-70) = +180$.

LO: LTR-5 c: Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators). • **Reference:** Rose and Hudgins, Bank Management and Financial Services, Ch.11, 5.5 Estimating Liquidity Needs, The Sources and Uses of Funds Approach, eq. (5.4), Tables 5.1 and 5.2

LTR-7 • Monitoring Liquidity • Vardeez: MODERATE

LTR-7 d

P1 BOTH

Interpret the term structure of expected cash flows and cumulative cash flows.

Workbook: A26 Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A26, recalled in the August 2026 sitting as "TSEF and TSAA numerical" (TSEF is the book's TSECF, the term structure of expected cash flows). Map #64: 2 of 16 recall sources.

Vardeez: a MODERATE reading: "you might see one question or you may not see at all from the calculations. So do not get hung up on these calculations", but "Tsef and tsaa numerical seems imp". His pointer for this LO is the term-structure (TSECF and TSECCF) numericals.

Reported exam shapes

- **S1** TSECF and TSAA numerical, the plain form of the August recall: which term structures move and by how much.
- **S4** TSECCF with initial cash and the asset sale that keeps the TSLe non-negative (workbook A26.2).

What the questions look like: a maturity table, or a small balance sheet with expiry dates and rates, from which you cumulate the net flows and read the TSECCF at a date; or a TSECF with initial cash, where you find the most negative cumulated point and the bond notional that covers it. The single-year flow is offered against the cumulated one, the initial cash is dropped, or the interest base ignores what has already matured. One or two statement questions ask what the TSECF contains and what a positive TSECCF does, and does not, guarantee. This block carries the background for the whole reading (causes and sources, LGC, expected liquidity); LTR-7 e, a and b point back to it.

1. Causes and sources of liquidity

Castagna and Fede (GARP Chapter 7) sort everything that moves a bank's cash into two classes, and the whole reading depends on keeping them apart.

	Definition	Direction	Term structures
Causes of liquidity (Def. 7.4.1)	All factors referring to existing and forecast future contracts originated by the ordinary business of the bank.	Positive and negative	TSECF, TSECCF
Sources of liquidity (Def. 7.4.2)	All factors capable of generating positive cash flows to manage and hedge liquidity risk that the bank can dispose of promptly. They determine the liquidity generation capacity (LGC).	Positive only	TSLGC, TSCLGC

⇒ Causes create the gaps; sources fill them. A source never enters the TSECF, which is why LGC is the tool for fixing a negative cumulated position and never the thing that creates one.

Notation (eqs. 7.1 to 7.3). $cf_e^+(t_0, t_i) = E[cf^+(t_0, t_i)]$ is the sum over all contracts $d_1, \dots, d_N$ of the expected positive cash flows on date t_i, seen from the reference date t_0; $cf_e^-(t_0, t_i)$ is the same for

negative flows. The cumulated flow from t_a to t_b is

$$CF(t_0, t_a, t_b) = \sum_{i=a}^{b} \left[cf_e^+(t_0, t_i) + cf_e^-(t_0, t_i) \right] \qquad \text{(eq. 7.3)}$$

Every entry is an **expected value**, because many flows are stochastic in time, amount or both (LTR-7 a).

2. The TSECF and the TSECCF

$$\text{TSECF}(t_0, t_b) = \{cf_e^+(t_0, t_0),\ cf_e^-(t_0, t_0),\ cf_e^+(t_0, t_1),\ cf_e^-(t_0, t_1), \dots,\ cf_e^-(t_0, t_b)\} \quad (7.4)$$

$$\text{TSECCF}(t_0, t_b) = \{CF(t_0, t_0, t_1),\ CF(t_0, t_0, t_2), \dots, CF(t_0, t_0, t_b)\} \quad (7.5)$$

- **TSECF:** the collection, ordered by date, of expected positive and negative cash flows, held separately at each date, up to the expiry of the contract with the longest maturity t_b. At the end, with an indefinite expiry at the end of business, equity is reimbursed to shareholders.
- **Maturity ladder:** the TSECF is often called the maturity ladder; the authors reserve the name for its first year. Up to one year is **short-term liquidity**, beyond one year **structural liquidity**.
- **TSECCF:** the running total of the TSECF. Banks care not only about the net balance on a date but about how past flows have built the cash position by that date: a net outflow can be met from cash accumulated from earlier inflows. The reverse also holds: a short cash position left by past outflows must be financed in the meantime (with new liabilities), and later inflows repay that debt.
- **Initial cash:** when the bank holds cash at t_0, the running total starts from it, $\text{TSECCF}(t_0, t_0) = \text{Cash}(t_0)$ (Section 5).

3. Building them: GARP Example 7.4.1

Order assets and liabilities by expiry, whatever the contract type. Assets bear no default risk and deposits carry no liquidity options, so every flow is deterministic: an expiring asset is an inflow of its notional, an expiring liability an outflow. Interest is paid yearly at the average yield of each maturity bucket on everything still outstanding during the year.

Expiry (years)	Assets	Liabilities	Asset yield	Liability yield
1	20		5.00%	
2		10		4.00%
5	50		6.00%	
7		70		4.50%
10	30		6.50%	
>10		20 (equity)		
Total	100	100		

Year	Notional in	Interest in	Notional out	Interest out	TSECF (net)	TSECCF
1	20	5.95	0	-3.55	22.40	22.40
2	0	4.95	-10	-3.55	-8.60	13.80
3	0	4.95	0	-3.15	1.80	15.60
4	0	4.95	0	-3.15	1.80	17.40
5	50	4.95	0	-3.15	51.80	69.20
6	0	1.95	0	-3.15	-1.20	68.00
7	0	1.95	-70	-3.15	-71.20	$\mathbf{-3.20}$
8	0	1.95	0	0	1.95	$\mathbf{-1.25}$
9	0	1.95	0	0	1.95	0.70
10	30	1.95	0	0	31.95	32.65
>10	0	0	-20	0	-20.00	12.65

Interest columns. Year 1 inflow: $20(5\%) + 50(6\%) + 30(6.5\%) = 1.00 + 3.00 + 1.95 = 5.95$; from year 2 the 20 has matured, so 4.95; from year 6 only the 30 remains, 1.95. Year 1 outflow: $10(4\%) + 70(4.5\%) = 0.40 + 3.15 = 3.55$, and still 3.55 in year 2 because the 10 pays interest in the year it expires; then 3.15 until the 70 matures in year 7.

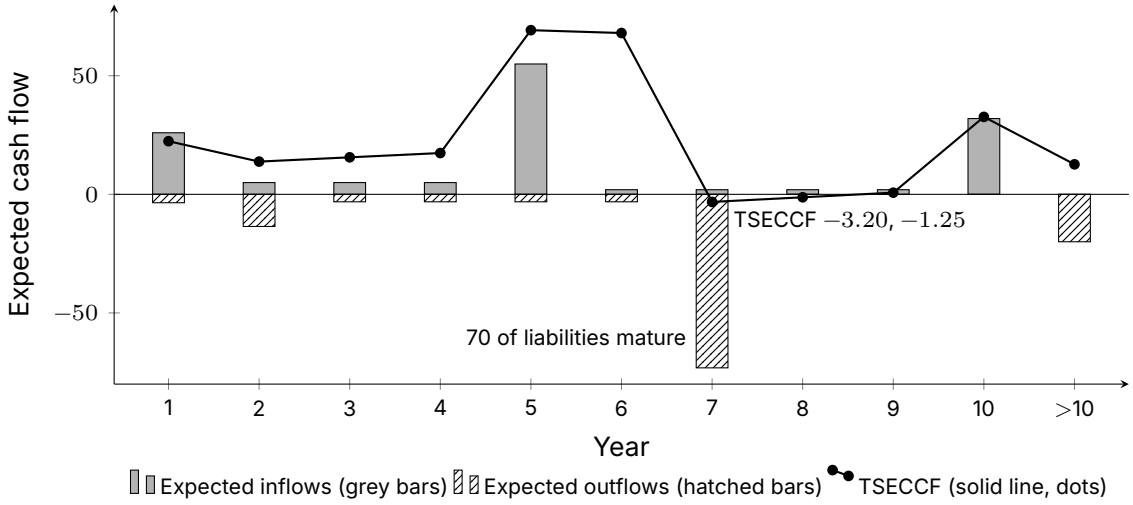


Figure 1. Example 7.4.1 with interest: the TSECF (bars, inflows above and outflows below zero) and the TSECCF (line). The year-7 outflow of 71.2 is mostly covered by the 68.0 accumulated by year 6; the cumulated position is negative only in years 7 and 8.

Correction to the source notes (GARP Table 7.3). GARP prints interest inflows of 1.69 in years 8 to 10, and so a TSECCF of -1.51, 0.18, 31.87 and 11.87. With all 30 of the 6.50% bonds still held, the coupon is $30 \times 6.5\% = 1.95$, giving -1.25, 0.70, 32.65 and 12.65 (recomputed). The 1.69 is $26 \times 6.5\%$, the coupon *after* the sale in Example 7.5.1 (Section 5), carried back into Table 7.3. The reading is unchanged: the TSECCF is negative in years 7 and 8.

Reading the picture.

- The single-year TSECF of year 7 (-71.2) is not the problem; the bank enters year 7 with 68.0 accumulated. The shortfall is the cumulated -3.2.
- The negative patch is short (two years) and small, then the year-10 redemption restores a large surplus. Such periods are accepted if they can be managed with the tools of Section 5.
- Beyond year 10 the equity of 20 is repaid. What remains (12.65) is the interest earned over the life of the balance sheet, net of interest paid.

Second example (Schweser and Vardeez: King Bank). Assets 125, liabilities 75, equity 50; 25 of assets expire at the end of year 1 and 15 of liabilities at the end of year 2; assets earn 6%, liabilities pay 5%.

$$\text{TSECCF}(1) = 25 + 0.06(125) - 0.05(75) = 25 + 7.50 - 3.75 = 28.75$$
$$\text{TSECCF}(2) = 28.75 + 0.06(100) - 15 - 0.05(75) = 28.75 + 6.00 - 15 - 3.75 = 16.00$$

$\Rightarrow$ Three ways to get it wrong: the year-2 TSECF alone (-12.75, the intermediate result); paying interest on 60 of liabilities in year 2 (16.75), although the 15 still pays in the year it expires; treating the expiring liability as an inflow (46.00).

4. What the TSECF contains, and what a positive TSECCF tells you

A real TSECF holds far more than a static balance sheet. It includes:

- cash flows from **all existing contracts**; many are stochastic because they are linked to market indices such as Libor or Euribor fixings, so interest rate models are needed to compute the expected flows;
- **credit adjustments**: credit models account for defaults on an aggregate basis, with the correlation between counterparties;
- **liquidity options**: behavioural models adjust flows for sight deposits, credit line usage and mortgage prepayment (LTR-7 b);
- flows from **new business** that increases assets, stochastic in both amount and time;
- the **rollover** of maturing liabilities and new bond issues planned to fund asset growth.

It excludes the **sources** of liquidity. Unsecured funding planned to roll over existing liabilities or to fund new business is a cause and stays in the TSECF; unsecured funding raised as a shorter-term, exceptional measure to cover a gap is a source and goes to the TSLGC.

The treasury's task is to monitor both structures. The perfect condition is a TSECCF positive at all times, so that positive flows cover negative ones. It cannot be guaranteed, for two reasons:

1. **Stochastic flows.** The TSECF forecasts only expected values, so the TSECCF does too. A TSECCF that is positive on average still comes from a distribution in which negative outcomes have some probability.
2. **Maturity mismatch.** The timing of asset and liability maturities can produce periods of negative cumulated flows, as in years 7 and 8 above.

A negative *expected* TSECCF means the bank may become insolvent and eventually go bankrupt. The unexpected side is measured with cash flow at risk (LTR-7 c); the authors note that cumulating the unexpected term structures makes little sense, because they diverge to unrealistic levels.

5. Covering negative cumulated flows: LGC and expected liquidity

Liquidity generation capacity (Def. 7.4.3) is the ability of a bank to generate positive cash flows, beyond contractual ones, from the sources of liquidity available on and off the balance sheet at a given date. It works in two ways, with repo as a third, "balance sheet neutral" case:

- **Balance sheet expansion:** new interbank deposits, withdrawals of credit lines received from other banks, new bond issues (secured or unsecured). It depends on external counterparties.

- **Balance sheet shrinkage:** selling assets, starting from the most liquid (Treasury bonds, corporate bonds, stocks), then less liquid ones (loans, buildings) over a longer horizon.

The three sources are asset sales (AS), secured funding and repo (RP) and unsecured funding (USF). AS and RP give **balance sheet liquidity** (BSL), the ground for liquidity buffers. Only unencumbered assets count, and a bond pledged for a period is excluded from the TSLGC for that period. The TSLGC collects AS, RP and USF by date; the TSCLGC is its running total (eq. 7.7). Both hold expected values, since prices, haircuts and the amount of unsecured funding available are uncertain.

Term structure of expected liquidity. Adding the two cumulated structures gives

$$\boxed{\text{TSLe}(t_0, t_b) = \{\text{TSECCF}(t_0, t_0),\ \text{TSECCF}(t_0, t_1) + \text{TSCLGC}(t_0, t_1),\ \ldots\}} \quad (7.10)$$

$$\text{TSECCF}(t_0, t_0) = \text{Cash}(t_0)$$

The TSLe must be **non-negative at every date** for the bank to stay solvent at all times. If negative expected cumulated flows cannot be excluded, insolvency cannot be prevented.

Example 7.5.1. In Example 7.4.1 the bank holds the 30 of 6.50% bonds unencumbered (TSAA of 30). In year 7, when the TSECCF turns negative, it sells 4 at the expected price of 99.00, raising 3.96, which sits in the TSCLGC from then on. The sale also changes the TSECF: coupons fall to $26 \times 6.5\% = 1.69$ and the year-10 redemption to 26.

Year	TSECF	TSECCF	TSCLGC	TSLe	TSAA	Price
6	−1.20	68.00	0	68.00	30	98.60
7	−71.20	−3.20	3.96	0.76	26	99.00
8	1.69	−1.51	3.96	2.45	26	99.50
9	1.69	0.18	3.96	4.14	26	99.75
10	27.69	27.87	3.96	31.83	0	100.00

⇒ The TSLe never goes below zero: the bank is in expected liquidity equilibrium. The minimum sale is $3.20/0.99 = 3.23$ notional; the book sells a round 4. Because the bond pays coupons, the lost coupons must be fed back into the TSECF and the later dates re-checked (LTR-7 e, Section 5). The TSLGC and the TSECF are intertwined, so building them is a recursive procedure.

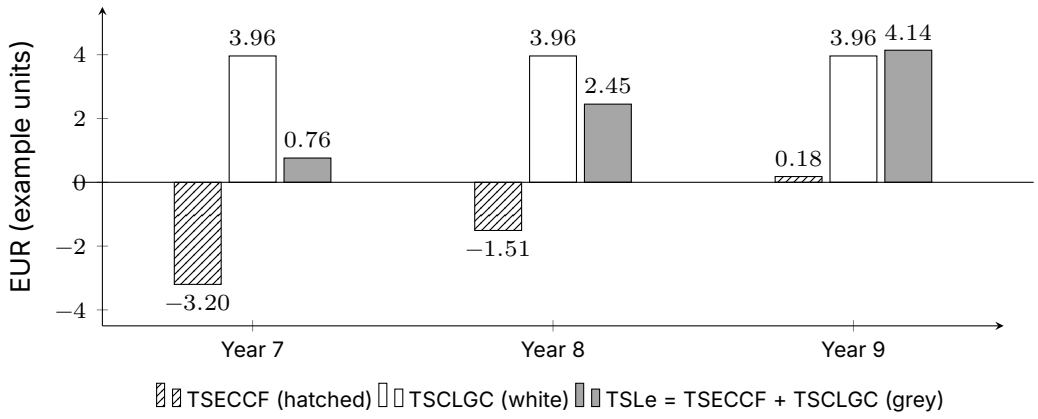


Figure 2. Example 7.5.1 in the critical years. The sale of 4 notional at 99.00 puts 3.96 into the TSCLGC; added to the negative TSECCF it keeps the TSLe at or above zero in every year.

The sale rule (workbook A26). For a bond whose sale removes no later flows (a zero-coupon bond), the minimum notional is |most negative TSECCF|/expected price. Start the TSECCF from initial cash, use the *most negative* cumulated point (not the last one, not one year's flow), and convert cash into notional through the price.

Structure	What it collects	Question it answers
TSECF	Expected inflows and outflows by date from the causes of liquidity	What arrives and leaves on each date?
TSECCF	Running total of the TSECF, from initial cash	How much cash has built up by each date?
TSLGC	Liquidity the sources (AS, RP, USF) can generate on each date	What could be raised on each date?
TSCLGC	Running total of the TSLGC	How much could have been raised by each date?
TSLe	TSECCF + TSCLGC	Is the bank solvent at every date?
TSAA	Quantity of each asset in possession (LTR-7 e)	What can still be sold or repoed?

Formula card: LTR-7 d

TSECF	Expected inflows cf_e^+ and outflows cf_e^- by date, held separately (eq. 7.4); causes of liquidity only
Cumulated flow	$CF(t_0, t_a, t_b) = \sum_{i=a}^{b} (cf_e^+ + cf_e^-)$ (eq. 7.3)
TSECCF	Running total of the TSECF; with initial cash, start from $\text{TSECCF}(t_0, t_0) = \text{Cash}(t_0)$
Interest in a year	Rate $\times$ balance outstanding during the year; an item pays or earns in the year it expires, not after
TSLe	$\text{TSECCF}(t_0, t_i) + \text{TSCLGC}(t_0, t_i) \geq 0$ at every date (eq. 7.10)
Sale to cover a gap	Notional $=$ $\lvert$most negative TSECCF$\rvert$ / expected price (zero coupon); for a coupon bond re-check later dates net of the lost coupons and redemption
Purchase	TSECF: $-N\times$ dirty price; TSAA: $+N$ until maturity; TSLGC unchanged
Maturity ladder	TSECF up to one year (short-term liquidity); beyond: structural liquidity

Exam traps

- **Intermediate result.** The TSECF of a date is the period flow; the TSECCF is the running total. GARP offers one for the other.
- **Initial cash.** $\text{TSECCF}(t_0, t_0) = \text{Cash}(t_0)$; leaving it out shifts every cumulated point down.
- **Most negative point.** The sale must cover the worst cumulated point, not the final one and not a single year's outflow.
- **Interest base.** An item pays (or earns) interest in the year it expires and stops afterwards; assets earn only on what is left.
- **Sign.** Expiring assets are inflows, expiring liabilities outflows; equity is repaid only at the end of business.
- **Notional versus cash.** Cash needed $\div$ price gives the notional to sell; a purchase moves the TSECF by price $\times$ notional but the TSAA by the notional.
- **Causes versus sources.** Planned rollovers and issues that fund new business are in the TSECF;

emergency funding, asset sales and repos are LGC.

- **Absolute qualifier.** A positive expected TSECCF does not guarantee that no shortfall occurs; a negative one warns of possible insolvency.

Practice questions: LTR-7 d

Question LTR-7 d.1

Bank Q277 • BT P2.T8.20.9.1 (rebuilt)

Geofinancial Bank has the following simplified balance sheet (in millions):

Assets		Liabilities and equity	
Bonds	30	Deposits	30
Loans	70	Bonds	50
		Equity	20

Assets expire as follows: bonds (30) in 1 year; loans: 40 in 5 years, 10 in 7 years, 20 in 10 years. Liabilities expire as follows: deposits: 10 in 1 year, 20 in 2 years; bonds: 10 in 5 years, 30 in 7 years, 10 beyond 10 years. Equity (20) is repaid to shareholders only at the end of the business, beyond 10 years. Ignore interest and assume the assets bear no default risk and no liquidity options exist. What are the net cash flow at the 2-year expiry and the TSECCF at the 7-year expiry?

A. Net -20; TSECCF $+10$

B. Net -20; TSECCF -20

C. Net 0; TSECCF $+10$

D. Net -20; TSECCF $+30$

Question LTR-7 d.2

Bank Q273 • AP Mock 3 Q74

The table gives the cash flows of institution XYZ, ordered by maturity. Assets bear no default risk and no liquidity options are embedded in the deposits.

Expiry	Notional +	Interest +	Notional −	Interest −	Net
1	20	6	0	-4	$+22$
2	0	5	-10	-4	-9
3	0	5	0	-3	$+2$
4	0	5	0	-2	$+3$
5	50	5	0	-3	$+52$
6	0	2	0	-3	-1

What is the term structure of expected cumulated cash flows (TSECCF) at the 6-year expiry?

A. 18

B. 22

C. 69

D. 70

Question LTR-7 d.3 New • built from GARP Ch.7 (Example 7.4.1)

A treasury analyst at a Canadian bank builds the TSECF from the simplified balance sheet below (CAD million). Assets earn 5.0% and liabilities cost 3.0% a year, paid at each year end on the balances outstanding during the year; an item pays or earns interest in the year in which it matures. Assets bear no default risk, the deposits carry no liquidity options, and the bank holds no cash at the reference date.

Item	Matures	Amount
Bonds held (asset)	End of year 1	60
Loans (asset)	End of year 4	140
Deposits (liability)	End of year 2	50
Bonds issued (liability)	End of year 5	110
Equity	End of business	40

Which of the following is closest to the TSECCF at the end of year 2?

A. CAD -47.8 million

B. CAD 17.4 million

C. CAD 18.9 million

D. CAD 20.4 million

Question LTR-7 d.4 New • built from GARP Ch.7 (7.4.1, 7.4.2)

The treasurer of a regional bank is updating next year's term structures. Four items are under discussion:

(i) the rollover of term deposits maturing in March, as planned in the budget;

(ii) a senior bond issue in June to fund the planned growth of the mortgage book;

(iii) the sale of unencumbered government bonds to cover a negative cumulated position in September;

(iv) a drawdown of a committed credit line received from another bank, kept for emergencies.

Which items belong in the term structure of expected cash flows (TSECF)?

A. (i) and (ii) only

B. (iii) and (iv) only

C. (i), (ii) and (iv) only

D. All four items

Question LTR-7 d.5 Bank Q274 • AnalystPrep Q.3930 (rebuilt)

Which of the following statements about the term structure of expected cash flows (TSECF) and the term structure of expected cumulated cash flows (TSECCF) is least accurate?

A. Behavioural models are used to adjust cash flows for liquidity options embedded in products such as sight deposits and prepayable mortgages.

B. The TSECF includes the cash flows of all current contracts; many are stochastic because they are linked to market indices such as Libor or Euribor.

C. Cash flows from new business are included; they are typically stochastic in both amount and timing and are handled with models that capture the related risks.

D. Because the TSECF is built from expected values, a TSECCF that is positive at every date guarantees that the bank faces no liquidity shortfall.

Question LTR-7 d.6 Bank Q269 • BT P2.T8.20.9.2 (rebuilt; filed under LTR-7 c, P4)

A treasury analyst must explain to the ALCO how the bank checks whether it can meet its obligations at every future date. Which of the following best describes the term structure of expected liquidity, TSLe?

A. The TSECF plus the TSCLGC at each date, starting from zero at the reference date.

B. The TSECCF plus the TSCLGC at each date, starting from the cash held at the reference date.

C. The TSECCF plus the term structure of available assets (TSAA) at each date, since the available assets measure the liquidity that can be raised.

D. The TSECCF less the negative cash flow at risk at each date, at a chosen confidence level.

Question LTR-7 d.7 Bank Q282 • GARP PE1 Q26 • Mock 3 Q54 (rebuilt)

A bank plans to purchase a newly issued 1-year bond with a face value of GBP 20 million. The bond has an annualized coupon rate of 8%, paid semi-annually, and is priced at 102% of par. A treasury analyst must describe the impact of the purchase on the cumulative expected cash flows (TSECCF) and the term structure of available assets (TSAA). Which of the following correctly describes the change in both, measured at a 7-month time horizon from the purchase date?

A. The cumulative cash flows will decrease by GBP 20 million and the TSAA will increase by GBP 20.4 million.

B. The cumulative cash flows will decrease by GBP 19.6 million and the TSAA will increase by GBP 20 million.

C. The cumulative cash flows will increase by GBP 19.6 million and the TSAA will decrease by GBP 20 million.

D. The cumulative cash flows will increase by GBP 20 million and the TSAA will decrease by GBP 20.4 million.

Question LTR-7 d.8 Workbook A26.2 • S4 • self-gen

A treasurer at a French bank has built the TSECF below (EUR million, expected values). Initial cash is EUR 1.5 million. The bank holds 20 notional of an unencumbered zero-coupon government bond whose expected price is 98.40 throughout. Which of the following is closest to the minimum notional of the bond that must be sold to keep the term structure of expected liquidity non-negative at every date?

Year	1	2	3	4	5	6
TSECF	15.0	−6.0	2.0	−16.5	1.2	−0.9

A. EUR 3.76 million

B. EUR 4.00 million

C. EUR 4.07 million

D. EUR 5.59 million

Solutions • LTR-7 d

Solution LTR-7 d.1

Answer: A

Bank Q277 • BT P2.T8.20.9.1

Step 1 – Net flow at each expiry (the TSECF)

Expiring assets are inflows, expiring liabilities outflows:

$$\text{Y1: } 30-10=+20, \quad \text{Y2: } 0-20=-20, \quad \text{Y5: } 40-10=+30$$

$$\text{Y7: } 10-30=-20, \quad \text{Y10: } +20, \quad >10\text{: } -(10+20)=-30$$

$\Rightarrow$ The 2-year entry of the TSECF is a net outflow of 20.

Step 2 – Cumulate to year 7 (the TSECCF)

$$20 \;\rightarrow\; 20-20=0 \;\rightarrow\; 0+30=30 \;\rightarrow\; 30-20=+10$$

$\Rightarrow$ By year 7 the bank has accumulated 10 net of everything paid out.

Interpretation

The year-2 outflow is met from the year-1 inflow, so the cumulated position never turns negative. It reaches 30 in year 10 and returns to zero once the long bonds and the equity are repaid, as it must when assets equal liabilities plus equity and interest is ignored.

Why the other options are wrong

B is incorrect. Reported the year-7 net flow (-20) as the TSECCF.
C is incorrect. Reported the year-2 TSECCF $(20-20=0)$ as the year-2 net flow.
D is incorrect. Stopped cumulating at year 5 $(+30)$ and missed the year-7 outflow.

LO: LTR-7 d: Interpret the term structure of expected cash flows and cumulative cash flows. • **Reference:** Castagna and Fede, Measuring and Managing Liquidity Risk, Ch.6 (GARP Ch.7), 7.4.1, Example 7.4.1, Table 7.1, Figure 7.4

Solution LTR-7 d.2

Answer: C

Bank Q273 • AP Mock 3 Q74

Step 1 – Net flow per expiry

Sum the four columns: $+22,\ -9,\ +2,\ +3,\ +52,\ -1$.

Step 2 – Running total

$$22, \quad 22-9=13, \quad 15, \quad 18, \quad 70, \quad 70-1=69$$

$\Rightarrow$ The TSECCF at the 6-year expiry is 69.

Interpretation

The only net outflows (years 2 and 6) are small against what has accumulated, so the cumulated position stays positive throughout: the treasury's "perfect condition". The TSECCF at a date includes every flow up to and including that date.

Why the other options are wrong

A is incorrect. Stopped at year 4: 18.
B is incorrect. Took the year-1 figure, 22, which is both the first TSECF and the first TSECCF entry.
D is incorrect. Stopped at year 5: 70, leaving out the year-6 net outflow of 1.

LO: LTR-7 d: Interpret the term structure of expected cash flows and cumulative cash flows. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.4.1, eqs. (7.3) to (7.5), Table 7.3

Solution LTR-7 d.3 — Answer: B

New • built from GARP Ch.7

Step 1 – Year 1

$$\text{TSECF}(1) = \underbrace{60}_{\text{bonds mature}} + \underbrace{0.05 \times 200}_{10.0} - \underbrace{0.03 \times 160}_{4.8} = 65.2 = \text{TSECCF}(1)$$

$\Rightarrow$ No initial cash, so the first cumulated value equals the first period flow.

Step 2 – Year 2: interest bases after the bond redemption

Assets now earn on the 140 of loans; the deposits still pay interest in the year they mature, so liabilities pay on the full 160:

$$\text{TSECF}(2) = 0.05 \times 140 - 50 - 0.03 \times 160 = 7.0 - 50 - 4.8 = -47.8$$

Step 3 – Cumulate

$$\text{TSECCF}(2) = 65.2 - 47.8 = \text{CAD } 17.4 \text{ million}$$

Interpretation

The deposit repayment in year 2 is funded by the bond redemption in year 1, so the cumulated position stays positive although the year-2 flow is a large outflow. Equity is a red herring here: it is repaid only at the end of business.

Why the other options are wrong

A is incorrect. Intermediate result: the year-2 TSECF alone, -47.8.

C is incorrect. Stopped paying interest on the deposits in their final year (liability base 110): $65.2 + 7.0 - 50 - 3.3 = 18.9$.

D is incorrect. Kept earning interest on the matured bonds (asset base 200): $65.2 + 10.0 - 50 - 4.8 = 20.4$.

LO: LTR-7 d: Interpret the term structure of expected cash flows and cumulative cash flows. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.4.1, Example 7.4.1, Tables 7.2 and 7.3

Solution LTR-7 d.4 — Answer: A

New • built from GARP Ch.7

Step 1 – Causes of liquidity go in the TSECF

Items (i) and (ii) are flows of existing and forecast contracts from ordinary business (Definition 7.4.1). GARP is explicit that debt "planned to roll over existing contracts or to fund new business" is included in the TSECF and the TSECCF.

Step 2 – Sources of liquidity go in the TSLGC

Item (iii) is an asset sale (AS, balance sheet shrinkage) and item (iv) unsecured funding (USF, balance sheet expansion). Both are sources (Definition 7.4.2): they generate liquidity to cover the gaps that the causes create, and they enter the TSLGC and the TSCLGC.

$\Rightarrow$ Only (i) and (ii) belong in the TSECF.

Interpretation

The split matters for the TSLe = TSECCF + TSCLGC: putting a source into the TSECF counts it twice, and hides how much of the bank's coverage depends on external counterparties.

Why the other options are wrong

B is incorrect. Reversed: (iii) and (iv) are the sources, (i) and (ii) the causes.

C is incorrect. The emergency drawdown of a received credit line is shorter-term exceptional

funding, a source in the TSLGC.
D is incorrect. Not every flow is a cause; sales and emergency funding are LGC.

LO: LTR-7 d: Interpret the term structure of expected cash flows and cumulative cash flows. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), Definitions 7.4.1 and 7.4.2; 7.4.2 Liquidity Generation Capacity

Solution LTR-7 d.5 — Answer: D

Bank Q274 • AnalystPrep Q.3930

Step 1 – Check A to C against the build of the TSECF

Behavioural models for liquidity options (A), all current contracts with index-linked, stochastic flows (B) and new business modelled as stochastic in amount and time (C) are all part of the TSECF.

Step 2 – Test D

The TSECF forecasts expected values only, so the TSECCF does too. A positive expected cumulated balance still comes from a distribution that can show negative outcomes with some probability.
$\Rightarrow$ D is the least accurate statement.

Interpretation

The converse is the warning sign: a negative expected TSECCF means the bank may become insolvent. Unexpected flows are measured separately with cash flow at risk (LTR-7 c).

Why the other options are wrong

A is incorrect. Accurate: behavioural models adjust flows for sight deposits, credit line usage and prepayments.
B is incorrect. Accurate: interest rate models give expected values of index-linked flows.
C is incorrect. Accurate: new business is included and modelled with all related risks.

LO: LTR-7 d: Interpret the term structure of expected cash flows and cumulative cash flows. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.4.1, discussion following Example 7.4.1

Solution LTR-7 d.6 — Answer: B

Bank Q269 • BT P2.T8.20.9.2

Step 1 – Equation 7.10

$$\text{TSLe}(t_0, t_b) = \{\text{TSECCF}(t_0, t_0),\ \text{TSECCF}(t_0, t_1) + \text{TSCLGC}(t_0, t_1), ...\}$$
$$\text{TSECCF}(t_0, t_0) = \text{Cash}(t_0)$$

$\Rightarrow$ Both building blocks are cumulated, and the first entry is the cash on hand.

Interpretation

The TSLe asks whether cash built up from business flows plus the liquidity that sources can have generated by each date covers the obligations. It must be non-negative at every date.

Why the other options are wrong

A is incorrect. Uses period flows instead of the cumulated TSECCF and drops the initial cash.
C is incorrect. The TSAA is a quantity of securities, not liquidity: price (and the haircut in a repo) converts it into LGC.
D is incorrect. Cash flow at risk belongs to the unexpected (at-risk) term structures, not to expected liquidity.

LO: LTR-7 d: Interpret the term structure of expected cash flows and cumulative cash flows. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.5 The Term Structure of Expected Liquidity, eq. (7.10)

Solution LTR-7 d.7

Answer: B

Bank Q282 • GARP PE1 Q26

Step 1 – Purchase outflow in the TSECF

$$20 \times 1.02 = \text{GBP } 20.4 \text{ million}$$

A new issue carries no accrued interest, so the dirty price is 102.

Step 2 – Coupon inside the 7-month horizon

$$20 \times \frac{8\%}{2} = \text{GBP } 0.8 \text{ million at month 6}$$

Step 3 – Change in the TSECCF at month 7

$$-20.4 + 0.8 = -\text{GBP } 19.6 \text{ million}$$

Step 4 – Change in the TSAA

The TSAA records the quantity in possession: +GBP 20 million notional until maturity, when it falls to zero.

Interpretation

The purchase is a cause of liquidity: the price and the coupons are contract flows in the TSECF. The TSLGC is not affected now, but the bond can later be sold or repoed, which is why the TSAA tracks it.

Why the other options are wrong

A is incorrect. Swapped the amounts (face value for the cash flow, price for the TSAA) and ignored the coupon.

C is incorrect. Signs reversed: a purchase is an outflow and raises the TSAA.

D is incorrect. Signs reversed and amounts swapped.

LO: LTR-7 d: Interpret the term structure of expected cash flows and cumulative cash flows. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.4.3 The Term Structure of Available Assets, Example 7.4.2, Table 7.5

Solution LTR-7 d.8

Answer: C

Workbook A26.2 • S4 • self-gen

Step 1 – Build the TSECCF, starting from initial cash

$$\text{Yr 1: } 1.5 + 15.0 = 16.5, \quad \text{Yr 2: } 16.5 - 6.0 = 10.5, \quad \text{Yr 3: } 10.5 + 2.0 = 12.5$$

$$\text{Yr 4: } 12.5 - 16.5 = -4.0, \quad \text{Yr 5: } -4.0 + 1.2 = -2.8, \quad \text{Yr 6: } -2.8 - 0.9 = -3.7$$

$\Rightarrow$ Expected cumulated cash turns negative in year 4 and stays negative; the worst point is -4.0.

Step 2 – Liquidity needed from the LGC

The TSCLGC must cover the most negative TSECCF: EUR 4.0 million (years 5 and 6 need less).

Step 3 – Notional to sell

$$\frac{4.0}{0.9840} = \text{EUR } 4.07 \text{ million notional}$$

Step 4 – Check the TSLe

TSCLGC $= 4.065 \times 0.984 = 4.0$; TSLe: $-4.0 + 4.0 = 0,\ -2.8 + 4.0 = 1.2,\ -3.7 + 4.0 = 0.3$.

Interpretation

Selling 4.07 notional keeps the bank in expected liquidity equilibrium; the TSAA of the bond drops from 20 to 15.93 from year 4. Being zero-coupon, the sale removes no later coupon inflows; for a coupon bond the lost flows must be fed back into the TSECF. (Covering the single-year TSECF of year 4 instead of the cumulated position would sell $16.5/0.984 = 16.77$.)

Why the other options are wrong

A is incorrect. Covered the final cumulated position instead of the most negative one: $3.7/0.984 = 3.76$, which leaves TSLe at -0.3 in year 4.

B is incorrect. Dropped the price conversion: treated notional as cash, selling 4.00 notional, which raises only $4.00 \times 0.984 = 3.94$.

D is incorrect. Ignored the initial cash in TSECCF(t_0, t_0): worst point -5.5, $5.5/0.984 = 5.59$.

LO: LTR-7 d: Interpret the term structure of expected cash flows and cumulative cash flows. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.5 The Term Structure of Expected Liquidity, eq. (7.10), Example 7.5.1, Table 7.10

LTR-7 • Monitoring Liquidity • Vardeez: MODERATE

LTR-7 e

P1 BOTH

Discuss the impact of available asset transactions on cash flows and liquidity generation capacity.

Workbook: A26 Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A26, the same August 2026 recall ("TSEF and TSAA numerical", Map #64). The TSAA half of the recall sits here: what each transaction on an available asset does to the cash flows, to the stock of assets the bank can still sell or repo, and to its liquidity generation capacity.
Vardeez: no pointer for this LO; the reading note applies (MODERATE; "Tsef and tsaa numerical seems imp", but do not get hung up on the calculations).

Reported exam shapes

- **S3** Repo of an available asset: cash into the TSCLGC, haircut and accrued interest (workbook A26.1).
- **S4** TSECCF with initial cash and the asset sale that keeps the TSLe non-negative; here with a coupon bond whose lost flows feed back into the TSECF.
- **S1** Which term structures move and by how much (workbook A26.3, the August recall itself, is filed here).

What the questions look like: a bond position with coupon, clean price, haircut and repo rate, and a transaction (sale, repo, reverse repo, buy/sellback, sell/buyback, lending or borrowing) a few months after a coupon date. You compute the cash and name the term structure it lands in. Distractors: the wrong term structure, a haircut applied where it does not belong (or left out where it does), the full semiannual coupon instead of the accrued part, and the repayment at the end offered for the cash at the start. Statement questions test ownership versus possession.

1. Where this LO starts

Recap from LTR-7 d (Sections 1 and 5): the causes of liquidity fill the TSECF and TSECCF; the sources (asset sales AS, secured funding and repo RP, unsecured funding USF) fill the TSLGC and TSCLGC; the bank is solvent while $\text{TSLe} = \text{TSECCF} + \text{TSCLGC} \geq 0$. Balance sheet liquidity (BSL) comes from selling or repoing **unencumbered** assets ("available for sale" here means not pledged, not the accounting category).

Most of the difficulty in building the TSLGC comes from these assets, mainly AFS bonds: the bank must keep track of how many are sold or repoed out, and the liquidity they can produce depends on several risk factors (the price, the haircut, the issuer's default probability). The tool for that tracking is the term structure of available assets.

2. The term structure of available assets (TSAA)

$$\text{TSAA}^{A_1}(t_0, t_b) = \{A^1(t_0), A^1(t_1), \ldots, A^1(t_b)\} \quad (7.8) \qquad \text{TSAA} = \sum_{j=1}^{M} \text{TSAA}^{A_j} \quad (7.9)$$

$A^1(t_i)$ is the **quantity of asset A_1 in the bank's possession** at t_i, **regardless of ownership**. Possession, not ownership, decides whether an asset can be sold or pledged to raise liquidity.

- The TSAA shows how much of each security is *available* for the LGC. It is a quantity, not cash: a sale converts it at the expected price, a repo at the price less the haircut. Both must be modelled as expected (or stressed) values.
- A purchase raises the TSAA by the notional until maturity, when it falls to zero; sales and repos reduce it; other repo-like contracts move it in either direction (Section 3).
- Stocks have no expiry: the TSECF records only the purchase price and the dividends, and the same contracts can be written on them.

3. Contracts on assets: who owns, who holds, what moves

Contract	Owner	Holder	TSECF and TSECCF	TSAA	TSLGC
Buy	Bank	Bank	Price out; coupons and redemption in	Up	No (possible)
Sell	Cpty.	Cpty.	Lost coupons and redemption	Down	Yes: proceeds
Repo	Bank	Cpty.	Coupons stay with the bank; repo interest out at the end	Down	Yes: cash after haircut
Reverse repo	Cpty.	Bank	Cash out (after haircut) at start; cash plus interest in at end	Up	No (possible)
Sell/buyback	Cpty.	Cpty.	Coupons during the contract go to the buyer	Down	Yes: sale proceeds
Buy/sellback	Bank	Bank	Purchase out; coupons in; sellback in	Up	No (possible)
Security lending	Bank	Cpty.	Coupons stay; fee in at the end	Down	No: decreases LGC
Security borrowing	Cpty.	Bank	Fee out at the end	Up	No (possible)

Based on GARP Table 7.4; Cpty. is the counterparty and Holder the party in possession. "Possible": the asset now in the bank's possession can be repoed during the contract, so it may increase the LGC later.

- **Repo.** The bank receives cash equal to the price less the haircut and delivers the asset. It still owns the asset, so coupons paid during the repo are its own and the TSECF is not affected; but the asset is encumbered and cannot be sold or pledged until returned. Cash received and repaid are both in the TSLGC; the repo interest is a contract flow in the TSECF at the end. A repo is a liability (a collateralised debt).
- **Reverse repo.** The bank pays cash (price less haircut) and receives the asset, which it may use as collateral until it must be returned. Coupons belong to the counterparty. Cash out and cash back with interest are TSECF flows. The TSLGC is not affected, though the asset can be repoed. An asset (a collateralised loan).

- **Sell/buyback.** Like a repo in cash and asset exchange, but ownership passes too, so the coupons paid before the buyback belong to the counterparty and are removed from the TSECF. The TSLGC is affected as in a repo. A commitment at the end.
- **Buy/sellback.** Like a reverse repo, but ownership passes to the bank: coupons are the bank's, and purchase and sale are TSECF flows. No haircut. An asset for the period.
- **Security lending.** Only possession passes and no cash is paid (except a fee). Coupons stay with the bank; the fee is in the TSECF at the end. The asset cannot be used to raise liquidity until returned: the **only contract that decreases LGC**.
- **Security borrowing.** Possession passes to the bank; only the fee (an outflow) enters the TSECF. The borrowed asset can be repoed until returned. A liability.

GARP adds a footnote: under the annex now in the standard GMRA, sell/buybacks and buy/sellbacks are treated like repos (payments returned to the seller, but at the end), so their TSECF effects match repo and reverse repo; the TSAA effects do not change.

4. Cash amounts: GARP Example 7.4.2

$$\begin{aligned}
\text{Sale, buy/sellback, sell/buyback:} &\quad N \times (P\% + c \times \tau) \\
\text{Repo, reverse repo:} &\quad N \times (P\% + c \times \tau) \times (1 - h) \\
\text{Repo repayment:} &\quad \text{cash} \times (1 + r_{\text{repo}} \times \text{term}) \\
\text{Lending or borrowing fee:} &\quad N \times \text{fee rate} \times \text{term}
\end{aligned}$$

N notional, $P\%$ the expected clean price, c the annual coupon rate, τ the year fraction **since the last coupon**, h the haircut. The haircut applies only to repo and reverse repo.

The bond. The bank buys 1,000,000 notional at 98.50 (settled after 3 days: $-985{,}000$ in the TSECF; TSAA 1,000,000). It pays a 10% coupon semiannually and matures in two years. Expected clean prices: 99.85 to 0.5 years, 99.90 from 0.75 to 1.25, 99.95 at 1.5 and 1.75; haircut 15%. Each operation below is a separate illustration on this position.

Operation	Cash	Term structures
Sale of 500,000 at 0.75	$500{,}000 \times (0.9990 + 0.025) = 512{,}000$	TSCLGC $+512{,}000$; TSAA 500,000; later coupons 25,000, redemption 525,000
Repo of 500,000 at 0.25 for 6 months	$500{,}000 \times (0.9985 + 0.025) \times 0.85 = 434{,}988$	TSCLGC $+434{,}988$; TSAA 500,000 to 0.75; coupon at 0.5 still 50,000
End of repo, rate 9%	$434{,}988 \times (1 + 0.09 \times 0.5) = 454{,}562$	TSCLGC back to 0; interest 19,574 in the TSECF; TSAA 1,000,000
Reverse repo of 500,000 at 1.25	$500{,}000 \times (0.9990 + 0.025) \times 0.85 = 435{,}200$	TSECF $-435{,}200$; TSAA 1,500,000; its coupon goes to the owner
End of reverse repo, rate 11%	$435{,}200 \times (1 + 0.11 \times 0.5) = 459{,}136$	TSECF $+459{,}136$; TSAA back to 1,000,000
Buy/sellback of 400,000 at 0.25	$400{,}000 \times (0.9985 + 0.025) = 409{,}400$	TSECF $-409{,}400$; TSAA 1,400,000; coupon at 0.5 is 70,000
Sellback at 0.75 (forward 99.90)	$400{,}000 \times (0.9990 + 0.025) = 409{,}600$	TSECF $+409{,}600$; TSAA 1,000,000
Sell/buyback of 300,000 at 1.25	$300{,}000 \times (0.9990 + 0.025) = 307{,}200$	TSCLGC $+307{,}200$; TSAA 700,000; coupon at 1.5 is 35,000
Lending 500,000 at 0.25, fee 3%	$500{,}000 \times 3\% \times 0.5 = 7{,}500$ at 0.75	TSECF $+7{,}500$; TSAA 500,000 meanwhile; coupon stays 50,000
Borrowing 300,000 at 1.25, fee 3%	$300{,}000 \times 3\% \times 0.5 = 4{,}500$ at 1.75	TSECF $-4{,}500$; TSAA 1,300,000 meanwhile

⇒ Cash raised by a sale or a sell/buyback, and by a repo, goes into the **TSCLGC**: these contracts generate liquidity. Cash paid in a purchase, reverse repo or buy/sellback is a contract flow in the **TSECF**: these contracts consume cash now and only add assets that *could* be repoed. Lending and borrowing move no principal at all.

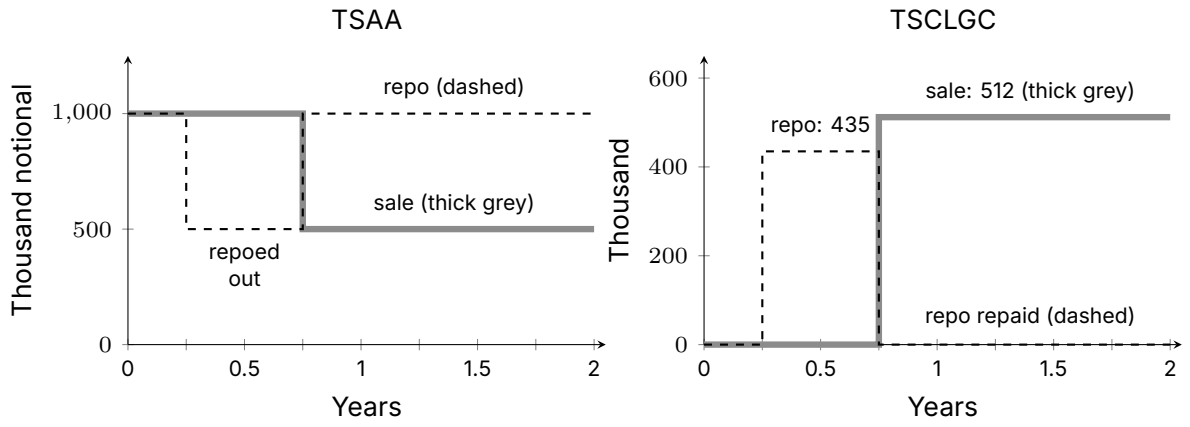


Figure 1. Example 7.4.2: selling 500,000 at 0.75 years (thick grey line) against repoing 500,000 from 0.25 to 0.75 years (dashed line). The sale removes the bond for good and leaves 512,000 of liquidity; the repo removes it only for six months, raises less (haircut) and gives the liquidity back at the end, when the interest of 19,574 enters the TSECF.

Corrections to the source notes.

- **Vardeez's impact grid** marks an asset sale as "TSECF up". GARP Table 7.6 puts the sale proceeds (512,000) in the TSCLGC; the TSECF goes *down*, by the lost coupons (25,000 a period) and redemption (500,000).
- **GARP Table 7.7** prints 424,469 and $-19{,}101$ for the repo, and $-424{,}681$ and 471,396 for the reverse repo. Its own text computes 434,988, 19,574, 435,200 and 459,136, which recompute correctly and are used here.
- **GARP's sell/buyback buyback**, $300{,}000 \times (99.95\% + 10\% \times 0.25)$, equals 307,350, not the printed 314,726.
- **GARP Table 7.9** leaves the TSAA at 1,000,000 during the security borrowing; by the book's own rule the TSAA rises by the 300,000 borrowed, to 1,300,000.

5. Selling a coupon bond to cover a negative TSECCF

Selling to cover a gap changes the TSECF in two ways: fewer coupons and a smaller final redemption (GARP Example 7.5.1, LTR-7 d Section 5). With a coupon bond the later cumulated points get worse, so the notional must be chosen and then the TSLe re-checked at every later date. This is the feedback that makes the build recursive.

Worked example. Initial cash 1.0. The TSECF, already including the coupons (4% annual) and redemption of 25 notional of an unencumbered bond maturing in year 5, is $+2.0$, -5.0, $+0.4$, -0.8 and $+31.0$ for years 1 to 5. The bank will sell at the end of year 2, after the coupon, at an expected price of 97.00.

$$\text{TSECCF}: \ 3.0,\ -2.0,\ -1.6,\ -2.4,\ 28.6$$

Selling N adds $0.97N$ to the TSCLGC from year 2 and removes $0.04N$ of coupon each year from year

3 (and N at year 5):

$$\text{TSLe}_2 = -2.0 + 0.97N, \qquad \text{TSLe}_3 = -1.6 + 0.93N, \qquad \text{TSLe}_4 = -2.4 + 0.89N$$

Year 4 binds: $N \geq 2.4/0.89 = 2.70$. Covering the most negative TSECCF with the zero-coupon rule, $2.4/0.97 = 2.47$, leaves the TSLe at -0.20 in year 4.

No double counting. A bond pledged as collateral for a loan from t_1 to t_2 must be excluded from the TSLGC for that period and for overlapping periods; the TSAA is what prevents the same bond from being counted twice.

Formula card: LTR-7 e

Accrued interest	$c \times \tau$, τ = year fraction since the last coupon (not since purchase)
Sale, buy/sellback, sell/buyback	Cash $= N \times (P\% + c\tau)$, no haircut
Repo, reverse repo	Cash $= N \times (P\% + c\tau) \times (1 - h)$
Repo repayment	Cash $\times(1 + r_{\text{repo}} \times \text{term})$; the interest part enters the TSECF at the end
Lending or borrowing fee	$N \times$ fee rate $\times$ term, in the TSECF at the end
TSAA	Quantity in possession, whoever owns it (eqs. 7.8, 7.9)
Where the start cash goes	Sale, repo, sell/buyback: TSCLGC. Purchase, reverse repo, buy/sellback: TSECF. Lending, borrowing: no principal
Coupon bond sale	TSCLGC $+N \times P$; TSECF $-c\,N$ per later coupon and $-N$ at maturity; re-check every later TSLe

Exam traps

- **Wrong term structure (sibling swap).** Repo and sell/buyback cash goes to the TSCLGC; reverse repo and buy/sellback cash goes to the TSECF.
- **Possession, not ownership.** The TSAA follows possession: a repo lowers it although the bank still owns the bond; a reverse repo raises it although the bank does not.
- **Coupons follow ownership.** Repo and security lending: the bank keeps them. Sell/buyback: the buyer gets them. Buy/sellback: the bank gets them.
- **Haircut misplaced.** Only repo and reverse repo cash is cut; a sale, buy/sellback or sell/buyback is not.
- **Accrual.** $c \times \tau$ since the last coupon, not the full semiannual coupon and not since purchase.
- **Intermediate result.** The repo repayment at maturity is not the cash raised at the start.
- **Security lending** is the only contract that decreases LGC; borrowing and reverse repos can increase it later.
- **Coupon bond sales.** The zero-coupon sale rule understates the notional once the lost coupons are fed back.

Practice questions: LTR-7 e

Question LTR-7 e.1

Bank Q283 • Schweser Module Quiz 72.3 Q1 (rebuilt)

Gerard Bank enters into a reverse repo transaction for six months. Which of the following statements is most accurate regarding the impact of the transaction?

A. The TSCLGC rises by the cash the bank pays at the start of the contract.

B. Both the TSAA and the TSECCF are impacted.

C. The TSECF is impacted, but the TSECCF will not be impacted.

D. Possession and ownership of the asset both pass to the bank.

Question LTR-7 e.2

Bank Q280 • AnalystPrep Q.3939 (rebuilt)

A liquidity division lends 600,000 notional of a bond, three months after buying it, for a period of six months. The TSECF records no cash flow at inception, and the TSAA falls by 600,000. When the bond is returned the bank receives a lending fee of 3.5% a year. The repo haircut on this bond is 15%. What fee enters the TSECF at the end of the contract?

A. 5,250

B. 8,925

C. 10,500

D. 21,000

Question LTR-7 e.3

Bank Q279 • AnalystPrep Q.3935 (rebuilt, key corrected)

A bank bought a bond with a notional of 1,000,000 on a coupon date at time 0. The bond pays a 10% annual coupon semiannually. After 10 months (0.833 years) the bank sells a notional of 300,000 at a clean price of 99.95 to generate liquidity. What inflow is recorded in the term structure of cumulated liquidity generation capacity (TSCLGC) for this sale?

A. 299,850

B. 309,850

C. 314,850

D. 324,850

Question LTR-7 e.4

Workbook A26.1 • S3 • self-gen

A liquidity analyst in the treasury of a German bank holds EUR 2,000,000 notional of an unencumbered bond paying an 8% annual coupon semiannually. Three months after a coupon date the bank repos EUR 800,000 notional for six months to generate balance sheet liquidity. The expected clean price is 99.60, the haircut is 12% and the repo rate is 7%. Which of the following is closest to the cash the bank records in the TSCLGC at the start of the repo?

A. EUR 715,264

B. EUR 729,344

C. EUR 740,298

D. EUR 812,800

Question LTR-7 e.5 New • built from GARP Ch.7 (Table 7.4)

A treasury analyst is summarising for the ALCO how contracts written on the bank's available assets affect its term structures. Which of the following statements is correct?

A. A reverse repo raises the TSCLGC at the start, because the bank takes possession of an asset it can repo.

B. In a repo, the coupons paid during the contract are removed from the TSECF, because possession passes to the counterparty.

C. In a sell/buyback, the TSAA is unchanged, because the bank has committed to buy the asset back.

D. Security lending is the only contract on assets that decreases LGC: possession passes, no cash is received, and the asset cannot raise liquidity until it is returned.

Question LTR-7 e.6 Workbook A26.3 • S1 • Aug-2026

A liquidity risk officer at a Swiss bank (reporting in USD) records a buy/sellback: three months after a coupon date the bank buys 600,000 notional of a bond paying a 6% annual coupon semiannually, and will sell it back after six months at the forward price. The expected clean price is 99.40 and the applicable repo haircut on this bond is 12%. Which of the following correctly describes the effect at the start of the contract?

A. TSECF outflow of USD 605,400; TSCLGC rises by USD 605,400

B. TSECF outflow of USD 605,400; TSAA rises by 600,000

C. TSECF outflow of USD 532,752; TSAA rises by 600,000

D. TSECF outflow of USD 614,400; TSAA rises by 600,000

Question LTR-7 e.7 New • built from GARP Ch.7 (Example 7.4.2, Table 7.8)

A treasury analyst at an Australian bank holds AUD 2,000,000 notional of an unencumbered bond paying a 6% annual coupon semiannually. Three months after a coupon date the bank enters a six-month sell/buyback on AUD 800,000 notional at an expected clean price of 98.80. The repo haircut on this bond is 12%. Under the contract, coupons paid before the buyback belong to the buyer. Which of the following gives the cash the bank records in the TSCLGC at the start and the coupon it records in the TSECF on the next coupon date?

A. TSCLGC AUD 706,112; coupon AUD 60,000

B. TSCLGC AUD 802,400; coupon AUD 36,000

C. TSCLGC AUD 802,400; coupon AUD 60,000

D. TSCLGC AUD 814,400; coupon AUD 36,000

Question LTR-7 e.8 New • built from GARP Ch.7 (Example 7.5.1)

A treasurer at a Singapore bank has initial cash of SGD 2.0 million and the TSECF below (SGD million, expected values). The TSECF already includes the coupons (5% annual, paid at each year end) and the year-6 redemption of the 40 notional of an unencumbered bond the bank holds. Because the cumulated position turns negative, the bank will sell part of the bond at the end of year 3, just after the year-3 coupon, at an expected price of 98.00.

Year	1	2	3	4	5	6
TSECF	3.0	−4.5	−6.0	1.0	−1.5	45.0

Which of the following is closest to the minimum notional the bank must sell to keep the term structure of expected liquidity non-negative at every date?

A. SGD 5.61 million

B. SGD 6.12 million

C. SGD 6.82 million

D. SGD 9.09 million

Solutions • LTR-7 e

Solution LTR-7 e.1

Answer: B

Bank Q283 • Schweser Module Quiz 72.3 Q1

Step 1 – Ownership and possession

In a reverse repo the counterparty keeps ownership and the bank takes possession of the asset against cash.

Step 2 – Term structures

Possession raises the TSAA by the notional. The cash paid at the start (price less haircut) and the cash plus interest received at the end are contract flows in the TSECF, so the TSECCF, their running total, moves too. The TSLGC is not affected, although the asset can be repoed until it is returned.

$\Rightarrow$ Both the TSAA and the TSECCF are impacted.

Interpretation

A reverse repo uses cash: it is a collateralised loan to the counterparty, an asset on the balance sheet. It adds liquidity potential only if the bank repos the asset on.

Why the other options are wrong

A is incorrect. The cash paid goes out through the TSECF; a reverse repo consumes liquidity rather than generating it.

C is incorrect. The TSECCF is the cumulation of the TSECF, so it moves whenever the TSECF does.

D is incorrect. Only possession passes; ownership staying with the counterparty is what separates a reverse repo from a buy/sellback.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.4.3 The Term Structure of Available Assets, Table 7.4

Solution LTR-7 e.2

Answer: C

Bank Q280 • AnalystPrep Q.3939

Step 1 – Fee formula

$$\text{Fee} = N \times \text{fee rate} \times \text{term} = 600{,}000 \times 3.5\% \times 0.5 = 10{,}500$$

$\Rightarrow$ The fee is the only cash flow of the contract and enters the TSECF when the bond is returned.

Interpretation

Only possession passes in security lending, so the coupons paid during the six months still belong to the bank. The haircut is a red herring: no cash is lent against the bond. While the bond is out it cannot be sold or repoed, which is why security lending is the one contract that reduces LGC.

Why the other options are wrong

A is incorrect. Used a 3-month term: $600{,}000 \times 3.5\% \times 0.25 = 5{,}250$.

B is incorrect. Applied the 15% haircut to the fee: $10{,}500 \times 0.85 = 8{,}925$.

D is incorrect. Did not scale the annual rate to six months: 21,000.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.4.3, Example 7.4.2, Table 7.9

Solution LTR-7 e.3 — Answer: B

Bank Q279 • AnalystPrep Q.3935

Step 1 – Accrued interest since the last coupon

Coupons fall at months 6 and 12, so at month 10 four months have accrued:

$$c \times \tau = 10\% \times \tfrac{4}{12} = 3.333\%$$

Step 2 – Sale proceeds at the dirty price

$$300{,}000 \times (0.9995 + 0.03333) = 309{,}850$$

$\Rightarrow$ The proceeds are new liquidity: they enter the TSCLGC.

Step 3 – Other term structures

The TSAA falls by 300,000 until maturity; the TSECF loses the coupons on 300,000 (15,000 a period) and 300,000 of redemption.

Interpretation

No haircut applies to an outright sale. The bank gives up future contract flows for cash now, the trade-off that the TSLe check has to capture.

Why the other options are wrong

A is incorrect. Clean price only: $300{,}000 \times 0.9995 = 299{,}850$.
C is incorrect. Accrued a full half-year: $300{,}000 \times (0.9995 + 0.05) = 314{,}850$.
D is incorrect. Accrued since purchase (0.833 years), ignoring the coupon paid at month 6: $300{,}000 \times (0.9995 + 0.0833) = 324{,}850$.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.4.3, Example 7.4.2, Table 7.6

Solution LTR-7 e.4 — Answer: A

Workbook A26.1 • S3 • self-gen

Step 1 – Accrued interest fraction

$$c \times \tau = 8\% \times 0.25 = 0.02$$

Step 2 – Dirty value of the repoed notional

$$800{,}000 \times (0.9960 + 0.02) = \text{EUR } 812{,}800$$

$\Rightarrow$ This is what the bond would raise if sold outright.

Step 3 – Apply the haircut

$$812{,}800 \times (1 - 0.12) = \text{EUR } 715{,}264$$

$\Rightarrow$ The counterparty lends only 88% of collateral value; this cash enters the TSCLGC.

Step 4 – Other term structures

TSAA falls from 2,000,000 to 1,200,000 for six months (possession passes); the TSECF is unaffected at the start because the bank still owns the bond and keeps the coupons. At the end the bank repays $715{,}264 \times (1 + 7\% \times 0.5) = 740{,}298$, and the interest of 25,034 enters the TSECF.

Interpretation

Repo extracts less than a sale (haircut) but keeps the bond on the bank's books; the treasurer must also track the reduced TSAA so the same bond is not counted twice in the LGC.

Why the other options are wrong

B is incorrect. Used the half-year coupon as accrued interest ($8\% \times 0.5$): $800{,}000 \times 1.036 \times 0.88 = 729{,}344$.

C is incorrect. Intermediate result: reported the repurchase amount at the end of the repo, 740,298.

D is incorrect. Dropped the haircut (treated as an outright sale): 812,800.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. • **Reference:** Castagna and Fede, Measuring and Managing Liquidity Risk, Ch.6, 7.4.3 The Term Structure of Available Assets, Example 7.4.2, Table 7.7

Solution LTR-7 e.5 Answer: D

New • built from GARP Ch.7

Step 1 – Why D is correct

In security lending only possession passes and no principal is paid; the bank receives a fee at the end. The asset cannot be sold or pledged until it comes back. GARP: "The only contract that does not increase the TSLGC is security lending, which actually decreases LGC related to BSL."

Interpretation

Every other contract either generates BSL (sale, repo, sell/buyback) or puts an asset in the bank's possession that could be repoed (purchase, reverse repo, buy/sellback, borrowing), so it increases the LGC at least potentially.

Why the other options are wrong

A is incorrect. The reverse repo cash goes out through the TSECF; the TSLGC is not affected, though the asset could later be repoed.

B is incorrect. In a repo the bank keeps ownership, so the coupons are still its own and the TSECF is unaffected; only possession passes.

C is incorrect. In a sell/buyback possession and ownership pass, so the TSAA falls by the notional until the buyback.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.4.3, Table 7.4 and the discussion preceding it

Solution LTR-7 e.6 Answer: B

Workbook A26.3 • S1 • Aug-2026

Step 1 – Accrued interest fraction

$$c \times \tau = 6\% \times 0.25 = 0.015$$

Step 2 – Cash paid at the start

$$600{,}000 \times (0.9940 + 0.015) = \text{USD } 605{,}400$$

$\Rightarrow$ An outright purchase price with accrued interest; no haircut applies to a buy/sellback.

Step 3 – Which term structures move

Ownership and possession pass to the bank, so the cash paid is a contract flow in the TSECF (and TSECCF), the TSAA rises by the notional of 600,000, and coupons received during the contract belong to the bank. The TSCLGC is not modified by the operation (the asset can, however, be repoed to raise liquidity until the sellback).

Interpretation

A buy/sellback consumes cash now but adds available assets; it does not by itself generate liquidity. At the sellback the forward price plus accrued interest returns to the TSECF and the TSAA falls back.

Why the other options are wrong

A is incorrect. Sibling swap: recorded the cash in the TSCLGC as if the operation generated liquidity; that is the sell/buyback treatment.
C is incorrect. Applied the repo haircut as if it were a reverse repo: $605{,}400 \times 0.88 = 532{,}752$.
D is incorrect. Used the full semiannual coupon as accrued interest: $600{,}000 \times (0.994 + 0.03) = 614{,}400$.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. •
Reference: Castagna and Fede, Ch.6, 7.4.3 The Term Structure of Available Assets, Table 7.4, Example 7.4.2, Table 7.8

Solution LTR-7 e.7

Answer: B

New • built from GARP Ch.7

Step 1 – Cash at the start (no haircut)

$$c \times \tau = 6\% \times 0.25 = 0.015, \qquad 800{,}000 \times (0.9880 + 0.015) = \text{AUD } 802{,}400$$

$\Rightarrow$ A sell/buyback generates balance sheet liquidity: the proceeds enter the TSCLGC, and the TSAA falls to 1,200,000 for six months.

Step 2 – The coupon three months later

Ownership has passed, so the coupon on the 800,000 goes to the buyer. The bank records the coupon only on what it still owns:

$$1{,}200{,}000 \times \tfrac{6\%}{2} = \text{AUD } 36{,}000 \quad (\text{instead of } 60{,}000)$$

Interpretation

This is the difference from a repo, where the bank keeps ownership and every coupon. GARP Table 7.8 shows the same effect: a coupon of 35,000 on the 700,000 retained after a 300,000 sell-/buyback. Under the GMRA annex, coupons would instead be passed back to the seller at the end, as in a repo.

Why the other options are wrong

A is incorrect. Applied the 12% repo haircut ($802{,}400 \times 0.88 = 706{,}112$) and kept the full coupon, as if it were a repo.
C is incorrect. Right cash, but kept the coupon on the full 2,000,000 (repo treatment of coupons).
D is incorrect. Used the full semiannual coupon as accrued interest: $800{,}000 \times (0.988 + 0.03) = 814{,}400$.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. •
Reference: Castagna and Fede, Ch.6 (GARP Ch.7), 7.4.3, Table 7.4, Example 7.4.2, Table 7.8

Solution LTR-7 e.8

Answer: C

New • built from GARP Ch.7

Step 1 – TSECCF from initial cash

$$2.0 \;\rightarrow\; 5.0, \quad 0.5, \quad -5.5, \quad -4.5, \quad -6.0, \quad 39.0 \quad (\text{years } 1 \text{ to } 6)$$

$\Rightarrow$ The cumulated position is negative from year 3 to year 5; the worst point before any sale is year 5.

Step 2 – What a sale of N at year 3 does

TSCLGC: $+0.98N$ from year 3. TSECF: $-0.05N$ in years 4, 5 and 6, and $-N$ of redemption in year 6.

Step 3 – TSLe at each date

$$\text{TSLe}_3 = -5.5 + 0.98N, \qquad \text{TSLe}_4 = -4.5 + 0.93N$$
$$\text{TSLe}_5 = -6.0 + 0.88N, \qquad \text{TSLe}_6 = 39.0 - 0.17N$$

Step 4 – Binding constraint

Year 3 needs $N \geq 5.61$, year 4 $N \geq 4.84$, year 5 $N \geq 6.0/0.88 = 6.82$.
$\Rightarrow$ Minimum notional SGD 6.82 million. Check: TSLe 1.18, 1.84, 0.00 and 37.84 in years 3 to 6.

Interpretation

The lost coupons make the year-5 point worse than the TSECCF suggests, so the zero-coupon rule is not enough. This is the feedback GARP describes between the TSLGC and the TSECF.

Why the other options are wrong

A is incorrect. Covered only the first negative point: $5.5/0.98 = 5.61$, leaving TSLe at -1.06 in year 5.
B is incorrect. Covered the most negative TSECCF but ignored the lost coupons: $6.0/0.98 = 6.12$, leaving TSLe at -0.61 in year 5.
D is incorrect. Ignored the initial cash: the worst point becomes -8.0, and $8.0/0.88 = 9.09$.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.5, Example 7.5.1, Table 7.10

LTR-12 • Managing Non-Deposit Liabilities • Vardeez: IMPORTANT

LTR-12 a

P1 CONCEPT

Distinguish between the various sources of non-deposit liabilities at a bank.

Workbook: A27 Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A27. The August 2026 sitting carried a "non-deposit liability theory question", and Map #47 (2 of 16 recall sources) reports "pooled cost approach, etc." for the same section. Shape S2 is the theory question: the sources, and the factors in choosing among them (LTR-12 c). Workbook A27.2 tags both LOs and is worked here.

Vardeez: no pointer for this LO; his reading note targets the two calculations (available funds gap, LTR-12 b; overall cost of funds, LTR-12 d). His notes list eight sources in one table and flag two sibling sets built for swapping: EuroCD against Yankee CD, and the three discount window rates (his ranking is corrected in Section 3).

Reported exam shapes

- **S2** Sources of non-deposit liabilities and the factors in choosing among them, theory as seen in August.

What the questions look like: statement questions. A treasurer or a funding desk describes the sources and one statement is right: who lends, whether collateral is pledged, how long the money stays, who can use the market, which discount window rate is highest. The distractors flip a polarity (secured or unsecured, most or least volatile), swap siblings (EuroCD and Yankee CD, primary and secondary credit) or add an absolute ("no credit availability risk").

1. Why banks borrow: liability management

Deposits are the traditional source of funds. When deposits cannot support all the loans and investments a bank would like to make, management can sell fee services, securitise or sell loans, sell securities it already holds, or, the subject of this reading, sell IOUs in the money and capital markets for periods from overnight to several years. These are the **nondeposit liabilities**.

The customer relationship doctrine

The first priority of a lending institution is to make loans to all customers from whom it expects **positive net earnings**. Turning down a good customer with "we don't have enough deposits" loses the account and often its future business; granting the loan brings in deposits and other services, sometimes from the firm's owners and employees too. So **lending decisions precede funding decisions**: make every loan whose return exceeds its cost and whose quality meets the standards, and if deposits fall short, find the **lowest-cost** borrowed funds. The limitation: the doctrine can produce scores of poor-quality loans, as in the subprime collapse of 2007 to 2009, when cheap money market funding supported a lending spree.

Liability management

In the 1960s and 1970s the doctrine spawned **liability management**: buying funds, mainly from other financial institutions, to cover good-quality credit requests and any legal reserve requirements.

- **Flexible.** The borrower decides how much and for how long; when a bank sells deposits, the depositor decides. To raise more the bank raises its offer rate, to raise less it lowers it: market

interest rates are the control lever.

- **Interest sensitive.** Money market lenders are highly rate elastic. A bank that will not pay the going rate sees the funds "gone in minutes". Using these sources raises interest rate risk relative to demand and savings deposits.
- **A daily job.** Much of the money raised today must be returned tomorrow and replaced.
- **Size.** Nondeposit borrowings support about 4% of assets at the smallest US banks (under $100 million) and about 16% at the largest (over $1 billion), four times as much.

Example (Rose and Hudgins, Table 13.1). A business customer asks for a $100 million loan and only $50 million of new deposits are expected today. The money market desk raises the other $50 million: Fed funds $19 million, Eurodollars from branches abroad $20 million, RPs $3 million, and $8 million borrowed from a holding-company affiliate that sold commercial paper.

⇒ Most of it is short term, deliberately: the customer is waiting and long-term debt takes time to arrange.

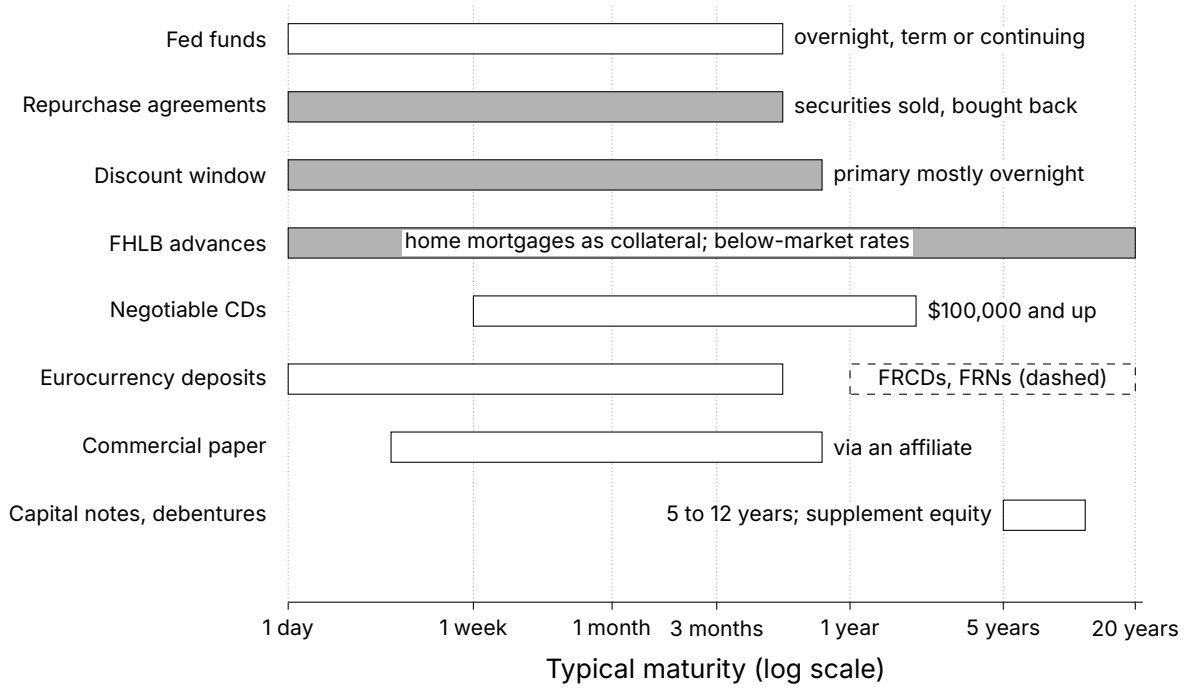


Figure 1. The nondeposit sources by typical maturity and security. Grey bars are secured (securities, acceptable collateral at the Fed, home mortgages); white bars are usually unsecured. The dashed extension is the floating-rate Euro market (FRCDs and FRNs, 1 to 20 years). Ranges are indicative, from Rose and Hudgins: Fed funds term loans run to months, primary credit occasionally to 90 days, CDs 7 days to 1 or 2 years, commercial paper 3 or 4 days to 9 months.

2. The sources, one by one

Federal funds (the most popular domestic source). Short-term borrowings of **immediately available money**: reserves at the Federal Reserve banks and interbank deposits, moved the same day over Fedwire, to meet reserve requirements, cover loan demand or park surpluses for a few hours. **No legal reserve requirement** on Fed funds borrowings. Three contracts: *overnight* (unwritten, by wire or phone, normally **unsecured**), *term* (days to months, often written) and *continuing* (renewed daily until either side ends it; small respondents with their correspondents). The borrower books "Federal

funds purchased", the lender "Federal funds sold". Troubled borrowers may pay tiered rates or be shut out.

Repurchase agreements (RPs). Less popular and more complex than Fed funds, often viewed as **collateralised Fed funds**: a temporary sale of high-quality securities such as T-bills (*starting leg*) with an agreement to buy them back at a predetermined price on a set date (*closing leg*). The collateral cuts the lender's credit risk but an RP may take longer to arrange. Interest for both Fed funds and RPs is amount × rate × days/360 (eq. 13.1): \$50 million for three days at 6% costs \$25,000. **GCF RPs** (1998) let any of an agreed set of eligible securities serve as collateral and settle with netting at the FICC. Bear Stearns and Lehman could not find adequate RP support in 2008 (repos in depth: LTR-13).

The Federal Reserve's discount window. The Fed credits the borrower's reserve account; every loan needs **acceptable collateral**.

- *Primary credit:* sound institutions, usually overnight (occasionally to 90 days), slightly **above the target Fed funds rate**; no need to exhaust other sources first.
- *Secondary credit:* institutions that do not qualify for primary, about 50 bp above primary, monitored, **not for asset expansion**.
- *Seasonal credit:* longer terms for small and medium institutions with intrayear swings (farm banks), priced at the average of the effective Fed funds rate and the 90-day CD rate.

Secondary credit posts the highest rate and seasonal credit the lowest. Before 2003 the discount rate was often the lowest money market rate; now primary credit is an above-market "Lombard" rate that tends to cap overnight rates. Undercapitalised banks are limited to 60 days in each 120 (FDIC Improvement Act, 1991).

> **Correction to the source notes.** Vardeez's notes rank the three discount window rates "primary is lowest, secondary is highest, seasonal is intermediate". GARP (Rose and Hudgins) says "secondary credit generally posting the highest interest rate and seasonal credit the lowest": seasonal credit is priced at the average of the effective Fed funds rate and the 90-day CD rate, below primary credit, which is set above the Fed funds target. Cheapest first: **seasonal, primary, secondary**. Schweser and Workbook A27 agree with GARP. His second point stands: the primary credit rate is among the highest short-term market rates.

Federal Home Loan Bank advances. Twelve regional banks chartered in 1932 to lend to institutions facing runs, against **home mortgages** as collateral. A **stable, below-market** source for community banks, thrifts, credit unions and insurers, overnight to more than 20 years, fixed or variable. The FHLB is **first in line, ahead even of the FDIC**, if a borrower fails.

Large negotiable CDs. A hybrid: **legally a deposit**, practically an IOU sold to corporations, governments and wealthy individuals. \$100,000 and up, usually in multiples of \$1 million; 7 days to 1 or 2 years, mostly 1 to 6 months; negotiable, in bearer form. Four types: *domestic* (US institutions inside the US), *EuroCDs* (dollar CDs issued by banks *outside* the US), *Yankee CDs* (large foreign banks such as Deutsche Bank and HSBC selling through their *US branches*) and *thrift CDs*. Fixed-rate CDs pay on a 360-day basis: $P + P \times \frac{\text{days}}{360} \times \text{rate}$ (eq. 13.2), so \$100,000 for 180 days at 8% returns \$104,000. The reserve requirement on CDs is currently **zero** (10% on the largest banks' checking accounts); the FDIC fee is charged on the full face amount.

Eurocurrency deposits. Eurodollars are dollar-denominated deposits in bank offices outside the US, centred in London, the largest unregulated market in the world. A US home office borrowing from its own branch records *liabilities to foreign branches*. Mostly fixed-rate time deposits maturing within six months, priced off LIBOR; large interbank *tap* CDs and smaller *tranche* CDs; FRCDs and FRNs run 1 to 20 years.

Commercial paper. Notes of 3 or 4 days to 9 months sold at a discount by well-known companies

(*industrial* paper) and finance companies or holding-company affiliates (*finance* paper). A bank uses it indirectly: the affiliate sells the paper and buys loans off the bank's books. **High volume, moderate cost, but volatile capacity and credit risk.**

Long-term nondeposit funds. Mortgages on buildings and **capital notes and debentures** of usually 5 to 12 years that **supplement equity**. Kept modest by regulation and by the maturity mismatch they create; their yields are a barometer of perceived default risk. Schweser and Vardeez: usually viewed as **secondary capital rather than funding**.

	Who lends	Term	Collateral	The fact GARP tests
Fed funds	Other depository institutions	Overnight; term; continuing	Usually none	Most popular; no reserve requirement; most volatile rate
RPs	Holders of surplus cash	Overnight to months	Securities	Collateralised Fed funds; slower to arrange; GCF substitution
Discount window	Federal Reserve bank	Primary usually overnight	Acceptable collateral	Secondary highest, seasonal lowest; primary above Fed funds target
FHLB advances	12 regional FHLBs	Overnight to 20+ years	Home mortgages	Stable, below market; FHLB ahead of the FDIC
Negotiable CDs	Corporations, governments, wealthy individuals	7 days to 1 or 2 years	None (a deposit)	$100,000+; domestic, Euro, Yankee, thrift; zero reserve requirement
Eurocurrency	Banks offshore (London)	Mostly under 6 months	None	Largest unregulated market; liabilities to foreign branches
Commercial paper	Money market investors	3 or 4 days to 9 months	None	Via an affiliate; volatile capacity, credit risk
Notes, debentures	Capital market investors	5 to 12 years	None (subordinated)	Supplement equity; barometer of default risk

Exam traps

- **Sibling swaps.** EuroCD: a dollar CD issued *outside* the US. Yankee CD: a *foreign bank* selling through its *US branch*. Eurodollars are dollars, not euros.
- **Collateral polarity.** RPs and discount window loans are secured, FHLB advances by mortgages; Fed funds (usually) and commercial paper are not.
- **Discount window ranks.** Secondary highest, seasonal lowest; primary is for sound banks and priced above the Fed funds target; secondary is not for asset expansion.
- **Wrong lender.** Fed funds come from other depository institutions, discount window loans from the Fed itself.
- **Popularity and complexity.** Fed funds are the most popular source; RPs are less popular and more complex, not simpler.
- **Absolute claims.** A CD's legal status as a deposit does not remove credit availability risk (LTR-12 c); commercial paper is high volume but its capacity is volatile.
- **Conventions.** Fed funds, RP and CD interest use a 360-day year; CDs and Fed funds borrowings carry no reserve requirement today.

Practice questions: LTR-12 a

Question LTR-12 a.1

Bank Q387 • AnalystPrep Q.3907 (rebuilt)

Late in the afternoon the treasurer of a US regional bank finds the bank short of reserves. She borrows reserve balances that another depository institution holds at the Federal Reserve, with no collateral pledged, to be returned the next business day. Which nondeposit source has she used?

A. A repurchase agreement

B. Primary credit at the discount window

C. A negotiable CD

D. Federal funds purchased

Question LTR-12 a.2

Bank Q399 • GARP book Ch.13 end-of-chapter (rebuilt)

A treasury analyst at a savings bank is preparing a briefing note on repurchase agreements (RPs) as a source of funds. Which statement in the note is correct?

A. An RP is an unsecured loan, so the lender relies solely on the borrower's creditworthiness.

B. The borrower temporarily sells high-quality securities (the starting leg) and agrees to buy them back at a predetermined price (the closing leg), and the collateral reduces the lender's credit risk.

C. RP interest is calculated on a 365-day year using the actual number of days borrowed.

D. In a General Collateral Finance (GCF) RP, the borrower must return exactly the securities pledged at the start of the loan.

Question LTR-12 a.3

Bank Q397 • Schweser Module Quiz 77.1 Q1

Which of the following nondeposit funding sources requires collateral?

A. Fed funds and commercial paper.

B. Commercial paper and discount window borrowing.

C. Fed funds and repurchase agreements.

D. Discount window borrowing and repurchase agreements.

Question LTR-12 a.4

Bank Q396 • Mock 2 Q76 (rebuilt)

A financially sound US bank with no seasonal funding pattern needs a short-term (overnight) loan from the Federal Reserve's discount window. It will provide acceptable collateral and wants to pay the lowest interest rate available to it. Which form of discount window borrowing is most appropriate?

A. Overnight credit

B. Primary credit

C. Seasonal credit

D. Secondary credit

Question LTR-12 a.5 Bank Q393 • AnalystPrep Q.4113 (rebuilt)

The treasurer of a US manufacturer buys dollar-denominated negotiable CDs that HSBC sells in New York through its US branch. Under Rose and Hudgins's classification of negotiable CDs, these are:

A. Domestic CDs

B. EuroCDs

C. Yankee CDs

D. Thrift CDs

Question LTR-12 a.6 Bank Q386 • GARP book Ch.13 end-of-chapter (rebuilt)

Which description of Eurodollar deposits as a nondeposit funding source is correct?

A. Euro-denominated deposits held at banks located in the euro area.

B. Dollar deposits held at US banks by foreign residents and subject to full Federal Reserve reserve requirements.

C. Dollar-denominated deposits placed in banks outside the United States; a US bank borrowing from its overseas branch records them as liabilities to foreign branches.

D. Mostly long-term fixed-rate deposits with maturities beyond five years, backed by Treasury collateral.

Question LTR-12 a.7 Bank Q398 • BT Mock Q69

In regard to the various sources of nondeposit funding, which of the following statements is true?

A. The Fed funds market is the least popular because its effective interest rate is higher than the alternatives.

B. Repurchase agreements are more popular than Fed funds because they are simpler; however, they incur greater credit risk.

C. The advantages of the commercial paper market are the lack of credit risk and consistently high (low volatility) credit availability; however, volume tends to be low and relative cost is high.

D. Advances from Federal Home Loan Banks are highly popular among thrifts and small banks as a stable source of funding, flexible in maturity, at below-market interest rates.

Question LTR-12 a.8 Workbook A27.2 • S2 • Aug-2026

The treasurer of a USD 400 million community bank in Kansas must cover a USD 12 million reserve deficit and a new loan commitment by the end of today, after deposits came in below forecast. She asks the funding desk which nondeposit source to tap. Which of the following statements made by the desk is correct?

A. Overnight Fed funds offer the most predictable cost, because the shorter the term of a money market borrowing, the less volatile its interest rate tends to be.

B. Seasonal credit carries the highest of the three discount window rates, so a sound bank needing funds today should request primary credit instead.

C. Eurodollar borrowings trade in multiples of USD 1 million and are usually available only to money-centre banks with top ratings, while Fed funds and the discount window can make small loans.

D. Negotiable CDs are legally deposits, so unlike Eurodollar borrowings and commercial paper they carry no credit availability risk for the issuing bank.

Solutions • LTR-12 a

Solution LTR-12 a.1 — Answer: D

Bank Q387 • AnalystPrep Q.3907

Step 1 – Who lends, what is lent, on what security

The lender is another depository institution, the money is reserve balances at the Fed, nothing is pledged and the funds come back the next day.
⇒ That is an overnight Fed funds loan: "short-term borrowings of immediately available money". The borrower books it as Federal funds purchased; the lender as Federal funds sold.

Step 2 – Check the siblings

An RP would need securities sold and repurchased; primary credit would come from the Fed; a CD is a deposit sold to investors.

Interpretation

Overnight Fed funds are unwritten, arranged by wire or telephone and normally unsecured, which is why they are the fastest source for a same-day reserve deficit. There is no legal reserve requirement on Fed funds borrowings.

Why the other options are wrong

A is incorrect. Wrong collateral: an RP is a temporary sale of securities with an agreement to buy them back; here nothing is pledged.
B is incorrect. Wrong lender: primary credit is a loan from the Federal Reserve bank, against acceptable collateral.
C is incorrect. Wrong instrument: a negotiable CD is a marketable deposit sold to corporations and other investors, not a loan of reserves from another bank.

LO: LTR-12 a: Distinguish between the various sources of non-deposit liabilities at a bank. • **Reference:** Rose and Hudgins, Bank Management and Financial Services 9e, Ch.13, 13.3 Federal Funds Market

Solution LTR-12 a.2 — Answer: B

Bank Q399 • GARP book Ch.13 end-of-chapter

Step 1 – The mechanics

An RP is the temporary sale of high-quality, easily liquidated assets (the starting leg) with an agreement to buy them back on a set date at a predetermined price (the closing leg).
⇒ Economically it is a collateralised Fed funds loan: the lender holds the securities, so its credit risk is lower.

Step 2 – Test the other statements

Interest on RPs and Fed funds is Amount × rate × days/360 (eq. 13.1). In a GCF RP the parties agree an array of eligible securities, any of which can be delivered.

Interpretation

The collateral has a cost in time: the lender must be satisfied with the quality and quantity of the securities, so an RP may take a bit longer to arrange than a Fed funds loan.

Why the other options are wrong

A is incorrect. Polarity flip: that describes an ordinary Fed funds loan; an RP is collateralised.
C is incorrect. Wrong convention: RP interest uses a 360-day year ($50 million for three days at 6% costs $25,000).
D is incorrect. Reversed: GCF RPs were created (1998) for low-cost collateral substitution; the securities pledged at the beginning do not have to be delivered at the end.

LO: LTR-12 a: Distinguish between the various sources of non-deposit liabilities at a bank. • **Reference:** Rose and Hudgins, Ch.13, 13.3 Repurchase Agreements as a Source of Funds, eq. (13.1)

Solution LTR-12 a.3 Answer: D

Bank Q397 • Schweser Module Quiz 77.1 Q1

Step 1 – Secured sources

RPs are collateralised by the securities sold and repurchased. Every discount window loan "must be backed by collateral acceptable to the Fed".

Step 2 – Unsecured sources

Overnight Fed funds are normally not secured by specific collateral; commercial paper is a short-term unsecured note.

$\Rightarrow$ Only D pairs two secured sources.

Interpretation

FHLB advances, not offered here, are also secured (by home mortgages). Collateral is the main thing that separates RPs from Fed funds, and the discount window from interbank borrowing.

Why the other options are wrong

A is incorrect. Neither Fed funds nor commercial paper requires collateral.

B is incorrect. Commercial paper is unsecured.

C is incorrect. Fed funds are generally unsecured; only the RP half is right.

LO: LTR-12 a: Distinguish between the various sources of non-deposit liabilities at a bank. • **Reference:** Rose and Hudgins, Ch.13, 13.3; SchweserNotes 2026 Book 4, Reading 77, LO 77.a

Solution LTR-12 a.4 Answer: B

Bank Q396 • Mock 2 Q76

Step 1 – The three facilities and who qualifies

Primary credit: sound institutions, usually overnight, slightly above the target Fed funds rate. Secondary credit: institutions that do not qualify for primary credit, at a higher rate. Seasonal credit: small and medium institutions with intrayear swings.

Step 2 – Apply the bank's facts

The bank is sound (primary is available), has no seasonal pattern (seasonal is not), and wants the cheapest rate it can get.

$\Rightarrow$ Primary credit.

Interpretation

Seasonal credit carries the lowest rate of the three, but only institutions with seasonal needs can use it. Secondary credit is the most expensive and is monitored; it is not meant to fund asset expansion.

Why the other options are wrong

A is incorrect. There is no "overnight credit" facility; primary credit is the facility that is usually overnight.

C is incorrect. Cheapest, but restricted to institutions with seasonal swings in deposits and loans.

D is incorrect. The highest of the three rates, for banks that do not qualify for primary credit.

LO: LTR-12 a: Distinguish between the various sources of non-deposit liabilities at a bank. • **Reference:** Rose and Hudgins, Ch.13, 13.3 Borrowing from Federal Reserve Banks

Solution LTR-12 a.5 — Answer: C

Bank Q393 • AnalystPrep Q.4113

Step 1 – Two discriminators: who issues, and where

The issuer is a foreign bank (HSBC) and the CDs are sold inside the United States through its US branch.

⇒ "The largest foreign banks active in the United States (such as Deutsche Bank and HSBC) sell CDs through their U.S. branches, called Yankee CDs."

Interpretation

Vardeez flags this pair as built for swapping: a EuroCD is a dollar CD issued by a bank *outside* the US; a Yankee CD is issued by a *foreign bank inside* the US. Both have a foreign element; the direction is the discriminator.

Why the other options are wrong

A is incorrect. Domestic CDs are issued by US institutions inside the US.

B is incorrect. EuroCDs are dollar CDs issued by banks outside the US; these are sold in New York.

D is incorrect. Thrift CDs are sold by nonbank savings institutions.

LO: LTR-12 a: Distinguish between the various sources of non-deposit liabilities at a bank. • **Reference:** Rose and Hudgins, Ch.13, 13.3 Development and Sale of Large Negotiable CDs

Solution LTR-12 a.6 — Answer: C

Bank Q386 • GARP book Ch.13 end-of-chapter

Step 1 – Currency plus location

"Eurodollars are dollar-denominated deposits placed in bank offices outside the United States."

Step 2 – How a US bank books them

When an overseas branch lends a Eurodeposit to its home office, the home office records it as *liabilities to foreign branches*.

Interpretation

Most Eurodollar deposits are fixed-rate time deposits maturing within six months and priced off LIBOR. The market is centred in London and is the largest unregulated marketplace in the world, which is why B's reserve requirement is wrong too.

Why the other options are wrong

A is incorrect. Confuses the currency: Eurodollars are dollars; "Euro" refers to the market's origin in Europe.

B is incorrect. Wrong location and wrong regulation: they are held outside the US and are not subject to US reserve requirements.

D is incorrect. Wrong term and security: the majority mature within six months; only FRCDs and FRNs run 1 to 20 years, and none is Treasury-backed.

LO: LTR-12 a: Distinguish between the various sources of non-deposit liabilities at a bank. • **Reference:** Rose and Hudgins, Ch.13, 13.3 The Eurocurrency Deposit Market

Solution LTR-12 a.7 — Answer: D

Bank Q398 • BT Mock Q69

Step 1 – Why D is correct

FHLB advances are "a stable source of funding at below-market interest rates", fully collateralised by home mortgages, with maturities from overnight to more than 20 years, and the number of

borrowers has grown dramatically "especially among smaller community banks and thrift institutions".

Interpretation

For a small bank funding mortgages, the FHLB solves the size and term problems that rule out Eurodollars and commercial paper (LTR-12 c). The FHLB's priority claim in a failure (ahead of the FDIC) is the price of that cheap, stable money.

Why the other options are wrong

A is incorrect. Polarity flip: Fed funds are the most popular domestic source of borrowed reserves and among the cheapest; their weakness is a volatile rate, not a high one.

B is incorrect. Reversed on both counts: RPs are less popular and more complex than Fed funds, and the collateral gives them less credit risk.

C is incorrect. Reversed: commercial paper is high in volume and moderate in cost, but volatile in available capacity and subject to credit risk.

LO: LTR-12 a: Distinguish between the various sources of non-deposit liabilities at a bank. • **Reference:** Rose and Hudgins, Ch.13, 13.3 Federal Funds Market; Repurchase Agreements; Advances from Federal Home Loan Banks; Commercial Paper Market

Solution LTR-12 a.8 Answer: C

Workbook A27.2 • S2 • Aug-2026

Step 1 – Why C is correct

Size of the borrowing institution is one of Rose and Hudgins's five factors. "The standard trading unit for most money market loans is USD 1 million", Eurodollar borrowings "usually are available only to money-center financial firms with the highest credit ratings", whereas "the central bank's discount window and the Fed funds market can make relatively small denomination loans that are suitable for smaller depository institutions".

Step 2 – The term factor points the same way

With funds needed this afternoon, the length-of-time factor also points to Fed funds or the discount window.

Interpretation

For a small bank the realistic menu today is Fed funds (cheapest, but a volatile rate) or primary credit at the window (collateral needed, priced above the Fed funds target). Size, term, cost, risk and regulation must be weighed together (LTR-12 c).

Why the other options are wrong

A is incorrect. Polarity flip: "the shorter the term of the loan, the more volatile the prevailing market interest rate tends to be"; overnight Fed funds have the most volatile rate of all, which is their key disadvantage.

B is incorrect. Sibling swap (rank of the three window facilities): secondary credit carries the highest rate and seasonal credit the lowest. The conclusion about primary credit is reasonable, but the stated reason is false.

D is incorrect. Absolute claim: experience shows that "the negotiable CD, Eurodollar, and commercial paper markets are especially sensitive to credit availability risks"; the legal status of a CD as a deposit does not protect the issuer from rationing.

LO: LTR-12 a: Distinguish between the various sources of non-deposit liabilities at a bank. • LTR-12 c: Discuss factors affecting the choice of non-deposit funding sources. • **Reference:** Rose and Hudgins, Ch.13, 13.3 Alternative Nondeposit Sources of Funds; 13.4 Choosing among Alternative Nondeposit Sources

LTR-12 • Managing Non-Deposit Liabilities • Vardeez: IMPORTANT

LTR-12 c

P1 CONCEPT

Discuss factors affecting the choice of non-deposit funding sources.

Workbook: A27 Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A27, recalled in August 2026 as a "non-deposit liability theory question" (Map #47). Shape S2 pairs the sources (LTR-12 a) with the factors in choosing among them. Workbook A27.2 tags both LOs and is worked under LTR-12 a (Question LTR-12 a.8): read it with this block.

Vardeez: no pointer for this LO. His notes list the five factors, state the cost ordering precisely and call the cheapest-source question "a shortcut question": it is answered from the ordering alone.

Reported exam shapes

- **S2** Sources of non-deposit liabilities and the factors in choosing among them, theory as seen in August.

What the questions look like: two kinds. Scenario questions give a bank of a stated size that needs a stated amount, by a stated time, for a stated term, and ask for the best source. Statement questions test the cost ranking and the two risks. The bank's effective-cost calculations (rate plus noninterest cost over net investable funds) are worked under LTR-12 d, Section 2; here the questions stay qualitative.

1. How much, then which

Funds managers must answer two questions: **how much** must be borrowed (the available funds gap, LTR-12 b) and **which** nondeposit sources are best, given the bank's goals at that moment. Under the customer relationship doctrine (LTR-12 a, Section 1) the lending decision comes first; the funding decision then looks for the lowest-cost source that fits. Rose and Hudgins name **five factors**:

1. the **relative costs** of raising funds from each source;
2. the **risk** (volatility and dependability) of each source;
3. the **length of time** (maturity or term) for which funds are needed;
4. the **size** of the institution that needs the funds;
5. the **regulations** limiting the use of each source.

2. Relative costs

Managers would prefer the cheapest source, so they track the posted rates every day. The ordering, as GARP states it:

- **Fed funds** (the effective overnight rate) is usually **among the cheapest**. It is not the absolute lowest: the shortest Treasury bills often yield slightly less.
- **Domestic CDs and Eurocurrency deposits** are in most cases slightly above Fed funds.
- **Commercial paper** normally issues slightly above the Fed funds and CD rates, depending on maturity and time of issue.

- The **primary credit rate** at the discount window is **among the highest** short-term rates, because the Fed prices it above its target for the Fed funds rate.

Vardeez's shortcut: Fed funds $<$ CDs $\approx$ Eurocurrency $<$ commercial paper $<$ primary credit. The counterintuitive end is the right-hand one: central bank credit is expensive by design (before 2003 it was the reverse, Figure 1).

- **Fed funds:** key advantage, ready availability by a phone call or online request, with maturities from a few hours to several months. Key disadvantage, a **volatile rate** that fluctuates around the Fed's target, widely on reserve settlement days, which makes planning difficult.
- **CDs and commercial paper:** rates somewhat more stable, but slightly higher (longer average maturity, marketing costs). Less popular than Fed funds and the discount window when money is needed right away; usually better for needs that stretch over several days or weeks.
- **Noninterest costs count.** Staff time, facilities, transaction costs, reserve requirements, deposit insurance fees and funds diverted into nonearning assets all raise the true cost, so sources are compared on an **effective cost** per dollar of net investable funds (calculated in LTR-12 d, Section 2: Fed funds 6.51% against CDs 8.07% in GARP's example).
- **Against deposits:** nondeposit funds are generally moderate in cost: more expensive than checkable deposits, less expensive than thrift (time and savings) deposits. But their costs and profits are **more volatile** year to year, and they are not as stable a funding source as time and savings deposits.

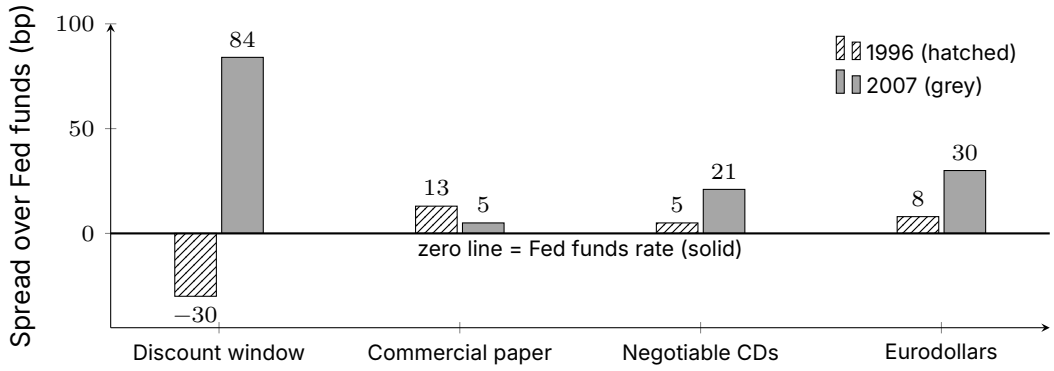


Figure 1. Average rates from Rose and Hudgins's Table 13.9, shown as spreads over the Fed funds rate (the zero line; 5.30% in 1996, 5.02% in 2007). In 1996 the Fed's discount rate sat 30 bp *below* Fed funds; since 2003 the primary credit rate is set above the target, and in 2007 it was the dearest source by far. Commercial paper is 1-month directly placed, CDs 1-month secondary market, Eurodollars 3-month; CP and CD rates sit close together and their order depends on maturity and time of issue.

3. Risk: two kinds

- **Interest rate risk**, the volatility of credit costs. Every rate in Table 13.9 except the central bank's discount rate is set by supply and demand in the open market and fluctuates. **The shorter the term of the loan, the more volatile the rate**: most Fed funds loans are overnight, so the Fed funds rate "tends to be the most volatile of all".
- **Credit availability risk.** No credit market guarantees that lenders will accommodate every borrower. When credit is tight they ration it to their soundest, most loyal customers; a borrower that looks risky may be refused or charged so much that earnings suffer. Experience shows that the

negotiable CD, Eurodollar and commercial paper markets are especially sensitive to credit availability risk. Funds managers must be ready to switch sources and, if necessary, pay more.

> **Correction to the source notes.** Vardeez's factor table reads "Risk: ... Fed funds are most reliable; CDs and commercial paper are riskier because they depend on market conditions." That is the *credit availability* half of the factor only (Schweser: Fed funds are "generally more reliable"). On *interest rate risk* GARP says the reverse: the shorter the term, the more volatile the rate, and the overnight Fed funds rate is the most volatile of all, its key disadvantage. A statement that Fed funds are the least risky source on both counts is false.

4. Length of time the funds are needed

Some sources are hard to tap immediately, notably commercial paper and long-term debt. A manager who needs loanable funds **this afternoon** turns to the **Fed funds market** (or the discount window); if the money is not needed for a few days, selling longer-term debt becomes viable. The term of the need also decides the rate risk: rolling overnight money for a six-week loan is cheap today but reprices every day.

5. Size of the borrowing institution

- The standard trading unit for most money market loans is **$1 million**, which may exceed the needs of the smallest institutions.
- Eurodollar borrowings come in multiples of $1 million and are usually available only to **money-centre** firms with the **highest credit ratings**.
- Investors prefer large CDs from the largest institutions, which have an active secondary market; small institutions may lack the credit standing to sell the largest CDs or commercial paper.
- The **discount window and the Fed funds market** make relatively small loans that suit smaller institutions; FHLB advances serve community banks and thrifts that fund mortgages.

6. Regulations

Federal and state rules may limit the **amount, frequency and use** of borrowed funds. In the US, CDs must have maturities of **at least seven days**; the Federal Reserve banks may limit discount window borrowing, especially by institutions with a significant risk of failure; and the central bank can impose legal reserve requirements. In the late 1960s and early 1970s, fighting inflation, the Fed imposed reserve requirements on Fed funds borrowings, RPs and commercial paper issued to buy assets from affiliated lenders. Those requirements are not in force today, but in a national emergency policymakers would move swiftly to impose new controls, changing both the cost and the risk of nondeposit borrowing.

Factor	The question to ask	Favours	Counts against
Relative cost	Effective cost per dollar of net investable funds?	Fed funds (among the cheapest)	Primary credit (above the Fed funds target); CP slightly above CDs
Interest rate risk	How volatile is the rate?	CDs, CP (more stable)	Overnight Fed funds (most volatile)
Credit availability risk	Will lenders still lend under stress?	Fed funds (generally more reliable)	CDs, Eurodollars, CP (especially sensitive)
Term	When and for how long?	Today: Fed funds, window. Weeks: CDs, CP	CP and long-term debt are not immediate
Size	Can the bank reach the market?	Small: Fed funds, window, FHLB	Eurodollars, large CDs, CP: $1 million units, top ratings
Regulation	Limits on amount, frequency, use?	Fed funds, CDs: no reserve requirement now	CDs at least 7 days; window limits for risky banks

Exam traps

- **Polarity flip on volatility.** The shorter the term, the *more* volatile the rate. Overnight Fed funds are cheap but the least predictable.
- **Rank swap.** The primary credit rate is among the *highest* short-term rates (above the Fed funds target), not the cheapest. Fed funds are among the cheapest, but T-bills can be lower.
- **Size swap.** Eurodollars and large CDs need money-centre size and top ratings; Fed funds and the window serve small banks.
- **Term mismatch.** Commercial paper and long-term debt cannot meet a need "this afternoon". Overnight money rolled for weeks carries rising-rate risk.
- **Absolute claims.** No source is free of credit availability risk; a CD's legal status as a deposit does not protect the issuer.
- **Deposit comparison.** Nondeposit funds cost more than checkable deposits and less than thrift deposits, and their cost is *more* volatile.
- **Order of decisions.** Under the customer relationship doctrine the lending decision comes first, the funding decision second.

Practice questions: LTR-12 c

Question LTR-12 c.1

Bank Q412 • Schweser Module Quiz 77.2 Q1

Kris Gaines, treasurer at Palm Air Bank and Trust, is considering ways to meet a funding gap created by greater than expected loan demand. Palm Air is a medium-sized bank located in Florida. The funding gap is approximately USD 850,000. Gaines is choosing between several nondeposit funding types. The funds are needed immediately. Which type of funding is most appropriate in this situation?

A. Commercial paper.

B. Negotiable certificates of deposit (CDs).

C. Federal funds borrowing.

D. Eurodollar deposits.

Question LTR-12 c.2

Bank Q411 • Mock 3 Q6

Two institutions face different funding issues. Bank A, a mid-size regional bank, has a shortfall in legal reserves for the day and needs an alternative to cover it. Bank B, a small community bank, has had a much greater than anticipated shortfall in long-term certificate of deposit (CD) renewals because of fierce local competition for retail deposits; it has traditionally used stable CDs to fund its home mortgage portfolio. Considering the timing and availability of nondeposit funds, what is the most appropriate funding response of each?

A. Bank A should borrow from the wholesale deposit market and Bank B should fund itself through the Eurocurrency deposit market.

B. Bank A should fund itself through the commercial paper market and Bank B should borrow from the Federal funds market.

C. Bank A should borrow from the Federal funds market and Bank B should borrow from the Federal Home Loan Banks.

D. Bank A should issue debentures and Bank B should fund itself through the commercial paper market.

Question LTR-12 c.3

New • built from GARP Ch.13 (13.4 Relative Costs, Table 13.9)

An analyst on the asset-liability committee of a US regional bank drafts notes on the relative cost of nondeposit sources for the bank's funding policy. Which of the following notes is correct?

A. The primary credit rate at the discount window is normally among the cheapest short-term borrowing rates, because the Federal Reserve prices it below its target for the Fed funds rate.

B. The effective Fed funds rate is usually among the cheapest short-term borrowing rates, although the rate on the shortest-term Treasury bills is often slightly lower.

C. Commercial paper is normally issued at rates below the Fed funds and CD rates, because it is sold at a discount from face value.

D. Nondeposit funds generally cost less than checkable deposits but more than time and savings deposits, and their cost is more stable from year to year.

Question LTR-12 c.4 New • built from GARP Ch.13 (13.4 The Risk Factor)

The treasurer of a mid-sized bank compares three ways of funding loan growth: overnight Fed funds, three-month negotiable CDs, and commercial paper issued through the bank's holding company. Which assessment of the risks she must weigh is correct?

A. Fed funds carry the least interest rate risk of the three, because overnight rates are set by the central bank and do not fluctuate around its target.

B. The negotiable CD and commercial paper markets are especially sensitive to credit availability risk, so the bank should be prepared to switch to other sources and pay more if lenders ration credit.

C. Credit availability risk is negligible for a profitable bank, because money market lenders accommodate every borrower willing to pay the market rate.

D. Interest rate risk rises with the term of the borrowing, so the three-month CD has the most volatile rate of the three sources.

Question LTR-12 c.5 Bank Q408 • GARP book Ch.13 Problem 10 (rebuilt)

Surfs-Up Security Savings must raise USD 80 million in money market funds to cover a six-week loan requested by one of its largest corporate customers. Current market rates are:

Source	Rate
Federal funds, average for the week just concluded	1.98%
Discount window of the Federal Reserve Bank	2.25%
CDs (prime rated, secondary market): one month / three months / six months	2.52% / 2.80% / 3.18%
Eurodollar deposits, three months	3.00%
Commercial paper (directly placed): one month / three months	2.33% / 2.70%

The bank's economics department forecasts a substantial rise in money market interest rates over the next six weeks. Which recommendation to the funds management department is most consistent with the factors in Rose and Hudgins?

A. Borrow overnight Fed funds at 1.98% and roll them daily, because Fed funds are the cheapest source and their rate is the least volatile.

B. Raise term money with a maturity close to the six-week need, such as directly placed commercial paper or CDs, locking in today's rates even though the starting rate is above Fed funds.

C. Take primary credit at the discount window for the whole six weeks, because it is designed as a routine, low-cost source of term funding.

D. Issue six-month CDs at 3.18%, because the longest available maturity always minimises interest rate risk, whatever the term of the need.

Question LTR-12 c.6 New • built from GARP Ch.13 (13.2 and 13.4 Regulations)

A board member asks the funding desk how regulation and the bank's lending philosophy shape its use of nondeposit funds. Which statement in the desk's reply is correct?

A. Under the customer relationship doctrine, the bank first decides how much it can raise in deposits and then approves only the loans those deposits can fund.

B. Liability management makes the bank's cost of funds less sensitive to market interest rates, because the bank rather than the depositor chooses the amount and the term.

C. Regulation can limit the amount, frequency and use of borrowed funds: US CDs must have a maturity of at least seven days, the Fed may limit window borrowing by risky banks, and reserve requirements were once imposed on Fed funds, RPs and commercial paper.

D. Because negotiable CDs are legally deposits, they carry the same 10% reserve requirement as checking accounts at the largest banks, which is why CDs cost more than Fed funds.

Solutions • LTR-12 c

Solution LTR-12 c.1 — Answer: C

Bank Q412 • Schweser Module Quiz 77.2 Q1

Step 1 – Size factor

The need is about USD 850,000. Commercial paper, negotiable CDs and Eurodollar deposits trade in units of USD 1 million or more, and Eurodollars and large CDs favour money-centre names.

Step 2 – Term factor

The money is needed immediately. Fed funds are immediately available; commercial paper is not.

$\Rightarrow$ Fed funds fit both the size and the timing.

Interpretation

The discount window would also make a small, immediate loan, but it is not offered; it is priced above the Fed funds target and needs collateral, so Fed funds would still be the first choice on cost.

Why the other options are wrong

A is incorrect. Commercial paper comes in large units, needs a credit standing a medium-sized bank may lack, and is not immediately available.

B is incorrect. Negotiable CDs are sold in multiples of USD 1 million and investors prefer those of money-centre banks.

D is incorrect. Eurodollars come in multiples of USD 1 million and are usually available only to money-centre firms with top ratings.

LO: LTR-12 c: Discuss factors affecting the choice of non-deposit funding sources. • **Reference:** Rose and Hudgins, Ch.13, 13.4 The Length of Time Funds Are Needed; The Size of the Borrowing Institution

Solution LTR-12 c.2 — Answer: C

Bank Q411 • Mock 3 Q6

Step 1 – Bank A: a same-day reserve shortfall

The need is overnight and immediate. Fed funds are the standard source for meeting legal reserve requirements and are available "through a simple phone call".

Step 2 – Bank B: long-term funding for mortgages at a small bank

The need is long term and mortgage related, and the bank is small. FHLB advances are stable, below market, collateralised by home mortgages, run from overnight to more than 20 years, and are widely used by community banks and thrifts.

$\Rightarrow$ Match the source to the timing, term and size of each need.

Interpretation

This is the five-factor framework applied twice: term (today versus years), size (a small bank cannot reach Eurodollars or commercial paper) and cost (both chosen sources are cheap).

Why the other options are wrong

A is incorrect. Wholesale deposits are not immediate, and Eurocurrency deposits are short term, come in USD 1 million units and are mainly for large institutions.

B is incorrect. Commercial paper is not immediately available for a same-day gap, and overnight Fed funds are too short and volatile to replace long-term mortgage funding.

D is incorrect. Debentures take time to issue and run 5 to 12 years; commercial paper is short

term and out of reach of a small bank.

LO: LTR-12 c: Discuss factors affecting the choice of non-deposit funding sources. • **Reference:** Rose and Hudgins, Ch.13, 13.3 Advances from Federal Home Loan Banks; 13.4 Nondeposit Funding Sources: Factors to Consider

Solution LTR-12 c.3 — Answer: B

New • built from GARP Ch.13

Step 1 – The ordering GARP gives

"Among the cheapest short-term borrowed funds source is usually the prevailing effective interest rate on Federal funds". Domestic CDs and Eurocurrency deposits are in most cases slightly higher; commercial paper slightly above Fed funds and CD rates; the primary credit rate is "generally among the highest short-term borrowing rates".

Step 2 – The qualifier

GARP's Factoid: the Fed funds rate is not the absolute lowest, because "the interest rate on the shortest-term U.S. Treasury bills is often slightly lower".
$\Rightarrow$ B states both halves correctly.

Interpretation

The ranking question is a shortcut question: Fed funds $<$ CDs $\approx$ Eurocurrency $<$ CP $<$ primary credit, with no arithmetic. Effective cost still needs the noninterest costs (LTR-12 d, Section 2).

Why the other options are wrong

A is incorrect. Polarity flip: since 2003 the primary credit rate is set *above* the Fed funds target, which puts it among the highest rates. Before 2003 the discount rate was often the lowest, which is what makes this statement tempting.
C is incorrect. Wrong direction: commercial paper normally issues slightly *above* Fed funds and CD rates. Selling at a discount is how it is quoted, not why it is cheap.
D is incorrect. Both halves reversed: nondeposit funds cost more than checkable deposits and less than thrift deposits, and their costs are *more* volatile from year to year.

LO: LTR-12 c: Discuss factors affecting the choice of non-deposit funding sources. • **Reference:** Rose and Hudgins, Ch.13, 13.4 Relative Costs, Table 13.9 and Factoids

Solution LTR-12 c.4 — Answer: B

New • built from GARP Ch.13

Step 1 – Credit availability risk

"There is no guarantee in any credit market that lenders will be willing and able to accommodate every borrower." When credit is tight lenders ration it, and "the negotiable CD, Eurodollar, and commercial paper markets are especially sensitive to credit availability risks. Funds managers must be prepared to switch to alternative sources of credit and, if necessary, pay more."
$\Rightarrow$ B is correct.

Step 2 – Interest rate risk

Market rates are set by supply and demand, and "the shorter the term of the loan, the more volatile the prevailing market interest rate tends to be": the overnight Fed funds rate is the most volatile of all.

Interpretation

The two risks pull in opposite directions: Fed funds score worst on rate volatility and best on availability; CDs and commercial paper are the reverse. That is why the treasurer must weigh both rather than rank sources on one.

Why the other options are wrong

A is incorrect. Polarity flip: the effective Fed funds rate fluctuates around the target, widely on settlement days; only the central bank's own discount rate is administered.

C is incorrect. Absolute claim: lenders may deny credit to a borrower that looks risky or price it so high that earnings suffer.

D is incorrect. Polarity flip: volatility falls as the term lengthens, so the overnight source is the most volatile.

LO: LTR-12 c: Discuss factors affecting the choice of non-deposit funding sources. • **Reference:** Rose and Hudgins, Ch.13, 13.4 The Risk Factor

Solution LTR-12 c.5

Answer: B

Bank Q408 • GARP book Ch.13 Problem 10

Step 1 – Term and rate outlook

The need lasts six weeks and rates are expected to rise substantially. Overnight money would reprice every day at ever higher rates, and the Fed funds rate is the most volatile of all.

Step 2 – Lock in the cost for (about) the life of the need

Term sources such as one-month directly placed commercial paper (2.33%, the cheapest term rate here, rolled once) or a CD placed for about six weeks fix most or all of the cost now. The starting rate is a little above Fed funds; that is the price of removing the rate risk.

$\Rightarrow$ B matches term, risk and cost together.

Interpretation

If the forecast were for falling rates the logic reverses: borrow short (Fed funds) and roll, so that each renewal reprices lower, and avoid locking in three- or six-month money. Reserve requirements and insurance fees, if any, raise effective cost by cutting net investable funds.

Why the other options are wrong

A is incorrect. Half true, half false: Fed funds are the cheapest today, but their rate is the *most* volatile, and rolling them into a rising market is exactly the risk the forecast warns about.

C is incorrect. Scope error: discount window credit is not meant for routine funding, the Fed may limit it, and primary credit is usually overnight (only occasionally up to 90 days).

D is incorrect. Absolute claim: six-month CDs overborrow by about four and a half months at the highest rate shown; the term of the need, not the longest maturity, sets the right source.

LO: LTR-12 c: Discuss factors affecting the choice of non-deposit funding sources. • **Reference:** Rose and Hudgins, Ch.13, 13.4 Relative Costs; The Risk Factor; The Length of Time Funds Are Needed; Problem 10

Solution LTR-12 c.6

Answer: C

New • built from GARP Ch.13

Step 1 – Why C is correct

"Federal and state regulations may limit the amount, frequency, and use of borrowed funds." In the US, CDs "must be issued with maturities of at least seven days"; the Federal Reserve banks "may limit borrowings from the discount window, particularly by depository institutions that appear to display significant risk of failure"; and in the late 1960s and early 1970s the Fed imposed reserve requirements on Fed funds borrowing, RPs and commercial paper issued to buy assets from affiliates.

Interpretation

Those reserve requirements are not in force today, but GARP warns that in a national emergency

policymakers would move swiftly to impose new controls, changing both the cost and the risk of each source.

Why the other options are wrong

A is incorrect. Reversed order: the doctrine says lending decisions *precede* funding decisions; if deposits fall short, the bank borrows the lowest-cost funds available.

B is incorrect. Half true: the borrower does choose amount and term (flexibility), but liability management is an *interest-sensitive* approach; money market lenders are highly rate elastic and leave in minutes if the bank will not pay the going rate.

D is incorrect. Wrong fact: the legal reserve requirement on CDs is currently zero, against 10% on checking accounts at the largest institutions; that is why switching deposits into CDs frees loanable funds.

LO: LTR-12 c: Discuss factors affecting the choice of non-deposit funding sources. • **Reference:** Rose and Hudgins, Ch.13, 13.2 Liability Management and the Customer Relationship Doctrine; 13.3 Negotiable CDs; 13.4 Regulations

LTR-12 • Managing Non-Deposit Liabilities • Vardeez: IMPORTANT

LTR-12 d

P1 CALC

Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach.

Workbook: A27 Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A27, recalled in August 2026; Map #47 reports "pooled cost approach, etc." (2 of 16 sources). The examinable contrast is the forward-looking pooled-funds approach against the backward-looking historical average cost approach.
Vardeez: "Cost of funds: historical average and pooled-funds approaches", half of his reading note (the other half is the available funds gap, LTR-12 b). His notes rebuild both chains from GARP and list the intermediate results GARP offers as distractors.

Reported exam shapes

- **S1** Pooled-funds approach (forward-looking hurdle rate) against the historical average cost approach, numerical.

What the questions look like: numerical. A table of funding sources with amounts, interest rates and noninterest costs, plus earning assets, equity, the shareholders' required return and a tax rate. The stem often carries inputs for *both* approaches (last year's totals and next year's plan) and only one belongs to the question. Each chain has two or three legitimate intermediate results, and GARP offers them. The effective cost rate of a single source, which the workbook files here, is the building block (Question LTR-12 d.1).

1. Two methods, two questions

Borrowing institutions draw on many sources at once: deposits, nondeposit borrowings and owners' equity. Rose and Hudgins give two methods for an **overall** cost of funds, and they answer different questions:

- **Historical average cost approach: looks at the past.** What funds has the institution raised to date, and what did they cost? The result is the rate existing earning assets must yield to cover the cost of funds already raised.
- **Pooled-funds approach: looks at the future.** What minimum rate of return must be earned on any *future* loans and investments just to cover the cost of all *new* funds raised? The result is a hurdle rate for new business.

Backward-looking against forward-looking is the discriminator GARP tests. Both approaches pool *all* sources, including equity; neither prices a single source.

2. The building block: effective cost of one source

Before pooling, Rose and Hudgins compare sources one at a time on the money that can actually be put to work:

$$\text{Effective cost rate} = \frac{\text{Interest cost on amounts borrowed} + \text{Noninterest costs to access the funds}}{\text{Net investable funds raised}}$$

(eq. 13.4). Interest cost = market rate × amount borrowed (13.5); noninterest cost = cost rate for staff time, facilities and transactions × amount borrowed (13.6); **net investable funds** = total borrowed less legal reserve requirements, deposit insurance assessments and funds placed in nonearning assets (13.7). Both costs are paid on the *whole* amount raised; only the net amount earns.

Example (GARP). The bank needs $25 million today, of which only $24 million can be fully invested.

- Fed funds at 6.00%, noninterest cost 0.25%: $(1.5 + 0.0625)/24 = 6.51\%$.
- Negotiable CDs at 7.00%, noninterest cost 0.75%: $(1.75 + 0.1875)/24 = 8.07\%$.
- The FDIC fee on the CDs, 0.27% of the full $25 million ($0.0675 million), also comes off net investable funds: $1.9375/(24 - 0.0675) = 1.9375/23.9325 = 8.10\%$.

$\Rightarrow$ The bank must earn at least 6.51% on loans funded with Fed funds just to break even. CDs cost more, but they stay for weeks, months or years while Fed funds must often be repaid in 24 hours (the term factor of LTR-12 c).

3. The historical average cost approach

Source of funds drawn upon	Raised ($m)	Average rate	Interest paid ($m)
Noninterest-bearing demand deposits	100	0%	0
Interest-bearing transaction deposits	200	7%	14
Savings accounts	100	5%	5
Time deposits	500	8%	40
Money market borrowings	100	6%	6
Total	1,000		65

Other operating costs (salaries, overhead) are $10 million, earning assets $750 million, stockholders' investment $100 million, the stockholders' required return 12% *after* tax, and the tax rate 35%.

1. **Weighted average interest expense**

$$= \frac{\text{All interest paid}}{\text{Total funds raised}} = \frac{65}{1{,}000} = 6.5\%.$$

   The cost of the money alone.

2. **Break-even cost rate on borrowed funds invested in earning assets**

$$= \frac{\text{Interest} + \text{Other operating costs}}{\text{All earning assets}} = \frac{65 + 10}{750} = 10.0\%.$$

   The institution must earn at least this on its earning assets (mainly loans and securities) just to meet the total operating costs of raising the funds. *The denominator switches* from funds raised to earning assets.

3. **Weighted average overall cost of capital**

$$= \text{Break-even} + \frac{\text{After-tax cost of stockholders' investment}}{1 - t} \times \frac{\text{Stockholders' investment}}{\text{Earning assets}}$$

$$= 10\% + \frac{12\%}{0.65} \times \frac{100}{750} = 10\% + 2.46\% = 12.46\%.$$

$\Rightarrow$ 12.5% (GARP rounds the equity term to 2.5%) is "the lowest rate of return over all fund-raising costs the borrowing institution can afford to earn on its assets" given $100 million of equity. The equity return is grossed up by $1 - t$ because it must be earned out of pre-tax income; skip that and you get 11.6%.

Inputs the formula does not want. Total assets (only earning assets), the equity cost in the break-even rate (equity is added in step 3, not step 2), and a before-tax equity cost grossed up a second time (if the stem says "before taxes", add it as it is; Question LTR-12 d.4).

4. The pooled-funds approach

New deposits and nondeposit funds	Raised ($m)	Share placed	Placed ($m)	Expenses ($m)
Interest-bearing transaction deposits	100	50%	50	8 (8%)
Time deposits	100	60%	60	9 (9%)
New stockholders' investment	100	90%	90	13 (13%)
Total	300		200	30

Expenses are interest plus other operating expenses, as a percentage of the amount raised; for new equity, its required return.

1. **Pooled deposit and nondeposit funds expense**

$$= \frac{\text{All expected operating expenses}}{\text{All new funds expected}} = \frac{30}{300} = 10\%.$$

   What the new money costs per dollar raised. Rose's end-of-chapter problem calls this the **pooled-funds marginal cost**.

2. **Hurdle rate of return over all earning assets**

$$= \frac{\text{All expected operating costs}}{\text{Dollars available to place in earning assets}} = \frac{30}{200} = 15\%.$$

$\Rightarrow$ Only two-thirds of the new money ($200 of $300 million) can acquire earning assets; the rest goes to reserves and other nonearning uses. The same $30 million must be recovered from a smaller base, so the firm must earn at least **15% before taxes**, on average, on all new funds it invests. The 15% is the decision number; 10% is the cost, not the hurdle.

Vardeez's warning: the unweighted average of the placement rates, $(50 + 60 + 90)/3 = 66.7\%$, turns 10% into 15% here only because each source raises the same $100 million. Weight by dollars placed, never by an average of percentages.

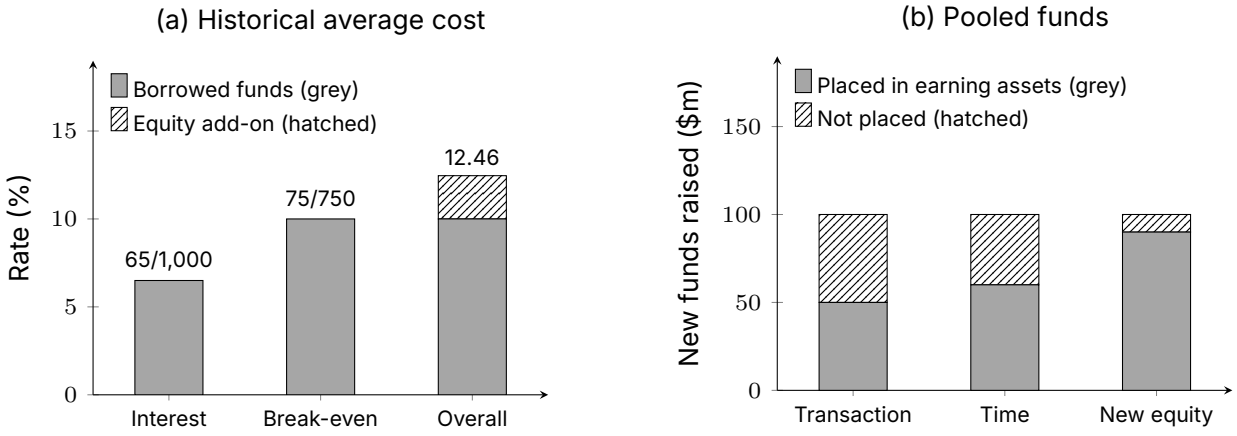


Figure 1. (a) GARP's historical chain: the interest cost on all funds raised (6.5%), the break-even rate on earning assets once operating costs are added (10.0%), and the overall cost of capital once the grossed-up equity return is added (hatched, 2.46%). (b) GARP's pooled-funds plan: $300 million raised, of which $200 million (grey) reaches earning assets. The $30 million of expected costs is 10% of all new funds but a 15% hurdle on the dollars placed.

5. Which approach, which number

- **Pricing new loans:** the pooled-funds hurdle rate. Last year's break-even rate describes funds already raised, at last year's rates.
- **Two different "pooled" ideas.** Rose and Hudgins's pooled-funds approach is forward looking and costs all *new* funds. Grant's *pooled average cost of funds* approach to liquidity transfer pricing (LTR-14 c) averages the interest expense of *existing* sources and so lags the market, which is closer in spirit to the historical average. A statement that swaps the two is a sibling swap.
- **Equity is a source of funds** in both approaches: it enters the historical chain in step 3 (grossed up, weighted by equity over earning assets) and the pooled plan as "new stockholders' investment" with its required return in the numerator and its placed dollars in the denominator.

Formula card: LTR-12 d

Effective cost, one source	$\dfrac{\text{rate} \times \text{borrowed} + \text{noninterest rate} \times \text{borrowed}}{\text{borrowed} - \text{reserves} - \text{insurance fee} - \text{nonearning uses}}$
Weighted avg interest expense	All interest paid $\div$ total funds raised
Break-even cost rate	(Interest + other operating costs) $\div$ all earning assets
Overall cost (historical)	$\text{Break-even} + \dfrac{\text{after-tax equity return}}{1-t} \times \dfrac{\text{equity}}{\text{earning assets}}$
Pooled funds expense	All expected operating expenses $\div$ all new funds raised
Hurdle rate (pooled)	All expected operating costs $\div$ dollars placed in earning assets
Interest conventions	Fed funds, RPs, CDs: amount $\times$ rate $\times$ days/360

	Historical average cost	Pooled funds
Looks	Backward: funds raised to date	Forward: all new funds
Question	What did our funds cost?	What must new loans earn?
Numerator	Interest + operating costs; then equity add-on	All expected costs, equity return included
Denominator	Funds raised (step 1), earning assets (steps 2, 3)	Funds raised (expense), dollars placed (hurdle)
Equity	Grossed up by $1 - t$, weighted by equity/earning assets	A source with its own cost and placement rate
GARP's numbers	6.5%, 10%, 12.5%	10%, 15%

Exam traps

- **Sibling swap.** Using last year's (historical) figures when the question asks what new loans must earn, or the reverse. Rose's pooled-funds approach is not Grant's pooled average cost of LTP.
- **Intermediate result.** 6.5% and 10% are correct answers to different questions in the historical chain; 10% (pooled expense) and 15% (hurdle) in the pooled chain. Read which one is asked.
- **Wrong denominator.** Interest expense over funds raised; break-even and overall cost over *earning assets* (not total assets, not funds raised); hurdle over *dollars placed*; effective cost over *net investable* funds.
- **Tax gross-up.** An after-tax equity return is divided by $1 - t$. A before-tax one is not; the tax rate is then a red herring.
- **Dropped costs.** Noninterest costs, reserve and collection costs, and the full-face deposit insurance fee all belong in the numerator (the fee also reduces net investable funds).
- **Equity in the wrong place.** No equity cost in the break-even rate; when new equity is a pooled source, keep both its cost and its placed dollars.

Practice questions: LTR-12 d

Question LTR-12 d.1

Bank Q407 • GARP book Ch.13 Problem 9 (rebuilt)

The holding company of Best-of-Times Financial plans to raise USD 775 million in short-term funds this week, of which USD 700 million will be available to meet new loan requests (net investable funds). Current market rates and noninterest costs, both stated as a percentage of the funds raised, are:

Source	Market rate	Noninterest cost
Fed funds	2.25%	0.25%
Negotiable CDs	2.40%	0.25%
Eurodollars	2.30%	0.35%
Commercial paper (one month, directly placed)	2.35%	0.50%
Primary credit, discount window	2.75%	0.25%

Which of the following is closest to the effective cost rate of raising the funds through commercial paper?

A. 2.60%

B. 2.85%

C. 3.16%

D. 3.32%

Question LTR-12 d.2

Bank Q421 • Mock 1 Q4 (rebuilt)

The Harper Valley Bank (HVB) uses the historical average cost approach to estimate its cost of funds. HVB estimates noninterest expenses, including salaries and overhead, at USD 20 million a year. HVB has total assets of USD 2.1 billion, earning assets of USD 1.5 billion and equity capital of USD 300 million. Total interest paid on borrowed funds is USD 52 million. The shareholders' after-tax required rate of return is 9% and the tax rate is 35%. HVB's weighted average overall cost of capital is closest to:

A. 6.24%

B. 6.60%

C. 6.78%

D. 7.57%

Question LTR-12 d.3 Bank Q415 • AnalystPrep Q.4109 (rebuilt; key as corrected in the bank)

A cost accountant at ABC Bank applies the historical average cost approach. The bank raised USD 1,530 million from the sources below, with costs as a percentage of each source:

Source	USD m	Interest	Noninterest	Reserve, coll.	Total
Money market borrowings	275	3.03%	0.24%	0.24%	3.51%
Time and savings deposits	580	2.27%	0.23%	0.60%	3.10%
Checkable deposits	385	0.52%	1.96%	0.80%	3.28%
Stockholders' equity	290		required return 12.75%		12.75%

Earning assets are USD 1,240 million. Which of the following is closest to the break-even cost rate on borrowed funds invested in earning assets?

A. 2.6%

B. 3.2%

C. 5.0%

D. 6.2%

Question LTR-12 d.4 Bank Q413 • GARP book Ch.13 Problem 11 (rebuilt to GARP's wording)

June Bug Bank and Trust has raised USD 750 million: USD 200 million of checkable deposits, USD 400 million of time and savings deposits, USD 100 million of money market borrowings and USD 50 million of stockholders' equity. Its costs, as a percentage of each source, are:

Source	Amount (USD m)	Interest	Noninterest	Reserve and collection
Checkable deposits	200	0.75%	2.00%	0.75%
Time and savings deposits	400	2.50%	0.50%	0.50%
Money market borrowings	100	2.75%	0.25%	0.25%

Management estimates the cost of stockholders' equity capital at 12% before taxes; the bank is in the 35% corporate tax bracket. Earning assets total USD 700 million. Using the historical average cost approach, which of the following is closest to June Bug's overall weighted average cost of capital?

A. 3.79%

B. 4.03%

C. 4.32%

D. 4.78%

Question LTR-12 d.5

Bank Q422 • Schweser Module Quiz 77.2 Q2

Blue Star Bank expects to raise USD 300 million, USD 250 million of which will be invested in earning assets. The total expected interest and overhead costs on the newly raised funds are forecast to be USD 22 million. The pooled deposit and nondeposit funds expense and the hurdle rate over all earning assets are, respectively:

A. 7.33%; 7.33%.

B. 7.33%; 8.80%.

C. 8.80%; 7.33%.

D. 8.80%; 8.80%.

Question LTR-12 d.6

Bank Q414 • GARP book Ch.13 Problem 12 (rebuilt)

Inspiration Savings Association is considering funding a package of new loans of USD 400 million. It projects that it must raise USD 450 million in order to have USD 400 million available for the loans. It expects to raise USD 325 million by selling time deposits at an average interest rate of 1.75%, with noninterest (operating) costs of 0.45% of the amount sold. The other USD 125 million will come from noninterest-bearing transaction deposits, whose noninterest costs are expected to be 2.00% of their total. Using the pooled-funds approach, which of the following is closest to the hurdle rate the Association must achieve on its earning assets?

A. 1.42%

B. 2.14%

C. 2.41%

D. 2.76%

Question LTR-12 d.7

Workbook A27.1 • S1 • Map #47

The treasurer of a UK retail bank is presenting next year's funding plan to the asset-liability committee. The bank expects to raise the following new funds:

New source	Raised (GBP m)	Placed in earning assets	Interest and operating expenses (% raised)
Interest-bearing transaction deposits	120	50%	7.5%
Time deposits	200	80%	6.0%
Interbank (money market) borrowings	100	95%	5.0%
New stockholders' investment	80	90%	15.0%

For reference, last year's accounts show total funds raised of GBP 2,000 million, all interest paid of GBP 100 million, other operating costs of GBP 40 million and earning assets of GBP 1,600 million. Which of the following is closest to the minimum average before-tax rate of return the bank must earn on the new loans and investments to cover the expected cost of all the new funds?

A. 7.60%

B. 8.75%

C. 9.82%

D. 12.06%

Solutions • LTR-12 d

Solution LTR-12 d.1

Answer: C

Bank Q407 • GARP book Ch.13 Problem 9

Step 1 – Costs on the full amount raised

$$\text{Interest} = 0.0235 \times 775 = 18.2125, \qquad \text{Noninterest} = 0.0050 \times 775 = 3.875$$

$$\text{Total} = \text{USD } 22.0875 \text{ million}$$

Step 2 – Divide by net investable funds

$$\text{Effective cost} = \frac{22.0875}{700} = 3.155\% \approx 3.16\%$$

$\Rightarrow$ Every source is grossed up by $775/700 = 1.107$, because costs are paid on all USD 775 million but only USD 700 million can be lent.

Interpretation

The same arithmetic ranks the menu: Fed funds 2.77%, CDs and Eurodollars 2.93% each, commercial paper 3.16%, primary credit 3.32%. On cost alone Fed funds win; the discount window, which the firm has used in each of the past two weeks, is the dearest and the Fed may limit repeated use (LTR-12 c).

Why the other options are wrong

A is incorrect. Dropped the noninterest cost: $2.35\% \times 775/700 = 2.60\%$.
B is incorrect. Added the two rates but kept the USD 775 million raised as the base: $2.35\% + 0.50\% = 2.85\%$.
D is incorrect. Sibling swap: the discount window's effective cost, $(2.75\%+0.25\%)\times 775/700 = 3.32\%$.

LO: LTR-12 d: Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach. • **Reference:** Rose and Hudgins, Ch.13, 13.4 Relative Costs, eqs. (13.4) to (13.7); Problem 9

Solution LTR-12 d.2

Answer: D

Bank Q421 • Mock 1 Q4

Step 1 – Break-even cost rate on earning assets

$$\frac{52 + 20}{1{,}500} = 4.80\%$$

Step 2 – Before-tax equity return

$$\frac{9\%}{1 - 0.35} = 13.846\%$$

Step 3 – Equity add-on, weighted by equity over earning assets

$$13.846\% \times \frac{300}{1{,}500} = 2.769\%$$

Step 4 – Overall cost of capital

$$4.80\% + 2.77\% = 7.57\%$$

$\Rightarrow$ HVB must earn at least 7.57% on its earning assets to cover its borrowing costs and give shareholders their 9% after tax.

Interpretation

Total assets (USD 2.1 billion) are a red herring: both the break-even rate and the equity weight use earning assets, because only earning assets generate the return.

Why the other options are wrong

A is incorrect. Dropped the USD 20 million of noninterest expenses: $52/1{,}500 + 2.769\% = 6.24\%$.

B is incorrect. No tax gross-up: $4.80\% + 9\% \times 300/1{,}500 = 6.60\%$.

C is incorrect. Weighted equity by total assets: $4.80\% + 13.846\% \times 300/2{,}100 = 6.78\%$.

LO: LTR-12 d: Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach. • **Reference:** Rose and Hudgins, Ch.13, Figuring the Overall Cost of Funds: The Historical Average Cost Approach

Solution LTR-12 d.3

Answer: B

Bank Q415 • AnalystPrep Q.4109

Step 1 – Dollar cost of each borrowed source (total cost column)

$$275 \times 3.51\% = 9.6525, \qquad 580 \times 3.10\% = 17.98, \qquad 385 \times 3.28\% = 12.628$$

$$\text{Interest + other operating costs} = \text{USD } 40.26 \text{ million}$$

Step 2 – Break-even rate on earning assets

$$\frac{40.26}{1{,}240} = 3.25\% \approx 3.2\%$$

$\Rightarrow$ Equity is not a borrowed fund: its 12.75% enters only in the next step, the overall cost of capital.

Interpretation

The noninterest and reserve and collection costs are operating costs of raising the funds, so they sit in the numerator with interest. The checkable deposits cost little in interest (0.52%) but most in total once their 1.96% handling cost is included.

Why the other options are wrong

A is incorrect. Divided by total funds raised instead of earning assets: $40.26/1{,}530 = 2.6\%$.

C is incorrect. Added the equity cost ($290 \times 12.75\% = 36.98$) and divided by funds raised: $77.24/1{,}530 = 5.0\%$.

D is incorrect. Added the equity cost to the break-even numerator: $77.24/1{,}240 = 6.2\%$.

LO: LTR-12 d: Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach. • **Reference:** Rose and Hudgins, Ch.13, The Historical Average Cost Approach

Solution LTR-12 d.4

Answer: C

Bank Q413 • GARP book Ch.13 Problem 11

Step 1 – Interest cost of borrowed funds

$$200(0.75\%) + 400(2.50\%) + 100(2.75\%) = 1.50 + 10.00 + 2.75 = \text{USD } 14.25 \text{ million}$$

That is 2.04% of the USD 700 million borrowed, the weighted average interest cost.

Step 2 – Other operating costs

Noninterest: $4.00 + 2.00 + 0.25 = 6.25$, Reserve and collection: $1.50 + 2.00 + 0.25 = 3.75$

Step 3 – Break-even cost rate on earning assets

$$\frac{14.25 + 6.25 + 3.75}{700} = \frac{24.25}{700} = 3.464\%$$

Step 4 – Add the equity cost, already before tax

$$3.464\% + 12\% \times \frac{50}{700} = 3.464\% + 0.857\% = 4.32\%$$

$\Rightarrow$ The bank must earn at least 4.32% on its earning assets to cover all fund-raising costs and its shareholders' return.

Interpretation

GARP states the equity cost "at 12 percent before taxes", so no $1 - t$ gross-up applies and the 35% tax bracket is a red herring. (The question bank's rebuild restates the 12% as after-tax, which makes 4.78% its answer; GARP's wording is followed here.)

Why the other options are wrong

A is incorrect. Left out the reserve and collection costs: $(14.25 + 6.25)/700 + 0.857\% = 3.79\%$.
B is incorrect. Divided by total funds raised (USD 750 million) instead of earning assets: $24.25/750 + 12\% \times 50/750 = 4.03\%$.
D is incorrect. Grossed up a return that is already before tax: $3.464\% + \frac{12\%}{0.65} \times \frac{50}{700} = 4.78\%$.

LO: LTR-12 d: Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach. • **Reference:** Rose and Hudgins, Ch.13, The Historical Average Cost Approach; Problem 11

Solution LTR-12 d.5 **Answer: B**

Bank Q422 • Schweser Module Quiz 77.2 Q2

Step 1 – Pooled funds expense: costs over all new funds

$$\frac{22}{300} = 7.33\%$$

Step 2 – Hurdle rate: costs over the dollars placed in earning assets

$$\frac{22}{250} = 8.80\%$$

$\Rightarrow$ The hurdle is higher because USD 50 million of the money raised never earns.

Interpretation

Same numerator, two denominators: the expense is what the new money costs per dollar raised; the hurdle is what each dollar actually lent must earn.

Why the other options are wrong

A is incorrect. Used USD 300 million for both; the hurdle must use the USD 250 million invested.
C is incorrect. Swapped the two denominators.
D is incorrect. Used USD 250 million for both; the pooled expense uses all USD 300 million raised.

LO: LTR-12 d: Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach. • **Reference:** Rose and Hudgins, Ch.13, The Pooled-Funds Approach; SchweserNotes 2026 Book 4, Reading 77, LO 77.d

Solution LTR-12 d.6

Answer: C

Bank Q414 • GARP book Ch.13 Problem 12

Step 1 – All expected costs of the new funds

$$\text{Time deposits: } 325 \times (1.75\% + 0.45\%) = 5.6875 + 1.4625 = 7.15$$

$$\text{Transaction deposits: } 125 \times 2.00\% = 2.50 \qquad \text{Total} = \text{USD } 9.65 \text{ million}$$

$\Rightarrow$ The transaction deposits pay no interest but still cost 2% to service.

Step 2 – Pooled-funds marginal cost (all new funds)

$$\frac{9.65}{450} = 2.14\%$$

Step 3 – Hurdle rate (dollars placed in earning assets)

$$\frac{9.65}{400} = 2.41\%$$

$\Rightarrow$ Every new loan must earn at least 2.41% before taxes, on average, to cover the cost of the funds raised for it.

Interpretation

The USD 400 million already is the net amount available; the USD 50 million difference goes to reserves and other nonearning uses and must not be deducted again.

Why the other options are wrong

A is incorrect. Interest only, dropping all noninterest costs: $5.6875/400 = 1.42\%$.
B is incorrect. Intermediate result: the pooled-funds marginal cost over all USD 450 million raised, $9.65/450 = 2.14\%$.
D is incorrect. Deducted the nonearning USD 50 million twice: $9.65/350 = 2.76\%$.

LO: LTR-12 d: Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach. • **Reference:** Rose and Hudgins, Ch.13, The Pooled-Funds Approach; Problem 12

Solution LTR-12 d.7

Answer: C

Workbook A27.1 • S1 • Map #47

Step 1 – Choose the approach

The question asks for the minimum return on future loans and investments to cover the cost of new funds: the pooled-funds approach. Last year's figures belong to the backward-looking historical average cost approach and are not needed.

Step 2 – All expected operating expenses on the new funds

$$120 \times 7.5\% + 200 \times 6.0\% + 100 \times 5.0\% + 80 \times 15.0\% = 9.0 + 12.0 + 5.0 + 12.0 = \text{GBP } 38.0 \text{ million}$$

$\Rightarrow$ The equity's required return counts as a cost of the new funds, exactly as the new stockholders' investment does in Rose's example.

Step 3 – Dollars available to place in earning assets

$$120 \times 0.50 + 200 \times 0.80 + 100 \times 0.95 + 80 \times 0.90 = 60 + 160 + 95 + 72 = \text{GBP } 387 \text{ million}$$

$\Rightarrow$ Only 387 of the 500 million raised can be invested; the rest goes to reserves and other nonearning uses.

Step 4 – Hurdle rate

$$\text{Hurdle rate} = \frac{\text{All expected operating costs}}{\text{Dollars placed in earning assets}} = \frac{38.0}{387} = 9.82\%$$

$\Rightarrow$ The bank must earn at least 9.82%, before taxes, on average, on the new earning assets to cover all expected fund-raising costs.

Interpretation

Loans priced below 9.82% before tax would not cover the cost of the funds that finance them. The pooled expense on all new funds, $38.0/500 = 7.60\%$, understates the hurdle because about 23% of the money raised never reaches earning assets. Last year's break-even rate (8.75%) describes the cost of funds already raised, which is why it should not set the price of new business.

Why the other options are wrong

A is incorrect. Intermediate result: pooled deposit and nondeposit funds expense on all new funds, $38.0/500 = 7.60\%$, without adjusting for the portion not placed in earning assets.

B is incorrect. Sibling swap: used the backward-looking historical average cost approach, break-even $= (100 + 40)/1{,}600 = 8.75\%$, which prices funds already raised, not new funds.

D is incorrect. Wrong denominator: kept the equity's cost in the numerator but dropped its GBP 72 million from the funds placed in earning assets, $38.0/(387 - 72) = 38.0/315 = 12.06\%$.

LO: LTR-12 d: Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach. • **Reference:** Rose and Hudgins, Ch.13, Figuring the Overall Cost of Funds: The Historical Average Cost Approach; The Pooled-Funds Approach

LTR-14 • Liquidity Transfer Pricing: A Guide to Better Practice • Vardeez: IMPORTANT

LTR-14 a

P1 CONCEPT

Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process.

Workbook: A27 Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A27, recalled in August 2026; Map #47 (2 of 16 sources) names "LTP's role in charging liquidity cost back to business lines" as the governance angle of the section. Workbook A27.3 tags this LO with LTR-14 c and d and is worked here.
Vardeez: "Process of LTP". His notes define LTP and lock one polarity pair first: **charge the asset side, credit the liability side**.

Reported exam shapes

- **S4** LTP's role in charging liquidity costs back to business lines, including the liquidity cushion (Workbook A27.3).

What the questions look like: statement questions. What LTP does (and does not do), why banks need it, and which recommendation in a consultant's or auditor's report is better practice: a central treasury, an LTP policy, granular trading book funding, independent oversight, LMIS, remuneration, and who pays for the liquidity cushion. Only four bank questions sit under this LO; two better-fitting ones are re-filed from LTR-14 b (implementation challenges, a P4 LO not in this book), and one near-duplicate (Schweser Quiz 79.1 Q1) is left out.

1. What LTP does, and why banks need it

Liquidity transfer pricing is a process that attributes the costs, benefits and risks of liquidity to the respective business units within a bank (it is part of funds transfer pricing, FTP). Its purpose is to transfer liquidity costs and benefits from the business units to a **centrally managed pool**. To do that, LTP

1. **charges users of funds** (assets, loans) for the cost of liquidity;
2. **credits providers of funds** (liabilities, deposits) for the benefit of liquidity;
3. **recoups the cost of carrying a liquidity cushion** by charging contingent commitments, such as lines of credit, on their **predicted (expected) use** of liquidity.

Why it matters. Banks make money by **maturity transformation**, funding long-term loans with short-term deposits, which makes them inherently vulnerable to liquidity risk (BCBS). LTP is not meant to stop maturity transformation but to price it. Banks with poor LTP under-price or (even worse) fail to price liquidity; they accrue illiquid assets and contingent exposures and under-value stable funding. Before the GFC funding was plentiful and cheap (the one-year LIBOR/swap spread was 0.5 bp in June 2005), and many banks treated liquidity as a **free good**. The Senior Supervisors Group (2009) found that the firms with the worst funding problems relied most on short-term financing of longer-term illiquid assets, and linked those business models to LTP that failed to penalise the liquidity risk booked. BCBS Principle 4 now asks banks to incorporate liquidity costs, benefits and risks in internal pricing, performance measurement and new product approval for all significant activities, on and off the balance sheet.

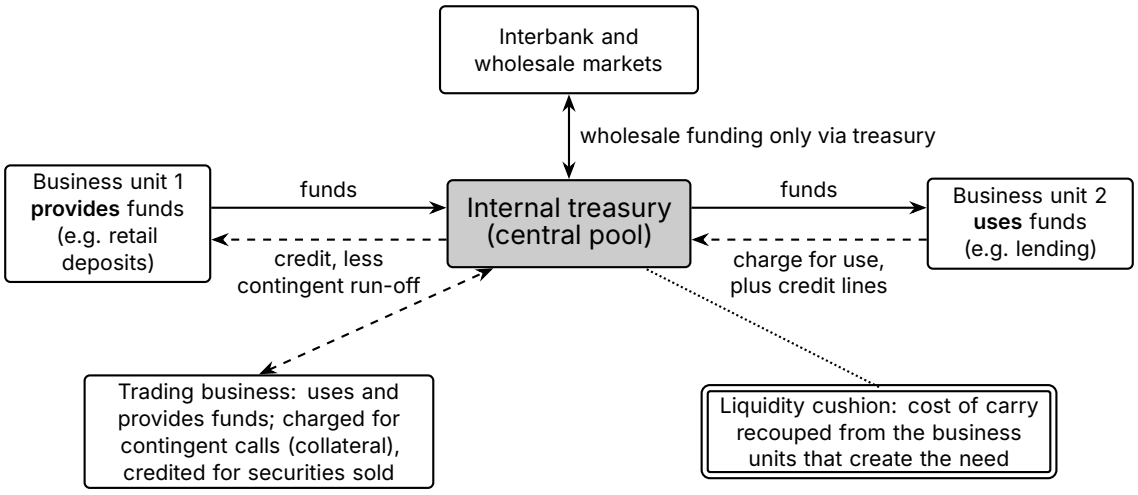


Figure 1. The LTP process, after Grant's Figure 15.1. Solid arrows carry funds; dashed arrows carry the internal charges and credits; the dotted line links the liquidity cushion (double border) to the charges that pay for it. All wholesale funding runs through the central treasury.

2. Better practice in governance and implementation

A 2009 survey of **38 banks in nine countries** found many LTP practices short of good practice (the detailed challenges are LTR-14 b). Grant's better practices for **governing LTP**:

- **An LTP policy** that defines LTP, states its purpose and gives principles or rules, applying to all business units that are material users or providers of funding liquidity. Most surveyed banks had none, so liquidity generators were underpaid and users got free or unduly cheap liquidity.
- **Central management.** LTP is run centrally, within group treasury or a subsidiary treasury, applied consistently, and **wholesale funding is confined to that function**. Decentralised funding centres produced inconsistent regimes, manual off-line updates and internal arbitrage (units raising wholesale funds and selling them to treasury at a higher rate for a "risk-free" profit).
- **Trading book funding policies** with funding provided at the **most granular level** (the trading desk, not the trading book), which needs treasury to have full line of sight to each business balance sheet; higher charges for positions likely to become stale; limits to curb over-trading. Before the GFC, policies assumed holding periods of 180 days or less, so long-term charges applied only when assets rolled into the banking book, and highly rated but illiquid CDO tranches were funded overnight to 90 days.
- **Independent oversight** by risk and financial control, with senior management involved and regular meetings (monthly rather than quarterly or semi-annually) of ALCO, senior management and treasury on funding costs.
- **LMIS** advanced enough to attribute all relevant costs, benefits and risks at a sufficiently granular level. Simplistic, inflexible systems mispriced products and distorted profit and performance.
- **Remuneration** sensitive to the risks taken: profit pools adjusted for the cost of liquidity through LTP, instead of a simple percentage of accrued revenues; staff in financial and risk control paid independently of the businesses they oversee (FSB Principle 3).
- **Attribution.** Charges for using liquidity and credits for providing it should **not be borne by central management** (**treasury**) but attributed, granularly, to the business activities that use or provide liquidity.

- **Method.** Liquidity is not a free good; the zero-cost approach should not be tolerated and banks should move to matched-maturity marginal cost (LTR-14 c).

3. The liquidity cushion: who pays

A cushion of highly liquid assets (or stand-by liquidity) lets a bank survive unexpected funding outflows (BCBS Principles 1 and 12; the LCR). Carrying it is costly, because the funding cost of the assets generally exceeds their return: the **cost of carry**.

- **What went wrong.** Few banks sized the cushion from stress tests, and those that did used idiosyncratic scenarios only; some "liquid" assets proved illiquid, correlated or encumbered; nearly all funded the cushion **short term (overnight)**, so the carry looked tiny; and most simply **averaged the cost across all assets**. If carry costs $30 million and total assets are $300 billion, every asset pays 1 bp, on top of its term liquidity premium:

$$\text{FTP} = \text{base rate} + \text{term liquidity premium} + \text{liquidity premium} \qquad \text{(eq. 15.2)}$$

  That averaging assumes all assets carry the same contingent risk, ignores the contingent risk in liabilities (deposit run-off) and off-balance-sheet items (credit-line drawdowns), ignores predicted usage, and is not granular enough to discourage the riskiest products.
- **Better practice.** Size the cushion from stress tests covering **idiosyncratic and systemic** scenarios, prolonged disruptions and a combination of the two; hold assets of the highest liquidity; charge cushion assets at **longer-term rates, not overnight rates** (if markets could stay stressed for two years, use the two-year term liquidity premium); treat the carry as **a cost of doing business**, not an opportunity cost, recouped from the business activities that create the need, according to their **predicted use** of contingent liquidity (Figure 2).
- **First step.** Before asking how much to charge, banks must accept that **all contingent commitments need to be charged**. At the most basic level of better practice, a line of credit is charged on its likelihood of drawdown: $\frac{\text{limit}-\text{drawn}}{\text{limit}} \times$ likelihood of drawdown $\times$ cost of funding the cushion. Grant's example, a $10 million line with $4 million drawn, a 60% likelihood and an 18 bp three-year premium, gives 6.48 bp, or $6,480 on the limit (the calculation is LTR-14 d, a P4 LO).

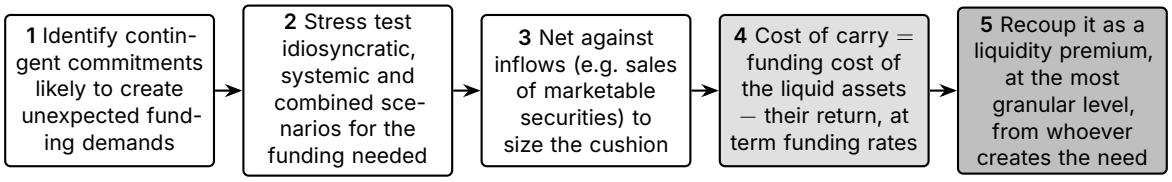


Figure 2. Toward better management of contingent liquidity risk, after Grant's Figure 15.10. Steps 4 (light grey) and 5 (darker grey) are the LTP part: the carry is charged to the business unit, product or transaction that makes the bank hold the cushion, not averaged over all assets and not absorbed by treasury.

Correction to the source notes. Grant's text adds that at an overnight funding rate of 0.5 bp "the dollar charge for the contingent liquidity risk would have been $1,800". The same formula gives $0.6 \times 0.6 \times 0.00005 \times \$10$ million $= \$180$ (0.18 bp); $1,800 would need a 5 bp rate. The point stands: overnight-funded cushions made contingent liquidity look almost free. Workbook A27 flags the same slip.

	Poor practice found in the survey	Better practice
Policy	No LTP policy; LTP undefined	Policy defining LTP, its purpose, rules for all material units
Structure	Decentralised funding centres; arbitrage	Central treasury; wholesale funding confined to it
Trading book	Funded on net requirement; 180-day assumption	Desk-level funding; higher charges for stale positions; limits
Oversight	Poor to nonexistent	Independent risk and financial control; monthly meetings
LMIS	Simplistic, not granular	Attributes all costs, benefits and risks granularly
Remuneration	Profit pools from accrued revenues	Profit pools adjusted for the cost of liquidity
Method	Zero cost or pooled average cost	Matched-maturity marginal cost (LTR-14 c)
Cushion	Idiosyncratic scenarios; overnight funding; cost averaged over all assets	Stress tests incl. systemic; term funding; charged by predicted use

Exam traps

- **Polarity of the transfer.** Users of funds (assets, loans) are *charged*; providers (liabilities, deposits) are *credited*.
- **Role misassignment.** The cost of carry is a cost of doing business recouped from the businesses that create the need; it is not absorbed by treasury, and treasury does not raise funds for business units to arbitrage.
- **Centralisation flip.** Decentralised funding caused the arbitrage; the fix is a *central* treasury with wholesale funding confined to it.
- **Granularity.** Charges at the desk, product or transaction level, not the bank or the trading book as a whole.
- **Timing of contingent charges.** Commitments are charged on predicted use from the start, not after the first drawdown.
- **Scope errors on the cushion.** Charged at longer-term rates, not overnight; sized on idiosyncratic *and* systemic scenarios.
- **Purpose.** LTP prices maturity transformation; it does not ban it, and it is not an allocation of general overheads.

Practice questions: LTR-14 a

Question LTR-14 a.1

Bank Q472 • AnalystPrep Q.4215 (rebuilt)

Which of the following is NOT a function of a bank's liquidity transfer pricing (LTP) process?

A. Charging users of funds (assets and loans) for the cost of liquidity

B. Crediting providers of funds (liabilities and deposits) for the benefit of liquidity

C. Recouping the cost of carrying a liquidity cushion by charging contingent commitments, such as lines of credit, based on their expected use of liquidity

D. Allocating the bank's general operating expenses, such as salaries and rent, to business units in proportion to their use of funding

Question LTR-14 a.2

Bank Q473 • BT Mock Q74

What is the primary (valid) reason that banks need proper liquidity transfer pricing (LTP)?

A. Because banks engage in maturity transformation

B. Because there is a lack of liquidity transformation products in the marketplace

C. Because the true economic cost of liquidity risk is approximately zero

D. Because markets are volatile, for example the volatility of the 1-year LIBOR/swap spread

Question LTR-14 a.3

Bank Q474 • GARP PE1 Q5 (rebuilt; key as corrected in the bank)

A risk consultant is asked to review a bank's liquidity transfer pricing (LTP) governance and make recommendations based on industry best practice. Which of the following would be a correct recommendation?

A. Wholesale funding should be restricted to a centralised group or subsidiary treasury.

B. LTP of contingent obligations, such as collateral calls and committed lines of credit, should be applied only after the first drawdown.

C. A pooled average cost of funds approach is the most accurate LTP method.

D. A liquidity transfer price should be set for the bank as a whole rather than attributed to individual business units.

Question LTR-14 a.4

Bank Q477 • Schweser Module Quiz 79.1 Q2 (re-filed from LTR-14 b)

Which of the following best describes one of the major challenges for banks in implementing an effective LTP process?

A. A decentralised LTP process is recommended to mitigate arbitrage opportunities between business units.

B. Illiquid long-term assets should be penalised for increasing liquidity risk.

C. Performance evaluations of business unit managers should be kept separate from the LTP process.

D. A liquidity management information system (LMIS) should produce and monitor high-quality reports on a quarterly basis.

Question LTR-14 a.5 Bank Q478 • bank question in GARP practice-exam style (re-filed from LTR-14 b)

A risk manager at a mid-sized bank is reviewing why the bank's liquidity profile deteriorated ahead of a funding stress. She finds that the bank's lending units concentrated on long-term loans, were charged only a nominal internal rate for the cost of funding liquidity, and reported high profits that drove large bonuses. Meanwhile, the retail deposit unit received little internal credit for gathering long-term deposits. Which implementation weakness of the bank's LTP process does this review most likely illustrate?

A. Funding costs were reviewed too frequently, causing business units to over-adjust their lending to short-term rate changes.

B. The remuneration policy for business unit executives did not account for the costs, benefits and risks of liquidity, so performance was mismeasured and incentives were distorted.

C. Charges for contingent liquidity risk, such as collateral calls, were applied only to the trading book rather than to the lending units.

D. The central treasury was too involved in pricing, which removed the incentive for business units to manage their own funding.

Question LTR-14 a.6 Workbook A27.3 • S3, S4 • Map #47

An internal auditor at a Hong Kong bank is reviewing its liquidity transfer pricing. Treasury charges every loan and credits every deposit a single pooled average cost of funds of 8 bps, whatever the maturity, and absorbs the cost of carrying the liquidity cushion centrally. The bank's current term liquidity premiums are 5, 10, 18, 28 and 40 bps for one to five years. Which of the following recommendations in the audit report is consistent with better LTP practice?

A. Adopt matched-maturity marginal cost, so a five-year bullet loan is charged the 40 bp five-year term liquidity premium and a five-year term deposit is credited the same premium.

B. Retain the pooled average cost, because the flat 8 bp rate overcharges long-term loans and so discourages business units from excessive maturity transformation.

C. Keep the cost of carrying the liquidity cushion in central treasury as an opportunity cost, rather than charging it to the business units that use contingent liquidity.

D. Price the funding of assets held in the liquidity cushion at overnight rates, because those assets can be sold immediately in any market disruption.

Solutions • LTR-14 a

Solution LTR-14 a.1

Answer: D

Bank Q472 • AnalystPrep Q.4215

Step 1 – The three things LTP does

"LTP charges users of funds (assets/loans) for the cost of liquidity, and credits providers of funds (liabilities/deposits) for the benefit of liquidity. LTP also recoups the cost of carrying a liquidity cushion by charging contingent commitments, such as lines of credit, based on their predicted (expected) use of liquidity."
⇒ A, B and C are functions; D is not.

Interpretation

LTP is confined to the costs, benefits and risks of *liquidity*. Allocating salaries and rent is general cost accounting; mixing it into the liquidity charge would blur the price signal LTP is meant to send.

Why the other options are wrong

A is incorrect. A core function: users of funds pay for the liquidity they consume.
B is incorrect. A core function: providers of funds are credited for the liquidity they supply.
C is incorrect. A core function: the cushion's carry is recouped from contingent commitments on predicted use.

LO: LTR-14 a: Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process. • **Reference:** Grant, FSI Occasional Paper No.10, 15.1 Introduction, Figure 15.1

Solution LTR-14 a.2

Answer: A

Bank Q473 • BT Mock Q74

Step 1 – The rationale

"Banks make money by funding long-term loans (assets) with short-term deposits (liabilities), a process that is commonly referred to as maturity transformation." This "makes banks inherently vulnerable to liquidity risk". Provided banks use LTP to account for the costs, benefits and risks of liquidity, "they should not be discouraged from engaging in maturity transformation".
⇒ LTP exists to price the liquidity risk that maturity transformation creates.

Interpretation

Without LTP, liquidity is effectively free inside the bank, so units pile up long illiquid assets funded short, the pattern the SSG tied to the worst funding problems of the GFC.

Why the other options are wrong

B is incorrect. Not the rationale: the issue is pricing the transformation banks already perform, not a shortage of products.
C is incorrect. The pre-GFC error itself: treating liquidity as a free good. Its true cost is significant.
D is incorrect. A symptom, not the reason: the near-zero LIBOR/swap spread of 2005 (0.5 bp) explains why banks *ignored* liquidity cost; the need for LTP comes from maturity transformation.

LO: LTR-14 a: Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process. • **Reference:** Grant, 15.3.1 Why Banks Need LTP; 15.3.2 An Example of What Can Go Wrong

Solution LTR-14 a.3 Answer: A

Bank Q474 • GARP PE1 Q5

Step 1 – Why A is correct

Better practice: "LTP should be managed centrally, within group treasury or a subsidiary treasury, and applied consistently across the group. Wholesale funding should be confined to this function." Banks with decentralised funding centres had inconsistent LTP regimes, manual offline updates and internal arbitrage between business units and treasury.

Interpretation

Centralising wholesale funding removes the "risk-free" profit a unit could make by raising money outside and selling it to treasury at a higher rate, and gives treasury the line of sight it needs to charge each business correctly.

Why the other options are wrong

B is incorrect. Timing error: contingent commitments are charged on their predicted use from the outset; waiting for a drawdown leaves the contingent risk unpriced.

C is incorrect. Polarity flip: the matched-maturity marginal cost approach is current best practice; pooled average cost undercharges long-term assets.

D is incorrect. Granularity reversed: costs, benefits and risks are attributed to the respective business units, at a sufficiently granular level, not set for the bank as a whole.

LO: LTR-14 a: Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process. • **Reference:** Grant, 15.2.1.2 Internal Funding Structure; 15.5 Conclusion, Governing LTP

Solution LTR-14 a.4 Answer: B

Bank Q477 • Schweser Module Quiz 79.1 Q2

Step 1 – Why B is correct

The crisis problem was short-term funding of long-term illiquid assets. A central challenge is to charge such assets properly for the liquidity they consume (and credit long-term deposits for the liquidity they provide), so that business units stop accumulating them on mispriced terms.

Interpretation

This question was filed under the implementation challenges (LTR-14 b) but tests the better-practice list: centralise, link pay to LTP, meet more often, charge long illiquid assets more.

Why the other options are wrong

A is incorrect. Reversed: a *centralised* treasury restricts arbitrage between business units and treasury, and between units.

C is incorrect. Reversed: profit and performance measures, and so remuneration, should include the cost of liquidity through LTP.

D is incorrect. Wrong frequency: funding-cost meetings are moving to monthly; quarterly or semi-annual reviews were the pre-crisis weakness.

LO: LTR-14 a: Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process. • **Reference:** Grant, 15.2.1 Management of the LTP Process; 15.2.3 Remuneration Practices

Solution LTR-14 a.5 Answer: B

Bank Q478 • re-filed from LTR-14 b

Step 1 – Read the symptoms

Long-term loans undercharged for liquidity, high reported profits, large bonuses, and little credit for long-term deposits.

Step 2 – Name the weakness

Profit measures that ignore the cost of liquidity overstate the lending units' performance, and bonuses built on them reward illiquid long-term lending. Grant: profit pools were "derived from a simple percentage of accrued revenues without any regard for the liquidity risk taken"; better practice adjusts them "for the cost of liquidity through the LTP process".
⇒ B.

Interpretation

Remuneration is one of Grant's governance items because it turns a pricing error into behaviour: the undercharged unit is paid to do more of the thing that is mispriced.

Why the other options are wrong

A is incorrect. Reversed: the fix is *more* frequent funding-cost meetings (monthly); nothing in the review points to over-adjustment.
C is incorrect. A different weakness (contingent charges for trading positions); the review is about on-balance-sheet loans and deposits.
D is incorrect. Reversed: a central treasury is the recommended structure; the review shows undercharging, not excess central control.

LO: LTR-14 a: Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process. • **Reference:** Grant, 15.2.3 Remuneration Practices; 15.5 Conclusion

Solution LTR-14 a.6 Answer: A

Workbook A27.3 • S3, S4 • Map #47

Step 1 – Why A is correct

Grant calls the matched-maturity marginal cost of funds approach current best practice. Charges for the use of funds and credits for their benefit are based on the term liquidity premium matching the transaction's maturity: a five-year bullet needs funding for its whole life and is charged 40 bp; a five-year term deposit provides five years of funding and is credited 40 bp.
⇒ Under the 8 bp average, the loan is undercharged 32 bp (3,200 per 1 million of principal) and the deposit under-credited by the same amount.

Interpretation

LTP's job is to send the cost of liquidity back to the business lines that use it, and the benefit to those that supply it, at market rates. The average approach lags the actual market cost of funds and pushes the bank toward long illiquid assets funded by short liabilities, the pattern the SSG linked to the worst funding problems in the GFC.

Why the other options are wrong

B is incorrect. Polarity flip: a single average rate *undercharges* long-term assets (8 bp against 40 bp) and so promotes maturity transformation; it ignores the heightened liquidity risk of longer-term assets.
C is incorrect. Role misassignment: the cost of carry is "a cost of doing business" that "should not be borne by central management (ie treasury)"; it should be recouped from the business activities that create the need, according to their predicted use of contingent liquidity.
D is incorrect. Scope error: charges on cushion assets "should not be based on short-term

overnight rates but should instead be derived from longer term rates to account for the possibility of longer than expected market disruptions".

LO: LTR-14 c: Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost). • LTR-14 a: Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process. • LTR-14 d: Describe the contingent liquidity risk pricing process and calculate the cost of contingent liquidity risk. • **Reference:** Grant, FSI Occasional Paper No.10, 15.3.4 to 15.3.6 (Tables 15.3, 15.4), 15.4.5 to 15.4.6, 15.5 Conclusion

LTR-14 • Liquidity Transfer Pricing: A Guide to Better Practice • Vardeez: IMPORTANT

LTR-14 c

P1 BOTH

Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost).

Workbook: A27 Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A27, recalled in August 2026; Map #47 reports "pooled cost approach, etc." and the examinable contrast includes the pooled average cost against the marginal cost of funds. Shape S3 is that contrast inside LTP. Workbook A27.3 tags this LO and is worked under LTR-14 a (Question LTR-14 a.6): read it with this block.
Vardeez: "Marginal cost of funds approach: calculation"; his reading note: "know the calculation of marginal cost of funds approach". The tag is BOTH (approved by the user): the LO says compare, Vardeez says calculations come.

Reported exam shapes

- **S3** Pooled average cost of funds against matched-maturity marginal cost in LTP.

What the questions look like: statement questions on the three (four) approaches: what each does to maturity transformation, why banks used the average, how the term liquidity premium is extracted. Numerical questions read a term liquidity premium table: the charge on a bullet loan, the tenor-weighted charge on an amortising loan, sometimes in dollars, with the average cost of funds planted as the sibling input.

1. One base rate, different liquidity spreads

LTP charges users of funds and credits providers (LTR-14 a, Section 1). Every approach in the survey prices **interest rate risk** off the **swap curve**; they differ in the **spread for liquidity** added on top. Grant describes three approaches, splitting the average one in two:

1. **zero cost** (no liquidity spread),
2. **pooled average cost** (one flat spread for everything), and its variant **separate average costs** (one flat spread for assets, another for liabilities),
3. **matched-maturity marginal cost** (a spread that rises with maturity and moves with the market).

The sequence is one axis of increasing granularity (Vardeez). Figure 1 puts the three on one chart.

2. Zero cost: liquidity as a free good

Some banks charged nothing to assets for using funding liquidity and credited nothing to liabilities for providing it: the rate charged to users came from the **swap curve only**. A zero spread over the swap curve is a zero charge for liquidity. Grant: "undoubtedly the worst practice identified in the survey".

- It **exacerbated maturity transformation to the largest degree possible**: banks hoarded long-term, highly illiquid assets and had very few long-term stable liabilities to meet funding demands as they came due.
- Why banks chose it: ideal funding conditions. In June 2005 the one-year LIBOR/swap spread was 0.5 bp, so spreads looked like pure credit adjustments; in the crisis it rose above 100 bp.

3. Pooled average cost of funds

An average rate is calculated from the interest expense of **all existing funding sources** (with deposits as the only source: total interest expense over average total deposits, adjusted for floats and reserve requirements) and applied to every asset and liability. With a 10 bp average, every $1 million loan is charged $1,000 and every $1 million deposit credited $1,000, whatever the maturity (Table 15.1). Much better than zero cost, but with **two major weaknesses**:

1. **It ignores the heightened liquidity risk of longer-term assets**: one rate assumes all maturities pose the same risk. Relative to the marginal curve it **undercharges long-term assets** and overcharges short-term ones, and **under-credits long-term liabilities**. Used for both sides, it also ties the two together: cutting the charge to fund users (10 bp to 5 bp) encourages loans but cuts the credit to deposits.
2. **It lags the market.** It reflects historical rates: if five-year funding rises 20 bp, the average moves much less, and only after the rise has lasted. It does not reflect market perceptions of risk for new business.

Implications. It **promotes maturity transformation**, unhealthy as well as healthy (units write long assets that are not charged more and are not paid to raise long liabilities), cross-subsidising long-dated liquidity risk; remuneration on net interest income and information asymmetries magnify the incentive. It **distorts profit assessment**, especially in volatile markets.

Why banks used it (a favourite "which is NOT a reason" question): (1) averaging across all assets is much simpler than charging each asset by its contractual or behavioural maturity; (2) simplicity makes it easier for business units to understand and comply; (3) basic LMIS suffice; (4) it is less sensitive to intermediate changes in the market cost of funds, so it **reduces net interest income volatility** and gives central management more control.

Separate average costs. One average for the cost of funds (say 10 bp: $1,000 per $1 million loan) and a different average for the benefit (say 4 bp: $400 per $1 million deposit), Table 15.2. Cutting the loan charge to 5 bp no longer directly discourages deposit gathering. It is better, but both lines are still flat, so it still ignores the extra liquidity value of long-term liabilities and the extra risk of long-term assets.

4. Matched-maturity marginal cost of funds: current best practice

From the bank's **actual market cost of funding**, this approach isolates the part that is attributable to liquidity:

- Unsecured wholesale term debt has **fixed-rate** costs from which the liquidity part is hard to strip. An **internal swap** converts them to **floating-rate** costs over a reference rate; the spread over the reference rate that values the internal swap at par is the **term liquidity premium** (TLP). It reflects idiosyncratic credit risk and market access premiums.
- The reference curve is the **swap curve** (LIBOR or Euribor up to one year, interest rate swaps beyond; Schweser notes SOFR replacing LIBOR). Swap curves reflect interbank lending risks and market conditions better than government curves.
- Assets are **charged** and liabilities **credited** the TLP for the transaction's maturity, or a blended TLP for amortising or indeterminate products.
- It must be **actively updated**: some advanced banks failed to update their premiums and mispriced assets as volatility rose.

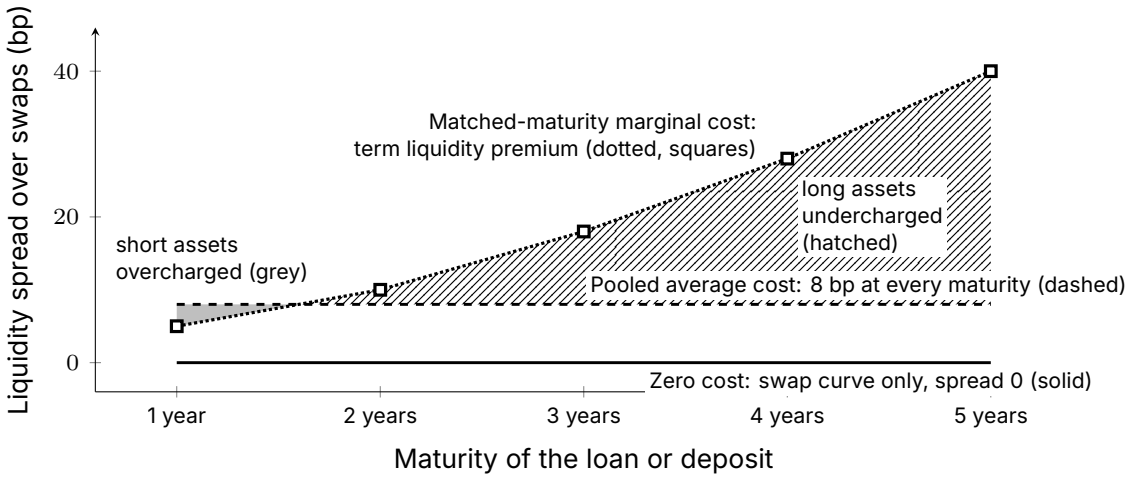


Figure 1. The three approaches on Grant's current premiums (Table 15.3, Panel B: 5, 10, 18, 28, 40 bp against an 8 bp average). The flat average sits above the marginal curve below about 1.6 years (grey: short assets overcharged, short deposits over-credited) and below it beyond (hatched: long assets undercharged, long deposits under-credited). Zero cost charges nothing at any maturity.

5. Pricing with the term liquidity premium

Term (years)	1	2	3	4	5
Panel A, pre-GFC: term liquidity premium (bp)	1	2	3	6	10
average cost of funds (bp)	2	2	2	2	2
difference, marginal − average	−1	0	1	4	8
Panel B, current: term liquidity premium (bp)	5	10	18	28	40
average cost of funds (bp)	8	8	8	8	8
difference, marginal − average	−3	2	10	20	32

Grant's Tables 15.3 and 15.4. A positive difference means the average approach undercharges the asset (and under-credits the liability).

Non-amortising bullet loans

All principal is repaid at maturity, so funding is needed for the whole term: **charge the TLP for the loan's maturity**. On $1 million: a one-year bullet pays 1 bp ($100) pre-GFC or 5 bp ($500) now; a five-year bullet 10 bp ($1,000) or 40 bp ($4,000). The average approach charges both 2 bp or 8 bp. A five-year loan is undercharged 8 bp ($800) pre-GFC and **32 bp ($3,200 per $1 million)** now; the one-year loan is overcharged. (Grant says the average charges more only at one year and less "for all other maturities"; in Panel A the two-year difference is exactly zero.) The gap widened in the crisis because the average cannot follow the market.

Amortising loans: the blended (tranching) premium

Repayments mean the loan becomes self-funding before maturity, so it is **not** charged the final-maturity premium. Treat a five-year linearly amortising $1 million loan as five annual loans of $200,000 maturing in years 1 to 5, and weight each premium by its term:

$$\text{Blended TLP} = \frac{\sum_i t_i \, \text{TLP}_i}{\sum_i t_i}$$

$$\text{Current: } \frac{1(5)+2(10)+3(18)+4(28)+5(40)}{1+2+3+4+5} = \frac{391}{15} = 26.1 \text{ bp}$$

$$\text{Pre-GFC: } \frac{1(1)+2(2)+3(3)+4(6)+5(10)}{15} = \frac{88}{15} = 5.9 \text{ bp}$$

Against the 8 bp (2 bp) average the loan is undercharged 18.1 bp (3.9 bp). In both cases the charge sits between the three- and four-year premiums: funding is needed for between three and four years, not five (Figure 2). An alternative is the IRR approach (Matz and Neu).

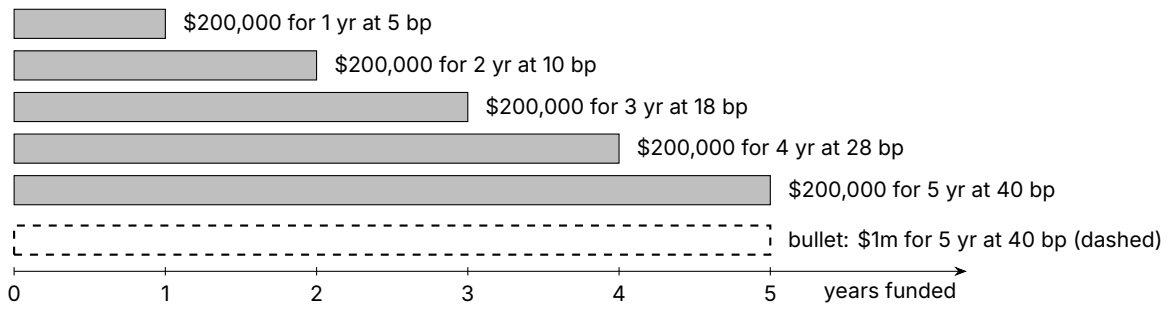


Figure 2. Tranching a five-year linearly amortising $1 million loan into five annual loans (grey bars), each charged the premium for its own term. Weighting by term gives 26.1 bp, well below the 40 bp a five-year bullet (dashed outline) pays.

Mortgages: behavioural maturity

A variable-rate mortgage may have a 25- or 30-year contractual term, but repayment timing is uncertain, so neither the 25-year premium (which overcharges and discourages lending) nor a simple tenor weighting works. Bundle mortgages into monthly vintages, model their run-off, and charge the whole portfolio the TLP of its **behavioural maturity**, usually the **weighted-average life**

$$\text{WAL} = \frac{\sum_i t_i P_i}{P} \qquad \text{(eq. 15.1; the average time to recoup \$1 of principal, not the median)}$$

A bank writing about $2 billion of mortgages a month with a WAL of about four years charges the four-year premium: 6 bp ($1.2 million) pre-GFC or 28 bp ($5.6 million) now, against 2 bp ($400,000) or 8 bp ($1.6 million) under the average.

Deposits: credits by stickiness

Deposits are credited for the liquidity they provide. **Sticky** money (term deposits) earns larger credits than **hot or volatile** money (demand, savings and transaction accounts). Term deposits are credited the TLP for their maturity: one year 1 bp or 5 bp, five years 10 bp or 40 bp. The average credits all of them 2 bp or 8 bp, over-compensating the one-year deposit by 1 bp (3 bp now) and under-compensating the five-year deposit by 8 bp (**32 bp** now), so units chase short deposits. For demand deposits, credit the **core** (stable) part on its modelled behavioural maturity and the volatile part less; a pure marginal approach would credit them only the overnight premium, close to zero, and discourage gathering them.

Formula card: LTR-14 c

Bullet loan or term deposit	Charge (credit) = TLP for its maturity × principal
Linear amortiser	Blended TLP $= \sum t_i \, \text{TLP}_i \big/ \sum t_i$ (5 years: weights 1 to 5, sum 15)
Mortgage portfolio	TLP at the behavioural maturity (WAL $= \sum t_i P_i / P$) × monthly volume
Under/overcharge vs average	TLP (or blended TLP) − average cost of funds
Funds transfer price	Base rate (swap curve) + TLP + liquidity premium for the cushion (LTR-14 a)

	Zero cost	Pooled average	Separate averages	Matched-maturity marginal
Liquidity spread	None	One flat rate, both sides	Flat cost rate; flat benefit rate	TLP by maturity
Long assets	Free	Undercharged	Undercharged	Charged the full premium
Long liabilities	No credit	Under-credited	Under-credited	Credited the full premium
Market changes	Ignored	Lag (historical)	Lag	Reflected at once (if updated)
Maturity transformation	Maximised	Promoted	Promoted	Priced
Verdict	Worst practice	Simple, smooth NII, basic LMIS	Better than pooled	Current best practice

Exam traps

- **Polarity flip.** The average *undercharges* long-term assets and *overcharges* short-term ones; it does not discourage maturity transformation.
- **Most transformation.** Zero cost, not average cost, produces the greatest maturity transformation.
- **Sibling input.** Using the average cost of funds row when the matched-maturity charge is asked, or the reverse.
- **Wrong maturity.** A bullet loan uses its own maturity's premium; an amortising loan the tenor-weighted blend, not the final-maturity premium, not the simple average.
- **Forgot to divide.** The weighted sum (391) is not the charge; divide by $\sum t_i$ (15).
- **Units.** 1 bp on $1 million is $100; scale by the principal in the stem.
- **Mechanics.** The TLP comes from swapping *fixed* to *floating*; deposits are *credited*, sticky ones more than hot ones.
- **"Which is NOT a reason for the average"**: anything about reflecting current market risk by maturity is a strength of the marginal approach.

Practice questions: LTR-14 c

Question LTR-14 c.1

Bank Q488 • Mock 2 Q55

A newly established bank is formulating its liquidity transfer pricing approach. Which method would most likely result in fewer long-term loans being underwritten?

A. Marginal cost of funds.

B. Pooled average cost of funds.

C. Separate average cost of funds.

D. Zero-cost approach.

Question LTR-14 c.2

Bank Q479 • AP Mock 1 Q54 (rebuilt)

Banks have historically used the pooled average cost of funds approach to liquidity transfer pricing (LTP) for several reasons. Which of the following is NOT one of those reasons?

A. All products are charged the same spread, so the approach is simple to implement and administer.

B. It is easier for business units to understand and comply with, and a basic liquidity management information system is sufficient.

C. It is less sensitive to intermediate changes in the bank's actual market cost of funds, reducing net interest income volatility across business units.

D. It reflects the current market perception of liquidity risk at each maturity, so long-term assets are charged more than short-term assets.

Question LTR-14 c.3

Bank Q487 • GARP PE2 Q26 • Mock 4 Q27

A treasurer of a bank is assessing the different methods of pricing liquidity and is concerned about the potential impact of applying each method. The treasurer asks an analyst to review and evaluate the various approaches to liquidity transfer pricing and to prepare a report with recommendations. Which of the following statements would be correct for the analyst to include in the report?

A. A zero-cost of funds approach tends to result in the bank holding long-term highly illiquid assets funded by long-term stable liabilities.

B. An average cost of funds approach tends to result in the greatest maturity transformation for a bank's balance sheet.

C. A matched-maturity marginal cost of funds approach converts the bank's fixed-rate borrowing cost to a floating-rate borrowing cost.

D. Both an average cost of funds approach and a zero-cost of funds approach appropriately align the maturity of the bank's lending and borrowing activities when management compensation is based on net income.

Question LTR-14 c.4 Bank Q486 • BT P2.T8.20.15.2 (rebuilt)

In regard to the various approaches to liquidity transfer pricing (LTP), each of the following statements is true EXCEPT which statement is false?

A. The zero cost of funds approach is poor practice because it views funding liquidity as free and funding liquidity risk as essentially zero.

B. The pooled average cost of funds approach is likely to undercharge long-term assets and therefore promote unhealthy maturity transformation.

C. The matched-maturity marginal cost of funds approach is best practice but requires the bank to keep its term liquidity premiums up to date.

D. The matched-maturity marginal cost of funds approach treats deposits as a use of funds, so sticky term deposits receive a larger charge than hot or volatile deposits.

Question LTR-14 c.5 Bank Q482 • AnalystPrep Q.4207 (rebuilt)

A bank recorded the following term liquidity premiums and average cost of funds (in bp) before the crisis and currently:

Term in years	1	2	3	4	5
Pre-GFC term liquidity premium	1	3	5	7	10
Pre-GFC average cost of funds	3	3	3	3	3
Current term liquidity premium	5	8	10	18	35
Current average cost of funds	10	10	10	10	10

The principal of a one-year non-amortising bullet loan is USD 5 million. Using the matched-maturity marginal cost of funds approach, what dollar charge is passed to the business unit writing the loan if it was originated pre-crisis and currently, respectively?

A. USD 100 and USD 500

B. USD 500 and USD 2,500

C. USD 1,500 and USD 5,000

D. USD 5,000 and USD 17,500

Question LTR-14 c.6 Bank Q480 • AP Mock 2 Q72 • AnalystPrep Q.4204 (rebuilt; key as corrected in the bank)

A treasury analyst prices a six-year linearly amortising loan that repays equal amounts of principal at the end of each year, so it can be viewed as six separate annual loans maturing in 1 through 6 years. The bank's term liquidity premiums are:

Term in years	1	2	3	4	5	6
Term liquidity premium (bp)	6	14	18	24	27	32

Using the matched-maturity marginal cost of funds approach, what is the charge for funding liquidity risk on the loan?

A. 20.17 bp

B. 24.33 bp

C. 32.00 bp

D. 511.00 bp

Solutions • LTR-14 c

Solution LTR-14 c.1

Answer: A

Bank Q488 • Mock 2 Q55

Step 1 – What drives long-term lending

Long-term loans are written in volume when the units that write them are charged too little for the long-term funding they use.

Step 2 – Compare the charges

Only the matched-maturity marginal approach charges a long loan the premium for its full term (40 bp at five years against 5 bp at one year in Grant's current panel). The pooled and separate averages charge one flat rate whatever the maturity, and zero cost charges nothing.

$\Rightarrow$ The marginal approach raises the hurdle for long-term loans, so fewer are written.

Interpretation

It does not forbid long-term lending; it makes each long loan pay for the liquidity it consumes, so only those that still earn a risk-adjusted profit get written.

Why the other options are wrong

B is incorrect. One rate for all maturities undercharges long loans, so more are written.

C is incorrect. Separate averages decouple the asset and liability rates but are still flat across maturities.

D is incorrect. Zero cost treats liquidity as free: the strongest incentive to write long-term loans.

LO: LTR-14 c: Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost). • **Reference:** Grant, FSI Occasional Paper No.10, 15.3.3 to 15.3.5

Solution LTR-14 c.2

Answer: D

Bank Q479 • AP Mock 1 Q54

Step 1 – Grant's four reasons for the average

Averaging is much simpler than charging each asset by its maturity; it is easier to understand and comply with; it can run on basic LMIS; and it is less susceptible to intermediate changes in the market cost of funds, reducing net interest income volatility.

$\Rightarrow$ A, B and C are reasons; D is not.

Interpretation

D describes the matched-maturity marginal approach. The average does the opposite: it applies one spread to every maturity and lags the market, which is its main weakness.

Why the other options are wrong

A is incorrect. A genuine reason: one spread for every product is simple to administer.

B is incorrect. A genuine reason: understanding, compliance and a basic LMIS.

C is incorrect. A genuine reason: smoothing reduces net interest income volatility and gives central management more control.

LO: LTR-14 c: Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost). • **Reference:** Grant, 15.3.4.2 Implications of Pooled "Average" Cost of Funds Approach

Solution LTR-14 c.3 — Answer: C

Bank Q487 • GARP PE2 Q26

Step 1 – Why C is correct

The matched-maturity approach works from the bank's actual market cost of funds "by converting fixed-rate borrowing costs to floating-rate borrowing costs through an internal swap transaction and observing the spread over the reference rate". That spread is the term liquidity premium, which charges assets and credits liabilities by maturity.

Interpretation

The swap removes the interest rate component, so what is left is the price of liquidity (idiosyncratic credit plus market access premiums).

Why the other options are wrong

A is incorrect. Half reversed: zero cost leaves the bank with long-term illiquid assets but "very few long-term stable liabilities".

B is incorrect. Wrong approach: zero cost "exacerbated maturity transformation to the largest degree possible"; the average promotes it, but less.

D is incorrect. Neither aligns maturities; with pay based on net interest income, both encourage long-term loans that are not charged more for liquidity.

LO: LTR-14 c: Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost). • **Reference:** Grant, 15.3.3, 15.3.4.2, 15.3.5

Solution LTR-14 c.4 — Answer: D

Bank Q486 • BT P2.T8.20.15.2

Step 1 – Check A, B and C

A: zero cost views funding liquidity as free and its risk as zero (true). B: one average spread undercharges long-term assets and promotes unhealthy maturity transformation (true). C: the marginal approach is current best practice, and banks that failed to update their premiums mispriced assets (true).

Step 2 – D is false

Deposits are a *source* of funds: they are *credited* for the liquidity they provide, and sticky (term) deposits receive *larger credits* than hot or volatile ones.

$\Rightarrow$ D is the false statement.

Interpretation

The direction of the transfer is the first thing to lock (Vardeez): assets charged, liabilities credited. Core demand deposits are credited on their modelled behavioural maturity.

Why the other options are wrong

A is incorrect. True: Grant's "worst practice identified in the survey".

B is incorrect. True: the average ignores the heightened liquidity risk of long assets.

C is incorrect. True: "some of the more advanced banks surveyed ... failed to actively update term liquidity premiums".

LO: LTR-14 c: Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost). • **Reference:** Grant, 15.3.3 to 15.3.6.3 Deposits

Solution LTR-14 c.5 — Answer: B

Bank Q482 • AnalystPrep Q.4207

Step 1 – The premium for a one-year bullet

A non-amortising bullet needs funding for its whole term, so it is charged the one-year TLP: 1 bp pre-crisis, 5 bp now.

Step 2 – Apply to the principal

$$0.0001 \times 5{,}000{,}000 = \text{USD } 500, \qquad 0.0005 \times 5{,}000{,}000 = \text{USD } 2{,}500$$

Interpretation

At one year the average approach would charge *more* (3 bp and 10 bp, USD 1,500 and USD 5,000): the flat average overcharges short assets and undercharges long ones (Figure 1).

Why the other options are wrong

A is incorrect. Used a USD 1 million principal, GARP's example size, instead of USD 5 million.
C is incorrect. Sibling input: the average cost of funds row (3 bp and 10 bp).
D is incorrect. Wrong maturity: the five-year premiums (10 bp and 35 bp).

LO: LTR-14 c: Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost). • **Reference:** Grant, 15.3.6.1 Non-Amortising Bullet Loans, Tables 15.3 and 15.4

Solution LTR-14 c.6 — Answer: B

Bank Q480 • AP Mock 2 Q72

Step 1 – Weight each tranche's premium by its term

$$1(6) + 2(14) + 3(18) + 4(24) + 5(27) + 6(32) = 6 + 28 + 54 + 96 + 135 + 192 = 511$$

Step 2 – Divide by the sum of the weights

$$\frac{511}{1+2+3+4+5+6} = \frac{511}{21} = 24.33 \text{ bp}$$

$\Rightarrow$ The loan is charged as if it needed funding for between four and five years (24 and 27 bp), not six.

Interpretation

Each annual tranche is outstanding for its own term, so longer tranches carry more weight; this is Grant's tranching approach (26.1 bp on his current five-year example).

Why the other options are wrong

A is incorrect. Simple average of the premiums, ignoring the term weights: $121/6 = 20.17$.
C is incorrect. The six-year bullet charge: ignores the amortisation.
D is incorrect. The weighted sum, not divided by 21.

LO: LTR-14 c: Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost). • **Reference:** Grant, 15.3.6.2 Amortising Loans

LTR-17 • Risk Management for Changing Interest Rates: ALM and Duration Techniques • Vardeez: VERY IMPORTANT

LTR-17 b

P1 BOTH

Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin.

Workbook: A16 Duration Gap and Interest-Sensitive Gap • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A16, recalled in the August 2026 sitting. Map #24 (5 of 16 recall sources) reports "IS gap" and "ISR gap LR" next to the duration gap questions of LTR-17 c; the August recall itself was the duration gap.

Vardeez: VERY IMPORTANT reading ("entire chapter and each learning objective is important"), ranked 3rd of 52 by return on study time; no pointer specific to this LO. His notes stress the earning-assets denominator of NIM, the polarity grid of the gap, and "do nothing" as a listed response.

Reported exam shapes

- **S4** IS gap and ISR: the cumulative gap over the planning horizon and the change in net interest income.

What the questions look like: a repricing schedule by maturity bucket, or totals of interest-sensitive assets and liabilities, sometimes with total assets and yields. You compute the dollar gap, relative gap, ISR, the cumulative gap over the stated horizon, and $\Delta\text{NII} \approx \Delta i \times$ cumulative gap. Statement versions ask which gap position a rate forecast calls for, or which balance-sheet action closes a gap. NIM itself comes as a one-line calculation with total assets planted next to earning assets.

1. Asset-liability management: two targets, two tools

This section is the background for the whole reading; LTR-17 a, c and d point back to it. Lending, investing and funding decisions inside a bank are linked: which loan requests can be met depends on the funds the bank can raise, and how much risk it can take depends on its capital (net worth). Rose and Hudgins call the coordinated, whole-balance-sheet approach to these decisions **asset-liability management (ALM)**. It grew up in three stages.

Approach	View of the balance sheet	Control lever
Asset management (historical)	Deposits and other funding taken as given: set by customers and, before deregulation, by regulation	Allocation of incoming funds: who receives the scarce loans and on what terms
Liability management	Funding can be controlled as assets are	**Price**: the offer rate on deposits and borrowings. Raise it when loan demand exceeds funds; hold or cut it when flush with funds
Funds management (today)	Both sides as one integrated whole	Volume, mix and return or cost of assets *and* liabilities, coordinated

Funds management, the approach that dominates today, has three objectives:

1. exercise as much control as possible over the **volume, mix, and return or cost** of both assets and liabilities, to achieve the institution's goals;

2. **coordinate** control over assets with control over liabilities, so the two are internally consistent and do not pull against each other;
3. recognise that **revenues and costs arise on both sides** of the balance sheet, and set policies that maximise returns and control costs. A bank sells a bundle of services (credit, payments, savings, advice), each priced to cover its cost.

The ALCO. Many banks run ALM through an asset-liability committee of key officers from different departments. A well-run ALCO meets regularly (quarterly, monthly or more often), works to goals usually centred on maximising shareholder wealth, adequate profitability and sufficient capitalisation (often numerical targets, such as an ROA of 1.5% and an equity-to-assets ratio of at least 10%), lays out how the bank is to be funded, the quality of loans it takes on and the limits on off-balance-sheet exposure, estimates the exposure of the net interest margin and net worth, keeps it within limits (often with simulation), and reports to the board.

Rates are set by the market, not the bank. The interest rate settles where the demand for and supply of loanable funds meet. Each bank is one of thousands of lenders and borrowers, so managers are **price takers, not price makers**. Each rate is built from components (a risk-free real rate plus premiums for inflation, default, term, liquidity and call risk), and any of them can move, so consistently accurate forecasting is virtually impossible. Changing rates create two kinds of interest rate risk:

- **Price risk:** rates *rise*, the market values of bonds and fixed-rate loans fall, and selling them means capital losses.
- **Reinvestment risk:** rates *fall*, and incoming funds must be invested in lower-yielding assets, cutting future income.

Two targets, two tools

Changing rates hit both the income statement (interest revenue and interest cost) and the balance sheet (the market values of assets and liabilities). Figure 1 is the map of the reading: identify the branch before answering any question.

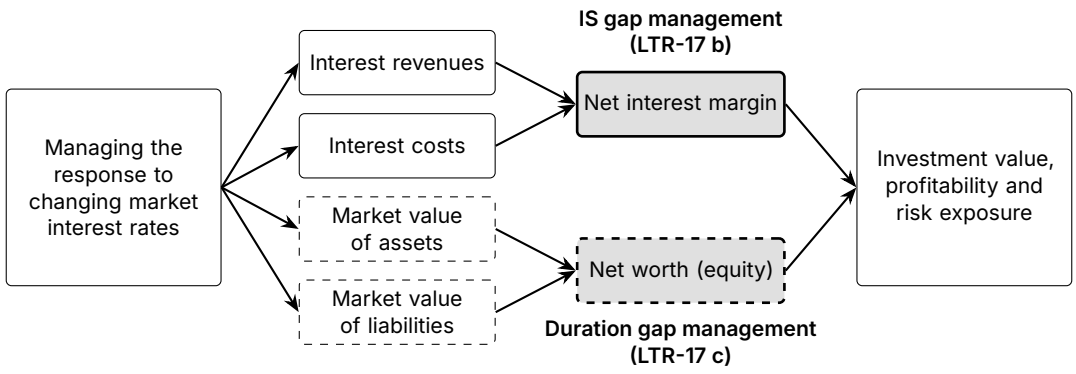


Figure 1. Asset-liability management, after Rose and Hudgins' Exhibit 18.1. The income branch (solid boxes) drives the net interest margin, which interest-sensitive gap management protects; the value branch (dashed boxes) drives net worth, which duration gap management protects. Protecting one does not protect the other.

2. The target: net interest margin

To insulate profits from fluctuating rates, management concentrates on the rate-sensitive parts of the balance sheet (earning assets and borrowings) and tries to hold the net interest margin fixed:

$$\boxed{\text{NIM} = \frac{\text{Interest income} - \text{Interest expense}}{\text{Total earning assets}} = \frac{\text{NII}}{\text{Total earning assets}}} \quad (18.5)$$

Example (GARP). Interest revenue \$4 billion, interest expense \$2.6 billion, earning assets \$40 billion: NIM $= 1.4/40 = 3.50\%$. This is not profit: noninterest expenses (salaries, overheads, taxes) have not been deducted, so the bank has little margin for error against interest rate risk. Net interest income has been about 60% to 80% of US banks' net earnings, which is why some banks have shifted towards fee income.

Example (Vardeez, Schweser). Gray Sky Bank: total assets \$50 billion, earning assets \$45 billion, interest income \$4 billion, interest expense \$2.2 billion. NIM $= 1.8/45 = 4.00\%$. Dividing by total assets gives $1.8/50 = 3.60\%$, the planted answer: the \$5 billion of cash, premises and reserves funds nothing that earns interest.

The margin is squeezed if the cost of borrowed funds rises faster than income from loans and securities, or if, when rates fall, that income falls faster than funding costs. Yield curves do not move in parallel, so the spread between borrowing costs and interest revenue is never constant. NIM is driven by five factors: (1) the level of rates; (2) the spread between asset yields and liability costs (the shape of the yield curve); (3) the volume of earning assets; (4) the volume of interest-bearing liabilities; (5) the mix (floating versus fixed rate, short versus long, higher versus lower yield).

3. What reprices: interest-sensitive assets and liabilities

Gap management needs an analysis of the maturities and *repricing* opportunities of interest-bearing assets and liabilities. An item is rate sensitive over a horizon if it matures, carries a variable rate, or is subject to expected but uncertain prepayment or withdrawal within the horizon.

Repriceable assets	Repriceable liabilities	Nonrepriceable assets	Nonrepriceable liabilities
Short-term securities about to mature	Money market borrowings (fed funds, RPs), repriced several times a day	Cash in the vault and reserves at the central bank	Demand deposits (no or a fixed rate)
Short-term loans about to mature or come up for renewal	Short-term savings accounts; maturing CDs	Long-term loans at a fixed rate	Long-term savings and retirement accounts
Variable-rate loans and securities (credit cards, ARMs)	Money market deposits; floating-rate deposits; NOW accounts	Long-term fixed-rate securities; buildings and equipment	Equity capital

4. Measuring the gap

$$\boxed{\text{Dollar gap} = \text{ISA} - \text{ISL}} \quad (18.7) \qquad \text{Relative gap} = \frac{\text{Dollar gap}}{\text{Size}} \qquad \boxed{\text{ISR} = \frac{\text{ISA}}{\text{ISL}}} \quad (18.11)$$

ISA and ISL are the interest-sensitive (repriceable) assets and liabilities in the period; size is usually total assets. The dollar and relative gaps are neutral at **zero**; the ISR is a ratio of levels, neutral at **one**.

- **Asset sensitive** (positive gap, ISR > 1): ISA \$500m, ISL \$400m, gap $+\$100$m. If rates rise, interest revenue rises more than interest cost and NII and NIM increase; if rates fall, NIM declines.
- **Liability sensitive** (negative gap, ISR < 1): ISA \$150m, ISL \$200m. Dollar gap $-\$50$m; ISR $= 150/200 = 0.75$. Rising rates lower NIM; falling rates raise it.
- **Size base.** The relative gap divides by the institution's size, "measured, for example, by total assets". GARP's own example divides $-\$50$m by \$150m (-0.33), the same figure as its ISA, which is why some question writers (BT Mock Q34) scale by ISA. Use total assets unless the stem names another base.
- **A zero gap** insulates NIM in principle, because revenues and costs move at the same rate. In practice it does not remove all risk: asset and liability rates are not perfectly correlated (loan rates lag money market rates), so revenues tend to grow more slowly than expenses in expansions and expenses fall faster in downturns (Section 7).

5. Maturity buckets, the cumulative gap and the change in NII

Every gapping method requires four decisions: (1) the period over which NIM is managed (six months, a year) and the subperiods or **maturity buckets** it is divided into; (2) a target NIM (freeze it or raise it); (3) to raise it, a correct rate forecast or a reallocation of assets and liabilities that widens the spread; (4) the volumes of ISA and ISL to hold.

The horizon decides the sign. GARP's computer run: the gap is $+\$10$m for the next 24 hours, but $-\$40$m for days 2 to 7, so the cumulative gap over the week is $-\$30$m. Rising rates would help over one day and hurt over one week. If a substantial rise is expected the bank might sell longer-term CDs at once or use futures to offset the margin loss.

GARP Table 18.1. A bank with \$4,100 million of assets buckets its balance sheet (Figure 2): liability sensitive through 90 days, asset sensitive after. Management has, consciously or not, positioned it for falling rates over the next three months and rising rates beyond. Net interest income in each bucket is

$$\text{NII} = y_{RSA}\,\text{RSA} + y_F\,(\text{TA} - \text{RSA}) - c_{RSL}\,\text{RSL} - c_F\,(\text{TA} - \text{RSL}) \qquad (18.12)$$

With rate-sensitive yields of 10% and costs of 8%, fixed assets at 11% and fixed liabilities at 9%, the first week gives $0.10(1{,}700) + 0.11(2{,}400) - 0.08(1{,}800) - 0.09(2{,}300) = \$83$m (NIM 2.02%). If the rate-sensitive rates rise two points to 12% and 10%, NII falls to \$81m (NIM 1.98%): a loss of \$2m, which is exactly $0.02 \times (-100)$.

> **Correction to the source notes.** In the second panel of GARP Table 18.1 the annualised NII for the 91 to 360 day column is printed as \$80.20 million, but the inputs give $0.10(480) + 0.11(3{,}620) - 0.08(150) - 0.09(3{,}950) = \$78.70$ million, the figure GARP itself divides by \$4,100 million on the next line to get 1.92%. Every other column checks (83, 84.9, 82.1, 81.3; after the rise 81, 79.1, 81.9, 85.3, 82.7).

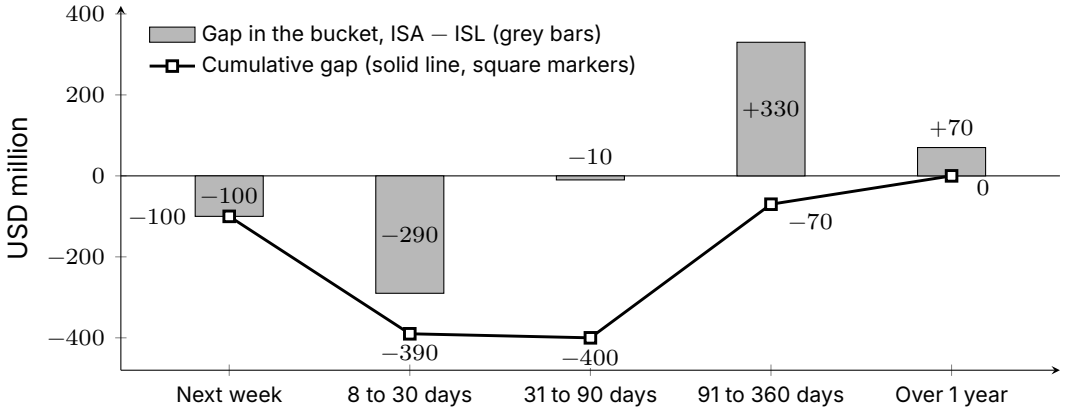


Figure 2. GARP Table 18.1 (USD million). ISA by bucket 1,700, 310, 440, 480, 1,170; ISL 1,800, 600, 450, 150, 1,100; ISR 94.4%, 51.7%, 97.8%, 320%, 106.4%. The bank is liability sensitive through 90 days (cumulative gap down to -400) and asset sensitive afterwards; over the whole balance sheet the cumulative gap returns to zero.

The cumulative gap and the change in NII

The **cumulative gap** is the total difference between assets and liabilities that can be repriced over a designated period: the sum of the bucket gaps out to the horizon. Given a rate change, it gives the approximate change in net interest income:

$$\boxed{\Delta\text{NII} \approx \Delta i \text{ (percentage points, as a decimal)} \times \text{Cumulative gap (in dollars)}} \qquad (18.13)$$

Example (GARP). A bank has \$100 million of earning assets and \$200 million of liabilities repricing each month for six months: cumulative gap $= 6 \times 100 - 6 \times 200 = -\$600$m. A one-point rise gives $\Delta\text{NII} \approx 0.01 \times (-600) = -\$6$ million.

- Use the gap **cumulated over the whole planning horizon**, not one bucket and not a shorter horizon.
- Δi is a change in *percentage points*. A 25% rise from a 3% level is 0.75 points, $\Delta i = 0.0075$.
- The formula assumes every repriceable rate moves by the same amount; Section 7 relaxes this.
- Negative cumulative gap: falling rates help, rising rates hurt. Positive: the reverse.

6. Managing the gap: aggressive and defensive

Aggressive gap management shades the gap towards asset or liability sensitivity according to management's confidence in its own forecast:

Forecast	Best IS gap position	Most likely action
Rising market rates	Positive IS gap	Increase ISA; decrease ISL
Falling market rates	Negative IS gap	Decrease ISA; increase ISL

This creates greater risk: consistently correct rate forecasting is impossible, and rates that move the wrong way magnify losses. Most managers have learned to rely on *hedging against*, not forecasting, rate changes. **Defensive gap management** sets the IS gap as close to zero as possible, to reduce the expected volatility of net interest income.

Eliminating a gap (GARP).

Position	The risk	Possible management responses
Positive gap, ISA > ISL (asset sensitive)	Losses if rates **fall**: NIM is reduced	1. Do nothing (rates may rise or be stable). 2. **Extend** asset maturities or **shorten** liability maturities. 3. Increase ISL or reduce ISA.
Negative gap, ISA < ISL (liability sensitive)	Losses if rates **rise**: NIM is reduced	1. Do nothing (rates may fall or be stable). 2. **Shorten** asset maturities or **lengthen** liability maturities. 3. Decrease ISL or increase ISA.

Two points from Vardeez. "Do nothing" is a listed response, so do not strike an option just because it proposes inaction. And maturity adjustments run *opposite* to the gap sign: a longer maturity makes an item *less* rate sensitive, so a positive gap is closed by lengthening assets (fewer of them reprice) or shortening liabilities (more of them reprice). Futures and other hedging instruments can do the same job off the balance sheet.

7. Basis risk and the weighted interest-sensitive gap

Rates on different assets and liabilities change by different amounts and at different speeds, relative to each other and over the business cycle: **basis risk**. Some banks therefore weight each repriceable item by its rate sensitivity relative to a reference rate (here fed funds, weight 1.0) and recompute the gap.

Item (USD million)	Balance	Weight	Weighted
Federal funds loans	50	1.00	50.00
Government securities and other investments	25	1.30	32.50
Loans and leases	125	1.50	187.50
Rate-sensitive assets	**200**		**270.00**
Interest-bearing deposits	159	0.86	136.74
Money market borrowings	64	0.91	58.24
Rate-sensitive liabilities	**223**		**194.98**
IS gap	**−23**		**+75**
ΔNII if fed funds rise 2 points	$-23 \times 0.02 = -0.46$		$+75 \times 0.02 = +1.50$

GARP rounds the weighted liabilities to 137 and 58. The sign of the gap **flips**: on the unweighted balance sheet the bank looks liability sensitive and would fear rising rates; on the weighted one it gains from them, because its loan rates move 1.5 times as far as fed funds while its deposit rates move only 0.86 times as far. "When it comes to assessing interest rate risk, things are not always as they appear." This is the response to the first limitation of IS gap management in LTR-17 d.

Formula card: LTR-17 b

Net interest margin	NIM $=$ (interest income $-$ interest expense)/total **earning** assets
Gap measures	Dollar gap $=$ ISA $-$ ISL; relative gap $=$ dollar gap/size (total assets); ISR $=$ ISA/ISL
Neutral values	Dollar and relative gap 0; ISR 1
Cumulative gap	Sum of bucket gaps out to the planning horizon
Change in NII	$\Delta\text{NII} \approx \Delta i \times$ cumulative gap (Δi in points, as a decimal)
Weighted gap	$\sum w_j\,\text{ISA}_j - \sum w_k\,\text{ISL}_k$, weights relative to the reference rate
NII by bucket	$y_{RSA}\text{RSA} + y_F(\text{TA} - \text{RSA}) - c_{RSL}\text{RSL} - c_F(\text{TA} - \text{RSL})$

	Positive gap (asset sensitive)	Negative gap (liability sensitive)
Definition	ISA $>$ ISL; relative gap > 0; ISR > 1	ISA $<$ ISL; relative gap < 0; ISR < 1
Rates rise	NII and NIM **up**	NII and NIM **down**
Rates fall	NII and NIM **down**	NII and NIM **up**
Aggressive choice when	Rates are forecast to rise	Rates are forecast to fall
To close it, maturities	Extend assets, shorten liabilities	Shorten assets, lengthen liabilities
To close it, volumes	Increase ISL or reduce ISA	Decrease ISL or increase ISA

Exam traps

- **Denominator.** NIM divides by *earning* assets; total assets is the planted alternative (4.00% against 3.60% for Gray Sky). Equity, funding (deposits plus borrowings) or loans alone are other wrong bases.
- **Polarity.** Anchor on one fact: a positive gap means assets reprice first, so rising rates help. Derive the other seven cells from it.
- **ISR inverted or misread.** ISR is ISA/ISL with neutral value 1; ISL/ISA reverses the diagnosis.
- **Bucket versus cumulative.** ΔNII uses the gap cumulated to the horizon in the stem; a single bucket or a shorter horizon is the planted error, and the sign can flip between horizons.
- **Rate change.** Points, not a relative change and not the level: 3% rising by 25% is $\Delta i = 0.0075$.
- **Maturity direction.** To close a positive gap *extend* assets or *shorten* liabilities; longer means less rate sensitive.
- **Inaction.** "Do nothing" is one of GARP's three listed responses.
- **Zero gap is not zero risk** (basis risk); the weighted gap can even flip the sign.
- **Sibling swap with LTR-17 c.** The IS gap protects NIM, a flow. A rising-rate forecast calls for a *positive* IS gap but a *negative* duration gap.

Practice questions: LTR-17 b

Question LTR-17 b.1

Bank Q555 • GARP book Ch.18 Problem 7 (rebuilt)

The CFO of New Comers State Bank is reviewing three years of data (USD million):

	Current year	Previous year	Two years ago
Interest revenues	82.00	80.00	78.00
Interest expenses	64.00	66.00	68.00
Loans (excluding nonperforming)	450.00	425.00	400.00
Investments	200.00	195.00	200.00
Total deposits	450.00	425.00	400.00
Money market borrowings	150.00	125.00	100.00

Taking earning assets as loans (excluding nonperforming) plus investments, which of the following gives the bank's net interest margin two years ago, in the previous year and in the current year, respectively?

A. 1.67%, 2.26%, 2.77% (rising)

B. 2.00%, 2.55%, 3.00% (rising)

C. 2.50%, 3.29%, 4.00% (rising)

D. 2.77%, 2.26%, 1.67% (falling)

Question LTR-17 b.2

Bank Q557 • GARP book Ch.18 Problem 10 (rebuilt)

Sparkle Savings Association has interest-sensitive assets of USD 400 million, interest-sensitive liabilities of USD 325 million and total assets of USD 500 million. Measuring the institution's size by its total assets, which of the following correctly describes Sparkle's interest rate position?

A. Dollar IS gap +USD 75 million; relative IS gap 15.0%; ISR 0.81; liability sensitive, so NII rises when rates fall.

B. Dollar IS gap +USD 75 million; relative IS gap 15.0%; ISR 1.23; asset sensitive, so NII rises when rates fall.

C. Dollar IS gap +USD 75 million; relative IS gap 15.0%; ISR 1.23; asset sensitive, so NII rises when rates rise.

D. Dollar IS gap +USD 75 million; relative IS gap 18.75%; ISR 1.23; asset sensitive, so NII rises when rates rise.

Question LTR-17 b.3

Bank Q571 • Schweser Module Quiz 82.1 Q3

A mortgage bank has the major part of its assets in 30-year, fixed-rate mortgages. The bank has a small amount of core deposits but is mostly funded with repurchase agreements and one-quarter negotiable certificates of deposit (CDs). The bank's 91-day interest-sensitive (IS) gap is most likely:

A. positive.

B. negative.

C. zero.

D. equal to its duration gap.

Question LTR-17 b.4

Bank Q556 • GARP book Ch.18 Problem 8 (rebuilt)

The First National Bank of Dogsville reports the following distribution of maturities and repricing opportunities (USD million):

	Coming week	Next 30 days	31 to 90 days	More than 90 days
Loans	200	300	475	525
Securities	21	26	40	70
Transaction deposits	320	0	0	0
Time accounts	100	290	196	100
Money market borrowings	136	140	100	65

Which of the following statements about the bank's exposure is correct?

A. The bank is asset sensitive over the coming week, so a rise in rates during the week would increase its net interest income.

B. Its largest exposure is the cumulative gap of −USD 439 million through the first 30 days; a 1 percentage point rise in rates over that horizon would cut annualised NII by about USD 4.39 million.

C. Its cumulative gap across all four intervals is +USD 210 million, so the bank benefits from rising rates at every horizon.

D. The 31 to 90 day interval has a gap of +USD 219 million, so the bank is liability sensitive there and benefits from falling rates.

Question LTR-17 b.5

Workbook A16.3 • S4 • Map #24

The ALCO secretary at a US community bank presents the following repricing schedule (USD million) for the one-year planning horizon:

Bucket	Next 7 days	8 to 30 days	31 to 90 days	91 to 360 days
Interest-sensitive assets	60	140	95	210
Interest-sensitive liabilities	45	190	120	225

Market rates on all interest-sensitive items are expected to rise by 125 basis points. Which of the following is closest to the approximate change in the bank's annual net interest income over the one-year horizon?

A. −USD 0.94 million

B. −USD 0.75 million

C. −USD 0.63 million

D. +USD 0.94 million

Question LTR-17 b.6 Bank Q554 • GARP book Ch.18 Problem 6 (rebuilt)

The treasurer of Poquoson Savings Bank reports that the bank's cumulative interest-sensitive gap has risen 60% from an initial USD 25 million. Market interest rates then rise by 25% from an initial level of 3% (that is, from 3.00% to 3.75%). Assuming all interest-sensitive assets and liabilities reprice by the same amount, what is the change in net interest income at the new gap?

A. +USD 0.19 million

B. +USD 0.30 million

C. +USD 1.20 million

D. +USD 10.00 million

Question LTR-17 b.7 Bank Q567 • AnalystPrep Q.5967 (rebuilt)

The ALCO of a bank holding company oversees four subsidiary banks, each with total assets of USD 1,000 million. The committee expects interest rates to rise over the coming year.

Bank	ISA (USD million)	ISL (USD million)	NIM
L	520	400	3.0%
M	450	560	3.5%
N	640	410	2.2%
O	500	500	3.8%

Which bank is best positioned to increase its net interest income?

A. Bank L

B. Bank M

C. Bank N

D. Bank O

Question LTR-17 b.8 Bank Q573 • Schweser Module Quiz 82.1 Q5

Paul Saunders, a risk manager and member of the asset-liability committee (ALCO) at National Bank and Trust, is considering strategies to hedge the bank's net interest margin (NIM) against expected increasing interest rates. The bank currently has negative cumulative interest-sensitive (IS) gap positions over the one-month, one-quarter, and one-year maturity buckets. Which would be the least effective strategy?

A. Selling long-term Treasury bonds and purchasing short-term Treasury bonds.

B. Shifting from a focus on making fixed-rate mortgages to a focus on making variable-rate mortgages.

C. Increasing the volume of overnight funding from the fed funds market and relying less on long-term funding from the Federal Home Loan Bank.

D. Shifting funding from overnight repurchase agreements to five-year retail certificates of deposit.

Question LTR-17 b.9 Bank Q594 • GARP book Ch.18 Problem 9 (rebuilt; filed under LTR-17 d in the bank)

The ALCO of Sunset Savings Bank weights its interest-sensitive balances by their rate sensitivity relative to the federal funds rate (USD million):

Interest-sensitive assets	Amount	Index	Interest-sensitive liabilities	Amount	Index
Federal funds loans	50.00	1.00	Interest-bearing deposits	250.00	0.75
Security holdings	50.00	1.20	Money market borrowings	90.00	0.95
Loans and leases	350.00	1.45			

If the federal funds rate rises by 50 basis points, which of the following is closest to the change in annual net interest income predicted by the weighted balance sheet?

A. +USD 0.55 million

B. +USD 1.39 million

C. +USD 1.72 million

D. +USD 17.23 million

Solutions • LTR-17 b

Solution LTR-17 b.1

Answer: A

Bank Q555 • GARP book Ch.18 Problem 7

Step 1 – Net interest income each year

$$78 - 68 = 10, \qquad 80 - 66 = 14, \qquad 82 - 64 = 18 \quad \text{(USD million, two years ago to current)}$$

Step 2 – Earning assets each year

$$400 + 200 = 600, \qquad 425 + 195 = 620, \qquad 450 + 200 = 650$$

Step 3 – NIM

$$\frac{10}{600} = 1.67\%, \qquad \frac{14}{620} = 2.26\%, \qquad \frac{18}{650} = 2.77\%$$

$\Rightarrow$ The margin has risen by about 110 basis points in two years.

Interpretation

Interest revenue per dollar of earning assets is almost flat (13.0%, 12.9%, 12.6%) while the interest cost per dollar of funding falls from 13.6% to 12.0% to 10.7%. The margin is widening because funding costs are falling faster than asset yields: the pattern of a liability-sensitive bank in a falling-rate period. If rates turn, the same position squeezes the margin, so management should check its gap before taking the trend for granted.

Why the other options are wrong

B is incorrect. Divided NII by funding (deposits plus money market borrowings: 500, 550, 600), giving 2.00%, 2.55%, 3.00%. NIM is over earning assets.

C is incorrect. Divided by loans only (400, 425, 450), leaving out investments, which also earn interest: 2.50%, 3.29%, 4.00%.

D is incorrect. Read the columns in the wrong order: the current year is the first column, so the trend is rising, not falling.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. • **Reference:** Rose and Hudgins, Bank Management and Financial Services 9e, Ch.7, 18.4 Protect the Net Interest Margin, eq. (18.5)

Solution LTR-17 b.2

Answer: C

Bank Q557 • GARP book Ch.18 Problem 10

Step 1 – Dollar gap

$$\text{ISA} - \text{ISL} = 400 - 325 = +\text{USD } 75 \text{ million}$$

Step 2 – Relative gap on total assets

$$\frac{75}{500} = 15.0\%$$

Step 3 – Interest sensitivity ratio

$$\frac{\text{ISA}}{\text{ISL}} = \frac{400}{325} = 1.23 > 1$$

Step 4 – Direction

All three measures say **asset sensitive**: more assets than liabilities reprice, so $\Delta\text{NII} \approx \Delta i \times 75$ is positive when rates rise and negative when they fall.

Interpretation

A 1 point rise adds about USD 0.75 million to NII; a 1 point fall costs the same. The ISR is the only one of the three measures whose neutral value is 1, not 0.

Why the other options are wrong

A is incorrect. Inverted the ratio, $325/400 = 0.81$, and then read the bank as liability sensitive, which contradicts the positive dollar gap.
B is incorrect. Correct measures, polarity flip: an asset-sensitive bank loses NII when rates fall.
D is incorrect. Scaled the gap by ISA, $75/400 = 18.75\%$, instead of by the institution's size as the stem defines it.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. • **Reference:** Rose and Hudgins, Ch.7, 18.4 Interest-Sensitive Gap Management, eq. (18.7), (18.11)

Solution LTR-17 b.3 Answer: B

Bank Q571 • Schweser Module Quiz 82.1 Q3

Step 1 – Which assets reprice within 91 days

Thirty-year fixed-rate mortgages neither mature nor reset within 91 days, so ISA is close to zero.

Step 2 – Which liabilities reprice

Repurchase agreements roll over within days and one-quarter CDs mature within the 91 days, so almost all funding is ISL.

Step 3 – Sign

ISA $\ll$ ISL, so the 91-day gap is negative: the bank is liability sensitive.

Interpretation

This is the classic thrift position: borrow short, lend long. Rising rates raise funding costs at once while mortgage income is locked, so NIM is squeezed. The same balance sheet has a large *positive* leverage-adjusted duration gap (LTR-17 c), so rising rates hurt both targets.

Why the other options are wrong

A is incorrect. Treats the long-term mortgages as the rate-sensitive side; it is the funding that reprices.
C is incorrect. A zero gap would need repriceable assets to match the repriceable funding, which this balance sheet rules out.
D is incorrect. The IS gap is a dollar repricing measure for a bucket; the duration gap is a market-value measure in years. They are different quantities.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. • **Reference:** Rose and Hudgins, Ch.7, 18.4, Examples of Repriceable and Nonrepriceable Assets and Liabilities

Solution LTR-17 b.4 Answer: B

Bank Q556 • GARP book Ch.18 Problem 8

Step 1 – ISA and ISL by interval

$$\text{ISA} = 221,\ 326,\ 515,\ 595 \qquad \text{ISL} = 556,\ 430,\ 296,\ 165$$

Step 2 – Interval gaps and cumulative gaps

$$\text{Gap} = -335,\ -104,\ +219,\ +430 \qquad \text{Cumulative} = -335,\ -439,\ -220,\ +210$$

Step 3 – Largest exposure

The cumulative gap is most negative through 30 days: $0.01 \times (-439) = -\text{USD } 4.39$ million of annualised NII for a 1 point rise.
$\Rightarrow$ Rising rates hurt over the first month; falling rates help.

Interpretation

The bank is liability sensitive in the first two intervals (USD 320 million of transaction deposits reprice this week) and asset sensitive in the last two. Over the full horizon it is asset sensitive, but the horizon the stem asks about decides the answer: the sign of the cumulative gap changes between 90 days (-220) and beyond $(+210)$.

Why the other options are wrong

A is incorrect. The coming-week gap is $221 - 556 = -335$: the bank is liability sensitive and a rise this week cuts NII.
C is incorrect. Absolute claim. The overall cumulative gap is $+210$, but the cumulative gaps through one week, 30 days and 90 days are all negative, so rising rates hurt at those horizons.
D is incorrect. A positive interval gap means asset sensitive, so rising rates help in that interval; "liability sensitive" is the polarity flip.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. • **Reference:** Rose and Hudgins, Ch.7, 18.4, Computer-Based Techniques and eq. (18.13)

Solution LTR-17 b.5 Answer: A

Workbook A16.3 • S4

Step 1 – Gap per bucket (ISA − ISL)

$$+15, \quad -50, \quad -25, \quad -15$$

Step 2 – Cumulative gap over the one-year horizon

$$15 - 50 - 25 - 15 = -\text{USD } 75 \text{ million}, \qquad \text{ISR} = \frac{505}{580} = 0.87 < 1$$

$\Rightarrow$ The bank is liability sensitive over the year: more liabilities than assets reprice. It is asset sensitive over the next 7 days only.

Step 3 – Change in net interest income

$$\Delta\text{NII} \approx \Delta i \times \text{Cumulative gap} = 0.0125 \times (-75) = -\text{USD } 0.94 \text{ million}$$

Interpretation

Rising rates squeeze NII because funding costs reprice faster than asset yields. A defensive manager would shorten asset maturities or lengthen liability maturities (or increase ISA, decrease ISL) to move the cumulative gap towards zero.

Why the other options are wrong

B is incorrect. Wrong horizon: used the cumulative gap only to 90 days, $0.0125\times(15-50-25) = -0.75$.

C is incorrect. Used the single largest bucket gap instead of the cumulative gap, $0.0125 \times (-50) = -0.63$.

D is incorrect. Polarity flip: treated a negative cumulative gap as benefiting from rising rates.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. • **Reference:** Rose and Hudgins, Ch.7, 18.4 Interest-Sensitive Gap Management, eq. (18.11), (18.13)

Solution LTR-17 b.6

Answer: B

Bank Q554 • GARP book Ch.18 Problem 6

Step 1 – New cumulative gap

$$25 \times 1.60 = +\text{USD } 40 \text{ million (asset sensitive)}$$

Step 2 – Rate change in percentage points

$$3.00\% \times 25\% = 0.75 \text{ points}, \qquad \Delta i = 0.0075$$

Step 3 – Change in NII

$$\Delta\text{NII} \approx 0.0075 \times 40 = +\text{USD } 0.30 \text{ million}$$

Interpretation

A positive gap gains when rates rise. The two percentage changes in the stem are traps of the same kind: the 60% applies to the gap and the 25% to the rate *level*, so each must be turned into a level or a point change before multiplying.

Why the other options are wrong

A is incorrect. Used the old gap: $0.0075 \times 25 = 0.19$.

C is incorrect. Multiplied by the rate level instead of the change: $0.03 \times 40 = 1.20$.

D is incorrect. Applied the 25% relative change directly to the gap: $0.25 \times 40 = 10.00$.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. • **Reference:** Rose and Hudgins, Ch.7, 18.4, eq. (18.13)

Solution LTR-17 b.7

Answer: C

Bank Q567 • AnalystPrep Q.5967

Step 1 – Dollar gap of each bank

$$\text{L} = +120, \qquad \text{M} = -110, \qquad \text{N} = +230, \qquad \text{O} = 0$$

Step 2 – Apply the forecast

With rising rates, $\Delta\text{NII} \approx \Delta i \times$ gap is largest for the largest positive gap: Bank N (relative gap $+23\%$, ISR 1.56).

$\Rightarrow$ N is the aggressive gap manager's position for a rising-rate forecast.

Interpretation

The level of NIM says nothing about its sensitivity. Bank O has the highest margin and a zero gap, so its NII barely moves; Bank N has the lowest margin but gains most if the forecast is right,

and loses most if it is wrong.

Why the other options are wrong

A is incorrect. Positive gap, but smaller $(+120)$ than N's $+230$.
B is incorrect. Highest NIM among the gapped banks, but liability sensitive (-110): its NII falls when rates rise.
D is incorrect. Highest NIM, but a zero gap protects NII rather than increasing it.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. • **Reference:** Rose and Hudgins, Ch.7, 18.4, Aggressive Interest-Sensitive GAP Management

Solution LTR-17 b.8 Answer: C

Bank Q573 • Schweser Module Quiz 82.1 Q5

Step 1 – What the hedge must do

A negative gap with rates expected to rise means NIM will be squeezed. The bank must move its gap towards zero: make assets more rate sensitive or liabilities less rate sensitive.

Step 2 – Test each action

A shortens asset maturity (more ISA); B moves lending to variable rates (more ISA); D locks funding for five years (less ISL). All three close the gap.

Step 3 – The odd one out

C replaces long-term funding with overnight fed funds: more ISL, so the negative gap *widens*.

Interpretation

This is GARP's table for eliminating a negative gap: shorten asset maturities, lengthen liability maturities, increase ISA or decrease ISL. Doing nothing is also on the list if the forecast is uncertain, but C is worse than doing nothing.

Why the other options are wrong

A is incorrect. Effective: short-term Treasuries reprice sooner, adding to ISA.
B is incorrect. Effective: variable-rate mortgages reprice with the market, adding to ISA.
D is incorrect. Effective: five-year CDs take funding out of the short repricing buckets, reducing ISL.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. • **Reference:** Rose and Hudgins, Ch.7, 18.4, Eliminating an Interest-Sensitive Gap

Solution LTR-17 b.9 Answer: C

Bank Q594 • GARP book Ch.18 Problem 9

Step 1 – Weighted interest-sensitive assets

$$50(1.00) + 50(1.20) + 350(1.45) = 50 + 60 + 507.5 = \text{USD } 617.5 \text{ million}$$

Step 2 – Weighted interest-sensitive liabilities

$$250(0.75) + 90(0.95) = 187.5 + 85.5 = \text{USD } 273.0 \text{ million}$$

Step 3 – Weighted gap and change in NII

$$617.5 - 273.0 = +344.5, \qquad \Delta\text{NII} \approx 0.005 \times 344.5 = +\text{USD } 1.72 \text{ million}$$

Interpretation

The unweighted gap is $450-340=+110$, which predicts only +USD 0.55 million. The weighted gap is 3.1 times larger because loans and leases move 1.45 times as far as fed funds while deposit rates move only 0.75 times as far. A 50 basis point fall would cost the same USD 1.72 million: the simple gap understates the exposure in both directions.

Why the other options are wrong

A is incorrect. Unweighted gap: $0.005 \times 110 = 0.55$.
B is incorrect. Weighted the assets but not the liabilities: $0.005 \times (617.5 - 340) = 1.39$.
D is incorrect. Took 50 basis points as 5%: $0.05 \times 344.5 = 17.23$.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. • **Reference:** Rose and Hudgins, Ch.7, 18.4, The Weighted Interest-Sensitive Gap: Dealing with Basis Risk

LTR-17 • Risk Management for Changing Interest Rates: ALM and Duration Techniques • Vardeez: VERY IMPORTANT

LTR-17 c

P1 BOTH

Describe duration gap management and apply this strategy to protect a bank's net worth.

Workbook: A16 Duration Gap and Interest-Sensitive Gap • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A16; your August 2026 recall had the duration gap twice. Map #24 (5 of 16 recall sources) reports "duration gap", "impact of asset and liability values on net worth given duration of asset and liability" and "duration of asset liability (2)", and calls the polarity of the leverage-adjusted gap against a rate move "the single most corruptible fact here". **Vardeez:** VERY IMPORTANT reading. His notes add the change-in-net-worth equation with a worked example (written from GARP, since his source layers print only the direction table), flag the unadjusted gap as the intermediate-result trap, and call the opposite IS gap and duration gap prescriptions for the same forecast the highest-value confusable pair in the reading.

Reported exam shapes

- **S1** The duration gap, the plain form of the August recall.
- **S2** Impact on net worth given asset and liability durations, with direction.
- **S3** The liability duration that immunises net worth.
- **S5** Polarity of the leverage-adjusted gap against a rate move, including a sign reversal from the leverage adjustment.

What the questions look like: the amounts and durations of assets and liabilities (sometimes item by item, so you build dollar-weighted averages first), a starting rate and a rate move. You give the leverage-adjusted gap, its sign and the direction of net worth, the dollar change in net worth, or the liability duration that immunises. Red herrings: an ISR, the discount rate in an immunisation question, equity in place of total assets.

1. Why net worth needs its own tool

IS gap management (LTR-17 b) protects the income branch of the ALM map (LTR-17 b, Figure 1). Changing rates also damage the value branch: **net worth**, the value of the stockholders' investment. Protecting NIM does not mean net worth is sheltered, and for most companies net worth is the more important of the two.

$$\text{NW} = A - L \quad (18.16) \qquad \Delta\text{NW} = \Delta A - \Delta L \quad (18.17)$$

A and L are market values. Portfolio theory gives two facts: (1) a rise in market rates lowers the market value of both fixed-rate assets and fixed-rate liabilities; (2) the longer their maturity, the more they fall. So the change in net worth depends on the *relative* maturities of assets and liabilities: a bank with longer-duration assets than liabilities loses more net worth when rates rise. Balancing the average maturity of cash inflows with that of outflows can **immunise** net worth.

2. Duration and price sensitivity

Duration is a value- and time-weighted measure of maturity that considers the timing of all cash inflows from earning assets and all cash outflows on liabilities: the average time needed to recover the funds committed.

$$D = \frac{\sum_{t=1}^{n} \text{CF}_t \times t/(1+\text{YTM})^t}{\sum_{t=1}^{n} \text{CF}_t/(1+\text{YTM})^t} = \frac{\sum_{t=1}^{n} \text{CF}_t \times t/(1+\text{YTM})^t}{\text{Current market value or price}} \qquad (18.14),\ (18.15)$$

Example (GARP). A five-year \$1,000 loan pays 10% a year and yields 10%, so its price is \$1,000. The present values of the cash flows (90.91, 82.64, 75.13, 68.30, 62.09 of interest and 620.92 of principal) times their dates sum to \$4,169.87, so $D = 4{,}169.87/1{,}000 = 4.17$ years, below the five-year maturity.

Price sensitivity. The percentage change in price equals minus duration times the relative change in the rate:

$$\boxed{\frac{\Delta P}{P} \approx -D \times \frac{\Delta i}{1+i}} \qquad (18.18)$$

A bond with $D = 4$ at 10%, if rates rise to 11%: $\Delta P/P = -4 \times 0.01/1.10 = -3.64\%$. Interest rate risk is **directly proportional to duration**: a duration of 2 is twice as risky as a duration of 1. The minus sign records that prices and rates move in opposite directions; i is the *original* rate.

Convexity. The price-rate relation is not linear. Price moves more when rates are low than when they are high; convexity rises with duration; holding maturity and yield constant it falls as the coupon rises. Duration is a straight-line approximation, so it works best for small rate changes (LTR-17 d).

3. Duration of a portfolio

The duration of a portfolio is the **market-value-weighted average** of the durations of its instruments: compute each instrument's duration, weight it by market value, and add. GARP's bank (USD million):

Asset	Value	Rate	D	Liability or equity	Value	Rate	D
US Treasury securities	90	10.00%	7.490	Negotiable CDs	100	6.00%	1.943
Municipal bonds	20	6.00%	1.500	Other time deposits	125	7.20%	2.750
Commercial loans	100	12.00%	0.600	Subordinated notes	50	9.00%	3.918
Consumer loans	50	15.00%	1.200	**Total liabilities**	**275**		**2.669**
Real estate loans	40	13.00%	2.250	Stockholders' equity	25		
Total assets	**300**		**3.047**	Total	300		

$$D_A = \frac{7.49(90) + 1.50(20) + 0.60(100) + 1.20(50) + 2.25(40)}{300} = \frac{914.1}{300} = 3.047 \text{ years} \qquad (18.24)$$

The CD figure comes from the same formula: \$100 million of two-year CDs paying 6%, $D = [6/1.06 + 2(6)/1.06^2 + 2(100)/1.06^2]/100 = 1.943$ years. Weighting the three liability durations by 100, 125 and 50 gives $D_L = 733.98/275 = 2.669$ years. Liabilities are shorter than assets, so this bank's net worth falls if rates rise; management is in effect betting on falling rates, and could hedge by lengthening liabilities, shortening assets, or using futures, options or swaps.

4. The leverage-adjusted duration gap

A first attempt at hedging sets $D_A \approx D_L$ (18.19), so the **duration gap** $D_A - D_L$ (18.20) is close to zero. But assets exceed liabilities (otherwise the bank would be insolvent), so equal *percentage* changes on a larger asset base and a smaller liability base do not cancel in dollars. The gap must be adjusted for leverage:

$$\boxed{\text{Leverage-adjusted duration gap} = D_A - D_L \times \frac{\text{Total liabilities}}{\text{Total assets}}} \qquad (18.21)$$

D_A and D_L are the dollar-weighted durations of the asset and liability portfolios. "The value of liabilities must change by slightly more than the value of assets" to eliminate the exposure. For GARP's bank:

$$3.047 - 2.669 \times \frac{275}{300} = 3.047 - 2.447 = +0.60 \text{ year} \qquad (\text{unadjusted: } 3.047 - 2.669 = 0.378)$$

The larger the leverage-adjusted gap, the more sensitive net worth is to rates. Figure 1 shows why the adjustment is needed: what must match is *dollar* duration, $D\times$ value.

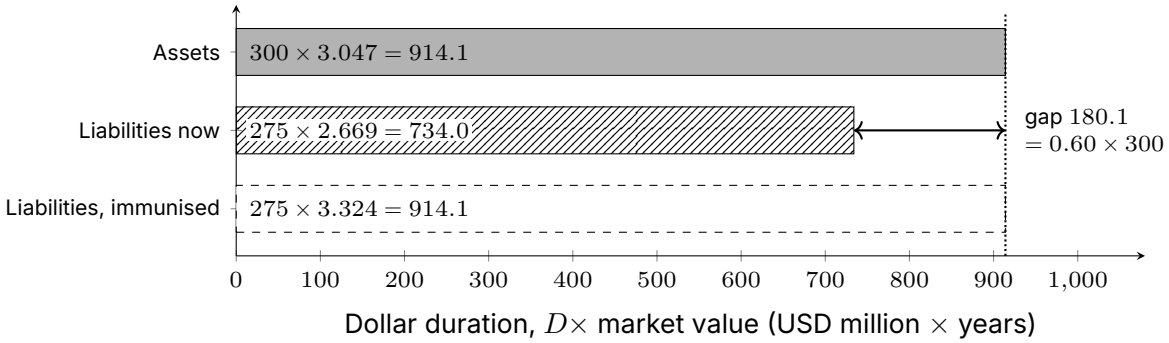


Figure 1. GARP's bank in dollar-duration terms. Assets (grey bar) carry 914.1, liabilities now (hatched bar) 734.0; the difference, 180.1, is the leverage-adjusted gap times total assets. Net worth is immunised when the liability bar reaches the dotted line (dashed white bar): $D_L = 3.047 \times 300/275 = 3.324$ years, longer than D_A.

The sign reversal

The sign that matters is the sign *after* the leverage adjustment:

$$\text{Gap} > 0 \iff \frac{D_A}{D_L} > \frac{L}{A}$$

A bank whose assets are *shorter* than its liabilities can still have a positive gap. Assets of 3.6 years, liabilities of 3.8 years and $L/A = 0.88$: the unadjusted gap is -0.20, but $3.6 - 3.8 \times 0.88 = +0.256$, because $3.6/3.8 = 0.947 > 0.88$. That bank loses net worth when rates rise (workbook A16.2). Solvency alone does not fix the sign either: with $A = 120$, $L = 100$, $D_A = 4$ and $D_L = 5$ the gap is $4 - 5(100/120) = -0.17$.

5. The change in net worth

Since $\Delta A/A \approx -D_A\,\Delta i/(1+i)$ and $\Delta L/L \approx -D_L\,\Delta i/(1+i)$:

$$\boxed{\Delta\text{NW} = \left[-D_A \times \frac{\Delta i}{1+i} \times A\right] - \left[-D_L \times \frac{\Delta i}{1+i} \times L\right]} \qquad (18.23)$$

$$= -\left(D_A - D_L\frac{L}{A}\right) \times A \times \frac{\Delta i}{1+i}$$

i is the **original** discount rate. The compact form says $\Delta\text{NW} = -\text{gap} \times A \times \Delta i/(1+i)$.

Example 1 (GARP). $D_A = 3$, $D_L = 2$, $A = \$120$m, $L = \$100$m; rates rise from 10% to 12%:

$$\Delta\text{NW} = \left[-3 \times \frac{0.02}{1.10} \times 120\right] - \left[-2 \times \frac{0.02}{1.10} \times 100\right] = -6.55 + 3.64 = -\$2.91 \text{ million}$$

Example 2 (GARP's bank above). Rates on assets and liabilities rise from 8% to 10%; $\Delta i/(1+i) = 0.02/1.08 = 0.018519$.

$$\Delta\text{NW} = -3.047(0.018519)(300) - [-2.669(0.018519)(275)] = -16.93 + 13.59 = -\$3.34 \text{ million}$$

If rates fall from 8% to 6%, the signs reverse: $+16.93 - 13.59 = +\$3.34$ million. Check: gap $+0.60$, $-0.60 \times 300 \times 0.018519 = -3.34$. Both sides lose value when rates rise; the bank is worse off only because assets lose more. The whole exposure is the \$3.34 million residual of two large, nearly offsetting legs.

GARP's three size factors: (A) the size of the duration gap; (B) the size of the institution (A and L); (C) the size of the rate change. Management can reduce the exposure by closing the gap (changing D_A, D_L or both) or by changing the relative amounts of A and L.

Leverage-adjusted gap	Rates rise	Rates fall
Positive, $D_A > D_L(L/A)$: assets change more than liabilities	Net worth **decreases**	Net worth **increases**
Negative, $D_A < D_L(L/A)$: liabilities change more than assets	Net worth **increases**	Net worth **decreases**
Zero, $D_A = D_L(L/A)$: immunised	No change	No change

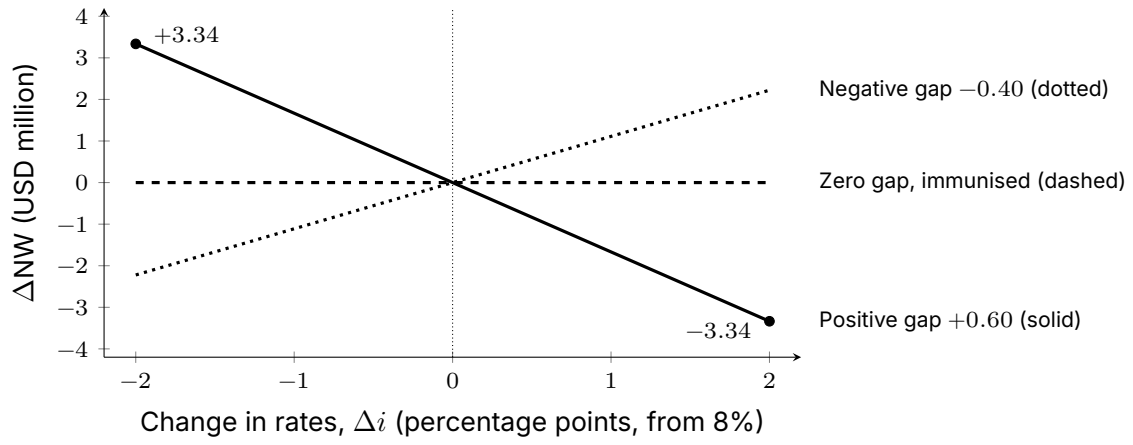


Figure 2. Change in net worth against the rate move for GARP's bank ($A = 300$, starting rate 8%) at three leverage-adjusted gaps. The solid line is the bank as it stands (± 2 points gives $\mp\$3.34$ million); the dotted line is the same bank with $D_L = 3.76$ years. The slope is $-\text{gap} \times A/(1+i)$: the sign of the gap decides which rate move hurts.

6. Immunisation

Setting the leverage-adjusted gap to zero gives the liability duration that immunises net worth:

$$0 = D_A - D_L\frac{L}{A} \implies \boxed{D_L = D_A \times \frac{A}{L}} \qquad 3.047 \times \frac{300}{275} = 3.324 \text{ years}$$

Check with rates up from 8% to 10%: $-3.047(0.018519)(300) - [-3.324(0.018519)(275)] = -16.93 + 16.93 = 0$; the same holds for a fall to 6%. GARP's footnote example: $D_A = 3$ years, $A = \$100$m, $L = \$92$m gives $D_L = 3 \times 100/92 = 3.261$ years. At immunisation the liabilities are *longer* than the assets, and the unadjusted gap is positive, not zero. The current discount rate plays no part in the target.

7. Defensive and aggressive duration gap management

The **defensive** strategy targets a zero leverage-adjusted gap and takes no view. More aggressive managers may find this "seemingly wimpy" and take chances to maximise the shareholders' position:

Expected rates	Management action	Possible outcome
Rates will rise	Reduce D_A and increase D_L (move towards a **negative** gap)	Net worth increases, if the forecast is correct
Rates will fall	Increase D_A and reduce D_L (move towards a **positive** gap)	Net worth increases, if the forecast is correct

Both rows promise the same outcome only on condition that the forecast is right: aggressive gap management turns interest rate risk into forecast risk; it does not remove risk.

The sibling with LTR-17 b. For the same rising-rate forecast the two tools prescribe opposite signs. The IS gap should be *positive* (assets reprice up first, so NIM rises); the duration gap should be *negative* (liabilities lose more value than assets, so net worth rises). One protects a flow, the other a value.

> **Correction to the source notes.** Vardeez's callout on this pair is headed "Rising rates call for a negative gap on both tools, but for different reasons". The body of the callout, and GARP's two tables, say the opposite of the heading: a rising-rate forecast calls for a *positive* IS gap and a *negative* duration gap. Learn the body, not the heading.

Real banks: Fannie Mae. Fannie Mae reported its duration gap monthly. As rates plummeted early in the 2000s and mortgage refinancing soared, its liabilities lengthened and its assets shortened, producing a **negative** gap of about 14 months at the close of 2002: assets were rolling over into cash more than a year sooner than liabilities. Had rates kept falling, net worth would have kept falling as the longer liabilities gained value relative to the assets; its stock price fell for a time when it announced the widening gap. Fannie's target band was -6 to $+6$ months; by September 2007 its monthly gaps were within ± 1 month.

Formula card: LTR-17 c

Duration	$D = \sum t\,\mathrm{PV}(\mathrm{CF}_t) / \sum \mathrm{PV}(\mathrm{CF}_t)$; portfolio D = value-weighted average
Price change	$\Delta P/P \approx -D\,\Delta i/(1+i)$, i = original rate
Leverage-adjusted gap	$D_A - D_L \times L/A$ (unadjusted $D_A - D_L$ is the intermediate result)
Change in net worth	$\Delta\mathrm{NW} = [-D_A \frac{\Delta i}{1+i} A] - [-D_L \frac{\Delta i}{1+i} L] = -\text{gap} \times A \times \frac{\Delta i}{1+i}$
Immunisation	Gap $= 0$: $D_L = D_A \times A/L$ (longer than D_A)
Sign of the gap	Positive iff $D_A/D_L > L/A$
Size factors	Gap, size of the institution, size of the rate change

	Interest-sensitive gap (LTR-17 b)	Leverage-adjusted duration gap
Protects	Net interest margin (a flow)	Net worth (a value)
Measure	Dollars repricing in a horizon	Years of duration, adjusted by L/A
Neutral value	0 (ISR 1)	0
Positive gap, rates rise	NIM **rises**	Net worth **falls**
Rising-rate forecast	Go positive: more ISA, less ISL	Go negative: reduce D_A, increase D_L
Dollar effect	$\Delta\mathrm{NII} \approx \Delta i \times$ cumulative gap	$\Delta\mathrm{NW} = -\text{gap} \times A \times \Delta i/(1+i)$

Exam traps

- **Polarity of the gap (the map's flag).** Positive leverage-adjusted gap: rates up, net worth down. Negative: rates up, net worth up. Read the sign only after the leverage adjustment: $D_A < D_L$ can still give a positive gap.
- **Leverage dropped or misplaced.** $D_A - D_L$ instead of $D_A - D_L(L/A)$; L/A applied to D_A; multiplying instead of dividing by L/A in the immunisation reverse-solve; setting $D_L = D_A$.
- **The $(1+i)$ divisor.** Dropped, or taken at the new rate: 10% instead of 8% gives -3.28 instead of -3.34.
- **The second minus sign.** $\Delta\mathrm{NW} = \Delta A - \Delta L$, and ΔL is itself negative when rates rise: adding the legs gives -30.52.
- **Intermediate results.** The asset leg (-16.93), the liability leg (-13.59) and the unadjusted gap (0.378) are all offered against -3.34 and $+0.60$.
- **Sibling swap with the IS gap.** A positive IS gap gains NII when rates rise; a positive duration gap loses net worth. A rising-rate forecast calls for a positive IS gap but a negative duration gap.
- **Red herrings.** An ISR or NIM figure in a net-worth question; the discount rate in an immunisation question; equity given in place of total assets $(A = L + E)$.

Practice questions: LTR-17 c

Question LTR-17 c.1

Bank Q581 • GARP book Ch.18 Problem 18 (rebuilt)

A treasury analyst at Carter National Bank notes that the bank holds USD 15 million of government bonds with a duration of 12 years. If interest rates suddenly rise from 6% to 7%, which of the following is closest to the percentage change in the bonds' market price?

A. -12.00%

B. -11.32%

C. -11.21%

D. $+11.32\%$

Question LTR-17 c.2

Bank Q585 • AnalystPrep Q.5391

Thomas Peters, the Chief Risk Officer at ABC Bank, is assessing the bank's net worth sensitivity to a potential interest rate change. By conducting duration analysis, Mr. Peters evaluates the impact of a 200-basis point change in interest rates on the bank's net worth. Which of the following statements accurately represents the conclusion made by the CRO?

A. A non-prepayable loan has a lower duration compared to a prepayable mortgage loan, making it less sensitive to overall interest rate changes.

B. Assets and liabilities exhibit higher sensitivity to changes in interest rates when the coupon is lower, the duration is higher, or the overall level of interest rates is lower.

C. The percentage change in the value of an asset or liability caused by a 200-basis point change in interest rates is directly related to its duration and remains unaffected by the corresponding percentage change in interest rates.

D. Paying a cash dividend of USD 50 million using cash will have no impact on the overall asset duration of the bank.

Question LTR-17 c.3

Bank Q579 • GARP book Ch.18 Problem 16 (rebuilt)

An ALM analyst at Blue Moon National Bank compiles the following average durations and amounts:

Item	Average duration (years)	Amount (USD million)
Investment-grade bonds	15.00	65.00
Commercial loans	3.00	400.00
Consumer loans	7.00	250.00
Deposits	1.25	600.00
Nondeposit borrowings	0.50	50.00

The first three items are the bank's assets and the last two its liabilities. Which of the following is closest to Blue Moon's leverage-adjusted duration gap?

A. 4.18 years

B. 4.30 years

C. 4.41 years

D. 7.54 years

Question LTR-17 c.4

Workbook A16.2 • S1, S5 • Aug-2026 • Map #24

A treasury analyst at an Indian private-sector bank is reporting the duration gap to the board risk committee. The bank's assets have a dollar-weighted duration of 3.6 years and its liabilities 3.8 years. Total assets are INR 2,000 billion and total liabilities INR 1,760 billion. A director remarks that because liabilities are longer than assets, the bank gains if rates rise. Which of the following correctly states the leverage-adjusted duration gap and the effect of a rise in rates on net worth?

A. -0.20 years; net worth rises when rates rise

B. $+0.26$ years; net worth falls when rates rise

C. $+0.26$ years; net worth rises when rates rise

D. -0.63 years; net worth rises when rates rise

Question LTR-17 c.5

Bank Q591 • Mock 1 Q55

For ABC Bank, the duration of its assets is 6.3 years and the duration of its liabilities is 9.2 years. Total equity is USD 1.2 billion and total liabilities are USD 2.1 billion. The economic consensus is that interest rates will decrease. Which statement regarding the bank's leverage-adjusted duration gap (LADG) and net worth is correct?

A. ABC Bank has a negative LADG; its net worth will decrease.

B. ABC Bank has a negative LADG; its net worth will increase.

C. ABC Bank has a positive LADG; its net worth will decrease.

D. ABC Bank has a positive LADG; its net worth will increase.

Question LTR-17 c.6

Workbook A16.1 • S2, S5 • Aug-2026 • Map #24

An ALM analyst at a eurozone savings bank is preparing a net-worth sensitivity for the asset-liability committee. Total assets are EUR 500 million with a dollar-weighted average duration of 4.2 years; total liabilities are EUR 450 million with an average duration of 1.8 years. All rates are currently 6% and are expected to rise by 150 basis points. The bank's interest-sensitivity ratio for the next 90 days is 0.92. Which of the following is closest to the change in the market value of the bank's net worth?

A. Decrease of EUR 29.72 million

B. Decrease of EUR 19.35 million

C. Decrease of EUR 18.25 million

D. Increase of EUR 18.25 million

Question LTR-17 c.7

Bank Q590 • GARP PE2 Q23 • Mock 4 Q24

The treasurer of a US bank is concerned about potential interest rate increases by the Federal Reserve and their impact on the bank's net worth. The treasurer asks a manager to add a scenario in which the Fed raises rates by 200 bps and to perform duration analysis. Balance sheet data:

Item	Amount (USD million)	Duration (years)
Cash	400	0
Federal funds loans	400	1.0
Government securities and mortgages	600	5.0
Loans and leases	1,100	3.0
Total assets	**2,500**	
Interest-bearing deposits (marketable)	1,000	0.5
Other borrowings	1,200	4.0
Total liabilities	**2,200**	

Assuming the current level of interest rates is 2%, which statement about this scenario is correct?

A. A 200-bps increase in interest rates will cause the bank's net worth to decrease by about USD 27.5 million.

B. A 200-bps increase in interest rates will cause the bank's net worth to decrease by about USD 13.3 million.

C. Compared to the bank's other balance sheet items, interest-bearing deposits will experience the smallest change in value given a 200-bps increase in interest rates.

D. Using USD 200 million of cash to pay off USD 200 million of other borrowings will cause the bank's net worth to increase in this scenario.

Question LTR-17 c.8

Workbook A16.4 • S3 • Map #24

The CFO of a Japanese regional bank wants to immunise net worth against parallel rate changes. Total assets are JPY 600 billion with an average duration of 2.8 years; total liabilities are JPY 540 billion. The current discount rate is 7%. Which of the following is closest to the average liability duration the bank must target?

A. 2.52 years

B. 2.80 years

C. 3.11 years

D. 3.33 years

Question LTR-17 c.9 Bank Q592 • Schweser Module Quiz 82.2 Q1 (rebuilt)

Bloomfield Bank has a positive leverage-adjusted duration gap and a negative interest-sensitive (IS) gap. A risk manager makes two statements about a rise in market interest rates.
Statement 1: The bank's net worth (capital) will increase.
Statement 2: The bank's net interest margin will increase.
Which is most correct?

A. Both statements are correct.

B. Both statements are incorrect.

C. Statement 1 is correct; Statement 2 is incorrect.

D. Statement 1 is incorrect; Statement 2 is correct.

Solutions • LTR-17 c

Solution LTR-17 c.1

Answer: B

Bank Q581 • GARP book Ch.18 Problem 18

Step 1 – Relative rate change at the original rate

$$\frac{\Delta i}{1+i} = \frac{0.01}{1.06} = 0.009434$$

Step 2 – Percentage price change

$$\frac{\Delta P}{P} \approx -12 \times 0.009434 = -11.32\%$$

$\Rightarrow$ The bonds lose about $0.1132 \times 15 =$ USD 1.70 million of value.

Interpretation

Price risk is proportional to duration: a 12-year duration turns a 1 point rise into an 11% loss. The USD 15 million is not needed for the percentage. Convexity makes the true loss slightly smaller than the duration estimate for a move this size.

Why the other options are wrong

A is incorrect. Dropped the $(1+i)$ divisor: $-12 \times 0.01 = -12.00\%$.
C is incorrect. Divided by one plus the *new* rate: $-12 \times 0.01/1.07 = -11.21\%$.
D is incorrect. Sign error: prices and rates move in opposite directions.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. • **Reference:** Rose and Hudgins, Ch.7, 18.5 The Concept of Duration as a Risk-Management Tool, eq. (18.18)

Solution LTR-17 c.2

Answer: B

Bank Q585 • AnalystPrep Q.5391

Step 1 – Duration

By eq. (18.18) the percentage price change is proportional to duration: higher duration, higher sensitivity.

Step 2 – Coupon and rate level (convexity)

Holding maturity and yield constant, a lower coupon means longer duration and more convexity; and a given rate change moves prices more when rates are low than when they are high. Both raise sensitivity, so B is correct.

Interpretation

These are the convexity facts in GARP's duration section. They matter for net worth because a bank's assets and liabilities differ in coupon and duration, so a 200 basis point move hits the two sides differently.

Why the other options are wrong

A is incorrect. Reversed: prepayments return principal earlier, so a prepayable mortgage has the *shorter* duration and the non-prepayable loan the longer one.
C is incorrect. $\Delta P/P \approx -D\,\Delta i/(1+i)$ depends on the size of the rate change as well as on duration.
D is incorrect. Cash has zero duration. Paying it out removes a zero-duration asset and *raises* the dollar-weighted asset duration (for example, USD 1,000 million of assets with dollar duration 3,000, of which 100 is cash: paying out 50 of cash moves D_A from 3.00 to $3{,}000/950 = 3.16$ years).

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. • **Reference:** Rose and Hudgins, Ch.7, 18.5, Price Sensitivity to Changes in Interest Rates and Duration; Convexity and Duration

Solution LTR-17 c.3

Answer: C

Bank Q579 • GARP book Ch.18 Problem 16

Step 1 – Dollar-weighted asset duration

$$A = 65 + 400 + 250 = 715, \qquad D_A = \frac{15(65) + 3(400) + 7(250)}{715} = \frac{3{,}925}{715} = 5.490 \text{ years}$$

Step 2 – Dollar-weighted liability duration

$$L = 600 + 50 = 650, \qquad D_L = \frac{1.25(600) + 0.5(50)}{650} = \frac{775}{650} = 1.192 \text{ years}$$

Step 3 – Leverage-adjusted gap

$$5.490 - 1.192 \times \frac{650}{715} = 5.490 - 1.084 = +4.41 \text{ years}$$

$\Rightarrow$ A large positive gap: net worth falls sharply if rates rise.

Interpretation

Long consumer loans and bonds are funded with short deposits. Equity is $715 - 650 =$ USD 65 million, and a 1 point rise from, say, 6% would cut net worth by about $4.41 \times 715 \times 0.01/1.06 =$ USD 29.7 million, almost half of it.

Why the other options are wrong

A is incorrect. Inverted the leverage ratio: $5.490 - 1.192 \times 715/650 = 4.18$.
B is incorrect. Unadjusted gap: $5.490 - 1.192 = 4.30$.
D is incorrect. Equal-weighted the item durations ($D_A = 8.33$, $D_L = 0.875$) instead of dollar-weighting: $8.33 - 0.875 \times 0.909 = 7.54$.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. • **Reference:** Rose and Hudgins, Ch.7, 18.6 Using Duration to Hedge Against Interest Rate Risk, eq. (18.21), (18.24)

Solution LTR-17 c.4

Answer: B

Workbook A16.2 • S1, S5 • Aug-2026

Step 1 – Leverage ratio

$$\frac{L}{A} = \frac{1{,}760}{2{,}000} = 0.88$$

Step 2 – Leverage-adjusted liability duration

$$D_L \times \frac{L}{A} = 3.8 \times 0.88 = 3.344 \text{ years}$$

$\Rightarrow$ Liabilities are a smaller dollar base, so their duration counts for only 88% against assets.

Step 3 – Leverage-adjusted duration gap

$$3.6 - 3.344 = +0.256 \approx +0.26 \text{ years}$$

Step 4 – Direction

Positive gap: a parallel rise in rates lowers asset values by more than liability values, so net worth falls.

Interpretation

The director's reading uses the unadjusted gap $3.6 - 3.8 = -0.20$. After the leverage adjustment the sign reverses: the bank is exposed to rising rates. To immunise it would need $D_L = 3.6 \times 2{,}000/1{,}760 = 4.09$ years.

Why the other options are wrong

A is incorrect. Dropped the leverage adjustment, $3.6 - 3.8 = -0.20$, and read the direction off that wrong sign.

C is incorrect. Right gap, polarity flip on the direction: a positive leverage-adjusted gap loses net worth when rates rise.

D is incorrect. Applied the leverage ratio to asset duration instead of liability duration: $3.6 \times 0.88 - 3.8 = -0.63$.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. • **Reference:** Rose and Hudgins, Ch.7, 18.6, eq. (18.21) and the table of net worth changes by gap sign

Solution LTR-17 c.5 **Answer: D**

Bank Q591 • Mock 1 Q55

Step 1 – Total assets from liabilities and equity

$$A = L + E = 2.1 + 1.2 = \text{USD } 3.3 \text{ billion}, \qquad \frac{L}{A} = 0.636$$

Step 2 – Leverage-adjusted gap

$$6.3 - 9.2 \times \frac{2.1}{3.3} = 6.3 - 5.855 = +0.445 \text{ years}$$

Step 3 – Direction for a rate fall

$$\Delta \text{NW} = -0.445 \times A \times \frac{\Delta i}{1+i} > 0 \quad \text{when } \Delta i < 0$$

$\Rightarrow$ Positive gap and falling rates: net worth increases.

Interpretation

Another sign reversal: liabilities are much longer than assets (9.2 against 6.3 years), but they fund only 64% of the balance sheet, and $6.3/9.2 = 0.685 > 0.636$. Equity is given in place of total assets, so A must be rebuilt first.

Why the other options are wrong

A is incorrect. Used the unadjusted gap $6.3 - 9.2 = -2.9$; a negative gap does lose when rates fall, but the true gap is positive.

B is incorrect. Internally inconsistent: a negative gap would lose net worth when rates fall; the gap is in any case positive.

C is incorrect. Right sign of gap, wrong direction: a positive gap gains when rates fall.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. • **Reference:** Rose and Hudgins, Ch.7, 18.6, eq. (18.21) and the table of net worth changes by gap sign

Solution LTR-17 c.6

Answer: C

Workbook A16.1 • S2, S5 • Aug-2026

Step 1 – Relative rate change

$$\frac{\Delta i}{1+i} = \frac{0.015}{1.06} = 0.014151$$

Step 2 – Change in asset value

$$-D_A \times \frac{\Delta i}{1+i} \times A = -4.2 \times 0.014151 \times 500 = -\text{EUR } 29.717 \text{ million}$$

$\Rightarrow$ Long-duration assets lose almost EUR 30 million of market value.

Step 3 – Change in liability value

$$-D_L \times \frac{\Delta i}{1+i} \times L = -1.8 \times 0.014151 \times 450 = -\text{EUR } 11.462 \text{ million}$$

$\Rightarrow$ Liabilities also fall in value, which is a gain to the owners.

Step 4 – Change in net worth

$$\Delta\text{NW} = -29.717 - (-11.462) = -\text{EUR } 18.25 \text{ million}$$

Step 5 – Check with the leverage-adjusted gap

$$4.2 - 1.8 \times \frac{450}{500} = 2.58 \text{ years}, \qquad -2.58 \times 0.014151 \times 500 = -18.25$$

Interpretation

A positive leverage-adjusted duration gap of 2.58 years means rising rates destroy net worth. The ALCO should shorten asset duration, lengthen liability duration, or hedge. The 90-day ISR concerns NIM, not net worth, and is irrelevant here.

Why the other options are wrong

A is incorrect. Dropped the liability side (asset value change only): 29.72.
B is incorrect. Omitted the $(1+i)$ divisor: $(4.2 \times 500 - 1.8 \times 450) \times 0.015 = 19.35$.
D is incorrect. Polarity flip: right size, wrong direction. A positive gap loses net worth when rates rise; this is the IS gap sign rule carried over to net worth.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. • **Reference:** Rose and Hudgins, Ch.7, 18.6 Using Duration to Hedge Against Interest Rate Risk, eq. (18.23)

Solution LTR-17 c.7

Answer: A

Bank Q590 • GARP PE2 Q23

Step 1 – Dollar-weighted durations

$$D_A = \frac{0(400) + 1(400) + 5(600) + 3(1{,}100)}{2{,}500} = \frac{6{,}700}{2{,}500} = 2.68$$

$$D_L = \frac{0.5(1{,}000) + 4(1{,}200)}{2{,}200} = \frac{5{,}300}{2{,}200} = 2.409$$

Step 2 – Leverage-adjusted gap

$$2.68 - 2.409 \times \frac{2{,}200}{2{,}500} = 2.68 - 2.12 = +0.56 \text{ years}$$

Step 3 – Change in net worth

$$\Delta \text{NW} = -0.56 \times 2{,}500 \times \frac{0.02}{1.02} = -\text{USD } 27.45 \text{ million} \approx -27.5$$

$\Rightarrow$ Equivalently, assets lose 131.4 and liabilities lose 103.9; the owners bear the 27.5 difference.

Interpretation

Checking C and D needs the item-by-item changes: cash 0, fed funds loans -7.8, deposits -9.8, government securities -58.8, loans -64.7, other borrowings -94.1. For D, after the paydown the asset dollar duration is still 6,700 (cash has none) but the liabilities' falls to $0.5(1{,}000) + 4(1{,}000) = 4{,}500$, so the loss grows to $-(6{,}700 - 4{,}500) \times 0.02/1.02 = -43.1$.

Why the other options are wrong

B is incorrect. Unadjusted gap: $(2.68 - 2.409) \times 2{,}500 \times 0.02/1.02 = 13.3$, ignoring that liabilities fund only 88% of assets.

C is incorrect. Cash (duration 0) has the smallest change, and fed funds loans (-7.8) also change less than deposits (-9.8).

D is incorrect. Retiring long-duration borrowings with zero-duration cash removes a liability whose fall in value was cushioning net worth; the loss grows to about 43.1 rather than net worth increasing.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. • **Reference:** Rose and Hudgins, Ch.7, 18.6, eq. (18.21), (18.23)

Solution LTR-17 c.8 **Answer: C**

Workbook A16.4 • S3

Step 1 – Zero leverage-adjusted gap condition

$$0 = D_A - D_L \times \frac{L}{A} \quad \Longrightarrow \quad D_L = D_A \times \frac{A}{L}$$

Step 2 – Solve

$$D_L = 2.8 \times \frac{600}{540} = 2.8 \times 1.1111 = 3.11 \text{ years}$$

$\Rightarrow$ Liabilities must be longer than assets because their smaller base must produce the same yen change in value.

Step 3 – Check

$$2.8 - 3.11 \times \frac{540}{600} = 2.8 - 2.8 = 0$$

Interpretation

With $D_L = 3.11$ years net worth is immunised: asset and liability value changes offset for any small parallel shift. The 7% discount rate does not affect the target.

Why the other options are wrong

A is incorrect. Multiplied by L/A instead of dividing: $2.8 \times 0.9 = 2.52$.

B is incorrect. Dropped the leverage adjustment, setting $D_L = D_A = 2.80$.

D is incorrect. Scaled by one plus the discount rate, an irrelevant input: $3.11 \times 1.07 = 3.33$.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. • **Reference:** Rose and Hudgins, Ch.7, 18.6, eq. (18.21) and the zero-gap example (3.324 years)

Solution LTR-17 c.9 — Answer: B

Bank Q592 • Schweser Module Quiz 82.2 Q1

Step 1 – Statement 1 (net worth)

Positive leverage-adjusted gap: $\Delta\text{NW} = -\text{gap} \times A \times \Delta i/(1+i) < 0$ when rates rise. Statement 1 is incorrect.

Step 2 – Statement 2 (NIM)

Negative IS gap (liability sensitive): funding costs reprice up faster than asset yields, so NII and NIM fall. Statement 2 is incorrect.

Interpretation

Both of Bloomfield's gaps point the same way: rising rates hurt net worth *and* the margin, the position of a bank that borrows short and lends long. To profit from a rising-rate forecast on both targets it would need the opposite signs on the two tools: a *positive* IS gap and a *negative* duration gap.

Why the other options are wrong

A is incorrect. Treats both gaps as benefiting from rising rates, which holds for a negative duration gap and a positive IS gap.

C is incorrect. Reverses the duration-gap effect on net worth.

D is incorrect. Reverses the IS-gap effect on NIM.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. • **Reference:** Rose and Hudgins, Ch.7, 18.4 and 18.6, the tables of NIM and net worth changes by gap sign

LIQUIDITY AND TREASURY RISK • PART

P2

Second priority

Tier B topics, or LOs Vardeez flags inside a Very Important chapter. 14 LOs from 6 readings.

LTR-3 • Early Warning Indicators • Vardeez: VERY IMPORTANT

LTR-3 a

P2 CONCEPT

Evaluate the characteristics of sound Early Warning Indicators (EWI) measures.

Workbook: D7 Early Warning Indicators and PIT vs TTC • Tier D

WHAT IS TESTED HERE

Why P2. Tier D workbook topic D7 (Map #45, 2 of 16 recall sources), reported as "E.W.I.S." in the same list as point-in-time versus through-the-cycle ratings. The PIT/TTC half belongs to Credit Risk (CR-6 b) and is not used here. Vardeez rates the reading VERY IMPORTANT: "Full chapter is important."

Vardeez: no LO-specific pointer. His notes build this LO on the M.E.R.I.T. framework and on two properties of a good measure: leading and sharp.

Reported exam shapes

- **S1** Characteristics of sound EWIs: leading, sharp, calibrated thresholds (workbook D7.1).

What the questions look like: statement questions, no numbers. A dashboard or a list of draft indicators is described and you pick the recommendation or statement that matches sound practice. The wrong options flip leading and lagging, treat sharpness as accuracy, drop the internal side or the stressed side, or push a threshold to an extreme. Workbook D7.1 also tests escalation, which LTR-3 c applies.

1. What an EWI is for

One of the critical aspects of a bank's liquidity risk management (LRM) is first *devising* and then *monitoring* a set of indicators that let the risk identification process spot new or increasing vulnerabilities. Negative trends in these indicators may warrant an assessment and a response by management. EWIs can be qualitative or quantitative (BCBS 2008; the supervisory lists are LTR-3 b).

Venkat and Baird compare EWIs to the warning lights on a car dashboard:

- **Turn signal:** an intentional risky activity, a purposeful change of direction. In LRM: running a deposit special to raise rate-sensitive liabilities.
- **Low oil light:** direct action may be needed in the near future. In LRM: the Liquidity Coverage Ratio (LCR) has dropped below a specified threshold.
- **Check engine light:** investigate, for a reason not yet known. In LRM: a dramatic increase in call centre volumes may portend a shift in the bank's liquidity position.

"Most importantly EWI triggers are used to initiate management discussions and actions that should be formally documented." This can be as simple as treasury acknowledging that a business line is running a deposit campaign, and adding to the high-quality liquid asset buffer in a cost-efficient way while staying LCR-compliant. Schweser puts the same idea as the main objective of EWIs: get management to acknowledge the situation and promote the necessary dialogue and action, with documentation.

2. The framework: M.E.R.I.T.

The EWI framework is summarised as **M.E.R.I.T.**: **M**easures, **E**scalation, **R**eporting, **I**ntegrated systems and **T**hresholds. An appropriate set of measures is the first essential building block, but the framework is "a mere academic exercise" unless it is linked to escalation. The journey from measures to escalation is made possible by timely reporting, integrated systems and data, and relevant, properly calibrated thresholds (Figure 1). This LO asks what makes each piece sound; LTR-3 c applies the chain inside the LRM process.

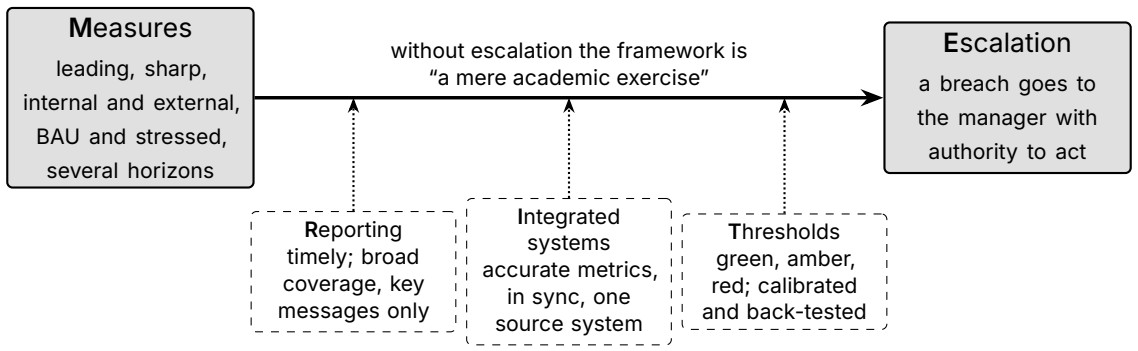


Figure 1. M.E.R.I.T. Measures (left) only reach escalation (right) through the three supports drawn as dashed boxes; the dotted arrows show each one carrying the signal along the solid arrow. A missing support breaks the chain as surely as a missing escalation plan.

3. Sound measures

Principle 5 of the BCBS Sound Principles sets out the hallmarks: "to obtain a forward looking view of liquidity risk exposures, a bank should use metrics that assess (a) the structure of the balance sheet, as well as metrics that (b) project cash flows and future liquidity positions, taking into account (c) off-balance sheet risks." The measures should span vulnerabilities across business-as-usual and stressed conditions, over various time horizons.

Leading and sharp: the forward-looking bias

A suite of EWIs with internal and external metrics should ideally be **leading** (forward looking) and **sharp** (sufficiently granular).

- **Leading.** Provides information and signals potential stress *before* an actual event; particularly important in preparing for systemic scenarios. Lagging indicators report events that have already happened, such as government-reported GDP figures. Instead, a bank can build proxies for the performance of the general economy from its own internal loan portfolio metrics.
- **Sharp.** The granularity and specificity of an indicator relative to the institution's profile: a signal that does not go unnoticed within the mass of data. A drop in overall deposit balances is an acceptable EWI; a drop in the deposits of volatile segments, such as high-net-worth customers or rate-sensitive products, brings into focus that important classes of customers are leaving the bank. Sharpness is about focus, not statistical accuracy.
- **Payoff.** Sharp, forward-looking EWIs give management more time for corrective or mitigating action, and a better understanding of risk drivers and trends than broad, lagging indicators.

Internal and external, in balance

Internal measures are customised to the bank's balance sheet and activities; external measures signal systemic changes in the economy or the market. Liquidity events can start inside the bank or be driven

by its environment: an idiosyncratic deposit run may follow the disclosure of poor performance or a systemic failure. Internal EWIs should catch internally driven stress *before it becomes public knowledge* (loan performance, planned significant accounting adjustments, operational losses). Externally, a systemic crisis such as a sovereign default or a banking-system crisis can dislocate the funding of every institution exposed to the affected markets. GARP's Figure 3.2 crosses "within the bank" and "outside the bank" with "Main Street" and "Wall Street"; its examples are deteriorating asset quality and increased debt spreads (within), asset class liquidity failure and country default (outside).

Normal and stressed environments, several horizons

- **Track in business as usual.** Deterioration in BAU metrics alerts leadership to weaknesses in the balance sheet or to difficult conditions in its markets. Monitoring that starts only once stress has arrived is too late to be early.
- **Add stressed measures and limits** to gauge whether the liquidity buffer is adequate for a stressed environment. Stress test results can also expose concentrations and risks nobody had identified.
- **Horizons.** Typical cycles are daily, weekly or monthly. "An indicator that is not updated for a period of more than a month is unlikely to satisfy the 'early' aspect of the EWI." Some EWIs are monitored intraday, as business conditions warrant or regulators require (Schweser adds hourly).

4. Reporting and integrated systems

- **Reporting** must be timely enough to give management lead time. A daily EWI dashboard is common practice; leading institutions are moving to intraday reporting of certain measures, and firms with a significant trading focus are the most likely to do so because of their exposure to external market conditions. Reports must be broad enough for wide coverage, yet specific enough to carry only the key messages.
- **Integrated systems.** Metrics are feasible and meaningful only if the bank can calculate and report them consistently. Integrated data and systems make reported metrics (a) accurate and (b) in sync with each other: deposit volumes feeding the liquidity stress tests should come from the same source system as non-stressed measures such as week-over-week declines in deposit funding. This matters more as EWI frameworks add internal indicators to the traditional market-based ones.

5. Thresholds: the stoplight and its calibration

Firms use a stoplight to show each EWI against its thresholds. **Green**: within normal bounds. **Amber**: investigate further. **Red**: a source of significant concern that may warrant an immediate response. The green-to-amber boundary is the calibration problem, and it fails in both directions (Figure 2):

- **Too wide:** movements go undetected and the metric keeps saying that conditions are healthy and unchanged.
- **Too narrow:** constant investigation desensitises management to the indicator, a "boy who cried wolf" circumstance.

How to calibrate. Use historical data to estimate the EWI's volatility. The series must be long enough to be significant, yet capture recent events so that it represents the current operating environment; if possible it should include a stress period such as 2007 to 2008; at least one year of history is used when available. Back-test the thresholds to see whether recalibration is needed, and allow a management override based on judgement: the historical volatility of the Fed funds rate is informative, but a trigger may be better set from the FOMC's target range.

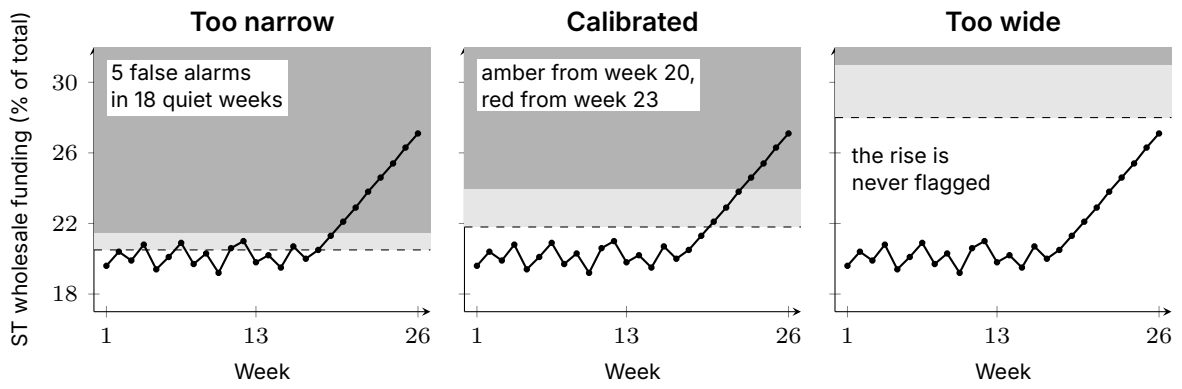


Figure 2. The same 26 weeks of one illustrative EWI (solid line with dots) under three threshold settings. White is the green zone, light grey amber, dark grey red; the dashed line is the green-to-amber boundary. Left: bands so narrow that ordinary noise turns the indicator amber five times before anything has happened. Middle: bands calibrated on the indicator's own history; the noise stays green and the genuine rise turns amber at week 20 and red at week 23. Right: bands so wide that the rise is never flagged.

	Sound EWI measure	Weak version (the distractor)
Timing	Leading: signals before the event (internal loan-portfolio proxies)	Lagging: reports what already happened (GDP releases)
Granularity	Sharp: volatile segments (high-net-worth, rate-sensitive deposits)	Broad aggregate lost in noise (total deposits only)
Source	Internal and external in balance	One side only; "external is more reliable"
Conditions	Tracked in BAU, plus stressed measures and limits	BAU only, or monitoring that starts once stress arrives
Frequency	Daily, weekly or monthly; intraday where warranted	Updated less often than monthly
Thresholds	Calibrated on a long, recent series with a stress period; back-tested; judgement override	As narrow as possible, or as wide as possible; never back-tested
Link	Tied to a defined escalation plan	Measures alone: "a mere academic exercise"

Exam traps

- **Sibling swap, leading versus lagging.** GDP releases are the book's lagging example. "Wait for confirmation before acting" is a lagging argument dressed up as prudence.
- **Sharp is not "more accurate".** Sharpness is granularity relative to the institution's profile: the signal is harder to miss, not more precise.
- **Absolute claims on sources.** Internal and external measures are both needed; neither is "more reliable". Internal ones catch stress before it is public.
- **BAU or stress only.** Track in business as usual *and* add stressed measures and limits. Tracking after a loss or only during stress is hindsight.
- **Absolute claims on thresholds.** Neither as narrow as possible (boy who cried wolf) nor so wide that movements go undetected. Recalibrate with data and back-testing; do not delete the amber band or ignore it.
- **Stale indicators.** Not updated for more than a month fails the "early" test.
- **Measures without escalation** have no effect on LRM.

Practice questions: LTR-3 a

Question LTR-3 a.1

Bank Q122 • AnalystPrep Q.3844

A liquidity risk analyst is classifying the indicators on her bank's early warning dashboard. An indicator that provides information on significant potential stress before the occurrence of an actual event is most likely known as a:

A. Sharp indicator

B. Granular indicator

C. Leading indicator

D. Lagging indicator

Question LTR-3 a.2

Bank Q124 • BT P2.T8.20.4.2

Venkat and Baird write that "The EWI framework can be summarized as M.E.R.I.T.," where the M refers to measures. Which of the following is a true characteristic of good early warning indicator (EWI) measures?

A. They include only internal measures and exclude external measures.

B. They are forward-looking and sharp, that is, granular and specific.

C. They are disassociated (de-linked) from the escalation process.

D. They are tracked in business-as-usual environments but are not affected by stressed environments.

Question LTR-3 a.3

Bank Q125 • Schweser Module Quiz 68.1 Q2

Which of the following statements regarding early warning indicators (EWIs) is correct?

A. The more granular the indicator, the more accurate it is.

B. To be effective, EWIs must be tied to a definitive escalation plan.

C. EWIs should focus on external metrics over internal metrics because the former are generally more reliable.

D. Lagging indicators are less prone to error than leading indicators; therefore, they are potentially more useful to liquidity risk managers.

Question LTR-3 a.4

Bank Q126 • AnalystPrep Q.3853

The head of treasury at a savings bank asks when tracking the bank's early warning indicator metrics is most valuable. In which of the following circumstances is it most important to track EWI metrics?

A. After an industry-wide decline in performance

B. During a business-as-usual environment

C. During periods of institution-specific stress

D. After the occurrence of a loss-causing event

Question LTR-3 a.5 Bank Q142 • BT P2.T8.25.2.3 (rebuilt; re-filed from LTR-3 c)

A bank's EWI dashboard uses green, amber and red thresholds. The Chief Risk Officer finds that the amber threshold for reliance on short-term wholesale funding is set so tightly that the indicator turns amber several times every month, and in nearly every case the follow-up shows no real problem. Which action is most appropriate?

A. Recalibrate the green-to-amber threshold using sufficiently long historical data that includes recent events, and back-test it, so that it limits false alarms without missing emerging issues.

B. Remove the amber level and keep only red, so that management is contacted only about serious problems.

C. Widen the amber threshold as far as possible, so that the indicator almost never changes colour.

D. Ignore amber signals in future, because only a red status requires any management follow-up.

Question LTR-3 a.6 Workbook D7.1 • S1, S2 • Map #45

A treasury risk manager at a mid-sized Canadian bank has been asked by the CFO to review the bank's liquidity early warning indicator dashboard and recommend improvements. The current dashboard tracks total deposit balances as a single line, uses quarterly GDP releases as its main macroeconomic signal, and has no documented escalation rules. Which of the following recommendations is consistent with sound EWI practice?

A. Set each amber threshold as narrowly as possible, so that every movement in an indicator triggers an investigation and no deterioration can go undetected.

B. Retain the quarterly GDP releases as the main macroeconomic signal, because lagging indicators confirm that stress has occurred before management commits to action.

C. Track deposit balances of high-net-worth and rate-sensitive segments separately from total deposits, so that the indicator becomes sharper and gives management more time to act.

D. Route every EWI breach to the corporate treasurer, including a signal of a potentially large cash outflow, since all EWI-triggered actions sit within the treasurer's delegated authority.

Solutions • LTR-3 a

Solution LTR-3 a.1

Answer: C

Bank Q122 • AnalystPrep Q.3844

Step 1 – Identify the property the stem describes

The stem is about *timing*: the signal arrives before the event.

Step 2 – Match it to the definition

Venkat and Baird: a leading indicator "will provide information and signal potential stress prior to the occurrence of an actual event"; lagging indicators "report on events that have already occurred".

⇒ Timing words (before, after) point to leading or lagging; detail words (granular, specific) point to sharpness.

Interpretation

Leading indicators matter most for systemic scenarios. Rather than wait for government-reported GDP, a bank can build proxies for the economy from its own loan portfolio metrics.

Why the other options are wrong

A is incorrect. Sharpness is granularity and specificity, so the signal is not lost in the mass of data. It says nothing about when the signal arrives.

B is incorrect. "Granular" is the same property as sharp ("sharp (sufficiently granular)"), not timing.

D is incorrect. Polarity flip: a lagging indicator reports after the event, the opposite of the stem.

LO: LTR-3 a: Evaluate the characteristics of sound Early Warning Indicators (EWI) measures. • **Reference:** Venkat and Baird, Liquidity Risk Management: A Practitioner's Perspective, Ch.6, Forward Looking Bias

Solution LTR-3 a.2

Answer: B

Bank Q124 • BT P2.T8.20.4.2

Step 1 – Recall what M.E.R.I.T. asks of the measures

A forward-looking view (BCBS Principle 5), leading and sharp indicators, internal and external metrics in balance, business-as-usual and stressed conditions, several horizons.

Step 2 – Test each option

Only B matches: "These measures ideally need to be leading (forward looking) and sharp (sufficiently granular)."

⇒ Leading buys time; sharp makes sure the signal is noticed.

Interpretation

The other three options each delete one of the requirements in Step 1, which is how GARP builds wrong statements on this LO.

Why the other options are wrong

A is incorrect. Absolute claim: "A suite of EWIs that includes internal and external metrics is required."

C is incorrect. Without escalation the framework is "a mere academic exercise"; measures must be linked to it.

D is incorrect. Polarity flip: institutions "should include stressed measures and limits into their EWI lists" to gauge the buffer in a stressed environment.

LO: LTR-3 a: Evaluate the characteristics of sound Early Warning Indicators (EWI) measures. • **Reference:** Venkat and Baird, Ch.6, Framework Components: M.E.R.I.T.; Measures; Forward Looking Bias

Solution LTR-3 a.3 — Answer: B

Bank Q125 • Schweser Module Quiz 68.1 Q2

Step 1 – The link to escalation

"EWIs will have a tangible and favorable impact on LRM governance only when they are linked to a clearly defined escalation plan." B is the reading's own claim.

Step 2 – Check the other three against the measure properties

Each mis-states one property of a sound measure: sharpness (A), the internal and external balance (C), and leading versus lagging (D).

⇒ An indicator that nobody is obliged to act on changes nothing.

Interpretation

Escalation is applied in LTR-3 c: the breach goes to the level of management with authority to act.

Why the other options are wrong

A is incorrect. Scope error: granularity makes an indicator sharper, harder to lose in the noise, not more accurate.

C is incorrect. Absolute claim: both kinds are needed, and internal measures are valuable precisely because they can catch stress before it becomes public. No ranking by reliability is given.

D is incorrect. Lagging indicators report events that have already happened. Lower error does not make up for lateness; leading indicators are preferred.

LO: LTR-3 a: Evaluate the characteristics of sound Early Warning Indicators (EWI) measures. • **Reference:** Venkat and Baird, Ch.6, Forward Looking Bias; Escalation

Solution LTR-3 a.4 — Answer: B

Bank Q126 • AnalystPrep Q.3853

Step 1 – What early warning needs

A signal is only worth having while there is still time to act, so the indicator must be running before trouble starts.

Step 2 – The reading's statement

"Tracking appropriate EWI metrics during a business as usual environment is essential as any deterioration in these metrics will alert the bank's leadership of weaknesses in the bank's balance sheet or of the emergence of challenging circumstances."

⇒ EWIs are watched continuously in normal times; that is what makes them early.

Interpretation

BAU tracking is not instead of stressed measures: sound practice also adds stressed measures and limits to the EWI list. The point is when monitoring has to be running, not which conditions it covers.

Why the other options are wrong

A is incorrect. After an industry-wide decline the warning window has passed; the indicator has become lagging.

C is incorrect. Monitoring that begins during institution-specific stress starts too late to warn of it.

D is incorrect. A loss-causing event is the outcome EWIs are meant to anticipate; tracking afterwards is hindsight.

LO: LTR-3 a: Evaluate the characteristics of sound Early Warning Indicators (EWI) measures. • **Reference:** Venkat and Baird, Ch.6, Environments, Both Normal and Stressed

Solution LTR-3 a.5 Answer: A

Bank Q142 • BT P2.T8.25.2.3

Step 1 – Diagnose the threshold

Several amber signals a month with nothing behind them means the band is too narrow: "constant investigation" that desensitises management, a "boy who cried wolf" circumstance.

Step 2 – Apply the calibration rule

Estimate the EWI's volatility from history: a series long enough to be significant, recent enough to reflect current conditions, ideally including a stress period, at least one year when available. Then back-test and recalibrate.

⇒ The fix is data-driven recalibration of the boundary, not removing or neutralising the amber level.

Interpretation

Calibration fails in both directions (Figure 2). The cure for false alarms must not create the opposite failure, a band so wide that genuine deterioration is missed.

Why the other options are wrong

B is incorrect. Removing amber deletes the "investigate further" stage, so signals arrive only once a problem is serious; the indicator stops being early.

C is incorrect. Over-correction: a band that "almost never" changes colour lets movements go undetected.

D is incorrect. Amber means further investigation by design; ignoring it breaks the escalation discipline instead of fixing the calibration.

LO: LTR-3 a: Evaluate the characteristics of sound Early Warning Indicators (EWI) measures. • **Reference:** Venkat and Baird, Ch.6, Thresholds

Solution LTR-3 a.6 Answer: C

Workbook D7.1 • S1, S2

Step 1 – Why C is correct

Venkat and Baird: "detecting a drop in overall deposit balances is an acceptable EWI; however, detecting drops in deposit balances of more volatile segments, such as high-net worth customers or rate-sensitive products balances, brings into focus that certain important classes of customers are leaving the bank." Sharp, forward-looking EWIs allow "more time for management to take corrective or mitigating actions".

⇒ Sharpness (granularity) and a leading bias are the two hallmarks of a sound measure; the bank's single deposit line has neither.

Interpretation

The dashboard has three weaknesses: a broad deposit line, a lagging macro signal and no escalation rules. Only the first is fixed by a correct option; B and D defend the other two.

Why the other options are wrong

A is incorrect. Absolute claim. Thresholds should not be so wide that movements go unnoticed, but institutions "should also be careful not to set boundaries that are so narrow as to force constant investigation", which desensitises management and creates a "boy who cried wolf" circumstance.

B is incorrect. Sibling swap. It credits a lagging indicator with the job of a leading one. The book names government-reported GDP as the lagging example and asks banks to build leading proxies from internal loan portfolio metrics instead.

D is incorrect. Role misassignment. Modest responses such as increasing the liquid asset buffer

usually sit within the treasurer's delegation, but an EWI that signals a potentially large cash outflow "may require the asset-liability committee (ALCO) to authorize a check on future asset growth", and the most extreme cases the contingency funding plan (LTR-3 c, Section 3).

LO: LTR-3 a: Evaluate the characteristics of sound Early Warning Indicators (EWI) measures. LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. • **Reference:** Venkat and Baird, Ch.6, Forward Looking Bias, Escalation and Thresholds

LTR-3 • Early Warning Indicators • Vardeez: VERY IMPORTANT

LTR-3 b

P2 CONCEPT

Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve).

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P2. Vardeez rates the reading VERY IMPORTANT ("Full chapter is important"). No workbook section is assigned to this LO, although D7's revision notes summarise the BCBS 2008, OCC 2012 and SR 10-6 lists.

Vardeez: no LO-specific pointer. His notes stress that there are four documents from three supervisors, because the BCBS appears twice, and that the main distractor attributes one document's content to another.

Reported exam shapes: none listed. Built from the LO wording and Vardeez's focus:

- which document says what (role misassignment across the four documents);
- the direction of an indicator (widening versus narrowing spreads, shorter versus longer liabilities);
- a bank-specific, unmitigated signal versus a peer signal or a mitigated one (GARP PE1 Q1).

What the questions look like: "which guideline focuses on", "which of the following is (not) an EWI", Roman-numeral lists, and GARP's "strongest warning signal" question.

1. Why supervisors wrote EWI guidance

The 2007 to 2008 crisis put the spotlight on liquidity risk management. The BCBS issued its Sound Principles in September 2008, followed closely by the standardised metrics, the LCR and the NSFR; this overarching framework includes the use of EWIs. National regulators have since benchmarked institutions against the Sound Principles, so expectations have risen, and the data needed for a modern LRM framework (quantity, quality, timeliness) is an expensive investment. Banks' responses range from minimising the cost of compliance to using it as a strategic tool for future decisions. The supervisory guidelines below "should act as one of the core guiding principles for EWI selection and monitoring at a bank."

2. Four documents, three supervisors

Document	What only this document says	Distractor to watch
OCC (2012), Liquidity booklet of the Comptroller's Handbook	EWIs that signal whether **embedded triggers** in certain products (callable public debt, OTC derivatives) are about to be breached, or whether contingent risks are likely to materialise; early recognition enhances readiness	"embedded options" attributed to BCBS or the Fed
BCBS (2008), Principles for Sound Liquidity Risk Management and Supervision	Indicators to identify the emergence of increased risk or vulnerabilities in the liquidity risk position or potential funding needs; EWIs **may be qualitative or quantitative**; the long general list	its items attributed to the intraday document
BCBS (2012), Monitoring Indicators for Intraday Liquidity	**Intraday only**	intraday items attributed to BCBS 2008 or OCC
SR 10-6, Interagency Policy Statement on Funding and Liquidity Risk Management (2010), the Federal Reserve	EWIs **and event triggers**, tailored to the institution's specific liquidity risk profile; early recognition lets it move into **progressive states of readiness** as the event evolves, with a framework to report or communicate inside and outside the institution	"progressive states of readiness" attributed to OCC

What each document lists

OCC (2012). Reluctance of traditional fund providers to continue funding at historic levels; pending regulatory action (formal or informal) or a CAMELS component or composite rating downgrade; widening spreads on senior and subordinated debt and credit default swaps, and stock price declines; difficulty accessing long-term debt markets; reluctance of trust managers, money managers, public entities and credit-sensitive funds providers to place funds; rising funding costs in an otherwise-stable market; counterparty resistance to off-balance-sheet products or increased margin requirements; elimination of committed credit lines by counterparties.

BCBS (2008), the full list (GARP Table 3.1). EWIs "may include but are not limited to":

- Rapid asset growth, especially when funded with potentially volatile liabilities
- Growing concentrations in assets or liabilities
- Increases in currency mismatches
- Decrease of weighted average maturity of liabilities
- Repeated incidents of positions approaching or breaching internal or regulatory limits
- Negative trends or heightened risk in a particular product line
- Significant deterioration in the bank's financial condition
- Negative publicity
- Credit rating downgrade
- Stock price declines
- Rising debt costs
- Widening debt or CDS spreads
- Rising wholesale or retail funding costs
- Counterparties requesting additional collateral or resisting new transactions
- Drop in credit lines
- Increasing retail deposit outflows
- Increasing redemptions of CDs before maturity
- Difficulty accessing longer-term funding
- Difficulty placing short-term liabilities

BCBS (2012). Daily maximum liquidity requirement; available intraday liquidity; total payments; time-specific and other critical obligations; value of customer payments made on behalf of financial

institution customers; intraday credit lines extended to financial institution customers; timing of intraday payments; intraday throughput. Intraday liquidity itself is LTR-6.

SR 10-6. Negative publicity concerning an asset class owned by the institution; increased potential for deterioration in the institution's financial condition; widening debt or credit default swap spreads; increased concerns over the funding of off-balance-sheet items.

> **Correction to the source notes.** Vardeez's BCBS-2008 list ends with "Increased concerns over the funding of off-balance-sheet items" and gives SR 10-6 only two early-warning signals. In GARP's Table 3.2 that item sits in the SR 10-6 column, which lists four signals: negative publicity concerning an asset class owned by the institution, increased potential for deterioration in its financial condition, widening debt or CDS spreads, and increased concerns over the funding of off-balance-sheet items (this matches the 2010 Interagency Policy Statement itself). It is not in the BCBS 2008 list (GARP Table 3.1). Schweser's SR 10-6 list also carries the spread item.

3. Reading the direction of an indicator

Every listed EWI describes *deterioration*. GARP's wrong options are usually the mirror image (a favourable change) or a listed signal with its condition removed:

Listed warning sign	Mirror image or mitigated version: not a warning
Widening debt or CDS spreads; rising debt or funding costs	Narrowing spreads; falling funding costs
Decrease of weighted average maturity of liabilities	Lengthening liability maturities
Rapid asset growth funded with potentially volatile liabilities	Asset growth funded with stable liabilities
Drop in credit lines; elimination of committed lines	Increase in available credit lines
The bank's own stock price declines	Peers' stock prices fall while the bank's does not
Growing concentrations in assets or liabilities	Greater diversification
Requests for additional collateral; increased margin requirements	Lower haircuts or margins
Credit rating or CAMELS downgrade	Upgrade

Exam traps

- **Role misassignment.** Intraday items (daily maximum liquidity requirement, intraday throughput) belong to BCBS 2012 and nowhere else; embedded triggers to OCC 2012; event triggers and progressive states of readiness to SR 10-6; the long general list, qualitative or quantitative, to BCBS 2008.
- **Four documents, three supervisors.** The BCBS appears twice, with completely different content.
- **Polarity flips.** Narrowing spreads, longer maturities, more credit lines and upgrades are improvements, not EWIs.
- **Dropped condition.** Asset growth is a warning when funded with *potentially volatile* liabilities; a stock price fall must be the bank's *own*.
- **Which Fed letter.** The LO's "Federal Reserve" guidance is SR 10-6. SR 12-7 is the end-to-end LRM framework expectation used in LTR-3 c.

Practice questions: LTR-3 b

Question LTR-3 b.1 Bank Q135 • Schweser Module Quiz 68.1 Q3

Which supervisory guideline for early warning indicators (EWIs) focuses on intraday liquidity monitoring indicators?

A. BCBS (2008)

B. BCBS (2012)

C. OCC (2012)

D. SR 10-6

Question LTR-3 b.2 Bank Q136 • Schweser Module Quiz 68.1 Q4

Which supervisory guideline for early warning indicators (EWIs) includes a specific provision on EWIs that deals with embedded options?

A. BCBS (2008)

B. BCBS (2012)

C. OCC (2012)

D. SR 10-6

Question LTR-3 b.3 Bank Q133 • BT Mock Q19 (rebuilt stem)

Liquidity early warning indicators can be compared to an automobile's dashboard, where a red light draws attention to something that may become a problem if not addressed. Each of the following is a credible red flag on a liquidity EWI dashboard EXCEPT:

A. Narrowing debt and CDS spreads

B. A sudden increase in debt costs

C. Sudden growth in assets accompanied by volatile liabilities

D. A rapid decline in the weighted average maturity of liabilities

Question LTR-3 b.4 Bank Q132 • AnalystPrep Q.5380 (rebuilt)

ABC Bank uses a set of early warning indicators to detect emerging risks and possible vulnerabilities in its liquidity position. Which of the following serves as an early warning indicator of a potential liquidity problem?

A. Rapid growth in leverage (assets relative to equity), funded mainly by short-term repo borrowing

B. A reduction in the collateral haircuts imposed on the bank's secured funding

C. An upward revision of the bank's credit rating

D. Increased diversification of the bank's assets

Question LTR-3 b.5 Bank Q128 • AnalystPrep Q.3848 (rebuilt)

A risk manager is compiling early warning indicators for a bank's liquidity risk dashboard using regulatory guidance (OCC 2012 and BCBS 2008). Which of the following are recognised EWIs of deteriorating liquidity?

I. A falling stock price of the bank

II. Higher margin requirements imposed on the bank

III. A longer average maturity of the bank's liabilities

IV. Frequent breaches or near breaches of internal limits

A. I and II only

B. I, II and IV only

C. II, III and IV only

D. I, II, III and IV

Question LTR-3 b.6 Bank Q134 • GARP PE1 Q1 • Mock 3 Q79

A risk consultant is assessing a small bank's liquidity risk profile. While reviewing a presentation produced by the bank, the consultant comes across a list of early warning indicators used to signal potentially heightened liquidity risk. Which of the following trends should the consultant consider as the strongest warning signal for potential liquidity risk at the bank?

A. A decrease in the stock price of some of the bank's peers but not in the stock price of the bank itself

B. An increase in available credit lines received from other financial institutions

C. Widening spreads on the bank's issued debt and credit default swaps

D. Significant asset growth funded by an increase in stable liabilities

Question LTR-3 b.7 New • built from GARP Ch.3, Table 3.2

A compliance officer at a US bank holding company is mapping each item on the bank's liquidity EWI dashboard to the supervisory document that recommends it. Which of the following pairings is correct?

A. BCBS (2008) Sound Principles: the daily maximum liquidity requirement and intraday throughput.

B. OCC (2012) Comptroller's Handbook: negative publicity concerning an asset class owned by the institution and concerns over the funding of off-balance-sheet items.

C. BCBS (2012) intraday indicators: rapid asset growth funded with potentially volatile liabilities and a decrease in the weighted average maturity of liabilities.

D. SR 10-6 Interagency Policy Statement: EWIs and event triggers tailored to the institution's liquidity risk profile, so that it can move into progressive states of readiness as an event evolves.

Solutions • LTR-3 b

Solution LTR-3 b.1 — Answer: B

Bank Q135 • Schweser Module Quiz 68.1 Q3

Step 1 – Recall each document's focus

OCC 2012: embedded triggers and advance notice. BCBS 2008: the general deterioration list. BCBS 2012: intraday indicators. SR 10-6: EWIs and event triggers tailored to the profile.

Step 2 – Match "intraday"

"Monitoring Indicators for Intraday Liquidity" (BCBS 2012): daily maximum liquidity requirement, available intraday liquidity, total payments, time-specific obligations, payments for financial institution customers, intraday credit lines to them, timing of payments, intraday throughput.
$\Rightarrow$ Anything intraday is BCBS 2012 and nothing else.

Interpretation

The BCBS appears twice in the LO's guidance. Confusing its two documents is the commonest error on this LO.

Why the other options are wrong

A is incorrect. BCBS 2008 is the general list (asset growth, concentrations, mismatches, shorter liabilities, limit breaches); it has no intraday indicators.
C is incorrect. OCC 2012 is the document on embedded triggers in products such as callable debt and OTC derivatives.
D is incorrect. SR 10-6 is about EWIs and event triggers tailored to the institution and progressive states of readiness, not intraday monitoring.

LO: LTR-3 b: Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve). •
Reference: Venkat and Baird, Ch.6, Key Supervisory Guidelines, Table 3.2

Solution LTR-3 b.2 — Answer: C

Bank Q136 • Schweser Module Quiz 68.1 Q4

Step 1 – Find the embedded-options provision

OCC 2012: "A bank should have EWIs that signal whether embedded triggers in certain products (i.e., callable public debt, OTC derivatives transactions) are about to be breached, or whether contingent risks are likely to materialize."
$\Rightarrow$ Embedded triggers or options point to the OCC's Comptroller's Handbook.

Interpretation

Schweser phrases it as EWIs for securities and derivatives with embedded options, showing when the options are likely to be exercised and any contingent liabilities they create.

Why the other options are wrong

A is incorrect. BCBS 2008 covers general deterioration and funding needs; it has no embedded-option provision.
B is incorrect. BCBS 2012 is intraday only.
D is incorrect. SR 10-6 covers EWIs and event triggers tailored to the institution's profile, not embedded options.

LO: LTR-3 b: Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve). •
Reference: Venkat and Baird, Ch.6, Key Supervisory Guidelines, Table 3.2

Solution LTR-3 b.3 — Answer: A

Bank Q133 • BT Mock Q19

Step 1 – Read each item's direction

B: rising debt costs are a BCBS 2008 EWI (and the OCC's "rising funding costs"). C: rapid asset growth funded with volatile liabilities is a BCBS 2008 EWI. D: a decrease in the weighted average maturity of liabilities is a BCBS 2008 EWI.

Step 2 – Find the exception

A: the listed EWI is *widening* debt or CDS spreads. Narrowing spreads mean the market sees the bank as less risky.

$\Rightarrow$ Narrowing spreads are the mirror image of a warning sign, so A is the exception.

Interpretation

On an "EXCEPT" question, check the direction of every item before the content: the exception is often a listed indicator with its sign flipped.

Why the other options are wrong

B is incorrect. A genuine red flag: rising debt costs (BCBS 2008), rising funding costs in an otherwise-stable market (OCC 2012).

C is incorrect. A genuine red flag: "Rapid asset growth, especially when funded with potentially volatile liabilities" (BCBS 2008).

D is incorrect. A genuine red flag: "Decrease of weighted average maturity of liabilities" (BCBS 2008).

LO: LTR-3 b: Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve). • **Reference:** Venkat and Baird, Ch.6, Table 3.1 and Table 3.2

Solution LTR-3 b.4 — Answer: A

Bank Q132 • AnalystPrep Q.5380

Step 1 – Match A to the BCBS list

Fast balance-sheet growth funded by short-term repo is "rapid asset growth, especially when funded with potentially volatile liabilities", and overnight or short-dated repo also shortens the maturity of liabilities. Both are BCBS 2008 EWIs.

Step 2 – Read the direction of the rest

Lower haircuts, a rating upgrade and diversification all describe improving funding conditions.

$\Rightarrow$ Only A describes deterioration.

Interpretation

The bank's original wording, "a rapid rise in the leverage ratio", was rebuilt: under Basel III a higher leverage ratio (capital over exposure) is an improvement, so the option now states leverage as assets relative to equity.

Why the other options are wrong

B is incorrect. Polarity flip: a reduction in haircuts makes secured funding easier; rising haircuts and collateral requests are the warning.

C is incorrect. Polarity flip: an upgrade is favourable; a credit rating or CAMELS downgrade is the EWI.

D is incorrect. Polarity flip: diversification reduces concentration; "growing concentrations in assets or liabilities" is the EWI.

LO: LTR-3 b: Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve). • **Reference:** Venkat and Baird, Ch.6, Table 3.1

Solution LTR-3 b.5 — Answer: B

Bank Q128 • AnalystPrep Q.3848

Step 1 – Classify each item

I: stock price declines (OCC 2012 and BCBS 2008), an EWI. II: increased margin requirements (OCC 2012), an EWI. III: the BCBS EWI is a *decrease* in the weighted average maturity of liabilities, so a longer maturity is a mitigant. IV: repeated incidents of positions approaching or breaching limits (BCBS 2008), an EWI.

Step 2 – Pick the set

I, II and IV.
$\Rightarrow$ B.

Interpretation

Item III is the BCBS indicator with its direction reversed, the usual way a non-EWI is planted in these lists.

Why the other options are wrong

A is incorrect. Leaves out IV, an explicit BCBS 2008 EWI.
C is incorrect. Includes III (favourable) and leaves out I.
D is incorrect. Treats III as an EWI; the warning is a shorter, not longer, liability maturity.

LO: LTR-3 b: Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve). • **Reference:** Venkat and Baird, Ch.6, Table 3.1 and Table 3.2

Solution LTR-3 b.6 — Answer: C

Bank Q134 • GARP PE1 Q1 • Mock 3 Q79

Step 1 – Test each trend against the lists

A: the EWI is a decline in the bank's *own* stock price; here the bank's price has not fallen. B: more credit lines means more available funding. D: asset growth is a warning when funded with potentially volatile liabilities; stable funding mitigates it.

Step 2 – Identify the listed, bank-specific signal

Widening spreads on the bank's own senior and subordinated debt and CDS appear in OCC 2012, BCBS 2008 and SR 10-6.
$\Rightarrow$ The market is pricing more risk into the bank's own debt: the strongest signal.

Interpretation

Wider spreads also raise the bank's funding cost directly, so the signal and the damage arrive together. Each wrong option is a listed EWI with its condition removed or its direction flipped.

Why the other options are wrong

A is incorrect. Dropped condition: only peers' prices fell, so the signal is not specific to this bank.
B is incorrect. Polarity flip: the EWIs are a drop in credit lines and the elimination of committed lines by counterparties.
D is incorrect. Dropped condition: the BCBS indicator requires funding by potentially volatile liabilities.

LO: LTR-3 b: Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve). • **Reference:** Venkat and Baird, Ch.6, Table 3.1 and Table 3.2

Solution LTR-3 b.7 **Answer: D**

New • built from GARP Ch.3, Table 3.2

Step 1 – Locate each pair of items

Daily maximum liquidity requirement and intraday throughput: BCBS 2012. Negative publicity about an owned asset class and off-balance-sheet funding concerns: SR 10-6. Rapid asset growth funded with volatile liabilities and a falling weighted average maturity: BCBS 2008.

Step 2 – Check D

SR 10-6: management "should monitor for potential liquidity stress events by using early-warning indicators and event triggers", tailored to "its specific liquidity risk profile"; early recognition lets the institution "position itself into progressive states of readiness as the event evolves". $\Rightarrow$ D is the only pairing that names the right document.

Interpretation

Each wrong option is a true list item placed under the wrong document, the role-misassignment shape Vardeez warns about. Learn one discriminator per document and the shape is easy.

Why the other options are wrong

A is incorrect. These are BCBS 2012 intraday indicators, not BCBS 2008.

B is incorrect. These are SR 10-6 signals. The OCC's distinctive guidance is embedded triggers in products.

C is incorrect. These are BCBS 2008 items; the 2012 document is intraday only.

LO: LTR-3 b: Identify EWI guidelines from banking regulators and supervisors (OCC, BCBS, Federal Reserve). • **Reference:** Venkat and Baird, Ch.6, Key Supervisory Guidelines, Table 3.2

LTR-3 • Early Warning Indicators • Vardeez: VERY IMPORTANT

LTR-3 c

P2 CONCEPT

Discuss the applications of EWIs in the context of the liquidity risk management process.

Workbook: D7 Early Warning Indicators and PIT vs TTC • Tier D

WHAT IS TESTED HERE

Why P2. Tier D workbook topic D7 (Map #45, 2 of 16 recall sources), and Vardeez rates the reading VERY IMPORTANT ("Full chapter is important").
Vardeez: no LO-specific pointer. His notes place EWIs inside the SR 12-7 chain (identify, inventory, assess, measure and monitor) and warn that EWIs sit alongside stress tests and limits, never instead of them.

Reported exam shapes

- **S2** Applications of EWIs in the liquidity risk process: escalation to the right authority (also option D of workbook D7.1, worked as Question LTR-3 a.6).

What the questions look like: a scenario (a cluster of indicators turning, a light changing colour, draft guidelines, a reporting cycle) and the response that fits the process. Wrong options wait for a regulatory minimum, jump to drastic or automatic actions, send everything to one authority, or let EWIs replace stress testing.

1. Where EWIs sit in the LRM process

EWIs must align with, and be a natural extension of, the enterprise LRM framework. Federal Reserve Supervisory Letter SR 12-7 requires that framework to be **end-to-end** and to "sufficiently capture the banking organization's exposures, activities, and risks." The chain runs as follows (Figure 1):

1. **Identify** the liquidity and funding risks inherent in the institution's business activities.
2. **Memorialise** them in the firm's liquidity risk inventory.
3. **Assess** them through a process that involves the businesses, corporate treasury, the independent risk management function and other stakeholders.
4. **Measure and monitor** them through stress tests, EWIs, and a limit and response framework.

EWIs are therefore an *output* of risk identification and assessment, not a substitute for it, and they work *alongside* stress testing and limits. GARP's Figure 3.1 then carries the signal on: limits and EWI thresholds, the EWIs themselves, escalations and notifications, and finally the funding playbook.

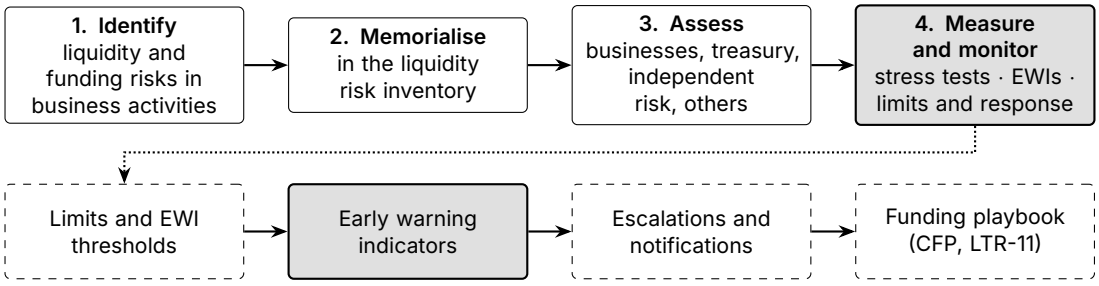


Figure 1. Top row: the end-to-end LRM chain of SR 12-7; EWIs enter only at the measure-and-monitor stage (shaded), next to stress tests and limits. Bottom row (dashed boxes, after GARP Figure 3.1): how a monitored EWI reaches action, through thresholds, escalation and the funding playbook. The dotted arrow joins the two rows.

GARP's conclusion adds that EWIs must stay current and aligned with the other core parts of the LRM framework: the risk inventory, liquidity stress testing assumptions, cash flow actuals and projections, business plans (short-term tactical and long-term strategic), the contingency funding plan, and the stakeholders (business lines, treasury, and risk as the second line of defence). Supervisors are likely to scrutinise idiosyncratic EWIs, which are tailored to the bank's own vulnerabilities, and an EWI list helps only if the data feeding it is of good quality and timely.

2. What EWIs are for: beyond compliance

- **A trigger for discussion and documented action** (LTR-3 a, Section 1), not an automatic remedy and not a risk measure in its own right.
- **Liquidity is a cost-benefit decision.** The protection of a buffer against the catastrophic impact of running short is offset by a significant drag on earnings if the buffer is too high. A minimum buffer set by risk appetite and then left on "set and forget" cannot manage the risk against a dynamic market.
- **The strategic payoff.** Increasing the buffer *before* stress extends the survival horizon cost-effectively, because a bank does not want to be caught short when the market freezes. A strong balance sheet in stress also opens market share and acquisitions that are not otherwise available. The EWI framework is the management tool that delivers this advantage.

3. Escalation to the right authority

Escalation criteria are bespoke to each bank, but an effective policy sends a limit breach "to the appropriate level of management with the authority to undertake corrective actions" (Figure 2):

- **Modest responses**, such as increasing the liquid asset buffer, usually sit within the **corporate treasurer's** delegation.
- An EWI that signals a **potentially large cash outflow** may require the **ALCO** to authorise a check on future asset growth.
- **The most extreme cases** may require management to initiate the bank's **contingency funding plan** (CFP).

EWIs have a tangible, favourable impact on LRM governance *only* when linked to a clearly defined escalation plan, and the governance of EWI-triggered escalation must be formalised, documented as part of the LRM framework, and included in the CFP.

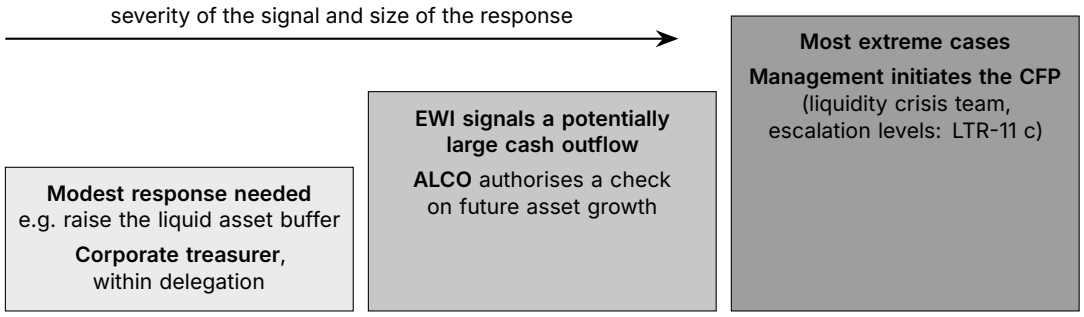


Figure 2. Escalation to the right authority. The steps darken from light grey to dark grey as the required response grows; the authority named in bold in each step is the one that can act. The stoplight colour of an EWI says how urgent a signal is; the size of the response it calls for decides who acts.

4. Reporting cadence in practice

Recap of LTR-3 a, Section 4. A daily dashboard is common practice; a trading-focused firm (a clearing member, a dealer) should report the relevant measures intraday; an indicator refreshed less often than monthly is not early. The stoplight is read as green (no further action), amber (investigate) and red (significant concern, possibly an immediate response, escalated under the plan).

5. EWIs inside the contingency funding plan

In the CFP (LTR-11 c, Section 4) the same indicators become the monitoring layer: market and business measures, external and internal, plus liquidity health measures, tied to escalation levels whose movements the liquidity crisis team approves. EWIs are most useful before or at the onset of a crisis and in the early escalation levels. This is the link between the two readings: LTR-3 explains what makes an indicator sound and how it is escalated; LTR-11 explains what the CFP does once it is.

6. Industry practice

Banks have increased the attention and resources devoted to EWI dashboards and their governance, largely because of supervisory demands (matters requiring attention, examinations), with credit also due to banks' own initiatives to improve liquidity management and risk reporting. GARP's illustrative lists (Tables 3.3 to 3.9, non-exhaustive) split into **market indicators** (agency ratings, equity indices including the BKX bank index, the VIX, money-market and rate spreads such as LIBOR/OIS, Treasury yields, FX rates, macro series such as unemployment and house prices) and **bank-specific indicators** (own CDS and ratings, CAMELS, counterparty and funding concentration, credit performance, funding variance to plan, liquidity metrics, share price relative to peers, maturity concentration, stress testing results including the LCR and NSFR, and activation of the CFP itself).

	What it does	What it is not
EWIs	Monitor actual metrics for early signs of stress; trigger discussion, documented action and escalation	A substitute for stress tests or for the risk inventory; an automatic remedy
Liquidity stress tests	Measure the effect of hypothetical but plausible stresses on the liquidity profile and buffer; expose hidden concentrations	A day-to-day monitoring tool
Limits and response framework	Fixed boundaries whose breaches are escalated	A judgement-free trigger for drastic action
Escalation plan	Sends each breach to the authority able to act: treasurer, ALCO, CFP	One authority for every signal
CFP (LTR-11)	Playbook of governance and contingent actions for severe events	The first response to every amber light

Exam traps

- **Waiting for the regulatory minimum.** EWIs are meant to act before a breach; an LCR still above 100% is no reason to sit still when several indicators are deteriorating.
- **Automatic or drastic first steps.** Triggers prompt discussion and judgement. Distressed asset sales or a first call on the central bank are not the opening response.
- **Role misassignment in escalation.** Treasurer for modest buffer moves, ALCO for an asset-growth check after a large-outflow signal, the CFP for the extreme.

- **EWIs instead of stress tests.** They sit alongside stress tests and limits, and they come after the risk inventory, not before it.
- **Slow updating.** Every two months, or even monthly with lags, fails the "early" test; trading firms need intraday.
- **Ignoring amber or red.** Amber means investigate; red means significant concern and escalation.

Practice questions: LTR-3 c

Question LTR-3 c.1

Bank Q137 • AnalystPrep Q.3849 (rebuilt)

The primary purpose of putting early warning indicators in place in a bank's liquidity risk management process is to:

A. measure precisely how risky each activity of the bank is.

B. ensure that the bank holds enough liquid assets to meet its financial obligations.

C. safeguard asset quality and ensure that liability limits are not breached.

D. prompt management discussion and necessary corrective action.

Question LTR-3 c.2

Bank Q139 • Schweser Module Quiz 68.1 Q1

Which component of the early warning indicator (EWI) framework represents the ultimate desired outcome when measures suggest a potential liquidity problem?

A. Escalation

B. Integrated systems

C. Protection

D. Reporting

Question LTR-3 c.3

Bank Q143 • AnalystPrep Q.3854 (rebuilt)

A firm reports its EWIs using a green, amber and red stoplight system. One EWI has just moved from green to red. Which interpretation and response is correct?

A. A serious problem exists; the issue requires immediate attention and remedy, escalated to the relevant management.

B. A potential problem exists; additional follow-up is required, but no urgent action.

C. The indicator is satisfactory; no further action is required.

D. The indicator is within its normal range, but its threshold should be tightened.

Question LTR-3 c.4 Bank Q144 • AnalystPrep Q.3855

Swift Investments, a US-based firm, is a clearing member at the Chicago Mercantile Exchange. It uses a set of early warning indicators to detect liquidity distress at a nascent stage. Given its substantial trading activity, which reporting schedule is most appropriate for the firm's EWI dashboard?

A. Daily

B. Weekly

C. Intraday

D. Monthly

Question LTR-3 c.5 Bank Q140 • AP Mock 1 Q42

Joe Hart is developing early warning indicators to help his bank monitor potential liquidity stress events. He has drafted the following background guidelines:

I. All EWIs must be updated every two months to detect potentially threatening events.

II. The bank should conduct a variety of short-term and protracted, bank-specific and market-wide liquidity stress tests using conservative and regularly reviewed assumptions.

III. The bank should establish a robust escalation policy so that critical decisions and transactions are handled at an appropriate level of management.

Which of these guidelines is incorrect?

A. I

B. II

C. III

D. None of the above

Question LTR-3 c.6 New • built from GARP Ch.3, Risk Identification and EWIs

The head of liquidity risk at a regional bank is drafting the section of its liquidity risk management (LRM) framework that explains where early warning indicators fit, following the end-to-end expectation of Federal Reserve SR 12-7. Which of the following descriptions is most appropriate?

A. EWIs are selected first, and the liquidity risk inventory is then built from the indicators that have breached their thresholds.

B. Risks inherent in the business activities are identified, recorded in the liquidity risk inventory and assessed; they are then measured and monitored through stress tests, EWIs, and a limit and response framework.

C. Once a sufficiently granular EWI dashboard is in place, separate liquidity stress testing becomes redundant, because the EWIs already capture stressed conditions.

D. Because EWIs are monitoring tools, their selection should be left to the independent risk function alone, without input from the businesses or corporate treasury.

Question LTR-3 c.7 Bank Q141 • BT P2.T8.25.2.2 (rebuilt)

Over the past week a large bank observed: a 12% decline in corporate deposits concentrated among large institutional clients; a 40 bp widening in its senior unsecured and CDS spreads; a fall in its LCR from 120% to 105%; higher margin demands from prime brokerage counterparties; and media speculation about its liquidity. The bank has a green, amber and red EWI framework with a defined escalation plan. Which response is most consistent with how EWIs should be applied in the liquidity risk management process?

A. Escalate the deteriorating EWIs to senior management under the escalation plan, run liquidity stress tests and review funding and counterparty positions, and document the decisions taken.

B. Take no escalation step, because the LCR of 105% is still above the 100% regulatory minimum.

C. Immediately sell non-core assets to rebuild liquidity, accepting distressed prices.

D. Request emergency liquidity assistance from the central bank before exploring any other funding measures.

Question LTR-3 c.8 New • built from GARP Ch.3, Escalation

The CRO of a mid-sized bank is writing the rules that link the bank's EWI dashboard to decision makers, and tests them on three situations:

I. A business line's deposit campaign calls for a modest increase in the liquid asset buffer.

II. An EWI signals a potentially large cash outflow from the wholesale funding base.

III. An extreme case: several indicators are red and funding markets are closing to the bank.

Which assignment of authority is most consistent with sound escalation practice?

A. I: ALCO; II: corporate treasurer, within delegation; III: contingency funding plan initiated by management.

B. I: corporate treasurer; II: corporate treasurer; III: ALCO, since the CFP is invoked only after a regulatory ratio has been breached.

C. I: corporate treasurer, within delegation; II: ALCO authorises a check on future asset growth; III: contingency funding plan initiated by management.

D. I, II and III: board of directors, since every EWI breach must be approved at the highest level of the bank.

Solutions • LTR-3 c

Solution LTR-3 c.1

Answer: D

Bank Q137 • AnalystPrep Q.3849

Step 1 – What an EWI does

"EWI triggers are used to initiate management discussions and actions that should be formally documented." Schweser: the main objective is to get management to acknowledge the situation and promote dialogue and action.

Step 2 – Eliminate the roles of other tools

EWIs signal; they do not measure risk precisely, hold the buffer, or set limits.
$\Rightarrow$ D.

Interpretation

Think of the dashboard light: it does not fix the engine, it makes the driver decide what to do. The action itself runs through the escalation plan.

Why the other options are wrong

A is incorrect. Describes a risk measure such as VaR, not a warning trigger.
B is incorrect. Holding enough liquid assets is the job of the buffer and the ratios; EWIs tell management when the buffer may need to change.
C is incorrect. Asset-quality controls and the limit framework are separate tools; EWIs alert management to deterioration.

LO: LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. • **Reference:** Venkat and Baird, Ch.6, Introduction: Dashboard and Beyond

Solution LTR-3 c.2

Answer: A

Bank Q139 • Schweser Module Quiz 68.1 Q1

Step 1 – Recall M.E.R.I.T.

Measures, Escalation, Reporting, Integrated systems, Thresholds. "The journey from measures to escalation is facilitated by timely reporting ... integrated systems and data ... and properly calibrated thresholds."

Step 2 – Identify the end point

Measures are the start; reporting, integrated systems and thresholds are the means; escalation is where the issue reaches someone who can act.
$\Rightarrow$ A.

Interpretation

Without escalation the framework is "a mere academic exercise", which is why escalation is the desired outcome.

Why the other options are wrong

B is incorrect. Integrated systems are an enabler that keeps metrics accurate and consistent, not the outcome.
C is incorrect. Not a component of M.E.R.I.T.; it is a plausible-sounding invented term.
D is incorrect. Reporting carries the signal toward escalation; it is a means, not the end.

LO: LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. • **Reference:** Venkat and Baird, Ch.6, Framework Components: M.E.R.I.T.; Escalation

Solution LTR-3 c.3 — Answer: A

Bank Q143 • AnalystPrep Q.3854

Step 1 – The three colours

Green: within normal bounds. Amber: investigate further. Red: "a source for significant concern and may warrant an immediate response".

Step 2 – Apply to a move to red

The measure has crossed the highest threshold, so the issue needs immediate attention and is escalated under the plan.

$\Rightarrow$ A.

Interpretation

Skipping amber is itself informative: either the move was sudden, or the amber band was too narrow or too wide to catch the drift (LTR-3 a, Section 5). Recalibration can be reviewed later; the red signal is acted on now.

Why the other options are wrong

B is incorrect. Describes amber, the intermediate state.

C is incorrect. Describes green.

D is incorrect. Red is a threshold breach, not the normal range; recalibration is a separate back-testing question.

LO: LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. • **Reference:** Venkat and Baird, Ch.6, Thresholds; Escalation

Solution LTR-3 c.4 — Answer: C

Bank Q144 • AnalystPrep Q.3855

Step 1 – The default cadence

"It is common practice to have the EWI dashboard reported on a daily basis."

Step 2 – Adjust for the business

"Firms with a significant trading focus are more likely to use intraday reporting as a result of their increased exposure to external market conditions." A clearing member faces margin calls and market moves within the day.

$\Rightarrow$ C.

Interpretation

Reporting has to give management lead time. For a clearing member a daily snapshot can be stale before the next margin call.

Why the other options are wrong

A is incorrect. Daily is the norm for most banks, but too slow for a trading-intensive clearing member.

B is incorrect. Weekly is too infrequent for fast-moving margin exposures.

D is incorrect. An indicator not updated for more than a month "is unlikely to satisfy the 'early' aspect"; monthly is the outer limit even for slow-moving metrics.

LO: LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. • **Reference:** Venkat and Baird, Ch.6, Reporting; Spanning Various Time Horizons

Solution LTR-3 c.5 — Answer: A

Bank Q140 • AP Mock 1 Q42

Step 1 – Guideline I

Typical cycles are daily, weekly or monthly, and "an indicator that is not updated for a period of more than a month is unlikely to satisfy the 'early' aspect." A two-month cycle fails.

Step 2 – Guidelines II and III

II: varied stress tests (short and protracted, bank-specific and market-wide) sit alongside EWIs in the measure-and-monitor stage. III: breaches must reach "the appropriate level of management with the authority to undertake corrective actions". Both are sound.
⇒ Only I is incorrect.

Interpretation

The guideline is also wrong in a second way: "all EWIs" on one fixed cycle ignores that horizons should match the balance sheet and the business, down to intraday for trading activity.

Why the other options are wrong

B is incorrect. II is consistent with using stress tests alongside EWIs in normal and stressed conditions.
C is incorrect. III restates the escalation requirement.
D is incorrect. I is incorrect, so "none" is wrong.

LO: LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. • **Reference:** Venkat and Baird, Ch.6, Spanning Various Time Horizons; Escalation

Solution LTR-3 c.6 — Answer: B

New • built from GARP Ch.3, Risk Identification and EWIs

Step 1 – The SR 12-7 chain

Identify risks inherent in the business activities, memorialise them in the liquidity risk inventory, assess them with the businesses, corporate treasury, independent risk management and other stakeholders, then measure and monitor them "through a framework consisting of stress tests, EWIs, and a limit and response framework."
⇒ B states the chain in order.

Interpretation

Two Vardeez traps meet here: EWIs come *after* the risk inventory, and they sit *alongside* stress tests rather than replacing them.

Why the other options are wrong

A is incorrect. Order reversed: the inventory comes from identifying the risks in business activities, and EWIs are chosen to monitor what it contains.
C is incorrect. Role error: EWIs monitor actual metrics; stress tests measure the effect of hypothetical stresses on the buffer and can expose concentrations no indicator shows.
D is incorrect. The assessment explicitly involves the businesses and corporate treasury as well as independent risk management.

LO: LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. • **Reference:** Venkat and Baird, Ch.6, Risk Identification and EWIs (FRB SR 12-7), Figure 3.1

Solution LTR-3 c.7 Answer: A

Bank Q141 • BT P2.T8.25.2.2

Step 1 – Read the indicators together

Internal (deposit outflows concentrated in institutional clients, a 15-point LCR fall) and external (wider spreads and CDS, margin calls, media speculation) EWIs are deteriorating at once.

Step 2 – Apply the process

EWIs are useful only when tied to escalation: the issue goes to the relevant management, remedies match the severity, and the actions are documented. Stress tests and a funding and counterparty review show whether the buffer holds under a stress that is now more plausible.
⇒ A.

Interpretation

This is early warning working as intended: action begins before any regulatory breach and before the bank is forced into fire sales.

Why the other options are wrong

B is incorrect. Treats the regulatory minimum as the trigger. EWIs are leading indicators meant to act before a breach, and five indicators are moving together.
C is incorrect. Distressed asset sales are a late-stage action; as an opening move they crystallise losses and can signal weakness (LTR-11 c, Section 3).
D is incorrect. Going first to the central bank skips escalation and the bank's own funding options and can itself prompt lenders to withdraw credit.

LO: LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. • **Reference:** Venkat and Baird, Ch.6, Escalation; Environments, Both Normal and Stressed

Solution LTR-3 c.8 Answer: C

New • built from GARP Ch.3, Escalation

Step 1 – The rule

Breaches go "to the appropriate level of management with the authority to undertake corrective actions."

Step 2 – Apply it to each situation

I: modest responses such as increasing the liquid asset buffer "may usually be done within the delegation of the corporate treasurer." II: a potentially large cash outflow "may require the asset-liability committee (ALCO) to authorize a check on future asset growth." III: the most extreme cases may require management to initiate the CFP.
⇒ Treasurer, ALCO, CFP: C.

Interpretation

Authority rises with the size of the response, not with the colour of the light alone (Figure 2).

Why the other options are wrong

A is incorrect. Swaps the first two authorities: the treasurer can raise the buffer; a large-outflow signal needs ALCO to authorise a check on asset growth.
B is incorrect. Under-escalates II and III. The CFP is for the most extreme cases and is not conditional on a regulatory breach.
D is incorrect. Absolute claim: sending every breach to the board ignores delegation and slows modest responses.

LO: LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. • **Reference:** Venkat and Baird, Ch.6, Escalation

LTR-4 • The Investment Function in Financial-Services Management • Vardeez: VERY IMPORTANT

LTR-4 c

P2 BOTH

Apply investment maturity strategies and maturity management tools based on the yield curve and duration.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P2. No workbook section reports it, but Vardeez flags it inside a VERY IMPORTANT chapter: his reading note names "learning objective 69C, apply investment maturity strategies", and his pointer is "Investment maturity strategies". The A30 revision notes summarise the five strategies and the two tools in a few lines.

Vardeez: the five strategies with his decision grid (no opinion, liquidity, return, both, money and ideas), the rate-direction traps, and the yield-curve and duration tools.

Reported exam shapes: none listed in the map. Built from the LO wording and Vardeez's focus:

- match a bank's goal or portfolio profile to one of the five maturity strategies;
- the rate expectations approach: which way to shift maturities for a rate forecast;
- reading the yield curve: its shape as a forecast, the carry trade, riding the curve, securities above or below the curve;
- duration: compute it, estimate a price change with $-D\,\Delta r/(1+r)$, and set it equal to the holding period to immunise.

What the questions look like: mostly scenarios and statements; two short duration calculations from GARP's own end-of-chapter problems. GARP PE1 Q11 (a tax swap on a flat curve) mixes this LO with LTR-4 b and is worked as Question LTR-4 b.9.

1. From what to hold to how long to hold it

LTR-4 a compared the instruments and LTR-4 b the factors that decide which to buy. Once the officer knows what kinds of securities to hold, the remaining question is how to spread them over maturities: mostly short bills and notes, only long bonds, or a mix. Rose and Hudgins describe five strategies and two tools for managing the result.

2. Five maturity strategies

- **The ladder, or spaced-maturity, policy.** Choose a maximum acceptable maturity and invest an equal share in each interval up to it: with a five-year limit, 20% in each of years 1 to 5. It does *not* maximise income, but it reduces income fluctuations, needs little expertise, and builds in flexibility: some securities are always rolling into cash for new opportunities. Popular with smaller institutions.
- **The front-end load maturity policy.** All investments short-term, for example everything three years or less from maturity (Schweser's example uses two years). The portfolio is used primarily as a **source of liquidity**, not of income; it avoids large capital losses if rates rise.
- **The back-end load maturity policy.** All investments long-term, for example only bonds in the 5 to 10-year range (Schweser: 7 to 10). The portfolio is used as a **source of income**, maximising income potential if rates fall; the institution relies heavily on money market **borrowing** for liquidity.
- **The barbell strategy.** A combination of the two: most funds in a short-term portfolio of highly

liquid securities and a long-term bond portfolio, with little in intermediate maturities. The short end provides liquidity, the long end income. Frequently used by smaller firms.

- **The rate expectations (total performance) approach.** The most aggressive: continually shift maturities with the interest rate forecast, **toward the short end when rates are expected to rise** and **toward the long end when rates are expected to fall**. Often used by the largest firms. Potential for large capital gains and for substantial losses; it needs in-depth market knowledge and carries higher transaction costs from frequent trading.

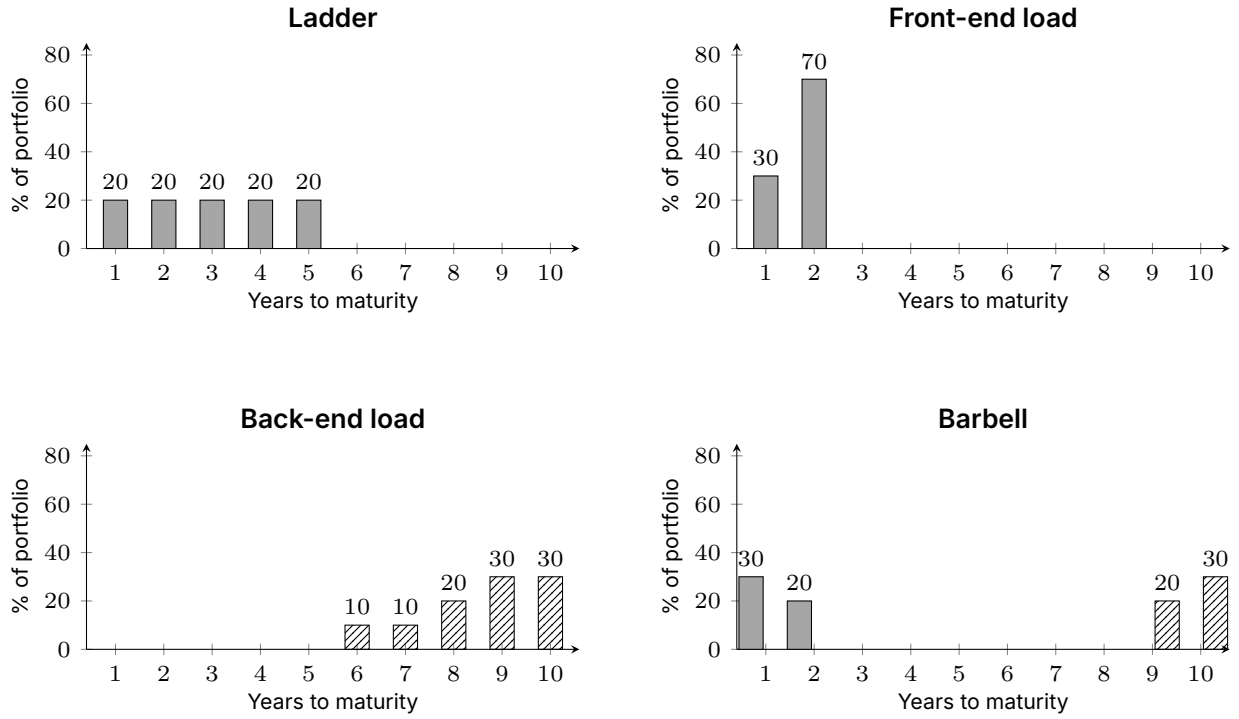


Figure 1. Four static maturity profiles with illustrative weights, after Rose and Hudgins' Exhibits 4.2 and 4.3. Grey bars: the short, liquidity end; hatched bars: the long, income end. The rate expectations approach has no fixed profile: it moves weight between the two ends as the rate forecast changes.

Correction to the source notes. In GARP's reproduction of Exhibit 4.2 the back-end load panel reads "STRATEGY: All security investments are short-term", copied from the front-end panel. The text, the chart (5 to 10-year bonds) and the stated advantage ("maximizes income potential ... if market interest rates fall") all describe an **all long-term** portfolio. Vardeez's notes already carry the corrected wording.

Vardeez's decision grid. The stem tells you what the institution has; the grid names the strategy.

What the institution has or wants	Strategy	Rose's typical user; the cost
No rate opinion, little expertise, steady income	Ladder (spaced maturity)	Smaller institutions; income not maximised
Liquidity first	Front-end load	Low income
Income first, accepts borrowing for liquidity	Back-end load	Relies on money market borrowing
Both liquidity and income	Barbell	Often smaller firms; little in the middle
Money, forecasting skill and a rate view	Rate expectations	Largest firms; big losses if wrong

When to trade. Losses on security sales cut pretax income, so managers trade when they can show the board the loss will be made up. Rose lists four conditions: (a) after-tax returns can be raised through

tax management; (b) higher yields can be locked in at the long end when rates are forecast to fall; (c) the trade improves asset quality ahead of a downturn; (d) the portfolio can move to higher-grade securities without an appreciable loss in yield. Banks are especially keen to sell securities that have risen in value when loan revenues are down.

3. Maturity management tool 1: the yield curve

The yield curve is a picture of how market rates differ across securities of different term to maturity, measured at one moment with all other rate-setting forces held fixed. It may slope up, slope down or be flat. Officers use it four ways (Figure 2).

- **Forecasting rates and the economy.** A positively sloped curve reflects the market's expectation that short-term rates will *rise*; investors shift away from longer-term securities, which would lose most. A downward-sloping curve points to falling short-term rates; the officer considers *lengthening* maturity for capital gains. Curves generally rise in expansions and fall in recessions.
- **Spotting mispriced securities.** A security whose yield lies **above** the curve has a yield that is temporarily too high and a price too low: a **buy**. One whose yield lies **below** the curve is overpriced: a sell or "don't buy".
- **Risk-return trade-off.** The slope shows the yield gained by extending maturity; a curve rising 100 basis points from 2 to 10 years offers a full point (less commissions and taxes) for switching, in exchange for more price volatility, a thinner resale market and a higher chance of a "cash out".
- **The carry trade.** Borrow at the short end (for example against the safest securities as collateral) and invest farther out. Borrowing for 30 days at 4% to buy five-year government bonds at 6% earns **carry income** of 2%; carry is greatest when the curve is steeply upward sloping, and positive only while the net yield exceeds the funding cost.
- **Riding the yield curve.** With a strong positive slope, securities approaching maturity see their yields fall and prices rise; the bank sells them for a capital gain and reinvests in longer-term securities at higher rates. It works only if the slope is steep enough to cover transaction costs and **does not fall** while the position is held.

Limits. Nobody knows exactly why the curve has its shape, the shape can change at any time, and the curve counts only clock time. It says nothing about *when* and *how much* cash each security will generate: that is what duration adds.

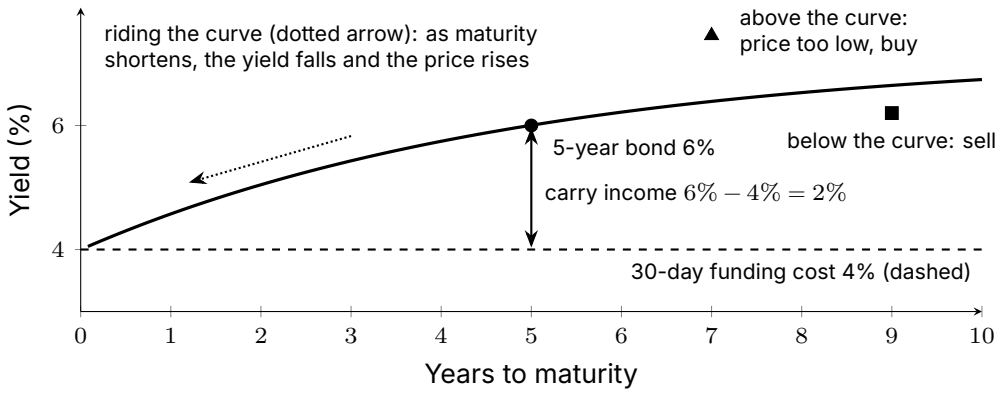


Figure 2. An upward-sloping yield curve (solid) and three of its uses: carry income against short-term funding (dashed), riding the curve (dotted arrow), and securities plotted above the curve (triangle, a buy) or below it (square, a sell). The carry and the ride both shrink if the curve flattens.

4. Maturity management tool 2: duration

Duration is a present-value-weighted measure of maturity: the average time needed for all of a security's cash flows to reach the investor, each weighted by its present value.

$$\boxed{D = \frac{\sum_{t=1}^{n} t \dfrac{CF_t}{(1+y)^t}}{\sum_{t=1}^{n} \dfrac{CF_t}{(1+y)^t}}} \qquad \boxed{\frac{\Delta P}{P} \approx -D \times \frac{\Delta r}{1+r}}$$

With m payments a year the denominator becomes $1 + r/m$. The minus sign carries the direction: rates up, prices down.

Example (GARP). A Treasury note has a duration of 4.26 years and rates rise from 10.73% to 12% (1.27 points):

$$\frac{\Delta P}{P} \approx -4.26 \times \frac{0.0127}{1.1073} = -4.89\%$$

⇒ "Almost a 5 percent decline in price"; the officer decides whether that sensitivity is acceptable.

Worked example (Rose's Concept Check 4.22). A bond pays \$100 a year for three years and \$1,000 at maturity; comparable yields are 12%.

t	CF_t	PV at 12%	$t\times$PV
1	100	89.29	89.29
2	100	79.72	159.44
3	1,100	782.96	2,348.87
		951.96	2,597.60

$$D = \frac{2{,}597.60}{951.96} = 2.73 \text{ years} \qquad \text{If rates fall to 10\%:} \quad \frac{\Delta P}{P} \approx -2.73 \times \frac{-0.02}{1.12} = +4.87\%$$

⇒ Duration is well inside the three-year maturity because the coupons arrive early. (Rose quotes a price of \$950; the cash flows are worth \$951.96 at 12%, and at the yield that gives \$950 the duration is still 2.73 years.)

Immunisation. Set

$$\boxed{\text{Duration of the security or portfolio} = \text{Length of the planned holding period}} \qquad \text{(eq. 4.9)}$$

Then the two effects of a rate change cancel: if rates rise, prices fall but coupons are reinvested at higher rates; if rates fall, reinvestment earns less but prices rise. The total return is approximately frozen whichever way rates move. Rose's example: a bank buying Treasury notes while loan demand is weak, which expects to sell them in a year to fund loans, picks notes with a duration of one year.

Formula card: LTR-4 c

Duration	$D = \sum t \cdot PV(CF_t) \,/\, \sum PV(CF_t)$, discounted at the *yield*, not the coupon rate
Price change	$\Delta P/P \approx -D\,\Delta r/(1+r)$; semiannual payer: $-D\,\Delta r/(1+r/2)$
Modified duration	$D/(1+r)$: the sensitivity, not the Macaulay duration itself
Immunisation	$D =$ planned holding period; price and reinvestment effects offset
Carry income	long-term yield − short-term borrowing rate; largest on a steep upward curve
Ladder	equal shares $1/n$ in each maturity up to the maximum ($n = 5$: 20% each)
Rate expectations	rates expected to rise: shorten; expected to fall: lengthen

Exam traps

- **Rate direction.** Expected rise: shorten (front-end protects). Expected fall: lengthen (back-end profits). An upward-sloping curve signals expected *rising* rates, not falling.
- **Ladder versus back-end.** The ladder spreads evenly over all maturities; the back-end load is the all-long strategy. The ladder smooths income and frees cash; it does not maximise income.
- **Front-end versus barbell.** Pure liquidity goal with no income goal: front-end. Liquidity *and* income: barbell.
- **Above the curve is a buy.** High yield means low price.
- **Carry and riding need slope.** Both fail on a flat curve, and riding fails if the slope falls during the ride.
- **Duration arithmetic.** Divide by $1+r$; keep the sign; 25 basis points is 0.0025. Macaulay is not modified duration; discount at the yield, include the principal.
- **Immunisation** sets duration equal to the *holding period*, not to the maturity and not to the current duration.

Practice questions: LTR-4 c

Question LTR-4 c.1

Bank Q195 • Schweser Module Quiz 69.3 Q1

A community bank's investments committee wants a maturity strategy that would most likely reduce income fluctuation in the medium and long term while generating sufficient ongoing liquidity for potential investment opportunities. Which strategy best fits?

A. Ladder

B. Barbell

C. Back-end load

D. Front-end load

Question LTR-4 c.2

New • built from GARP Ch.4, Concept Check 4.20

Bacone National Bank has structured its investment portfolio, which extends out to four-year maturities, so that it holds about USD 11 million each in one-year, two-year, three-year and four-year securities. Dunham National Bank and Trust holds USD 36 million in one- and two-year securities and about USD 30 million in 8- to 10-year maturities, with almost nothing in between. Which maturity strategies are the two banks following?

A. Bacone: ladder; Dunham: barbell.

B. Bacone: front-end load; Dunham: back-end load.

C. Bacone: ladder; Dunham: rate expectations approach.

D. Bacone: barbell; Dunham: ladder.

Question LTR-4 c.3

Bank Q190 • GARP book Ch.4 end-of-chapter Q11 (rebuilt)

A bank's economics department forecasts GDP growth of 4.5% a year for at least the next two years, which management expects to push interest rates higher. Under a rate expectations approach, what should the investments officer do?

A. Shift toward shorter-maturity or floating-rate securities and consider selling long-term fixed-rate bonds, accepting that some sales may realise losses.

B. Extend the portfolio to longer maturities to lock in current yields before they fall.

C. Make no change, since interest rate movements affect only the bank's liabilities.

D. Sell short-term securities and buy long-term bonds to maximise capital gains.

Question LTR-4 c.4

Bank Q187 • GARP book Ch.4 end-of-chapter Q8 (rebuilt)

Treasury yields by maturity are: 3-month 1.90%, 6-month 2.10%, 1-year 2.25%, 2-year 2.51%, 3-year 2.82%, 5-year 3.28%, 7-year 3.56%, 10-year 3.98%, 20-year 4.69%, 30-year 5.25%. An institution holds 75% of its investment portfolio in 7- to 30-year Treasury bonds and 25% in bills and notes maturing in under one year. Which assessment and recommendation is most appropriate?

A. The curve is upward sloping; the long-heavy portfolio earns more yield but faces large capital losses if rates rise and has limited liquidity, so management should shorten maturities, for example toward a ladder or a barbell with a shorter long end.

B. The curve is upward sloping, which signals falling rates; management should extend further into 20- to 30-year bonds to lock in capital gains.

C. The curve is downward sloping, signalling falling short-term rates; the long-heavy portfolio is well positioned and no change is needed.

D. The curve is flat, so maturity choice does not matter and the portfolio carries no interest rate risk.

Question LTR-4 c.5

New • built from GARP Ch.4, Section 4.9

An investments officer at a regional bank observes a steep, upward-sloping Treasury yield curve. Which of the following statements about using the curve is correct?

A. A security whose yield lies above the yield curve is overpriced and is a candidate for sale.

B. Carry income from borrowing short and investing farther out along the curve tends to be greatest when the curve is flat.

C. Riding the yield curve sells securities nearing maturity whose prices have risen and reinvests in longer-term securities, and it works only if the curve's slope does not fall.

D. Because the yield curve orders securities by time to maturity, it also tells the officer when and how much cash each security will generate.

Question LTR-4 c.6

Bank Q185 • GARP book Ch.4 end-of-chapter Q2 (rebuilt)

A bond has a USD 1,000 face value, a 6% coupon paid annually, 10 years to maturity and a yield to maturity of 5%. What is the bond's (Macaulay) duration?

A. 5.10 years

B. 7.52 years

C. 7.89 years

D. 8.31 years

Question LTR-4 c.7 Bank Q188 • GARP book Ch.4 end-of-chapter Q9 (rebuilt)

A bond held by a bank has a duration of 8.89 years. Market rates on comparable bonds fall from 7.50% to 7.25%. What is the approximate percentage change in the bond's value?

A. -2.07%

B. $+2.07\%$

C. $+2.22\%$

D. $+20.7\%$

Question LTR-4 c.8 Bank Q189 • GARP book Ch.4 end-of-chapter Q10 (rebuilt)

A savings institution's bond portfolio has an average duration of 4.5 years. Its investments officer wants to protect the portfolio's return against interest rate changes over a planned one-year holding period. What should the officer do?

A. Reduce the portfolio's duration to about one year.

B. Lengthen the portfolio's duration to above 4.5 years.

C. Leave the duration at 4.5 years, because duration already measures the interest rate risk.

D. Raise the duration to about 9 years so that price gains offset reinvestment losses.

Solutions • LTR-4 c

Solution LTR-4 c.1

Answer: A

Bank Q195 • Schweser Module Quiz 69.3 Q1

Step 1 – Match the two goals

Smooth income over the medium and long term, and steady liquidity.

Step 2 – Which strategy delivers both?

The ladder puts an equal share in each maturity, so income is averaged across the term structure, and one rung matures every year, freeing cash for new opportunities.

⇒ The ladder: Rose says it "has the advantage of reducing income fluctuations" and "tends to build in investment flexibility".

Interpretation

Vardeez's grid places the ladder at "no opinion": it needs no rate forecast and little expertise, which is why smaller institutions like it.

Why the other options are wrong

B is incorrect. The barbell takes its liquidity from the short end and its income from the long end; its income swings with rates at both extremes.

C is incorrect. All long maturities: little ongoing liquidity, and the bank relies on borrowing.

D is incorrect. Highly liquid, but everything reprices quickly, so income moves with short rates.

LO: LTR-4 c: Apply investment maturity strategies and maturity management tools based on the yield curve and duration. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.8 Investment Maturity Strategies

Solution LTR-4 c.2

Answer: A

New • built from GARP Ch.4

Step 1 – Bacone

Equal amounts in each one-year interval up to the four-year maximum: the definition of the ladder, or spaced-maturity, policy.

Step 2 – Dunham

Heavy at both ends (USD 36 million short, USD 30 million long) and almost nothing in the middle: the barbell.

⇒ Bacone ladders; Dunham runs a barbell.

Interpretation

Rose asks why each bank might have chosen its strategy: Bacone wants steady income with little expertise; Dunham wants liquidity from the short end and income from the long end.

Why the other options are wrong

B is incorrect. Bacone holds securities out to four years in equal shares, not only short ones; Dunham holds a large short block, so it is not back-end loaded.

C is incorrect. The rate expectations approach is a policy of shifting maturities over time; one snapshot with two ends and an empty middle is a barbell.

D is incorrect. Reverses the two profiles.

LO: LTR-4 c: Apply investment maturity strategies and maturity management tools based on the yield curve and duration. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.8, Exhibits 4.2 and 4.3; Concept Check 4.20

Solution LTR-4 c.3 — Answer: A

Bank Q190 • GARP book Ch.4 Q11

Step 1 – Translate the forecast

Strong growth, so rates are expected to rise.

Step 2 – Apply the rate expectations rule

Bond prices fall when rates rise, longer maturities the most. The approach shifts "toward the short end of the maturity spectrum when interest rates are expected to rise".

$\Rightarrow$ Shorten maturities or move to floating-rate securities; selling long fixed-rate bonds may realise losses, which tax management can partly offset.

Interpretation

Rate expectations is the most aggressive strategy; if the forecast is wrong, the bank gives up the long-end income for nothing.

Why the other options are wrong

B is incorrect. Extending to lock in yields is the response to *falling* rates.

C is incorrect. Rising rates cut the market value of the bank's bond portfolio directly.

D is incorrect. The falling-rate trade; it would magnify losses if rates rise.

LO: LTR-4 c: Apply investment maturity strategies and maturity management tools based on the yield curve and duration. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.8, The Rate Expectations Approach

Solution LTR-4 c.4 — Answer: A

Bank Q187 • GARP book Ch.4 Q8

Step 1 – Read the curve

Yields rise steadily from 1.90% at 3 months to 5.25% at 30 years, about 335 basis points: upward sloping.

Step 2 – Read the portfolio

75% in 7 to 30-year bonds means a long duration: a rate rise causes large capital losses, and only 25% sits in liquid short-term paper.

Step 3 – Recommend

A positive slope rewards extension but reflects the market's expectation of rising rates; reduce duration and add liquidity, for example by laddering or by shortening the long end of a barbell.

$\Rightarrow$ A long-heavy portfolio on an upward-sloping curve is exposed to exactly the move the curve forecasts.

Interpretation

The curve's slope is both the reward (higher yield for longer maturity) and the warning (the expected rise in short-term rates).

Why the other options are wrong

B is incorrect. Polarity flip: an upward slope signals expected rising, not falling, rates.

C is incorrect. Misreads the curve: it slopes up, and a long-heavy book is the one a rate rise hurts.

D is incorrect. Yields clearly differ by maturity, and longer bonds always carry more rate risk.

LO: LTR-4 c: Apply investment maturity strategies and maturity management tools based on the yield curve and duration. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.9, The Yield Curve, Forecasting Interest Rates and the Economy

Solution LTR-4 c.5 — Answer: C

New • built from GARP Ch.4

Step 1 – Why C is correct

Rose: the officer "looks for a situation in which some securities are soon to approach maturity and their prices have risen significantly while their yields to maturity have fallen", sells them for the gain and "reinvest[s] the proceeds of that sale in longer-term securities"; "if the riding maneuver works (i.e., the slope of the yield curve does not fall)" the bank gains current and future income.

Interpretation

Carry and riding are both bets that a steep curve stays steep; a flattening curve takes away both.

Why the other options are wrong

A is incorrect. Polarity flip: a yield above the curve is too high for its maturity, so the price is too low: a buy.

B is incorrect. Polarity flip: carry income "tends to be greatest when the yield curve has a steep upward slope".

D is incorrect. A limitation of the curve, reversed: it counts only clock time, not the timing and size of cash flows, which is why duration is needed.

LO: LTR-4 c: Apply investment maturity strategies and maturity management tools based on the yield curve and duration. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.9, Pursuing the Carry Trade; Riding the Yield Curve

Solution LTR-4 c.6 — Answer: C

Bank Q185 • GARP book Ch.4 Q2

Step 1 – Price at the 5% yield

$$P = \sum_{t=1}^{10} \frac{60}{1.05^t} + \frac{1{,}000}{1.05^{10}} = 1{,}077.22$$

Step 2 – Time-weighted present values

$$\sum_{t=1}^{10} t \times PV_t = 8{,}501.6$$

Step 3 – Duration

$$D = \frac{8{,}501.6}{1{,}077.22} = 7.89 \text{ years}$$

$\Rightarrow$ Below the 10-year maturity because coupons arrive before the principal.

Interpretation

A 1% rise in yield would cut the price by about $7.89 \times 0.01/1.05 = 7.5\%$.

Why the other options are wrong

A is incorrect. Weighted only the coupon stream and left out the principal: 5.10 years.

B is incorrect. Modified duration, $7.89/1.05 = 7.52$, the sensitivity rather than the Macaulay duration.

D is incorrect. Weighted the times by undiscounted cash flows: $(60 \times 55 + 1{,}000 \times 10)/1{,}600 = 8.31$.

LO: LTR-4 c: Apply investment maturity strategies and maturity management tools based on the yield curve and duration. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.9, Duration

Solution LTR-4 c.7 — Answer: B

Bank Q188 • GARP book Ch.4 Q9

Step 1 – Rate change

$\Delta r = 7.25\% - 7.50\% = -0.0025$; $1 + r = 1.075$.

Step 2 – Duration estimate

$$\frac{\Delta P}{P} \approx -8.89 \times \frac{-0.0025}{1.075} = +0.0207 = +2.07\%$$

$\Rightarrow$ Rates fall, so the price rises by about 2.07%.

Interpretation

The division by $1 + r$ turns Macaulay duration into the price sensitivity; for small moves the estimate is close to the exact repricing.

Why the other options are wrong

A is incorrect. Sign error: a fall in rates raises bond prices.
C is incorrect. Forgot to divide by $1 + r$: $8.89 \times 0.0025 = 2.22\%$.
D is incorrect. Decimal slip: entered 25 basis points as 0.025.

LO: LTR-4 c: Apply investment maturity strategies and maturity management tools based on the yield curve and duration. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.9, Duration (price-change relationship)

Solution LTR-4 c.8 — Answer: A

Bank Q189 • GARP book Ch.4 Q10

Step 1 – The immunisation rule

Set the portfolio's duration equal to the planned holding period (eq. 4.9).

Step 2 – Compare

The horizon is one year and the duration 4.5 years, far too long.

Step 3 – Act

Move toward shorter-maturity or higher-coupon securities until the duration is about one year.
$\Rightarrow$ Price risk and reinvestment risk then offset over the year, whichever way rates move.

Interpretation

With a duration of 4.5 years and a one-year horizon, a rate rise would cut the sale price by far more than higher reinvestment rates could recover in a year.

Why the other options are wrong

B is incorrect. Lengthening raises price sensitivity and moves further from the horizon.
C is incorrect. A 4.5-year duration immunises a 4.5-year horizon, not a one-year one.
D is incorrect. A duration above the horizon leaves the portfolio exposed to price losses if rates rise.

LO: LTR-4 c: Apply investment maturity strategies and maturity management tools based on the yield curve and duration. • **Reference:** Rose and Hudgins, Ch.10 (GARP Ch.4), 4.9, Immunization, eq. (4.9)

LTR-11 • Contingency Funding Planning • Vardeez: VERY IMPORTANT

LTR-11 a

P2 CONCEPT

Discuss the relationship between contingency funding planning and liquidity stress testing.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P2. Vardeez rates the reading VERY IMPORTANT: "contingency funding planning entire chapter is important. Refer the crash course notes and know this chapter." No workbook section is assigned to this LO, although D26 (Tier D, notes only) opens with it.
Vardeez: no LO-specific pointer. His notes reduce the LO to a two-row table, business as usual for low-impact, high-probability events and the CFP for high-impact, low-probability ones, and warn that each of the four words can be flipped on its own.

Reported exam shapes: none listed. Built from the LO wording and Vardeez's focus:

- stress test results as inputs to the CFP's governance, menu of contingent actions and decision framework;
- the impact and probability split between BAU management and the CFP;
- the explicit link through limits and escalation levels, with BAU measures as the base of the CFP's EWIs;
- CFP scenarios that match the stress tests and may extend them.

What the questions look like: "which statement is correct" about the two frameworks. Supply was thin (three bank questions), so one GARP practice-exam question is re-filed from LTR-11 c and three new questions are built from GARP Ch.11.

1. What a contingency funding plan is

Shared background for LTR-11 b and c. A contingency funding plan (CFP) provides "a structured approach for developing and implementing the institution's financial and operational strategies for effectively managing such contingent liquidity events during periods of severe market and financial stress."

- **Evolution.** Formalised CFPs have gained traction over the past decade as market disruptions have become more common. Smaller institutions typically include their CFP in their broader business continuity plans; larger ones have formal CFPs, and complex firms several, for different subsidiaries and legal entities. A 2008 European survey of 84 institutions ("EU Banks' Liquidity Stress Testing and Contingency Funding Plans") recommended enhancements.
- **US requirement.** The Federal Reserve's Enhanced Prudential Standards require US banking organisations with USD 50 billion or more in assets to establish and maintain a CFP. It must have policies and procedures for governance, roles and responsibilities, liquidity measures and triggers, the menu of contingent actions and communication protocols; it must be tailored to the institution's business and risk profile (businesses, subsidiaries, legal entities, products, geographies and FX); and it must be tested regularly.
- **Linkages.** The Fed's Comprehensive Liquidity Assessment and Review (CLAR) and daily liquidity reporting link to the CFP's governance and measures. CFPs are referenced in, and aligned to, recovery and resolution planning, because liquidity is often a key driver of failure. The CFP is the **bridge between BAU liquidity risk management and recovery and resolution planning**.

2. Companion frameworks

"A contingency funding plan (CFP) serves as a logical connection to its companion, the liquidity stress testing framework, by linking the stress test results and other related information as *inputs* to the CFP governance, menu of contingent liquidity actions, and decision framework."

Contingent liquidity events are sorted by estimated adverse impact and probability (Figure 1). Institutions manage the **low-impact, high-probability** end in business-as-usual (BAU) funding and liquidity risk management, and use the CFP for the other end, the **high-impact, low-probability** events.

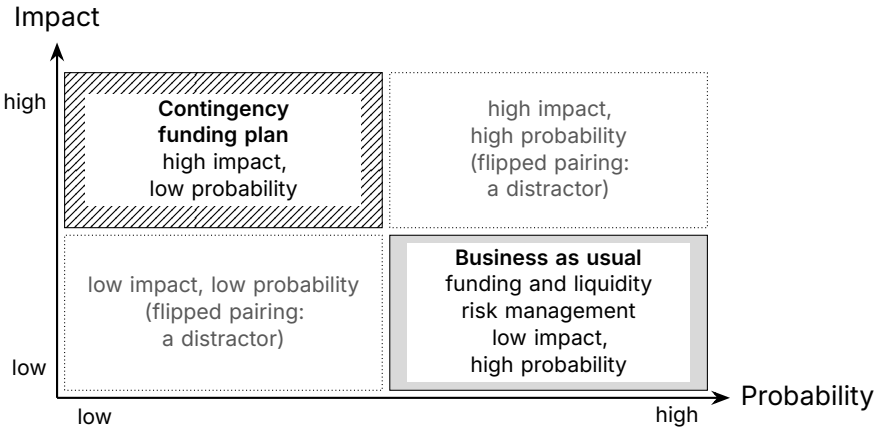


Figure 1. The two ends of the spectrum of contingent liquidity events. The hatched cell is the CFP's territory; the grey cell is business as usual. The dotted cells are the pairings GARP builds by flipping one word: each of "impact" and "probability" can be corrupted on its own, so say both words before choosing.

The two frameworks also divide the work. The stress testing framework "should ensure that effects of these stresses on the institution's liquidity profile is appropriately measured and monitored." The CFP "in turn should provide a tactical mechanism for escalating a developing crisis to management's attention and ensuring actionable responses are available." The stress test *measures*; the CFP *escalates and acts*.

3. Where the two frameworks connect

1. **Inputs.** Stress results feed the CFP's governance (the treasurer may invoke the CFP after reviewing markets, industry, institution-specific conditions *and liquidity stress testing results*), its menu of contingent actions (sized against the liquidity gap) and its decision framework.
2. **Limits and escalation levels.** "The CFP should be explicitly linked to the liquidity risk measurement framework and the liquidity stress test, in particular through its limit structure and escalation levels." The liquidity risk measures used in BAU risk management are the **foundation from which the CFP defines its EWIs** (normal-times measures, not stressed ones). Level 1 of the CFP can itself be triggered by stress test results that show a greater decline in liquidity than the risk appetite targets (LTR-11 c, Section 4).
3. **Scenarios.** CFP stress scenarios are aligned to the liquidity stress testing scenarios and to the recovery and resolution plans. The stress tests cover systemic (general market) and idiosyncratic (institution-specific) risks, market (asset) and funding liquidity, over short-term and prolonged periods. The CFP *may add its own scenarios*, outside the broader monitoring and limit structure: the Fedwire intraday debit cap is exceeded, specific counterparties fail, or Federal Home Loan Bank funding becomes unavailable. The CFP's scenario set matches the stress tests and extends them.
4. **Capital.** Capital and liquidity management were historically siloed. An integrated approach

brings a common set of EWIs and a comprehensive view of how liquidity stress results and proposed response actions affect capital positions, and vice versa, aligned with recovery and resolution, capital management and ERM.

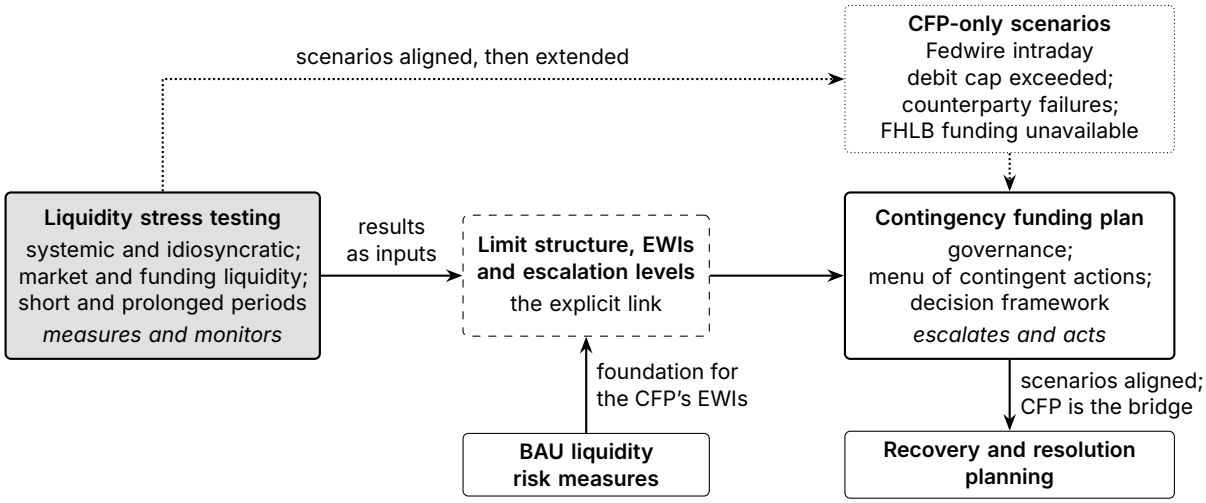


Figure 2. How the two frameworks connect. Solid arrows: stress results flow into the CFP through the limit structure and escalation levels (dashed box), whose EWIs are built on BAU measures. Dotted arrows: the CFP takes the stress test scenarios and adds its own. The CFP in turn links BAU management to recovery and resolution planning.

	Liquidity stress testing framework	Contingency funding plan
Question	How much liquidity would we lose, and when, under plausible stresses?	Who decides, what do we do, and in what order?
Events	A range of stresses, systemic and idiosyncratic	The high-impact, low-probability end; BAU handles the rest
Scenarios	The stress test set	The same set, aligned to RRP, plus its own extra scenarios
Role	Measures and monitors the effects on the liquidity profile	Tactical mechanism to escalate a developing crisis and keep actionable responses ready
Link	Stress results are inputs to CFP governance, contingent actions and decisions; the explicit link is the limit structure and escalation levels; BAU measures are the base of the CFP's EWIs	

Exam traps

- **Polarity on the four words.** CFP: high impact *and* low probability. BAU: low impact *and* high probability. Each word can be flipped on its own.
- **Superset, not a copy.** "The CFP uses exactly the stress test scenarios" drops the extension; "the CFP designs its scenarios independently" breaks the link.
- **EWI baseline.** BAU (normal-times) measures are the foundation of the CFP's EWIs, not stressed measures.
- **Stress results are inputs**, not just a buffer-sizing tool, and not "hypothetical, so ignore them": they can trigger Level 1.
- **The link mechanism** is the limit structure and escalation levels, not the communication plan or an individual ratio.

- **Coverage.** Stress tests cover systemic *and* idiosyncratic risks, market *and* funding liquidity.

Practice questions: LTR-11 a

Question LTR-11 a.1

Bank Q356 • BT Mock Q59 • BT P2.T8.20.12.1

Which of the following provides the best explicit link between a bank's contingency funding plan (CFP) and its liquidity stress testing framework?

A. The communication plan

B. The limit structure and escalation levels

C. Funding source concentration and non-core funding

D. Industry factors such as profitability trends in the financial sector

Question LTR-11 a.2

Bank Q358 • bank new question, GARP practice-exam style

The head of liquidity risk at a mid-sized bank is redesigning the bank's contingency funding plan (CFP) so that it is properly connected to the bank's liquidity stress testing framework. A colleague proposes several design principles. Which of the following principles is most consistent with the relationship between the CFP and liquidity stress testing?

A. Stress test results should be used only to size the liquidity buffer, while the CFP should be built independently from management judgement about likely crisis actions.

B. Liquidity measures used in business-as-usual risk management should be the foundation for the CFP's early warning indicators, and the two frameworks should be linked through limits and escalation levels.

C. Early warning indicators should be calibrated only from liquidity measures observed under stressed scenarios, because the CFP is used exclusively in periods of extreme stress.

D. The CFP should be activated for the low-impact, high-probability liquidity events that arise regularly, because these are the events that stress tests identify most reliably.

Question LTR-11 a.3

New • built from GARP Ch.11, Scenarios and Liquidity Gap Analysis

A bank's asset-liability committee (ALCO) is reviewing how its liquidity stress testing framework and its contingency funding plan (CFP) divide the work between them. Which statement best describes the relationship?

A. The CFP replaces the stress testing framework once a crisis starts, so its scenarios are designed independently and need not be aligned with the stress tests.

B. Liquidity stress testing covers only market (asset) liquidity, while the CFP covers funding liquidity, so the two frameworks address different risks.

C. The stress testing framework measures and monitors the effect of systemic and idiosyncratic stresses on the liquidity profile, while the CFP provides the tactical mechanism for escalating a developing crisis and ensuring actionable responses are available.

D. The CFP must use exactly the stress test scenarios and no others, because a scenario outside the limit structure would make the two frameworks inconsistent.

Question LTR-11 a.4 Bank Q373 • GARP PE1 Q71 (re-filed from LTR-11 c)

The CFO of a bank has asked for a review of the bank's contingency funding plan (CFP) and would like to ensure that key components are incorporated in alignment with best practices. Which of the following is a correct statement regarding the key components to be found in an effective CFP?

A. Liquidity stress testing scenarios are designed to focus solely on institution-specific risks and address both market (asset) liquidity and funding liquidity over short-term and prolonged stress periods.

B. Institutions should align their CFP stress scenarios to those in their liquidity stress testing framework, as well as to other frameworks such as recovery and resolution plans.

C. Contingent actions, such as maintaining investment strategies to reinvest maturing securities, should be identified in order to maximize and maintain bank profitability during stressed periods.

D. The liquidity crisis team may invoke the CFP based on a review of the markets, industry, bank-specific conditions, and liquidity stress testing results.

Question LTR-11 a.5 New • built from GARP Ch.11, Governance; Escalation Levels

The quarterly liquidity stress test of a regional bank shows that, under the combined idiosyncratic and market-wide scenario, its liquidity would fall further than the bank's risk appetite targets. No early warning indicator on deposits or funding spreads has moved yet. Under the bank's three-level contingency funding plan (CFP), which response is most consistent with best practice?

A. Take no CFP action until an actual outflow or a market indicator confirms the stress, since stress test results describe hypothetical scenarios only.

B. Move the bank directly to Level 3 and begin marketing a business unit for sale, since the stress test has already identified a shortfall.

C. Move automatically to Level 2 without review by the liquidity crisis team, since a quantitative breach of risk appetite leaves no room for management judgement.

D. Treat the result as a potential Level 1 trigger: the treasurer, in consultation with the CFO, may invoke the CFP and convene the liquidity crisis team, which considers and approves the move to elevated monitoring.

Question LTR-11 a.6 New • built from GARP Ch.11, Liquidity and Capital

The CRO of a bank holding company notes that its liquidity and capital stress tests, and the contingency plans built on them, have always been run by separate teams. Which change is most consistent with the practice described for contingency funding plans?

A. Bring them together: use a common set of EWIs and assess the effect of liquidity stress results and proposed CFP actions on capital, and of capital stress on liquidity, in line with recovery and resolution plans.

B. Keep them separate, because a liquidity crisis unfolds too quickly for contingent actions to affect capital positions.

C. Replace the liquidity stress test with the capital stress test, because a well-capitalised bank cannot face a liquidity shortfall.

D. Fold the CFP into the business continuity plan, as large and complex institutions typically do, so that one crisis document covers both.

Solutions • LTR-11 a

Solution LTR-11 a.1

Answer: B

Bank Q356 • BT Mock Q59

Step 1 – Where the reading puts the link

"The CFP should be explicitly linked to the liquidity risk measurement framework and the liquidity stress test, in particular through its limit structure and escalation levels."

Step 2 – Classify the other options

The communication plan is a design consideration of the CFP itself; funding concentration and non-core funding are liquidity health measures; sector profitability is an industry measure among the market and business EWIs.

$\Rightarrow$ B.

Interpretation

Limits are set with the stress results in view; a breach moves the institution up the escalation levels, which is how stress results reach CFP decisions. The BAU measures behind those limits are also the base of the CFP's EWIs.

Why the other options are wrong

A is incorrect. A component of the CFP (design consideration V), not the connection to stress testing.

C is incorrect. Liquidity health measures monitored under the CFP: inputs to the escalation framework, not the link itself.

D is incorrect. An industry measure among the market and business EWIs.

LO: LTR-11 a: Discuss the relationship between contingency funding planning and liquidity stress testing. • **Reference:** Venkat and Baird, Liquidity Risk Management: A Practitioner's Perspective, Ch.7 (Lai and Tuosto), Design Considerations II

Solution LTR-11 a.2

Answer: B

Bank Q358 • bank new question

Step 1 – Stress results are inputs

The CFP links "the stress test results and other related information as inputs to the CFP governance, menu of contingent liquidity actions, and decision framework".

Step 2 – The link and the EWI baseline

The link is the limit structure and escalation levels, and "the liquidity risk measures used in the institution's BAU risk management activities serve as a foundation from which the CFP defines its early warning indicators".

$\Rightarrow$ B states both.

Interpretation

The CFP is for severe events, but its indicators must pick up deterioration *before* the crisis, so they are built from normal-times measures.

Why the other options are wrong

A is incorrect. Treats the frameworks as separate. Stress results feed CFP governance and the menu of actions, not just the buffer.

C is incorrect. Inference error: because the CFP is used in stress, its EWIs must be stress-calibrated. The reading builds them from BAU measures.

D is incorrect. Polarity flip: low-impact, high-probability events are managed in BAU; the CFP

covers the high-impact, low-probability end.

LO: LTR-11 a: Discuss the relationship between contingency funding planning and liquidity stress testing. • **Reference:** Venkat and Baird, Ch.7, 11.1 Actions in a Liquidity Crisis; Design Considerations II

Solution LTR-11 a.3 — Answer: C

New • built from GARP Ch.11

Step 1 – What stress testing does

"The liquidity stress testing framework should ensure that effects of these stresses on the institution's liquidity profile is appropriately measured and monitored."

Step 2 – What the CFP does

"The CFP in turn should provide a tactical mechanism for escalating a developing crisis to management's attention and ensuring actionable responses are available."

⇒ The stress test measures; the CFP escalates and acts: C.

Interpretation

The two share their scenarios: the CFP's are aligned with the stress tests (and with recovery and resolution plans) and may go beyond them.

Why the other options are wrong

A is incorrect. Breaks the link: the CFP is the stress test's companion, and its scenarios should be aligned to the stress tests.

B is incorrect. Sibling split: the stress tests address both market (asset) liquidity and funding liquidity.

D is incorrect. Drops the extension: the CFP "may contain additional liquidity-related stress scenarios" outside the limit structure, such as a breached Fedwire intraday debit cap, counterparty failures or lost FHLB funding.

LO: LTR-11 a: Discuss the relationship between contingency funding planning and liquidity stress testing. • **Reference:** Venkat and Baird, Ch.7, Scenarios and Liquidity Gap Analysis

Solution LTR-11 a.4 — Answer: B

Bank Q373 • GARP PE1 Q71

Step 1 – The scenario rule

"Institutions should align their CFP stress scenarios to those in its liquidity stress testing framework, as well as to other frameworks such as the recovery and resolution plans."

⇒ B quotes it.

Step 2 – Check the other three

A narrows the stress test coverage; C reverses a contingent action and its purpose; D gives the invocation to the wrong body.

Interpretation

GARP pairs this question with PE2 Q67 (Question LTR-11 c.7): one tests who *invokes* the CFP, the other who *coordinates* it. The treasurer invokes; the liquidity crisis team coordinates.

Why the other options are wrong

A is incorrect. "Solely institution-specific" is wrong: the scenarios cover systemic (general market) and idiosyncratic (institution-specific) risks.

C is incorrect. The listed action is to roll off, not reinvest, maturing securities, and contingent actions aim to strengthen liquidity, not to protect profitability.

D is incorrect. Role misassignment: "The treasurer, in consultation with the CFO and others,

may invoke the CFP and convene the liquidity crisis team (LCT)". The LCT is convened; it does not invoke.

LO: LTR-11 a: Discuss the relationship between contingency funding planning and liquidity stress testing. • **Reference:** Venkat and Baird, Ch.7, Scenarios and Liquidity Gap Analysis; Governance and Oversight

Solution LTR-11 a.5 — Answer: D

New • built from GARP Ch.11

Step 1 – Stress results are a named trigger

The treasurer may invoke the CFP "based on a review of the markets, industry, institution-specific conditions, and liquidity stress testing results." Level 1 "could be triggered by stress test results indicating a greater decline in liquidity than the institution's risk appetite targets".

Step 2 – Who approves the move

"The movement from one escalation level to another, whether up or down, should be explicitly considered and approved by the LCT."
$\Rightarrow$ D.

Interpretation

No indicator has moved, but the stress test has already shown that a plausible scenario breaches appetite. The CFP exists to escalate a developing crisis, and Level 1 means monitoring and communication, not drastic action.

Why the other options are wrong

A is incorrect. Stress results are inputs to the CFP and can trigger Level 1; waiting for actual outflows wastes the lead time the test provides.
B is incorrect. Level 3 is the late, survival stage; selling businesses early can be read as distress and precipitate the crunch (LTR-11 c, Section 3).
C is incorrect. Moves are approved by the LCT, and "the importance of management's expert judgment ... should not be understated."

LO: LTR-11 a: Discuss the relationship between contingency funding planning and liquidity stress testing. • **Reference:** Venkat and Baird, Ch.7, Governance and Oversight; Escalation Levels

Solution LTR-11 a.6 — Answer: A

New • built from GARP Ch.11

Step 1 – The reading's diagnosis

"Historically, capital and liquidity management practices have been highly siloed; however, most recently, institutions have started to bring a more integrated approach."

Step 2 – What integration brings

"A common set of EWIs, a more comprehensive view of the impact of liquidity specific stress test results, and proposed response actions on capital positions and vice versa", aligned with recovery and resolution plans, capital management and ERM.
$\Rightarrow$ A.

Interpretation

The two interact both ways: selling assets to raise cash can crystallise losses that reduce capital, and a capital or rating hit can bring collateral calls that drain liquidity.

Why the other options are wrong

B is incorrect. Contingent actions such as asset sales, equity raising and cuts in distributions do change capital; the reading asks for the joint view.

C is incorrect. Liquidity is not solvency: a well-capitalised bank can still run short of cash (LTR-1).

D is incorrect. Polarity flip and wrong fix: it is *smaller* institutions that include their CFP in business continuity plans, while larger ones formalise it; and the business continuity plan does not address capital.

LO: LTR-11 a: Discuss the relationship between contingency funding planning and liquidity stress testing. • **Reference:** Venkat and Baird, Ch.7, 11.5 Additional Considerations, Liquidity and Capital

LTR-11 • Contingency Funding Planning • Vardeez: VERY IMPORTANT

LTR-11 b

P2 CONCEPT

Describe best practices in the design of a sound contingency funding plan.

Workbook: D26 Contingency Funding Planning • Tier D • notes only

WHAT IS TESTED HERE

Why P2. Tier D workbook topic D26 (Map #62, 1 of 16 recall sources, Vardeez notes only: "treat as thin-evidence-but-high-rating"), and Vardeez rates the reading VERY IMPORTANT.
Vardeez: no LO-specific pointer. His notes list the five design considerations and stress that the third is two-sided: operational and actionable, *but* a flexible playbook.

Reported exam shapes

- **S4** Testing requirements and operational readiness (D26 notes: testing).

What the questions look like: name the five considerations, the frameworks the CFP integrates with and the stakeholders it involves, then pick the testing plan or design recommendation that matches best practice. GARP places testing inside the governance component; the map files the testing shape here, so the testing notes sit in Section 3 and LTR-11 c points back. Bank supply was thin (four questions, one a near duplicate), so three new questions are built from GARP Ch.11.

1. Five design considerations

"No universal CFP exists that can cover all types of institutions and situations", but five considerations apply when designing or refreshing one.

Consideration	What it requires	The distractor
I. Aligned to business and risk profiles	Fit the scope of businesses, products and asset classes, geographic and FX coverage, legal entities. Align the risk appetite statement to the CFP through *quantifiable* EWIs, limits and escalation levels. Refresh as the profile changes, internally (strategy, new products) and externally (economy, markets); leaders build the CFP into strategic planning, making it forward-looking and flexible	"Aligned to the stress testing framework" as the replacement; updating only when a supervisor asks
II. Integrated with broader risk management frameworks	Not a stand-alone tool: part of LRM and firm-wide risk management, namely **ERM, capital management, and business continuity and crisis management**, so it can lean on established controls. Explicitly linked to the stress test through limits and escalation levels (LTR-11 a)	"Financial management" or an economic value added framework
III. Operational and actionable, but a flexible playbook	Its value lies in operational readiness: a menu of contingency actions for different scenarios at graduated levels of severity, aligned to EWIs, triggers and actions. Because no CFP can anticipate every crisis, it balances recommended actions with management's flexibility and discretion	A "theoretical" or "precoded instruction set"; a fixed sequence of actions
IV. Inclusive of appropriate stakeholder groups	Management committees (ALCO, risk and capital committee, investment committee), business units, finance, corporate treasury, risk, operations and technology; a forum where issues are discussed openly	Internal audit; "all shareholders"
V. Supported by a communication plan	Coordinated, timely communication inside the institution and to clients, analysts, counterparties and regulators, to sustain confidence and stop rumours spreading	"A backup plan"; "approved by the national supervisor"

2. One plan or many: structure and institution type

- **Portfolio of CFPs.** A single CFP document is unlikely to cover every business, geography and legal entity clearly. Institutions run a portfolio of CFPs by business segment, product or asset class, geography and FX, and legal entity. Ring-fenced entities supervised by a different regulator from the holding company have separate CFPs; some firms align CFP coverage to their recovery and resolution legal entities; firms with large FX businesses may build a **currency CFP**. The policy is applied consistently across all of them.
- **Institution type shapes the design.** Depositories focus on deposit runoff; wholesale-funded institutions on credit lines and funding markets; insurers fund from policy liabilities and have historically emphasised cash flow matching. Revolver lenders must model drawdowns of unfunded commitments; prime brokers and dealers face clients withdrawing accounts and counterparties making new collateral calls; wholesale borrowers face higher margins and haircuts as their credit is questioned.
- **Size.** Smaller institutions often include their CFP in business continuity plans; larger ones formalise it.

3. Testing and operational readiness (shape S4)

"On a periodic basis, institutions should evaluate their CFP operational readiness and test targeted elements of their CFPs to ensure relevance and execution effectiveness in times of stress, particularly given changing market dynamics and the institution's business and risk profiles."

May be appropriate to test	May be impractical or unavailable to test	Operational effectiveness to evaluate
Debt issuances; brokered deposit issuances; limited securities sales from the investment portfolio	Business divestitures; large asset sales; use of Federal Reserve borrowing	Governance, escalation process, communication, coordination and reporting

Leading institutions run frequent exercises that best simulate a liquidity crisis. These improve the CFP's operational effectiveness and **response times**, and can expose gaps and improvement opportunities "that would otherwise be undetected if the CFP were left purely as a theoretical design exercise."

4. What an effective CFP looks like

GARP's conclusion lists the attributes: strong management involvement and participation from the enterprise; alignment to other capital and risk management frameworks; evaluation of a wide range of possible scenarios; a clearly documented management action plan; and a communication plan coordinated with internal and external stakeholders. "The CFP is a playbook" and needs regular updating; a strong one also improves the institution's strategic, business and risk management capabilities and its credibility with external stakeholders.

Exam traps

- **Playbook, two-sided.** Operational and actionable *and* flexible. "Fixed script", "precoded instruction set" and "theoretical" are all wrong; so is "pure discretion, no recommended actions".
- **Integration list.** ERM, capital management, business continuity and crisis management. Not "financial management" or EVA.

- **Stakeholder list.** ALCO, risk and capital committee, investment committee, business units, finance, treasury, risk, operations and technology. Internal audit and "all shareholders" are not on it.
- **Testing scope.** Divestitures, large asset sales and Fed borrowing may be impractical or unavailable to test; debt issuance, brokered deposits and limited portfolio sales may be tested. Governance, escalation and communication are tested too.
- **One document for everything** (absolute claim): a portfolio of CFPs is the practice.
- **Size polarity.** Smaller institutions embed the CFP in business continuity plans; larger ones formalise it.
- **Refresh.** The CFP is updated as the business and the market change, not only on supervisory request.

Practice questions: LTR-11 b

Question LTR-11 b.1

Bank Q361 • BT P2.T8.20.12.2

When Acme Bank developed its contingency funding plan (CFP), it adopted five design considerations:

I. Aligned to business and risk profiles

II. Integrated with broader risk management frameworks

III. Theoretical basis for a precoded instruction set

IV. Inclusive of appropriate stakeholder groups

V. Supported by a communication plan

Although no single universal CFP exists, all of the above are correct EXCEPT one. Which statement identifies the incorrect consideration and gives the correct replacement?

A. I is incorrect; better is: Aligned to the liquidity stress testing framework.

B. II is incorrect; better is: Integrated with the economic value added framework.

C. III is incorrect; better is: Operational and actionable, but a flexible playbook.

D. V is incorrect; better is: Approved by the national supervisor.

Question LTR-11 b.2

Bank Q360 • AnalystPrep Q.4067

A key design consideration is that a contingency funding plan (CFP) be integrated with broader risk management frameworks. Which of the following is NOT one of the frameworks with which a CFP should be integrated?

A. Enterprise risk management (ERM)

B. Business continuity and crisis management

C. Financial management

D. Capital management

Question LTR-11 b.3 Bank Q359 • AnalystPrep Q.4068

Roy Jakes, the financial risk manager for Pathway, is designing a contingency funding plan (CFP). One design consideration is to include appropriate stakeholder groups. Which of the following is NOT among the stakeholder groups that should be involved in the CFP design?

A. Internal audit committee

B. Risk and capital committee

C. Management committee

D. Operations and technology

Question LTR-11 b.4 New • built from GARP Ch.11, Design Considerations I; Organizational Structure of the CFPs

Over three years a bank holding company has added a prime brokerage business, a ring-fenced bank subsidiary supervised by a different regulator, and a large foreign exchange business. Its single contingency funding plan (CFP) has not been revised since. Which recommendation is most consistent with best practice in CFP design?

A. Keep one CFP for the whole group, since a single document guarantees consistency and the group stress test already covers every business.

B. Leave the CFP unchanged until the next supervisory examination, since a CFP needs updating only when a regulator asks for it.

C. Fold the CFP into the group's business continuity plan, as is common practice at large and complex institutions.

D. Refresh the CFP for the new business and risk profile, tie it to the risk appetite statement through quantifiable EWIs, limits and escalation levels, and move to a portfolio of CFPs, including one for the ring-fenced subsidiary and possibly a currency CFP.

Question LTR-11 b.5 New • built from GARP Ch.11, Testing and Readiness Assessment

The treasurer of a large US bank is drawing up next year's programme for testing the bank's contingency funding plan (CFP). Which programme is most consistent with best practice?

A. Review only the CFP document each year, since any market transaction undertaken as a test would signal distress to counterparties.

B. Test targeted market activities that are practical to test, such as a debt issuance, a brokered deposit issuance and a limited sale of securities from the investment portfolio.

C. Test a business divestiture, a large asset sale and Federal Reserve borrowing each year, since these are the actions most likely to be needed in a severe crisis.

D. Test the CFP only when an early warning indicator turns red, so that every test runs under genuinely stressed conditions.

Question LTR-11 b.6 New • built from GARP Ch.11, Testing and Readiness Assessment

After a contingency funding plan (CFP) simulation exercise, the head of liquidity risk summarises the lessons for the board risk committee. Which statement about CFP testing and operational readiness is correct?

A. The main purpose of testing is to confirm the liquidity value of each contingent action, so testing asset sales is enough and governance and communication can be left untested.

B. Because a CFP must be a fixed script, testing should check that management follows the pre-set sequence of actions exactly, without discretion.

C. Frequent exercises that simulate a liquidity crisis improve the CFP's operational effectiveness and response times, and can reveal gaps that would go undetected if the CFP were left as a purely theoretical design.

D. Testing is needed once, when the CFP is first approved; later changes in market dynamics or in the bank's business and risk profile do not call for re-testing.

Solutions • LTR-11 b

Solution LTR-11 b.1

Answer: C

Bank Q361 • BT P2.T8.20.12.2

Step 1 – List the five

Aligned to business and risk profiles; integrated with broader risk management frameworks; operational and actionable, but flexible playbook; inclusive of appropriate stakeholder groups; supported by a communication plan.

Step 2 – Find the impostor

I, II, IV and V match. III does not: the CFP is a practical playbook whose "effectiveness ... lies in its operational readiness", balanced with management discretion.

$\Rightarrow$ C.

Interpretation

"Precoded" fails twice: a CFP is not theoretical, and it is not a fixed script, because no CFP "can anticipate all possible situations".

Why the other options are wrong

A is incorrect. I is already correct. The link to stress testing is part of integration (II), not a replacement for alignment to the business and risk profile.

B is incorrect. II is already correct; the named frameworks are ERM, capital management, and business continuity and crisis management, not EVA.

D is incorrect. V is already correct; supervisory approval is not a design consideration.

LO: LTR-11 b: Describe best practices in the design of a sound contingency funding plan. • **Reference:** Venkat and Baird, Ch.7, 11.3 Design Considerations

Solution LTR-11 b.2

Answer: C

Bank Q360 • AnalystPrep Q.4067

Step 1 – The integration list

"The CFP is not a stand-alone tool, but rather, an integrated part of the institution's liquidity risk management and firm-wide risk management frameworks, including enterprise risk management (ERM), capital management, and business continuity and crisis management."

Step 2 – Find the one not named

"Financial management" is not on the list.

$\Rightarrow$ C.

Interpretation

Integration lets the CFP "leverage and reference established controls and processes", and the business continuity link reinforces operational and communication protocols in a crisis.

Why the other options are wrong

A is incorrect. ERM is named.

B is incorrect. Business continuity and crisis management is named.

D is incorrect. Capital management is named.

LO: LTR-11 b: Describe best practices in the design of a sound contingency funding plan. • **Reference:** Venkat and Baird, Ch.7, Design Considerations II

Solution LTR-11 b.3 Answer: A

Bank Q359 • AnalystPrep Q.4068

Step 1 – The stakeholder list

"Various management committees (e.g., asset-liability committee (ALCO), risk and capital committee, investment committee), business units, finance, corporate treasury, risk, operations and technology."

Step 2 – Find the outsider

B, C and D are named; internal audit is not.
$\Rightarrow$ A.

Interpretation

The stakeholders are the groups needed to design the CFP and to execute it. Internal audit provides independent assurance over the framework afterwards; it is not a design stakeholder in the reading.

Why the other options are wrong

B is incorrect. The risk and capital committee is named among the management committees.
C is incorrect. Management committees such as ALCO are named.
D is incorrect. Operations and technology are named; they run collateral and cash management in a crisis.

LO: LTR-11 b: Describe best practices in the design of a sound contingency funding plan. • **Reference:** Venkat and Baird, Ch.7, Design Considerations IV

Solution LTR-11 b.4 Answer: D

New • built from GARP Ch.11

Step 1 – Alignment to the business and risk profile

The CFP must reflect the scope of businesses, products, geographic and FX coverage and legal entities, with the risk appetite statement aligned "through quantifiable early warning indicators, limits, and escalation levels", and it "should be refreshed accordingly as the institution's business and risk profile changes over time".

Step 2 – Structure

"It is unlikely that a single CFP document can properly cover all these elements"; institutions run a portfolio of CFPs. "Ring-fenced entities that are supervised by a different regulator ... have separate CFPs", and firms with large FX businesses "may develop a currency CFP".
$\Rightarrow$ D.

Interpretation

The prime brokerage business also changes the stresses the CFP must plan for: clients withdrawing their accounts and counterparties making collateral calls.

Why the other options are wrong

A is incorrect. Absolute claim: one document is unlikely to cover every business and entity clearly, and a ring-fenced entity needs its own CFP.
B is incorrect. The CFP is refreshed as internal strategy and products and external conditions change; leading institutions build it into strategic planning.
C is incorrect. Polarity flip: smaller institutions include their CFP in business continuity plans; larger ones formalise it.

LO: LTR-11 b: Describe best practices in the design of a sound contingency funding plan. • **Reference:** Venkat and Baird, Ch.7, Design Considerations I; 11.5 Organizational Structure of the CFPs

Solution LTR-11 b.5 Answer: B

New • built from GARP Ch.11

Step 1 – What may be tested

"There are some market activities such as debt issuances, brokered deposits issuances, and limited securities sale of the investment portfolio to generate additional liquidity that may be appropriate to test."

Step 2 – What may not

"Certain contingent activities, such as business divestitures, large asset sales, and use of Federal Reserve borrowing, may be impractical or unavailable for testing."
⇒ B.

Interpretation

Small, real transactions show that the channels work (documentation, counterparties, settlement) without the cost or signalling of a large action. Governance, escalation and communication are tested alongside them (Question LTR-11 b.6).

Why the other options are wrong

A is incorrect. A document review does not test execution, and the reading names market activities that may be tested.
C is incorrect. These are exactly the actions the reading calls impractical or unavailable to test.
D is incorrect. Testing is periodic and ahead of stress, so that gaps are found before a crisis rather than in one.

LO: LTR-11 b: Describe best practices in the design of a sound contingency funding plan. • **Reference:** Venkat and Baird, Ch.7, Testing and Readiness Assessment

Solution LTR-11 b.6 Answer: C

New • built from GARP Ch.11

Step 1 – What testing achieves

"Leading institutions that perform frequent exercises that best simulate the potential liquidity crisis environment will improve the CFP's operational effectiveness and response times", and test simulations "may also identify potential gaps ... that would otherwise be undetected if the CFP were left purely as a theoretical design exercise."
⇒ C.

Step 2 – Scope and timing

Operational effectiveness covers "governance, escalation process, communication, coordination, and reporting", evaluated periodically as markets and the business change.

Interpretation

Readiness is the point of design consideration III: a playbook only works in a crisis if people have rehearsed it.

Why the other options are wrong

A is incorrect. Scope error: governance, escalation, communication, coordination and reporting are tested too.
B is incorrect. The CFP is a flexible playbook that balances recommended actions with management discretion; a fixed script is the wrong design.
D is incorrect. Testing is periodic, "particularly given changing market dynamics and the institution's business and risk profiles".

LO: LTR-11 b: Describe best practices in the design of a sound contingency funding plan. • **Reference:** Venkat and Baird, Ch.7, Testing and Readiness Assessment; Design Considerations III

LTR-11 • Contingency Funding Planning • Vardeez: VERY IMPORTANT

LTR-11 c

P2 CONCEPT

Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting).

Workbook: D26 Contingency Funding Planning • Tier D • notes only

WHAT IS TESTED HERE

Why P2. Tier D workbook topic D26 (Map #62, notes only), and Vardeez rates the reading VERY IMPORTANT.

Vardeez: no LO-specific pointer. His traps for this LO: four governance bodies with four verbs (treasury invokes, the LCT designs, monitors and coordinates, the management committee oversees, the board advises), three escalation levels with the LCT as gatekeeper, and contingent actions that are held back because using them early signals weakness.

Reported exam shapes

- **S1** Governance of the CFP: who designs, invokes, oversees and approves.
- **S2** Trigger design and escalation: EWIs, liquidity health measures, escalation levels.
- **S3** The menu of contingent actions and their reliability under stress, including signalling effects.

What the questions look like: role-matching, level descriptions, measure classification and "which recommendation fits best practice". GARP's practice exams test "invoke versus coordinate" (PE1 Q71, worked as Question LTR-11 a.4; PE2 Q67 here).

1. Governance and oversight (shape S1)

The CFP framework covers people, process, data and reporting, tailored to the institution. Its five components are **(1) governance and oversight, (2) scenarios and liquidity gap analysis, (3) contingent actions, (4) monitoring and escalation, (5) data and reporting**. Governance needs well-defined roles and a strong communication strategy, both supported by policies and procedures and reinforced through periodic testing (LTR-11 b, Section 3).

Body	Role in the CFP	The verb
Corporate treasury	In BAU monitors the business, risk, funding and liquidity profile. "The treasurer, in consultation with the CFO and others, may invoke the CFP and convene the liquidity crisis team (LCT)" after reviewing markets, industry, institution-specific conditions and stress test results	**invokes**, convenes
Liquidity crisis team (LCT)	C-level executives and heads of segments, geographies and legal entities. Central point of contact; continuous monitoring of the liquidity profile; recommends CFP actions with treasury and the management committee; coordinates and communicates internally and externally. "Generally, the LCT is responsible for designing the CFP and submitting it to the senior management group for review and approval." Approves moves between escalation levels	**designs**, monitors, **coordinates**
Management committee	Senior management. During a crisis provides oversight of the LCT, consults the board, monitors the liquidity risk profile, reviews recommendations for and coordination of CFP actions	**oversees** the LCT
Board of directors	Actively engaged with the management committee and LCT; "advisor and counsel", especially on strategic actions such as large asset or subsidiary sales	**advises**

Complex groups need a well-defined structure at parent and subsidiary level, so the chain of command is clear across the institution.

> **Correction to the source notes.** Vardeez and Schweser say the board "provides leadership to the LCT and management committee during crises", and Vardeez's table has the management committee "approving CFP recommendations". GARP's verbs are narrower: the board is "an advisor and counsel" to the management committee and the LCT, while direct oversight of the LCT in a crisis belongs to the management committee, which reviews recommendations for, and coordination of, CFP actions. The approval GARP names is of the CFP itself, which the LCT submits "to the senior management group for review and approval". Use GARP's verbs: the management committee oversees; the board advises.

Communication and coordination

Groups across the institution must work in concert so information is available on time for decisions. Communication promotes **confidence**, which is critical: loss of confidence "can quickly spiral downward when rumors start to take over news headlines", whether or not they are true. Messaging and content are **centrally managed** ("everyone should be working off the same page"), but each two-way relationship stays with the team that already owns it:

Owner	Counterparts
Business units	Clients, counterparties
Corporate treasury	Regulators and supervisors, rating agencies, clearing banks
Investor relations	Investors, analyst community, public media
Legal and compliance	Regulators and supervisors

Policies and procedures

Document the CFP and align it with risk management, business continuity and recovery planning policies. GARP's illustrative outline: introduction (purpose, related policies and CFPs); governance (roles, review and approval, periodic review); stress testing and scenarios; methodology (scenario design, inputs and calculations, key assumptions, data control, model validation); monitoring and escalation (regular monitoring, liquidity gap analysis, contingent actions); reporting (frequency, briefing

decks). Review and update it periodically.

2. Scenarios and liquidity gap analysis

Recap of LTR-11 a, Section 3: CFP scenarios are aligned to the liquidity stress testing scenarios and to recovery and resolution plans, cover systemic and idiosyncratic, market and funding stresses over short and prolonged periods, and may be extended with CFP-only scenarios (Fedwire intraday debit cap exceeded, counterparty failures, FHLB funding unavailable). The liquidity gap analysis from these scenarios sizes the contingent actions in Section 3.

3. Contingent actions and their reliability (shape S3)

Based on the liquidity gap analysis, the institution develops contingent actions (business scoping, pricing, disposition and expense control), chosen in light of the **nature and severity** (**amount**) of the shortfall, its **timing and pattern**, the **estimated relief** from the action, and the institution's **ability to execute** it. GARP's examples:

- Maintain lines of credit with few restrictions and reasonable rates
- Raise underwriting standards; dial back lending
- Pay higher deposit rates to attract deposits
- Roll off, rather than reinvest, maturing securities
- Shift from short-term to longer-term funding
- Issue more brokered CDs or direct-to-consumer deposits
- Securitise retail assets (mortgages, card receivables, loans, auto leases)
- Pledge assets at the Fed discount window
- Sell liquid assets
- Draw on securitisation conduits
- Issue subordinated debt
- Slow asset growth (fewer balance transfers)
- Issue at-call loans, recallable for cash
- Sell consumer loans or card receivables
- Sell businesses or business units
- Raise equity through asset sales or issuance
- Reduce capital distributions
- Curtail discretionary spending

Reliability under stress

Availability and impact depend on whether the stress is systemic, idiosyncratic or both, and on its severity. A general freeze can block access to existing credit lines; lenders may refuse credit on perceptions of the institution's strength; asset liquidity may collapse. Factors that can impair contingent actions: securitisation markets shut; repo access restricted by solvency concerns, downgrades or reputational damage; downgrade-driven collateral and margin calls; predatory margin and collateral practices by counterparties; higher cash deposit requirements at custodian banks; higher funding costs; deposit runoff; collapse of the interbank market and wholesale funding; counterparties unwilling to roll over funding. The CFP should anticipate these and document mitigating actions.

Market signals and reputational impact

Early defensive steps (curtailing businesses, tightening credit, limiting higher-risk counterparties) improve liquidity but "may send inadvertent signals to the market", damaging perception and limiting the availability or effectiveness of *later* contingent actions:

- spurned borrowers go to other lenders; hedge funds relying on prime brokerage credit move to other trading partners;
- debt holders may "force" a buyback: agreeing drains liquidity, refusing raises questions about viability;

- accessing a sovereign or emergency facility may make lenders "immediately withdraw existing credit";
- selling businesses or suspending reverse repo or other customer funding can be read as distress.

Such actions may be necessary in the later stages, but "at earlier stages, perceived signals of weakness may actually precipitate a liquidity crunch." Management weighs the short-term relief against what each action tells markets, lenders, clients and counterparties.

4. Monitoring and escalation (shape S2)

The CFP leverages the institution's liquidity risk monitoring and measurement framework (the EWIs of LTR-3), with measures of both the current liquidity profile and the anticipated effects of potential events. Together these measures form the CFP's EWIs, in two families. Internal and external EWIs are selected in concert so the institution can tell whether an emerging risk is idiosyncratic, systemic or both. EWIs are most useful before or at the onset of a crisis and in the early escalation levels.

	Market and business measures	Liquidity health measures
Source	External and internal	Internal
Nature	Broad: the market environment and the institution's business strategy	Targeted: deterioration has a direct, adverse effect on current or projected liquidity
Sub-groups	Macroeconomic (repo spreads, haircut trends, CDS moves); industry (sector profitability, rating agency actions, industry capital adequacy, competitor analysis); institution-specific	Ratios with associated limits; their importance "is related to their associated limits"
Examples	Sharp fall in stock indices; sovereign downgrade; VIX spike; catastrophic events; rapid asset growth funded by volatile liabilities; negative publicity; falling asset quality or earnings; rating downgrades; cancelled or unrenewed loan commitments; wider debt and CDS spreads; more collateral demanded; counterparties refusing unsecured or longer-term deals; early withdrawal requests; debt buyback calls; FX volatility	Projected net funding requirements to unused funding capacity; non-core funding to long-term assets; overnight borrowings to total assets; short-term liabilities to total assets; funding source concentration; funding maturity profile; used to total borrowing capacity; liquid assets to volatile liabilities; unpledged eligible collateral to total assets; loans to commitments; for advanced firms, daily liquidity position reporting, the LCR and the NSFR

Three of the health ratios are tested by definition: **used capacity to total borrowing capacity** shows the borrowing headroom left across all channels; **liquid assets to volatile liabilities** the cushion of liquid assets over volatile funding; **loans to commitments** the exposure to facilities that may be drawn. As health measures approach their predefined limits, management starts to evaluate which contingent actions are appropriate.

Escalation levels

Escalation levels are aligned to the scenarios, contingent actions and measures. There is no required number; **three to five** is common practice. GARP's three-level illustration is in Figure 1. Trigger events are analysed for their cause and their association with capital adequacy, and findings are communicated with recommendations. "The movement from one escalation level to another, whether up or down, should be explicitly considered and approved by the LCT", and "the importance of management's expert judgment ... should not be understated."

every move, up or down, is explicitly considered and approved by the LCT

Business as usual normal times	Level 1 elevated monitoring	Level 2 adverse impact	Level 3 survival
Corporate treasury monitors the business, risk, funding and liquidity profile. EWIs and liquidity health measures tracked against limits; stress tests run. **Approvals:** normal delegations.	**Trigger:** e.g. stress results show a larger fall in liquidity than risk appetite targets, or market perception shifts. LCT convened; communication plan executed. Watch forward-looking health measures and market perception. **Approvals:** notification to senior management, CFO, CRO; actions may need ALCO and/or the Treasurer.	Market and/or idiosyncratic events are hurting the business and liquidity profile. Closer monitoring of current liquidity, causes of deterioration, peers and counterparties. Focus: recovery, short horizon. Curtail activity, limit new credit, sell less liquid assets; evaluate larger actions. **Approvals:** senior management and/or board.	Dramatic steps *have already been taken*: significant curtailing of liquidity-intensive businesses, sales of businesses. Focus primarily on survival; conditions resemble recovery and resolution planning. **Approvals:** senior management and/or board.

Figure 1. GARP's three-level illustration of CFP escalation (three to five levels is common practice). Header fills darken from white (business as usual) to dark grey (Level 3). Level 1 watches, Level 2 acts, Level 3 survives; the past tense in Level 3 is its tell. Approvals widen from ALCO and the Treasurer at Level 1 to senior management and the board at Levels 2 and 3.

5. Data and reporting

- Consider how often each liquidity measure is generated, assessed and reported. Daily reporting to treasury and the funding desks is prevalent, but many institutions would benefit from **more frequent reporting to other areas**: senior management, ALCO and other risk committees, the board, with context and qualitative guidance.
- Reports should convey the methods used to determine liquidity coverage of upcoming liabilities and funding needs, and the level of coverage predicted.
- Reports should capture **intraday liquidity positions**, track exposure to **contingent liabilities**, and monitor **capacity usage in funding sources**.
- Consider how reports are disseminated, how management and the board use them, and whether their content suits the audience.

Exam traps

- **Invoke versus coordinate.** The treasurer (with the CFO) invokes the CFP and convenes the LCT; the LCT designs the CFP, monitors, coordinates across all levels and approves level moves; the management committee oversees the LCT; the board advises.
- **Automatic escalation.** No move between levels without the LCT's explicit consideration and approval; a single EWI breach does not activate the CFP mechanically.
- **Approvals by level.** Level 1 may need only ALCO and/or the Treasurer; senior management and/or the board come in at Levels 2 and 3. "All levels need board approval" is wrong.
- **Level descriptions.** Level 1 watches (elevated monitoring); Level 2 acts and analyses causes; Level 3 survives, after dramatic steps have already been taken. Large sales are evaluated at Level 2, not done the moment it begins.

- **Sibling swap of measure families.** Market and business measures are broad, external and internal; liquidity health measures are targeted internal ratios.
- **Communication owners.** Investor relations faces investors, analysts and media, not regulators or rating agencies; treasury faces regulators, rating agencies and clearing banks.
- **"Deploy everything early".** Some actions are held back because early use signals weakness; emergency facilities do not reassure lenders.
- **Direction of actions.** Roll off maturing securities, shift to longer-term funding, cut distributions; the reverses drain liquidity.
- **Reporting.** Intraday positions must be captured; end-of-day balances alone are not enough.

Practice questions: LTR-11 c

Question LTR-11 c.1

Bank Q372 • BT P2.T8.20.12.3

Which of the following is correctly specified as a key component of a typical contingency funding plan (CFP)?

A. Governance and oversight include stakeholder involvement, roles, and responsibilities.

B. Contingent actions include overnight borrowings to total assets.

C. Monitoring and escalation include adjusting pricing strategies and issuing subordinated debt.

D. Data and reporting include Level 1, Level 2, and Level 3.

Question LTR-11 c.2

Bank Q364 • AnalystPrep Q.4069

Within the governance and oversight component of a contingency funding plan (CFP), which of the following best describes the role of the corporate treasury function?

A. It serves as adviser and counsel to the management committee and the liquidity crisis team (LCT).

B. As part of business-as-usual activities, it monitors the ongoing business, risk, funding and liquidity profile of the institution and may invoke the CFP.

C. It engineers the CFP and presents it to the management committee for approval.

D. It manages the LCT in periods of crisis and approves recommendations on CFP implementation.

Question LTR-11 c.3 Bank Q375 • Mock 3 Q10 (rebuilt, key corrected in the bank)

A treasury manager at a regional bank is drafting the bank's contingency funding plan (CFP) and considers the circumstances in which it should be activated. Which of the following statements would be most appropriate for the manager to include in the plan?

A. The CFP should be activated automatically whenever any single early warning indicator breaches its threshold.

B. The CFP should focus exclusively on bank-specific stress events, since market-wide events are addressed through regulatory facilities.

C. The CFP should specify a single fixed sequence of funding actions to be executed in every stress scenario.

D. The treasurer, in consultation with the CFO, may invoke the CFP and convene the liquidity crisis team based on a review of markets, industry, bank-specific conditions, and liquidity stress testing results.

Question LTR-11 c.4 New • built from GARP Ch.11, Communication and Coordination

During a liquidity squeeze, a bank's contingency funding plan (CFP) sets out who speaks to whom. Which arrangement is consistent with best practice?

A. Investor relations handles rating agencies, clearing banks and regulators, so that every external message comes from a single team.

B. Each business unit drafts its own crisis messages to its clients and counterparties without central coordination, since business units know their clients best.

C. Messaging is centrally managed, but each relationship stays with its existing owner: corporate treasury with regulators, rating agencies and clearing banks; investor relations with investors, analysts and the media; business units with clients and counterparties.

D. Legal and compliance handles all external communications, including with investors and the media, since every crisis statement carries legal risk.

Question LTR-11 c.5 Bank Q369 • AnalystPrep Q.4077 (rebuilt)

In the monitoring and escalation component of a contingency funding plan, which of the following best describes liquidity health measures (for example, short-term funding as a proportion of total funding, or the firm's credit rating)?

A. They are broad macroeconomic indicators of systemic distress in the financial system.

B. They are targeted indicators of the institution's liquidity base and strength, where deterioration directly reflects an adverse impact on its liquidity profile.

C. They are indicators that become relevant only after the highest escalation level is reached.

D. They are qualitative judgements of management policy that carry no quantitative limits.

Question LTR-11 c.6 Bank Q368 • AnalystPrep Q.4076 (rebuilt)

A contingency funding plan typically defines escalation levels (three to five are common, and no specific number is required). Using a three-level illustration, which statement correctly describes Level 1, the first escalation level?

A. The institution is focused on survival and remaining a going concern.

B. It represents elevated monitoring of market conditions and their impact on the business segments and performance.

C. Market and/or idiosyncratic events are clearly affecting the business and the liquidity risk profile adversely.

D. Drastic contingent actions are taken and require approval from the board.

Question LTR-11 c.7 Bank Q374 • GARP PE2 Q67

The treasurer of a large bank is reviewing the bank's contingency funding plan (CFP), including the three escalation levels and the reporting requirements under the plan, and asks a manager to review the CFP against best practices. Which of the following recommendations would be appropriate for the manager to make?

A. Intraday liquidity position reporting is not required under the CFP, as the bank's triggers for all escalation levels should be based on its end-of-day cash balances.

B. As soon as the CFP is escalated to Level 2, the bank should quickly take significant actions to stabilize its liquidity position, including the sale of large portfolios or business units.

C. CFP actions taken under all escalation levels should require approval by senior management and the board of directors.

D. The bank's liquidity crisis team should be responsible for coordinating CFP actions taken under all escalation levels.

Question LTR-11 c.8 Bank Q365 • AnalystPrep Q.4070 (rebuilt)

A contingency funding plan should prepare for events that could adversely affect liquidity. Which of the following does NOT represent a negative liquidity event, but is instead a contingent action that strengthens liquidity?

A. Predatory margin and collateral practices by counterparties

B. Unavailability of Federal Home Loan Bank funding

C. Shifting the funding mix from short-term toward long-term sources

D. The intraday debit cap with Fedwire being exceeded

Question LTR-11 c.9 Bank Q366 • AnalystPrep Q.4072 (rebuilt)

Several market factors can impair an institution's ability to take contingent actions that strengthen its liquidity position. Which of the following would NOT be expected to impair that ability?

A. A shutdown of securitization markets

B. Deposit runoff

C. Counterparties declining to roll over funding

D. A decline in the institution's cost of funding

Question LTR-11 c.10 New • built from GARP Ch.11, Market Signals and Reputational Impact

In the first week of an idiosyncratic liquidity squeeze, a bank's liquidity crisis team reviews the menu of contingent actions in its contingency funding plan. Which proposal is most consistent with the CFP guidance on market signals?

A. Draw on the central bank's emergency lending facility at once, since showing that official support is in place will reassure lenders and counterparties.

B. Weigh what each action signals to lenders, clients and counterparties, use less visible actions first, and keep actions that may be read as distress, such as emergency facilities or business sales, for later stages if they are needed.

C. Cut prime brokerage credit to hedge fund clients at once; because the action raises liquidity, it has no effect on the availability of later contingent actions.

D. Deploy every available contingent action as early as possible, since unused actions provide no liquidity and their effectiveness does not depend on timing.

Solutions • LTR-11 c

Solution LTR-11 c.1

Answer: A

Bank Q372 • BT P2.T8.20.12.3

Step 1 – The five components and their contents

Governance and oversight (roles, communication, policies, testing); scenarios and liquidity gap analysis; contingent actions (the menu); monitoring and escalation (EWIs, health measures, levels); data and reporting.

Step 2 – File each option

Stakeholder involvement, roles and responsibilities sit under governance and oversight.
$\Rightarrow$ A.

Interpretation

The three wrong options each take a real item and file it under the wrong component, the same sibling-swap shape GARP uses for the governance roles.

Why the other options are wrong

B is incorrect. Overnight borrowings to total assets is a liquidity health measure (monitoring and escalation), not a contingent action.
C is incorrect. Adjusting deposit pricing and issuing subordinated debt are contingent actions, not monitoring.
D is incorrect. Levels 1 to 3 are escalation levels (monitoring and escalation), not data and reporting.

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, 11.4 Framework and Building Blocks

Solution LTR-11 c.2

Answer: B

Bank Q364 • AnalystPrep Q.4069

Step 1 – Treasury's role

"As part of its BAU activities, corporate treasury monitors the ongoing business, risk, funding, and liquidity profile of the institution. The treasurer, in consultation with the CFO and others, may invoke the CFP and convene the liquidity crisis team (LCT)."
$\Rightarrow$ B.

Step 2 – Assign the other three

A is the board; C is the LCT; D is the management committee.

Interpretation

Four bodies, four verbs: treasury invokes, the LCT designs and coordinates, the management committee oversees, the board advises.

Why the other options are wrong

A is incorrect. The board of directors serves "as an advisor and counsel".
C is incorrect. Designing the CFP and submitting it for approval is the LCT's role.
D is incorrect. Overseeing the LCT during a crisis and reviewing CFP recommendations is the management committee's role.

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, Governance and Oversight

Solution LTR-11 c.3 — Answer: D

Bank Q375 • Mock 3 Q10

Step 1 – Who activates, and on what

The treasurer, with the CFO and others, "may invoke the CFP and convene the liquidity crisis team (LCT) based on a review of the markets, industry, institution-specific conditions, and liquidity stress testing results."
⇒ D.

Step 2 – Test the other three

A removes judgement; B narrows the scenarios; C turns the playbook into a fixed script.

Interpretation

Invocation is a judgement informed by several inputs, not a mechanical switch. The bank's question had its key corrected to D; the rebuilt version is used here.

Why the other options are wrong

A is incorrect. A single breach may be a false alarm. Escalation runs through levels whose moves the LCT approves, and expert judgement "should not be understated".
B is incorrect. CFP scenarios cover systemic (market-wide) as well as idiosyncratic stress.
C is incorrect. The CFP is a flexible playbook; the feasible actions depend on the nature and severity of the stress.

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, Governance and Oversight; Design Considerations III

Solution LTR-11 c.4 — Answer: C

New • built from GARP Ch.11

Step 1 – Central content

"Communications with respect to messaging and content should be centrally managed. Everyone should be working off the same page."

Step 2 – Existing owners

"The bidirectional communication and coordination with the stakeholders should reside with those executive and management teams that have existing working relationships" (the owner table in Section 1).
⇒ C.

Interpretation

Central content keeps the story consistent; existing owners keep the relationships credible. Both matter because loss of confidence "can quickly spiral downward when rumors start to take over".

Why the other options are wrong

A is incorrect. Role misassignment: rating agencies, clearing banks and regulators sit with corporate treasury (and regulators also with legal and compliance), not investor relations.
B is incorrect. Drops central management of the message; business units own the relationships, not the content.
D is incorrect. Role misassignment: legal and compliance faces regulators and supervisors; investors and the media belong to investor relations.

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, Communication and Coordination

Solution LTR-11 c.5 — Answer: B

Bank Q369 • AnalystPrep Q.4077

Step 1 – The two families

Market and business measures are broad, external and internal. Liquidity health measures "serve as indicators of the institution's liquidity base and strength" and "are more targeted in that deterioration in these metrics reflect a direct and adverse impact on the institution's current and/or projected liquidity profile and strength."
⇒ B.

Interpretation

Health measures are ratios with limits; as they approach those limits, management starts to evaluate contingent actions.

Why the other options are wrong

A is incorrect. Sibling swap: broad macro indicators are market and business measures.
C is incorrect. They are monitored continuously over the course of the crisis, not only at the top level.
D is incorrect. They are quantitative ratios whose importance "is related to their associated limits".

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, Monitoring and Escalation; Liquidity Health Measures

Solution LTR-11 c.6 — Answer: B

Bank Q368 • AnalystPrep Q.4076

Step 1 – Level 1 in GARP's words

"The initial escalation level and represents elevated monitoring over market conditions and the impact to the institution's business segments and performance." It can be triggered by stress results or a shift in market perception; the LCT is convened and the communication plan executed.
⇒ B.

Step 2 – Place the other descriptions

C is Level 2; A is Level 3; D describes the later levels.

Interpretation

Level 1 watches, Level 2 acts, Level 3 survives.

Why the other options are wrong

A is incorrect. Survival as a going concern is Level 3.
C is incorrect. Events already hurting the business and liquidity profile define Level 2.
D is incorrect. Drastic actions needing senior management or board approval belong to Levels 2 and 3; Level 1 actions may need only ALCO and/or the Treasurer.

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, Escalation Levels

Solution LTR-11 c.7 — Answer: D

Bank Q374 • GARP PE2 Q67

Step 1 – The LCT's role

The LCT "serves as the central point of contact and is responsible for the continuous monitoring of the institution's liquidity profile" and "helps ensure effective coordination and communication across the organization". It approves every move between levels.
⇒ D: the LCT coordinates at every level.

Step 2 – Test the other three

A contradicts the reporting component; B moves Level 3 actions into Level 2; C ignores tiered approvals.

Interpretation

This is the "coordinate" half of GARP's pair; the "invoke" half is PE1 Q71 (Question LTR-11 a.4), where the treasurer, not the LCT, invokes the CFP.

Why the other options are wrong

A is incorrect. Reports should "capture intraday liquidity positions"; end-of-day balances alone miss intraday stress.
B is incorrect. At Level 2 the bank sells less liquid assets and *evaluates the feasibility* of larger actions; dramatic steps such as selling businesses characterise Level 3.
C is incorrect. Level 1 may require only notification, with actions approved by ALCO and/or the Treasurer; senior management and/or the board come in at Levels 2 and 3.

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, Governance and Oversight; Escalation Levels; Data and Reporting

Solution LTR-11 c.8 — Answer: C

Bank Q365 • AnalystPrep Q.4070

Step 1 – Sort events from actions

Predatory margin and collateral practices impair contingent actions. Lost FHLB funding and a breached Fedwire intraday debit cap are scenarios the CFP itself may add.

Step 2 – The action

"Shifting allocation from short-term funding to longer-term funding sources" is on the menu of contingent actions.
⇒ C.

Interpretation

Lengthening funding reduces rollover risk. Its reverse, shortening funding, would add to the stress.

Why the other options are wrong

A is incorrect. A negative factor that drains liquidity and limits contingent actions.
B is incorrect. A CFP stress scenario: a funding source disappears.
D is incorrect. A CFP stress scenario: payments are constrained during the day.

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, Scenarios and Liquidity Gap Analysis; Contingent Actions

Solution LTR-11 c.9 — Answer: D

Bank Q366 • AnalystPrep Q.4072

Step 1 – The impairing factors

Shutdown of securitisation markets; restricted repo access; downgrade-driven collateral calls; predatory margin practices; higher custodian cash deposits; increased cost of funding; deposit runoff; collapse of interbank and wholesale funding; counterparties unwilling to roll over funding.

Step 2 – Find the opposite

A *decline* in the cost of funding is the reverse of a listed factor and makes contingent funding easier.

$\Rightarrow$ D.

Interpretation

Whether an action is available depends on the nature (systemic or idiosyncratic) and severity of the stress: a general freeze can close even existing lines of credit.

Why the other options are wrong

A is incorrect. Without securitisation markets, securitising assets raises no cash.

B is incorrect. Deposit runoff drains funding and the capacity to act.

C is incorrect. Funding that is not rolled over opens gaps and removes room to act.

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, Contingent Actions

Solution LTR-11 c.10 — Answer: B

New • built from GARP Ch.11

Step 1 – The signalling rule

Early defensive steps "may send inadvertent signals to the market" that limit "the availability and/or effectiveness of future contingent actions"; management must weigh what each action conveys to markets, lenders, clients and counterparties.

Step 2 – Timing of the visible actions

Accessing a sovereign or emergency facility, selling businesses or suspending reverse repo "may be necessary during the later stages of the crisis; however, at earlier stages, perceived signals of weakness may actually precipitate a liquidity crunch."

$\Rightarrow$ B.

Interpretation

An action that is available is not automatically the right one to use now: using it early can destroy the options that would be needed later.

Why the other options are wrong

A is incorrect. Polarity flip on the signal: emergency borrowing "may lead counterparties and lenders to immediately withdraw existing credit out of concern that the institution will fail".

C is incorrect. Hedge funds that rely on prime brokerage credit "could cease their relationship with the institution"; curtailing business has side effects on later actions.

D is incorrect. Absolute claim and the intuitive wrong answer: effectiveness depends heavily on timing, and some actions are deliberately deferred.

LO: LTR-11 c: Assess the key components of a contingency funding plan (governance and oversight, scenarios and liquidity gap analysis, contingent actions, monitoring and escalation, and data and reporting). • **Reference:** Venkat and Baird, Ch.7, Market Signals and Reputational Impact

LTR-13 • Repurchase Agreements and Financing • Vardeez: VERY IMPORTANT

LTR-13 e

P2 CONCEPT

Compare the use of general and special collateral in repo transactions.

Workbook: D6 Repurchase Agreements • Tier D • notes only • Vardeez focus

WHAT IS TESTED HERE

Why P2. Vardeez focus in a VERY IMPORTANT reading. His reading note for LTR-13 reads: "understand general collateral, special collateral and LO78F" (LO 78.f is LTR-13 f, the next block). No recall source reports this LO on its own; workbook D6 (Tier D, Map #43) covers it in its revision notes, and its question D6.1 (worked under LTR-13 a) plants the GC-versus-special rate as a wrong-input twin.

Vardeez: General vs special collateral.

Reported exam shapes: none listed. The questions below are built from the LO wording and Vardeez's focus: what "trading special" means, which rate is higher and why, who uses which collateral, the fed funds-GC spread, and how the holder of a special bond finances it.

What the questions look like: statement questions ("which is correct", "EXCEPT", I/II/III). No calculations: the TYPE is CONCEPT.

This block also carries the repo background for the whole reading (Section 1). LTR-13 f, a and c recap it and point back here.

1. Repo in one page (background for the reading)

A **repurchase agreement** or repo is "a contract in which a security is traded at some initial price with the understanding that the trade will be reversed at some future date at some fixed price." Economically it is a short-term loan of cash secured by the security.

- **Two legs.** At initiation counterparty A sells the bond to B at the invoice (full) price; at the unwind A buys it back at that price plus interest at the repo rate, actual/360. GARP's example: EUR 100 million face of the DBR 4s of 4 January 2037, invoice EUR 111.772 million, 92 days at 0.23%, repurchase price EUR 111,837,697.10 (eq. 14.1). The settlement calculation, with accrued interest and a haircut, is LTR-13 a.
- **Two names for one trade.** The party delivering the bond and taking cash does a **repo**: it "repos out" the bond or "sells the repo". The party delivering cash does a **reverse repo**: it "reverses in" the bond or "buys the repo". A hedge fund that borrows a bond to short it therefore does a reverse, while the lender of that bond does a repo, so GARP's advice is to say whether *cash* or *securities* are being borrowed or lent.
- **Protection for the cash lender.** Collateral (sell it if the borrower defaults), a **haircut** (lend less than the collateral value: EUR 106 million against EUR 111.772 million) and **margin calls** (extra collateral in falling markets, withdrawals in rising ones). The risks are LTR-13 c.
- **Terms.** Overnight (most common); **open** repo, a one-day repo that renews itself until either party cancels; **term** repo, any maturity longer than one day, out to several months, with demand falling rapidly with term.

Who	Side	Motive	Collateral
Money market funds, municipalities, mutual funds, insurers, pension funds, some corporates	B: lends cash (reverse repo)	Cash management: a short-term rate "without sacrificing much liquidity or incurring significant default risk"	**General**: any issue in an accepted class
Dealer financing inventory, proprietary or customer positions	A: borrows cash (repo)	Long financing: fund a bond until it is sold, without using scarce capital	Usually general; specials market if the bond trades special
Hedge fund or trader shorting a bond	B: lends cash to borrow the bond (reverse)	Short financing: obtain a particular bond to deliver	**Special**: one named issue

⇒ General collateral is what you take when you want to *lend cash*; special collateral is what you ask for when you want to *borrow a bond*.

2. General collateral (GC)

- **What the cash lender accepts.** Repo investors "can be very particular about which classes of securities they will take as collateral, e.g., Euro-area government bonds with less than five years to maturity", but "they will not insist on receiving any particular security within that delineated class." They are said to accept general collateral.
- **The GC rate.** Each day there is a GC rate for each bucket of collateral and each term. "The" GC rate is the overnight rate for U.S. Treasury collateral.
- **Who uses it.** GC trades suit repo investors: "they obtain the highest rate for the collateral they are willing to accept." Funding trades are predominantly GC.
- **GC versus fed funds.** Fed funds is the *unsecured* overnight rate between banks in the Federal Reserve system; a repo secured by Treasuries is safer, so the GC rate is **typically close to, but below**, fed funds (about 16 bp below on average in GARP's sample to 1 July 2010).
- **The fed funds-GC spread widens** (GC falls relative to fed funds) when Treasuries become **scarce**, as in the late 1990s and early 2000s when U.S. budget surpluses were paying down debt, and in **financial stress**: a flight to quality raises the demand to hold Treasuries and to lend cash against them, while holders become less willing to lend Treasuries in repo for fear the collateral will not be returned. After Bear Stearns, with the fed funds target cut to 2.25%, GC traded below 0.50%; spreads were extremely wide again after Lehman.

3. Special collateral

- **Who initiates.** In specials trading "the lender of cash initiates the repo in order to take possession of a particular security." It is then more natural to say that A *lends a security and takes cash as collateral* than that B lends cash against a security; the two descriptions are economically equivalent.
- **Vocabulary.** Any request for specific collateral is a specials trade. The bonds most in demand to be borrowed are said to be **trading special**.
- **The special rate.** There can be one for each security for each term. "Every special rate is typically less than the GC rate": the chance to borrow cash cheaply induces holders of in-demand bonds to lend them, and being forced to lend cash cheaply allocates those bonds to the traders who most want to borrow them. The **special spread** is the GC rate minus the special rate (LTR-13 f).

- **The cash lender's trade-off.** Traders who intend to short a particular bond have to do specials trades and "must decide whether they are willing to lend money at rates below GC rates in order to borrow those securities." They are paid in collateral, not in yield.
- **The holder of a special bond.** An institution that wants to borrow money against a bond that is trading special "will lend that security in the specials market and borrow cash at a rate below GC, rather than financing that security as part of a GC trade." The value of that cheap funding over the time the bond stays special is its financing advantage (LTR-13 g, outside P1 to P3).
- **What sets a special rate.** The demand to *borrow* the issue to that date relative to the supply *available for borrowing*. That "is not the same as the demand and supply to buy and sell issues": some owners, for institutional reasons, do not lend in repo, so lendable supply can be much less than the amount outstanding. A bond that trades rich to its neighbours shows high demand to *own* it, but "may or may not trade very special" depending on how much traders want to *short* it. Individual specials are hard to predict; the one predominant explanatory factor in the U.S. is the auction cycle (LTR-13 f).

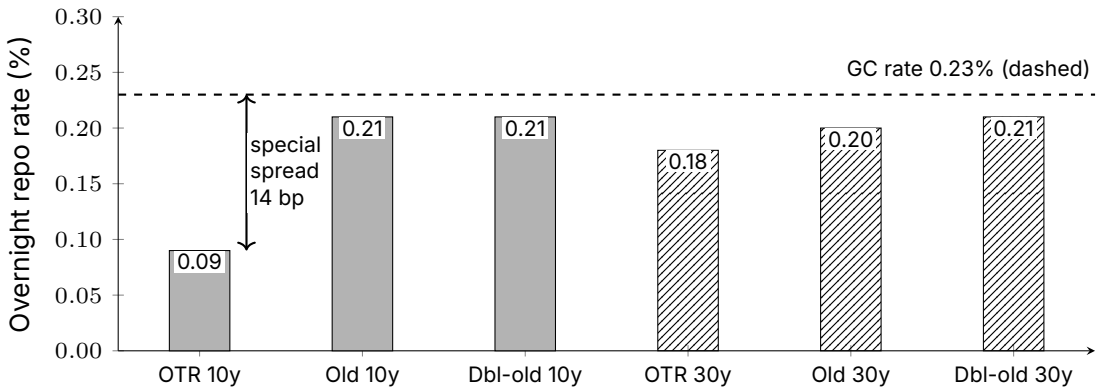


Figure 1. Overnight special repo rates for recent 10-year (grey bars) and 30-year (hatched bars) U.S. Treasury issues on 28 May 2010, against the GC rate of 0.23% (dashed line), from GARP Table 14.1. Each special spread is the gap from a bar's top to the dashed line: 14, 2 and 2 bp for the on-the-run (OTR), old and double-old 10-year; 5, 3 and 2 bp for the 30-year. Every special rate sits below GC, and the newest issue at each maturity is the most special.

Correction to the source notes.

- Vardeez writes that lenders in specials trades "seek specific securities: to short a bond, or to fund inventory", and Schweser that the cash lender wants a particular security "to finance the purchase of a bond (for shorting), or to finance its inventory or proprietary positions". Only shorting is a cash lender's motive. GARP: "Funding trades are predominantly GC"; a dealer that wants to fund a bond that is trading special is the *borrower* of cash, and it lends that bond in the specials market to borrow cash below GC. Inventory financing appears in specials trades only on the cash-borrowing side.
- Vardeez calls the ordering fed funds $>$ GC $>$ special "strict" and any other order "wrong by construction"; the workbook D6 trap says the special rate is "always below GC". GARP's word is **typically**, for both gaps. An issue requested but not in demand repos at or within a basis point or two of GC (2 bp for the double-old issues in Figure 1), and the fed funds-GC gap moves with Treasury supply and stress. Reject options that say "always"; accept "typically".

	General collateral (GC)	Special collateral (specials)
Cash lender wants	Any issue within an accepted class (classes can be precise)	One particular issue
Who initiates	A cash investor, or a dealer funding a position	The cash lender who needs the bond (a short seller)
Typical users	Money market funds, municipalities, other cash investors; funding trades	Short sellers and hedgers; holders of special bonds funding them
Repo rate	GC rate: the highest rate for the collateral accepted; typically just below fed funds	Special rate: typically below GC, one per issue per term
Set by	Bucket of collateral and term; Treasury supply and stress	Demand to borrow the issue relative to lendable supply
Natural description	B lends cash and takes a security as collateral	A lends a security and takes cash as collateral
Spread that measures it	Fed funds − GC (the fed funds-GC spread)	GC − special rate (the special spread)

Exam traps

- **Polarity of the rate.** Demand for a specific bond *lowers* its repo rate: the special rate is below GC, not above. The cash lender is compensated with the bond it wants, not with yield.
- **Sibling swap of spreads.** Fed funds minus GC is the fed funds-GC spread; GC minus the special rate is the special spread.
- **Which side.** The specials trade is initiated by the *lender of cash* who wants the bond. A dealer funding a special bond is the *borrower* of cash and should use the specials market, not GC.
- **Absolute qualifiers.** "Always" below GC, "never" above fed funds: GARP says typically. GC lenders *can* restrict the class of collateral; they only decline to insist on an issue.
- **Own versus borrow.** A bond that trades rich is in demand to own; whether it trades special depends on demand to borrow it against lendable supply.
- **Direction in stress.** Scarce Treasuries and flights to quality push GC *down* and *widen* the fed funds-GC spread.

Practice questions: LTR-13 e

Question LTR-13 e.1

Bank Q460 • AnalystPrep Q.2253 (rebuilt)

A junior trader on Alameda Bank's repo desk is told that the on-the-run 10-year U.S. Treasury note is "trading special". Which of the following best describes what this means?

A. The note is acceptable as general collateral in any repo backed by U.S. Treasuries.

B. The note is in unusually high demand to be borrowed in the repo market.

C. The note is a government bond, so it is financed in repo without a haircut.

D. The note can be financed in repo at a rate above the general collateral rate.

Question LTR-13 e.2 Bank Q461 • BT P2.T7.514.2

A treasury analyst compares special repo rates with general collateral (GC) repo rates. Which of the following statements about special repo rates is true?

A. Special rates are typically less than general collateral rates.

B. If the counterparty's primary motivation is to lend cash rather than borrow a security, the special rate applies.

C. Special rates are well suited to repo investors who are looking to obtain the highest rate for the collateral they are willing to accept.

D. The most commonly cited special rates are for overnight repos where any U.S. Treasury collateral is acceptable.

Question LTR-13 e.3 Bank Q448 • BT P2.T7.513.3 (rebuilt) • re-filed from LTR-13 b

Consider a simplified repurchase agreement between Counterparty A (the party delivering securities) and Counterparty B (the party delivering cash), at initiation of the trade. Consider three primary motivations for repo trades:

I. Lending funds short-term on a secured basis: a money market mutual fund that holds cash for liquidity or safekeeping would buy the repo (as Counterparty B) and is willing to accept general collateral.

II. Financing a long position in a security: a trading desk that wants to finance the purchase of a bond would sell the repo (repo out, as Counterparty A) using the purchased bond as collateral.

III. Borrowing a security to sell it short: a hedge fund that needs to borrow a bond in order to short it would do a reverse repo (as Counterparty B) and, because it needs that bond, requires delivery of a particular security rather than general collateral.

Which of the statements is accurate?

A. None are accurate.

B. I only is accurate.

C. I and II only are accurate.

D. All are accurate.

Question LTR-13 e.4

Bank Q447 • BT P2.T7.513.2 • re-filed from LTR-13 b

Rankcon is a repo investor in the money market mutual fund industry with the exclusive motive of cash management: it lends money while taking collateral and, at maturity, collects the loan plus interest and returns the collateral. Given this motivation, Rankcon would employ each of the following preferences with respect to its repo investments EXCEPT which is the LEAST likely preference?

A. Rankcon insists on a sufficient (i.e., greater rather than lesser) collateral haircut.

B. Rankcon only accepts securities with the highest credit quality, e.g., debt of government-sponsored entities (GSEs).

C. Rankcon refuses to accept general collateral (at general collateral repo rates) and instead insists on particular securities within delineated asset classes.

D. Rankcon places a premium on liquidity, so it tends to lend overnight rather than for term; or, if it wants to lend cash for an extended period, it engages in an open repo.

Question LTR-13 e.5

Bank Q462 • Schweser Module Quiz 78.2 Q1 (rebuilt)

In a presentation to management, a bond trader makes the following statements about repo collateral:

Statement 1: The difference between the federal funds rate and the general collateral (GC) rate is called the special spread.

Statement 2: During financial crises, the difference between the federal funds rate and the GC rate widens as the willingness to lend Treasury securities declines, lowering the GC rate.

Which of the trader's statements are accurate?

A. Both statements are incorrect.

B. Only Statement 1 is correct.

C. Only Statement 2 is correct.

D. Both statements are correct.

Question LTR-13 e.6

New • built from GARP Ch.14, 14.3

A relative value analyst at a pension fund notices that a seasoned 7-year Treasury issue trades rich to its neighbouring issues in the cash market, yet its overnight repo rate is within 1 basis point of the general collateral (GC) rate. Which of the following statements best explains the observation?

A. Issues that trade rich in the cash market always trade special in repo, so the quoted repo rate must be stale.

B. Special rates are set by the demand to buy and sell an issue, so a rich issue should repo well above the GC rate.

C. Only on-the-run issues can be requested in a specials trade, so a seasoned issue is always financed at the GC rate.

D. A rich price shows strong demand to own the issue, but its repo rate depends on the demand to borrow it relative to the amount available for borrowing, which can be quite different.

Question LTR-13 e.7 New • built from GARP Ch.14, 14.3

The treasurer of a primary dealer must finance overnight a USD 500 million long position in the on-the-run 10-year Treasury note, which the desk is holding for market making. Overnight quotes are:

Quote	Rate
General collateral repo rate (U.S. Treasury collateral)	0.23%
Special repo rate, on-the-run 10-year note	0.09%
Unsecured overnight borrowing	0.35%

Which of the following is the cheapest appropriate way to finance the position?

A. Lend the note in the specials market at 0.09%, because the demand to borrow the note lets its holder borrow cash below the GC rate.

B. Repo the note out as general collateral at 0.23%, because funding trades are done at the GC rate whatever the collateral.

C. Borrow unsecured at 0.35%, because a note that is trading special cannot also be pledged as repo collateral.

D. Lend the note in the specials market at 0.37%, because a cash lender who requires a specific issue must earn the GC rate plus the special spread.

Solutions • LTR-13 e

Solution LTR-13 e.1 — Answer: B

Bank Q460 • AnalystPrep Q.2253

Step 1 – What "special" refers to

A repo that requires delivery of a particular issue is a specials trade. GARP: "bonds most in demand to be borrowed are said to be trading special, although any request for specific collateral is a specials trade."

Step 2 – What follows for the rate

Because many traders want to borrow the note, its holders can lend it and borrow cash below the GC rate.

$\Rightarrow$ "Special" describes the demand to borrow one issue. It says nothing about the issuer or the haircut.

Interpretation

On 28 May 2010 the on-the-run 10-year repoed at 0.09% against GC at 0.23%, 14 bp of specialness; the older 10-year issues, equally government bonds, were within 2 bp of GC (Figure 1).

Why the other options are wrong

A is incorrect. Describes general collateral, the opposite case: the cash lender accepts any issue in a class, so no particular issue is in demand.

C is incorrect. Issuer type does not make a bond special, and haircuts depend on collateral risk and the counterparty, not on specialness.

D is incorrect. Polarity flip: demand for a specific bond lowers its repo rate. Special rates are typically below GC.

LO: LTR-13 e: Compare the use of general and special collateral in repo transactions. • **Reference:** Tuckman and Serrat, Fixed Income Securities 3e, Ch.12, 14.3 General and Special Repo Rates

Solution LTR-13 e.2 — Answer: A

Bank Q461 • BT P2.T7.514.2

Step 1 – Why special rates sit below GC

GARP: "every special rate is typically less than the GC rate: being able to borrow cash at a relatively low rate induces holders of securities that are in great demand to lend those securities, while being forced to lend cash at a relatively low rate allocates securities that are in great demand to potential borrowers of that security."

Step 2 – Check the other three

B, C and D each describe general collateral: a cash-first motive, the highest rate for the collateral accepted, and "the" overnight Treasury GC rate.

$\Rightarrow$ A is the only statement about special rates that is true.

Interpretation

The special rate is the price that clears the market for borrowing one bond: the cash lender pays for access to the bond by accepting less interest. GARP's "typically" matters; an issue requested but not in demand repos at or near GC.

Why the other options are wrong

B is incorrect. Sibling swap: a counterparty whose main motive is to lend cash, with no need for a particular bond, does a GC trade.

C is incorrect. That is GARP's description of GC trades, which suit repo investors because "they

obtain the highest rate for the collateral they are willing to accept."
D is incorrect. That is the definition of "the" GC rate. Special rates are quoted for one security and one term.

LO: LTR-13 e: Compare the use of general and special collateral in repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.3 General and Special Repo Rates

Solution LTR-13 e.3 — Answer: D

Bank Q448 • BT P2.T7.513.3

Step 1 – Statement I, cash management

A money market fund lends cash against collateral (B's side, buying the repo). It cares about the class of collateral but not the issue, so it accepts general collateral. True.

Step 2 – Statement II, long financing

A desk that buys bonds from a client finances them until a buyer is found by repoing them out (A's side, selling the repo), with the bonds as collateral. True.

Step 3 – Statement III, short financing

A hedge fund that has sold a bond short must deliver that bond, so it lends cash in a reverse repo and requires the particular issue: special collateral at the special rate. True.
⇒ All three are accurate. General collateral serves cash investment and long financing; special collateral serves the borrowing of a specific bond.

Interpretation

The trade looks the same in all three cases (A delivers securities, B delivers cash). What differs is who initiates and why. In I the cash investor wants yield and takes GC; in III the cash lender wants the bond and accepts a rate below GC to get it.

Why the other options are wrong

A is incorrect. Rejects three standard motivations that GARP describes in turn.
B is incorrect. Rejects II: repoing out a newly bought bond is the standard way a dealer finances inventory.
C is incorrect. Rejects III: borrowing a bond through a reverse to short it is the motivation that creates the specials market.

LO: LTR-13 e: Compare the use of general and special collateral in repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.1 Repurchase Agreements: Structure and Uses; 14.3

Solution LTR-13 e.4 — Answer: C

Bank Q447 • BT P2.T7.513.2

Step 1 – What a cash-management investor wants

Safety and liquidity, in exchange for a relatively low return.

Step 2 – Match each preference

Safety: "only securities of the highest credit quality are typically accepted" (B) and haircuts so that borrowers deliver more collateral than the loan (A). Liquidity: investors "tend to lend overnight, rather than for term", or use an open repo (D).

Step 3 – Collateral type

Investors can be very particular about classes, but "they will not insist on receiving any particular security within that delineated class": they accept general collateral.
⇒ Insisting on particular securities is the behaviour of a cash lender who needs a specific bond, not of a cash investor.

Interpretation

Insisting on specific issues would also cost Rankcon yield: special collateral repos below GC, while GC gives an investor the highest rate for the collateral it accepts.

Why the other options are wrong

A is incorrect. A genuine preference: the haircut protects the lender against a fall in collateral value.

B is incorrect. A genuine preference: government and GSE debt are the most common collateral.

D is incorrect. A genuine preference: repo investors place a premium on liquidity.

LO: LTR-13 e: Compare the use of general and special collateral in repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.1 Repos and Cash Management

Solution LTR-13 e.5 Answer: C

Bank Q462 • Schweser Module Quiz 78.2 Q1

Step 1 – Statement 1

Fed funds minus GC is the fed funds-GC spread. The special spread is the GC rate minus the special rate for a particular security and term. Incorrect.

Step 2 – Statement 2

In stress, a flight to quality raises demand to hold Treasuries and to lend cash against them, and "willingness to lend Treasury bonds in repo declines as market participants fear that collateral may not be returned." GC falls relative to fed funds and the spread widens. Correct.

⇒ Only Statement 2 is correct.

Interpretation

After the Fed cut its target to 2.25% following the collapse of Bear Stearns, GC traded below 0.50%. The same spread also widens when Treasuries are scarce, as during the budget surpluses of the late 1990s and early 2000s.

Why the other options are wrong

A is incorrect. Rejects Statement 2, which matches GARP.

B is incorrect. Accepts the mislabelled spread in Statement 1 and rejects the correct Statement 2.

D is incorrect. Accepts Statement 1 by confusing the fed funds-GC spread with the special spread.

LO: LTR-13 e: Compare the use of general and special collateral in repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.3 General and Special Repo Rates

Solution LTR-13 e.6 Answer: D

New • GARP Ch.14, 14.3

Step 1 – Two different markets

GARP: special rates are "determined by the demand of borrowing that issue to that date relative to the supply available", and that demand and supply "is not the same as the demand and supply to buy and sell issues."

Step 2 – Why they diverge

Some owners, for institutional reasons, do not lend in repo, so lendable supply may be "somewhat less or very much less than the amount outstanding". And a bond that trades rich "may or may not trade very special in repo depending on the extent traders want to short it relative to the supply available for borrow."

⇒ Buy-and-hold demand makes this issue rich, but few traders want to short it, so it repos almost

as general collateral.

Interpretation

This is why GARP finds individual special spreads hard to predict, and why the auction cycle, which concentrates short demand in the newest issue, is the one predominant explanatory factor (LTR-13 f).

Why the other options are wrong

A is incorrect. Absolute claim: richness in the cash market does not guarantee specialness in repo.

B is incorrect. Wrong driver and wrong direction: specials are driven by demand to borrow, and high demand pushes the repo rate below GC, never above.

C is incorrect. Any request for a specific issue is a specials trade, whatever its age; on-the-run issues are simply the ones most often in demand.

LO: LTR-13 e: Compare the use of general and special collateral in repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.3 General and Special Repo Rates

Solution LTR-13 e.7 Answer: A

New • GARP Ch.14, 14.3

Step 1 – Which market

GARP: "Funding trades are predominantly GC. Should an institution find itself wanting to borrow money against a security that is trading special, however, it will lend that security in the specials market and borrow cash at a rate below GC, rather than financing that security as part of a GC trade."

Step 2 – What the dealer saves

The special rate is 14 bp below GC. On USD 500 million overnight that is $500\text{m} \times 0.0014/360 =$ USD $1{,}944$ a night (USD 1,250 at 0.09% against USD 3,194 at 0.23%).

$\Rightarrow$ The specialness belongs to the holder of the bond: cash lenders accept less interest to get the note.

Interpretation

The converse case, a bond that is not trading special and so is financed at GC, is the trap in Question LTR-13 a.7 (workbook D6.1). The value of this cheap funding over the period the note is expected to stay special is its financing advantage (LTR-13 g). Unsecured borrowing is the red herring: secured repo is cheaper.

Why the other options are wrong

B is incorrect. A sound rule for ordinary bonds, but for a special bond it gives away 14 bp.

C is incorrect. Special collateral is perfectly good repo collateral; lending it is exactly what a specials trade is. Unsecured funding is also the most expensive quote.

D is incorrect. Polarity flip: the cash lender in a specials trade accepts *less* than GC; $0.23\% + 0.14\% = 0.37\%$ adds the spread instead of subtracting it.

LO: LTR-13 e: Compare the use of general and special collateral in repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.3 General and Special Repo Rates

LTR-13 • Repurchase Agreements and Financing • Vardeez: VERY IMPORTANT

LTR-13 f

P2 CONCEPT

Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle.

Workbook: D6 Repurchase Agreements • Tier D • notes only • Vardeez focus

WHAT IS TESTED HERE

Why P2. Vardeez focus in a VERY IMPORTANT reading: his reading note names it ("LO78F. Special spreads and the auction cycle"). No recall source reports it and there is no workbook question; workbook D6 covers it in its revision notes.
Vardeez: Special spreads and the auction cycle (his "LO 78.f").

Reported exam shapes: none listed. The questions are built from the LO wording and Vardeez's focus: the characteristics (which issues trade special, how volatile and how large), the pattern over the auction cycle and its reasons, the term structure of special spreads, and the limits on the spread before and after the 2009 fails penalty.

What the questions look like: statement questions, one with a term-structure table to interpret. No calculations (TYPE CONCEPT). Bank Q464, which asks for a spread computed from a negative special rate, is left out under the TYPE rule; its point is tested conceptually in Question LTR-13 f.6.

1. Which bonds trade special, and why

Recap: the special spread is the GC rate minus the special rate for one security and one term (LTR-13 e, Section 3).

- **The auction schedule.** The U.S. Treasury sells each maturity on a fixed schedule (when GARP wrote, a new 10-year every three months, with re-openings in between). The most recent issue is the **on-the-run (OTR)** or current issue; all others are **off-the-run (OFR)**. The second most recent is the **old** issue, the third the **double-old**.
- **Ranking.** At each maturity the OTR trades the most special, then the old, then the double-old. The OTR 10-year trades more special than the OTR 30-year, "a regularity that has been true for some time" (Figure 1 of LTR-13 e: 14 bp against 5 bp).
- **Why liquidity makes them special.** Current issues have the lowest bid-ask spreads and take large trades quickly, partly self-fulfilling (those needing liquidity flock to them) and partly because dealers hold a new issue until it seasons. So the OTR suits shorts as well as longs. Most Treasury shorts are brief: a desk hedging the rate risk of its position, a corporation or underwriter hedging an upcoming sale of its own bonds, an investor betting that rates will rise. They want to cover quickly and cheaply, so they short the most liquid issue. Holders of the OTR need compensation to give up its liquidity by lending it, and shorts are willing to pay for it. Result: the OTR trades special.

2. Characteristics: lessons from 13 years of data

GARP charts the overnight special spread of the OTR 10-year from July 1997 to July 2010, with each 10-year auction marked by a vertical line (Figures 14.4 to 14.6). Four lessons:

1. Special spreads are **quite volatile day to day**, reflecting each day's supply of and demand for special collateral.
2. They can be **quite large**: spreads of hundreds of basis points are common.
3. They reach **higher levels in some periods than others**, partly because of the level of interest rates (Section 4).
4. Although far from regular, they follow the **auction cycle**: small right after an auction and peaking before the next one (Section 3).

3. Behaviour over the auction cycle

- **Just after an auction: small.** "It takes some time for a short base to develop." Right after a new OTR is issued, shorts can stay in the previous OTR or move to the new one, and this **substitutability** depresses spreads. After a **re-opening** (an auction that adds to an existing issue), the **extra supply** depresses spreads, and re-opened issues "do not get as special as do new issues".
- **During the cycle: rising.** As time passes, shorts migrate toward the OTR and its special spread rises.
- **Into the next auction: a spike.** Many participants **short the OTR to hedge their purchases of the to-be-issued next OTR**, so demand to short the OTR, and its special spread, "can increase dramatically or spike going into the subsequent auction."
- **On average.** Averaged over all cycles from 1997 to 2010, with time measured as the fraction of the way from one issue date to the next, the special spread rises across the cycle and spikes as the next auction approaches (GARP Figure 14.7).
- **Other maturities.** The OTR 5-year behaves much like the 10-year. Shorter maturities show a similar cycle with much narrower spreads, because frequent issuance stops any one issue from becoming "far and away the most liquid bond or most-favored short". The 30-year was once as special as the 10-year, but not since the years leading up to the end of 30-year issuance in 2001; apart from active specials trading around the 2005 announcement of its return and its 2006 sale, its spreads have been muted.

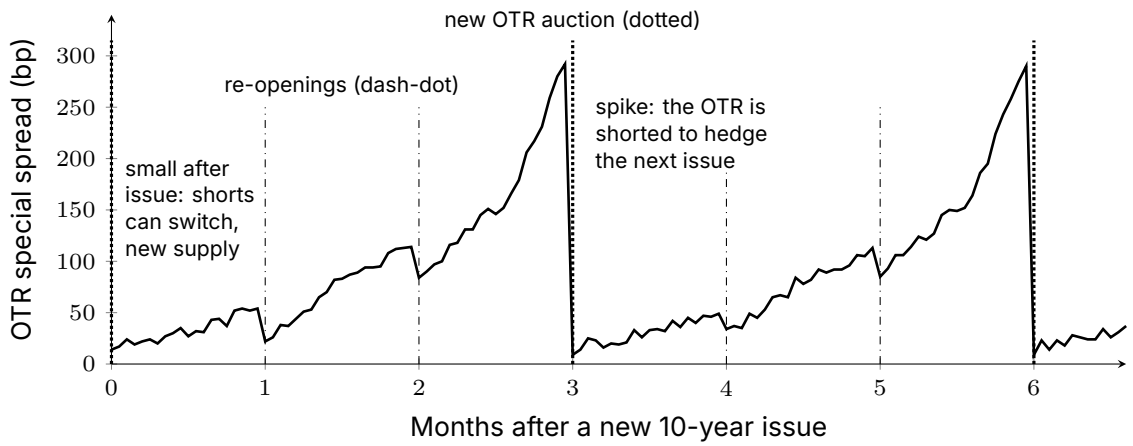


Figure 2. Stylized OTR 10-year special spread (solid line) over two quarterly cycles, after the pattern in GARP Figures 14.4 to 14.7; the levels are illustrative, not data. Dotted vertical lines mark auctions of a new OTR issue, dash-dot lines re-openings. The spread starts small, dips after each re-opening, rises as shorts migrate to the OTR, spikes into the next new issue and collapses when it arrives.

The same pattern in the term structure of special spreads

Term special rates are quoted to fixed dates and to futures expiries (the OTR is traded against the futures contract into which it is deliverable). GARP Table 14.2 gives the OTR 10-year, the $3\frac{1}{2}$s of 15 May 2020, on 28 May 2010. Term spreads are term GC minus term special; each forward spread is the spread for the period between two term dates, implied from the term spreads (eq. 14.2: $33 \times 0.32\% + (61 - 33)\, s_{\text{fwd}} = 61 \times 0.28\%$, so $s_{\text{fwd}} = 0.23\%$ from 30 June to 28 July). The Treasury was expected to re-open the $3\frac{1}{2}$s in mid-June and mid-July and to issue a new OTR in mid-August.

Term	Ends	Days	Special rate	GC rate	Term spread	Forward spread
Overnight	1 Jun 2010	4	0.09%	0.23%	0.14%	0.14%
1 week	4 Jun 2010	7	0.00%	0.23%	0.23%	0.35%
June futures (TYM0)	30 Jun 2010	33	−0.10%	0.22%	0.32%	0.34%
2 months	28 Jul 2010	61	−0.05%	0.23%	0.28%	0.23%
3 months	30 Aug 2010	94	0.00%	0.25%	0.25%	0.20%
Sept futures (TYU0)	30 Sep 2010	125	0.04%	0.26%	0.22%	0.13%
Dec futures (TYZ0)	31 Dec 2010	217	0.17%	0.30%	0.13%	0.01%

$\Rightarrow$ The market expects the spread to rise into the June re-opening (forwards of 0.35% and 0.34%), the $3\frac{1}{2}$s to stay special into and somewhat past the July and August auctions (0.23%, 0.20%, 0.13%), and then to lose its specialness: from 30 September to 31 December the forward is 1 bp. Once the new 10-year has been outstanding for about a month, shorts have moved to it. The negative term special rates are explained in Section 4.

4. Limits on special spreads

- **Before May 2009: special rate ≥ 0, spread $\leq$ GC.** There was no explicit penalty for a **fail** (not delivering a bond that had been sold). A trader who shorted the OTR 10-year and failed to deliver did not receive the sale proceeds and so lost a day's interest on them. Borrowing the bond at a 0% repo rate, that is lending the proceeds at 0%, has exactly the same economics: no interest on the proceeds. "Therefore, because no trader would borrow the bond if the special rate were 0% or less, the special rate should never be less than 0%. Equivalently, the special spread should not exceed

the GC rate." The bound was not strict (fails also carry non-monetary costs such as regulatory capital, and the OTR 10-year traded at negative special rates in the second half of 2003).

- **The level of rates.** GARP Figure 14.8 shows that over all but the most recent period the OTR 10-year special spread was limited by the level of rates (the fed funds target): part of the explanation for periods of high and low spreads.
- **The penalty from 1 May 2009.** After the Lehman bankruptcy, fails to deliver the OTR 10-year hit a record USD 5.311 trillion in the week ending 22 October 2008 (pre-crisis average USD 165 billion). With regulators' prodding, the Treasury Market Practices Group, an industry group, adopted a penalty rate for fails equal to the greater of **3% minus the fed funds target rate, or zero**. With fed funds near zero the penalty is near 3%: failing to deliver USD 100 million costs $100\text{m} \times 3\%/360 = \text{USD } 8{,}333$ a day. As fed funds rises the penalty falls, because higher rates already raise the opportunity cost of failing.
- **The new limit.** "The new upper limit for the special spread is the penalty rate rather than the GC rate." In June 2009 demand to short the OTR drove the spread up to this limit. Special rates can now be negative: in Table 14.2 the OTR 10-year repos at -0.10% to 30 June 2010, a term spread of 0.32%, above the 0.22% GC rate.

Precision. Strictly, a failing trader now loses the interest *and* pays the penalty, so the spread can reach GC plus the penalty; with rates near zero, as in 2009 and 2010, that is about 3%, the penalty rate itself. On the exam use GARP's words: the cap is the penalty rate.

Correction to the source notes.

- **Why spreads widen before an auction.** Schweser says spreads widen "due to the substitutability of the special collateral as shorts change to the new OTR security". GARP says the reverse: substitutability right after an auction *depresses* spreads; they rise as shorts migrate into the OTR and spike because participants short the OTR to hedge purchases of the to-be-issued next OTR. Vardeez rightly rejects his Lecture layer's reason (falling demand for the old issue) but keeps "demand for the soon-to-be-issued OTR bond rises". The spread that spikes is that of the *current* OTR, which is shorted as the hedge; demand for the new issue is the cause of the hedge, not the cause of the spread.
- **Who set the penalty, and on which rate.** Vardeez and Schweser say "regulators adopted" a penalty of 3% minus "the federal funds rate". GARP: the Treasury Market Practices Group, an industry group prodded by regulators, from 1 May 2009, at $\max(3\% - \text{fed funds } \textit{target} \text{ rate}, 0)$.
- **The 0% floor is pre-2009.** Vardeez's "Floor: 0% on the special rate" and his trap "the special rate floors at 0%" hold only before the penalty. Since 2009 special rates can be negative (-0.10% in Table 14.2).

Point in the cycle	What drives it	OTR special spread
Just after a new issue	Short base not yet built; shorts can stay in the old OTR (substitutable)	Small
Just after a re-opening	Extra supply of the same issue; re-opened issues get less special than new ones	Dips
Mid-cycle	Shorts migrate to the OTR	Rising
Into the next auction	Shorting the OTR to hedge purchases of the next issue	Spikes (peak)
After the next new issue	Shorts move to the new OTR; the old issue drifts to GC	Collapses
Limit before May 2009	No fails penalty: failing = borrowing at 0%	Special rate ≥ 0; spread $\leq$ GC
Limit from May 2009	Penalty $\max(3\% - \text{FF target}, 0)$	Cap = penalty rate; negative special rates possible

Exam traps

- **Cycle polarity.** Narrow *just after* an auction, widest *just before* the next. "Peaks after the auction and decays" is the planted reversal.
- **Reason swap.** Substitutability and new supply *narrow* the spread; migration of shorts and hedging shorts *widen* it.
- **Re-openings.** They add supply: re-opened issues get *less* special than new issues.
- **Ranking.** OTR > old > double-old; OTR 10-year > OTR 30-year; shorter maturities narrower because of frequent issuance.
- **Size and stability.** Volatile day to day and often hundreds of bp, not small and stable.
- **Floor and cap by date.** Before 2009: special rate floored at 0%, spread capped at GC. After: cap is the penalty rate, and special rates can be negative.
- **Penalty polarity.** $\max(3\% - \text{FF target}, 0)$ *falls* as rates rise; it is largest (3%) when rates are zero.
- **Own versus borrow.** Specialness is about demand to *borrow* (short), not demand to own.

Practice questions: LTR-13 f

Question LTR-13 f.1

Bank Q463 • BT P2.T7.514.3

A repo analyst lists four observations about special spreads. Each of the following is true EXCEPT which is false?

A. On-the-run (OTR) issues tend to trade "more special" than off-the-run (OFR, i.e., old or double-old) issues, where "more special" refers to special spreads that are larger.

B. The special spread equals the general collateral (GC) repo rate minus the special collateral (aka, specifically requested collateral) repo rate.

C. On-the-run special spreads peak immediately after an auction, and tend to decrease over the cycle, reaching their lowest level immediately before the next auction.

D. Special spreads tend to be volatile on a daily basis (reflecting supply and demand for special collateral) and special spreads can be quite large (e.g., hundreds of basis points).

Question LTR-13 f.2

New • built from GARP Ch.14, 14.3

A portfolio manager asks why the on-the-run (OTR) 10-year Treasury note usually trades special while older issues of similar maturity repo close to the general collateral rate. Which explanation is most consistent with the reading?

A. The Treasury restricts how much of each new issue may be lent in repo until the next auction, which creates an artificial scarcity of the OTR note.

B. Most Treasury shorts are held briefly and prefer the most liquid issue so they can be covered quickly and cheaply, while holders of that issue require compensation to give up its liquidity by lending it.

C. The OTR note carries less credit risk than older Treasury issues, so cash lenders accept a lower repo rate to hold it as collateral.

D. Older issues stop being eligible as repo collateral once a new issue is auctioned, so all demand for Treasury collateral moves to the OTR note.

Question LTR-13 f.3

Bank Q465 • bank-written (GARP style)

A fixed income strategist studies the repo market for the on-the-run (OTR) 10-year U.S. Treasury note over a full auction cycle. Which of the following best describes the typical behavior of its special spread?

A. The spread tends to be small just after an auction because shorts can remain in the previous OTR issue, and it rises as shorts migrate to the new OTR, spiking ahead of the next auction.

B. The spread tends to peak just after an auction because dealers hold large inventories of the new issue, and it declines steadily as the issue becomes seasoned.

C. The spread tends to stay roughly constant over the cycle because the OTR issue keeps its liquidity premium until the next auction replaces it.

D. The spread tends to rise over the cycle, and re-opened issues become more special than newly issued securities because the added supply increases borrowing demand.

Question LTR-13 f.4

New • built from GARP Ch.14, Table 14.2

On 28 May 2010 a strategist studies the term structure of special spreads for the on-the-run 10-year note, the $3\frac{1}{2}$s of 15 May 2020. The Treasury is expected to re-open the $3\frac{1}{2}$s in mid-June and mid-July and to issue a new on-the-run 10-year in mid-August.

Term to	Special rate	GC rate	Term spread	Forward spread
1 Jun (overnight)	0.09%	0.23%	0.14%	0.14%
4 Jun (1 week)	0.00%	0.23%	0.23%	0.35%
30 Jun	−0.10%	0.22%	0.32%	0.34%
28 Jul	−0.05%	0.23%	0.28%	0.23%
30 Aug	0.00%	0.25%	0.25%	0.20%
30 Sep	0.04%	0.26%	0.22%	0.13%
31 Dec	0.17%	0.30%	0.13%	0.01%

Which interpretation is most consistent with the data?

A. The $3\frac{1}{2}$s are expected to become more special after mid-August, because the new issue makes the older note scarcer.

B. Forward spreads are highest for the longest term, showing that specialness builds steadily until the note matures.

C. The 0.00% one-week special rate shows that the note is not trading special during its first week.

D. The $3\frac{1}{2}$s are expected to stay special into and somewhat past the July and August auctions, and to trade close to GC once the new 10-year has been outstanding for about a month.

Question LTR-13 f.5

New • built from GARP Ch.14, 14.3

A repo trader explains why, before May 2009, the special spread on the on-the-run 10-year Treasury rarely exceeded the general collateral (GC) rate. At that time there was no explicit penalty for failing to deliver a bond that had been sold. Which statement correctly explains the limit?

A. The special rate could not fall below the GC rate, because a short seller who failed to deliver could always lend its cash at GC instead.

B. The special rate could not fall below the fed funds rate, because failing to deliver cost the trader one day's interest at the unsecured rate.

C. The special rate could not fall below 0%, because borrowing the bond at a 0% repo rate left a short seller no better off than failing; the spread was therefore capped at the GC rate.

D. The special spread could not fall below the GC rate, because short sellers had to compensate holders for the liquidity they gave up by lending.

Question LTR-13 f.6 New • built from GARP Ch.14, 14.3 and Table 14.2

On 28 May 2010 the term special rate on the on-the-run 10-year note to 30 June was -0.10%, against a term general collateral (GC) rate of 0.22%. The fed funds target range was 0% to 0.25%. A risk manager asks how a negative special rate is possible. Which statement is correct?

A. It is a data error: a short seller would always rather fail to deliver than lend cash at a negative rate to borrow the note.

B. Since May 2009 a failing trader pays a penalty of $\max(3\% - \text{fed funds target}, 0)$, about 3% here, so borrowing at -0.10% beats failing; the penalty rate, not GC, now caps the spread.

C. The penalty for failing rises with the fed funds target, so with rates near zero it was at its minimum and cannot explain a negative special rate.

D. The special spread is capped at the GC rate, so the 0.32% spread shows the note was mispriced and holders could earn a riskless profit by lending it.

Solutions • LTR-13 f

Solution LTR-13 f.1

Answer: C

Bank Q463 • BT P2.T7.514.3

Step 1 – Check the true statements

A: at each maturity the OTR trades most special, then the old, then the double-old. B: the special spread is the GC rate minus the special rate. D: GARP's first two lessons from the 1997 to 2010 history: spreads "are quite volatile on a daily basis" and "spreads of hundreds of basis points are quite common".

Step 2 – The false one

GARP: "spreads tend to be small immediately after auctions and to peak before auctions." C states the cycle backwards.
$\Rightarrow$ C is the exception.

Interpretation

After an auction the short base takes time to build and shorts can stay in the previous OTR; the spread then rises as shorts migrate and spikes as the OTR is shorted to hedge purchases of the next issue (Figure 2).

Why the other options are wrong

A is incorrect. True: the OTR's liquidity makes it the favoured short, so it trades more special than older issues.
B is incorrect. True: this is the definition. Choosing it comes from writing the spread as special minus GC.
D is incorrect. True: these are the first two lessons GARP draws from its charts.

LO: LTR-13 f: Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle. • **Reference:** Tuckman and Serrat, Ch.12, 14.3 Special Spreads in the United States and the Auction Cycle

Solution LTR-13 f.2

Answer: B

New • GARP Ch.14, 14.3

Step 1 – Why the OTR is the favoured short

Current issues are the most liquid: low bid-ask spreads and large trades done quickly. Most Treasury shorts are brief (hedging a position, hedging an upcoming bond sale, betting on higher rates), and "holders of these relatively brief short positions prefer to sell particularly liquid Treasuries so that, when necessary, they can cover their short positions quickly and at relatively low transaction costs."

Step 2 – Why that makes it special

"Investors and traders who are long an OTR bond for liquidity reasons require compensation if they are to sacrifice that liquidity by lending that bond in the repo market. At the same time, investors and traders wanting to short the OTR securities are willing to pay for the liquidity". The payment is a repo rate below GC.
$\Rightarrow$ Demand to short the most liquid issue, met by holders who must be paid to lend it, makes the OTR trade special.

Interpretation

Older issues are the same credit and remain good collateral; they simply are not where the shorts are, so they repo near GC (2 bp for the old and double-old 10-year in Figure 1 of LTR-13 e).

Why the other options are wrong

A is incorrect. There is no such Treasury rule. Lendable supply is limited because some owners choose not to lend for institutional reasons.

C is incorrect. All Treasury issues carry the same issuer's credit. Specialness reflects liquidity and demand to borrow, not credit quality.

D is incorrect. Older issues remain eligible general collateral; they repo close to GC because few want to borrow them specifically.

LO: LTR-13 f: Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle. • **Reference:** Tuckman and Serrat, Ch.12, 14.3 Special Spreads in the United States and the Auction Cycle

Solution LTR-13 f.3

Answer: A

Bank Q465 • bank-written

Step 1 – Just after the auction

Shorts can stay in the previous OTR or move to the new one; "this substitutability tends to depress special spreads". A re-opening's extra supply depresses them too.

Step 2 – Through the cycle

"As time passes after an auction, shorts tend to migrate toward the OTR security, and its special spread tends to rise."

Step 3 – Into the next auction

Participants short the OTR to hedge purchases of the to-be-issued next OTR, so the spread "can increase dramatically or spike going into the subsequent auction."

$\Rightarrow$ Small, rising, then spiking: option A.

Interpretation

GARP's average across 13 years of cycles (Figure 14.7) shows the same shape, which is why the term structure of special spreads for an OTR issue also rises into the auctions and collapses after a new issue (Section 3, table).

Why the other options are wrong

B is incorrect. Reverses the cycle: it takes time for a short base to develop, so the spread is small, not at its peak, right after an auction.

C is incorrect. The average spread clearly rises over the cycle; liquidity alone does not hold it constant.

D is incorrect. The rising pattern is right, but re-opened issues "do not get as special as do new issues": extra supply depresses the spread.

LO: LTR-13 f: Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle. • **Reference:** Tuckman and Serrat, Ch.12, 14.3 Special Spreads in the United States and the Auction Cycle

Solution LTR-13 f.4

Answer: D

New • GARP Ch.14, Table 14.2

Step 1 – Read the forward column against the schedule

Forward spreads of 0.35% and 0.34% run into the June re-opening; 0.23% and 0.20% through the July re-opening and the August auction; 0.13% to 30 September; then 0.01% from 30 September to 31 December.

Step 2 – Link it to the cycle

The $3\frac{1}{2}$s stay the favoured short until a new OTR arrives in August. Shorts then migrate to the new note, and once it has been outstanding for about a month the $3\frac{1}{2}$s are expected to trade as GC.
⇒ GARP: "by the time the then-current 10-year has been around for a month, the specialness of the $3\frac{1}{2}$s is projected to have dissipated."

Interpretation

The term structure carries the market's expectation of the auction-cycle pattern. GARP uses it to value the note's financing advantage (LTR-13 g) over the 125 days to 30 September only, because the forward spread after that is 1 bp. The negative term special rates are possible only because of the fails penalty (Question LTR-13 f.6).

Why the other options are wrong

A is incorrect. Polarity flip: once a new OTR is issued, shorts move to it and the old issue's spread collapses (1 bp forward).
B is incorrect. The longest forward spread, 30 September to 31 December, is the smallest in the table (0.01%).
C is incorrect. Polarity flip: a 0.00% special rate against a 0.23% GC rate is a 23 bp special spread, so the note is very special.

LO: LTR-13 f: Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle. • **Reference:** Tuckman and Serrat, Ch.12, 14.3, Table 14.2 and eq. (14.2)

Solution LTR-13 f.5 Answer: C

New • GARP Ch.14, 14.3

Step 1 – The cost of failing

A short seller who fails to deliver does not receive the sale proceeds, so it loses a day's interest on that cash. Before May 2009 there was no other penalty.

Step 2 – The cost of borrowing the bond at 0%

To deliver, the trader borrows the bond by lending the sale proceeds at the special rate. At 0% it earns nothing on the proceeds: "in both cases no interest is earned on the proceeds from selling the bond." Below 0% failing is cheaper, so no one borrows.
⇒ The special rate is floored at 0%, so the special spread, GC minus special, cannot exceed the GC rate.

Interpretation

This tied the largest possible spread to the level of rates: over most of 1997 to 2009 the OTR 10-year spread was bounded by the fed funds target (GARP Figure 14.8), part of the reason spreads were high in some periods and low in others. "Rarely" because the bound was not strict: fails also carry non-monetary costs such as regulatory capital.

Why the other options are wrong

A is incorrect. A failing trader never receives the cash, so it cannot lend it at GC; and the floor is on the special rate at 0%, not at GC.
B is incorrect. Wrong rate: the floor came from earning nothing on the proceeds, which gives 0%, not fed funds.
D is incorrect. Sibling swap of floor and cap: the GC rate was the *cap* on the spread, not a floor under it.

LO: LTR-13 f: Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle. • **Reference:** Tuckman and Serrat, Ch.12, 14.3 Special Spreads in the United States and the Level of Rates

Solution LTR-13 f.6 **Answer: B**

New • GARP Ch.14, 14.3 and Table 14.2

Step 1 – The cost of failing since May 2009

The Treasury Market Practices Group's penalty is $\max(3\% - \text{fed funds target}, 0)$. With the target near zero it is about 3% a year: USD 8,333 a day on USD 100 million.

Step 2 – Compare with borrowing the note

Borrowing at -0.10% costs the short seller 0.10% a year on the proceeds; failing costs about 3%. The trader borrows.

$\Rightarrow$ The special rate can now fall below zero. GARP: "the new upper limit for the special spread is the penalty rate rather than the GC rate."

Interpretation

The term spread here is $0.22\% - (-0.10\%) = 0.32\%$, above the GC rate, which the pre-2009 limit would not have allowed. In June 2009 demand to short the OTR drove the spread all the way up to the new limit. The penalty is designed to fall as rates rise, because higher rates already make failing costly.

Why the other options are wrong

A is incorrect. True only before May 2009, when failing cost no more than lending at 0%.

C is incorrect. Polarity flip: the penalty *falls* as fed funds rises and is at its 3% maximum when rates are near zero.

D is incorrect. Applies the pre-2009 cap. Since the penalty, a spread above GC is possible and is no evidence of mispricing.

LO: LTR-13 f: Identify the characteristics of special spreads and explain the typical behavior of U.S. Treasury special spreads over an auction cycle. • **Reference:** Tuckman and Serrat, Ch.12, 14.3 Special Spreads in the United States and the Level of Rates; Table 14.2

LTR-15 • The US Dollar Shortage in Global Banking and the International Policy Response • Vardeez: VERY IMPORTANT

LTR-15 a

P2 CONCEPT

Identify the causes of the U.S. dollar shortage during the Great Financial Crisis.

Workbook: D8 US Dollar Shortage and the Cross-Currency Basis • Tier D

WHAT IS TESTED HERE

Why P2. Vardeez rates the reading VERY IMPORTANT: "Short Reading and anything can be tested here - so know it well". It is also Tier D workbook topic D8 (Map #46, 2 of 16 recall sources), reported as "why non-US banks faced dollar shortage during GFC".
Vardeez: no LO-level pointer; the reading note applies. This block carries the shared background of the reading (the three ways to fund a foreign asset, funding risk, the funding gap); LTR-15 b and c recap it and point back here.

Reported exam shapes

- **S1** Why non-US, chiefly European, banks faced a US dollar shortage in the GFC (workbook D8.1).

What the questions look like: statement questions, no calculation. A board, a panel or a consultant asks which statement best explains the shortage, or which factor did *not* contribute. The distractors repeat: the domestic bank-run story, net *short* instead of net long dollars, a falling dollar, foreign central banks "issuing" fewer dollars, and writedowns (solvency) as the cause. A few numbers appear as recognition facts, never as inputs.

1. Not a bank run

McGuire and von Peter (BIS Working Paper 291, 2009; GARP Ch.16) ask what caused the global shortage of US dollars, which banking systems it hit, and how it developed so quickly when dollar liquidity had looked plentiful. Their answer needs "a departure from the familiar domestic bank run story". In the open-economy run model, depositors convert domestic deposits into foreign currency and the demand exhausts the country's FX reserves; that story still fits emerging markets. The GFC shortage came instead from the **international operations of the major banking systems** and from the **global interbank and FX swap markets** they relied on. "Dollar shortage" here means the difficulty banks faced in securing *short-term US dollar funding*, not the postwar scarcity of dollars that Kindleberger and Triffin described.

2. The build-up, 2000 to mid-2007

Balance sheet expansion. Banks' foreign claims grew from USD 10 trillion at the start of 2000 to USD 34 trillion by end-2007, a large rise even when scaled by world GDP; year-on-year growth went from about 10% in 2001 to nearly 30% by mid-2007. The period of financial innovation behind it: **structured finance**, the spread of **universal banking** (commercial and investment banking plus proprietary trading) and the growth of **hedge funds**, served with prime brokerage. European banks stand out: Swiss banks' foreign claims went from about five to more than seven times Swiss GDP, and US dollar (and other non-euro) positions were more than half of European banks' foreign asset growth.

Appetite for dollar assets. The assets were US dollar claims on *non-banks*: retail and corporate loans, loans to hedge funds, and structured finance products on US mortgages. Their high ratings made the perceived risk low during the build-up; in the crisis they became the main source of mark-to-market losses.

Three ways to fund a foreign currency asset

1. **Spot:** borrow domestic currency, convert it spot, buy the asset. No later foreign currency need, but an unhedged currency mismatch.
2. **FX swap:** borrow domestic currency and swap it into the foreign currency (spot leg now, promise to swap back at maturity).
3. **Direct:** borrow the foreign currency from the interbank market, from non-bank participants or from central banks.

Banks are assumed to hedge on-balance sheet currency mismatches with swaps and forwards, so route 1 in practice becomes route 2. The second leg of the swap in route 2 "is not that different" from route 3: in both, the bank must *deliver foreign currency* when the contract comes due. The balance sheet identity (eq. 16.1) says a net long position in foreign currencies is mirrored by net borrowing in the domestic currency: $\sum_{i\neq 0}(A_i - L_i) = L_0 - A_0$, with L_0 including equity.

- **Cross-currency funding:** the extent to which banks fund in one currency and invest in another.
- **Funding risk:** "the risk that funding liabilities cannot be rolled over". Its size depends on the maturity transformation in the balance sheet.
- **Foreign currency funding gap:** exists when the investment horizon of the foreign currency assets exceeds the maturity of the foreign currency funding or FX swaps. If the funding cannot be rolled, assets meant to be held must be sold, possibly in distressed markets. How it is measured is LTR-15 b.

The long dollar book

The Japanese and major European banking systems ran large **net long** positions in foreign currencies, mostly US dollars, mirrored by net borrowing in their own currency from home residents and bridged by FX swaps. European banks' combined long dollar position reached about **USD 700 billion** by mid-2007 (USD 1.3 trillion with Japanese banks' USD 600 billion). UK banks, for example, were long USD 200 billion on about USD 2 trillion of gross dollar claims, funded by net sterling borrowing of some USD 550 billion. The pattern fits European universal banks using their retail arms to fund dollar-heavy investment banking. Exceptions: Belgian and French banks were roughly neutral, and **Spanish banks were short dollars**, borrowing dollars to fund euro lending at home until mid-2006. The swaps were very short: 78% of FX swap turnover was under seven days.

European banks' lower-bound dollar funding gap was **USD 1.0 to 1.2 trillion** by mid-2007, met by interbank borrowing, deposits from central banks and FX swaps, with about USD 1 trillion more

from US money market funds (Figure 1).

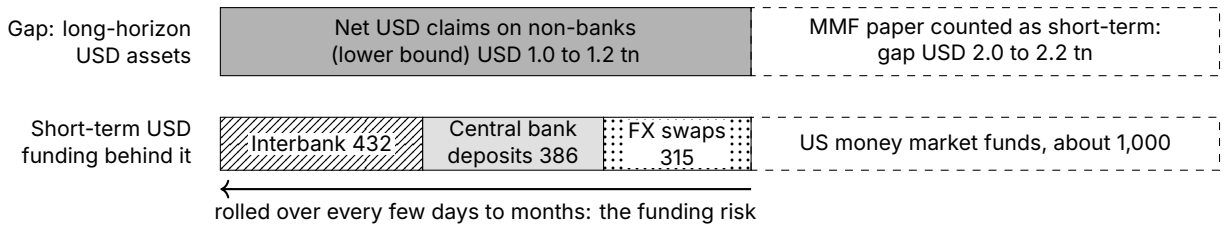


Figure 1. European banks' US dollar funding gap at mid-2007 (USD billion; GARP Section 16.3.4). Top row: the long-horizon dollar investment in non-banks (solid grey) and the extra gap if money market fund paper is treated as short-term (dashed outline). Bottom row: the short-term sources that filled the lower-bound gap: interbank (hatched), central bank deposits (light grey) and FX swaps (dotted), which sum to 1,133.

3. How the gap became a shortage

Funding shortened.

- From **August 2007** heightened counterparty risk and liquidity concerns compromised short-term **interbank** funding.
- **FX swap dislocations** made swapped dollars even more expensive, as European banks' dollar needs exceeded other entities' needs in other currencies (the cross-currency basis of LTR-16).
- After **Lehman** (September 2008), **money market funds** facing redemptions withdrew from bank-issued paper, a wholesale run.
- **Central banks withdrew dollar reserves** placed with commercial banks: their deposits fell USD 257 billion between mid-2007 and end-2008, partly as emerging-market authorities supported their own banks.

Assets lengthened.

- Apart from liquid US government securities, dollar assets such as **structured finance products** could not be sold into illiquid markets without large losses.
- **Off-balance sheet vehicles** were brought back onto balance sheets and **prearranged credit commitments** were drawn.
- So European banks' dollar claims *rose* by USD 248 billion (3%) between Q2 2007 and Q1 2008, despite writedowns; deleveraging appeared only in Q4 2008.

"**The effective holding period of assets lengthened just as the maturity of funding shortened.**" This endogenous rise in maturity mismatch, difficult to hedge ex ante, generated the shortage (Figure 2). Home central banks could not fix it: they can be lender of last resort in their own currency but **cannot create dollars** (LTR-15 c). Before Lehman, banks with US offices bridged part of the gap themselves: those offices raised local dollar liabilities (including Fed facilities) by USD 329 billion (13%) and sent the funds to head office through inter-office transfers.

The authors' verdict: "What pushed the system to the brink was not cross-currency funding per se, but rather too many large banks employing funding strategies in the same direction, the funding equivalent of a **crowded trade**." Every cause is about where the dollars came from and how long they lasted, not about the credit quality of the assets: a funding-structure problem, not a solvency problem.

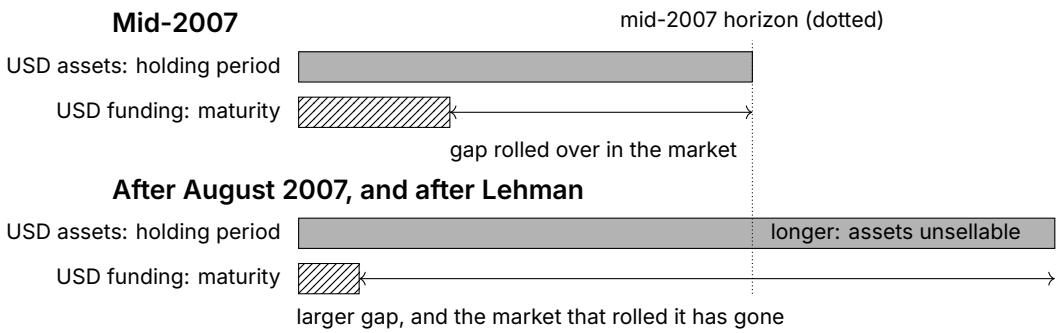


Figure 2. The endogenous rise in maturity mismatch (schematic, lengths illustrative). Grey bars: effective holding period of dollar assets; hatched bars: maturity of dollar funding. In the crisis the asset bar stretched (illiquid structured products, vehicles back on balance sheet, drawn credit lines) while the funding bar shrank (interbank, FX swaps, money market funds and central bank deposits all pulled back).

Cause	Stage	What to remember
Balance sheet expansion	Build-up	USD 10 to 34 trillion; structured finance, universal banking, hedge funds
Appetite for USD assets	Build-up	Claims on non-banks; long-horizon, later illiquid
Cross-currency funding	Build-up	Long dollars funded in home currency, bridged by short FX swaps
Maturity transformation	Build-up	Funding gap USD 1.0 to 1.2 trillion (lower bound)
Funding sources withdrew	Trigger	Interbank, FX swaps, money market funds, central bank deposits
Assets could not be shed	Trigger	Structured products, vehicles, drawn commitments
No home lender of last resort	Amplifier	Central banks cannot create dollars
Crowded trade	Amplifier	Many large banks, same funding strategy

Exam traps

- **Bank-run sibling.** Depositors converting home deposits into dollars and draining FX reserves is the emerging-market story the paper departs from.
- **Long, not short.** The shortage hit banks *long* dollar assets funded short-term. Borrowing dollars to fund home lending is Spain, the exception.
- **Direction of the mismatch.** Asset holding period *lengthened*, funding maturity *shortened*: the mismatch rose. Reversing either half is the distractor.
- **Demand rose.** Dollars were scarce because demand to roll funding surged; the dollar came under *upward* pressure, not depreciation.
- **Who withdrew.** Central banks withdrew dollar *reserve placements* from commercial banks; only the Fed can "issue" dollars, so "foreign central banks issued fewer dollars" is never a cause.
- **Funding, not solvency.** Writedowns hurt, but banks holding liquid Treasuries still had to roll the short-term funding behind their dollar books.
- **Build-up versus proximate cause.** The shift into non-bank dollar assets built the need; the proximate cause is the endogenous rise in maturity mismatch.

Practice questions: LTR-15 a

Question LTR-15 a.1

Bank Q497 • AP Mock 1 Q55 (rebuilt)

In the analysis of the US dollar shortage, the extent to which banks fund their assets in one currency and invest in another (for example, using FX swaps to convert euro funding into US dollars) is best described as:

A. Interest rate swapping

B. Cross-currency funding

C. Carry trading

D. Currency hedging of the net open position

Question LTR-15 a.2

Bank Q499 • AnalystPrep Q.4167 (rebuilt)

Banks' foreign claims grew from USD 10 trillion at the start of 2000 to USD 34 trillion by end-2007, and year-on-year growth in foreign claims rose from around 10% in 2001 to nearly 30% by mid-2007. This acceleration took place during a period of financial innovation. That period included all of the following EXCEPT:

A. Auctions of US dollars to banks by central banks

B. Significant growth in the hedge fund industry

C. The emergence of structured finance

D. The spread of universal banking

Question LTR-15 a.3

Bank Q498 • AP Mock 2 Q23 • AnalystPrep Q.4166

In a discussion panel on preventing a repeat of the Great Financial Crisis, a delegate lists the following causes of the US dollar shortage during the crisis:

I. Increased appetite of banks for foreign currency assets, notably US dollar claims on non-banks

II. Cross-currency funding

III. US dollar funding gap

Which of the above causes are correct?

A. I and II

B. I and III

C. II and III

D. I, II and III

Question LTR-15 a.4 Bank Q501 • BT Mock Q79

According to McGuire and von Peter, which of the following best summarizes the proximate cause of the US dollar shortage during the global financial crisis?

A. Depositors initiated a classic run on the banks and converted their domestic deposits into foreign currency.

B. The effective maturity of banks' dollar funding shortened while the holding period of their dollar assets lengthened, increasing their maturity mismatch.

C. The effective maturity of banks' dollar funding lengthened while the holding period of their dollar assets shortened, decreasing their maturity mismatch.

D. Banks shifted their portfolios toward non-bank assets, including retail and corporate loans, loans to hedge funds and holdings of securities.

Question LTR-15 a.5 Bank Q503 • Schweser Module Quiz 80.2 Q1

Which of the following factors is least likely a contributor to the US dollar shortage during the Global Financial Crisis (GFC)?

A. Banks' prearranged credit commitments were drawn.

B. Structured finance products became more difficult to sell.

C. Central banks globally reduced their issuance of US dollars.

D. Banks brought off-balance sheet vehicles back onto their balance sheets.

Question LTR-15 a.6 Bank Q502 • BT P2.T8.25.1.1 (rebuilt)

In 2008, a European bank relied on short-term US dollar funding to finance its US dollar-denominated assets. As interbank lending froze and liquidity dried up, it struggled to roll over its obligations. Which of the following is LEAST likely to explain the global US dollar shortage during the Global Financial Crisis?

A. Banks relied heavily on short-term US dollar wholesale and FX swap funding, and could not roll it over when interbank markets froze.

B. Monetary authorities and other suppliers of funds withdrew dollar placements from commercial banks, reducing the supply of dollar funding.

C. Banks could not easily cut their dollar assets because structured finance products were hard to sell, so their dollar funding needs persisted.

D. A sharp fall in demand for US dollars depreciated the currency, reducing its availability to foreign banks.

Question LTR-15 a.7 Workbook D8.1 • S1 • Map #46

A risk consultant is presenting to the board risk committee of a German universal bank on lessons from the US dollar shortage of 2007 to 2009, drawing on the BIS international banking statistics. Which of the following statements best explains why European banks faced a US dollar shortage during the crisis?

A. They had built large net long US dollar positions in non-bank assets funded by short-term interbank borrowing, FX swaps and money market fund paper, so when those sources withdrew, their funding shortened just as the assets became harder to sell.

B. Their domestic retail depositors ran on them and converted home currency deposits into US dollars, so the resulting demand for foreign currency exhausted their home central banks' foreign exchange reserves within a few weeks.

C. They had built large net short US dollar positions, borrowing dollars short-term to fund long-term lending in their home currencies, so when dollar lenders withdrew they had no dollar assets left to sell to repay them.

D. The shortage arose only at banks with large losses on US structured finance products, since banks whose dollar assets were mainly US Treasury and agency securities could roll over their dollar funding throughout the crisis.

Solutions • LTR-15 a

Solution LTR-15 a.1

Answer: B

Bank Q497 • AP Mock 1 Q55

Step 1 – Separate the currency of assets from the currency of funding

The paper breaks banks' assets and liabilities down by currency "to examine cross-currency funding, or the extent to which banks fund in one currency and invest in another".

Step 2 – Match the example

Euro funding converted into dollars with FX swaps to buy dollar assets is exactly that.
⇒ The term is cross-currency funding.

Interpretation

The original mock key gave carry trading; the bank corrected it to B, which is the paper's own term. Cross-currency funding is the bridge between the long dollar book and the funding gap (Section 2).

Why the other options are wrong

A is incorrect. An interest rate swap exchanges interest flows, usually in one currency; it does not describe funding and investing in different currencies.
C is incorrect. A carry trade is a speculative position in a high-yielding asset funded by a low-yielding liability; the banks were largely hedged through swaps, not speculating on the currency.
D is incorrect. Hedging is how the open currency position is neutralised (the FX swap), not the funding structure itself.

LO: LTR-15 a: Identify the causes of the U.S. dollar shortage during the Great Financial Crisis. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.1 Introduction and 16.3.3 Cross-Currency Funding Positions

Solution LTR-15 a.2

Answer: A

Bank Q499 • AnalystPrep Q.4167

Step 1 – The innovation period, 2000 to 2007

GARP lists three features: the emergence of structured finance, the spread of universal banking (commercial and investment banking plus proprietary trading), and significant growth in the hedge fund industry, to which banks offered prime brokerage.

Step 2 – Place dollar auctions in time

Central bank dollar auctions were the policy response of 2008 to 2009 (LTR-15 c), after the shortage had appeared.
⇒ Auctions are the exception.

Interpretation

The question tests the build-up versus response timeline. The innovations built the dollar book; the auctions, fed by Fed swap lines, relieved the shortage.

Why the other options are wrong

B is incorrect. Hedge fund growth, with banks as prime brokers, was part of the innovation period.
C is incorrect. Structured finance emerged then, and later became the main source of mark-to-market losses.
D is incorrect. Universal banking spread then, and European universal banks used retail funding to expand dollar investment banking.

LO: LTR-15 a: Identify the causes of the U.S. dollar shortage during the Great Financial Crisis. • **Reference:** McGuire

and von Peter (2009), GARP Ch.16, 16.3.2 Balance Sheet Expansion since 2000

Solution LTR-15 a.3 Answer: D

Bank Q498 • AP Mock 2 Q23

Step 1 – Statement I

As balance sheets grew, so did banks' appetite for foreign currency assets, notably US dollar claims on non-banks (loans, hedge fund lending, structured products). True.

Step 2 – Statement II

These assets were funded in other currencies and converted with FX swaps: cross-currency funding. True.

Step 3 – Statement III

Dollar assets with longer horizons than the dollar funding behind them left a funding gap that had to be rolled over. True.

$\Rightarrow$ All three are causes: the need (I), the route (II) and the exposure (III).

Interpretation

The three statements are the paper's chain: accumulate dollar assets, fund them across currencies, carry a short-term gap. The crisis then made the gap impossible to roll.

Why the other options are wrong

A is incorrect. Leaves out the funding gap, the quantity that could not be rolled over.

B is incorrect. Leaves out cross-currency funding, which tied non-dollar funding to dollar assets through short swaps.

C is incorrect. Leaves out the accumulation of dollar assets that created the funding need in the first place.

LO: LTR-15 a: Identify the causes of the U.S. dollar shortage during the Great Financial Crisis. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.1 and 16.3

Solution LTR-15 a.4 Answer: B

Bank Q501 • BT Mock Q79

Step 1 – Funding side

Interbank funding was compromised from August 2007, swapped dollars became dearer, and after Lehman money market funds and central banks withdrew. The effective maturity of funding shortened.

Step 2 – Asset side

Structured products could not be sold without large losses, vehicles came back on balance sheet and credit lines were drawn. The effective holding period of assets lengthened.

$\Rightarrow$ "The effective holding period of assets lengthened just as the maturity of funding shortened": an endogenous rise in maturity mismatch.

Interpretation

"Proximate" is the key word. The portfolio shift of D built the exposure; the proximate cause is what turned the exposure into a shortage.

Why the other options are wrong

A is incorrect. The domestic bank-run story, which the paper says the dollar shortage requires a departure from.

C is incorrect. Reverses both movements, and with them the conclusion: the mismatch rose, it did not fall.

D is incorrect. The pre-crisis build-up that created the funding need, not the proximate cause.

LO: LTR-15 a: Identify the causes of the U.S. dollar shortage during the Great Financial Crisis. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.4 The US Dollar Shortage

Solution LTR-15 a.5 Answer: C

Bank Q503 • Schweser Module Quiz 80.2 Q1

Step 1 – What hampered deleveraging

GARP names three factors that stopped banks from shrinking their dollar assets: structured finance products were hard to sell into illiquid markets, off-balance sheet vehicles were brought back onto balance sheets, and prearranged credit commitments were drawn. A, B and D are all on the list.

Step 2 – Test C

Only the Federal Reserve issues US dollars; other central banks cannot reduce an issuance they never had.

⇒ C is not a contributor.

Interpretation

The true fact behind the distractor: central banks *withdrew dollar reserves they had placed with commercial banks* (deposits fell USD 257 billion between mid-2007 and end-2008). That is a withdrawal of funding, not a cut in issuance. The same fact explains why home central banks needed the Fed's swap lines (LTR-15 c).

Why the other options are wrong

A is incorrect. A real contributor: drawn commitments added dollar assets just when funding was scarce.

B is incorrect. A real contributor: illiquid structured products lengthened the effective holding period.

D is incorrect. A real contributor: consolidating vehicles brought their dollar assets and funding needs onto the balance sheet.

LO: LTR-15 a: Identify the causes of the U.S. dollar shortage during the Great Financial Crisis. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.4 The US Dollar Shortage

Solution LTR-15 a.6 Answer: D

Bank Q502 • BT P2.T8.25.1.1

Step 1 – The mechanism

Long-horizon dollar assets were funded with short-term dollar liabilities (interbank, wholesale, FX swaps). When counterparty and liquidity concerns froze those markets, the liabilities could not be rolled (A). Official holders also withdrew placements (B), and assets could not be cut quickly (C).

Step 2 – Test D

Banks needed dollars urgently to roll their funding, so demand for dollars rose. GARP notes that the swap lines later eased the *upward* pressure on the US dollar.

⇒ D reverses the direction of the shock: it is the least likely explanation.

Interpretation

A shortage is excess demand at the going price. Any option built on falling dollar demand or a weaker dollar contradicts the paper.

Why the other options are wrong

A is incorrect. The core cause: the rollover failure of short-term dollar funding.
B is incorrect. A genuine cause: monetary authorities' deposits with commercial banks fell USD 257 billion.
C is incorrect. A genuine cause: illiquid structured assets kept dollar funding needs alive.

LO: LTR-15 a: Identify the causes of the U.S. dollar shortage during the Great Financial Crisis. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.4 and 16.5

Solution LTR-15 a.7 — Answer: A

Workbook D8.1 • S1

Step 1 – The position

By mid-2007 European banks' combined long US dollar position was about USD 700 billion, mirrored by net borrowing in their own currencies from home residents. Their lower-bound dollar funding gap (net dollar claims on non-banks) was USD 1.0 to 1.2 trillion, covered by interbank borrowing, central bank deposits and FX swaps, plus about USD 1 trillion from US money market funds.

Step 2 – The trigger

From August 2007 interbank funding was compromised; FX swap dislocations made swapped dollars dearer; after Lehman, money market funds and central bank reserve managers withdrew. Structured assets could not be sold without large losses, vehicles came back on balance sheet and credit lines were drawn.
$\Rightarrow$ "The effective holding period of assets lengthened just as the maturity of funding shortened": an endogenous rise in maturity mismatch that generated the shortage, and that home central banks could not fix because they cannot create dollars.

Interpretation

The consultant's point for the board: the vulnerability was visible only on the consolidated global balance sheet (LTR-15 b), and became systemic because many large banks ran the same funding strategy, "the funding equivalent of a crowded trade".

Why the other options are wrong

B is incorrect. Sibling swap. This is the open-economy domestic bank run story (depositors converting to foreign currency and draining FX reserves) that the paper says the dollar shortage requires "a departure from"; it remains relevant to emerging markets.
C is incorrect. Polarity flip. The major European systems were net long dollars (dollar assets funded in home currency and swapped). Borrowing dollars to fund home-currency lending describes Spanish banks until mid-2006, the exception.
D is incorrect. Scope/condition error. Writedowns worsened the picture, but the shortage was a funding mismatch across the system: banks with liquid US government securities still had to roll the short-term dollar funding behind all their dollar assets as suppliers withdrew.

LO: LTR-15 a: Identify the causes of the U.S. dollar shortage during the Great Financial Crisis. LTR-15 b: Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities. • **Reference:** McGuire and von Peter, BIS Working Paper 291 (2009), sections 16.2 to 16.4 and 16.6

LTR-15 • The US Dollar Shortage in Global Banking and the International Policy Response • Vardeez: VERY IMPORTANT

LTR-15 b

P2 CONCEPT

Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P2. Vardeez flags the whole reading VERY IMPORTANT ("anything can be tested here"). The map carries no workbook section for this LO, but D8's revision notes have a bullet on the consolidated view, and workbook D8.1 tags LTR-15 a and b (it is worked as Question LTR-15 a.7).

Vardeez: no LO-level pointer. His notes for this LO stress two points: the unit of analysis is the global consolidated balance sheet, not nationality or residency, and the funding gap is a *range*, not a point.

Reported exam shapes: none listed. The questions below follow the LO wording.

What the questions look like: definitions (maturity mismatch, funding gap, the lower and upper bounds), the three funding routes and which risk each leaves, the asymmetry between domestic and foreign currency funding risk, and Roman-numeral or "which comment is correct" questions on residency versus consolidated data. No calculations.

1. Recap

Banks fund foreign currency assets spot, through FX swaps or by borrowing the currency; funding risk is the risk that the funding cannot be rolled; a funding gap exists when the assets' horizon exceeds the funding's maturity (LTR-15 a, Section 2). This LO asks *where* and *how* to measure those mismatches, and why it matters.

2. Why foreign currency mismatches deserve special attention

Maturity transformation is an essential function of banking, and banks would find it unprofitable to eliminate it. In the domestic currency the central bank can act as **lender of last resort** and supply enough liquidity to end a funding shortage, which is both time-honoured practice (Bagehot) and optimal policy. By contrast, "**central banks cannot create foreign currencies**"; their ability to meet banks' demand for foreign currency is constrained by the exchange rate regime or limited to available FX reserves. A foreign currency funding gap may therefore have to be met from international sources. Vardeez calls this asymmetry the reading's central point; any option that treats domestic and foreign funding risk as equivalent has lost it. It is also why the policy response (LTR-15 c) had to come from the Fed.

3. Nationality is not residency

Internationally active banks manage currency and maturity positions **across the consolidated global entity, not office by office**. A large measured mismatch in one office may be hedged off-balance sheet or offset by positions booked by offices elsewhere, leaving a matched book for the bank as a whole (Figure 3). The BIS data show how far apart the two views are:

- In most systems less than half of banks' foreign claims are booked by home offices. Swiss banks are the extreme: foreign claims over 80% of total assets, yet only **18%** of them booked in Switzerland.
- Seen from host countries, banks' cross-border positions were almost half of external assets in Belgium, Switzerland, Germany, Japan and the UK at end-2007. The offices of *foreign* banks alone held nearly **40% of UK external assets**.

So a residency-based "national balance sheet" (balance of payments, domestic credit data, BIS locational statistics by residency) is a poor indicator of vulnerability, for two reasons:

1. **Host-country data can show a mismatch that does not exist**, or swing with foreign banks' activity. Swiss, German and Dutch banks grew largely through their UK offices; foreign banks' offices in the UK changed their net positions by USD -490 billion before the crisis against -190 billion for UK banks.
2. **Home-country data can miss a mismatch that does exist**, because a quarter or more of each system's assets (except Japan's) sit in offices abroad.

"Vulnerabilities can only be measured by taking into account the entire balance sheet of the consolidated global entity."

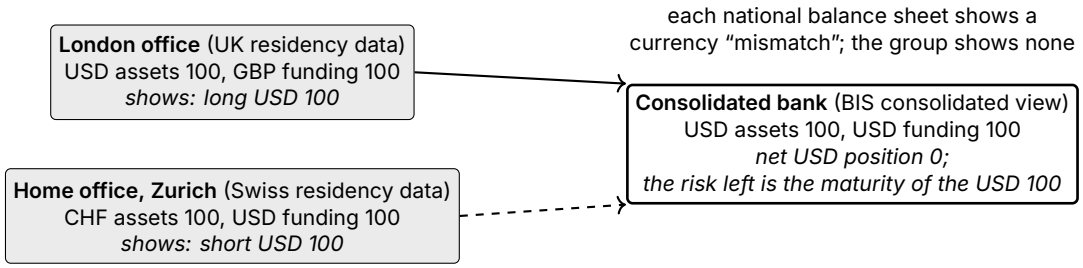


Figure 3. One bank, three views (illustrative numbers). Each office looks mismatched in its host country's residency data, but the positions offset on the consolidated balance sheet. The converse also holds: a maturity mismatch booked abroad is invisible in home-country data. Only the consolidated entity shows the true currency and maturity position.

4. Measuring the maturity mismatch

Published bank accounts lack the currency, maturity and counterparty breakdowns needed, and national statistics are residency-based. The authors splice the BIS consolidated statistics (by nationality) with the locational statistics by nationality to rebuild each banking system's global balance sheet by currency and counterparty. Maturity is not observed, so **counterparty type proxies for maturity**: interbank positions and FX swaps are short (78% of swap turnover under seven days), while claims on non-banks are the *desired investment portfolio* (loans, hedge fund lending, Treasuries through to structured products).

With full hedging, $A_i = L_i + S_i$ (S_i the FX swaps). Short-term funding $(L_i - L_i^{LT}) + S_i = A_i - L_i^{LT}$ can be met by liquid assets $A_i - A_i^{LT}$, so

$$\boxed{\text{Funding gap}_i = (A_i - L_i^{LT}) - (A_i - A_i^{LT}) = A_i^{LT} - L_i^{LT}} \qquad \text{(eq. 16.2)}$$

the long-term foreign currency assets not supported by long-term foreign currency liabilities: the amount to be rolled over before the assets mature. The estimate is a **range**, because the maturity of liabilities to non-banks is unknown:

Non-bank liabilities assumed	Measure	European banks, mid-2007
Long-term (they match non-bank claims)	**Lower bound:** net USD claims on non-banks	USD 1.0 to 1.2 trillion
Money market fund paper short-term	Lower bound plus MMF liabilities	USD 2.0 to 2.2 trillion
All short-term	**Upper bound:** gross USD claims on non-banks	USD 6.5 trillion

Dutch, German, Swiss and UK banks had the largest gaps; German banks leaned more on interbank funding, UK banks on cross-currency funding. Later readings are clouded by writedowns: FX swaps are backed out as a residual, so a writedown mechanically shrinks the measured gap. The lower bound fell nearly 50% by end-Q1 2009, to USD 583 billion if banks closed out the funding behind written-down assets, or about USD 880 billion if they rolled it over.

> **Correction to the source notes.** (1) Vardeez (and Schweser) write that "Europe primarily funded foreign claims with domestic currency" while "other countries funded foreign claims with foreign currencies, mostly US dollars". GARP's sentence is about the *currency of the claims*: Belgian, Dutch, French and German banks' foreign claims were mostly in euros (intra-euro area), while Japanese, Swiss and UK banks' foreign claims were mostly in foreign currencies, mainly dollars. On funding GARP says the opposite of the second half: the long dollar books (UK, Swiss, German, Dutch, Japanese) were funded by borrowing *domestic* currency from home residents and swapping it. (2) Vardeez's "USD 1 to 2.2 trillion" merges two estimates: the lower bound is USD 1.0 to 1.2 trillion, USD 2.0 to 2.2 trillion counts money market fund paper as short-term, and the upper bound is USD 6.5 trillion. Question LTR-15 b.7 tests point (1).

5. Why it matters: the macroprudential view

Stresses build up across the balance sheet as mismatches in maturity, currency and counterparty, and the relevant decision-makers (banks) transcend national borders. Measured this way, the data show that (i) stresses can build up in a national banking system without showing in its home country's residency-based statistics, and (ii) banks' cross-border positions are large relative to countries' external positions, clouding what a "national balance sheet" means for residents. Aggregation also reveals what a single-bank view misses: the **crowded trade**, many large banks funding in the same direction. Banks do not disclose these breakdowns, which is why supervisors and the BIS data matter.

	Residency-based view	Consolidated (nationality) view
Unit	Offices located in a country	The banking group worldwide
Data	Balance of payments, domestic credit, BIS locational by residency	BIS consolidated, spliced with locational by nationality
Currency mismatch	Can show one that is offset elsewhere	Nets offices: shows the real open position
Maturity mismatch	Misses gaps booked abroad	Captures the dollar funding gap (a range)
Use	Host-country external position	Funding vulnerability of a banking system

Exam traps

- **Asymmetry.** Domestic funding risk has a home lender of last resort; foreign currency funding risk does not. Treating them as equivalent is wrong.
- **Bounds swapped.** Long-term non-bank liabilities give the *lower* bound (net position); all short-term gives the *upper* bound (gross position).
- **A range, not a point.** USD 1.0 to 1.2 trillion is the lower bound; 2.0 to 2.2 adds money market funds; 6.5 is the upper bound.
- **Currency versus maturity mismatch.** An FX swap hedges currency risk but leaves maturity (rollover) risk; direct dollar borrowing matches currency but can still leave a gap; spot conversion leaves currency risk but no later dollar need.
- **Office by office.** Positions are managed on the consolidated global entity; a local mismatch can be hedged elsewhere.
- **Swiss 18%.** Only 18% of Swiss banks' foreign claims are booked in Switzerland, not "most".
- **Nationality is not residency.** Bank nationality does not tell you where positions are booked, and residency data do not tell you whose they are.

Practice questions: LTR-15 b

Question LTR-15 b.1

Bank Q506 • AnalystPrep Q.4177 (rebuilt)

Which of the following is correct about a maturity mismatch in a bank's foreign currency positions?

A. It is not visible on the balance sheet and therefore has no effect on the bank's liquidity.

B. It exists when the investment horizon of the bank's assets exceeds the maturity of the liabilities funding them, creating the need to roll over funding.

C. It arises only when assets and liabilities are denominated in different currencies.

D. It earns the bank a spread when the yield curve is inverted, because short-term funding then costs more than long-term lending.

Question LTR-15 b.2

Bank Q504 • AnalystPrep Q.4176 (rebuilt)

Which of the following best defines a bank's US dollar funding gap?

A. The amount of short-term US dollar funding a bank requires because its dollar assets have longer horizons than the funding behind them, that is, the dollars to be rolled over before the assets mature

B. The excess of a non-US bank's dollar-denominated liabilities over its dollar-denominated assets

C. A country's inability to obtain enough dollars to finance its international trade

D. A high-yielding currency asset financed with a low-yielding currency liability

Question LTR-15 b.3

Bank Q508 • BT P2.T8.25.1.2 (rebuilt)

The treasurer of a bank whose domestic currency is the WK wants to acquire USD 1 million of 10-year US Treasuries and compares three funding choices:

Choice	Transactions
1. Borrow domestic, convert spot	Borrow WK 900,000 domestically; convert to USD 1 million spot; buy the Treasuries
2. FX swap, rolled	Borrow WK 900,000 domestically; enter a 1-year FX swap of WK 900,000 for USD 1 million (reversed in one year, then rolled); buy the Treasuries
3. Borrow USD directly	Borrow USD 1 million from foreign sources for one year; buy the Treasuries

Which statement best describes the implications?

A. In Choice 1, no subsequent currency transactions are required and no exchange rate exposure remains.

B. In Choice 2, the currency risk is hedged by the swap, so the maturity mismatch is also eliminated.

C. In Choice 3, there is no currency mismatch because assets and liabilities are both in USD, but USD must be delivered when the borrowing matures, so funding risk remains.

D. All three choices require delivery of USD at maturity and therefore face the same foreign currency rollover risk.

Question LTR-15 b.4

Bank Q509 • Schweser Module Quiz 80.1 Q1

If the maturity of a bank's US dollar liabilities to non-banks is longer term, a(n):

A. lower bound of the US dollar funding gap can be established as the net US dollar position to non-banks.

B. upper bound of the US dollar funding gap can be established as the net US dollar position to non-banks.

C. lower bound of the US dollar funding gap can be established as the gross US dollar position to non-banks.

D. upper bound of the US dollar funding gap can be established as the gross US dollar position to non-banks.

Question LTR-15 b.5 New • built from GARP Ch.16, Section 16.2

A liquidity supervisor at a euro area national central bank explains to a banking committee why, after 2007 to 2009, supervisors examine banks' foreign currency maturity mismatches separately, even though maturity transformation in euros is ordinary banking business. Which of the following is the best explanation?

A. Foreign currency assets were of lower credit quality than domestic assets, so foreign currency mismatches carry more solvency risk than domestic ones.

B. Maturity transformation is not a normal banking function in foreign currencies, so any foreign currency funding gap breaches prudential rules.

C. A national central bank can act as lender of last resort in any currency its banks borrow in, so foreign currency gaps only need to be reported to it more often.

D. A central bank can supply liquidity without limit in its own currency but cannot create foreign currency, so a foreign currency funding gap can be met only from FX reserves or from international sources.

Question LTR-15 b.6 Bank Q505 • AnalystPrep Q.4171 (rebuilt)

Consider the following statements about the structure of large international banks' balance sheets:

I. Banks manage currency and maturity positions on a consolidated global basis rather than office by office.

II. A mismatch at one office may be left unhedged locally because it is hedged off-balance sheet or offset by on-balance sheet positions booked at the bank's offices elsewhere.

III. A country's national banking balance sheet is a reliable indicator of the vulnerabilities of the banks headquartered there.

Which statements are correct?

A. I only

B. II only

C. I and II only

D. I, II and III

Question LTR-15 b.7 New • built from GARP Ch.16, Sections 16.3.1 and 16.3.3

An economist at an international organisation uses residency-based statistics (the balance of payments and national external positions) to judge which banking systems were exposed to a US dollar funding shortage before the crisis. A colleague who works with the BIS consolidated banking statistics comments on the approach. Which comment is correct?

A. Residency data are adequate, because large banks manage currency and maturity positions office by office, so each host country's statistics capture the mismatches of the offices located there.

B. A dollar mismatch booked by a foreign bank's London office can be offset elsewhere in its group, so UK data can show a mismatch the bank does not have, while home-country data can miss mismatches its banks book abroad.

C. Residency data miss little for Swiss banks, because the great majority of Swiss banks' foreign claims are booked by their offices in Switzerland.

D. Japanese, Swiss and UK banks funded their dollar claims mainly by borrowing dollars, so their dollar books were matched and needed no consolidated analysis.

Solutions • LTR-15 b

Solution LTR-15 b.1 — Answer: B

Bank Q506 • AnalystPrep Q.4177

Step 1 – Define the mismatch

A maturity mismatch means assets are held for longer than the liabilities (or FX swaps) that fund them.

Step 2 – Its consequence

The liabilities must be rolled before the assets mature; if rollover fails, assets are sold, possibly at a loss. In a foreign currency this is the funding gap.

$\Rightarrow$ Only B states the definition and its consequence.

Interpretation

The mismatch is a maturity concept. Currency enters separately: the same maturity gap is more dangerous in a foreign currency because no home central bank can create that currency (Section 2).

Why the other options are wrong

A is incorrect. Mismatches can be estimated from balance sheet data, and they are the source of funding liquidity risk.

C is incorrect. Confuses maturity mismatch with currency mismatch: a maturity mismatch can exist within one currency.

D is incorrect. Maturity transformation earns a spread when the curve slopes upward; with an inverted curve short-term funding costs more than long-term lending earns.

LO: LTR-15 b: Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.2 Banks' International Positions: Concepts and Data

Solution LTR-15 b.2 — Answer: A

Bank Q504 • AnalystPrep Q.4176

Step 1 – The paper's definition

The funding gap is "the amount of short-term US dollar funding banks require": long-horizon dollar assets not supported by long-term dollar liabilities, $A^{LT} - L^{LT}$ (eq. 16.2).

Step 2 – Meaning

It is the amount to be rolled over before the investments mature.

$\Rightarrow$ A maturity concept inside one currency, estimated from net (lower bound) to gross (upper bound) dollar claims on non-banks.

Interpretation

European banks' gap was USD 1.0 to 1.2 trillion by mid-2007 on the lower-bound measure. A bank can be fully hedged against currency risk and still have a large funding gap.

Why the other options are wrong

B is incorrect. Reverses the position: the gap arises from long-dated dollar *assets* exceeding stable dollar funding, not from excess dollar liabilities.

C is incorrect. The postwar, trade-based "dollar shortage", not a bank funding gap.

D is incorrect. Describes a carry trade.

LO: LTR-15 b: Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.2, eq. (16.2); 16.3.4 Maturity Trans-

formation across Banks' Balance Sheets

Solution LTR-15 b.3 Answer: C

Bank Q508 • BT P2.T8.25.1.2

Step 1 – Choice 1, spot

No later USD need, but a WK liability funds a USD asset with no hedge: currency risk remains.

Step 2 – Choice 2, FX swap

The swap hedges the currency, but a 1-year swap funds a 10-year asset, so the swap must be rolled: a maturity mismatch (funding gap) remains.

Step 3 – Choice 3, direct USD

USD asset against USD debt: no currency mismatch. The 1-year borrowing must be repaid or rolled in USD: funding risk remains.

$\Rightarrow$ C is the only accurate statement.

Interpretation

This is GARP's three funding routes in one table. The swap's second leg in Choice 2 "is not that different" from Choice 3: both must deliver dollars when the contract comes due, and both depend on dollar markets staying open.

Why the other options are wrong

A is incorrect. Confuses "no future currency transactions" with "no exposure": the unhedged USD asset against a WK liability is exposed to the exchange rate.

B is incorrect. Conflates currency risk with maturity mismatch: hedging with a 1-year swap does not extend the funding to 10 years.

D is incorrect. Over-generalises: Choice 1 has a single spot conversion and no dollars to deliver later.

LO: LTR-15 b: Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.2, funding options 1 to 3

Solution LTR-15 b.4 Answer: A

Bank Q509 • Schweser Module Quiz 80.1 Q1

Step 1 – Long-term non-bank liabilities

If liabilities to non-banks are long-term, they support the non-bank claims, so only the *net* dollar position vis-a-vis non-banks has to be rolled: the smallest possible gap, the lower bound.

Step 2 – Short-term non-bank liabilities

If they are all short-term, none of them supports the long-horizon assets, so the *gross* claims on non-banks are the gap: the upper bound.

$\Rightarrow$ Longer-term liabilities give the lower bound, the net position.

Interpretation

For European banks at mid-2007: lower bound USD 1.0 to 1.2 trillion, USD 2.0 to 2.2 trillion if money market fund paper is short-term, upper bound USD 6.5 trillion. The figure is a range because the maturity of the non-bank liabilities is not observed.

Why the other options are wrong

B is incorrect. Pairs the net position with the upper bound; the net position is the smallest gap.

C is incorrect. Pairs the gross position with the lower bound; gross applies when liabilities are short-term.

D is incorrect. Right measure for the opposite case: the gross upper bound applies when non-

bank liabilities are short-term.

LO: LTR-15 b: Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.3.4 Maturity Transformation across Banks' Balance Sheets

Solution LTR-15 b.5 — Answer: D

New • built from GARP Ch.16, Section 16.2

Step 1 – Domestic funding risk

Maturity transformation is an essential banking function. In the home currency the central bank can act as lender of last resort and eliminate a domestic funding shortage.

Step 2 – Foreign currency funding risk

"Central banks cannot create foreign currencies"; their help is constrained by the exchange rate regime or limited to FX reserves, so foreign currency needs may have to be met internationally.
$\Rightarrow$ The asymmetry in the lender of last resort, not the assets, makes foreign currency mismatches special.

Interpretation

This asymmetry is why the policy response had to come from the issuer of the currency: euro area, Swiss and UK FX reserves (USD 294 billion) were far below the dollar funding gap, so the Fed supplied dollars through swap lines (LTR-15 c).

Why the other options are wrong

A is incorrect. Scope error: the issue is funding (rollover) risk, not the credit quality of the assets.
B is incorrect. Absolute claim: maturity transformation is a normal banking function in any currency; the problem is the missing backstop.
C is incorrect. Role misassignment: a national central bank cannot create foreign currency, which is the whole point.

LO: LTR-15 b: Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.2 Banks' International Positions: Concepts and Data

Solution LTR-15 b.6 — Answer: C

Bank Q505 • AnalystPrep Q.4171

Step 1 – Statement I

True: currency and maturity positions are managed across the consolidated global entity, so stresses build up on the global balance sheet.

Step 2 – Statement II

True: a large measured mismatch at one office may be hedged off-balance sheet or offset by positions booked elsewhere, leaving a matched book for the bank as a whole.

Step 3 – Statement III

False: positions booked in a country are a small part of a system's global balance sheet, and assets booked abroad are a quarter or more of most systems' assets, so national data can miss vulnerabilities.
$\Rightarrow$ I and II only.

Interpretation

Statement III is the residency trap: "bank nationality" and "bank residency" are fundamentally

different concepts. Vulnerabilities show up only on the consolidated global entity.

Why the other options are wrong

A is incorrect. Omits II, which is true and is the reason local mismatches can be left open.
B is incorrect. Omits I, the premise of the whole analysis.
D is incorrect. Includes III, which is false.

LO: LTR-15 b: Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.3.1 The Structure of Banks' Operations

Solution LTR-15 b.7 Answer: B

New • built from GARP Ch.16, Sections 16.3.1 and 16.3.3

Step 1 – How positions are managed

Currency and maturity positions are managed on the consolidated global entity, so an office's mismatch can be offset elsewhere in the group.

Step 2 – The two failures of residency data

Host-country data can show a mismatch that does not exist for the bank (foreign banks' offices held nearly 40% of UK external assets); home-country data can miss one booked abroad.
$\Rightarrow$ B states both halves.

Interpretation

The consolidated view is also how the paper finds the long dollar books of UK, Swiss, German and Dutch banks, funded in home currency and swapped. That pattern was invisible in any one country's residency data.

Why the other options are wrong

A is incorrect. Premise reversed: banks manage positions globally, not office by office.
C is incorrect. Factual error: only 18% of Swiss banks' foreign claims are booked in Switzerland, so Swiss residency data miss most of the book.
D is incorrect. The misreading corrected in the fixbox: Japanese, Swiss and UK banks' *claims* were mainly in dollars, but their long dollar positions were funded by borrowing domestic currency and swapping it, the cross-currency funding that created the gap.

LO: LTR-15 b: Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.3.1 The Structure of Banks' Operations; 16.3.3 Cross-Currency Funding Positions

LTR-15 • The US Dollar Shortage in Global Banking and the International Policy Response • Vardeez: VERY IMPORTANT

LTR-15 c

P2 CONCEPT

Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P2. Vardeez flags the whole reading VERY IMPORTANT ("anything can be tested here"). The map carries no workbook section for this LO; D8's revision notes cover the swap lines and their effectiveness.

Vardeez: no LO-level pointer. He notes that the 2026 wording adds "assess its effectiveness" to the old "how swap agreements overcame the challenges of international lending of last resort", so the assessment half is examinable.

Reported exam shapes: none listed. The questions below follow the LO wording.

What the questions look like: statement questions on the mechanism (who lends to whom, against what), the timeline, the two lender-of-last-resort challenges the network overcomes, and the evidence on effectiveness. Favourite distractors: uncollateralized lines, the Fed lending directly to foreign commercial banks or bearing their credit risk, and the wrong central banks receiving unlimited lines. The bank has five questions here; two new ones cover the form of the response and the assessment.

1. Why the response had to be international

European central banks could relieve their banks' pressures in domestic currency, but they could not provide enough dollars (LTR-15 b, Section 2). Their combined FX reserves at mid-2007, USD 294 billion for the euro area, Switzerland and the UK, were about a quarter of the lower-bound dollar funding gap of USD 1.0 to 1.2 trillion (the paper says "an order of magnitude smaller"). The dollars had to come from the issuer of the currency, the Federal Reserve.

2. The swap line network

- **Mechanism.** Temporary **reciprocal currency arrangements** (swap lines): the Fed lends dollars to a foreign central bank, **collateralised by the foreign currency**; that central bank lends the dollars on through **US dollar auctions** in its jurisdiction. Dollars reached banks with no US subsidiary or without enough collateral eligible at the Fed (Figure 4). In providing dollars on a global scale, the Fed "effectively engaged in **international lending of last resort**".
- **Timeline.** Lines with the ECB and the Swiss National Bank from **December 2007**. After Lehman (September 2008) the existing lines were doubled and new ones opened with the Bank of Canada, the Bank of England and the Bank of Japan, a total of **USD 247 billion**. As disruption spread, lines were extended to Australia and New Zealand, Scandinavia, and several countries in Asia and Latin America, forming a global network, and central banks also set up regional swap arrangements in their own currencies. On **13 October 2008** the lines with the **Bank of England, the ECB and the SNB became unlimited** (Figure 5).
- **Where the dollars went.** Once the lines were unlimited, the Eurosystem, the Bank of England and the SNB took 81% of the Fed's international dollar provision (15 October 2008), and 50% to 60% from December 2008: the pressure was concentrated on European banks.

- The structure is not new: a Fed-centred swap network was set up in 1962 to support Bretton Woods parities. A further set of lines (April 2009) let the Fed obtain foreign currencies for US banks if needed.

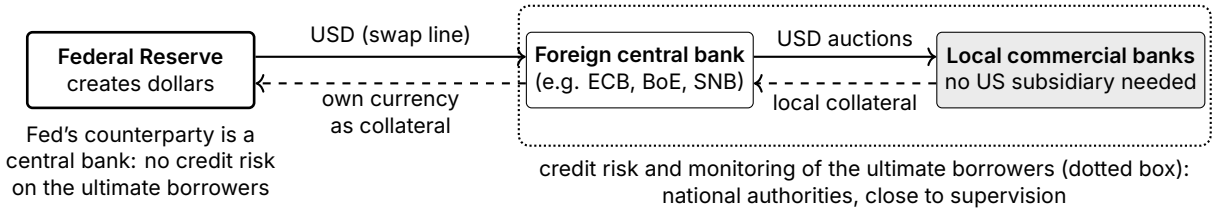


Figure 4. How a dollar swap line works. Solid arrows: dollars; dashed arrows: collateral. The Fed lends only to central banks against their own currency; each central bank auctions the dollars locally and carries the credit risk and the monitoring of its banks (dotted box). This is the structure that answers the moral hazard objection.

3. Assessing effectiveness

"While it is too soon for conclusive answers, the immediate effects have been largely positive."

- **Usage tracked the need, then receded.** Dollars provided through the swap lines surged in October 2008, peaked at **USD 583 billion in December 2008**, and fell gradually to **USD 50 billion by early October 2009**.
- **Prices normalised.** The level and spreads of US dollar interest rates, notably Libor, receded from their autumn 2008 peak. Baba and Packer find that the dollar auctions reduced the **level and volatility of FX swap spreads**.
- **Fire sales averted.** The policy helped avert more extensive distress selling of dollar assets, and possibly mitigated **interbank rate volatility** and **upward pressure on the US dollar**.
- **Caveat.** Banks' increased reliance on public dollar funding "**may have delayed the necessary restructuring**" of their balance sheets.

The institutional assessment: two challenges overcome

International lending of last resort usually faces two objections. The swap network overcomes both.

1. **Resources.** Effective lending of last resort needs enough resources to reassure markets. The Fed and its counterparts can, in principle, **create any amount of money**, unlike international financial institutions (the IMF) that administer limited resources. Demand for other currencies can be met the same way, by bringing their issuing central banks into the network.
2. **Moral hazard and information.** The network "does not compound the informational problems that can give rise to moral hazard". Lending against collateral to central banks, the Fed **assumes no credit risk vis-a-vis the ultimate borrowers**, and delegates monitoring (and collateralising) to the national authorities closer to bank supervision. Supervision stays national, which an international institution lending to banks outside its reach could not match.

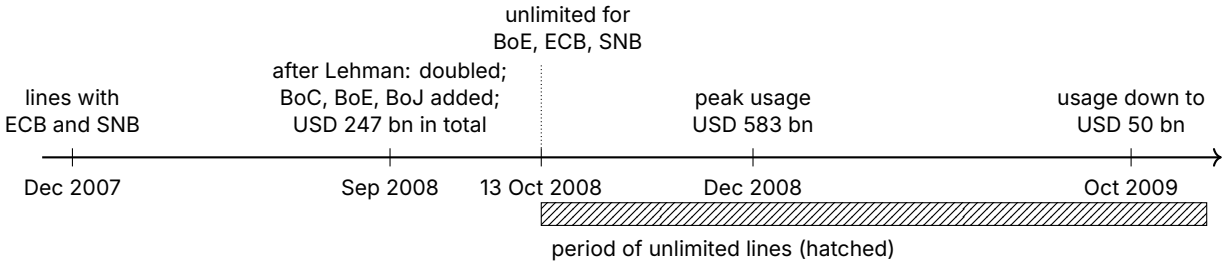


Figure 5. Timeline of the Fed's dollar swap lines (GARP Section 16.5). Lines were extended to central banks in Australia, New Zealand, Scandinavia, Asia and Latin America during autumn 2008. Usage rose with the post-Lehman funding need and fell as markets normalised.

Correction to the source notes. (1) Vardeez writes that "in late 2008 the Fed made the swaps unlimited". Only the lines with the Bank of England, the ECB and the SNB became unlimited, on 13 October 2008; the others kept announced sizes. (2) His effectiveness table lists "mitigated pressure on other currencies" because other central banks "did not have to sell reserves". GARP's point is institutional, not an observed effect: central banks can create money without limit, so demand for *other* currencies can be met the same way by adding their issuing central banks to the network. The observed effects are lower Libor levels and spreads, lower swap-spread level and volatility, fewer distress sales, and possibly lower interbank rate volatility and less upward pressure on the dollar.

Question	What happened	Exam phrasing to recognise
Who lends to whom	Fed to central banks; they auction to banks	Fed's counterparty is never a commercial bank
Against what	Foreign currency as collateral	Collateralised, not unsecured
Size	USD 247 bn after Lehman; unlimited for 3 from 13 Oct 2008	BoE, ECB, SNB (not BoC, BoJ)
Resources challenge	Central banks create money without limit	Unlike the IMF's limited resources
Moral hazard challenge	No credit risk on ultimate borrowers; national monitoring	Does not compound informational problems
Effectiveness	Peak USD 583 bn (Dec 2008), USD 50 bn (Oct 2009); Libor and swap spreads down	"Largely positive", "too soon" to be conclusive
Caveat	May have delayed balance sheet restructuring	A cost, not a failure

Exam traps

- **Collateralised.** The lines were secured by the borrowing central bank's own currency. "Uncollateralized" is always wrong.
- **Fed to central banks only.** The Fed did not lend to foreign commercial banks through the lines and bore no credit risk on them; national central banks did.
- **Which lines went unlimited.** BoE, ECB and SNB on 13 October 2008, not BoC and BoJ, and not "all lines".
- **Moral hazard polarity.** The network *avoids* compounding moral hazard; an option saying it increased moral hazard inverts the assessment.
- **Falling usage is success.** Usage fell from USD 583 billion to USD 50 billion because markets

normalised, not because the facility failed.

- **Two USD 583 billions.** Peak swap-line usage (December 2008) and the closed-out estimate of the funding gap at end-Q1 2009 are both USD 583 billion. Read which one the stem means.
- **Reserves were not enough.** USD 294 billion of reserves against a USD 1.0 to 1.2 trillion gap is why the lines were needed.

Practice questions: LTR-15 c

Question LTR-15 c.1

Bank Q515 • AnalystPrep Q.4175 (rebuilt)

The Federal Reserve extends loans, collateralised by foreign currencies, to foreign central banks. These central banks then make the US dollars available to banks in their own jurisdictions through US dollar auctions. Which of the following best describes this arrangement?

A. Maturity transformation across banks' balance sheets

B. A consolidated-entity currency mismatch

C. International lending of last resort

D. Direct Fed discount window lending to foreign commercial banks

Question LTR-15 c.2

Bank Q517 • Schweser Module Quiz 80.2 Q2

Which of the following statements about the US dollar swap lines extended by the Fed to global central banks during the 2007 to 2009 financial crisis is least accurate?

A. Swap lines could be unlimited.

B. Swap lines were primarily uncollateralized.

C. The Fed can be seen as a lender of last resort.

D. Central banks typically made funds available locally through auctions.

Question LTR-15 c.3

New • built from GARP Ch.16, Section 16.5

After Lehman Brothers failed, the governing council of a European central bank faced acute US dollar funding needs at banks in its jurisdiction. A staff paper compares the options. Which statement best explains why the international response took the form of a network of central bank swap lines with the Federal Reserve?

A. Euro area, Swiss and UK central banks held more than enough dollar reserves to cover their banks' needs, but swap lines were cheaper than selling reserves.

B. Swap lines allowed the Fed to lend directly to European commercial banks, which otherwise had no access to dollars.

C. The IMF lacked the legal power to lend in dollars, so central banks were the only institutions able to act.

D. Home central banks cannot create dollars, and their FX reserves of about USD 294 billion were far below the dollar funding gap, so the dollars had to come from the Fed through the central banks that supervise the banks.

Question LTR-15 c.4 Bank Q514 • AnalystPrep Q.4170 (rebuilt)

The Federal Reserve's swap network is seen as overcoming two challenges commonly associated with international lending of last resort. Consider the following statements:

I. The Fed and its foreign counterparts can in principle create any amount of money, unlike international financial institutions administering limited resources.

II. The swap lines were made unlimited only with the central banks of Canada, England and Japan.

III. The network does not compound the informational problems that give rise to moral hazard.

IV. The Fed bears the credit risk of the ultimate bank borrowers and monitors them directly.

Which statements correctly identify the two challenges overcome?

A. I and III

B. II and IV

C. I and IV

D. II and III

Question LTR-15 c.5 Bank Q513 • AnalystPrep Q.4165 (rebuilt)

During the global financial crisis, the Federal Reserve established reciprocal swap lines with foreign central banks in response to the acute US dollar shortage faced by banks outside the United States. Which of the following was an effect of this international policy response?

A. It increased banks' demand for foreign currency assets.

B. It mitigated interbank rate volatility and upward pressure on the US dollar.

C. It created the US dollar funding gap of banks.

D. It forced banks to unwind their US dollar assets in distress sales.

Question LTR-15 c.6 New • built from GARP Ch.16, Sections 16.5 and 16.6

A board member of a central bank asks its economists for an assessment of the Fed's dollar swap lines in line with McGuire and von Peter. Which statement in the assessment is correct?

A. Usage of the swap lines kept rising through 2009 as the crisis spread, which shows that the facility failed to stabilise dollar funding markets.

B. The swap lines eliminated European banks' US dollar funding gap by the end of 2008, so no balance sheet adjustment by the banks was needed.

C. Usage peaked at about USD 583 billion in December 2008 and fell to about USD 50 billion by early October 2009 as Libor levels and spreads receded, although reliance on public dollar funding may have delayed balance sheet restructuring.

D. The swap lines were unlimited for every participating central bank from December 2007, which is why the Fed's lending did not need to be collateralised.

Solutions • LTR-15 c

Solution LTR-15 c.1 — Answer: C

Bank Q515 • AnalystPrep Q.4175

Step 1 – Who provides the dollars

The Fed, the issuer of the currency, lends dollars on a global scale to foreign central banks against their currencies.

Step 2 – How they reach banks

Those central banks auction the dollars locally, reaching banks without US subsidiaries or eligible collateral.

⇒ The Fed "effectively engaged in international lending of last resort".

Interpretation

The foreign central banks are intermediaries, not lenders of last resort in dollars in their own right: they cannot create dollars (LTR-15 b).

Why the other options are wrong

A is incorrect. Maturity transformation is the banks' funding practice that created the vulnerability, not the policy response.

B is incorrect. A currency mismatch is a balance sheet vulnerability, not a central bank facility.

D is incorrect. The Fed lent to central banks, not to foreign commercial banks; reaching banks without access to the Fed's own facilities was the point.

LO: LTR-15 c: Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.5 The International Policy Response

Solution LTR-15 c.2 — Answer: B

Bank Q517 • Schweser Module Quiz 80.2 Q2

Step 1 – Check each feature

A: true, the lines with the BoE, ECB and SNB became unlimited on 13 October 2008. C: true, global dollar provision made the Fed an international lender of last resort. D: true, the borrowing central banks auctioned the dollars locally.

Step 2 – Test B

The Fed's loans were collateralised by the foreign currencies pledged by the borrowing central banks.

⇒ B is the least accurate statement.

Interpretation

Collateral is what lets the Fed take no credit risk on the ultimate borrowers, the basis of the moral hazard argument. GARP notes the pledged currencies "are best regarded as collateral".

Why the other options are wrong

A is incorrect. Accurate: three lines became unlimited in October 2008.

C is incorrect. Accurate: unlimited, global dollar provision is international lending of last resort.

D is incorrect. Accurate: local US dollar auctions were the distribution channel.

LO: LTR-15 c: Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.5 The International Policy Response

Solution LTR-15 c.3 — Answer: D

New • built from GARP Ch.16, Section 16.5

Step 1 – Why not the home central bank alone

European central banks could address pressures in their own currencies, but cannot create dollars. Their FX reserves at mid-2007 were USD 294 billion for the euro area, Switzerland and the UK combined.

Step 2 – Compare with the need

The lower-bound dollar funding gap was USD 1.0 to 1.2 trillion, so reserves covered about a quarter of it at most.

Step 3 – Why central banks as intermediaries

Routing the dollars through the national central banks reached banks with no US presence and left credit risk and monitoring with the authorities that supervise them.

$\Rightarrow$ D captures both halves: the source of dollars had to be the Fed, and the channel had to be the home central banks.

Interpretation

This is the paper's own framing: the size of the funding gap "shed light on why the international policy response was necessary, and why it took the form of a global network of central bank swap lines".

Why the other options are wrong

A is incorrect. Reserves were "an order of magnitude smaller" than the gap in the paper's words, not more than enough.

B is incorrect. Role misassignment: the Fed lent to central banks, not to European commercial banks; those with US offices could already use Fed facilities.

C is incorrect. Wrong reason: the issue is that an international institution administers *limited resources*, not a legal bar; the swap lines solve the resource problem because central banks can create money.

LO: LTR-15 c: Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.1 and 16.5

Solution LTR-15 c.4 — Answer: A

Bank Q514 • AnalystPrep Q.4170

Step 1 – Challenge 1, resources

Central banks can create any amount of their own currency, unlike international financial institutions with limited resources. I is true.

Step 2 – Challenge 2, moral hazard

The Fed lends against collateral to central banks, assumes no credit risk on the ultimate borrowers and delegates monitoring to national authorities, so informational problems are not compounded. III is true.

Step 3 – Reject II and IV

II names the wrong central banks: the unlimited lines were with the ECB, the BoE and the SNB. IV is the opposite of the design.

$\Rightarrow$ I and III.

Interpretation

Both challenges are answered by the structure itself: the right issuer (resources) and the right counterparty (information and moral hazard).

Why the other options are wrong

B is incorrect. Both statements are false.
C is incorrect. I is right, but IV reverses who carries the credit risk.
D is incorrect. III is right, but II misstates which lines became unlimited.

LO: LTR-15 c: Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.5 The International Policy Response

Solution LTR-15 c.5 — Answer: B

Bank Q513 • AnalystPrep Q.4165

Step 1 – What the lines did

They supplied dollars, against collateral, through central banks to banks that needed to roll dollar funding.

Step 2 – The observed effects

Usage peaked and receded, Libor levels and spreads fell, the auctions lowered the level and volatility of swap spreads, distress sales of dollar assets were averted, and interbank rate volatility and upward pressure on the dollar were possibly mitigated.
$\Rightarrow$ B is an effect of the policy.

Interpretation

Upward pressure on the dollar is the price signature of the shortage (LTR-15 a.6): banks scrambling for dollars push its value up, and an elastic supply of dollars relieves it.

Why the other options are wrong

A is incorrect. The build-up of foreign currency assets is a cause of the shortage, not a result of the policy.
C is incorrect. The funding gap predates the policy; it is what the policy addressed.
D is incorrect. The opposite: the lines helped banks avoid further distress sales of dollar assets.

LO: LTR-15 c: Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.5 The International Policy Response

Solution LTR-15 c.6 — Answer: C

New • built from GARP Ch.16, Sections 16.5 and 16.6

Step 1 – Usage

Dollars provided through the lines surged in October 2008, peaked at USD 583 billion in December 2008 and fell to USD 50 billion by early October 2009.

Step 2 – Prices and the caveat

Libor levels and spreads receded from the autumn 2008 peak; the auctions reduced the level and volatility of swap spreads. The paper adds that reliance on public dollar funding "may have delayed the necessary restructuring" of balance sheets.
$\Rightarrow$ C is the paper's balanced assessment: immediate effects largely positive, with a cost.

Interpretation

"Assess its effectiveness" expects both sides: the evidence of success and the caveat. The authors also say it is "too soon for conclusive answers".

Why the other options are wrong

A is incorrect. Usage fell after December 2008; falling usage reflects normalising markets, not failure.

B is incorrect. The gap was still large in 2009 (the lower bound was about USD 583 to 880 billion at end-Q1 2009 depending on the treatment of writedowns), and the caveat is precisely about delayed adjustment.
D is incorrect. Timeline and collateral are both wrong: the December 2007 lines (ECB, SNB) had set sizes, only three lines became unlimited (13 October 2008), and all lines were collateralised.

LO: LTR-15 c: Describe the policy response by international central banks to alleviate the U.S. dollar shortage and assess its effectiveness. • **Reference:** McGuire and von Peter (2009), GARP Ch.16, 16.4, 16.5 and 16.6 Concluding Remarks

LTR-17 • Risk Management for Changing Interest Rates: ALM and Duration Techniques • Vardeez: VERY IMPORTANT

LTR-17 a

P2 CONCEPT

Discuss how asset-liability management strategies can help a bank hedge against interest rate risk.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P2. No workbook questions, but Vardeez rates the whole reading VERY IMPORTANT ("entire chapter and each learning objective is important"). The A16 revision notes give this LO one line: asset, liability and funds management, the ALCO, and price versus reinvestment risk. **Vardeez:** his ALM map (two channels, two tools: identify the branch first) and a three-line table of strategies: matching maturities, interest rate layering (the mix of fixed and variable rates) and off-balance-sheet solutions.

Reported exam shapes: none listed; built from the LO wording and Vardeez's focus.

What the questions look like: definitional and statement questions. Which approach is ALM, and how it differs from asset or liability management; why rates cannot be forecast; the two kinds of interest rate risk and the two channels through which rates hit a bank; which hedge offsets a stated exposure. No calculations.

1. Recap: the ALM frame

The background is in LTR-17 b, Section 1 (and its Figure 1). In short: banks moved from **asset management** (funding taken as given) to **liability management** (the offer rate as the lever over funding) to **funds management**, which controls the volume, mix and return or cost of both sides together, keeps asset and liability decisions consistent, and treats both sides as sources of revenue and cost. An **ALCO** runs it. Rates are set by the market, so managers are price takers who face **price risk** (rates rise, asset values fall) and **reinvestment risk** (rates fall, incoming funds earn less). Rate changes reach the bank through two channels: the net interest margin and net worth.

2. Why hedge rather than forecast

Rates come from the interaction of thousands of suppliers and demanders of loanable funds, and each rate is built from a risk-free real rate plus premiums for inflation, default, term (maturity), liquidity and call risk, any of which can change. Consistently accurate forecasting is "virtually impossible". GARP's cautionary case: First Bank System of Minneapolis bought unusually large quantities of government bonds on a forecast that rates would fall; rates rose, bond prices fell, and it reported a loss of about $500 million, which led to the sale of its headquarters building.

The maturity gap and the yield curve

Lending institutions' loans and securities tend to be longer than their sources of funds, so most have a **positive maturity gap**. With an **upward-sloping** yield curve, revenues from long assets outstrip the cost of short liabilities and the net interest margin is positive. A flat or downward-sloping curve gives a small or even negative margin to firms that borrow short and lend long (Figure 1). Yield curves rarely shift in parallel: short rates rise and fall faster than long rates. In recessions short rates tend to fall relative to long rates and the spread widens; in a prosperous period the spread narrows and can turn negative.

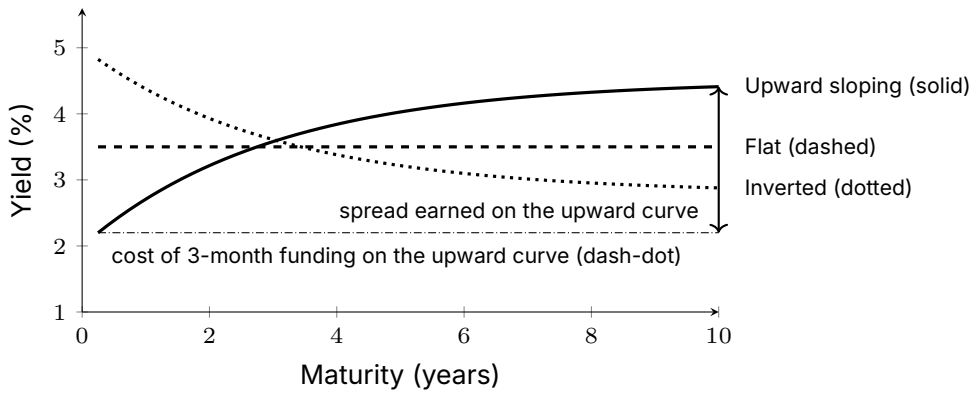


Figure 1. Stylised yield curves. A bank that funds at 3 months and lends at 10 years earns about 2.2 points on the upward-sloping curve (solid), nothing on the flat curve (dashed) and loses about 1.9 points on the inverted curve (dotted). This maturity gap is exactly the exposure ALM strategies are used to manage.

3. How ALM strategies hedge interest rate risk

ALM hedges by shaping the balance sheet so that a rate move changes both sides by similar amounts, or by adding positions off the balance sheet that gain when the balance sheet loses.

Strategy	What the bank does	What it protects
Match repricing (IS gap management)	Hold equal volumes of assets and liabilities repricing in each maturity bucket, so revenue and cost move together	NIM; detail in LTR-17 b
Match durations (duration gap management)	Set the leverage-adjusted duration gap to zero, so asset and liability values change by the same dollar amount	Net worth; detail in LTR-17 c
Adjust maturities	Lengthen or shorten assets or liabilities: a longer maturity is less rate sensitive	Both
Adjust the mix ("interest rate layering")	Shift between fixed-rate and variable-rate assets and liabilities, and change volumes	Both
Off-balance-sheet hedges	Futures, options and swaps (Rose and Hudgins' next chapter) that gain when the balance sheet loses, without selling assets	Both
Take a view (aggressive gapping)	Deliberately hold a gap that profits if the forecast is right	Raises expected NIM or net worth at the cost of risk

The defensive strategies (matched repricing, matched durations, hedges) aim to stabilise NIM and net worth whichever way rates move. The aggressive one converts interest rate risk into forecast risk, which is why GARP says most managers have learned to hedge against, not forecast, rate changes. Note what ALM does *not* require: it does not need a correct forecast, and it does not need assets and liabilities matched one for one.

4. The ALCO in practice

- Key officers from different departments, meeting regularly (quarterly, monthly or more often).
- Goals: maximising shareholder wealth, adequate profitability and sufficient capital, often as numerical targets (an ROA of 1.5%, an equity-to-assets ratio of at least 10%).
- Lays out how the bank is funded, the quality of loans it takes on and the limits on off-balance-sheet exposure.
- Estimates the exposure of the net interest margin and net worth ratios to rate changes, keeps it within limits, and may use simulation to test alternative strategies.
- Reports the bank's condition to the board with suggestions for correcting weaknesses.

	Price risk	Reinvestment risk
Arises when	Rates **rise**	Rates **fall**
What happens	Market values of bonds and fixed-rate loans fall; selling them means capital losses	Incoming funds must be invested in lower-yielding earning assets
Hits	Net worth (and realised gains on sale)	Future interest income and NIM
ALM response	Duration matching, shorter assets, hedges that gain when rates rise	Repricing and duration matching, longer fixed-rate assets, hedges that gain when rates fall

Exam traps

- **Sibling swap of the three approaches.** Asset management takes funding as given; liability management uses price (the offer rate) to control funding; funds management (ALM) coordinates both sides. "Asset-liability and owners' equity management" is not a term.
- **Polarity of the two risks.** Price risk comes with rising rates, reinvestment risk with falling rates.
- **Two channels.** Rates hit NII *and* the market value of assets and liabilities; operating costs, fees and loan-loss provisions are not the direct channels.
- **One-sided management.** Watching NII alone misses losses in net worth; watching net worth alone misses the margin. ALM uses both tools.
- **Forecasting.** Managers are price takers; rates come from the market, not from one regulator, and cannot be forecast consistently.
- **Hedge direction.** A bank hurt by rising rates needs a position that gains when rates rise (pay fixed, receive floating), not more short funding or longer fixed-rate assets.

Practice questions: LTR-17 a

Question LTR-17 a.1

Bank Q550 • Schweser Module Quiz 82.1 Q1

An integrated approach to managing balance sheet interest rate risk is called:

A. asset management.

B. liability management.

C. asset-liability management (ALM).

D. asset-liability and owners' equity management.

Question LTR-17 a.2

Bank Q546 • GARP book Ch.18 Internet Exercise 1 (rebuilt)

The chair of a bank's asset-liability committee is briefing new board members on the two major kinds of interest rate risk the bank faces. Which of the following statements is correct?

A. Price risk arises when rates fall, because incoming funds must be invested at lower yields; reinvestment risk arises when rates rise and fixed-rate assets lose value.

B. Price risk arises when rates rise and the market values of bonds and fixed-rate loans fall; reinvestment risk arises when rates fall and incoming funds must be invested at lower yields.

C. Both risks arise only when rates rise, so a bank that expects falling rates faces no interest rate risk.

D. Price risk affects only the liability side and reinvestment risk only the asset side, so each is managed separately from the other.

Question LTR-17 a.3 Bank Q547 • AnalystPrep Q.4233 (rebuilt)

Financial managers have long tried, with little success, to forecast interest rate movements. Each of the following is a reason why consistently accurate interest rate forecasting is so difficult EXCEPT:

A. Interest rates are determined by the interaction of thousands of suppliers and demanders of credit, so no single institution sets them.

B. Any particular interest rate is built from several components, such as a risk-free real rate and various risk premiums, and a change in any component moves the rate.

C. Market interest rates are set by a single regulator and moved only at announced dates, so banks cannot observe the rate decisions that matter.

D. Because rates are largely beyond their control, most financial managers must act as price takers and plan around rate levels they cannot influence.

Question LTR-17 a.4 Bank Q548 • AnalystPrep Q.6345

Changes in market interest rates have a direct impact on a bank's economic value. Which two components of a bank's financial performance are most significantly affected by these changes?

A. Net interest income and operating costs.

B. Loan origination fees and cost of customer deposits.

C. Net interest income and the value of financial instruments.

D. Operating profit and profit per share.

Question LTR-17 a.5 Bank Q551 • AnalystPrep Q.5575 (rebuilt)

A regional bank has grown rapidly by lending to a narrow group of clients whose deposits and loans are highly sensitive to interest rates. Management has so far monitored only the effect of rate changes on near-term net interest income, and a large unrealised decline in the market value of its long-term fixed-rate assets has gone unnoticed. Under an integrated asset-liability management (ALM) approach, which of the following is the most appropriate way to strengthen the bank's interest rate risk management?

A. Continue to focus on net interest income alone, since earnings are the primary objective and equity is only a residual.

B. Manage both sides of the balance sheet together, using the interest-sensitive gap to monitor net interest income and the duration gap to monitor the market value of equity (net worth).

C. Focus solely on the market value of equity using duration gap, because earnings effects of rate changes are temporary.

D. Stop measuring rate risk with gap tools and rely only on matching each asset's maturity to a single funding source.

Question LTR-17 a.6 Bank Q549 • AnalystPrep Q.6352 (rebuilt)

A bank holds a large portfolio of long-term, fixed-rate mortgages funded mainly with short-term liabilities and is worried that market interest rates will rise. Which action would most directly hedge the interest rate exposure of the existing portfolio without selling the loans?

A. Raise the rate charged on new mortgages.

B. Enter a pay-fixed, receive-floating interest rate swap.

C. Increase the share of mortgage funding raised through short-term borrowing.

D. Shift new lending toward even longer-maturity fixed-rate loans.

Solutions • LTR-17 a

Solution LTR-17 a.1

Answer: C

Bank Q550 • Schweser Module Quiz 82.1 Q1

Step 1 – What "integrated" means

Treating both sides of the balance sheet as one whole, for profitability and for risk, is asset-liability management, run by the ALCO.

Step 2 – Place the alternatives

Asset management and liability management are the two one-sided stages that preceded it.

Interpretation

Rose and Hudgins call today's integrated approach funds management: control of the volume, mix and return or cost of assets and liabilities, coordinated so the two sides do not pull against each other.

Why the other options are wrong

A is incorrect. Asset management controls only the allocation of funds; funding is taken as given.

B is incorrect. Liability management controls funding through the offer rate; it is one side only.

D is incorrect. Not a recognised term; equity is the residual that ALM protects, not a separately managed side.

LO: LTR-17 a: Discuss how asset-liability management strategies can help a bank hedge against interest rate risk. • **Reference:** Rose and Hudgins, Bank Management and Financial Services 9e, Ch.7, 18.1 and 18.2 Asset-Liability Management Strategies

Solution LTR-17 a.2

Answer: B

Bank Q546 • GARP book Ch.18 Internet Exercise 1

Step 1 – Price risk

When rates rise, the market values of most bonds and fixed-rate loans fall; selling them means accepting capital losses.

Step 2 – Reinvestment risk

When rates fall, incoming funds must be reinvested in lower-yielding earning assets, lowering expected future income.

Interpretation

The two risks sit on opposite rate moves, so no rate path is risk free. A big part of ALM is finding ways to deal with both: matching repricing and durations, adjusting maturities and mix, and hedging with futures and swaps.

Why the other options are wrong

A is incorrect. Swaps the two definitions.

C is incorrect. Absolute claim: reinvestment risk is a falling-rate risk, so a bank expecting falling rates still faces it.

D is incorrect. Both risks can hit either side, and ALM exists precisely to manage the two sides together.

LO: LTR-17 a: Discuss how asset-liability management strategies can help a bank hedge against interest rate risk. • **Reference:** Rose and Hudgins, Ch.7, 18.3 Interest Rate Risk: One of the Greatest Management Challenges, Forces Determining Interest Rates

Solution LTR-17 a.3 — Answer: C

Bank Q547 • AnalystPrep Q.4233

Step 1 – The reading's reasons

Rates come from the interactions of thousands of credit suppliers and demanders (A); each rate is a set of building blocks, a real rate plus risk premiums, any of which can move (B); managers are therefore price takers (D).

Step 2 – The odd one out

C says one regulator sets market rates. The reading says the opposite: market rates emerge from the marketplace.

Interpretation

The practical lesson is the reading's: since rates cannot be forecast consistently, most managers hedge against rate changes rather than bet on them.

Why the other options are wrong

A is incorrect. A genuine reason (many participants); choosing it misses the EXCEPT.
B is incorrect. A genuine reason (rate components); choosing it misses the EXCEPT.
D is incorrect. A genuine point (price takers, not price makers); choosing it misses the EXCEPT.

LO: LTR-17 a: Discuss how asset-liability management strategies can help a bank hedge against interest rate risk. •
Reference: Rose and Hudgins, Ch.7, 18.3, Forces Determining Interest Rates; The Components of Interest Rates

Solution LTR-17 a.4 — Answer: C

Bank Q548 • AnalystPrep Q.6345

Step 1 – Income channel

Rate changes move the yield on earning assets and the cost of funds, so interest income, interest expense and net interest income change.

Step 2 – Value channel

Rate changes move the market value of fixed-rate securities, loans and liabilities, and so net worth.

Interpretation

These are the two branches of the ALM map (LTR-17 b, Figure 1): the margin, protected by IS gap management, and net worth, protected by duration gap management.

Why the other options are wrong

A is incorrect. Operating costs are noninterest expenses, not driven directly by market rates.
B is incorrect. Fees are not primarily rate driven, and deposit cost is only one piece of the income channel; the value channel is missing.
D is incorrect. Broad outcomes that depend on many factors, not the two direct channels.

LO: LTR-17 a: Discuss how asset-liability management strategies can help a bank hedge against interest rate risk. •
Reference: Rose and Hudgins, Ch.7, 18.3; Exhibit 18.1

Solution LTR-17 a.5 — Answer: B

Bank Q551 • AnalystPrep Q.5575

Step 1 – Diagnose the failure

The bank watched only the income channel and missed the fall in the market value of its long fixed-rate assets: the value channel.

Step 2 – Apply integrated ALM

Manage both sides together and both targets: the IS gap for net interest income, the leverage-adjusted duration gap for net worth.

Interpretation

GARP's warning in so many words: "just because the net interest margin is protected against interest rate risk doesn't mean an institution's net worth is also sheltered from loss". Effective ALM targets desirable levels of both net interest income and net worth.

Why the other options are wrong

A is incorrect. Repeats the original mistake: NII-only monitoring misses the net worth loss.
C is incorrect. Overcorrects: duration gap alone ignores the earnings effect controlled by IS gap management.
D is incorrect. Abandons the measuring tools; one-for-one maturity matching is neither practicable nor integrated ALM.

LO: LTR-17 a: Discuss how asset-liability management strategies can help a bank hedge against interest rate risk. •
Reference: Rose and Hudgins, Ch.7, 18.2 Funds Management Strategy; 18.4 and 18.5

Solution LTR-17 a.6 Answer: B

Bank Q549 • AnalystPrep Q.6352

Step 1 – The exposure

Fixed mortgage income is locked in while short-term funding reprices upward if rates rise, and the mortgages lose value: the bank is hurt by rising rates on both branches.

Step 2 – A hedge that gains when rates rise

Paying fixed and receiving floating: the floating receipts rise with rates and offset the higher funding cost, while the fixed payments do not change. The swap also gains value as rates rise, offsetting the fall in the loans' value.

Interpretation

Swaps and futures are the off-balance-sheet tools the reading names: they change the bank's effective repricing and duration position without selling the loans or restructuring the funding.

Why the other options are wrong

A is incorrect. Affects only new loans; the existing portfolio's income and value exposure is unchanged.
C is incorrect. More short-term funding widens the repricing mismatch and increases the exposure.
D is incorrect. Longer fixed-rate loans raise asset duration and the exposure.

LO: LTR-17 a: Discuss how asset-liability management strategies can help a bank hedge against interest rate risk. •
Reference: Rose and Hudgins, Ch.7, 18.4 and 18.6 (hedging tools such as financial futures, options and swaps)

LTR-17 • Risk Management for Changing Interest Rates: ALM and Duration Techniques • Vardeez: VERY IMPORTANT

LTR-17 d

P2 CONCEPT

Discuss the limitations of interest-sensitive gap management and duration gap management.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P2. No workbook questions, but Vardeez rates the whole reading VERY IMPORTANT. The A16 revision notes list both sets of limitations.
Vardeez: four limitations of the IS gap and five of the duration gap (the Schweser lists), and one callout: both tools fail on the same instruments, demand deposits and savings accounts, but for different reasons. He calls this "the diagnostic one".

Reported exam shapes: none listed; built from the LO wording and Vardeez's focus.

What the questions look like: which limitation belongs to which tool (the sibling swap), NOT or EXCEPT lists, and scenarios in which a hedge that looks complete still loses. The bank has only four questions here and one is a weighted-gap calculation, worked under LTR-17 b as Question LTR-17 b.9; three new questions built from GARP fill the gap.

1. Limitations of interest-sensitive gap management

"While interest-sensitive gap management works beautifully in theory, practical problems in its implementation always leave financial institutions with at least some interest-rate risk exposure."

a) **Asset and liability rates do not move together.** Rates on liabilities, mostly short term, tend to move faster than rates on assets, many of which are long term; and the rates a bank sets do not move at the same speed as open-market rates (deposit rates lag loan rates). So a zero gap does not remove all risk: this is **basis risk**. The response is the weighted gap of LTR-17 b, Section 7, which can even flip the sign of the gap ($-\$23$ to $+\$75$ in GARP's example).

b) **Rates change at different speeds within a bucket.** Schweser's example: variable-rate loans that reprice once a year, funded with short-term money that reprices daily. The cumulative gap for the year is zero and the bank looks immunised, yet NII varies over the year as funding costs follow market rates months before loan rates reset.

c) **The repricing point is hard to identify.** Checkable deposits and savings accounts can have their rates changed at any time, so they cannot be placed in a bucket with confidence.

d) **The planning periods are arbitrary.** Some items always "fall between the cracks" of the chosen buckets and cause trouble if rates move against the bank, so wise managers use several bucket lengths.

e) **It ignores net worth.** IS gap management does not consider the effect of rates on the owners' position. An aggressive gap may widen NIM, but at the cost of more volatile earnings and a lower value of the stockholders' investment (Schweser: earnings volatility can hurt the stock price). Effective ALM needs acceptable levels of *both* net interest income and net worth.

2. Limitations of duration gap management

a) **Matching durations is hard.** Finding assets and liabilities of the same duration that fit the portfolio is "often a frustrating task". Duration is *always less* than maturity for instruments that pay out over time; it equals maturity only for zero-coupon securities, single-payment loans and Treasury bills. The more often an instrument pays, the shorter its duration; the *shorter* its maturity, the closer its duration is to its maturity.

b) **Cash flows can be ill defined.** Checkable deposits and passbook savings have no well-defined cash flow pattern, so their duration is hard to calculate. Customer prepayments and defaults also distort expected loan cash flows (prepayments shorten duration).

c) **The price-rate relation is assumed linear.** It is convex (Figure 1), so a big rate change (one or two points) gives a distorted reading of the risk. Convexity must be added for precision on large moves.

d) **Small, parallel shifts are assumed.** The method works reasonably well if the yield curve moves by small amounts and in parallel. Real curves do not: short rates move over a wider range than long rates, and different rates move at different speeds.

e) **Duration itself drifts.** Durations shift as rates move, and different instruments' durations change at different speeds with the passage of time, so a zero gap does not stay zero without rebalancing.

The other side. Research suggests duration balancing "can still be effective, even with moderate violations" of its assumptions, and it remains valuable because it manages the value of the firm to its shareholders.

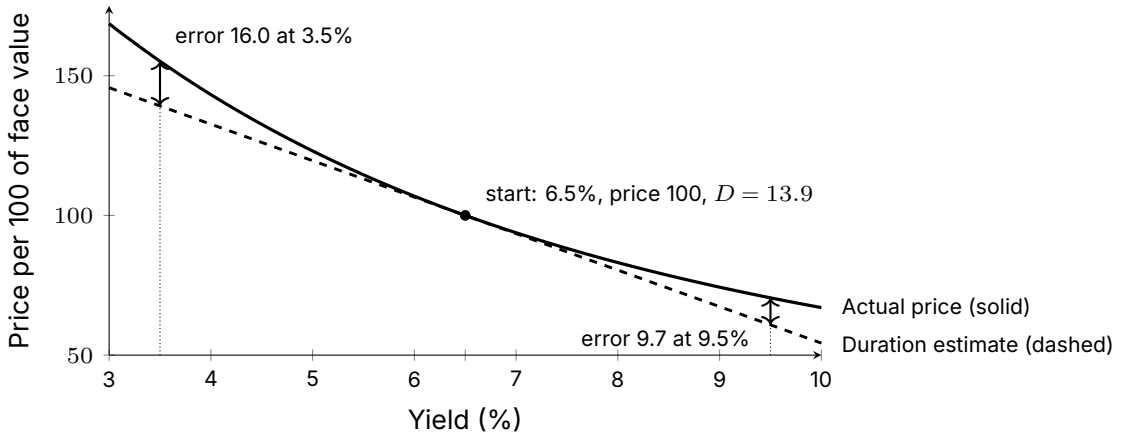


Figure 1. A 30-year 6.5% annual-coupon bond priced at par. The dashed line is the duration estimate $100\,[1 - D\,\Delta i/(1+i)]$; the solid curve is the true price. For a 50 basis point move the two are within 0.35, but a 3 point move misses by 16.0 (rates down) and 9.7 (rates up): duration understates gains and overstates losses. The same bond rises 6.88 when its yield falls from 6.5% to 6%, but only 3.49 when it falls from 10% to 9.5%: price risk is greater when rates are low.

GARP's own illustration: when the yield on a 30-year bond falls from 6.5% to 6% its price rises by about 8 points, but a fall from 10% to 9.5% raises it by just over 4 points.

> **Correction to the source notes.** GARP glosses "about 8 points" as "about \$8 per \$1,000 in face value" and "slightly more than 4 points" as "about \$4 per \$1,000". A point is 1% of face value, so 8 points is about \$80 per \$1,000 and 4 points about \$40. The convexity lesson (bigger price moves at lower rates) is unaffected.

3. The shared failure, and which limitation belongs to which tool

Both tools founder on the same instruments, demand deposits and savings accounts, but for different reasons: for the IS gap their **repricing date** cannot be identified; for the duration gap their **cash flow pattern** cannot be defined. A distractor that attaches the right failure to the wrong tool is the build this pairing invites.

	IS gap management	Duration gap management
Demand and savings deposits	Repricing point cannot be identified	Cash flow pattern cannot be defined
How rates move	Asset and liability rates do not move together or at the same speed (basis risk)	Assumes small, parallel shifts; short rates move more than long
Time structure	Arbitrary buckets; timing differences inside a bucket	Duration drifts with rates and time
Shape of the model	ΔNII linear in Δi	Linear price-rate relation; convexity on large moves
Implementation	Choice of horizons and buckets	Hard to find matching durations; duration $<$ maturity
What it ignores	Net worth; aggressive gapping raises earnings volatility	(It is the tool that protects net worth)

Exam traps

- **Sibling swap.** "Repricing point hard to identify", "rates do not move together" and "different speeds within a bucket" are IS gap limitations; "cash flow patterns hard to define", convexity, parallel shifts and duration drift are duration gap limitations.
- **"Ignores net worth"** is a limitation of IS gap management, never of duration gap management.
- **Zero is not immune.** A zero cumulative IS gap can still leave NII exposed (basis risk, timing within buckets); a zero duration gap protects net worth only for small, parallel moves.
- **Duration and maturity.** Duration is less than maturity for any instrument paying over time; equal only for zero-coupon securities, single-payment loans and T-bills; *shorter* instruments are closer to their maturity.
- **Prepayments shorten duration**; they do not lengthen it.
- **Unadjusted-gap fix.** Setting $D_A = D_L$ is not a remedy for any limitation; it is the leverage mistake of LTR-17 c.
- **Absolute qualifiers.** Duration balancing still works under moderate violations of its assumptions; it is not useless.

Practice questions: LTR-17 d

Question LTR-17 d.1 Bank Q597 • Schweser Module Quiz 82.2 Q3

Which of the following is a limitation associated with the duration gap?

A. Rates on assets and liabilities may not move together.

B. Rates may change at different speeds within a maturity bucket.

C. The point at which assets and liabilities reprice is not easy to identify.

D. It may be difficult to define the pattern of cash flows for instruments such as demand deposits.

Question LTR-17 d.2 Bank Q595 • AnalystPrep Q.4238 (rebuilt)

Which of the following is NOT a limitation of duration gap management?

A. The pattern of cash flows for instruments such as checkable deposits and savings accounts is hard to define, making their durations difficult to calculate.

B. Duration is less accurate for large changes in interest rates because of convexity.

C. Duration assumes parallel shifts of the yield curve, so it only approximates the sensitivity of asset and liability values.

D. It ignores the effect of interest rate changes on the market value of the bank's equity.

Question LTR-17 d.3 New • built from GARP Ch.18, 18.4, and Schweser LO 82.d

The ALCO of a regional bank reports a one-year cumulative interest-sensitive gap of zero: USD 400 million of variable-rate business loans, whose rates reset once a year in December, are funded by USD 400 million of money market borrowings that reprice daily. The CFO tells the board that net interest income over the coming year is therefore immune to changes in market rates. Which of the following is the most accurate assessment of the CFO's claim?

A. The claim is correct: with equal volumes of interest-sensitive assets and liabilities in the planning period, interest revenue and interest cost change by the same amount whatever the path of rates.

B. The claim is incorrect: the borrowings reprice daily while the loans reset only in December, so a rise in rates early in the year raises interest cost months before it raises loan income, and NII falls.

C. The claim is incorrect: a zero interest-sensitive gap implies a positive leverage-adjusted duration gap, so net interest income falls whenever rates rise.

D. The claim is incorrect only if rates fall: falling rates always squeeze the margin of a bank with a zero gap because deposit rates cannot fall below zero.

Question LTR-17 d.4 New • built from GARP Ch.18, 18.7

An ALM analyst proposes to immunise net worth by matching the maturities of the bank's assets and liabilities, arguing that each instrument's duration can be read from its maturity date. Which of the following statements about duration and maturity is correct?

A. For any loan or security that pays out gradually over time, duration equals calendar maturity, so matching maturities matches durations.

B. Duration equals calendar maturity only for instruments such as zero-coupon securities, single-payment loans and Treasury bills; for instruments that pay out gradually, duration is always less than maturity.

C. The longer an instrument's maturity, the closer its duration is to its maturity, so the approximation is most reliable for long-term mortgages.

D. Expected prepayments lengthen a loan's duration, because principal is received earlier than scheduled.

Question LTR-17 d.5 New • built from GARP Ch.18, 18.7

At the start of the year the ALCO of a savings bank set its leverage-adjusted duration gap to zero. Over the following months short-term rates rose by 250 basis points while long-term rates rose by 60 basis points, and the bank's net worth fell. Which of the following best explains why the hedge failed?

A. Duration gap management assumes small, parallel shifts in the yield curve; a large move in which short and long rates change by different amounts violates both assumptions, and durations themselves drift as rates change.

B. A zero leverage-adjusted duration gap protects net worth against any change in rates, so the loss must have come from the bank's interest-sensitive gap.

C. The bank should have set the unadjusted duration gap, $D_A - D_L$, to zero, because the leverage adjustment overstates the liability duration needed.

D. Duration gap management ignores the effect of interest rate changes on the market value of equity, which is why net worth was not protected.

Question LTR-17 d.6 Bank Q596 • BT answers P2.T8.20.18.3 (rebuilt)

In regard to asset-liability management and duration techniques, each of the following statements is true EXCEPT which is false?

A. A solvent bank (i.e., total assets exceed total liabilities) must, by definition, have a positive leverage-adjusted duration gap.

B. A limitation of duration gap management is that it is often difficult to find assets and liabilities of the same duration to fit into the bank's portfolio.

C. A disadvantage of interest-sensitive gap management is that it analyzes the impact of interest rate changes on net income but neglects the impact on the market value of the bank's equity.

D. According to duration gap analysis, a bank is fully hedged (immunized) when its leverage-adjusted duration gap is zero, that is, when the asset duration equals the liability duration multiplied by the ratio of liabilities to assets.

Solutions • LTR-17 d

Solution LTR-17 d.1

Answer: D

Bank Q597 • Schweser Module Quiz 82.2 Q3

Step 1 – What duration needs

Duration is computed from the timing of cash flows, so it fails where the cash flow pattern is undefined: demand deposits and savings accounts.

Step 2 – Sort the rest

A, B and C concern repricing behaviour and buckets: limitations of IS gap management.

Interpretation

The same instruments defeat both tools: for the IS gap the problem is the repricing date, for the duration gap the cash flow pattern. The question tests whether you attach each failure to the right tool.

Why the other options are wrong

A is incorrect. Basis risk: an IS gap limitation (the simple gap assumes rates move together).
B is incorrect. Timing inside a bucket undermines bucket analysis, an IS gap tool.
C is incorrect. The repricing point matters for bucketing; duration does not use buckets.

LO: LTR-17 d: Discuss the limitations of interest-sensitive gap management and duration gap management. • **Reference:** Rose and Hudgins, Ch.7, 18.4 Problems with Interest-Sensitive GAP Management; 18.7 The Limitations of Duration Gap Management

Solution LTR-17 d.2

Answer: D

Bank Q595 • AnalystPrep Q.4238

Step 1 – List the duration gap limitations

Ill-defined cash flows for checkable and savings deposits (A); inaccuracy for large moves because of convexity (B); the assumption of parallel yield-curve shifts (C).

Step 2 – The exception

D describes the opposite of duration gap management, which is built to measure and protect the market value of equity. Ignoring equity value is the weakness of IS gap management.

Interpretation

The bank's version of this question carried a corrected key; D is the only defensible answer. Expect the same statement as a distractor on the other side: "IS gap management ignores net worth" is *true*.

Why the other options are wrong

A is incorrect. A genuine duration gap limitation; not the exception.
B is incorrect. A genuine limitation: duration is a linear approximation of a convex relation.
C is incorrect. A genuine limitation: real yield curves do not shift in parallel.

LO: LTR-17 d: Discuss the limitations of interest-sensitive gap management and duration gap management. • **Reference:** Rose and Hudgins, Ch.7, 18.7; 18.4 (IS gap ignores net worth)

Solution LTR-17 d.3

Answer: B

New • built from GARP Ch.18, 18.4, and Schweser LO 82.d

Step 1 – What the zero gap measures

The one-year cumulative gap counts only *whether* items reprice within the year, not *when*. Here

both sides are in the bucket, so the gap is zero.

Step 2 – What happens to NII during the year

If rates rise in, say, February, the borrowing cost rises the next day, but the loan rates do not reset until December. For ten months interest expense is higher while interest income is unchanged, so NII falls.

⇒ Rates changing at different speeds within a bucket is a limitation of IS gap management.

Interpretation

This is Schweser's example, and GARP's point that items "fall between the cracks" of the planning periods. The remedy is finer buckets (several planning periods) and, for basis risk, a weighted gap. The CFO's claim confuses a zero gap with an immune margin.

Why the other options are wrong

A is incorrect. Absolute claim: equal volumes in the period do not mean equal timing of repricing.

C is incorrect. Non sequitur: the IS gap and the duration gap measure different things, and NII is not governed by the duration gap.

D is incorrect. Wrong mechanism and wrong scope: the problem arises for rises as well as falls, and it is about timing, not a zero floor on deposit rates.

LO: LTR-17 d: Discuss the limitations of interest-sensitive gap management and duration gap management. • **Reference:** Rose and Hudgins, Ch.7, 18.4 Problems with Interest-Sensitive GAP Management; SchweserNotes LO 82.d

Solution LTR-17 d.4

Answer: B

New • built from GARP Ch.18, 18.7

Step 1 – When duration equals maturity

Only when all the cash comes at the end: zero-coupon securities, single-payment loans and Treasury bills.

Step 2 – Everything else

Interest and principal paid along the way pull the weighted average time forward, so duration is always less than maturity, and the more frequent the payments the shorter the duration.

Interpretation

This is the first limitation GARP lists: finding assets and liabilities of matching duration is "often a frustrating task" precisely because maturities do not reveal durations. The analyst's maturity-matching plan would leave a duration gap, and it would ignore the leverage adjustment as well.

Why the other options are wrong

A is incorrect. Polarity flip: for gradual payers duration is less than maturity, never equal.

C is incorrect. Reversed: the *shorter* the maturity, the closer the match; long mortgages are where the two differ most.

D is incorrect. Reversed: receiving principal earlier *shortens* duration. Prepayments are a limitation because they make cash flows, and so duration, uncertain.

LO: LTR-17 d: Discuss the limitations of interest-sensitive gap management and duration gap management. • **Reference:** Rose and Hudgins, Ch.7, 18.7 The Limitations of Duration Gap Management

Solution LTR-17 d.5

Answer: A

New • built from GARP Ch.18, 18.7

Step 1 – What the zero gap guarantees

A zero leverage-adjusted gap offsets asset and liability value changes for a *small*, *parallel* change

in rates.

Step 2 – What happened

The move was large and not parallel: short rates rose four times as much as long rates. Liabilities and assets sit at different points on the curve, so their values changed by different proportions; the linear approximation also misses convexity on a move of this size, and durations drift as rates move.

Interpretation

GARP: duration gap analysis is reasonably effective if the yield curve changes by small amounts in parallel steps, but "if there are major changes in interest rates and different interest rates move at different speeds, the accuracy and effectiveness of duration gap management decreases somewhat". The fix is regular rebalancing and adding convexity, not abandoning the tool.

Why the other options are wrong

B is incorrect. Absolute claim: the zero gap protects only against small parallel shifts, and the IS gap governs NII, not net worth.

C is incorrect. Reversed: immunisation requires $D_L = D_A \times A/L$; dropping the leverage adjustment would have left a positive gap.

D is incorrect. False: protecting the market value of equity is what duration gap management is for; ignoring it is the IS gap's weakness.

LO: LTR-17 d: Discuss the limitations of interest-sensitive gap management and duration gap management. • **Reference:** Rose and Hudgins, Ch.7, 18.7 The Limitations of Duration Gap Management

Solution LTR-17 d.6 Answer: A

Bank Q596 • BT answers P2.T8.20.18.3

Step 1 – Test A with the gap formula

The leverage-adjusted gap is $D_A - D_L \times L/A$. Solvency makes $L/A < 1$, but the sign still depends on the durations: with $A = 120$, $L = 100$, $D_A = 4$ and $D_L = 5$, the gap is $4 - 5 \times 100/120 = -0.17$, negative for a solvent bank.

Step 2 – Check B, C and D

B is a limitation GARP lists; C is the stated weakness of IS gap management; D is the immunisation condition. All true.

Interpretation

Solvency lowers the weight on liability duration but does not fix the sign of the gap. Fannie Mae reported a *negative* gap of about 14 months at the close of 2002 (LTR-17 c, Section 7).

Why the other options are wrong

B is incorrect. True: matching durations is "often a frustrating task".

C is incorrect. True: IS gap management focuses on NII and neglects equity value.

D is incorrect. True: zero gap means $D_A = D_L \times L/A$.

LO: LTR-17 d: Discuss the limitations of interest-sensitive gap management and duration gap management. • **Reference:** Rose and Hudgins, Ch.7, 18.6 and 18.7

LIQUIDITY AND TREASURY RISK • PART

P3

Third priority

Tier C and D topics, or LOs Vardeez flags in other chapters. 17 LOs from 9 readings.

LTR-1 • Liquidity Risk • Vardeez: IMPORTANT

LTR-1 b

P3 CONCEPT

Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft.

Workbook: D27 Intraday Liquidity and Dealer Bank Failure Mechanics • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic: the three cases sit in section D27 (Map #65, 1 of 16 recall sources), which has revision notes but no questions. The recall source expects "case↔lesson matching against Northern Rock".
Vardeez: no pointer for this LO; his reading note picks LVaR (LTR-1 a) and black holes (LTR-1 d). His notes carry the case table that Section 3 checks against GARP.

Reported exam shapes

- **S4** Case↔lesson matching against Northern Rock (D27's other shapes, S1 to S3, belong to LTR-6 and LTR-8).

What the questions look like: no numbers. A four-statement audit or a matching table: which case taught which lesson. Distractors swap Ashanti and Metallgesellschaft (the price moved in opposite directions), pin another case's story on Northern Rock (a hedge, a clearing bank's right of offset), or turn the lesson into "do not hedge". Expect one question on the sources of funding and how they behave under stress.

1. Liquidity funding risk

Liquidity funding risk is the risk relating to a financial institution's ability to meet its cash needs as they arise. The trading/funding split and "liquidity is not solvency" are set out in LTR-1 a, Section 1. Hull's illustration: a bank holding mostly illiquid mortgages, financed 90% by deposits and 10% by equity, is comfortably solvent, yet it fails if 25% of its depositors withdraw at once.

The key to managing it is **predicting cash needs** and making sure they can be met in adverse scenarios. Some needs are predictable (coupons on bonds the bank has issued); others are not (retail withdrawals, drawdowns on credit lines the bank has granted). Complex instruments make prediction harder: downgrade triggers, guarantees and counterparty defaults on derivatives can hit cash without warning. Hull names three causes of funding problems, and in a severe case all three arrive together:

1. **Liquidity stresses in the economy**, such as the flight to quality of 2007 to 2009: investors will not fund anything with any credit risk.
2. **Overly aggressive funding decisions**: short-term instruments funding long-term needs, a liquidity mismatch. "How much of a mismatch is too much?"
3. **Poor financial performance**, a loss of confidence: deposits leave and funding cannot be rolled over.

2. Sources of liquidity, and how each behaves under stress

Hull's six sources	What happens in a stress
Cash and Treasuries	Convertible at once, but *relatively expensive*: they earn less than loans.
Trading book sales	Only as liquid as the book in *stressed* conditions: plan on stressed spreads (LTR-1 a).
Borrowing at short notice	Stress brings higher rates, shorter terms or refusal. Lines of credit help, at a cost (Countrywide's USD 11.5 billion facility in 2007).
Wholesale and retail deposits	Wholesale funding is volatile; retail savers now move money online. Stress is market-wide: if one bank raises rates, others do too.
Securitization	Dried up almost overnight in August 2007; ABCP backstops forced banks to buy the paper.
Central bank borrowing	Lender of last resort: haircut on collateral, a relatively high rate, and a stigma that banks keep secret.

Hull also flags **hedging**: liquidity problems arise when a company hedges an *illiquid* asset with contracts subject to margin, the story of Ashanti and Metallgesellschaft.

3. The three cases

Northern Rock (2007). A top-five UK mortgage lender funding much of its book with short-term debt and securitization (Granite). After August 2007 investors would not lend to mortgage-heavy banks and maturing debt could not be replaced, although the FSA judged it solvent. It asked the Bank of England for support on 12 September; the BBC broke the news on 13 September; the first British bank run in 150 years followed (about GBP 2 billion withdrawn by 17 September) and ended only when the government guaranteed all deposits. Emergency borrowing at a penalty rate reached GBP 25 billion by February 2008; the bank was nationalized.

Ashanti Goldfields (1999). A Ghanaian gold miner that **sold gold forward** against *falling* prices, with daily margin. When 15 European central banks limited their gold sales (26 September 1999), gold **jumped over 25%**; Ashanti could not meet the margin calls, because its offsetting gain was gold in the ground, an illiquid asset. It sold a mine, diluted its shareholders and restructured its hedges.

Metallgesellschaft (early 1990s). MG sold 5- to 10-year fixed-price heating oil and gasoline contracts above market prices and hedged with **long, rolled short-dated futures**. Oil **fell**: the futures paid margin now, the supply contracts would pay back only over years (Figure 1), and its large trades were anticipated by others. Management and its bankers, alarmed by the cash drain, closed the hedges and abandoned the contracts: a loss of USD 1.33 billion.

The lesson GARP draws from Ashanti and MG is not that companies should avoid forwards and futures for hedging, but that they should **ensure access to funding to handle the cash flow mismatches** that might arise in extreme circumstances. In all three cases the economics were arguably sound and the cash timing was not.

Matching against Northern Rock. D27 sets Northern Rock beside the dealer-bank failures of LTR-8: Bear Stearns (repo not rolled, prime brokerage flight), Lehman (its clearing bank's right of offset), AIG (collateral calls on downgrade). None of these is Northern Rock's story.

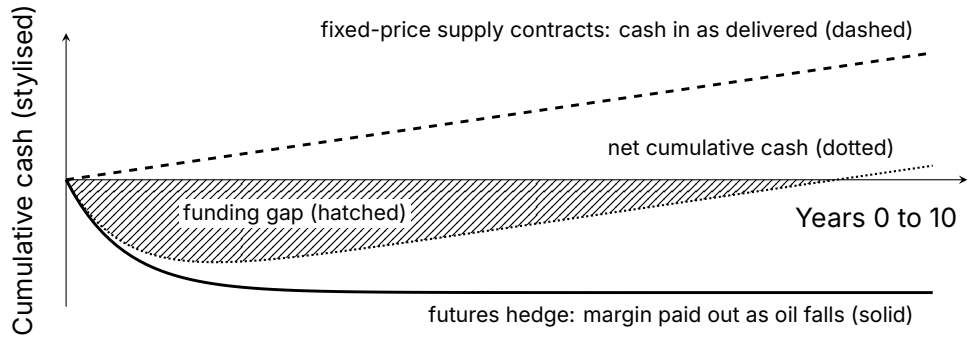


Figure 1. Why Metallgesellschaft's hedge drained cash: a stylised picture, not MG's actual flows. The hedge loses (solid line) in the first years while the hedged contracts pay back slowly over a decade (dashed). The net position (dotted) ends positive, but only after years of negative cumulative cash: the hatched area is the funding the hedge required. Ashanti's picture is the mirror image, with a rising gold price.

	Northern Rock	Ashanti Goldfields	Metallgesellschaft
Position	Short-term debt funding long mortgages	**Short** gold forwards	**Long** rolled short-dated futures
Shock	Wholesale funding closes, 2007	Gold **rises** over 25%	Oil **falls**
Cash drain	Debt not rolled, then a retail run	Margin calls	Margin calls
Outcome	GBP 25 billion support; nationalized	Restructured; mine sold	Hedges closed; USD 1.33 billion loss
Lesson	Liquidity is not solvency; wholesale funding flees; central bank news can start a run	Hedging is not the error: secure funding for the cash flow mismatch of a margined hedge on an illiquid or long-dated asset	

Exam traps

- **Ashanti/MG swap.** Ashanti: short forwards, gold *rose*. MG: long futures, oil *fell*. Say the direction and the actor before locking an answer.
- **Wrong lesson.** The lesson is funding access for cash flow mismatches, never "avoid hedging" and never "futures beat forwards".
- **Case cross-mapping.** A hedge or a clearing bank's right of offset is not Northern Rock; positive feedback trading is LTR-1 d (1987, LTCM), not MG.
- **Sequence at Northern Rock.** Wholesale funding failed first; the retail run followed the news of Bank of England support.
- **Polarity on sources.** Wholesale deposits are the *less* stable source; Treasuries are liquid but *expensive*; central bank money is costly and carries stigma; securitization *failed* in 2007.

Practice questions: LTR-1 b

Question LTR-1 b.1

Bank Q32 • BT P2.T8.20.3.1

A treasury analyst is preparing training notes on liquidity funding risk, the risk relating to a firm's ability to meet its cash needs as they arise. Which of the following statements in the notes is correct?

A. Wholesale deposits are the most stable source of funding.

B. A lack of confidence can contribute to an institution's liquidity funding problems.

C. A solvent firm cannot experience a liquidity funding problem; only an insolvent firm can.

D. Northern Rock and Ashanti Goldfields show that a hedged position is the best protection against funding problems.

Question LTR-1 b.2

Bank Q33 • Schweser Module Quiz 66.2 Q1

Which of the following statements accurately reflects liquidity management for a financial institution?

A. Investments in Treasury securities help a bank increase its risk and return profile.

B. Stability in retail deposits over the years has enhanced bank liquidity.

C. Central bank borrowings are relatively cheap ways for banks to stay afloat.

D. Stressed markets can limit the effectiveness of trading book liquidation strategies.

Question LTR-1 b.3

New • built from GARP Ch.1, 1.2.1 Sources of Liquidity

The treasurer of a mid-sized bank is reviewing the contingency section of its liquidity plan, which lists the sources it would draw on in a market-wide funding squeeze. Which of the following statements in the plan is consistent with Hull's analysis of funding sources?

A. Securitizing new mortgage loans is a dependable source of cash in a squeeze, since demand for asset-backed securities does not depend on funding conditions.

B. If wholesale funding dries up, the bank can quickly replace it by offering above-market rates on retail and wholesale deposits.

C. Committed lines of credit arranged in advance, at a cost, can be drawn when other borrowing becomes expensive or unavailable.

D. Holding a large stock of Treasury securities is a cheap way to secure liquidity, since they convert to cash quickly without any loss of return.

Question LTR-1 b.4

Bank Q30 • AP Mock 1 Q52 (rebuilt)

A risk manager preparing a case review for the asset-liability committee notes that Northern Rock, a British mortgage lender, was nationalized after the 2007 crisis. Which of the following best describes the funding weakness behind its liquidity failure?

A. It funded long-term mortgages with short-term debt and securitization, both of which dried up when investors stopped lending in 2007.

B. It relied mainly on retail deposits, the least stable funding source, and the retail run began before wholesale markets closed.

C. It sold gold forward to protect against falling prices and could not meet margin calls when the gold price rose sharply.

D. It hedged long-dated fixed-price supply contracts with rolled short-dated futures, and falling prices triggered margin calls.

Question LTR-1 b.5

Bank Q31 • GARP PE1 Q42 • Mock 3 Q38

A risk manager at a bank is seeking to better understand recent liquidity risk failures. Several real-life cases are reviewed. Which of the following lessons would be best illustrated by the case of Metallgesellschaft in 1993?

A. Negative public perception of emergency borrowing from the central bank can cause a bank run.

B. Positive feedback trading in illiquid instruments can cause excessive losses.

C. Hedging liabilities by rolling forward short-term futures contracts may create cash flow mismatches.

D. Futures provide a better effective hedge for commodities exposure than forwards.

Question LTR-1 b.6

New • built from GARP Ch.1, Business Snapshots 1.1 to 1.3; workbook D27 notes

A liquidity risk trainer asks participants to match three cases with the lesson each best illustrates.

Cases: 1 Northern Rock 2 Ashanti Goldfields 3 Metallgesellschaft

P A forward-sale hedge lost value when the commodity's price rose sharply; the offsetting gain sat in an illiquid asset that could not be sold to meet margin calls.

Q A hedge of rolled short-dated futures lost value when the commodity's price fell, while gains on long-dated fixed-price contracts would arrive only over many years.

R Long-term assets were funded with short-term debt that could not be rolled over; news of emergency central bank support then triggered a run on a solvent firm.

S The firm's clearing bank invoked its right of offset and stopped processing its payments, so it could no longer settle.

Which of the following matches is correct?

A. 1 with R; 2 with P; 3 with Q

B. 1 with R; 2 with Q; 3 with P

C. 1 with S; 2 with P; 3 with Q

D. 1 with S; 2 with Q; 3 with P

Solutions • LTR-1 b

Solution LTR-1 b.1 — Answer: B

Bank Q32 • BT P2.T8.20.3.1

Step 1 – Recall what causes funding problems

Hull's three causes are liquidity stresses in the economy, overly aggressive funding (short-term money for long-term needs), and poor performance that leads to a lack of confidence, a loss of deposits and difficulty rolling over funding.

⇒ Loss of confidence is one of the named causes: B is correct.

Step 2 – Test the other statements against the reading

Wholesale deposits are the volatile source; liquidity and solvency differ; the hedging cases show hedges *creating* cash needs.

Interpretation

Northern Rock pairs B with the solvency point: the FSA judged it solvent with a good-quality loan book, and it still needed GBP 25 billion of emergency borrowing once confidence went.

Why the other options are wrong

A is incorrect. Polarity flip: Hull calls wholesale deposits a more volatile source than retail deposits; they "can disappear quickly in stressed market conditions".

C is incorrect. Confuses liquidity with solvency: solvent institutions can, and sometimes do, fail because of liquidity problems.

D is incorrect. Reverses the lesson: Ashanti's hedge produced the margin calls it could not meet, and Northern Rock was not a hedging case. The lesson is to secure funding for the cash flow mismatches a hedge can create.

LO: LTR-1 b: Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft. • **Reference:** Hull, Risk Management and Financial Institutions 5e, Ch.24, 1.2 Liquidity Funding Risk; 1.2.1 Sources of Liquidity

Solution LTR-1 b.2 — Answer: D

Bank Q33 • Schweser Module Quiz 66.2 Q1

Step 1 – Trading book liquidation depends on the market

Selling part of the trading book is a funding source only if the book can be sold. In stressed markets spreads widen and buyers retreat, so a strategy that works in normal conditions may fail when it is needed.

⇒ D is correct.

Step 2 – Check the other three sources

Treasuries, retail deposits and central bank borrowing are each misdescribed (see below).

Interpretation

Hull: the analysis of trading book liquidity "should therefore be based on stressed market conditions, not normal market conditions", which is why the stressed cost of liquidation of LTR-1 a matters for funding plans.

Why the other options are wrong

A is incorrect. Treasuries *lower* risk and return: they are liquid but "relatively expensive sources of liquidity", because they earn less than loans.

B is incorrect. Polarity flip: retail deposits are less stable than they used to be, since customers compare rates and move money over the Internet.

C is incorrect. Central bank borrowing is costly: collateral with a haircut, a relatively high rate (a penalty rate for Northern Rock), and a signal of distress that banks try to keep secret.

LO: LTR-1 b: Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft. • **Reference:** Hull, Ch.24, 1.2.1 Sources of Liquidity; Schweser-Notes 2026 Book 4, Reading 66, LO 66.b

Solution LTR-1 b.3 — Answer: C

New • built from GARP Ch.1, 1.2.1

Step 1 – Ask which sources survive a market-wide squeeze

Hull: liquidity problems "tend to be market-wide rather than something that affects one or two financial institutions". A source that depends on other institutions' appetite at the time of the squeeze is the weakest.

Step 2 – Lines of credit

A bank "can (at a cost) mitigate its funding risks somewhat by arranging lines of credit". Countrywide drew on its USD 11.5 billion syndicated facility in the 2007 crisis.
⇒ C is the statement consistent with Hull.

Interpretation

Lines of credit buy time rather than certainty: Countrywide still had significant problems and was taken over by Bank of America in January 2008. Hull also asks banks to monitor the assets they could pledge as collateral at short notice.

Why the other options are wrong

A is incorrect. Securitization "dried up almost overnight in August 2007" as investors decided the products were too risky; banks with ABCP backstops had to buy the paper themselves.
B is incorrect. When one institution offers better rates to raise deposits, "others usually want to do the same thing", so the extra funding is hard to achieve; wholesale deposits are the first to leave.
D is incorrect. Treasuries are liquid but relatively expensive: they earn less than loans, so there is a trade-off between liquidity and return and a limit to how much can be held.

LO: LTR-1 b: Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft. • **Reference:** Hull, Ch.24, 1.2.1 Sources of Liquidity

Solution LTR-1 b.4 — Answer: A

Bank Q30 • AP Mock 1 Q52

Step 1 – Identify the funding model

Northern Rock relied on selling short-term debt instruments for much of its funding and securitized mortgages through Granite, while its assets were long-term mortgages.

Step 2 – Identify the trigger and the sequence

After August 2007 investors became nervous about lending to banks heavily involved in mortgages, so maturing debt could not be replaced. The bank asked the Bank of England for support on 12 September; the news broke on 13 September; the run started on 14 September.
⇒ The weakness was the funding mismatch; the retail run was its consequence.

Interpretation

Hull's verdict: had it been managed a little more conservatively, paying more attention to its access to funding, it might have survived. It was solvent throughout.

Why the other options are wrong

B is incorrect. Reverses the sequence: the retail run followed the news of emergency borrowing, after wholesale funding had failed. Hull also ranks wholesale, not retail, deposits as the less stable source.
C is incorrect. Case cross-mapping: this is Ashanti Goldfields (1999).
D is incorrect. Case cross-mapping: this is Metallgesellschaft (early 1990s).

LO: LTR-1 b: Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft. • **Reference:** Hull, Ch.24, 1.2 Liquidity Funding Risk, Business Snapshot 1.1 Northern Rock

Solution LTR-1 b.5 — Answer: C

Bank Q31 • GARP PE1 Q42

Step 1 – MG's position

MG sold 5- to 10-year fixed-price supply contracts on heating oil and gasoline and hedged them with long positions in short-dated futures, rolled forward.

Step 2 – The cash timing

When oil fell, the futures were marked to market daily and generated margin calls at once; the offsetting gains on the fixed-price contracts would be realized only as product was delivered over the years.

Step 3 – The outcome

Senior management and its bankers, alarmed by the cash drain, closed out the hedges and abandoned the contracts: a loss of USD 1.33 billion.
⇒ The lesson is the cash flow mismatch created by a short-dated hedge of long-dated commitments: C.

Interpretation

GARP's conclusion from Ashanti and MG is not that companies should avoid forwards and futures, but that they should make sure they have access to funding for the cash flow mismatches that can arise in extreme circumstances (Figure 1).

Why the other options are wrong

A is incorrect. That is the Northern Rock lesson: the run followed news of Bank of England support.
B is incorrect. Positive feedback trading belongs to liquidity black holes (LTR-1 d: the 1987 crash, LTCM), not to MG.
D is incorrect. Not a lesson of the case: Hull draws no ranking of futures against forwards, and it was the futures' daily margining that drained MG's cash.

LO: LTR-1 b: Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft. • **Reference:** Hull, Ch.24, 1.2.1 Sources of Liquidity, Hedging Issues, Business Snapshot 1.3 Metallgesellschaft

Solution LTR-1 b.6 — Answer: A

New • built from GARP Ch.1 and workbook D27

Step 1 – Place Northern Rock

R describes Northern Rock: short-term funding that could not be rolled, then a run on news of Bank of England support, although the FSA judged it solvent. S is the Lehman endgame (its clearing bank's right of offset), a dealer-bank case of LTR-8.
⇒ Case 1 goes with R, which removes C and D.

Step 2 – Separate the two hedgers by actor and direction

Ashanti *sold* gold forward, so a *rise* in gold hurt it: P. MG was *long* rolled oil futures against fixed-price sales, so a *fall* in oil hurt it: Q.

$\Rightarrow$ 1 with R, 2 with P, 3 with Q: option A.

Interpretation

Both hedgers faced margin calls, so the direction word is the only discriminator between P and Q. State the actor and the direction ("Ashanti short gold, gold rose; MG long oil futures, oil fell") before locking the answer.

Why the other options are wrong

B is incorrect. Swaps Ashanti and MG: Ashanti used forwards and was hurt by a rising gold price; MG used futures and was hurt by a falling oil price.

C is incorrect. Pins the Lehman endgame on Northern Rock. Northern Rock's lesson is wholesale funding plus a retail run on news of central bank support.

D is incorrect. Makes both errors: Northern Rock cross-mapped to Lehman, and Ashanti swapped with MG.

LO: LTR-1 b: Identify examples of liquidity funding risk, funding sources, and lessons learned from real cases: Northern Rock, Ashanti Goldfields, and Metallgesellschaft. • **Reference:** Hull, Ch.24, Business Snapshots 1.1 to 1.3; Duffie, The Failure Mechanics of Dealer Banks (LTR-8)

LTR-1 • Liquidity Risk • Vardeez: IMPORTANT

LTR-1 d

P3 CONCEPT

Explain liquidity black holes and identify the causes of positive feedback trading.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P3. No workbook section covers it and no recall source reports it, but Vardeez flags it: "liquidity black holes and positive feedback traders" is the second of his two pointers for this IMPORTANT reading (LVaR, LTR-1 a, is the first). The A11 revision notes carry one paragraph on it.

Vardeez: the negative versus positive feedback table and the causes of positive feedback trading.

Reported exam shapes: none listed. The questions are built from the LO wording and Vardeez's focus.

What the questions look like: statement questions. Classify a behaviour as positive or negative feedback (stop-loss, trend following, option hedging, profit-taking at a target), pick the warning sign of a black hole, or judge a claim about 1987, LTCM, deleveraging or regulation. The usual distractors flip the polarity (buy on falls is *negative* feedback) or confuse hedging a short option with hedging a long one.

1. What a liquidity black hole is

A **liquidity black hole** is a market in which liquidity has dried up because everyone wants to sell and no one wants to buy, or the reverse; it is also called a **crowded exit** (LTR-1 a, Section 1). In a well-functioning market the price adjusts to news without overreacting: if a fall goes too far, traders buy and a new equilibrium forms. A black hole is created when a price decline makes *more* participants want to sell, driving the price well below where it will eventually settle; the asset can then be sold only at a fire-sale price (Figure 1). Black holes usually come with falls, though in theory they can occur on the way up. Average spreads have narrowed over time, but Hull notes that situations in which almost everyone wants the same trade at the same time have become more frequent.

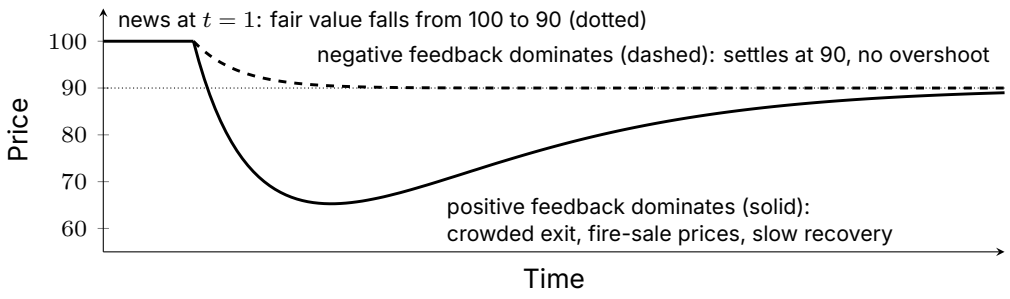


Figure 1. The same bad news with two trader populations, stylised. Buying into the fall stops the dashed path at the new value; selling into the fall drives the solid path far below 90 before it recovers, as the market did within four months of October 1987.

2. Negative versus positive feedback traders

	Negative feedback traders	Positive feedback traders
Behaviour	Buy when prices fall, sell when they rise	Sell when prices fall, buy when they rise
Typical label	Value, contrarian	Momentum, trend following
Effect	Stabilising	Destabilising
When they dominate	Liquid market, reasonable prices, buyers and sellers balanced	Unstable prices; the market may become one-sided and illiquid

Hull calls the split a simplification (some strategies are neither). Positive feedback traders are not wrong about value: their reaction has the *same sign* as the price move, so each round of selling manufactures the next.

3. Causes of positive feedback trading

Hull lists seven. Each *mechanically* sells after a fall; none is a judgement about value.

1. **Trend trading**: buy when the price trends up, sell when it trends down. *Breakout trading* is a variant: after six months between USD 25 and USD 30, buy above 30 and sell below 25.
2. **Stop-loss rules**: sell automatically when the price falls below a set level; "always present".
3. **Dynamic hedging of short option positions** (calls or puts): the hedge buys after a rise and sells after a fall. Hedging a *long* option does the reverse and is negative feedback. Banks hedging options sold to British insurers moved UK long-term rates.
4. **Creating options synthetically**: a synthetic long option uses the trades that hedge a short one. In 1987 portfolio insurers ran synthetic puts on over USD 60 billion of equities. The market fell about 10% from 14 to 16 October; the rules called for at least USD 12 billion of sales but only USD 4 billion were done. On Monday 19 October the Dow fell more than 20%; three insurers made almost 10% of NYSE sales, and others sold ahead of them. Lesson: it is dangerous to follow a strategy, even a hedging strategy, when many others do the same.
5. **Margins**: big moves bring margin calls that leveraged traders cannot meet, so they close out; volatility rises and exchanges raise margins, which makes it worse.
6. **Predatory trading**: traders who know an investor must sell short ahead of it (LTR-1 a, Section 1).
7. **LTCM (1998)**: short liquid bonds, long similar illiquid ones. After the Russian default the gap widened; highly leveraged and unable to meet margin calls, LTCM and its imitators had to buy the liquid and sell the illiquid bonds, widening the gap further.

> **Correction to the source notes.** Vardeez lists three causes (trend trading, stop-loss rules, dynamic hedging); GARP lists the seven above, as do the A11 notes and Schweser. His "dynamic hedging" means hedging *short* options: hedging a long option is negative feedback. Schweser's "contrast this with dynamic hedging, which involves long positions" can suggest the term means long options only; in GARP it covers both, and only the short-option hedge is positive feedback.

4. Why traders end up on the same side

- **Leveraging and deleveraging** (Figure 2). Easy credit raises asset and collateral values and borrowing grows; in reverse, falling collateral values cut credit lines and force sales. A hedge fund told to cut leverage from 20 to 5 times equity can only sell assets; prices and its equity fall, and more sales follow.

- **Irrational exuberance** (Greenspan, 1996): most black holes trace back to it. Many institutions build the same exposure, rising prices make it look profitable, and year-end bonuses keep traders building positions even when they expect a correction next year ("as long as the music is playing, you've got to get up and dance", Chuck Prince, 2007).
- **Regulation.** Uniform rules make banks react alike: when volatility rises, VaR and capital rise everywhere and banks with similar positions all cut exposure at once. Insurers and pension funds have longer horizons and should *not* be regulated like banks.
- **Diversity is the cure.** The lack of independence between traders causes black holes. Hedge funds add diversity but are leveraged and unwind together; a contrarian strategy has merit but is hard to follow under short-term VaR-based risk management.

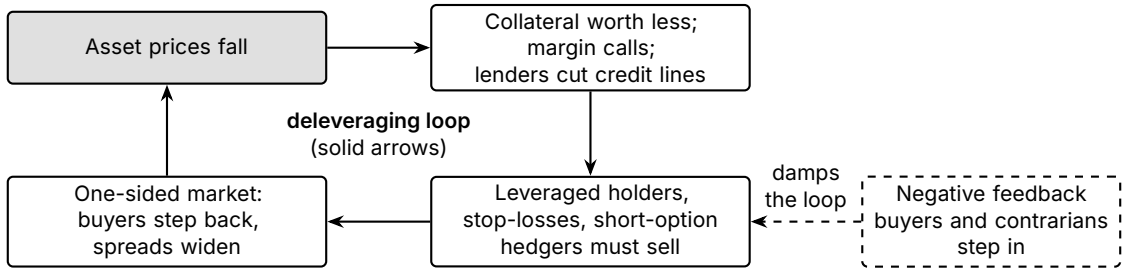


Figure 2. The self-reinforcing loop behind a liquidity black hole, after Hull's deleveraging cycle (his Figure 1.3) with the forced sellers of Section 3 added. Higher leverage turns it faster; only traders who buy into the fall (dashed box) break it.

Exam traps

- **Polarity flip.** Negative feedback = buy on falls = stabilising; positive feedback = sell on falls = destabilising. Profit-taking at a price target (selling after a rise) is negative feedback.
- **Short versus long option.** Hedging a short option, or creating one synthetically, is positive feedback; hedging a long option is negative feedback.
- **Outcome versus cause.** A black hole is the outcome, not a trader type; narrower spreads signal liquidity, not a black hole.
- **Direction words.** LTCM's close-out made liquid and illiquid prices diverge *further*; 1987 insurers could *not* execute in time; uniform regulation *raises* black-hole risk.

Practice questions: LTR-1 d

Question LTR-1 d.1

Bank Q47 • Schweser Module Quiz 66.3 Q1

A highly liquid market tends to result from situations where:

A. stop-loss rules take effect as prices decline.

B. negative feedback traders sell shares as prices rise.

C. positive feedback traders purchase shares as prices rise.

D. breakout trading occurs because of prices moving outside of a range.

Question LTR-1 d.2

Bank Q44 • AnalystPrep Q.3949 (rebuilt)

A risk analyst is reviewing behaviours that can cause positive feedback trading and, in extreme cases, a liquidity black hole. Which of the following is an example of a behaviour that gives rise to positive feedback trading?

A. A fund rebalances to its target weights by buying an equity whose price has fallen.

B. A trader sells automatically when an asset's price falls below a pre-set stop-loss level.

C. A holder of a long option position delta-hedges by selling the underlying after a price increase.

D. A market maker narrows its bid-ask spread as trading volume increases.

Question LTR-1 d.3

New • built from GARP Ch.1, 1.3.1 and Business Snapshot 1.4

The CRO of a bank asks the head of equity derivatives which of the desk's activities could add to a liquidity black hole in a falling market. Which of the following statements in the reply is correct?

A. Delta-hedging a large long position in index puts requires selling the index after it falls, so it adds to selling pressure.

B. Hedging a large short call position requires buying the underlying after it falls, so the hedge supports prices in a decline.

C. Portfolio insurance contributed little to the 1987 crash, because insurers had executed the sales their models required before Monday 19 October.

D. Creating a long put synthetically requires the same trades as hedging a short put, selling after declines and buying after rises, so it is positive feedback trading.

Question LTR-1 d.4

Bank Q46 • GARP PE2 Q28 • Mock 4 Q29

A national regulator is developing a set of indicators to serve as warning signs that the liquidity of the stock market could be decreasing. The regulator is particularly concerned about the potential for a "liquidity black hole" to develop which could destabilize the stock market. Which of the following observations would be of greatest concern to the regulator as a signal of a potential liquidity black hole occurring?

A. A majority of large investors increasingly follow a trend-following trading strategy.

B. Brokerage firms purchase call options as a hedge against the sale of a large number of put options to institutional investors.

C. Portfolio managers sell stocks that have increased in value and reached their price target.

D. Bid-ask spreads for less liquid stocks become narrower.

Question LTR-1 d.5

New • built from GARP Ch.1, 1.3.1 and 1.3.2

A hedge fund that has borrowed 20 times its equity is told by its prime broker to cut its leverage to 5 times. Several other funds with similar positions receive the same instruction. Which of the following best describes the likely market dynamics, according to Hull?

A. Falling asset prices raise the value of the collateral supporting the funds' loans, so lenders extend more credit and the selling stops quickly.

B. As relative-value traders, the funds narrow the gap between liquid and illiquid bonds when they unwind, as LTCM's forced close-out did in 1998.

C. Exchanges typically cut margin requirements after a large price move because volatility falls, which lets leveraged traders keep their positions.

D. The funds can cut leverage only by selling assets; the sales lower prices and the funds' equity, so further sales are needed and a liquidity black hole can develop.

Question LTR-1 d.6

New • built from GARP Ch.1, 1.3.3 to 1.3.5

A policy adviser at a central bank is drafting a note on how to make markets less prone to liquidity black holes. Which of the following recommendations is consistent with Hull?

A. Regulate pension funds and life insurers exactly like banks, so that no institution gains an advantage by holding illiquid assets.

B. Require all banks to respond to rising volatility in the same way, since a uniform VaR-based rule makes deleveraging orderly.

C. Encourage diversity in trading strategies and regulate different types of institutions differently, since black holes arise when traders act alike.

D. Tie traders' bonuses more closely to year-end profits, so that traders exit irrationally exuberant markets before the correction comes.

Solutions • LTR-1 d

Solution LTR-1 d.1

Answer: B

Bank Q47 • Schweser Module Quiz 66.3 Q1

Step 1 – Who stabilises a market

Negative feedback traders buy when prices fall and sell when they rise. In liquid markets they dominate the trading.

⇒ Selling into a rise supplies stock to buyers and keeps the price near value: B.

Step 2 – Classify the other three

Stop-loss selling, buying into a rise and breakout trading all trade *with* the move: positive feedback.

Interpretation

Liquidity is a balance of buyers and sellers. Trades against the move provide the other side; trades with the move remove it.

Why the other options are wrong

A is incorrect. Stop-loss rules sell after a fall; Hull calls them "a source of positive feedback trading that is always present".

C is incorrect. Polarity flip: buying after a rise is positive feedback, pushing the price further and draining liquidity.

D is incorrect. Breakout trading is a form of trend trading (buy above the top of the range, sell below the bottom): positive feedback.

LO: LTR-1 d: Explain liquidity black holes and identify the causes of positive feedback trading. • **Reference:** Hull, Risk Management and Financial Institutions 5e, Ch.24, 1.3.1 Positive and Negative Feedback Traders

Solution LTR-1 d.2

Answer: B

Bank Q44 • AnalystPrep Q.3949

Step 1 – Test each behaviour: does it trade with the price move?

A buys after a fall (against the move); B sells after a fall (with the move); C sells after a rise (against the move); D is not a trading rule at all.

⇒ Only B is positive feedback.

Step 2 – Name the cause

A stop-loss rule is one of Hull's seven causes of positive feedback trading, and the one that is always present.

Interpretation

The option case is the trap. Hedging a *short* option buys after rises and sells after falls (positive feedback); hedging a *long* option, as in C, does the reverse and "should not interfere with market liquidity".

Why the other options are wrong

A is incorrect. Rebalancing by buying a fallen asset is negative feedback: it stabilises prices.

C is incorrect. Polarity flip: dynamic hedging of a long option position is negative feedback.

D is incorrect. Narrower spreads with higher volume signal more liquidity; it is not a behaviour that amplifies price moves.

LO: LTR-1 d: Explain liquidity black holes and identify the causes of positive feedback trading. • **Reference:** Hull, Ch.24, 1.3.1 Positive and Negative Feedback Traders

Solution LTR-1 d.3 — Answer: D

New • built from GARP Ch.1, 1.3.1

Step 1 – Hedging a short option

A short call is hedged with a long position in the underlying; after a rise the call's delta rises, so the hedger buys, and after a fall it sells. A short put is hedged with a short position, which also grows after a fall. Either way: buy after rises, sell after falls.

Step 2 – Synthetic options

Hull: hedging a short position in an option is equivalent to creating a long position in the same option synthetically, so the synthetic long put trades exactly like a short-put hedge.
⇒ Positive feedback trading: D.

Interpretation

This was portfolio insurance in 1987: rules to create synthetic puts on more than USD 60 billion of equities. Hull's lesson is that it is dangerous to follow a strategy, even a hedging strategy, when many others are doing the same.

Why the other options are wrong

A is incorrect. Polarity flip: as the index falls, a long put's delta becomes more negative, so its hedge *buys* the index. Hedging a long option is negative feedback.
B is incorrect. Polarity flip: after a fall a short call's hedge *sells* the underlying, adding to the decline.
C is incorrect. Factually wrong: by Friday 16 October insurers had sold only USD 4 billion of the USD 12 billion their rules required, and on 19 October three of them made almost 10% of NYSE sales.

LO: LTR-1 d: Explain liquidity black holes and identify the causes of positive feedback trading. • **Reference:** Hull, Ch.24, 1.3.1 Positive and Negative Feedback Traders (dynamic hedging; creating options synthetically), Business Snapshot 1.4 The Crash of 1987

Solution LTR-1 d.4 — Answer: A

Bank Q46 • GARP PE2 Q28 • Mock 4 Q29

Step 1 – What a black hole needs

Most participants on the same side of the market at the same time, which happens when positive feedback traders dominate.

Step 2 – Find the observation that shows positive feedback spreading

If most large investors follow trends, they all sell into declines and buy into rallies, leaving no counterparties.
⇒ A is the strongest warning sign.

Interpretation

The regulator is watching the *composition* of the market. Hull's concern is exactly this: an increasing tendency for almost everyone to want to do the same trade at the same time.

Why the other options are wrong

B is incorrect. Buying calls against sold puts is an option-for-option trade; it does not force the brokers to trade the underlying in the direction of price moves, as dynamic hedging of short options would.
C is incorrect. Selling after a rise to a price target is negative feedback: stabilising.
D is incorrect. Narrower spreads mean more liquidity, not less.

LO: LTR-1 d: Explain liquidity black holes and identify the causes of positive feedback trading. • **Reference:** Hull, Ch.24, 1.3 Liquidity Black Holes; 1.3.1 Positive and Negative Feedback Traders

Solution LTR-1 d.5 Answer: D

New • built from GARP Ch.1, 1.3.1 and 1.3.2

Step 1 – How leverage is cut

Hull's own example: a fund that could borrow 20 times its equity before 2007 is told to cut to 5 times. "It can only do this by selling assets."

Step 2 – Why it feeds on itself

Prices fall because of what this fund and the others are doing; their equity falls; further sales are necessary. Falling prices also lower collateral values, so banks cut lines of credit and force more sales.

⇒ A self-reinforcing deleveraging spiral, the setting for a black hole: D.

Interpretation

This is Hull's deleveraging cycle (Figure 2). Hedge funds add diversity in calm markets, but because they are highly leveraged they all unwind together when liquidity tightens, as in the second half of 2007.

Why the other options are wrong

A is incorrect. Polarity flip: falling prices *reduce* collateral values, and lenders reduce lines of credit.

B is incorrect. Wrong direction: LTCM's forced close-out (buying liquid bonds, selling illiquid ones) made the prices diverge further and reinforced the flight to quality.

C is incorrect. Polarity flip: big moves raise volatility, and exchanges *increase* margin requirements, which makes forced selling worse.

LO: LTR-1 d: Explain liquidity black holes and identify the causes of positive feedback trading. • **Reference:** Hull, Ch.24, 1.3.1 (margins, LTCM); 1.3.2 Leveraging and Deleveraging

Solution LTR-1 d.6 Answer: C

New • built from GARP Ch.1, 1.3.3 to 1.3.5

Step 1 – Find the root cause

Hull: models assume market participants act independently, but "it is this lack of independence that causes liquidity black holes".

Step 2 – Find the remedy

"To solve the problem of liquidity black holes, we need more diversity in financial markets." One way is to recognise that different types of institutions have different risks and to regulate them differently.

⇒ C.

Interpretation

Hull adds that a contrarian strategy, buying when everyone else is selling, has merit if markets overreact, but it is hard for an institution to follow under short-term VaR-based risk management.

Why the other options are wrong

A is incorrect. Hull's answer is "no": insurers and pension funds have longer horizons and should not be penalized for investing in illiquid assets.

B is incorrect. Uniform rules are the problem: when volatilities rise, VaR and capital rise at every bank, banks with similar positions all cut exposure, and a black hole can develop.

D is incorrect. Year-end bonuses *feed* irrational exuberance: a trader who expects the correction next year keeps building the position to maximise short-term pay.

LO: LTR-1 d: Explain liquidity black holes and identify the causes of positive feedback trading. • **Reference:** Hull, Ch.24, 1.3.3 Irrational Exuberance; 1.3.4 The Impact of Regulation; 1.3.5 The Importance of Diversity

LTR-6 • Intraday Liquidity Risk Management • Vardeez: IMPORTANT

LTR-6 a

P3 CONCEPT

Identify and explain the uses and sources of intraday liquidity.

Workbook: D27 Intraday Liquidity and Dealer Bank Failure Mechanics • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D27 (Map #65, reported by 1 of 16 recall sources). D27 has revision notes and traps but no questions. Vardeez rates the reading IMPORTANT, and this LO is his only pointer for it.
Vardeez: "Uses and sources of intraday liquidity".

Reported exam shapes

- **S2** The payment-system angle: uses and sources of intraday liquidity, daylight overdraft policy, and the defences and tools of financial market utilities (FMUs).

What the questions look like: no calculations. You may have to classify an item as a use or a source, or name the largest of each. Other questions describe an item by its timing, forecastability or who starts it and ask you to name it. Expect also: which source carries an explicit cost, how to respond to a daylight overdraft, and statement audits on central bank intraday credit and FMU risk management. Governance (LTR-6 b) is P4 and is not built here. The monitoring measures are LTR-6 c.

1. Why banks run daylight overdrafts

A **daylight (intraday) overdraft** is a negative balance during the business day: debits posted so far exceed the opening balance plus credits posted so far. **Intraday liquidity** is cash the bank can reach at any point in the day to keep processing transactions; **intraday liquidity risk** (settlement risk) is the risk of not meeting a payment or settlement obligation at the expected time.

In the high-rate Volcker era, reserves earned nothing, so banks ran balances to the minimums and went negative intraday. Central banks feared (1) **inadvertent credit** to a member that fails during the day, and (2) a **"shadow" money supply**: peak overdrafts were 9.4% of M1 in 1986 and 12.4% in 1991, against 0.7% in 2013. Fed policy since then:

1985	Net debit cap: self-assessed maximum overdraft, approved by the board.
1994	24 bp fee above a deductible of 10% of risk-based capital (36 bp from 1995). 2001: collateral lets a few banks exceed their caps.
2008	50 bp on *unsecured* overdrafts, zero if *fully collateralised*, no deductible, 150 bp penalty without discount window access.

Overdrafts persist even with about USD 1.5 trillion at the Fed (2012) for two reasons. **Market conventions**: fed funds are returned first thing next morning, and FMUs settle in one end-of-day window, so a bank expecting a late credit runs a deficit until it arrives. **Intraday credit to clients**: payroll paid before an afternoon wire arrives, or a custody client buying before a maturing asset is credited.

2. Uses and sources

Uses (GARP Table 6.2):

- **Outgoing wire transfers** on LVPSs (Fedwire, CHIPS), for clients or the bank: **typically the largest use**. Steady all day; near the debit cap, banks "throttle" payments, and most LVPSs have hard limits.
- **Settlements at PCS systems**: mostly **once a day, late afternoon**; a **use or a source** by net position; forecastable for multi-day settlements (T+3), not same-day.
- **Funding of nostro accounts**: correspondent balances kept at a **target average monthly balance** as compensation for services; a **use or a source** day to day.
- **Collateral pledging**: OTC trading, public deposits, margin calls; known about a day ahead.
- **Asset purchases and funding**: securities, client loans (hard to forecast), fixed assets.

Sources (GARP Table 6.3):

- **Cash balances**: opening balances at the central bank and correspondents.
- **Incoming funds flow**: payments and FMU credits, **the largest source in periods of normal market function**; real-time (LVPS) or batch.
- **Intraday credit**: a line or overdraft covered by the close of business. **Fed**: **unsecured committed** line (the net debit cap), **charged**. **Bank of England**: top-quality government **collateral**, **no interest**. **Interbank daylight lines**: generally **uncommitted and free**. FMUs let settlement positions build until the window.
- **Liquid assets**: money market instruments, time deposits, banker's acceptances, short-term government debt; low volatility, predictable; a client repo turns them into cash.
- **Overnight borrowings**: fed funds, LIBOR, Eurodollar; repaid *next* day. Cost of excess cash at day end versus the risk of breaching a daylight limit.
- **Other term funding**: FHLB, term repo; **incremental**, not a consistent source.

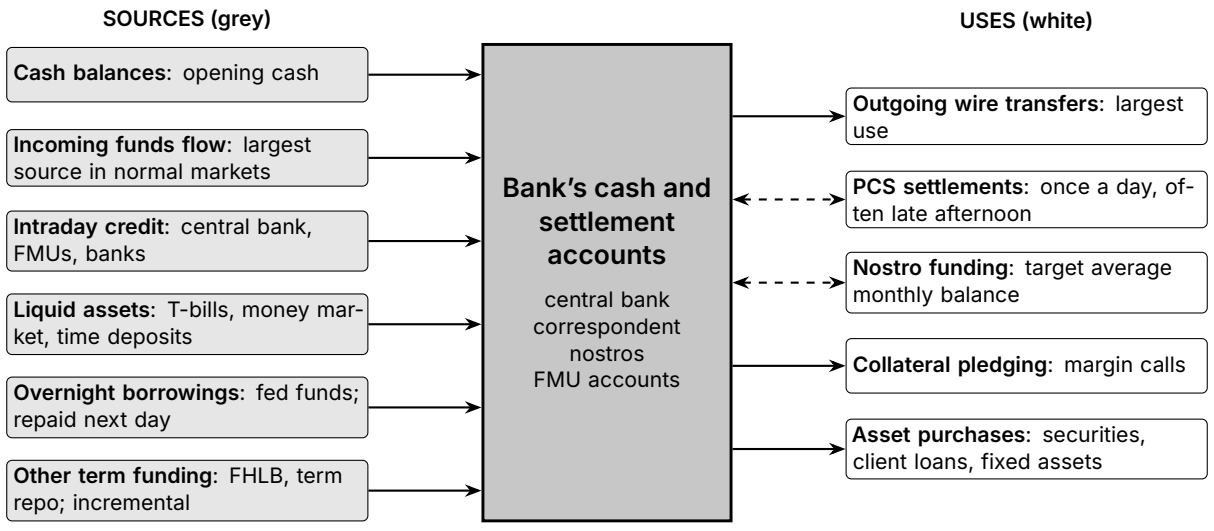


Figure 1. Sources (grey boxes, left) feed the bank's accounts and uses (white boxes, right) drain them, after GARP Tables 6.2 and 6.3. Solid arrows run one way; the two dashed double-headed arrows mark the uses that can be a source on a given day, depending on the net position.

The **treasurer's challenge** is to stay within daylight limits and end the day at target balances despite (1) variable cash flows, (2) market forces (unexpected margin calls, central bank money desks) and

(3) no single source of real-time data.

3. How FMUs protect themselves

FMUs have three **lines of defence** against a participant that cannot settle (not the bank's own Treasury, Risk Management and Internal Audit lines of LTR-6 b):

1. **Membership criteria** (size, creditworthiness). **Tiering**, indirect members routed through direct ones, *adds* risk: a direct member's failure hits those behind it, others hesitate to pay a failing member's clearing bank, and interdependence grows. CHAPS had 18 direct members in 2010 (Target2 866, Fedwire over 7,000); the Bank of England set out to reduce its tiering.
2. **Monitoring participants**: financial statements, settlement positions, margin collateral. For a weakening member: intraday margin calls, a cut or removed net debit cap, **prefunding**, extra collateral, a letter of credit or performance bond.
3. **Risk mutualisation**: a **prefunded** guarantee fund, losses shared by survivors; or central bank underwriting of the LVPS, as for Fedwire, which gives **immediate finality**.

Tools: net debit caps, collateral (composition, advance rates, concentration limits), **liquidity savings mechanisms** (netting, throughput requirements, prioritisation, intra-system lending), which *reduce* the liquidity needed; staggered settlement windows; contingent backup lines for emergencies.

> **Correction to the source notes.** Vardeez says intraday credit lines are uncommitted and free, so that calling them "committed, or interest-bearing" is a corruption. That is true of interbank daylight lines (and of GARP's one-line Table 6.3 summary), not of the Fed: GARP's text says the Fed provides an *unsecured committed* line and *charges* for it, while the Bank of England charges no interest but takes collateral. Also, a footnote in GARP Ch.8 quotes a 36 bp Fed fee; that is the 1995 level, replaced in 2008.

Comparison card: LTR-6 a

Largest	Use: outgoing wires. Source: incoming funds flow, *in normal markets*
Always a use	Outgoing wires, collateral pledging, asset purchases and funding
Use *or* source	PCS settlements, nostro funding (net position that day)
Sources	Cash balances, incoming funds flow, intraday credit, liquid assets, overnight borrowings, other term funding
Intraday credit terms	Fed: unsecured, committed, charged. BoE: collateralised, free. Interbank: uncommitted, free
Explicit cost	Intraday credit (interest or collateral); own cash and liquid assets: opportunity cost only
FMU defences	1 membership (tiering adds risk); 2 monitoring; 3 mutualisation (prefunded fund, central bank)

Exam traps

- **Polarity flip and dropped scope.** Wires are the largest *use*; incoming funds flow the largest *source*, *in normal markets* only.
- **Sibling swap.** Fed (unsecured, committed, charged) versus Bank of England (collateralised, free).
- **Overnight is not intraday.** Intraday credit is covered by the close; overnight borrowing is

repaid next day.

- **Wrong year.** The 10% deductible was 1994; since 2008 there is none and collateralised overdrafts are free.
- **FMU defences.** Tiering *raises* risk; liquidity savings mechanisms *lower* liquidity needs; the guarantee fund is the *third* line and prefunded.

Practice questions: LTR-6 a

Question LTR-6 a.1

Bank Q241 • Schweser Module Quiz 71.1 Q1 (rebuilt)

A new treasurer at a large US bank asks an analyst which intraday flows matter most for the bank's daylight overdraft at the Federal Reserve. Which of the following statements should the analyst make?

A. Outgoing wire transfers are typically the largest use of intraday liquidity, and incoming funds flow is the largest source in periods of normal market function.

B. Outgoing wire transfers are typically the largest source of intraday liquidity, and incoming payment flows from FMU settlements are the largest use.

C. Intraday credit from the central bank is the largest source of intraday liquidity in all market conditions, and collateral pledging is the largest use.

D. Settlements at PCS systems are the largest use of intraday liquidity, because most of them settle continuously throughout the business day.

Question LTR-6 a.2

Bank Q235 • AnalystPrep Q.3908 (rebuilt)

A bank treasurer reviews an item in the bank's intraday funding plan that has the following characteristics:

1. There is a single settlement per day, mainly in the late afternoon.
2. It may be either a source or a use of funds, depending on the participant's net position on the day.
3. It can be forecast for securities with multi-day settlements, but is harder to forecast for same-day settlement activity.

Which item do these characteristics describe?

A. Outgoing wire transfers

B. Funding of nostro accounts

C. Collateral pledging

D. Settlements at payment, clearing and settlement (PCS) systems

Question LTR-6 a.3 Bank Q236 • AnalystPrep Q.3919 (rebuilt)

XYZ Bank borrows federal funds in the afternoon to cover a funding shortfall and returns them the next morning. It uses the borrowed funds to pay a correspondent bank abroad for the services it provides. Which of the following correctly identifies the source of intraday liquidity and the use to which XYZ puts the funds?

A. Cash balances; outgoing wire transfers

B. Incoming funds flow; settlements at PCS systems

C. Overnight borrowings; funding of nostro accounts

D. Liquid assets; outgoing wire transfers

Question LTR-6 a.4 Bank Q242 • Schweser Module Quiz 71.1 Q2

A bank treasurer is attempting to minimise liquidity funding costs. Which of the following sources of intraday liquidity is most likely to have the greatest explicit financial cost?

A. Cash balances

B. Incoming funds flow

C. Intraday credit

D. Liquid assets

Question LTR-6 a.5 Bank Q239 • GARP PE1 Q50

A junior treasury analyst at a US-based regional bank evaluates the bank's intraday transactions and their effect on its liquidity profile. Which of the following observations would be correct?

A. Intraday overdrafts are prohibited by regulators, but the bank can borrow from the Federal Reserve before the opening of business to resolve expected liquidity needs.

B. The bank's clients can serve as a source of intraday liquidity, which allows the bank to convert liquid assets such as US Treasury bills to cash.

C. The balance of money market assets in the bank's investment portfolio tends to be volatile and is therefore difficult to forecast on any given day.

D. Funding of nostro accounts at correspondent banks is typically initiated by clients entering into loans or repo transactions.

Question LTR-6 a.6

Bank Q240 • GARP PE2 Q22

A junior treasury analyst at a large US bank assesses how the bank manages intraday liquidity. In a scenario, the bank has an intraday overdraft of its account at a financial market utility (FMU). Which of the following correctly describes a cause of an intraday overdraft and the bank's most likely response?

A. The bank pledges cash collateral against a purchase of OTC securities, which leads it to borrow intraday funds from the US Federal Reserve to eliminate the overdraft.

B. Customers transfer funds to the bank in advance of an expected purchase of real estate, which leads the bank to deposit funds into a special account to eliminate the overdraft.

C. Interest rates fall while the bank has a positive duration gap, which leads the bank to sell liquid assets to eliminate the overdraft.

D. The FMU increases the bank's net debit cap, which triggers an overdraft and leads the bank to initiate a reverse repurchase agreement to provide the necessary intraday liquidity.

Question LTR-6 a.7

New • built from GARP Ch.6 and workbook D27

A payments risk analyst at a global bank compares central bank intraday credit in the United States and the United Kingdom for a board briefing. Which of the following statements in the briefing is correct?

A. The Federal Reserve requires intraday overdrafts to be collateralised and charges no interest, while the Bank of England provides an unsecured committed line and charges for its use.

B. The Federal Reserve provides an unsecured committed line through its net debit cap and charges for its use, while the Bank of England requires top-quality government collateral and charges no interest.

C. Since 2008 the Federal Reserve has charged 50 bp on all daylight overdrafts, collateralised or not, above a deductible equal to 10% of the bank's risk-based capital.

D. Peak daylight overdrafts have grown as a share of the M1 money supply since the 1980s, which is why central banks now worry that they act as a shadow money supply.

Question LTR-6 a.8

Bank Q233 • AP Mock 1 Q50 (rebuilt, key corrected in the bank)

Molino Bank's clearing bank stops processing Molino's daylight overdrafts, so Molino can no longer count on intraday credit. Molino starts the day with limited cash at the central bank and has large outgoing client wire transfers scheduled. Which of the following is the most appropriate way for Molino to manage its intraday position?

A. Delay or throttle non-critical outgoing payments so that payments are released only up to cash balances plus incoming funds received.

B. Release all payments on schedule, because any overdraft will be cleared by the end of the business day.

C. Declare insolvency, because the loss of intraday credit means that obligations cannot be met.

D. Take no action until the end-of-day PCS settlement, because intraday shortfalls are resolved at that point.

Question LTR-6 a.9 New • built from GARP Ch.6, 6.4

The risk committee of a large-value payment system reviews how the system protects itself against a participant that cannot meet its settlement obligation. Which of the following statements about FMU risk management is correct?

A. Routing indirect members' payments through a few direct members (tiering) reduces intraday settlement risk, because it concentrates payments at the most creditworthy banks.

B. The guarantee fund is the FMU's first line of defence and is raised after a default from contributions by the surviving participants.

C. When a participant's credit weakens, the FMU may cut or remove its net debit cap, require it to prefund its account, or call for intraday margin and extra collateral.

D. Liquidity savings mechanisms such as multilateral netting raise the liquidity each participant must hold, in exchange for lower settlement risk.

Solutions • LTR-6 a

Solution LTR-6 a.1

Answer: A

Bank Q241 • Schweser Module Quiz 71.1 Q1

Step 1 – The largest use

GARP: outgoing wire payments, on behalf of clients or the bank's own account, "are typically the largest use of intraday liquidity". They run all day, and large banks throttle them as the overdraft nears the debit cap.

Step 2 – The largest source

Incoming flows from payments and FMU settlements "are the largest source of intraday funding in periods of normal market function".

$\Rightarrow$ The two superlatives are a matched pair, one use and one source, and the source carries a scope condition.

Interpretation

In a stress, incoming flows are the first thing to dry up: counterparties delay payments, and the treasurer falls back on intraday credit, liquid assets and overnight borrowing. That is why the qualifier "in normal markets" is what GARP tests.

Why the other options are wrong

B is incorrect. Polarity flip: wires are outflows, a use; incoming flows are a source.

C is incorrect. Absolute qualifier: central banks are "a large source" of intraday credit, but not the largest in all conditions. Collateral pledging is a frequent use, not the largest.

D is incorrect. Most PCS systems settle *once* a day, many in the late afternoon, and the net settlement can be a source as well as a use.

LO: LTR-6 a: Identify and explain the uses and sources of intraday liquidity. • **Reference:** Venkat and Baird, Liquidity Risk Management: A Practitioner's Perspective, Ch.4 (GARP Ch.6), 6.2 Uses and Sources of Intraday Liquidity

Solution LTR-6 a.2

Answer: D

Bank Q235 • AnalystPrep Q.3908

Step 1 – Match each characteristic

One settlement a day, mostly late afternoon: PCS systems. Source or use depending on the net position: a net settlement with a clearinghouse. Forecastable for multi-day settlements (T+3), harder for same-day activity: GARP's Table 6.2 entry for PCS settlements.

$\Rightarrow$ All three point to settlements at PCS systems.

Interpretation

Nostro funding shares the "source or use" feature but is managed to a target average monthly balance, not settled once a day. Recognising an item from its profile is GARP's way of testing these lists.

Why the other options are wrong

A is incorrect. Wires run throughout the day in a steady pattern, and they are only a use.

B is incorrect. Nostro funding can be a source or a use, but it is a daily cash management of correspondent balances, not a single late-afternoon settlement.

C is incorrect. Collateral pledging is always a use and is forecast from trading volumes and price changes.

LO: LTR-6 a: Identify and explain the uses and sources of intraday liquidity. • **Reference:** GARP Ch.6, 6.2, Table 6.2 Uses of Intraday Credit

Solution LTR-6 a.3 Answer: C

Bank Q236 • AnalystPrep Q.3919

Step 1 – The source

Fed funds borrowed today and repaid the next morning are an **overnight borrowing**. They are not intraday credit, which must be covered by the close of business.

Step 2 – The use

A cash transfer to a correspondent bank for services it provides is **funding of a nostro account**.
⇒ Source: overnight borrowings. Use: funding of nostro accounts.

Interpretation

"Borrow in the afternoon, return first thing next morning" is also one of GARP's two reasons why daylight overdrafts arise: after repaying, the borrower has the funds for less than 24 hours, and its account can go negative mid-morning.

Why the other options are wrong

A is incorrect. Cash balances are the bank's opening balances, not borrowed funds; and the payment is compensation to a correspondent, not a wire for a client.
B is incorrect. Incoming funds flow comes from payments received and FMU credits, not from borrowing; a payment for services is not a PCS settlement.
D is incorrect. Liquid assets are near-cash the bank owns and sells or repos, not funds it borrows; the use is again misidentified.

LO: LTR-6 a: Identify and explain the uses and sources of intraday liquidity. • **Reference:** GARP Ch.6, 6.2, Tables 6.2 and 6.3

Solution LTR-6 a.4 Answer: C

Bank Q242 • Schweser Module Quiz 71.1 Q2

Step 1 – Separate explicit from implicit costs

Cash balances and liquid assets are the bank's own assets: using them costs only the return given up. Incoming funds flow is money received: no charge.

Step 2 – Intraday credit is borrowed

The Fed charges for use of its net debit cap (50 bp on unsecured overdrafts since 2008). The Bank of England charges no interest but requires the highest-quality government collateral.
⇒ Intraday credit is the only listed source with an explicit charge, in interest or in collateral that must be posted.

Interpretation

Interbank daylight lines are generally uncommitted and free, so the explicit cost comes from central bank credit. Overnight borrowing also carries explicit interest, but it is not offered here.

Why the other options are wrong

A is incorrect. The bank's own cash: only an opportunity cost.
B is incorrect. Receipts from counterparties and FMUs carry no charge; their risk is timing, not cost.
D is incorrect. Owned near-cash assets carry an opportunity cost (the yield given up), not an explicit one.

LO: LTR-6 a: Identify and explain the uses and sources of intraday liquidity. • **Reference:** GARP Ch.6, 6.1 and 6.2, Intraday credit

Solution LTR-6 a.5 — Answer: B

Bank Q239 • GARP PE1 Q50

Step 1 – Check each observation against GARP's Table 6.3

Liquid assets: "client may be sources of liquidity in converting liquid assets to cash (e.g. repo transaction)". A client that takes the bank's Treasury bills in a repo gives the bank same-day cash.

⇒ B restates the table.

Interpretation

The table also tells you who drives each item and how predictable it is. Use those two columns to dismiss C and D.

Why the other options are wrong

A is incorrect. Intraday overdrafts are permitted, within the net debit cap, and the Fed charges for them; the bank does not have to pre-borrow before the opening.

C is incorrect. GARP: liquid assets in the investment portfolio and other money market investments "tend to have low volatility and as a result are predictable".

D is incorrect. Nostro funding is not directly initiated by clients: it is the bank's correspondent cash, monitored by its operations groups.

LO: LTR-6 a: Identify and explain the uses and sources of intraday liquidity. • **Reference:** GARP Ch.6, 6.2, Table 6.3 Sources of Intraday Credit

Solution LTR-6 a.6 — Answer: A

Bank Q240 • GARP PE2 Q22

Step 1 – A cause of the overdraft

Collateral pledging is a use of intraday liquidity: posting cash collateral against an OTC securities purchase drains the account and can take it negative.

Step 2 – The response

Intraday credit from the central bank is a standard source to cover a temporary shortfall, repaid by the close of business.

⇒ A pairs a genuine use with a genuine source.

Interpretation

Any of the five uses can create the overdraft; any of the six sources can cover it. The wrong options each break one half of that pairing.

Why the other options are wrong

B is incorrect. Customer transfers *into* the bank are incoming funds, a source: they reduce an overdraft rather than cause one.

C is incorrect. A duration-gap effect on net worth is interest rate risk (ALM), not an intraday cash outflow; with a positive gap, falling rates even raise the value of equity.

D is incorrect. A higher net debit cap permits a larger overdraft; it does not trigger one. A reverse repo lends cash out, so it would deepen the shortfall.

LO: LTR-6 a: Identify and explain the uses and sources of intraday liquidity. • **Reference:** GARP Ch.6, 6.2, Collateral pledging and Intraday credit

Solution LTR-6 a.7 Answer: B

New • built from GARP Ch.6 and workbook D27

Step 1 – Central bank terms

GARP: "The Federal Reserve provides an unsecured committed line of credit (in the form of its Net Debit Cap program) and charges interest for tapping the line. The Bank of England requires intraday overdrafts to be collateralized by the highest quality government securities, but does not charge interest."
⇒ B is GARP's sentence.

Step 2 – Check the history

The 2008 schedule is 50 bp for unsecured overdrafts, zero for fully collateralised ones, and no deductible. Peak overdrafts were 9.4% of M1 in 1986 and 12.4% in 1991, against 0.7% in 2013.

Interpretation

D27 lists A as the sibling-swap distractor for this point. Interbank daylight lines are a third model: generally uncommitted and free.

Why the other options are wrong

A is incorrect. Sibling swap: the Fed and Bank of England terms are reversed.
C is incorrect. Mixes two years: collateralised overdrafts have been free since 2008, when the deductible was eliminated; the 10% deductible applied to the 24 bp fee of 1994.
D is incorrect. Reversed trend: the share fell from about 10% in the late 1980s to 0.7% in 2013. The shadow money supply was the 1980s concern.

LO: LTR-6 a: Identify and explain the uses and sources of intraday liquidity. • **Reference:** GARP Ch.6, 6.1 Introduction, Table 6.1; 6.2, Intraday credit

Solution LTR-6 a.8 Answer: A

Bank Q233 • AP Mock 1 Q50

Step 1 – What is left without intraday credit

Intraday credit is one source. Without it, the funds available during the day are the opening cash balances plus incoming funds flow, the largest source in normal markets.

Step 2 – The lever the bank controls

Outgoing wires are the largest use, and their timing can be managed: GARP describes banks "throttling" outgoing payments and closely monitoring incoming credits so as not to exceed the cap.
⇒ Molino should release critical payments first and hold the rest until receipts rebuild its balance.

Interpretation

If the gap persists, Molino can add sources: overnight borrowing, or a repo of liquid assets. Delays carry reputational risk, so time-sensitive obligations go first.

Why the other options are wrong

B is incorrect. Treats the overdraft as harmless: the clearing bank has refused to fund it, and most LVPSs have hard controls that stop a participant exceeding its limit.
C is incorrect. Overstates the problem: losing one source of intraday liquidity is a constraint to manage, not insolvency.
D is incorrect. PCS settlement happens once, near day end; waiting leaves the day's payments unmade.

LO: LTR-6 a: Identify and explain the uses and sources of intraday liquidity. • **Reference:** GARP Ch.6, 6.2, Outgoing wire transfers and the treasurer's challenge

Solution LTR-6 a.9 **Answer: C**

New • built from GARP Ch.6, 6.4

Step 1 – The FMU's three lines of defence

(1) Membership criteria; (2) monitoring participants; (3) risk mutualisation through a pre-funded guarantee fund or central bank underwriting.

Step 2 – Actions against a weakening participant

GARP lists more frequent margining (an intraday margin call), reducing the member's net debit cap or requiring it to prefund its account, and requiring more collateral or a third-party letter of credit or performance bond.

$\Rightarrow$ C lists second-line actions.

Interpretation

The net debit cap appears twice in this reading: as the Fed's limit on a bank's overdraft (Section 1) and as an FMU tool that can be cut when a member weakens.

Why the other options are wrong

A is incorrect. Tiering *introduces* risk: a direct member's failure hits the indirect members behind it, others hesitate to pay a failing member's clearing bank, and operational interdependence grows. The Bank of England tried to reduce it in CHAPS.

B is incorrect. Membership criteria are the first line; the guarantee fund is the third, and it is prefunded before any default.

D is incorrect. Liquidity savings mechanisms raise throughput for a given amount of liquidity, so they *reduce* the liquidity required as well as settlement positions.

LO: LTR-6 a: Identify and explain the uses and sources of intraday liquidity. • **Reference:** GARP Ch.6, 6.4 Risk Management, Measurement, Monitoring Tools for FMUs

LTR-6 • Intraday Liquidity Risk Management • Vardeez: IMPORTANT

LTR-6 c

P3 CONCEPT

Differentiate between methods for tracking intraday flows and monitoring risk levels.

Workbook: D27 Intraday Liquidity and Dealer Bank Failure Mechanics • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D27 (Map #65, 1 of 16 recall sources), notes only. Vardeez rates the reading IMPORTANT but has no pointer on this LO.

Reported exam shapes

- **S1** Intraday liquidity monitoring tools: measures of flows versus measures of risk levels.

What the questions look like: classification. Which list an item belongs to, which pair holds only risk-level measures, which measure matches a description, and which one answers a stated concern (systemic risk, an FMU payment deadline, a client that leans on its line). No calculations: the measures are ratios, but the LO asks you to tell them apart.

1. Recap: two perspectives, two sets of measures

Daylight overdrafts come from market conventions and from intraday credit to clients (LTR-6 a, Section 1). Leading banks monitor **the intraday credit they extend to clients** and **the intraday credit they use themselves**; the second is harder, because FMU and correspondent data are patchy and few banks can aggregate all PCS activity in one place. GARP's measures follow BCBS, *Monitoring Tools for Intraday Liquidity Management* (April 2013): a first set describes **the profile of intraday flows**; a second set gives **ratios for monitoring risk levels**, against which the bank may set limits.

2. Measures for understanding intraday flows

- **Total payments**: per-payment data (amount, times, processing steps, routing, payer and payee, system, suspensions), summarised into payments sent and received by client type and for the bank, the net settlement-account position at any time of day, and volume trends.
- **Other cash transactions**: intraday and end-of-day settlement positions at *every* FMU, with full transaction detail; at securities settlement networks the bank snapshots its account and collateral positions through the day (the Fed tracks collateral positions every minute).
- **Settlement positions**: the minimum if positions cannot be rebuilt at any time of day; deadline-specific, often large FMU positions, correlated with market factors.
- **Time-sensitive obligations**: must complete by a set time (return of borrowings, margin payments, a client's acquisition closing); record amounts and deadlines.
- **Total intraday credit lines to clients and counterparties**: potential and actual credit extended; **most lines are uncommitted and undisclosed**; keep average and peak usage.
- **Total bank intraday credit lines available and usage**: the credit the bank uses in normal conditions against the most it can draw, committed and uncommitted, across all accounts; expected in resolution planning as a gauge of the systemic risk it poses.

3. Measures for quantifying and monitoring risk levels

- **Daily maximum intraday liquidity usage** = the day's **largest net negative balance** ÷ the committed or uncommitted credit line (Figure 1). No real-time monitoring needed if the day's positions can be rebuilt ex post. Track peak and average (e.g. monthly) for every central bank, FMU and correspondent account, and ideally consolidated across accounts between which liquidity moves freely.
- **Intraday credit relative to Tier 1 capital**, for total and unsecured credit, available and used (collateral mitigates settlement risk). **Available, unsecured credit** ÷ **Tier 1** is "a rough measure of the inadvertent systemic risk" the bank poses.
- **Client intraday credit usage** = a client's **peak** daily intraday overdraft ÷ its line. In aggregate: the liquidity needed to support clients. Client by client: frequent overdrafters to reprice or change.
- **Payment throughput**: the **percentage of outgoing payments** made by each time of day, to (1) process the day's payments in time, (2) meet FMU rules on a target share by a deadline, and (3) find peak periods and their link to liquidity and credit usage.

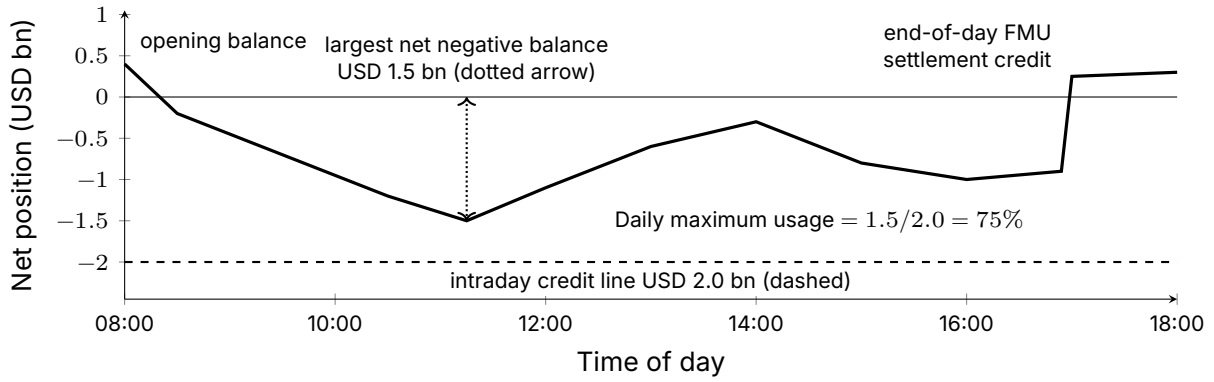


Figure 1. An illustrative day for one account (net position, solid line): borrowed fed funds go back first thing and wires run ahead of receipts until the end-of-day settlement credit. The risk measure takes the largest net negative balance (dotted) relative to the line (dashed); the small positive end-of-day balance hides all of it.

4. Telling the two sets apart

Flow measures record what moved and what is available; risk-level measures scale a peak or a share against a line, Tier 1 capital or the clock. Settlement positions, time-sensitive obligations and other cash transactions sound like risk but are *flow* measures. End-of-day figures (net position, LCR, HQLA share) are not intraday measures at all.

Stress testing. The indicators describe business as usual. BCBS suggests at least four intraday stress categories: **own financial stress**, **counterparty stress**, **customer stress** (a correspondent's customer bank) and **market-wide credit or liquidity stress**, plus reverse stress tests. Benefits include management discussion, playbooks and a data aggregation capability.

> **Correction to the source notes.** Vardeez's flow column has five measures and folds settlement positions into "other cash transactions". GARP and Schweser list six: *settlement positions* is its own measure, the minimum kept when positions cannot be rebuilt at any time of day. Question LTR-6 c.4 turns on this.

Comparison card: LTR-6 c

Tracking intraday flows (amounts, positions, lines)	**Monitoring risk levels** (a peak or share, scaled)
Total payments	Payment throughput: % of *outgoing* payments by time of day
Total intraday credit lines to clients and counterparties	Client intraday credit usage: *peak* overdraft ÷ line
Total bank intraday credit lines available and usage	Daily maximum usage: largest net negative balance ÷ line
Other cash transactions; settlement positions; time-sensitive obligations	Intraday credit ÷ Tier 1; available unsecured ÷ Tier 1 = systemic risk

Exam traps

- **Sibling swap.** Each row of the card is a swap pair; the left item is the flow measure.
- **Peak, not average or end-of-day,** for client usage and daily maximum usage.
- **Systemic risk** is *available*, *unsecured* credit ÷ Tier 1, not used or secured credit.
- **No real-time requirement** for daily maximum usage if positions can be rebuilt ex post.
- **Throughput is outgoing**, measured against the clock and FMU deadlines.

Practice questions: LTR-6 c

Question LTR-6 c.1

Bank Q248 • AnalystPrep Q.3846 (rebuilt)

An intraday liquidity manager is assembling the bank's risk-level dashboard from the BCBS monitoring tools. Which of the following is *not* a measure for quantifying and monitoring intraday risk levels?

A. Client intraday credit usage

B. Payment throughput

C. Time-sensitive obligations

D. Intraday credit relative to Tier 1 capital

Question LTR-6 c.2

Bank Q249 • AnalystPrep Q.3911 (rebuilt)

Which of the following pairs consists *only* of measures for quantifying and monitoring intraday risk levels?

A. Daily maximum intraday liquidity usage and client intraday credit usage

B. Client intraday credit usage and total bank intraday credit lines available and usage

C. Total bank intraday credit lines available and usage and total intraday credit lines to clients and counterparties

D. Time-sensitive obligations and payment throughput

Question LTR-6 c.3 Bank Q251 • AnalystPrep Q.3914

A bank keeps, for each electronic payment, the payer and payee, the payment system used, any suspensions, the time received or originated, and routing information. Under which measure for understanding intraday flows do these statistics most accurately fall?

A. Other cash transactions

B. Total payments

C. Settlement positions

D. Total intraday credit lines to clients and counterparties

Question LTR-6 c.4 Bank Q253 • AnalystPrep Q.3916 (rebuilt, key corrected in the bank)

A bank tracks its intraday and end-of-day settlement positions at every financial market utility (FMU) in which it participates, captures as much transaction-level detail as possible and, at securities settlement networks, takes snapshots of its account and collateral positions through the day. Which measure for understanding intraday flows does this describe?

A. Total payments

B. Total bank intraday credit lines available and usage

C. Other cash transactions

D. Settlement positions

Question LTR-6 c.5 Bank Q252 • AnalystPrep Q.3915 (rebuilt)

The head of payments at a bank wants to identify the corporate clients that lean hardest on the daylight overdraft lines the bank grants them, so that it can reprice the lines or ask the clients to change their practices. Which measure, with its definition, should she use?

A. Client intraday credit usage: each client's peak daily intraday overdraft relative to its committed or uncommitted credit line.

B. Client intraday credit usage: each client's end-of-day overdraft balance relative to the bank's Tier 1 capital.

C. Daily maximum intraday liquidity usage: the bank's own largest net negative balance relative to its intraday credit line.

D. Total intraday credit lines to clients and counterparties: the sum of the committed and uncommitted lines granted.

Question LTR-6 c.6

Bank Q258 • Mock 2 Q24

A treasury manager at a large clearing bank is establishing indicators to monitor the bank's intraday liquidity position. The bank settles a high volume of payments through a real-time gross settlement system and relies on incoming payments to fund a portion of its outgoing obligations. Which of the following best describes an appropriate intraday liquidity monitoring indicator?

A. The bank's end-of-day net settlement position across all payment systems.

B. The bank's daily maximum intraday liquidity usage, measured as the largest net cumulative negative position during the day relative to the credit line.

C. The bank's liquidity coverage ratio calculated at the close of each business day.

D. The proportion of the bank's total assets that qualify as Level 1 high-quality liquid assets.

Question LTR-6 c.7

Bank Q259 • Schweser Module Quiz 71.2 Q2

Which of the following metrics is most appropriate in determining the systemic risk that a bank poses through its intraday activity?

A. Client intraday credit usage

B. Intraday credit relative to Tier 1 capital

C. Daily maximum intraday liquidity usage

D. Total intraday credit lines to clients and counterparties

Question LTR-6 c.8

Bank Q257 • BT answers P2.T8.25.3.3 (rebuilt)

During a market-wide liquidity crunch, a bank's incoming payments are delayed and it is concerned that its own outgoing payments may be sent late in the day, breaching a financial market utility's (FMU) requirement to settle a minimum share of payments by set times. Which risk-level monitoring metric is most directly designed to address this concern?

A. Payment throughput, the proportion of outgoing payments made and when they occur during the day

B. Intraday credit relative to Tier 1 capital

C. Client intraday credit usage

D. Total intraday credit lines extended to clients and counterparties

Question LTR-6 c.9 New • built from GARP Ch.6, 6.3

An internal auditor reviews the intraday liquidity measurement standard drafted by a bank's treasury. Which of the following statements in the draft is correct?

A. Daily maximum intraday liquidity usage can be computed only for accounts with real-time balance reporting, because a day's positions cannot be rebuilt after the fact.

B. Used, secured intraday credit relative to Tier 1 capital is the rough measure of the inadvertent systemic risk that the bank poses to the financial system.

C. Daily maximum usage should be tracked for every central bank, FMU and correspondent account, and ideally on a consolidated basis across accounts between which liquidity moves freely.

D. Most intraday credit lines that the bank grants to clients are committed and disclosed, so the flow measure needs only contractual limits, not average and peak usage.

Solutions • LTR-6 c

Solution LTR-6 c.1

Answer: C

Bank Q248 • AnalystPrep Q.3846

Step 1 – List the risk-level measures

Daily maximum intraday liquidity usage, intraday credit relative to Tier 1 capital, client intraday credit usage, payment throughput.

Step 2 – Place the odd one out

Time-sensitive obligations (return of borrowings, margin payments, a client's acquisition closing) are tracked by amount and deadline to understand flows.

$\Rightarrow$ C is a flow measure, not a risk-level measure.

Interpretation

Time-sensitive obligations sound like risk, but nothing is scaled: the bank records amounts and deadline times. Every risk-level measure is a peak or a share set against a line, capital or the clock.

Why the other options are wrong

A is incorrect. A risk-level measure: a client's peak overdraft relative to its line.

B is incorrect. A risk-level measure: the share of outgoing payments by time of day.

D is incorrect. A risk-level measure: intraday credit (total and unsecured, available and used) relative to Tier 1, the systemic risk gauge.

LO: LTR-6 c: Differentiate between methods for tracking intraday flows and monitoring risk levels. • **Reference:** Venkat and Baird, Ch.4 (GARP Ch.6), 6.3, Measurement of Intraday Liquidity

Solution LTR-6 c.2

Answer: A

Bank Q249 • AnalystPrep Q.3911

Step 1 – Classify each item

Risk levels: daily maximum usage, client usage, Tier 1 ratio, throughput. Flows: total payments, other cash transactions, settlement positions, time-sensitive obligations, total lines to clients and counterparties, total bank lines available and usage.

Step 2 – Find the pure pair

Only A pairs two risk-level measures.

$\Rightarrow$ A.

Interpretation

B and C are built on the credit-line sibling pair. "Total bank intraday credit lines available and usage" records what the bank can draw and what it uses; it does not scale a peak against the line, as daily maximum usage does.

Why the other options are wrong

B is incorrect. Pairs a risk-level measure with a flow measure (total bank lines available and usage).

C is incorrect. Both items are flow measures about credit lines.

D is incorrect. Throughput is a risk-level measure, but time-sensitive obligations is a flow measure.

LO: LTR-6 c: Differentiate between methods for tracking intraday flows and monitoring risk levels. • **Reference:** GARP Ch.6, 6.3, Measures for Understanding Intraday Flows; Measures for Quantifying and Monitoring Risk Levels

Solution LTR-6 c.3 — Answer: B

Bank Q251 • AnalystPrep Q.3914

Step 1 – What the data are

Per-payment attributes stored in a data warehouse: amount, timing, processing steps, routing, payer and payee, system, suspensions.

Step 2 – Which measure stores them

GARP: for every payment, the bank stores this information and summarises it into payments sent and received by client type, the net settlement position at any time of day, and volume trends.
⇒ Total payments.

Interpretation

Total payments is the raw material for payment throughput; the two are different measures in different sets.

Why the other options are wrong

A is incorrect. Other cash transactions are settlement positions at FMUs and collateral snapshots, not payment-level records.
C is incorrect. Settlement positions are deadline-specific FMU positions, the fallback when full reconstruction is impossible.
D is incorrect. Credit lines to clients record credit granted and used, not payment details.

LO: LTR-6 c: Differentiate between methods for tracking intraday flows and monitoring risk levels. • **Reference:** GARP Ch.6, 6.3, Total Payments

Solution LTR-6 c.4 — Answer: C

Bank Q253 • AnalystPrep Q.3916

Step 1 – Match GARP's wording

"A bank should also track its intraday and end-of-day settlement positions at all financial market utilities in which it participates", maximise transaction-level detail, and at securities settlement networks "capture snapshots of its account and collateral positions throughout the trading day". This is *Other Cash Transactions*.

Step 2 – Rule out the near sibling

Settlement Positions is the minimum fallback: "if complete data to reconstruct account positions at any time of day is not available", keep data on settlement positions with all FMUs.
⇒ The full-detail, all-FMU, collateral-snapshot description is other cash transactions.

Interpretation

The bank's own key was corrected to C. Vardeez folds settlement positions into other cash transactions (see the correction box), which makes D look tempting.

Why the other options are wrong

A is incorrect. Total payments is per-payment data, not FMU positions and collateral snapshots.
B is incorrect. Concerns intraday credit available and used, not FMU positions.
D is incorrect. Settlement positions is the narrower minimum when full reconstruction is not possible; it has no collateral-snapshot element.

LO: LTR-6 c: Differentiate between methods for tracking intraday flows and monitoring risk levels. • **Reference:** GARP Ch.6, 6.3, Other Cash Transactions; Settlement Positions

Solution LTR-6 c.5 Answer: A

Bank Q252 • AnalystPrep Q.3915

Step 1 – The concern is client by client

The bank wants to see which clients use their daylight lines most, which is client intraday credit usage.

Step 2 – The definition

GARP: derived "by comparing a client's peak daily intraday overdraft to the established (committed or uncommitted) credit line". Tracked client by client, it pinpoints clients "that run frequent overdrafts" and whether they need to change practices or the bank needs to charge more.
$\Rightarrow$ A.

Interpretation

In aggregate the same measure shows how much intraday liquidity the bank must hold to support its clients.

Why the other options are wrong

B is incorrect. Right name, wrong numerator and denominator: daylight overdrafts are covered by the close, so the end-of-day balance misses them, and Tier 1 belongs to the bank's own systemic-risk measure.
C is incorrect. Measures the bank's own usage, not each client's.
D is incorrect. A flow measure that counts lines granted; it does not compare usage with the line.

LO: LTR-6 c: Differentiate between methods for tracking intraday flows and monitoring risk levels. • **Reference:** GARP Ch.6, 6.3, Client Intraday Credit Usage

Solution LTR-6 c.6 Answer: B

Bank Q258 • Mock 2 Q24

Step 1 – What intraday risk is

The risk lies in the path of the balance during the day, so the indicator must capture the intraday extreme.

Step 2 – Which option does

Daily maximum intraday liquidity usage takes the day's largest net negative balance relative to the committed or uncommitted line, tracked per account and consolidated.
$\Rightarrow$ B.

Interpretation

A bank that funds outgoing payments from incoming ones runs its deepest deficit when outflows run ahead of receipts (Figure 1). Every other option is measured at the close or is a stock.

Why the other options are wrong

A is incorrect. The end-of-day position can be flat or positive after a deep mid-day overdraft.
C is incorrect. The LCR is a 30-day stress ratio measured at a point in time; it says nothing about intraday peaks.
D is incorrect. The Level 1 HQLA share is a balance-sheet composition measure, not a measure of intraday usage.

LO: LTR-6 c: Differentiate between methods for tracking intraday flows and monitoring risk levels. • **Reference:** GARP Ch.6, 6.3, Daily Maximum Intraday Liquidity Usage

Solution LTR-6 c.7 — Answer: B

Bank Q259 • Schweser Module Quiz 71.2 Q2

Step 1 – Which measure is scaled to capital

Intraday credit relative to Tier 1 capital is "a broad representation of the intraday settlement risk posed by a bank".

Step 2 – The systemic risk version

"Available, unsecured intraday credit relative to an institution's tier 1 capital is a rough measure of the inadvertent systemic risk that the institution poses to the financial system."
$\Rightarrow$ B.

Interpretation

Unsecured credit counts because posting high-quality collateral mitigates settlement risk. Available credit counts because it is what the bank could draw. Viewed over time and against peers, the ratio shows the relative systemic risk of the business model.

Why the other options are wrong

A is incorrect. Shows the liquidity the bank provides to clients, not the risk it poses to the system.
C is incorrect. The bank's own peak usage against its line: a liquidity-usage measure, not scaled to loss-absorbing capital.
D is incorrect. A flow measure of potential exposure to clients, not scaled to capital.

LO: LTR-6 c: Differentiate between methods for tracking intraday flows and monitoring risk levels. • **Reference:** GARP Ch.6, 6.3, Intraday Credit Relative to Tier 1 Capital

Solution LTR-6 c.8 — Answer: A

Bank Q257 • BT answers P2.T8.25.3.3

Step 1 – The concern

The timing of the bank's own outgoing payments against an FMU's rule on settling a share of payments by set times.

Step 2 – The measure built for it

Payment throughput tracks the percentage of outgoing payments relative to time of day. GARP gives three reasons, one of which is "to meet FMU requirements for submitting a target percentage of payments by a deadline".
$\Rightarrow$ A.

Interpretation

Throughput also identifies peak periods and their correlation with liquidity on hand, which helps the bank plan around delayed receipts.

Why the other options are wrong

B is incorrect. A systemic-risk gauge scaled to capital, not a timing measure.
C is incorrect. The clients' peak borrowing against their lines, not the bank's own payment timing.
D is incorrect. A flow measure of credit extended, unrelated to when payments are released.

LO: LTR-6 c: Differentiate between methods for tracking intraday flows and monitoring risk levels. • **Reference:** GARP Ch.6, 6.3, Payment Throughput

Solution LTR-6 c.9 **Answer: C**

New • built from GARP Ch.6, 6.3

Step 1 – Why C is correct

GARP: "At a minimum, this measure should be tracked for every cash account held at the central bank, FMUs, and correspondent banks. Ideally, a bank should also monitor its consolidated position across all accounts between which liquidity can be readily transferred intraday without restrictions."

Interpretation

Each wrong statement corrupts one word of a GARP sentence: "does not require" becomes "only", "available, unsecured" becomes "used, secured", and "most are uncommitted and undisclosed" is reversed.

Why the other options are wrong

A is incorrect. GARP: the calculation "does not require real-time monitoring of an account (provided the bank has the ability to recreate all of a day's positions ex-post)".

B is incorrect. The systemic-risk gauge is *available*, *unsecured* intraday credit relative to Tier 1; secured credit is mitigated by collateral.

D is incorrect. Most client lines are uncommitted and undisclosed, and the bank should keep average and peak usage data.

LO: LTR-6 c: Differentiate between methods for tracking intraday flows and monitoring risk levels. • **Reference:** GARP Ch.6, 6.3, Measures for Quantifying and Monitoring Risk Levels

LTR-7 • Monitoring Liquidity • Vardeez: MODERATE

LTR-7 a

P3 CONCEPT

Differentiate between deterministic and stochastic cash flows and provide examples of each.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P3. No workbook question tests it (the A26 revision notes summarise the taxonomy in one bullet). Vardeez flags it: his reading note opens with "deterministic and stochastic cash flows and liquidity option concept".
Vardeez: deterministic versus stochastic cash flows. A cash flow is classified on two dimensions, time and amount, and can be deterministic on one and stochastic on the other.

Reported exam shapes: none listed. The questions are built from the LO wording and the Vardeez focus.

What the questions look like: definitions and classification. Which of four flows is deterministic; which is the exception among flows stochastic in both dimensions; where a given flow sits in the time-by-amount grid. No calculations. The background on the TSECF, where these flows end up, is in LTR-7 d (Sections 2 and 4).

1. Two dimensions: time and amount

Castagna and Fede (GARP Chapter 7) classify every future cash flow on two separate dimensions.

- **Time.** A flow is **deterministic** if it occurs at an instant known with certainty at the reference date, **stochastic** if it occurs at a random future instant.
- **Amount.** A flow is **deterministic** ("fixed", set by the terms of a contract) if its amount is known with certainty at the reference date, **stochastic** if it cannot be fully determined.

Only a flow known on *both* dimensions is deterministic without qualification. Stochastic amounts fall into four subcategories according to the source of the uncertainty:

- **credit related**: credit events, such as the default of one or more clients;
- **indexed/contingent**: market variables, such as Libor fixings or the price of an option's underlying;
- **behavioural**: decisions of the bank's clients or counterparties, only loosely predictable from rational behaviour and sometimes based on information the bank does not have;
- **new business**: contracts to be dealt in the future, more or less planned by the bank.

The classification depends on the point of view: flows are classified from a given reference date, and a flow can change category when the reference date moves.

2. The taxonomy, with GARP's examples

Amount ↓ Time →	Deterministic time	Stochastic time
Fixed (deterministic)	Coupons of risk-free fixed-rate bonds; capital amortisation of a risk-free fixed-rate mortgage	Payout of a one-touch option when the barrier is hit; the bank's withdrawals from credit lines it has received
Stochastic: credit related		Missing cash flows after a client's default; recovery after default
Stochastic: indexed/contingent	Coupons of a risk-free floating-rate bond; payout at exercise of a European option	Payout at exercise of an American option
Stochastic: behavioural		Withdrawals from (sight and saving) deposits; withdrawals of credit lines by clients; prepayments of mortgages
Stochastic: new business	New debt issuance by the bank; renewal of expiring contracts	New deposits; new loans; new assets

Figure 1. Castagna and Fede's taxonomy of cash flows (after GARP Figure 7.1), redrawn as a grid: amount down the side, time across the top. Empty cells have no example in the reading.

3. Reading the grid

- **Deterministic in both** needs a risk-free obligor. Coupons and amortisation of fixed-rate bonds, mortgages and loans, on either side of the balance sheet (bonds issued and loans received too), are deterministic only if the counterparty cannot default; otherwise credit events can change the schedule.
- **Fixed amount, stochastic time.** A one-touch option pays a known sum, but when the barrier will be breached is unknown. A credit line the bank has *received* can be drawn in an amount the bank chooses (between zero and the limit), but at whatever time its liquidity needs arise.
- **Credit related.** Once the default time is fixed the amount follows, but the default time is unknown, so these flows are stochastic in time as well. A missing flow is a negative flow that alters the contract schedule; the recovery value belongs here too.
- **Indexed/contingent.** A floating coupon or a European payout has a known date and an unknown amount. An American option's payout is stochastic in both, because exercise depends on future market conditions, whether the bank is long or short the option.
- **Behavioural.** Prepayments, deposit withdrawals and drawdowns of credit lines *granted* to clients depend on client decisions: stochastic in amount and time. These are the liquidity options of LTR-7 b.
- **New business.** Replacing expiring contracts and planned bond issues have known dates (old contracts expire at known maturities; issues can be well defined in time) but uncertain amounts. Completely new contracts (new retail deposits, new mortgages) are stochastic in both, and are the hardest to model.

⇒ Why it matters: deterministic flows go straight into the TSECF; stochastic flows enter only as expected values, produced by interest rate models, credit models with default correlation, behavioural models and new-business models (LTR-7 d, Section 4). Their volatility is what cash flow at risk measures (LTR-7 c).

Exam traps

- **One dimension only.** A floating coupon is deterministic in time but stochastic in amount; calling it "deterministic" or "stochastic" without saying which dimension is the trap.
- **Credit lines appear twice.** Lines the bank has *received*: fixed amount, stochastic time. Lines the bank has *granted*: behavioural, stochastic in both.
- **Known expiry is not known exercise.** A European payout has a deterministic time; an American payout does not.
- **New business splits.** New debt issuance and renewals: deterministic time, stochastic amount. New loans, deposits and assets: stochastic in both.
- **Contract schedule is not enough.** Fixed-rate coupons are deterministic only if the obligor is risk-free.
- **Not fixed for life.** The category depends on the reference date and can change as it moves.

Practice questions: LTR-7 a

Question LTR-7 a.1

Bank Q260 • AnalystPrep Q.3920 (rebuilt)

A treasury analyst is classifying the bank's future cash flows to build its monitoring tools, following the taxonomy of Castagna and Fede. Along which two dimensions are the cash flows classified?

A. Time and credit quality of the counterparty

B. Amount and type of contract

C. Time and amount

D. Maturity and currency

Question LTR-7 a.2

Bank Q261 • AnalystPrep Q.3921

Which of the following statements accurately describes deterministic cash flows?

A. They occur at random future instants, and their amount cannot be fully determined.

B. They occur at predictable future instants and in an amount known with certainty.

C. Their amount cannot be predicted, but they occur at predictable future instants.

D. They are unpredictable in both amount and timing.

Question LTR-7 a.3

Bank Q263 • BT P2.T8.25.4.1 (rebuilt)

A commercial bank classifies its cash flows over the next year to assess potential liquidity shortfalls:

1. Fixed-rate bond coupon payments: $10 million due every six months.
2. Floating-rate loan interest receipts: about $5 million per quarter, with the rate indexed to 3-month SOFR.
3. Customer deposit withdrawals: on average $20 million per month, but fluctuating between $15 million and $30 million depending on economic conditions.
4. Committed corporate credit lines of $100 million, with an estimated drawdown of 25% to 50% in periods of economic stress.

Which of the above cash flows is deterministic?

A. 1

B. 2

C. 3

D. 4

Question LTR-7 a.4

Bank Q264 • Schweser Module Quiz 72.1 Q1

Each of the following cash flows is stochastic in both timing and amount EXCEPT:

A. new debt issued by a bank.

B. a bank's recovery on a client default.

C. the payout of an American-style call option.

D. mortgage prepayments due to interest rate declines.

Question LTR-7 a.5

New • built from GARP Ch.7 (7.1, Figure 7.1)

A liquidity analyst notices that credit line drawdowns appear in two different cells of Castagna and Fede's cash flow taxonomy. Which of the following correctly classifies them?

A. Both are behavioural cash flows, stochastic in amount and time, because any credit line can be drawn at any time up to its limit.

B. Drawdowns by clients are deterministic in amount because they cannot exceed the line's limit; drawdowns by the bank are stochastic in amount.

C. Drawdowns by the bank on lines it has received are new business cash flows, deterministic in time and stochastic in amount.

D. Drawdowns by clients on lines the bank has granted are behavioural, stochastic in amount and time; drawdowns by the bank on lines it has received are of fixed amount at a stochastic time.

Question LTR-7 a.6 New • built from GARP Ch.7 (7.1)

A risk manager reviews a junior analyst's notes on the classification of cash flows. Which of the following statements in the notes is correct?

A. The payout of an American option is deterministic in time, because the option has a fixed expiry date.

B. A coupon on a risk-free floating-rate bond is stochastic in amount but deterministic in time, and belongs to the indexed/contingent subcategory.

C. The coupons of a fixed-rate loan are deterministic in amount and time even if the borrower is risky, because the schedule is set by the contract.

D. Once a cash flow has been classified at the reference date, it keeps that category until it is paid.

Solutions • LTR-7 a

Solution LTR-7 a.1

Answer: C

Bank Q260 • AnalystPrep Q.3920

Step 1 – The two dimensions

Castagna and Fede classify each flow by *when* it occurs (time) and *how much* it is (amount); on each dimension a flow is deterministic or stochastic.

Interpretation

Credit quality, contract type and market variables are *drivers* of a stochastic amount (the four subcategories), not classification axes.

Why the other options are wrong

A is incorrect. Credit quality is the driver of credit-related flows, one subcategory of stochastic amount.

B is incorrect. Contract type is deliberately disregarded: the TSECF orders flows by date "disregarding which kind of contract they are".

D is incorrect. Maturity is part of time, but currency is not a dimension of the taxonomy.

LO: LTR-7 a: Differentiate between deterministic and stochastic cash flows and provide examples of each. • **Reference:** Castagna and Fede, Measuring and Managing Liquidity Risk, Ch.6 (GARP Ch.7), 7.1 A Taxonomy of Cash Flows

Solution LTR-7 a.2

Answer: B

Bank Q261 • AnalystPrep Q.3921

Step 1 – Apply both dimensions

Deterministic without qualification means known on both dimensions: a known instant and a known amount.

Interpretation

Statements that fix only one dimension describe mixed cases, such as a floating coupon (known date, unknown amount).

Why the other options are wrong

A is incorrect. Describes flows stochastic in both time and amount.

C is incorrect. Known timing with an unknown amount is stochastic in amount (for example a floating coupon).

D is incorrect. Fully stochastic flows, such as a recovery after default.

LO: LTR-7 a: Differentiate between deterministic and stochastic cash flows and provide examples of each. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.1

Solution LTR-7 a.3

Answer: A

Bank Q263 • BT P2.T8.25.4.1

Step 1 – Test each flow on time and amount

Item 1: fixed coupon on fixed dates, known amount and timing. Item 2: known dates, amount indexed to SOFR (indexed/contingent). Item 3: depositor behaviour drives size and timing (behavioural). Item 4: drawdowns of lines granted to clients (behavioural, both dimensions stochastic).

$\Rightarrow$ Only item 1 is deterministic.

Interpretation

Item 1 is deterministic in the taxonomy's sense only if the issuer of the bond is risk-free; the stem implies no credit risk.

Why the other options are wrong

B is incorrect. Timing is known but the amount floats with SOFR: stochastic in amount.
C is incorrect. Withdrawals vary in size and timing with customer behaviour.
D is incorrect. A committed line is a liquidity option sold to clients; amount (25% to 50%) and timing are uncertain.

LO: LTR-7 a: Differentiate between deterministic and stochastic cash flows and provide examples of each. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.1, Figure 7.1

Solution LTR-7 a.4 Answer: A

Bank Q264 • Schweser Module Quiz 72.1 Q1

Step 1 – Locate each flow in the grid

New debt issuance by the bank: new business, deterministic in time (it can be well defined in time) but stochastic in amount. Recovery on default: credit related, stochastic in both. American option payout: indexed/contingent at a stochastic exercise time. Mortgage prepayment: behavioural, stochastic in both.
$\Rightarrow$ New debt issuance is the exception.

Interpretation

New business is the subcategory that splits: renewals and planned issues have known dates; new loans, deposits and assets do not.

Why the other options are wrong

B is incorrect. The default time and the recovered amount are both unknown at the reference date.
C is incorrect. Exercise can come at any time and the payoff depends on the market.
D is incorrect. Prepayment timing and amount depend on rates and borrower behaviour.

LO: LTR-7 a: Differentiate between deterministic and stochastic cash flows and provide examples of each. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.1, Figure 7.1

Solution LTR-7 a.5 Answer: D

New • built from GARP Ch.7

Step 1 – Lines granted to clients

Withdrawals "may occur at any time until the expiry of the contract and in an uncertain amount, although within the limits of the line": behavioural, stochastic in time and amount.

Step 2 – Lines received by the bank

The bank chooses the amount it draws (between zero and the limit), but its liquidity needs arise at a stochastic time: fixed amount, stochastic time, alongside the one-touch option payout.

Interpretation

The same product sits in two cells because the decision maker differs: a client's decision is behavioural from the bank's point of view; the bank's own decision is not.

Why the other options are wrong

A is incorrect. Ignores who draws: the bank's own drawdowns are not behavioural from its point of view.
B is incorrect. Reversed: a limit caps a client's drawdown but does not make it known.
C is incorrect. Withdrawals from lines received are not new business, and their time, not their

amount, is stochastic.

LO: LTR-7 a: Differentiate between deterministic and stochastic cash flows and provide examples of each. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.1, Figure 7.1

Solution LTR-7 a.6 Answer: B

New • built from GARP Ch.7

Step 1 – Why B is correct

A floating coupon is paid on a known date, but its amount depends on a market fixing: indexed/-contingent, stochastic in amount and deterministic in time (GARP Figure 7.1).

Interpretation

Each of the other statements breaks one of the reading's conditions: optimal exercise timing, a risk-free obligor, and the reference point of view.

Why the other options are wrong

A is incorrect. An American option can be exercised at any time up to expiry, so the payout time is stochastic.

C is incorrect. Contract flows are deterministic only if the obligor is risk-free; a risky borrower's default creates credit-related missing flows.

D is incorrect. Classification is made from a given reference date, and flows "can also change category when we shift the point of view" to a later date.

LO: LTR-7 a: Differentiate between deterministic and stochastic cash flows and provide examples of each. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.1

LTR-7 • Monitoring Liquidity • Vardeez: MODERATE

LTR-7 b

P3 CONCEPT

Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P3. No workbook question tests it (the A26 revision notes cover it in one bullet). Vardeez flags it: his reading note names the "liquidity option concept".
Vardeez: liquidity options. The discriminator is why they are exercised (for the cash, not for profit), and the impact must be signed from the bank's side.

Reported exam shapes: none listed. The questions are built from the LO wording and the Vardeez focus.

What the questions look like: definitions and identification (which position is a liquidity option), the two impacts of exercise, the sign of the financial impact for the bank, and how the bank manages the liquidity impact. No calculations. Where liquidity options enter the TSECF is in LTR-7 d (Section 4); the cash flow categories are in LTR-7 a.

1. What a liquidity option is

A **liquidity option** is the right of a holder to receive cash from, or to give cash to, the bank at predefined times and terms. Exercising it does not directly entail a profit or a loss; it follows from the holder's *need for*, or *surplus of*, liquidity. Financial effects may still be linked to the exercise, sometimes strongly.

	Standard (financial) option	Liquidity option
Why exercised	When there is a profit, independently of the cash flows that follow	Because of the cash flows produced after exercise, even when it is not convenient financially
Cash after exercise	Typically positive, but irrelevant to the decision (a holder may exercise and keep the stock)	The point of the exercise
Modelling	Standard option pricing, rational exercise	Behavioural and statistical models jointly with the financial side

2. Examples

- **Committed credit line sold to a client.** The obligor may draw up to the notional whenever she wants at the contract terms (say 3-month Libor plus a spread). She may draw because it makes financial sense (her credit spread has widened, so the contract terms are cheap), or for other reasons even when her spread has narrowed and a new loan would be cheaper.
- **Sight and savings deposits.** Clients can withdraw all or part with no or short notice: to buy higher-yielding assets, or simply for transaction needs and many other hard-to-predict reasons.
- **Prepayment of fixed-rate mortgages or loans** (exercise "the other way round"). The client repays early: for exogenous reasons (divorce, retirement), or more often to refinance after rates fall.

The bank can also be *long* liquidity options: committed credit lines it has received from other banks.

3. Two impacts of exercise

1. **Liquidity impact** on the balance sheet: the amount withdrawn or repaid.
2. **Financial impact**, positive or negative: the difference between the contract's interest rate and credit spread and the market levels of the same variables at exercise, applied to the amount withdrawn or repaid.

Exercise	Liquidity impact	Financial impact on the bank
Credit line drawn after the client's spread widened	Outflow	Negative: the bank lends at a spread below the market spread for this client
Credit line drawn for convenience after the spread narrowed	Outflow	Positive: the client pays more than it would on a new loan
Large savings account closed	Outflow, possibly large	Small loss: the lost margin between deposit rate and reinvestment rate, or the cost of replacing the deposit at a higher rate
Mortgage prepaid for exogenous reasons, rates flat or higher	Early inflow	None, or a gain: funds can be reinvested at better rates
Mortgage prepaid to refinance after rates fell	Early inflow	Negative: the funds are relent at lower rates than the old contract

$\Rightarrow$ More liquidity is not automatically good news: an early inflow from prepayment usually comes with a financial loss. Always state whose point of view: the client's gain is typically the bank's loss.

> **Correction to the source notes (Vardeez, LTR-7 b).** Vardeez's second worked direction says a customer who withdraws from a savings account paying *above* the market rate finds it "good for the customer to leave". It is not: the customer gives up an above-market rate, which is exactly why such an exercise is a liquidity option (driven by the need for cash, not by profit). The impact on the bank, positive because it stops paying above market, is right. GARP's own savings example is the usual case: a small *loss* to the bank (the lost margin or a costlier replacement deposit) with a liquidity impact that can be large.

4. Impact on liquidity management

- **Hedging.** The financial effects can be hedged directly, though only partially, with a mixture of standard and statistical techniques. The **liquidity impact can only be managed** with cash reserves, a constrained allocation of assets in liquid assets, or easy access to credit lines (a long position in liquidity options for the bank).
- **Cost and pricing.** All of these tools are costly, and the cost must be accounted for when pricing client contracts; so banks need models to price both the liquidity options they sell and those they buy.
- **Joint models.** Because exercise also produces financial gains or losses, models must consider the liquidity and financial aspects together.
- **Monitoring.** Behavioural models adjust the TSECF for sight deposits, credit line usage and prepayments (LTR-7 d, Section 4).

Exam traps

- **Definition.** A standard option is exercised for profit; a liquidity option for the cash flows after exercise, even at a financial loss.
- **Two impacts, not one.** A liquidity impact (amount moved) and a financial impact (contract versus market terms on that amount). Cash reserves are a hedging tool, not an impact.
- **Whose side.** Sign the financial impact from the bank's side: a drawdown after the client's spread has widened hurts the bank; a prepayment after rates fall hurts the bank.
- **What can be hedged.** The financial effect, partially, with derivatives and statistics; the liquidity effect only with reserves, liquid assets or credit lines.
- **Not every option.** An equity or rate option the bank sells is a financial option, even though it produces cash flows.

Practice questions: LTR-7 b

Question LTR-7 b.1

Bank Q265 • AnalystPrep Q.3924

A bank's liquidity options (for example, credit lines and demand deposits) are exercised by customers. Which of the following correctly describes the two primary effects that the exercise of a liquidity option has on the bank?

A. An impact on the balance sheet from the amount repaid or withdrawn, and a (positive or negative) financial impact.

B. A (positive or negative) financial impact and an effect on the contract deposit rate.

C. An effect on the contract deposit rate and an impact on the balance sheet.

D. An effect on available cash reserves and a financial impact.

Question LTR-7 b.2

Bank Q266 • Schweser Module Quiz 72.1 Q2 (rebuilt)

Which of the following statements comparing liquidity options to standard financial options is most accurate?

A. Financial options typically produce negative cash flows after exercise.

B. Liquidity options are exercised only when exercise is financially profitable to the holder.

C. Both types of option are typically exercised only when they are in the money.

D. Liquidity options may or may not be financially profitable upon exercise.

Question LTR-7 b.3

New • built from GARP Ch.7 (7.2)

A liquidity risk manager is listing the liquidity options the bank has sold. Which of the following positions does NOT give its holder a liquidity option against the bank?

A. A committed credit line the bank has opened for a corporate client

B. A savings account from which the client can withdraw with no or short notice

C. A European call option on an equity index that the bank has sold to an investor

D. A fixed-rate mortgage that the borrower may prepay before maturity

Question LTR-7 b.4 New • built from GARP Ch.7 (7.2)

A corporate client draws USD 50 million on a committed credit line priced at 3-month SOFR plus 1.20%, at a time when its credit spread in the market has widened to 2.50%. Which of the following describes the effect on the bank?

A. A liquidity outflow of USD 50 million and a positive financial impact, because the client pays a spread above the risk-free rate.

B. A liquidity outflow of USD 50 million and a negative financial impact, because the bank lends at a contract spread below the current market spread for this client.

C. No liquidity impact, because the drawn amount becomes a loan asset, and a negative financial impact.

D. A liquidity outflow of USD 50 million and no financial impact, because a liquidity option is exercised for liquidity and not for profit.

Question LTR-7 b.5 New • built from GARP Ch.7 (7.2)

Which of the following statements about managing the impact of liquidity options is correct?

A. The liquidity impact can be fully hedged with standard interest rate derivatives, while the financial impact needs cash reserves.

B. Because liquidity options are exercised for non-financial reasons, they can be priced with standard option pricing models that assume rational exercise.

C. The financial impact can be partially hedged with standard and statistical techniques, while the liquidity impact can only be managed with cash reserves, liquid assets or access to credit lines.

D. Holding liquid assets to cover liquidity options carries no cost, so it need not be reflected in the pricing of client contracts.

Question LTR-7 b.6 New • built from GARP Ch.7 (7.2)

A risk manager reviews liquidity options exercised against the bank last quarter. Which assessment is correct?

A. A client closed a large savings account: the financial impact on the bank was small (the lost margin, or the cost of replacing the deposit), but the liquidity impact was large.

B. A borrower prepaid a fixed-rate mortgage after a divorce while market rates were rising: the bank suffered a financial loss, since it received the funds earlier than expected.

C. A borrower prepaid a fixed-rate mortgage because market rates had fallen: the bank gained, since it can relend the funds at current market rates.

D. A client drew on a committed credit line for convenience although its credit spread had narrowed: the financial impact on the bank was negative.

Solutions • LTR-7 b

Solution LTR-7 b.1

Answer: A

Bank Q265 • AnalystPrep Q.3924

Step 1 – The two impacts

GARP: exercise has (1) a liquidity impact on the balance sheet, the amount withdrawn or repaid, and (2) a positive or negative financial impact, the difference between the contract's rate and spread and the market levels at exercise, applied to that amount.

Interpretation

The contract deposit rate is an *input* to the financial impact; cash reserves are a *tool* to manage the liquidity impact.

Why the other options are wrong

B is incorrect. Omits the balance sheet effect; the contract rate is a contract term, not an effect of exercise.

C is incorrect. Omits the financial impact.

D is incorrect. Cash reserves are a hedging tool, not a primary effect; the balance sheet effect is missing.

LO: LTR-7 b: Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process. • **Reference:** Castagna and Fede, Measuring and Managing Liquidity Risk, Ch.6 (GARP Ch.7), 7.2 Liquidity Options

Solution LTR-7 b.2

Answer: D

Bank Q266 • Schweser Module Quiz 72.1 Q2

Step 1 – Exercise rule of each type

Standard options are exercised when profitable; the cash flows after exercise do not drive the decision. A liquidity option is exercised because of the cash flows after exercise, "even if it is sometimes not convenient to exercise it from a financial perspective".

$\Rightarrow$ Exercise of a liquidity option may or may not be profitable.

Interpretation

This is why behavioural models, not standard option pricing, drive the TSECF adjustments for liquidity options.

Why the other options are wrong

A is incorrect. Reversed: the cash flows after exercising a financial option are typically positive.

B is incorrect. Applies the financial-option rule to liquidity options.

C is incorrect. Only financial options are exercised when in the money; liquidity options follow liquidity needs.

LO: LTR-7 b: Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.2

Solution LTR-7 b.3

Answer: C

New • built from GARP Ch.7

Step 1 – Test each position against the definition

A liquidity option is the right to receive cash from, or give cash to, the bank at predefined times

and terms, exercised for a need or surplus of liquidity. The credit line (A), the savings account (B) and the prepayable mortgage (D) are the three examples in the reading.

Step 2 – The equity index call

It is exercised at expiry if it is in the money, for profit: a standard financial option, even though the bank pays cash on exercise.

Interpretation

Cash moving on exercise is not the test; the reason for exercise is.

Why the other options are wrong

A is incorrect. A committed line gives the client the right to draw cash at contract terms: the reading's first example.
B is incorrect. Withdrawal with no or short notice is a liquidity option.
D is incorrect. Prepayment is a liquidity option exercised "the other way round": the client gives cash to the bank early.

LO: LTR-7 b: Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.2

Solution LTR-7 b.4 Answer: B

New • built from GARP Ch.7

Step 1 – Liquidity impact

The drawn USD 50 million leaves the bank: a balance sheet outflow it must fund.

Step 2 – Financial impact from the bank's side

The client borrows at SOFR + 1.20% when the market would charge it 2.50%. The bank lends 1.30% below the spread that matches the client's current credit risk on the amount drawn.
$\Rightarrow$ Negative financial impact for the bank (a saving for the client).

Interpretation

This is GARP's case of a drawdown that "also makes sense under a financial perspective" for the obligor. The contract terms are fixed until expiry, so the bank bears the widening.

Why the other options are wrong

A is incorrect. A spread above the risk-free rate is not the test; the comparison is with the current market spread.
C is incorrect. The cash leaves the bank, so there is a liquidity impact.
D is incorrect. Exercise for liquidity reasons does not remove the financial effect; it can be large.

LO: LTR-7 b: Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.2

Solution LTR-7 b.5 Answer: C

New • built from GARP Ch.7

Step 1 – Why C is correct

GARP: "While the financial effects of liquidity options can be directly, although partially, hedged by a mixture of standard and statistical techniques, the liquidity impact can only be managed by" cash reserves, a constrained allocation in liquid assets, or easy access to credit lines.

Interpretation

These tools cost money, so their cost must be included when pricing client contracts, and models

must treat the liquidity and financial sides jointly.

Why the other options are wrong

A is incorrect. Reversed: derivatives address the financial impact; cash and liquid assets address the liquidity impact.

B is incorrect. Non-rational exercise is the reason standard option pricing is not enough.

D is incorrect. Liquidity buffers have a cost that must be priced into the contracts.

LO: LTR-7 b: Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.2

Solution LTR-7 b.6 Answer: A

New • built from GARP Ch.7

Step 1 – Why A is correct

GARP: when a client closes a savings account the bank's financial loss is "the missing margin between the contract deposit rate and the rate it earns on the reinvestment" or the cost of a replacement deposit, often small; "the liquidity impact, on the other hand, can be quite substantial if the deposit has a big notional".

Step 2 – Sign the other cases from the bank's side

Prepayment for exogenous reasons with rates rising: no loss, the bank can reinvest at better rates. Prepayment after rates fell: a loss, the funds are relent at lower rates. Drawdown after the client's spread narrowed: the client pays more than market, a gain for the bank.

Interpretation

The direction of the financial impact depends on the market at exercise relative to the contract, not on whether the bank receives or pays cash.

Why the other options are wrong

B is incorrect. Rising rates make the early funds worth more to the bank: no financial loss.

C is incorrect. Falling rates make the prepayment a loss: the old contract rate is replaced at lower market rates.

D is incorrect. With a narrower spread the contract terms are above market for this client, so the impact on the bank is positive.

LO: LTR-7 b: Describe and identify examples of liquidity options and explain the impact of liquidity options on a bank's liquidity position and its liquidity management process. • **Reference:** Castagna and Fede, Ch.6 (GARP Ch.7), 7.2

LTR-8 • The Failure Mechanics of Dealer Banks • Vardeez: IMPORTANT

LTR-8 a

P3 CONCEPT

Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business.

Workbook: D27 Intraday Liquidity and Dealer Bank Failure Mechanics • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Vardeez focus: he rates the reading IMPORTANT ("One question might come up from dealer banks. So know it well, maybe go through crash course notes"). The map reports no exam shape for this LO; the Tier D workbook section D27 covers it in its revision notes only.

Reported exam shapes: none listed. The questions here follow the LO wording and Vardeez's table of line of business, function and risk factor.

What the questions look like: statement audits on what dealers do (primary versus secondary markets, matched book, repo financing, prime brokerage, internal funds), on off-balance-sheet vehicles, on economies and diseconomies of scope, and on which risk each line brings when solvency is doubted. No calculations.

1. What a dealer bank is

A **dealer bank** intermediates the markets for securities and derivatives: it makes markets, lends securities, runs repo books and is a counterparty in OTC derivatives. Large dealer banks sit inside large complex financial groups under **holding companies**; the 14 dealers the New York Fed invited to its April 2009 OTC derivatives meeting overlap heavily with the primary dealers in US government securities. Most also run commercial banking, investment banking, asset management and prime brokerage, and trade on their own account (**proprietary trading**). Much of this sat outside traditional bank-failure resolution mechanisms; since the crisis, all large dealers operate as regulated banks or within regulated bank holding companies. Their failures can damage the economy, so they are sometimes considered "too big to fail".

2. The lines of business

- **Securities dealing, underwriting and trading.** *Primary market*: the dealer, often as underwriter, buys equities or bonds from the issuer and sells them to investors over time. *Secondary market*: it stands ready to have its bids hit by sellers and its offers hit by buyers. Dealers dominate OTC bond trading (corporate, municipal, some sovereign, securitised products) and intermediate large equity block trades. Investment banking adds M&A advice and merchant banking (oil, metals, foodstuffs).
- **Repo.** A repo is a short-term cash loan collateralised by securities, mostly overnight and renewed daily; a haircut protects the lender (10%: USD 100m of securities secures USD 90m). Dealers lend to hedge funds that lever up securities, and finance their own inventories the same way: pre-crisis, overnight repo at average haircuts **under 2%** meant little incremental capital, and Bear Stearns and Lehman ran leverage above 30. Dealers settle through clearing accounts at JPMorgan Chase and BNY Mellon, which also dominate **tri-party repo** (USD 2.5 trillion in 2007), where the clearing bank holds the collateral.
- **OTC derivatives.** One party to most OTC trades is a dealer, which runs a **matched book**: it lays off client risk with offsetting trades and earns the bid-offer difference. About USD 600 trillion notional (exchange-traded about USD 400 trillion), mostly interest rate swaps. The total market value of

all derivatives is zero; they transfer wealth and cause net losses only through the frictional costs of distress. ISDA master agreements **net** exposures and collateral (A owes B 100, B owes A 80: B posts 20). Dealer exposures of USD 25,372 billion fell to USD 3,744 billion after netting (June 2009); USD 23 trillion of USD 28 trillion CDS notional was dealer-to-dealer, partly because hedge funds exit by **novating** trades to another dealer.

- **Prime brokerage.** Custody and management of securities, clearing, cash management, securities lending, financing and reporting for hedge funds; often also their derivatives counterparty. Morgan Stanley, Goldman Sachs and Bear Stearns had most of the business in 2007.
- **Asset management.** For institutions and wealthy clients, including **internal hedge funds** and private equity partnerships with the bank as general partner. Dealers supported failing internal funds to protect reputation: Bear Stearns offered USD 3.2 billion to its High-Grade Structured Credit Fund (June 2007); Citigroup put USD 500 million into Falcon (February 2008).
- **Off-balance-sheet financing.** Loans are sold to a **special purpose entity** that pays with debt issued to third-party investors and services it from the loans' cash flows. Being contractually remote, its assets could stay off the sponsor's books for accounting and capital (Citigroup: over USD 800 billion in such entities, June 2008). **SIVs** financed mortgages with short-term debt sold to money market funds; when their creditors stopped rolling in 2007, HSBC (about USD 35 billion) and Citigroup (USD 49 billion) brought SIV assets onto their balance sheets.
- **Commercial banking.** Deposits and corporate and consumer lending: credit risk and deposit runs, as at any bank. Insured deposits are less runnable, but Rule 23A bars using them to fund broker-dealer affiliates.

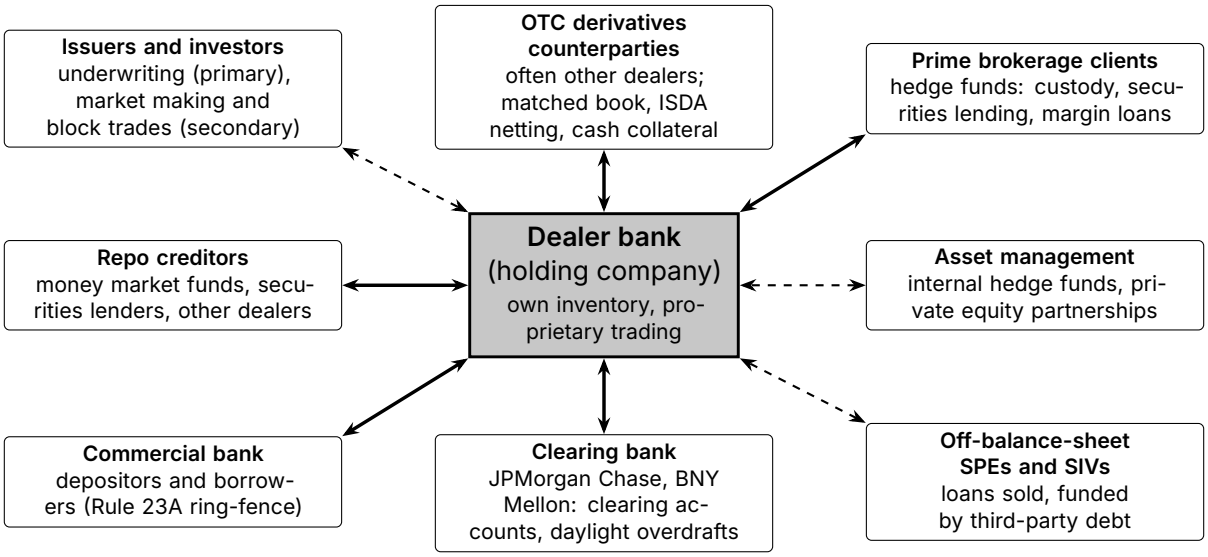


Figure 1. Who is on the other side of each line of business. Thick solid lines: relationships that finance the dealer (cash, collateral, client balances, daylight credit) and can be pulled in a run, as in LTR-8 b. Dashed lines: lines whose stress comes back through lost revenue or reputational support.

3. Economies and diseconomies of scope

Research does not find a strong case for the net benefits of large diversified conglomerates. There may be **economies of scope in information technology, marketing and financial innovation**. The crisis suggested **diseconomies of scope in risk management and corporate governance**: some executives and boards found it too hard to comprehend or control the risk-taking inside their own firms (UBS's 2008 shareholder report is GARP's example).

Correction to the source notes. Vardeez (after Schweser) says that before 2007 to 2009 dealer banks "lacked access to central bank borrowing and federal deposit insurance", and that repo leverage came from "the lack of capital requirements". GARP is narrower: dealers *not regulated as banks* had no discount window (Morgan Stanley and Goldman Sachs became bank holding companies right after Lehman), while dealers inside banking groups did have it, though Rule 23A stops insured deposits funding their broker-dealers. Repo leverage came from overnight haircuts under 2%, which let dealers hold securities with little incremental capital.

Comparison card: LTR-8 a

Line	What the dealer does	Risk factor when solvency is doubted
Securities, underwriting	Buys from issuers, sells to investors; makes markets	Inventory market risk; issuers go elsewhere, fees fall, issue liquidity suffers
Repo	Lends to clients; funds own inventory overnight	Creditors stop rolling; fire sales; high leverage on thin haircuts
OTC derivatives	Matched book, earns bid-offer; ISDA netting	Counterparty default; dealer-to-dealer concentration; cash drained by exits
Prime brokerage	Custody, financing, securities lending to hedge funds	Clients leave with cash and securities; margin loan demands
Asset management	Internal hedge funds, private equity	Reputational support of failing funds
Off-balance-sheet	SPEs and SIVs funded by third-party debt	Risk not truly transferred; assets come back (HSBC, Citigroup)
Commercial banking	Deposits and loans	Credit risk, deposit runs; Rule 23A ring-fence

Exam traps

- **Primary versus secondary.** Primary: issuer to investors (underwriting). Secondary: among investors (bids hit, offers lifted).
- **Matched book is not speculation.** It lays off client risk and earns the bid-offer spread; directional positions are proprietary trading.
- **Netting is real.** Master agreements reduce exposures and collateral, not just paperwork; the total market value of all derivatives is zero.
- **Dealers do finance clients** through repo, and support their *internal* funds and sponsored SIVs, even off balance sheet.
- **Scope.** Economies in IT, marketing, innovation; diseconomies in risk management and governance.

Practice questions: LTR-8 a

Question LTR-8 a.1

Bank Q285, Q286 • AnalystPrep Q.2294, Q.2295 (merged and rebuilt)

Bens Bank, a large dealer bank, is active in both the primary and the secondary market for corporate bonds. Which of the following correctly describes its role in each?

A. Primary: it intermediates between investors who sell and investors who buy existing bonds. Secondary: it buys new bonds from issuers and sells them to investors over time.

B. Primary: it buys new bonds from issuers, often as underwriter, and sells them to investors over time. Secondary: it stands ready to have its bids hit by sellers and its offers hit by buyers.

C. Primary: it lends to corporate clients against their bonds through repurchase agreements. Secondary: it advises issuers on mergers and acquisitions.

D. Primary: it acts only as an agent, matching issuers with investors for a commission. Secondary: it trades only on exchanges, never over the counter.

Question LTR-8 a.2

Bank Q287 • AnalystPrep Q.2298 (rebuilt)

A dealer bank runs a "matched book" in over-the-counter derivatives. Which statement best describes how this practice works for the dealer?

A. It keeps client-initiated positions unhedged and profits from favourable market moves.

B. It charges each client a fixed monthly premium regardless of trading activity.

C. It lays off the risk of client trades through offsetting trades and earns the difference between bid and offer terms.

D. It delivers the underlying commodities to counterparties at a markup over cost.

Question LTR-8 a.3

Bank Q289 • BT answers P2.T7.809.1

Duffie describes the main business lines of large dealer banks as (1) securities dealing, underwriting and trading, (2) over-the-counter derivatives, and (3) prime brokerage and asset management. Each of the following statements about these business lines is true EXCEPT:

A. A tri-party repo, in which a clearing bank holds the collateral and returns the cash to the creditor, helps mitigate counterparty risk and eases collateral management.

B. For most OTC derivatives, one counterparty is a dealer, which typically lays off much or all of the risk of client-initiated trades by running a matched book.

C. The best measure of the systemic risk of OTC derivatives is the total market value of all outstanding contracts, and master swap agreements reduce only administrative burden, not actual risk.

D. Some large dealers are prime brokers offering securities management, clearing, cash management, securities lending, financing and reporting services to clients.

Question LTR-8 a.4

Bank Q290 • GARP PE2 Q24 • Mock 4 Q25

A treasury manager presents to new analysts on the activities of dealer banks, comparing their major lines of business and the risks in each. Which of the following would be a correct statement for the manager to make?

A. Dealer banks often intermediate large equity block trades and trades in municipal and corporate bonds.

B. Dealer banks fund their own trading positions through repos but typically refrain from providing repo financing to fund investors' purchases of securities.

C. Dealer banks usually prefer to support external hedge fund counterparties over internally managed hedge funds in times of financial stress.

D. Dealer banks are prohibited by regulators from operating asset-management activities for institutional and wealthy clients.

Question LTR-8 a.5

Bank Q291 • Schweser Module Quiz 73.1 Q2 (rebuilt)

Before 2007, a dealer bank sold residential mortgages to a trust it had set up for the purpose; the trust paid for them with debt issued to third-party investors. Which of the following statements about this arrangement is correct?

A. The trust is a special purpose entity whose debt is serviced from the mortgage cash flows and, being contractually remote, could be kept off the bank's balance sheet for accounting and capital purposes.

B. The trust is a prime brokerage client, so the bank may re-hypothecate the mortgages it holds up to 140% of any margin loan it has made to the trust.

C. Because the trust's debt is contractually remote from the bank, the sponsoring bank bore no economic or reputational exposure when the trust's short-term creditors stopped renewing.

D. Holding the mortgages in the trust required the bank to hold more regulatory capital than if it had kept the loans on its own balance sheet.

Question LTR-8 a.6

Bank Q292 • Schweser Module Quiz 73.1 Q3 (rebuilt)

A board member asks whether combining commercial banking, dealing, prime brokerage and asset management in one holding company made the 2007 to 2009 losses worse. Which of the following best reflects Duffie's view?

A. The combination created diseconomies of scope in information technology and marketing, but economies of scope in risk management.

B. The combination may create economies of scope in information technology, marketing and innovation, but the crisis suggests diseconomies of scope in risk management and corporate governance.

C. Research finds a strong case for the net benefits of large diversified conglomerates, so the crisis losses were unrelated to the breadth of the firms.

D. Senior executives and boards understood the risks well, and the failures came only from operational errors in trade processing.

Question LTR-8 a.7 New • built from GARP Ch.8, 8.1 and 8.2

A credit analyst covering a large dealer bank lists, for each line of business, how the bank's liquidity would suffer if its solvency were questioned. Which pairing is correct?

A. OTC derivatives: insured depositors withdraw their balances, because derivatives counterparties are covered by deposit insurance.

B. Prime brokerage: hedge fund clients move cash and securities to other prime brokers or custodian banks, removing a source of financing the bank relied on.

C. Repo financing: overnight repo creditors become more willing to roll their loans, because the haircut fully protects them against the dealer's default.

D. Underwriting and advisory: issuers keep their mandates with the bank because fee income is contractual, so this line adds no liquidity strain.

Solutions • LTR-8 a

Solution LTR-8 a.1

Answer: B

Bank Q285, Q286 • AnalystPrep Q.2294, Q.2295

Step 1 – Primary market

Duffie: "In the primary market, the dealer bank, sometimes acting as an underwriter, effectively buys equities or bonds from an issuer and then sells them over time to investors."

Step 2 – Secondary market

"In secondary markets, a dealer stands ready to have its bid prices hit by sellers and its ask prices hit by buyers."

$\Rightarrow$ B.

Interpretation

In both markets the dealer is a principal that holds inventory, which is why it carries market risk and needs financing (repo) for that inventory.

Why the other options are wrong

A is incorrect. Swaps the two roles.

C is incorrect. Repo lending is financing and M&A advice is investment banking; neither defines the dealer's role in these markets.

D is incorrect. The dealer buys and resells as principal, and it dominates the OTC bond market rather than trading only on exchanges.

LO: LTR-8 a: Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business. • **Reference:** Duffie, The Failure Mechanics of Dealer Banks, Journal of Economic Perspectives 24:1 (2010) (GARP Ch.8), 8.1 Securities Dealing, Underwriting, and Trading

Solution LTR-8 a.2

Answer: C

Bank Q287 • AnalystPrep Q.2298

Step 1 – What a matched book is

Duffie: the dealer "usually lays off much or all of the risk of its client-initiated derivatives positions by running a 'matched book', that is, by aiming for offsetting trades, profiting on the differences between bid and offer terms".

$\Rightarrow$ C. The bid-offer difference is also the answer to "what is the dealer's primary source of profit" (bank Q288).

Interpretation

A matched book removes market risk, not counterparty risk: the dealer still faces each counterparty, and many of them are other dealers. That is the risk factor of this line.

Why the other options are wrong

A is incorrect. Describes proprietary trading, a separate activity.

B is incorrect. Dealers earn spreads on trades, not subscription premiums.

D is incorrect. OTC derivatives dealing is risk transfer, not physical delivery at a markup.

LO: LTR-8 a: Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business. • **Reference:** GARP Ch.8, 8.1 Over-the-Counter Derivatives

Solution LTR-8 a.3 — Answer: C

Bank Q289 • BT answers P2.T7.809.1

Step 1 – Total market value is not a risk measure

"It is an accounting identity that the total market value of all derivatives contracts must be zero": purchased positions equal sold positions.

Step 2 – Netting reduces actual exposure

Under a master swap agreement, exposures and collateral are netted: if A has an exposure of USD 100 million to B and B one of USD 80 million to A, B posts USD 20 million. Dealer exposures fell from USD 25,372 billion to USD 3,744 billion after netting.
$\Rightarrow$ C is false on both counts, so it is the exception.

Interpretation

Counterparty risk is gauged by exposure to default, reduced by collateral and netting. A hedge fund that has posted USD 60 million and defaults on a portfolio worth USD 100 million to the dealer leaves a USD 40 million loss.

Why the other options are wrong

A is incorrect. True: in a tri-party repo the clearing bank holds the collateral and returns the cash, insulating the lender somewhat.
B is incorrect. True: Duffie's description of the matched book.
D is incorrect. True: Duffie's list of prime brokerage services.

LO: LTR-8 a: Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business. • **Reference:** GARP Ch.8, 8.1, Table 8.2

Solution LTR-8 a.4 — Answer: A

Bank Q290 • GARP PE2 Q24

Step 1 – Securities dealing

Duffie: dealers "dominate the intermediation of over-the-counter securities markets, covering bonds issued by corporations, municipalities" and others, and "often intermediate large block trades" in equities.
$\Rightarrow$ A.

Interpretation

GARP builds the three wrong options by reversing what Duffie says about repo, internal funds and asset management.

Why the other options are wrong

B is incorrect. Reversed: a hedge fund can "finance the purchase of a large quantity of securities with a small amount of capital by placing purchased securities into repurchase agreements with a dealer".
C is incorrect. Reversed: dealers support their *internal* hedge funds to protect reputation (Bear Stearns' USD 3.2 billion offer, Citigroup's USD 500 million to Falcon).
D is incorrect. False: dealers "often have large asset-management divisions" for institutional and wealthy clients.

LO: LTR-8 a: Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business. • **Reference:** GARP Ch.8, 8.1 What Large Dealer Banks Do

Solution LTR-8 a.5 — Answer: A

Bank Q291 • Schweser Module Quiz 73.1 Q2

Step 1 – Name the vehicle

A trust set up to buy the bank's loans with the proceeds of debt sold to third parties is a special purpose entity; its debt is paid from the cash flows of the assets bought.

Step 2 – Why it is off balance sheet

"Because the debt obligations of a special purpose entity are usually contractually remote from the sponsoring bank, under certain conditions banks have not been required to treat the assets and debt obligations of such entities as their own" for accounting and minimum capital.
$\Rightarrow$ A.

Interpretation

This let some dealers run much larger loan businesses on a given amount of capital (Citigroup reported over USD 800 billion in such entities in June 2008). The risk was not truly gone, as C's error shows.

Why the other options are wrong

B is incorrect. Mixes in prime brokerage: re-hypothecation of up to USD 140 per USD 100 of margin loan applies to a prime brokerage client's securities, not to a sponsored SPE.
C is incorrect. When SIV creditors stopped rolling in 2007, HSBC (about USD 35 billion) and Citigroup (USD 49 billion) brought SIV assets onto their balance sheets to protect reputation and market share.
D is incorrect. Reversed: off-balance-sheet treatment *reduced* the capital the bank had to hold.

LO: LTR-8 a: Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business. • **Reference:** GARP Ch.8, 8.1 Off-Balance-Sheet Financing

Solution LTR-8 a.6 — Answer: B

Bank Q292 • Schweser Module Quiz 73.1 Q3

Step 1 – The possible benefits

GARP: "There may exist economies of scope in information technology, marketing, and financial innovation."

Step 2 – The crisis lesson

"Some of the risk-management failures discovered during the financial crisis are associated with diseconomies of scope in risk management and corporate governance": executives and boards found it too hard to comprehend or control the risk-taking inside their firms.
$\Rightarrow$ B.

Interpretation

This is a two-sided claim: benefits in IT, marketing and innovation; costs in risk management and governance. Each wrong option corrupts one side.

Why the other options are wrong

A is incorrect. Swaps the two sides.
C is incorrect. Reversed: the research "does not find a strong case for the net benefits" of such conglomerates.
D is incorrect. Reversed: the lesson is that boards did *not* understand the risks; the problem was governance, not trade processing.

LO: LTR-8 a: Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business. • **Reference:** GARP Ch.8, 8.1 What Large Dealer Banks Do

Solution LTR-8 a.7 **Answer: B**

New • built from GARP Ch.8, 8.1 and 8.2

Step 1 – The prime brokerage risk

Prime brokers finance themselves in part with the cash and securities clients leave in their accounts. When the dealer weakens, hedge funds "begin to shift their cash and securities to better capitalized prime brokers or, safer yet, custodian banks", which reduces the dealer's financial flexibility.

$\Rightarrow$ B pairs the line with its risk factor.

Interpretation

Every line fails through the same channel: someone who doubts solvency pulls back. The lines differ in who withdraws and what they take (LTR-8 b).

Why the other options are wrong

A is incorrect. Wrong line: derivatives counterparties are not insured depositors. They cut exposure by borrowing from the dealer, trading so that it pays cash, or novating.

C is incorrect. Polarity flip: despite haircuts, repo creditors "may have little or no incentive to renew repos in the face of concerns over the dealer bank's solvency".

D is incorrect. Issuers can take their business elsewhere when solvency is in doubt, cutting fee inflows and disrupting the liquidity of the issues the dealer supports.

LO: LTR-8 a: Compare and contrast the major lines of business in which dealer banks operate and the risk factors they face in each line of business. • **Reference:** GARP Ch.8, 8.1 and 8.2; SchweserNotes 2026 Book 4, Reading 73, LO 73.a

LTR-8 • The Failure Mechanics of Dealer Banks • Vardeez: IMPORTANT

LTR-8 b

P3 CONCEPT

Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks.

Workbook: D27 Intraday Liquidity and Dealer Bank Failure Mechanics • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D27 (Map #65, 1 of 16 recall sources), notes only, and Vardeez rates the reading IMPORTANT.

Reported exam shapes

- **S3** Dealer-bank runs on repo, on prime-brokerage balances and by OTC derivatives counterparties, ending with the clearing bank.

What the questions look like: which action accelerates (or does not accelerate) a crisis, which event is the final step, which mitigant fits which run, and what a worried derivatives counterparty does. One GARP practice exam question (LTR-8 b.3) asks which bank is most exposed to a repo run from a table of pledged and unpledged securities; it needs a ratio, so it sits outside the CONCEPT tag and is kept because GARP set it.

1. Same economics, different plumbing

A dealer-bank run has the same cause as a depositor run: **fear over solvency** leads others to cut their potential losses. But the **institutional mechanisms and the systemic impact differ**. Many of those exposed to a dealer have no default insurance, or do not want the frictional costs of its failure procedures, so they leave early. Duffie's four mechanisms are the flight of short-term creditors, the departure of prime brokerage clients, cash-draining actions by derivatives counterparties, and "finally and most decisively" the **loss of clearing-bank privileges**. His composite **Alpha Bank** runs through them in turn (Figure 1).

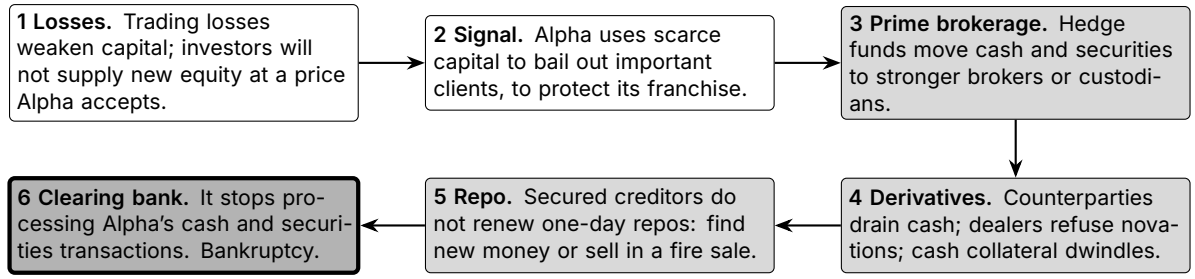


Figure 1. Duffie's Alpha Bank, read clockwise from top left. White boxes: the prelude. Grey boxes: the runs, each taking cash or collateral out of the dealer. Darkest box with the thick border: the final and most decisive step.

2. The four runs and their mitigants

Short-term secured creditors (repo). Dealers financed inventories with overnight repo at haircuts under 2%; Bear Stearns and Lehman had leverage above 30. Despite the collateral, creditors "may have little or no incentive to renew": money market funds under Rule 2a-7 may have to sell the collateral at once, disposal can bring litigation, and they can simply reinvest with another dealer. A mass refusal forces a **fire sale**; low prices cut the value of unsold collateral and so repo capacity, a **death spiral**

that can spread to other banks. Haircuts on investment-grade corporate bonds rose from under 5% to about 20% after Lehman; corporate bond repo fell about 60% (March 2008 to March 2009). *Mitigants*: lines of bank credit, a buffer of cash and liquid securities, and **laddered liability maturities**, so that only a small fraction must be refinanced at once. The central bank response is lender-of-last-resort financing, which buys time for orderly liquidation (Fed special facilities such as the Primary Dealer Credit Facility; ECB repo auctions; Bank of England facilities). Insured deposits run less, but Rule 23A keeps them from funding broker-dealer affiliates.

Prime brokerage clients. In the UK, client cash and securities are commingled with the broker's own, "equivalent to uninsured deposits". US Rule 15c3-3 lets a broker pool clients' free credit balances apart from its own funds, so one client's cash meets another's needs: with Hedge Fund A's USD 150 million of cash, a USD 100 million loan to Hedge Fund B and USD 50 million in reserve, A's exit leaves the broker to find USD 100 million. Brokers also re-hypothecate up to USD 140 of client assets per USD 100 of margin loan (equity advance rate 50%). **Even without a run**, clients can keep demanding margin loans that a doubted broker cannot refinance with their securities. Re-pledgeable client collateral at Morgan Stanley fell from USD 832 billion to USD 294 billion (August to November 2008); Lehman's London clients could not reclaim re-hypothecated assets. *Mitigant*: hedge funds diversify across prime brokers and custodian banks.

Derivatives counterparties. They cut exposure by borrowing from the dealer, entering trades that make it pay cash, harvesting in-the-money positions, or **novating** to another dealer; dealers began refusing Bear Stearns novations, which spread alarm. A dealer cannot refuse trades at market terms without signalling weakness. Collateral-on-downgrade triggers drain cash (Morgan Stanley: USD 498.3 million for one notch, a further USD 1,456.2 million for two) and were the proximate cause of AIG's bailout. Most OTC derivatives are **qualifying financial contracts**, exempt from the automatic stay, so close-out at replacement cost drains the bankrupt estate with priority, and others run first. *Mitigant*: **central clearing**, for standard contracts only.

Loss of cash settlement privileges. A clearing bank normally gives creditworthy dealers daylight overdrafts against securities in the clearing account; the peak, around 10 a.m., "easily exceeds several hundred billion dollars". When liquidity is in doubt it uses its **right of offset** to stop payments that would take the account below zero. JPMorgan Chase's refusal to process Lehman's wires left Lehman unable to settle, and it filed. *Mitigants* are policy proposals (LTR-8 c): an emergency bank or a tri-party repo utility.

> **Correction to the source notes.** GARP (Duffie) and workbook D27 say Morgan Stanley's re-pledgeable client collateral "declined by 69 percent, from $832 billion to $294 billion". Those two figures give a fall of 64.7%; a 69% fall to USD 294 billion would need a start near USD 948 billion. Quote the dollar figures, or "about two thirds". The 51% (Merrill Lynch) and 30% (Goldman Sachs) falls are given without dollar amounts and cannot be checked.

Comparison card: LTR-8 b

Run	Who leaves and what it drains	Mitigant
Repo creditors	Money market funds, securities lenders, other dealers stop rolling; fire sales, death spiral	Credit lines, cash and liquid buffer, laddered maturities; lender of last resort
Prime brokerage	Hedge funds take cash and securities; margin loans still demanded	Hedge funds diversify brokers and custodians
Derivatives	Counterparties borrow, trade for cash, harvest, novate; downgrade triggers	Central clearing (standard contracts only)
Clearing bank	Right of offset stops payments: the final step (Lehman)	Emergency bank, tri-party repo utility (LTR-8 c)

Exam traps

- **Same cause, different institutions.** Solvency fear, as in a retail run; different mechanisms and larger systemic impact.
- **Haircuts do not keep repo creditors.** They have little or no incentive to renew; haircuts *rise* in a crisis.
- **No run needed.** Prime brokerage clients drain a weak dealer through margin loan demands even if they stay.
- **QFC exemption** speeds the run: derivatives counterparties close out with priority, so others go first.
- **The last step is the clearing bank**, through its right of offset (Lehman). Northern Rock's story (wholesale funding, retail run after news of Bank of England support) is LTR-1 b.
- **Match the mitigant to the run**: repo → laddering and buffers; derivatives → central clearing; prime brokerage → diversification.

Practice questions: LTR-8 b

Question LTR-8 b.1

Bank Q301 • Schweser Module Quiz 73.1 Q1 (rebuilt, key corrected in the bank)

A large dealer bank's counterparties begin to doubt its solvency. Which of the following actions by counterparties or clients is least likely to accelerate a liquidity crisis at the dealer bank?

A. Repurchase agreement creditors refuse to roll over their short-term funding.

B. Prime brokerage clients withdraw their cash and securities from the dealer.

C. A hedge fund requests a novation of its derivative to another dealer bank.

D. A hedge fund agrees to leave additional unencumbered cash balances in its prime brokerage account.

Question LTR-8 b.2

Bank Q297 • AnalystPrep Q.2984 (rebuilt)

Duffie's composite dealer, Alpha Bank, suffers a sequence of withdrawals after trading losses weaken its capital. Which event does Duffie describe as the final and most decisive step before Alpha declares bankruptcy?

A. Its short-term secured creditors fail to renew their overnight repos, forcing fire sales.

B. Its prime brokerage clients move cash and securities to better-capitalised prime brokers.

C. Other dealers begin to refuse novations that would place them between Alpha and its counterparties.

D. Its clearing bank exercises its right to stop processing Alpha's cash and securities transactions.

Question LTR-8 b.3

Bank Q293 • GARP PE1 Q29 • Mock 3 Q51

Large dealer banks have often financed significant fractions of their assets using short-term (overnight) repurchase agreements in which creditors hold bank securities as collateral against default losses. The table shows the quarter-end financing of four A-rated broker-dealer banks (USD billion).

Financial instruments	Bank P	Bank Q	Bank R	Bank S
Owned	656	750	339	835
Pledged as collateral	258	472	139	209
Not pledged	398	278	200	626

If repo creditors become equally nervous about each bank's solvency, which bank is most vulnerable to a liquidity crisis?

A. Bank P

B. Bank Q

C. Bank R

D. Bank S

Question LTR-8 b.4 Bank Q294 • AP Mock 1 Q60 (rebuilt)

Capital Bank is a dealer bank with investment banking, deposit-taking and corporate lending segments. The CRO is wary of a run by short-term creditors triggered by poor performance or an external shock. Which of the following steps could the bank take to mitigate the risk of losing liquidity in such a run?

I. Establishing lines of bank credit

II. Holding a buffer stock of cash and liquid securities

III. Laddering the maturities of its liabilities so that only a small fraction must be refinanced in a short period

A. I only

B. II and III only

C. I and II only

D. I, II and III

Question LTR-8 b.5 Bank Q300 • Mock 1 Q15

An institutional trader is a counterparty to numerous OTC derivatives transactions with a dealer bank. In recent months the trader has lost confidence in the solvency of the dealer bank. Which of the following actions is the trader LEAST likely to take?

A. Borrow funds from the dealer

B. Request a novation of the contract to another dealer

C. Enter into new offsetting derivatives contracts with the dealer

D. Request to have at-the-money options revised to in-the-money strike prices

Question LTR-8 b.6 Bank Q296 • AnalystPrep Q.2305 (rebuilt)

A dealer bank is perceived to face a solvency crisis. Its OTC derivatives counterparties are trying to reduce their exposure, partly by trading with the dealer in ways that make it pay out cash. If the dealer wants to avoid sending an adverse signal of weakness to the market, what should it do?

A. Refuse to quote competitive bid prices on options, so that no cash is paid out

B. Continue to let counterparties trade at terms prevailing elsewhere in the market, even though this drains cash

C. Widen its bid-ask spreads sharply to discourage counterparties from trading with it

D. Decline novation and offsetting-trade requests from its smaller counterparties only

Question LTR-8 b.7 Bank Q299 • BT answers P2.T7.809.2

Duffie explains the similarities between a classic depositor run at a commercial bank and the failure mechanisms of dealer banks: the flight of short-term creditors, departures of prime brokerage clients, cash-draining actions by derivatives counterparties, and, most decisively, the loss of clearing-bank privileges. Which of the following statements about these mechanisms is TRUE?

A. The contractual "right of offset" guarantees that a dealer bank will retain its cash settlement privileges.

B. During the crisis, the exit of prime brokerage clients reduced the stress on dealers such as Morgan Stanley by shrinking their balance sheets and so their exposure.

C. A dealer bank that finances long-term assets with overnight repos can face funding liquidity problems if haircuts suddenly decrease.

D. In Lehman's case, OTC derivatives were exempt as "qualifying financial contracts" from the automatic stay at bankruptcy, which caused a large post-bankruptcy drain on the defaulting dealer.

Solutions • LTR-8 b

Solution LTR-8 b.1

Answer: D

Bank Q301 • Schweser Module Quiz 73.1 Q1

Step 1 – List the accelerators

Repo creditors not rolling, prime brokerage clients leaving, derivatives counterparties cutting exposure (offsetting trades, novations), and finally the clearing bank.

Step 2 – Test D

A client that leaves *more* unencumbered cash with its prime broker adds to the pool the broker can use; it does the opposite of a run.

$\Rightarrow$ D.

Interpretation

The bank's original Schweser key (deposit withdrawals) was corrected: dealers with commercial banks can also suffer deposit runs, so "depositors withdraw" is not a safe answer.

Why the other options are wrong

A is incorrect. The flight of short-term secured creditors is the first mechanism Duffie names.

B is incorrect. Departing clients take cash and securities the dealer was using as a source of liquidity.

C is incorrect. A novation request moves exposure, and possibly cash collateral, to another dealer; refusals at Bear Stearns spread alarm.

LO: LTR-8 b: Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks. • **Reference:** Duffie (GARP Ch.8), 8.2 Failure Mechanisms for Dealer Banks

Solution LTR-8 b.2

Answer: D

Bank Q297 • AnalystPrep Q.2984

Step 1 – Duffie's ordering

"The key mechanisms that lead to the failure of a dealer bank are the flight of short-term creditors, the departures of prime-brokerage clients, various cash-draining actions by derivatives counterparties ..., and finally and most decisively, the loss of clearing-bank privileges."

Step 2 – The Alpha Bank ending

Alpha's clearing bank "exercised its right to stop processing Alpha's cash and securities transactions ...Unable to execute trades or to send cash to meet its obligations, Alpha declares bankruptcy."

$\Rightarrow$ D.

Interpretation

Lehman is the real case: JPMorgan Chase invoked its right of offset and refused to process Lehman's wires.

Why the other options are wrong

A is incorrect. The repo run comes just before the end; it forces fire sales but the dealer can still settle.

B is incorrect. The first run in the Alpha story, not the last.

C is incorrect. Part of the derivatives run; it damages reputation and cash but is not the final step.

LO: LTR-8 b: Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks. • **Reference:** GARP Ch.8, introduction (Alpha Bank) and 8.2 Loss of Cash Settlement Privileges

Solution LTR-8 b.3 — Answer: B

Bank Q293 • GARP PE1 Q29

Step 1 – What a repo run takes away

If creditors stop rolling, the bank must replace the cash using assets that are not already pledged. The measure is the share of owned instruments already encumbered.

Step 2 – Pledged share

$$\text{P} = \frac{258}{656} = 39.3\%, \quad \text{Q} = \frac{472}{750} = 62.9\%, \quad \text{R} = \frac{139}{339} = 41.0\%, \quad \text{S} = \frac{209}{835} = 25.0\%$$

$\Rightarrow$ Bank Q relies most on repo and has the smallest unpledged cushion relative to its book (278 of 750).

Interpretation

Kept outside the CONCEPT tag because GARP set it. The size of the balance sheet is the red herring: S is the largest but the least encumbered.

Why the other options are wrong

A is incorrect. P is 39.3% pledged.
C is incorrect. R is 41.0% pledged; its small size is irrelevant.
D is incorrect. S has the most unpledged securities (626) and the lowest pledged share (25.0%): the least vulnerable.

LO: LTR-8 b: Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks. • **Reference:** GARP Ch.8, 8.2 The Flight of Short-Term Creditors

Solution LTR-8 b.4 — Answer: D

Bank Q294 • AP Mock 1 Q60

Step 1 – Duffie's three mitigants

"A dealer bank can mitigate the risk of a loss of liquidity from a run by short-term creditors in various ways: by establishing lines of bank credit; by dedicating a buffer stock of cash and liquid securities for emergency liquidity needs; and by 'laddering' the maturities of its liabilities."
$\Rightarrow$ All three: D.

Interpretation

Credit lines and the buffer supply cash when funding leaves; laddering limits how much must be replaced at once. Major dealers have teams that manage the maturity profile and pools of collateral acceptable to secured creditors.

Why the other options are wrong

A is incorrect. Omits the buffer and laddering.
B is incorrect. Omits credit lines.
C is incorrect. Omits laddering.

LO: LTR-8 b: Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks. • **Reference:** GARP Ch.8, 8.2 The Flight of Short-Term Creditors

Solution LTR-8 b.5 — Answer: D

Bank Q300 • Mock 1 Q15

Step 1 – What a worried counterparty wants

To reduce its exposure to the dealer, ideally by pulling cash out of it.

Step 2 – Test each action

Borrowing from the dealer creates an offsetting payable; novation moves the exposure to another dealer; new trades can make the dealer pay cash. Moving at-the-money options to in-the-money strikes *raises* what the dealer owes with no cash received.
⇒ D increases exposure, so it is least likely.

Interpretation

The real move runs the other way: revise in-the-money options to at-the-money strikes and take the difference in cash (Schweser), Duffie's "harvest cash" from positions that have moved in the counterparty's favour.

Why the other options are wrong

A is incorrect. A standard exposure-reducing step named by Duffie.
B is incorrect. Novation is the classic exit; refusals at Bear Stearns spread alarm.
C is incorrect. New trades that make the dealer pay cash reduce exposure.

LO: LTR-8 b: Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks. • **Reference:** GARP Ch.8, 8.2 When Derivatives Counterparties Duck for Cover

Solution LTR-8 b.6 Answer: B

Bank Q296 • AnalystPrep Q.2305

Step 1 – Duffie on the dealer's choice

"If the dealer wants to avoid an adverse signal of its weakness, the dealer cannot afford to refuse its counterparties the opportunity to make these trades at terms prevailing elsewhere in the market."
⇒ B: keep trading at market terms and accept the cash drain.

Interpretation

This is why the derivatives run is so damaging: the dealer must pay out cash to look healthy. Central clearing (LTR-8 c) removes much of the reason to run.

Why the other options are wrong

A is incorrect. Refusing competitive bids saves cash but signals liquidity trouble.
C is incorrect. Off-market spreads are a refusal by another name and are read the same way.
D is incorrect. Selective refusal still signals weakness, and news spreads to all counterparties.

LO: LTR-8 b: Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks. • **Reference:** GARP Ch.8, 8.2 When Derivatives Counterparties Duck for Cover

Solution LTR-8 b.7 Answer: D

Bank Q299 • BT answers P2.T7.809.2

Step 1 – Why D is correct

"Most over-the-counter derivatives contracts are exempted by law as 'qualifying financial contracts' from the automatic stay at bankruptcy …The effect of unwinding the dealer's derivatives portfolio is a large post-bankruptcy drain on the defaulting dealer, with priority to derivatives counterparties."

Interpretation

The exemption raises the incentive of other creditors to run before default. Close-out at replacement cost adds the bid-offer: 0.2% on Citibank's USD 30 trillion notional would be about USD 60 billion.

Why the other options are wrong

A is incorrect. Reversed: the right of offset is how the clearing bank *withdraws* settlement privileges.

B is incorrect. Reversed: client exits removed cash and re-pledgeable collateral (Morgan Stanley's fell from USD 832 billion to USD 294 billion).
C is incorrect. Polarity flip: problems arise when haircuts *increase*, as they did after Lehman.

LO: LTR-8 b: Identify situations that can cause a liquidity crisis at a dealer bank and explain responses that can mitigate these risks. • **Reference:** GARP Ch.8, 8.2 Failure Mechanisms for Dealer Banks

LTR-8 • The Failure Mechanics of Dealer Banks • Vardeez: IMPORTANT

LTR-8 c

P3 CONCEPT

Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.

Workbook: D27 Intraday Liquidity and Dealer Bank Failure Mechanics • Tier D • notes only

WHAT IS TESTED HERE

Why P3. Vardeez focus (IMPORTANT reading). The map reports no exam shape for this LO; D27 covers it in its revision notes only.

Reported exam shapes: none listed. The questions follow the LO wording: which measures are proposed, which run each measure targets, and Duffie's own assessments.

What the questions look like: "which is not a proposed measure", "which measure addresses this vulnerability", "with which statement would Duffie disagree", moral hazard from implicit support, and the design of PPIP. The bank had only five questions; Question LTR-8 c.2 is re-filed from LTR-8 b.

1. The problem the policies address

Standard tools for bank failures are supervision and risk-based capital, deposit insurance, and resolution mechanisms. Large dealers sat partly outside them, and their runs (LTR-8 b) work through repo, prime brokerage, derivatives and the clearing bank rather than deposits. Duffie calls **dealer-bank fire sales**, and their impact on prices and portfolios, "perhaps the most important source of systemic risk". Government support limits catastrophic failure but creates a **perverse incentive**: firms take inefficient risks and creditors lend to them cheaply because of the implicit backstop (moral hazard).

2. The measures

- **Crisis support (used).** Lender-of-last-resort financing (Fed facilities including the Primary Dealer Credit Facility; Bank of England facilities; ECB repo auctions) and capital injections (Bank of England; US Treasury TARP) significantly mitigated fire-sale risk, but they are costly to taxpayers and increase moral hazard.
- **PPIP (used; Schweser and Vardeez).** The 2009 Public-Private Investment Program under TARP subsidised *buyers* of "toxic" assets with below-market financing and absorbed losses beyond a set level, to counter **adverse selection** in those markets.
- **Regulation (envisioned).** Higher capital requirements, which would cover off-balance-sheet positions and charge more for uncleared derivatives; new supervisory councils; special powers to resolve failing firms (Schweser adds bridge banks for dealers that are too big to fail); risk retention by securitisation sponsors; and trade repositories. Duffie's caveat: because most repos and OTC derivatives are exempt from the automatic stay, these alone are unlikely to make the failure of repo-funded, derivatives-heavy dealers safe.
- **Tri-party repo.** A **tri-party repo utility** (Bernanke) with less discretion in rolling dealers' repos, high standards and fewer conflicting incentives; **central bank insurance** of tri-party repo (Abate); or an **emergency bank**, financed by repo market participants and with discount window access, to manage orderly unwinds of a weak dealer's repos and insulate the clearing banks from losses.
- **Central clearing of OTC derivatives.** Multilateral netting lowers total exposure to any one dealer, so counterparties have less reason to flee. Limits: standard contracts only (not AIG's customised

CDS); most OTC derivatives were not yet cleared; the CCP's own strength and implicit government backing are crucial.

- **Recapitalisation before a run. Distress-contingent convertible debt**, complemented by **mandatory rights offerings** triggered by leverage or liquidity thresholds (Duffie's proposal). Both recapitalise before a run starts and reduce incentives for excessive risk taking.

Correction to the source notes. (1) PPIP and bridge banks are not in the 2026 GARP text of Duffie's reading; they come from Schweser, which Vardeez follows. GARP mentions TARP only as capital injections. (2) Vardeez says naming the ECB as a provider of crisis secured lending is the "role-misassignment build". GARP says "other central banks have taken similar steps" and describes the ECB's repo auctions accepting a wide range of collateral, so the ECB is not wrong. (3) Vardeez counts central clearing and higher capital as proposals only; GARP notes that a large quantity of interest rate swaps was already cleared and that CDS central counterparties had been set up, though most OTC derivatives were not cleared.

Comparison card: LTR-8 c

Measure	Targets	Assessment
Lender of last resort, capital injections	Fire sales (systemic)	Worked in the crisis; costly; moral hazard
PPIP	Toxic asset markets	Subsidised buyers against adverse selection; did not force sales
Higher capital, resolution powers	Leverage, failure (firm)	Cannot alone resolve repo- and OTC-funded dealers (stay exemption)
Tri-party repo utility, CB insurance, emergency bank	Repo and clearing bank runs	Remove clearing-bank discretion; insulate clearing banks
Central clearing	Derivatives runs	Multilateral netting; standard contracts only; CCP strength crucial
Contingent convertibles, mandatory rights offerings	Recapitalisation (firm)	Act before a run; curb risk taking (Duffie favours them)

Exam traps

- **PPIP helped buyers**, with cheap financing and loss absorption; forcing sales was not an objective.
- **Central clearing** needs standard terms: it would not have covered AIG's customised CDS.
- **Repo utility** means *less* discretion for whoever rolls a weak dealer's repos, not more.
- **Contingent capital** is Duffie's proposal, designed to stabilise, not a destabilising instrument.
- **Implicit support** creates moral hazard: more inefficient risk taking and cheaper funding.
- **Capital proposals** add off-balance-sheet positions and uncleared derivatives; excluding them is the distractor.

Practice questions: LTR-8 c

Question LTR-8 c.1 Bank Q303 • AnalystPrep Q.2983 (rebuilt)

Which of the following is NOT among the policy measures discussed for alleviating the firm-specific and systemic risks of large dealer banks?

A. Creating a tri-party repo utility with less discretion in rolling over a dealer's repo positions

B. Central clearing of standardised OTC derivatives

C. Higher capital requirements that include off-balance-sheet positions

D. Excluding off-balance-sheet positions from capital requirements to preserve the dealer's cash-management flexibility

Question LTR-8 c.2 Bank Q302 • Schweser Module Quiz 73.2 Q1 (re-filed from LTR-8 b)

One potential solution for mitigating the liquidity risk caused by derivatives counterparties exiting their exposures to a large dealer bank is most likely the:

A. use of central clearing.

B. use of a novation through another dealer bank.

C. requirement that dealer banks pay out cash to reduce counterparty exposure.

D. creation of new offsetting contracts by counterparties.

Question LTR-8 c.3 Bank Q305 • BT answers P2.T8.24.3.3 (rebuilt)

In a crisis of confidence in a dealer bank's solvency, tri-party clearing banks have an incentive to limit their exposure by restricting the dealer's repo financing and clearing functions. Which proposed policy measure most directly addresses this vulnerability?

A. A tri-party repo utility with less discretion in rolling over the dealer's repo positions

B. Central clearing of standardised OTC derivatives

C. Distress-contingent convertible debt

D. Giving clearing banks full discretion to cut repo financing to weakened dealers

Question LTR-8 c.4 Bank Q304 • BT answers P2.T7.809.3

Duffie examines several policy measures that might alleviate firm-specific and systemic risks related to large dealer banks. With which of the following statements would he most DISAGREE?

A. The threat posed by the flight of over-the-counter derivatives counterparties can be lowered by central clearing.

B. Distress-contingent convertible debt is likely to be destabilising during financial distress, and regulators should consider curtailing its use.

C. Short-term tri-party repos are a particularly unstable source of financing; remedies include a tri-party repo utility, central bank insurance of tri-party repos, or an emergency bank financed by repo market participants.

D. Perhaps the most important source of systemic risk is the impact of dealer-bank fire sales on market prices and investor portfolios; in the crisis this risk was significantly mitigated by lender-of-last-resort financing and capital injections.

Question LTR-8 c.5 Bank Q307 • AnalystPrep Q.2292

Coliseu Bank is a large dealer bank considered "too big to fail" that can expect support from the government and the central bank in distress. This implicit support is associated with behaviour that may have disastrous consequences. Which of the following is an example?

A. Manipulation of financial statements

B. Taking more inefficient risks

C. Offering products at exorbitant prices

D. Doing business only with high-net-worth firms and individuals

Question LTR-8 c.6 Bank Q306 • Schweser Module Quiz 73.2 Q2 (rebuilt)

The 2009 Public-Private Investment Program (PPIP), set up under the US Treasury's Troubled Asset Relief Program (TARP), was designed to help dealer banks and the wider industry recover from the crisis. Which of the following was NOT one of its features or objectives?

A. Providing below-market financing rates for bidders of "toxic" assets

B. Absorbing losses beyond a pre-specified level

C. Forcing the sale of illiquid assets in order to establish their "true" value

D. Mitigating the effect of adverse selection in markets for toxic assets

Solutions • LTR-8 c

Solution LTR-8 c.1 — Answer: D

Bank Q303 • AnalystPrep Q.2983

Step 1 – Check each against the proposals

A tri-party repo utility (Bernanke), central clearing of standard OTC derivatives, and higher capital that captures off-balance-sheet positions are all proposed.

Step 2 – The odd one out

Excluding off-balance-sheet positions is the pre-crisis practice that let dealers run larger businesses on less capital.

$\Rightarrow$ D.

Interpretation

Schweser adds that separating tri-party repo clearing from other clearing functions would cut leverage by tightening the dealer's cash-management flexibility. The proposals reduce that flexibility; D preserves it.

Why the other options are wrong

A is incorrect. A proposed repo fix.

B is incorrect. The proposed fix for derivatives runs.

C is incorrect. A proposed capital measure.

LO: LTR-8 c: Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.

• **Reference:** Duffie (GARP Ch.8), 8.3 Policy Responses; SchweserNotes 2026 Book 4, Reading 73, LO 73.c

Solution LTR-8 c.2 — Answer: A

Bank Q302 • Schweser Module Quiz 73.2 Q1

Step 1 – How counterparties exit

They novate to another dealer, enter offsetting trades that make the dealer pay cash, or harvest in-the-money positions; each drains the dealer.

Step 2 – The fix

"The threat posed by the flight of over-the-counter derivatives counterparties can be lowered by central clearing": the CCP stands between the parties, and multilateral netting lowers total exposure to any one dealer.

$\Rightarrow$ A.

Interpretation

It works only for derivatives with relatively standard terms, so it would not have covered AIG's customised credit derivatives, and the CCP must itself be strong.

Why the other options are wrong

B is incorrect. Novation is one of the exit channels, not a fix.

C is incorrect. Paying out cash is the drain itself.

D is incorrect. Offsetting contracts that make the dealer pay cash are another exit channel.

LO: LTR-8 c: Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.

• **Reference:** GARP Ch.8, 8.2 When Derivatives Counterparties Duck for Cover; 8.3 Policy Responses

Solution LTR-8 c.3 Answer: A

Bank Q305 • BT answers P2.T8.24.3.3

Step 1 – Name the vulnerability

"Because tri-party clearing banks have an incentive to limit their exposures to a dealer bank, they are likely to limit the access of a weakened dealer bank to repo financing and to clearing account functions."

Step 2 – The matching proposal

A tri-party repo utility "would have less discretion in rolling over a dealer's repo positions, meet high standards, and suffer from fewer conflicting incentives".
⇒ A.

Interpretation

The emergency bank and central bank insurance of tri-party repo target the same vulnerability.

Why the other options are wrong

B is incorrect. Targets the derivatives run, a different channel.
C is incorrect. Recapitalises the dealer but does not change clearing-bank incentives.
D is incorrect. Increases the discretion that causes the problem.

LO: LTR-8 c: Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.
• **Reference:** GARP Ch.8, 8.3 Policy Responses

Solution LTR-8 c.4 Answer: B

Bank Q304 • BT answers P2.T7.809.3

Step 1 – Duffie's own proposal

"Duffie (2009) proposes that distress-contingent convertible debt be complemented with regulations favoring mandatory rights offerings of equity …These two new instruments can be designed to recapitalize a financial institution before a destructive run is likely to commence, and to reduce a financial institution's incentives for socially excessive risk taking."
⇒ He favours contingent convertibles, so he would disagree with B.

Interpretation

A, C and D paraphrase his text, including the trade-off in D: support worked but is costly and raises moral hazard.

Why the other options are wrong

A is incorrect. Duffie's own sentence on central clearing.
C is incorrect. His list of tri-party repo remedies.
D is incorrect. His view of fire sales and of crisis support.

LO: LTR-8 c: Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.
• **Reference:** GARP Ch.8, 8.3 Policy Responses

Solution LTR-8 c.5 Answer: B

Bank Q307 • AnalystPrep Q.2292

Step 1 – Duffie on implicit support

"The common knowledge that large financial institutions will receive support when they are sufficiently distressed …provides an incentive to large financial institutions to take inefficient risks, and for their creditors to cooperate by financing them at a lower cost."
⇒ B: moral hazard.

Interpretation

This is why Duffie argues for alternatives to last-resort support: central clearing, repo utilities and automatic recapitalisation.

Why the other options are wrong

A is incorrect. Fraud is not the incentive effect of a guarantee.
C is incorrect. Pricing power is unrelated to the backstop.
D is incorrect. Client selection is unrelated to moral hazard.

LO: LTR-8 c: Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.
• **Reference:** GARP Ch.8, introduction

Solution LTR-8 c.6 Answer: C

Bank Q306 • Schweser Module Quiz 73.2 Q2

Step 1 – What PPIP did

Schweser: PPIP aimed to "mitigate the effect of adverse selection in the market for 'toxic' assets" by subsidising bidders with below-market financing and absorbing losses beyond a predetermined level.

Step 2 – The odd one out

Forcing sales into thin markets would deepen the discounts and losses the program was meant to avoid.
⇒ C.

Interpretation

Adverse selection: buyers who cannot value the assets bid only at deep discounts, so a dealer forced to sell takes extra losses. PPIP comes from Schweser; the 2026 GARP text mentions only TARP capital injections (see the correction box).

Why the other options are wrong

A is incorrect. A feature: cheap financing for buyers.
B is incorrect. A feature: loss absorption beyond a set level.
D is incorrect. The stated objective.

LO: LTR-8 c: Assess policy measures that can alleviate firm-specific and systemic risks related to large dealer banks.
• **Reference:** SchweserNotes 2026 Book 4, Reading 73, LO 73.c

LTR-9 • Liquidity Stress Testing • Vardeez: MODERATE

LTR-9 b

P3 CALC

Estimate contingent liquidity via the liquid asset buffer.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P3. No workbook question tests it (the A26 revision notes give the formula in one bullet). Vardeez flags it: "Know the calculation of contingent liquidity asset buffer. Calculation is important"; the rest of the reading can be read from the crash course notes.
Vardeez: contingent liquidity via the liquid asset buffer; the calculation is important.

Reported exam shapes: none listed. The questions are built from the LO wording and the Vardeez focus.

What the questions look like: a normal liquid asset buffer, a list of stressed outflows and inflows, and red herrings (operational cash, pledged securities, a contractual-versus-contingent split already inside a total). You compute the stressed buffer. Variations add haircuts on buffer securities, a realisation rate on inflows, or the Basel III cap of inflows at 75% of outflows. One question checks what contingent liquidity is.

1. Contingent liquidity and the liquid asset buffer

Baird (GARP Chapter 9) defines liquidity for stress testing as **funding liquidity**, used for four purposes. Only one of them can meet general obligations under stress:

- **Contingent liquidity**: available to meet general financial obligations under a stress scenario, held as the **liquid asset buffer** (liquid investments, liquidity facility availability and unrestricted deposits). Measuring the contingent liquidity required to cover stressed outflows is the principal objective of the liquidity stress test.
- **Operational** liquidity (cash for daily business and payment clearing) is unavailable under stress: *excluded*.
- **Restricted** liquidity (for example collateral for wholesale borrowings) is applied only to the outflows it supports, never to general obligations: *excluded from the buffer*.
- **Strategic** liquidity (for acquisitions or capital expenditure) may be redirected to contingent needs, in practice only the part held in highly liquid assets.

What qualifies for the buffer. Market characteristics: low credit and market risk, ease and certainty of valuation, an active and sizable market, and low concentration of buyers and sellers, so that the securities keep their liquidity-generating capacity in severe idiosyncratic and market stress. Operational requirement: the liquidity must be under the control of the central treasury of the entity being tested. Securities enter at their value after stress haircuts, which widen in systemic stress (under the LCR, for example, Level 2A agency MBS take a 15% haircut and Treasuries none).

2. The stress test model

With LAB for the liquid asset buffer (normal, before the scenario):

$$\text{Stressed LAB} = \text{LAB} - \text{stressed cash outflows} + \text{stressed cash inflows}$$

- **Stressed outflows:** a mix of contractual and contingent outflows, from settling non-contractual obligations early and failing to refund contractual ones that would normally roll over. Categories: retail deposit outflows, unsecured wholesale funding outflows, secured funding runoff, derivative transaction funding (collateral and margin calls), loss of funding on asset-backed issuances, and drawdowns of credit and liquidity facilities.
- **Stressed inflows:** assumed to *partially* offset the outflows: secured funding maturities, loan repayments from customers, and drawdowns on liquidity facilities available to the bank. They may be reduced by market conditions or capped by regulation (not to exceed 75% of outflows under the Basel III LCR).
- **Stressed buffer:** the buffer net of both legs, indicating whether the current buffer is adequate under the scenario. A shortfall may point to contractual maturity mismatch, insufficient asset quality, or overconcentration in certain funding sources.

Both legs must be *stressed* figures. A facility available to the bank is counted once: either in the buffer (facility availability) or as an inflow (a drawdown), not both.

Worked example. A bank holds unencumbered Treasuries of 300 (no haircut), agency MBS of 200 (15% haircut) and unrestricted central bank deposits of 40. It also holds 120 of Treasuries pledged against a wholesale loan and 25 of operational cash. Stressed outflows: retail deposits 150, unsecured wholesale 110, secured funding runoff 30, derivative collateral calls 20, facility drawdowns by clients 50. Stressed inflows: loan repayments 60, maturing reverse repos 40.

$$\begin{aligned} \text{LAB} &= 300 + 200(1 - 0.15) + 40 = 510 && \text{(pledged 120 and operational 25 excluded)} \\ \text{Stressed outflows} &= 150 + 110 + 30 + 20 + 50 = 360 \\ \text{Stressed inflows} &= 60 + 40 = 100 && \text{(cap } 0.75 \times 360 = 270\text{, not binding)} \\ \text{Stressed LAB} &= 510 - 360 + 100 = 250 \end{aligned}$$

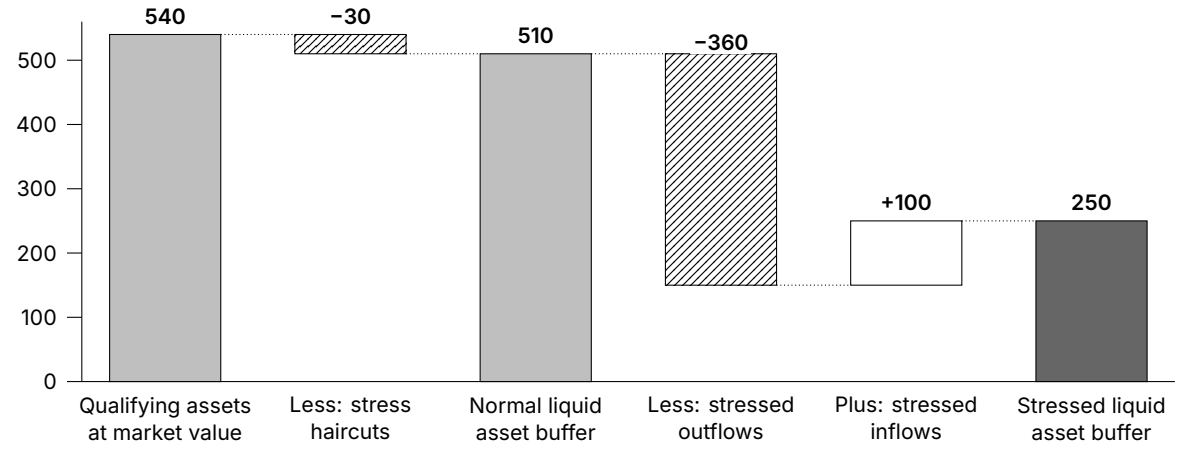


Figure 1. The worked example as a waterfall, after GARP Figure 9.2. Stock bars are grey (the stressed buffer dark grey), deductions hatched, the inflow white; dotted lines carry each level across. Pledged securities and operational cash never enter the first bar.

Formula card: LTR-9 b

Stressed buffer	Liquid asset buffer − stressed outflows + stressed inflows
Liquid asset buffer	$\sum$ market value $\times(1-h)$ of unencumbered qualifying securities + unrestricted deposits + facility availability
Inflows used	Stressed inflows $\times$ realisation rate, then $\min(\text{inflows},\ 0.75 \times \text{outflows})$ if the LCR cap applies
Excluded	Operational cash; restricted (pledged) assets; strategic liquidity not held in highly liquid assets
Reading the result	Positive: buffer adequate under the scenario. Negative: shortfall (maturity mismatch, asset quality, funding concentration)

Exam traps

- **Sign.** Outflows are subtracted, inflows added; derivative margin and collateral calls are outflows.
- **Mixed basis.** Both legs are stressed; normal inflows against stressed outflows overstate the buffer.
- **Operational and restricted liquidity.** Neither is in the buffer; adding operational cash is the commonest red herring.
- **Haircuts.** Buffer securities enter after the stress haircut.
- **The cap is on outflows.** Inflows are limited to 75% of *outflows*, not cut to 75% of themselves; and a realisation rate applies to inflows, not to outflows.
- **Double counting.** A facility counts once, in the buffer or as an inflow.

Practice questions: LTR-9 b

Question LTR-9 b.1 Bank Q309 • BT Mock Q49 (filed under LTR-9 a, P4)

Which of the following types of liquidity is available in the form of the institution's liquid asset buffer (LAB) and represents liquidity available to meet general financial obligations under a stress scenario?

A. Operational

B. Strategic

C. Contingent

D. Restricted

Question LTR-9 b.2

Bank Q313 • BT P2.T8.20.10.2

Acme Bank has a normal liquid asset buffer (LAB) of $60.0 million. To estimate its contingent liquidity, it runs stress scenarios. Stressed outflows (a mix of 60% contractual and 40% contingent outflows, covering retail deposit outflows, unsecured wholesale funding outflows, secured funding runoff and drawdowns of credit and liquidity facilities) produce a net estimate of $28.0 million. Stressed inflows (secured funding maturities, customer loan repayments and drawdowns on liquidity facilities available to Acme) produce a net estimate of $15.0 million. Acme also holds $16.0 million of operational liquidity. What is the bank's estimate of contingent liquidity (the stressed liquid asset buffer)?

A. $32.0 million

B. $47.0 million

C. $63.0 million

D. $103.0 million

Question LTR-9 b.3

Bank Q314 • Schweser Module Quiz 74.1 Q2 (rebuilt)

Firm A has a normal liquid asset buffer of $1,000 million in highly liquid assets. Separately, it holds $120 million of operational liquidity that is not part of the buffer. In a sudden stress scenario, retail customers withdraw $150 million of deposits and make $80 million of loan repayments. The firm must also meet $60 million of margin and collateral calls on its derivatives because of falling collateral values and higher volatility. What is the estimate of Firm A's stressed liquid asset buffer?

A. $0.71 billion

B. $0.87 billion

C. $0.93 billion

D. $0.99 billion

Question LTR-9 b.4

New • built from GARP Ch.9 (9.1, 9.2, 9.10)

A treasury analyst at a mid-sized US bank is building the normal liquid asset buffer for the liquidity stress test (USD million). The holdings are:

Holding	Market value	Stress haircut
Unencumbered US Treasuries	420	0%
Unencumbered agency MBS	200	15%
US Treasuries pledged as collateral for a wholesale borrowing	150	0%
Cash held to fund daily payment clearing	60	
Unrestricted deposits at the central bank	80	

Which of the following is closest to the normal liquid asset buffer?

A. USD 670 million

B. USD 700 million

C. USD 730 million

D. USD 820 million

Question LTR-9 b.5 Bank Q315 • new question written for the bank (GARP practice-exam style)

A treasury analyst at a regional bank is estimating the bank's contingent liquidity using the liquid asset buffer. The bank's normal liquid asset buffer is $240 million. Under the stress scenario, modeled outflows are retail deposit withdrawals of $90 million, unsecured wholesale funding losses of $70 million, secured funding runoff of $35 million, and drawdowns of credit and liquidity facilities of $25 million. Contractual inflows of $30 million of customer loan repayments and $20 million of secured funding maturities are expected, but the scenario assumes market conditions limit realized inflows to 60% of these amounts. What is the bank's stressed liquid asset buffer?

A. $20 million

B. $50 million

C. $70 million

D. $158 million

Question LTR-9 b.6 New • built from GARP Ch.9 (Figure 9.2)

The head of liquidity risk at a European bank runs a 30-day stress scenario (EUR million). The normal liquid asset buffer is 520. Stressed outflows are retail deposits 220, unsecured wholesale funding 180, derivative collateral calls 60 and drawdowns of committed facilities by clients 40. Stressed inflows are customer loan repayments 260 and maturing reverse repos 160. Following the Basel III LCR, stressed inflows may not exceed 75% of stressed outflows. Which of the following is closest to the stressed liquid asset buffer?

A. EUR 20 million

B. EUR 335 million

C. EUR 395 million

D. EUR 440 million

Solutions • LTR-9 b

Solution LTR-9 b.1

Answer: C

Bank Q309 • BT Mock Q49

Step 1 – Match the definition

Contingent liquidity is "the liquidity that is available to meet general financial obligations under a stress scenario. This liquidity is available in the form of the institution's liquid asset buffer."

Interpretation

Estimating contingent liquidity therefore means estimating the buffer, and the stressed buffer shows whether it is enough.

Why the other options are wrong

A is incorrect. Operational liquidity funds daily business and payment clearing; it is unavailable under stress.

B is incorrect. Strategic liquidity is held for future business needs; only its highly liquid part can be redirected.

D is incorrect. Restricted liquidity is reserved for defined purposes such as collateral and applied only to the outflows it supports.

LO: LTR-9 b: Estimate contingent liquidity via the liquid asset buffer. • **Reference:** Venkat and Baird, Liquidity Risk Management, Ch.3 (GARP Ch.9), 9.1 Measuring Contingent Liquidity Requirements, Figure 9.1

Solution LTR-9 b.2

Answer: B

Bank Q313 • BT P2.T8.20.10.2

Step 1 – Stressed buffer

$$\text{Stressed LAB} = 60.0 - 28.0 + 15.0 = \$47.0 \text{ million}$$

$\Rightarrow$ After the scenario's net outflows the bank still holds \$47 million of contingent liquidity.

Interpretation

Two red herrings: operational liquidity (\$16 million) funds day-to-day operations and cannot meet general obligations, so it is neither added nor subtracted; the 60/40 contractual/contingent mix is already inside the \$28.0 million.

Why the other options are wrong

A is incorrect. Ignored the stressed inflows: $60.0 - 28.0 = 32.0$.

C is incorrect. Added operational liquidity: $47.0 + 16.0 = 63.0$.

D is incorrect. Added the outflows instead of subtracting them: $60.0 + 28.0 + 15.0 = 103.0$.

LO: LTR-9 b: Estimate contingent liquidity via the liquid asset buffer. • **Reference:** Venkat and Baird, Ch.3 (GARP Ch.9), 9.2 Overview of the Model, Figure 9.2

Solution LTR-9 b.3

Answer: B

Bank Q314 • Schweser Module Quiz 74.1 Q2

Step 1 – Stressed outflows

Deposit withdrawals and derivative margin and collateral calls: $150 + 60 = \$210$ million.

Step 2 – Stressed inflows

Loan repayments: \$80 million.

Step 3 – Stressed buffer

$$1{,}000 - 210 + 80 = \$870 \text{ million} = \$0.87 \text{ billion}$$

Interpretation

Operational liquidity ($120 million) sits outside the buffer and cannot meet general obligations. Schweser's original version also has the firm with $10 million undrawn on its own $100 million liquidity facility; GARP lists drawdowns on facilities available to the bank as inflows, so that version adds $10 million and gets $0.88 billion.

Why the other options are wrong

A is incorrect. Subtracted the loan repayments as if they were outflows: $1{,}000 - 150 - 60 - 80 = 710$.
C is incorrect. Omitted the margin calls: $1{,}000 - 150 + 80 = 930$.
D is incorrect. Added operational liquidity: $870 + 120 = 990$.

LO: LTR-9 b: Estimate contingent liquidity via the liquid asset buffer. • **Reference:** Venkat and Baird, Ch.3 (GARP Ch.9), 9.2, Figure 9.2

Solution LTR-9 b.4

Answer: A

New • built from GARP Ch.9

Step 1 – Keep only what qualifies

Unencumbered Treasuries, unencumbered agency MBS and unrestricted central bank deposits qualify. The pledged Treasuries are restricted liquidity; the payment-clearing cash is operational liquidity.

Step 2 – Apply the stress haircuts

$$420 + 200 \times (1 - 0.15) + 80 = 420 + 170 + 80 = \text{USD } 670 \text{ million}$$

$\Rightarrow$ This is the contingent liquidity in place before any stressed flows.

Interpretation

The pledged securities are applied to the outflow they support (the secured borrowing), not to general obligations. Haircuts would widen further in a systemic scenario.

Why the other options are wrong

B is incorrect. Ignored the MBS haircut: $420 + 200 + 80 = 700$.
C is incorrect. Added the operational cash: $670 + 60 = 730$.
D is incorrect. Counted the pledged Treasuries: $670 + 150 = 820$.

LO: LTR-9 b: Estimate contingent liquidity via the liquid asset buffer. • **Reference:** Venkat and Baird, Ch.3 (GARP Ch.9), 9.1, Figure 9.1; 9.2; 9.10 Liquidity Optimization

Solution LTR-9 b.5

Answer: B

Bank Q315 • new (bank)

Step 1 – Stressed outflows

$$90 + 70 + 35 + 25 = \$220 \text{ million}$$

Step 2 – Stressed inflows after the market limit

$$0.60 \times (30 + 20) = \$30 \text{ million}$$

Step 3 – Stressed buffer

$$240 - 220 + 30 = \$50 \text{ million}$$

Interpretation

GARP: "the level of inflows may be reduced or limited by market conditions". The realisation rate applies to inflows only.

Why the other options are wrong

A is incorrect. Added nothing back for inflows: $240 - 220 = 20$.
C is incorrect. Used the full contractual inflows: $240 - 220 + 50 = 70$.
D is incorrect. Applied the 60% to the outflows instead: $240 + 50 - 0.6 \times 220 = 158$.

LO: LTR-9 b: Estimate contingent liquidity via the liquid asset buffer. • **Reference:** Venkat and Baird, Ch.3 (GARP Ch.9), 9.2, Figure 9.2

Solution LTR-9 b.6 **Answer: C**

New • built from GARP Ch.9

Step 1 – Stressed outflows

$$220 + 180 + 60 + 40 = \text{EUR } 500 \text{ million}$$

Step 2 – Cap the inflows

$$\text{Inflows} = 260 + 160 = 420; \qquad \text{cap} = 0.75 \times 500 = 375; \qquad \min(420, 375) = 375$$

Step 3 – Stressed buffer

$$520 - 500 + 375 = \text{EUR } 395 \text{ million}$$

Interpretation

The cap binds: 45 of expected inflows cannot be counted. The rule stops a bank from relying on inflows to offset all of its outflows.

Why the other options are wrong

A is incorrect. Ignored the inflows: $520 - 500 = 20$.
B is incorrect. Cut the inflows to 75% of themselves: $520 - 500 + 0.75 \times 420 = 335$.
D is incorrect. Ignored the cap: $520 - 500 + 420 = 440$.

LO: LTR-9 b: Estimate contingent liquidity via the liquid asset buffer. • **Reference:** Venkat and Baird, Ch.3 (GARP Ch.9), 9.2, Figure 9.2 (cap of 75% of outflows under the Basel III LCR)

LTR-10 • Liquidity Risk Reporting and Stress Testing • Vardeez: IMPORTANT

LTR-10 b

P3 CONCEPT

Compare and interpret different types of liquidity risk reports.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P3. No workbook question tests it (the A26 revision notes list the reports in one bullet). Vardeez flags it in an IMPORTANT reading: "Just know the liquidity risk reporting, what are the different reports specifically for liquidity? ... Just know the basics, the interpretation is all that is needed."

Vardeez: know the different liquidity reports; basics and interpretation only.

Reported exam shapes: none listed. The questions are built from the LO wording and the Vardeez focus.

What the questions look like: which report shows what (the deposit tracker against the concentration report, the mismatch report against the daily liquidity report), what a reading means (an LTD above its ceiling, a liquidity gap turning negative, a survival horizon under 30 days, a funding curve above peers) and what management should do. No calculations: where a report extract appears, you read it rather than compute it.

1. The reports at a glance

A bank produces liquidity reports daily, weekly, monthly and quarterly; the format of liquidity management information (MI) must be transparent and accessible. The main reports are required by the regulator, who sets their frequency, and ALCO reviews them (LTR-10 a). Choudhry (GARP Chapter 10) gives a benchmark set:

Report	What it shows	Frequency, user	Adverse reading
Deposit tracker	Month-end deposits by customer type, change from each month-end, total customer loans and so the loan-to-deposit (LTD) ratio, forecast to year end; a second part shows the liability increase or asset reduction needed to reach a target LTD	Weekly and monthly; the LTD is a key management ratio	LTD above the board ceiling (GARP: 86.08% against 85%, with a year-end forecast of 80.26%)
Daily liquidity report	Liquid and marketable securities; assets and liabilities by tenor up to 1 year and beyond; cumulative liquidity report; 1-week and 1-month ratios; liquidity risk factor	End of day, for Treasury and Finance; one per branch and subsidiary	Liquidity gap negative in a bucket; ratios through their limits
Funding maturity gap (mismatch)	Maturity gap per time bucket for all assets and liabilities, with an adjustment for liquid securities; includes the cumulative liquidity cash flow	Can be combined with the daily report; the same report generates the cash flow survival horizon report	Cumulative mismatch negative in early buckets; survival horizon below 30 days
Funding concentration (large depositors)	Reliance on each source or sector of funds, including intra-group; depositors above USD 50 million, or defined by share of total liabilities	Key MI for senior Treasury and relationship managers	One depositor above the single-source limit (GARP: 16.5% against 10%)

Report	What it shows	Frequency, user	Adverse reading
Undrawn commitments	Aggregate undrawn committed facilities (liquidity lines, revolving credit, letters of credit, guarantees) and the trend of drawn and undrawn	Aggregate level, with detailed breakdowns per customer	Rising undrawn amounts: they are drawn in stress, exactly when funding is short
Liability profile	Share of each liability type: individuals, large enterprises, repo of highly liquid, high-quality or other assets, ABS, unsecured wholesale, conditional liabilities	Format set by the user	Rising share of short-term, wholesale or secured funding
Wholesale pricing and volume	Firm-specific funding yield curve; wholesale funding by product and tenor	Compare with peers at least quarterly; the regulator compiles it too	Own curve rising materially above the peer group: early warning of funding stress
Summary and qualitative	1-page monthly snapshot of key metrics; monthly qualitative highlights from branches and subsidiaries	Monthly or as ALCO directs, more often in stress	Unexplained moves in ratios, gaps, the LRF, inter-group positions or deposits

2. Reading the daily liquidity report

The daily report lists the liquid securities (government bonds, central-bank-eligible securities, bank CDs), the assets and liabilities by tenor, and a **cumulative liquidity report**:

- **Cumulative cash gap**: cumulative inflows less outflows by bucket.
- **Counterbalancing capacity**: the securities available to cover sudden outflows (sale or repo).
- **Liquidity gap** = cumulative cash gap + counterbalancing capacity, compared with a limit.

In GARP's example (Figure 1) the cash gap is negative from day 1; liquid securities turn the liquidity gap positive from 1 week to 1 month, but from 2 months it is negative again and widens to about EUR 1 billion. The report also gives a 1-week ratio (11.13% against a 0% limit) and a 1-month ratio (2.03% against −5%), and a **liquidity risk factor** of 3.85: the average remaining term of assets, adjusted for marketable securities (34.88 months), over the combined average remaining term of liabilities (9.06 months). The higher the LRF, the wider the term mismatch.

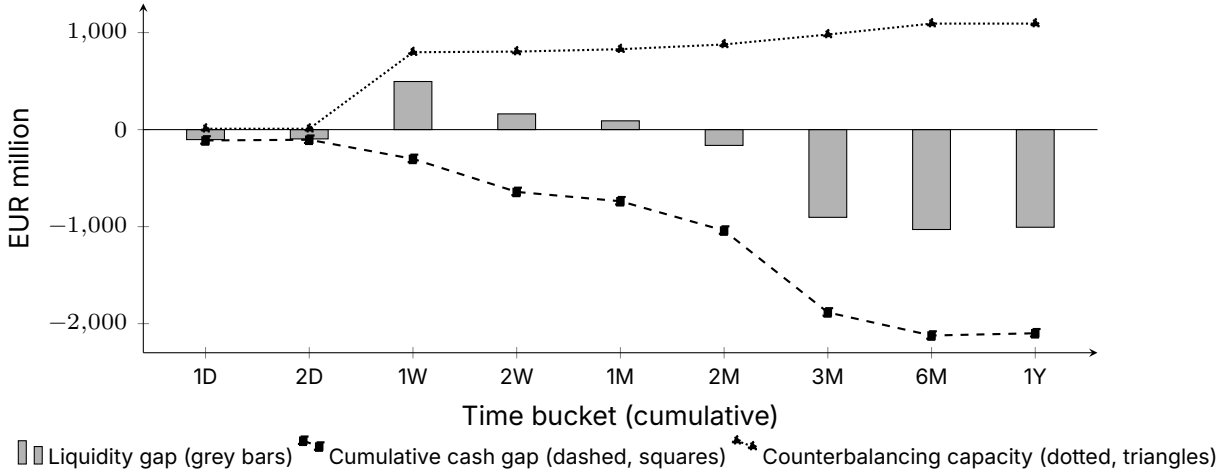


Figure 1. Cumulative liquidity report from GARP's daily liquidity report example (Table 10.2B). Each grey bar is the dashed line plus the dotted line: liquid securities cover the cash gap from 1 week to 1 month only.

3. The mismatch report and the survival horizon

The funding gap report shows the mismatch by bucket for all assets and liabilities, adjusted for liquid securities, and the same report generates the **cash flow survival horizon** report: the day on which the cumulative cash flow turns negative, first on contractual flows, then after adding the cash value of liquid securities and other adjustments. In GARP's example the horizon is only 7 days unadjusted and 27 days adjusted: still below the Basel III requirement (a 30-day stress period), so on the strength of the report the bank must act to address the shortage. Read the horizon against the regulatory minimum, not against the unadjusted figure.

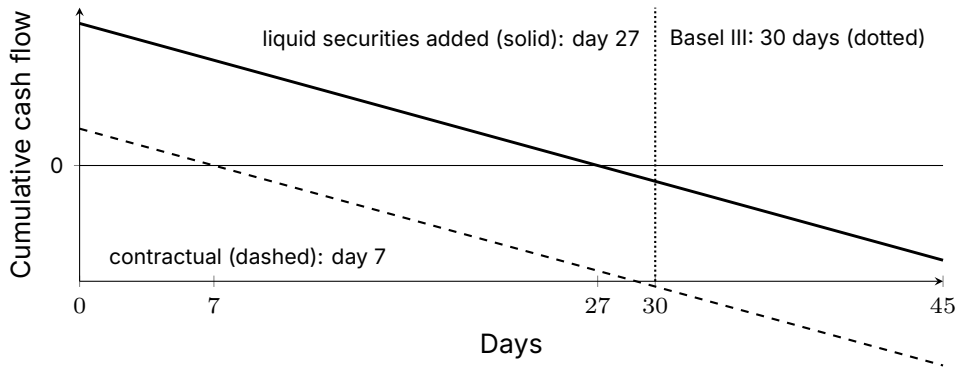


Figure 2. Cash flow survival horizon, schematic, with the horizons of GARP's example. The adjusted line (solid) is the contractual line (dashed) shifted up by the cash value of liquid securities; both cross zero before the 30-day line.

Correction to the source notes (Vardeez, LTR-10 c). Vardeez writes that "mitigating actions extend the survival period from 49 days to 27 days". GARP has two separate examples: the funding gap report's survival horizon of 7 days, extended to 27 days by liquid securities (still below Basel III, this LO); and the stress test report's 49 days, extended to 93 days (LTR-10 c). An extension cannot shorten the horizon.

4. What management does with the readings

- **LTD above the ceiling:** the tracker's second part shows the extra liabilities or the asset reduction needed (GARP, at the end of May: 31,564 more deposits or 26,829 fewer loans, in the report's units of thousands, to bring the LTD from 86.08% to 85%). Current accounts and rolling deposits count as short-term (1-day) money for regulators unless they behave as long-term funds; moving customers into fixed-term or notice accounts improves the metrics.
- **Large depositor above the limit:** ALCO should generally treat a deposit of 5% of total liabilities as large. If one depositor exceeds the single-source limit and the relationship matters, the bank grows its liability base to bring the share back within limit, rather than ask the client to remove funds.
- **Undrawn commitments:** reported separately and included in liquidity metrics, because they are drawn down in stress as customers face funding difficulties of their own.
- **Funding curve above peers:** the market charges the bank more than its peers; regulators read it as an early warning of funding stress.
- **Summary report:** kept to one side of A4 so that senior management actually reads it.

Exam traps

- **Sibling swap.** The deposit tracker serves the LTD ratio; the daily report the end-of-day position; the mismatch report the bucket gap and the survival horizon; the concentration report the reliance on single sources.
- **Polarity of the LTD limit.** A high LTD is the risk, so the board limit is a ceiling, not a floor.
- **Liquidity gap.** Cumulative cash gap *plus* counterbalancing capacity, not minus.
- **Survival horizon.** Judged against the 30-day requirement; the adjusted horizon is the relevant one, and an improvement on the unadjusted figure is not a pass.
- **"Large" depositor.** 5% of total liabilities for ALCO; USD 50 million is the report's illustration.
- **Demand deposits** are 1-day money for regulators, not one-year money.

Practice questions: LTR-10 b

Question LTR-10 b.1

Bank Q335 • AnalystPrep Q.4056

The liability profile is a simple breakdown of the share of each type of liability at the bank. Which of the following is not a part of the liability profile?

A. Customers: individuals

B. Repo: highly liquid securities

C. Asset-backed securities

D. Highly liquid assets

Question LTR-10 b.2

Bank Q349 • BT Mock Q54 • BT P2.T8.20.11.1

Umbrella Street Bank, a commercial bank, produces several liquidity risk reports on a daily, weekly, monthly and quarterly basis, including a deposit tracker report. Which of the following metrics is MOST LIKELY to appear in the deposit tracker report?

A. Loan-to-deposit (LTD) ratio (current and forecast) versus a board-approved lower floor (limit) of 120.0%

B. Loan-to-deposit (LTD) ratio (current and forecast) versus a board-approved upper ceiling (limit) of 80.0%

C. Market-to-book ratio of common equity (current and forecast) versus an investor-communicated target of 1.30

D. The leverage-adjusted duration gap (current and forecast) versus a board-approved upper ceiling of 3.5 years

Question LTR-10 b.3 Bank Q337 • AnalystPrep Q.4050 (rebuilt)

Which of the following details would not be obtained from a deposit tracker report?

A. Month-end actuals for deposits by customer type

B. Each month-end change

C. Each month-end forecast for the position to the end of the year

D. Each large depositor's share of total liabilities, measured against a single-source concentration limit

Question LTR-10 b.4 Bank Q339 • AnalystPrep Q.4053 (rebuilt)

The treasury of MCT Bank is preparing its funding maturity gap ("mismatch") report. Which of the following statements about this report is accurate?

A. It shows the maturity gap for all assets and liabilities per time bucket, with an adjustment for liquid securities.

B. It is unrelated to the cash flow survival horizon report, which is built from separate data.

C. Its main content is the list of the bank's marketable securities by central bank eligibility and time to liquidity.

D. It reports the end-of-day liquidity position for Treasury and Finance, completed by each branch and subsidiary.

Question LTR-10 b.5 Bank Q346 • AnalystPrep Q.4063 (rebuilt as interpretation; data from GARP Table 10.2B)

A treasury analyst reviews the cumulative liquidity section of the bank's daily liquidity report (EUR million):

Bucket	1 day	1 week	1 month	3 months	1 year
Cumulative cash gap	−112	−302	−738	−1,884	−2,099
Counterbalancing capacity	9	798	829	980	1,093
Liquidity gap	−103	496	91	−904	−1,006

Which of the following interpretations is correct?

A. Liquidity is adequate at all horizons, because the counterbalancing capacity grows in every bucket.

B. The liquidity gap is the cumulative cash gap less the counterbalancing capacity, so the bank's position is weaker than the cash gap alone suggests.

C. The bank is short only at the 1-day bucket; from 1 week onward the counterbalancing capacity covers the cumulative cash gap.

D. Liquid securities that can be sold or repoed cover the cumulative cash gap from 1 week to 1 month, but the liquidity gap is negative again by 3 months and the shortfall widens to 1 year.

Question LTR-10 b.6 Bank Q351 • Schweser Module Quiz 75.1 Q2 (rebuilt)

A national bank produces a cash flow survival report from its maturity gap (mismatch) report. The report shows that cumulative cash flow under normal conditions turns negative on day 9, and that after including marketable securities and other adjustments it turns negative on day 24. Which statement best describes the bank's position relative to the Basel III requirement?

A. It meets Basel III because the adjusted survival horizon of 24 days exceeds the unadjusted horizon of 9 days.

B. It does not meet Basel III because even after adjustments the survival horizon is shorter than the required 30 days.

C. It meets Basel III because the 24-day adjusted horizon falls within the 30-day stress period.

D. It does not meet Basel III because the unadjusted horizon is shorter than 30 days, and the adjusted horizon is irrelevant.

Question LTR-10 b.7 Bank Q340 • AnalystPrep Q.4054 (rebuilt with GARP Table 10.4)

Group treasury's large depositor concentration report shows that the bank's largest depositor provides 16.5% of its funding, against an internal single-source limit of 10%. The bank values the relationship. Which of the following statements is consistent with Choudhry's guidance on this report?

A. A depositor counts as large only if it deposits USD 50 million or more; a share-of-liabilities test is not used.

B. The bank should ask the depositor to withdraw enough funds to bring its share below 10%, because the limit takes priority over the relationship.

C. ALCO should generally treat a deposit of 5% of total liabilities as large, and the bank should grow its liability base to bring the depositor's share back within the limit.

D. Funding concentration matters only for unsecured wholesale funding; intra-group funds and retail depositors are excluded from the report.

Question LTR-10 b.8 Bank Q352 • AnalystPrep Q.4062 (rebuilt)

A regulatory authority compares the firm-specific wholesale funding yield curve of Bank ABC with the curves of its peers. It finds that ABC's curve has risen significantly above the peer curves. What is the most likely implication?

A. ABC is experiencing funding stress.

B. ABC has an unusually high level of liquidity.

C. ABC has more liabilities than assets.

D. ABC has an unusually strong solvency position.

Question LTR-10 b.9 Bank Q350 • BT P2.T8.20.11.2

In regard to the types of liquidity risk reports generated by a typical bank, each of the following statements is true EXCEPT which is inaccurate?

A. Wholesale funding reports that break down the bank's funding by price, volume, product type and tenor allow users to evaluate indicators of liquidity strength.

B. Liquidity metrics should include undrawn commitments because the existence of undrawn commitments can exacerbate funding shortages at exactly the wrong time.

C. In the liquidity coverage ratio (LCR), a strategic and regulatory liquidity metric, retail demand deposits are treated as one-year money with a run-off rate in the range of 0.28% (1/365 days) to 0.40% (1/255 days).

D. The funding concentration report matters to senior managers because a central principle of liquidity management is funding diversity: a bank should not become over-reliant on a single source or sector of funds.

Solutions • LTR-10 b

Solution LTR-10 b.1

Answer: D

Bank Q335 • AnalystPrep Q.4056

Step 1 – What the profile covers

The liability profile breaks down the funding side: customers (individuals, large enterprises), repo against highly liquid, high-quality or other assets, asset-backed securities, unsecured wholesale funding, conditional liabilities.

Step 2 – Find the outsider

Highly liquid assets sit on the asset side.

Interpretation

Read the profile for the mix of funding: a growing share of short-term wholesale or secured funding weakens the bank's liquidity even when total funding is stable.

Why the other options are wrong

A is incorrect. Individual customer deposits are a liability and usually the largest share.
B is incorrect. Repo against liquid securities is secured borrowing, a liability; "liquid" misleads.
C is incorrect. Asset-backed securities issued by the bank are liabilities; "asset" misleads.

LO: LTR-10 b: Compare and interpret different types of liquidity risk reports. • **Reference:** Choudhry, The Principles of Banking, Ch.14 (GARP Ch.10), 10.1 Liquidity Risk Reporting, Liability Profile, Figure 10.7

Solution LTR-10 b.2

Answer: B

Bank Q349 • BT Mock Q54

Step 1 – Purpose of the deposit tracker

It reports current and forecast deposits against customer loans, and so the LTD ratio, a key management liquidity ratio, tracked weekly and monthly.

Step 2 – Direction of the limit

A higher LTD means more loans per unit of deposits, a weaker liquidity position, so the board limit is an upper ceiling (GARP's example: 85%).

Interpretation

The tracker's second part translates a breach into the deposit increase or asset reduction needed to get back under the ceiling.

Why the other options are wrong

A is incorrect. A floor points the wrong way: a high LTD is the risk.
C is incorrect. Market-to-book is an equity valuation measure, not a deposit metric.
D is incorrect. Duration gap is an interest rate (ALM) measure.

LO: LTR-10 b: Compare and interpret different types of liquidity risk reports. • **Reference:** Choudhry, Ch.14 (GARP Ch.10), 10.1, Deposit Tracker Report, Tables 10.1A and 10.1B

Solution LTR-10 b.3

Answer: D

Bank Q337 • AnalystPrep Q.4050

Step 1 – Content of the deposit tracker

Month-end actual deposits by customer type, the change from each month-end, aggregate customer loans and the LTD ratio, and a forecast for each month-end to year end.

Step 2 – Where depositor concentration lives

A single depositor's share of liabilities against a single-source limit is the content of the funding (large depositor) concentration report.

Interpretation

The two reports answer different questions: "are deposits funding our loans?" versus "do we depend on too few depositors?".

Why the other options are wrong

A is incorrect. Deposits by customer type are the core of the tracker.
B is incorrect. Month-end changes are shown.
C is incorrect. The forecast to year end is a defining feature.

LO: LTR-10 b: Compare and interpret different types of liquidity risk reports. • **Reference:** Choudhry, Ch.14 (GARP Ch.10), 10.1, Deposit Tracker Report; Funding Concentration Report

Solution LTR-10 b.4 — Answer: A

Bank Q339 • AnalystPrep Q.4053

Step 1 – Definition

The funding gap report "shows the maturity gap (also known as the maturity mismatch) per time bucket, for all assets and liabilities, with an adjustment for liquid securities".

Interpretation

It includes the cumulative liquidity cash flow, can be combined with the daily report, and is the source of the survival horizon report.

Why the other options are wrong

B is incorrect. Reversed: the same report generates the cash flow survival horizon report.
C is incorrect. The list of liquid securities is the input to the daily liquidity report.
D is incorrect. The end-of-day position by branch and subsidiary is the daily liquidity report.

LO: LTR-10 b: Compare and interpret different types of liquidity risk reports. • **Reference:** Choudhry, Ch.14 (GARP Ch.10), 10.1, Funding Maturity Gap ("Mismatch") Report, Table 10.3

Solution LTR-10 b.5 — Answer: D

Bank Q346 • rebuilt from GARP Table 10.2B

Step 1 – How the rows relate

Liquidity gap = cumulative cash gap + counterbalancing capacity: for example $-302 + 798 = 496$ at 1 week.

Step 2 – Read the sign of the liquidity gap

Negative at 1 day (-103), positive at 1 week and 1 month (496, 91), negative from 3 months (-904) to 1 year $(-1{,}006)$.
$\Rightarrow$ Liquid securities bridge the short buckets but not the longer ones.

Interpretation

The counterbalancing capacity grows far more slowly than the cash gap after 1 month, so the bank depends on funding it does not yet have; this is the shortage that the survival horizon report (Section 3) turns into a number of days.

Why the other options are wrong

A is incorrect. The capacity grows, but the cash gap grows faster; the liquidity gap is negative from 3 months.
B is incorrect. The capacity is added, not subtracted; it improves the position relative to the cash gap.

C is incorrect. The gap turns negative again from 3 months onward.

LO: LTR-10 b: Compare and interpret different types of liquidity risk reports. • **Reference:** Choudhry, Ch.14 (GARP Ch.10), 10.1, Daily Liquidity Report, Table 10.2B

Solution LTR-10 b.6 Answer: B

Bank Q351 • Schweser Module Quiz 75.1 Q2

Step 1 – The benchmark

Basel III requires the bank to survive a 30-day stress period.

Step 2 – Compare the horizons

Unadjusted 9 days and adjusted 24 days: both below 30.
⇒ The bank does not meet the requirement and must act to address the shortage.

Interpretation

The adjusted horizon, which includes the cash value of marketable securities, is the relevant measure; GARP's own example (7 days, then 27 days) reaches the same verdict.

Why the other options are wrong

A is incorrect. The improvement from 9 to 24 days is not the test; the benchmark is 30 days.
C is incorrect. Reverses the logic: cash must last at least 30 days, so running out inside the window is a failure.
D is incorrect. Right conclusion, wrong reason: the adjusted horizon is the relevant one, and it also fails.

LO: LTR-10 b: Compare and interpret different types of liquidity risk reports. • **Reference:** Choudhry, Ch.14 (GARP Ch.10), 10.1, Funding Maturity Gap Report, Figure 10.5

Solution LTR-10 b.7 Answer: C

Bank Q340 • AnalystPrep Q.4054

Step 1 – What "large" means

The report's illustration uses USD 50 million, but a bank may define large as a share of total liabilities: "generally speaking, a deposit of 5% of total liabilities should be treated as large by ALCO".

Step 2 – What to do about the breach

In GARP's report the largest depositor (16.5%) exceeds the 10% limit; for a close relationship the bank "will need to increase its liabilities base to bring the share down to limit, or otherwise risk damaging the relationship by asking the depositor to remove some of the funds".

Interpretation

The principle behind the report is funding diversity: no over-reliance on a single source or sector, including intra-group funds.

Why the other options are wrong

A is incorrect. The absolute amount is one option; a percentage-of-liabilities definition is also used.
B is incorrect. Asking the client to withdraw is the alternative that risks damaging the relationship, not the guidance.
D is incorrect. The report covers all sources and sectors, including intra-group funding; GARP's list includes a retail depositor.

LO: LTR-10 b: Compare and interpret different types of liquidity risk reports. • **Reference:** Choudhry, Ch.14 (GARP Ch.10), 10.1, Funding Concentration Report, Table 10.4

Solution LTR-10 b.8 — Answer: A

Bank Q352 • AnalystPrep Q.4062

Step 1 – What the curve measures

The firm-specific funding yield curve shows what the bank pays for wholesale funds by tenor.

Step 2 – Interpret the gap to peers

A curve rising materially above the peer group means lenders charge the bank a premium: the regulator reads it as an early warning of funding stress.

Interpretation

Bank managers cannot see other banks' curves directly, but can get a peer average from the regulator and should compare at least quarterly.

Why the other options are wrong

B is incorrect. A highly liquid bank would fund at or below its peers.

C is incorrect. The curve is a price signal; it says nothing directly about balance sheet composition.

D is incorrect. A strong solvency position would lower, not raise, funding costs.

LO: LTR-10 b: Compare and interpret different types of liquidity risk reports. • **Reference:** Choudhry, Ch.14 (GARP Ch.10), 10.1, Wholesale Pricing and Volume, Figure 10.8

Solution LTR-10 b.9 — Answer: C

Bank Q350 • BT P2.T8.20.11.2

Step 1 – Check each statement

A: wholesale pricing and volume reports give indications of liquidity strength, true. B: undrawn commitments are included because they can exacerbate funding shortages at the wrong time, true. D: funding diversity is the principle behind the concentration report, true.

Step 2 – The inaccurate one

Callable and demand deposits are treated as **1-day** money for regulatory purposes, not one-year money; a behavioural adjustment (for example 50% treated as longer term) is allowed only where they are shown to be stable.

$\Rightarrow$ C is inaccurate.

Interpretation

This is why a deposit base dominated by current accounts and rolling deposits does not help the regulatory liquidity metrics, even when it is stable in practice.

Why the other options are wrong

A is incorrect. True: GARP's wholesale funding breakdown by product and tenor.

B is incorrect. True: undrawn lines are drawn in stress and are reported separately.

D is incorrect. True: concentration reports guard against over-reliance on one source.

LO: LTR-10 b: Compare and interpret different types of liquidity risk reports. • **Reference:** Choudhry, Ch.14 (GARP Ch.10), 10.1, Undrawn Commitment Report; Wholesale Pricing and Volume; Example 10.1

LTR-12 • Managing Non-Deposit Liabilities • Vardeez: IMPORTANT

LTR-12 b

P3 BOTH

Describe and calculate the available funds gap.

Workbook: A27 Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing • Tier A • notes only

WHAT IS TESTED HERE

Why P3. No workbook question and no reported exam shape target this LO; Workbook A27 carries the available funds gap (AFG) only in its revision notes and formula list. It is in the book because Vardeez flags it: "Available funds gap calculation" is half of his reading note for LTR-12, in a reading he rates IMPORTANT, and the formula sits on his formula-recall watchlist.
Vardeez: AFG calculation. His notes rebuild the formula from GARP (the source notes stop at "It is calculated as") and list three ways the number is corrupted.

Reported exam shapes: none listed; built from the LO wording and Vardeez's focus.
What the questions look like: one-line arithmetic dressed in a list of items: new loans, securities purchases, credit-line drawdowns, deposits received today and expected, withdrawals, securities sales. The skill is putting each item on the right side. GARP's own practice-exam version turns it round: given a zero-AFG target, how large a withdrawal can the bank absorb? One statement question covers the "describe" half. The sources themselves are LTR-12 a; the choice among them is LTR-12 c.

1. What the gap measures

Funds managers face two questions: **how much** must be borrowed, and **which** sources to use (LTR-12 c). The first is answered by the available funds gap. The demand for nondeposit funds is determined by the gap between the institution's **total credit demands** and its **deposits and other available monies**. Management must be ready to meet not only today's credit requests but all those it can reasonably anticipate, so the projections of loans and of deposits and withdrawals rest on the needs of customers, **especially the largest**, and on economic conditions and interest rates: not wild guesses.

$$\begin{aligned}\text{AFG} = {} & \text{Current and projected loans and investments desired} \\ & - \text{Current and expected deposit inflows and other available funds}\end{aligned}$$

(Rose and Hudgins, eq. 13.3). A **positive** AFG is a shortfall to be covered from nondeposit sources; a negative one is a surplus. Most institutions **add a small amount** to the estimate to cover unexpected credit demands or shortfalls in deposits and other inflows: the AFG is a planning figure, not a floor.

2. GARP's example

A commercial bank has new loan requests that meet its quality standards of $150 million, wishes to buy $75 million of new Treasury securities this week, and expects drawings on credit lines from its best corporate customers of $135 million. Deposits and other customer funds received today total $185 million, and those expected in the coming week another $100 million.

$$\text{AFG} = (150 + 75 + 135) - (185 + 100) = 360 - 285 = \$75 \text{ million}$$

$\Rightarrow$ The bank must raise about $75 million from nondeposit sources this week, plus a small cushion.

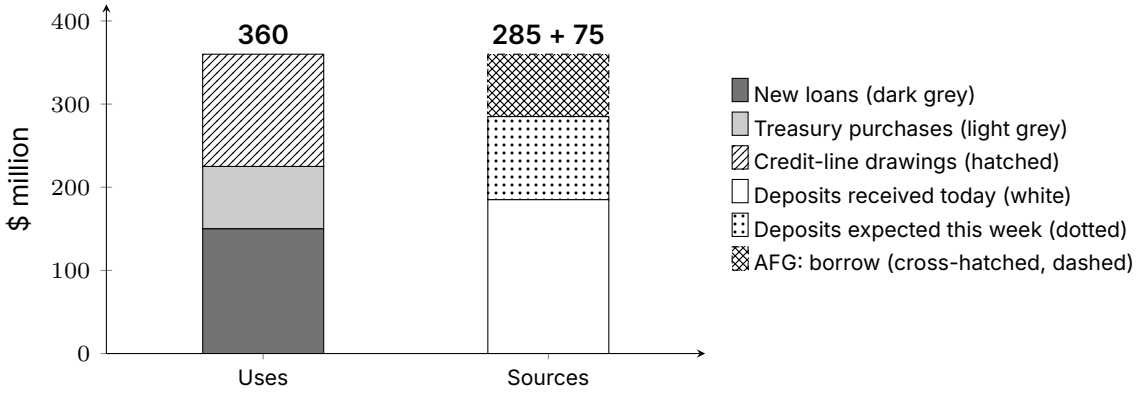


Figure 1. GARP's available funds gap. Uses (left) are the loans and investments the bank wants to make, including expected drawings on existing credit lines; sources (right) are today's and the week's expected deposits. The cross-hatched block with a dashed border is the \$75 million gap to be borrowed in the money and capital markets.

Correction to the source notes. GARP's worked line prints "AFG $= (\$150 + \$75 + \$135) - (\$185 + \$100) = \$306 - \$285 = \$75$". The sum of the uses is \$360 million, not \$306 million; the stated answer of \$75 million confirms \$360 is intended. Vardeez flags the same transposed digit.

3. Which side each item goes on

- **Uses (loans and investments desired):** new loans meeting quality standards, planned securities purchases, and **expected drawdowns on committed credit lines**. The drawdowns are a use even though no new lending decision is made.
- **Sources:** deposits received today **and** deposits expected over the period, and other available funds such as the proceeds of securities the bank plans to sell.
- **Withdrawals** reduce the net deposit inflow, so they raise the gap. In the bank's versions of GARP's Thyme Bank problem they are added to the uses; netting them against inflows gives the same answer.
- **Cushion.** Schweser's example: \$100 million of new loans, \$15 million of MBS and \$30 million of line drawdowns against \$75 million of new deposits gives AFG $= \$70$ million; with a 10% safety margin the bank plans to borrow $1.10 \times 70 = \$77$ million. The margin is applied to the gap, not to the uses.

Formula card: LTR-12 b

AFG	(new loans + securities purchases + expected credit-line drawdowns + expected withdrawals) − (deposits received + expected deposit inflows + other available funds, e.g. securities sales)
Sign	AFG > 0: shortfall, borrow nondeposit funds; AFG < 0: surplus
Planned borrowing	AFG $\times(1 +$ safety margin$)$
Withdrawal capacity at AFG $= 0$	−AFG when the gap is a surplus

Exam traps

- **Dropped credit-line drawings.** In GARP's example, $150 + 75 - 285 = -\$60$ million: a clean, sign-flipped answer that reads as a surplus.
- **Dropped securities purchase.** $150 + 135 - 285 = 0$, and zero is an offered answer.
- **Only today's deposits.** $360 - 185 = \$175$ million. Current *and* expected inflows are subtracted.
- **Withdrawals on the wrong side.** Withdrawals reduce deposit funding; adding them to inflows understates the gap.
- **Cushion on the wrong base.** The small addition is made to the gap, not to the total uses.
- **Uses or sources alone.** Reporting total uses or total sources as the gap; GARP plants both.

Practice questions: LTR-12 b

Question LTR-12 b.1 New • built from GARP Ch.13 (Measuring a Financial Firm's Total Need for Nondeposit Funds)

A junior analyst on a bank's asset-liability committee writes a note on how the bank estimates its need for nondeposit funds. Which statement in the note is correct?

A. Expected drawdowns on credit lines already committed to customers are left out of the available funds gap, because no new lending decision is required for them.

B. A positive available funds gap is a surplus of deposits that the bank can lend in the Fed funds market.

C. The available funds gap compares current and projected loans and investments with current and expected deposit inflows and other available funds, and most institutions add a small amount to cover unexpected credit demands or deposit shortfalls.

D. Only deposits already received count as a source, because expected inflows are uncertain and would understate the gap.

Question LTR-12 b.2 Bank Q403 • BT P2.T8.20.14.2 (rebuilt)

A commercial bank's situation today includes the following:

- New loan requests meeting its quality standards of USD 250.0 million
- Plans to purchase USD 100.0 million of Treasuries in the coming week
- Anticipated drawings on credit lines from its best corporate customers of USD 140.0 million
- Deposits and other customer funds received today of USD 110.0 million
- Expected deposit inflows of USD 230.0 million in the coming week

What is the bank's available funds gap (AFG)?

A. USD 50.0 million

B. USD 150.0 million

C. USD 260.0 million

D. USD 490.0 million

Question LTR-12 b.3 Bank Q405 • Schweser Module Quiz 77.1 Q3

Barbara Friedman, a bank manager on the asset-liability committee, must estimate the amount of money market funding the bank will need in the coming week. She estimates that the bank will make USD 60 million in new loans. The bank does not plan any security investments but expects additional drawdowns on credit lines of USD 10 million. The bank is in a highly competitive deposit market and expects only USD 15 million in new deposits. Based on previous years' experience, she also expects two of the bank's largest customers to withdraw USD 1 million each in the coming week. Friedman should estimate the available funds gap for the coming week to be:

A. USD 43 million

B. USD 45 million

C. USD 53 million

D. USD 57 million

Question LTR-12 b.4 Bank Q400 • GARP book Ch.13 Problem 6 (rebuilt)

Thyme Bank of New York expects new deposit inflows next month of USD 265 million and deposit withdrawals of USD 425 million. Its economics department projects new loan demand of USD 400 million, and customers with approved credit lines will need USD 175 million in cash. The bank will sell USD 450 million of securities but plans to add USD 60 million of new securities to its portfolio. Which of the following is closest to its projected available funds gap?

A. USD -80 million (a surplus)

B. USD 285 million

C. USD 345 million

D. USD 795 million

Question LTR-12 b.5 New • built from GARP Ch.13 and the Schweser Reading 77 example

The treasurer of Coastal Bank prepares next month's funding plan. The bank expects USD 220 million of new loans meeting its standards, a USD 40 million purchase of agency mortgage-backed securities and USD 60 million of drawdowns on existing credit lines. Deposits received today total USD 90 million and further deposit inflows of USD 110 million are expected during the month. Policy is to add a 10% safety margin to the estimated gap. How much should the treasurer plan to raise from nondeposit sources?

A. USD 66 million

B. USD 120 million

C. USD 132 million

D. USD 152 million

Question LTR-12 b.6 Bank Q404 • GARP PE2 Q59 • Mock 4 Q60 (rebuilt; key as corrected in the bank)

A manager at a financial institution is assessing the cash flow requirements of the institution's largest customers using the available funds gap (AFG) model. The manager obtains the following current and projected inflows and outflows of funds:

Item	Amount
Deposits received today	USD 60 million
Approved new loan requests	USD 57 million
Expected drawdown on corporate clients' credit lines	USD 16 million
Deposit inflows expected within the next week	USD 12 million
10-year Treasury securities expected to be purchased this week	USD 35 million
Other customer funds available today	USD 45 million

Assuming all else is held constant, if the institution's objective is to maintain an AFG of zero, what is the maximum amount of unexpected customer withdrawals it can meet within the next week?

A. USD 0 million

B. USD 3 million

C. USD 9 million

D. USD 25 million

Solutions • LTR-12 b

Solution LTR-12 b.1

Answer: C

New • built from GARP Ch.13

Step 1 – Why C is correct

Rose and Hudgins: "The difference between current and projected outflows and inflows of funds yields an estimate of each institution's available funds gap", AFG = current and projected loans and investments desired − current and expected deposit inflows and other available funds; and "most institutions will add a small amount to this available funds gap estimate to cover unexpected credit demands or unanticipated shortfalls in deposits".

Interpretation

The gap is a planning estimate built from projections of the largest customers' needs; once it is known, the choice of source follows the five factors of LTR-12 c.

Why the other options are wrong

A is incorrect. Scope error: GARP's own example includes "drawings on credit lines from its best corporate customers" among the uses. Committed lines become funded loans when drawn.
B is incorrect. Polarity flip: a positive gap is a *shortfall* to be borrowed in the money and capital markets; a surplus is a negative gap.
D is incorrect. Wrong side of the trade-off: both current and expected inflows are subtracted. Leaving out expected inflows would *overstate* the gap (in GARP's example, $175 million instead of $75 million).

LO: LTR-12 b: Describe and calculate the available funds gap. • **Reference:** Rose and Hudgins, Ch.13, 13.4 Measuring a Financial Firm's Total Need for Nondeposit Funds: The Available Funds Gap, eq. (13.3)

Solution LTR-12 b.2

Answer: B

Bank Q403 • BT P2.T8.20.14.2

Step 1 – Loans and investments desired

$$250 + 100 + 140 = \text{USD } 490 \text{ million}$$

Step 2 – Deposit inflows and other available funds

$$110 + 230 = \text{USD } 340 \text{ million}$$

Step 3 – Gap

$$\text{AFG} = 490 - 340 = \text{USD } 150 \text{ million}$$

$\Rightarrow$ The bank must borrow about USD 150 million of nondeposit funds, plus a small cushion.

Interpretation

The structure is GARP's example exactly: loans, a Treasury purchase and line drawings against today's and the week's deposits.

Why the other options are wrong

A is incorrect. Left out the USD 100 million Treasury purchase: $390 - 340 = 50$.
C is incorrect. Subtracted only the expected inflow: $490 - 230 = 260$.
D is incorrect. Reported total uses as the gap: 490.

LO: LTR-12 b: Describe and calculate the available funds gap. • **Reference:** Rose and Hudgins, Ch.13, 13.4, eq. (13.3)

Solution LTR-12 b.3

Answer: D

Bank Q405 • Schweser Module Quiz 77.1 Q3

Step 1 – Loans and investments desired

$$60 + 10 = \text{USD } 70 \text{ million}$$

Step 2 – Net deposit inflow

$$15 - 2 = \text{USD } 13 \text{ million}$$

$\Rightarrow$ The expected withdrawals reduce the deposit funding available.

Step 3 – Gap

$$\text{AFG} = 70 - 13 = \text{USD } 57 \text{ million}$$

Interpretation

Equivalently, put the USD 2 million of withdrawals with the uses: $72 - 15 = 57$. Either way they raise the gap.

Why the other options are wrong

A is incorrect. Ignored the drawdowns and added the withdrawals to deposits: $60 - 17 = 43$.
B is incorrect. Ignored both the drawdowns and the withdrawals: $60 - 15 = 45$.
C is incorrect. Kept the drawdowns but added the withdrawals to deposits: $70 - 17 = 53$.

LO: LTR-12 b: Describe and calculate the available funds gap. • **Reference:** Rose and Hudgins, Ch.13, 13.4, eq. (13.3); SchweserNotes 2026 Book 4, Reading 77, LO 77.b

Solution LTR-12 b.4

Answer: C

Bank Q400 • GARP book Ch.13 Problem 6

Step 1 – Outflows: loans, investments, drawdowns and withdrawals

$$400 + 175 + 60 + 425 = \text{USD } 1{,}060 \text{ million}$$

Step 2 – Inflows: deposits and other available funds

$$265 + 450 = \text{USD } 715 \text{ million}$$

$\Rightarrow$ The securities sale is an "other available funds" item; the purchase is an investment desired.

Step 3 – Gap

$$\text{AFG} = 1{,}060 - 715 = \text{USD } 345 \text{ million}$$

Interpretation

Deposits alone run off by USD 160 million $(265 - 425)$, and the bank is selling a net USD 390 million of securities to help fund USD 575 million of loans and drawings; USD 345 million must still come from nondeposit sources.

Why the other options are wrong

A is incorrect. Ignored the USD 425 million of withdrawals: $635 - 715 = -80$, a false surplus.
B is incorrect. Left out the USD 60 million of new securities: $1{,}000 - 715 = 285$.
D is incorrect. Left out the USD 450 million securities sale: $1{,}060 - 265 = 795$.

LO: LTR-12 b: Describe and calculate the available funds gap. • **Reference:** Rose and Hudgins, Ch.13, 13.4, eq. (13.3); Problem 6

Solution LTR-12 b.5

Answer: C

New • built from GARP Ch.13 and Schweser

Step 1 – Uses

$$220 + 40 + 60 = \text{USD } 320 \text{ million}$$

Step 2 – Sources

$$90 + 110 = \text{USD } 200 \text{ million}$$

Step 3 – Gap and planned borrowing

$$\text{AFG} = 320 - 200 = 120, \qquad 1.10 \times 120 = \text{USD } 132 \text{ million}$$

$\Rightarrow$ The safety margin covers unexpected loan demand or a shortfall in deposits.

Interpretation

GARP says only that "most institutions will add a small amount"; Schweser's example uses 10% of the gap. The margin scales the gap, the amount actually to be borrowed.

Why the other options are wrong

A is incorrect. Left out the credit-line drawdowns: $(260 - 200) \times 1.10 = 66$.
B is incorrect. The gap without the safety margin: 120.
D is incorrect. Applied the margin to total uses: $320 \times 1.10 - 200 = 152$.

LO: LTR-12 b: Describe and calculate the available funds gap. • **Reference:** Rose and Hudgins, Ch.13, 13.4, eq. (13.3); SchweserNotes 2026 Book 4, Reading 77, Example: Available funds gap

Solution LTR-12 b.6

Answer: C

Bank Q404 • GARP PE2 Q59

Step 1 – Uses

$$57 + 16 + 35 = \text{USD } 108 \text{ million}$$

Step 2 – Sources

$$60 + 12 + 45 = \text{USD } 117 \text{ million}$$

Step 3 – Current gap

$$\text{AFG} = 108 - 117 = -\text{USD } 9 \text{ million}$$

$\Rightarrow$ A surplus of USD 9 million.

Step 4 – Withdrawal capacity

Unexpected withdrawals reduce deposit funding and raise the AFG one for one, so up to USD 9 million can be met before the AFG rises above zero.

Interpretation

The question reverses the usual direction: instead of the borrowing need, it asks how much of the surplus is available to absorb outflows while keeping the bank out of the nondeposit market.

Why the other options are wrong

A is incorrect. Assumes a zero-AFG target leaves no room, ignoring the surplus.

B is incorrect. Left out the USD 12 million of expected deposits: $108 - 105 = +3$, which is a shortfall, not a capacity.

D is incorrect. Left out the USD 16 million of drawdowns: $117 - 92 = 25$.

LO: LTR-12 b: Describe and calculate the available funds gap. • **Reference:** Rose and Hudgins, Ch.13, 13.4, eq. (13.3)

LTR-13 • Repurchase Agreements and Financing • Vardeez: VERY IMPORTANT

LTR-13 a

P3 BOTH

Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction.

Workbook: D6 Repurchase Agreements • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D6, reported in 2 of 16 recall sources (Map #43): "Closed-book settlement amount: $[(\text{price} + \text{accrued}) \times (1 - \text{haircut})] \times (1 + \text{repo}\cdot t)$. This template is confirmed to recur across GARP practice papers."
Vardeez: no pointer of its own; his reading note ("understand general collateral, special collateral and LO78F") covers LTR-13 e and f.

Reported exam shapes

- **S1** Repo settlement amount from the closed-book template, with accrued interest and haircut, and the repurchase price at the unwind.

What the questions look like: mostly numerical. A face amount, a clean price or an invoice price, sometimes accrued interest and a haircut, a repo rate and a term; you find the cash at the start, the interest or the repurchase price. GARP's practice-exam version plants three distractors, one per omitted input (accrued interest, price, haircut). Expect one or two questions on the vocabulary: who does the repo and who the reverse, and why the repurchase price is fixed.

1. The two legs

Recap (LTR-13 e, Section 1): a repo is a sale of a security with an agreement to buy it back at a fixed price on a fixed date. The party that delivers the security and takes cash does the **repo** (borrows cash); its counterparty does the **reverse repo** (lends cash). Economically it is a short-term secured loan.

- **Start leg.** The cash lent is the **invoice (full) value** of the collateral, clean price plus accrued interest, less any **haircut**. The face amount is not the base.
- **Unwind leg.** The repurchase price is fixed at the start: the cash lent plus interest at the repo rate. Repo rates are quoted annualised on the money-market **actual/360** basis, so count the actual days.
- **GARP's example (eq. 14.1).** A sells EUR 100 million face of the DBR 4s of 4 January 2037 to B for settlement on 31 May 2010 at an invoice price of EUR 111.772 million, and agrees to buy it back on 31 August 2010 (92 days) at a repo rate of 0.23%:

$$111{,}772{,}000 \times \left(1 + 0.23\% \times \frac{92}{360}\right) = \text{EUR } 111{,}837{,}697.10$$

  $\Rightarrow$ 92 days of secured funding cost A EUR 65,697.10.
- **With a haircut.** B may lend only EUR 106 million against the EUR 111.772 million of bonds, a haircut of $5.772/111.772 = 5.16\%$, and collect only EUR 106 million plus interest, $106{,}000{,}000 \times (1 + 0.23\% \times 92/360) = \text{EUR } 106{,}062{,}304.44$. A must find the EUR 5.772 million difference from its own capital: **the haircut is funded by the borrower of cash.**
- **Margin calls.** If the bunds fall from EUR 111.772 million to EUR 110 million, A must post the EUR 1.772 million difference in extra collateral, so B still holds EUR 111.772 million against its EUR 106 million loan. In a rising market A may withdraw collateral.

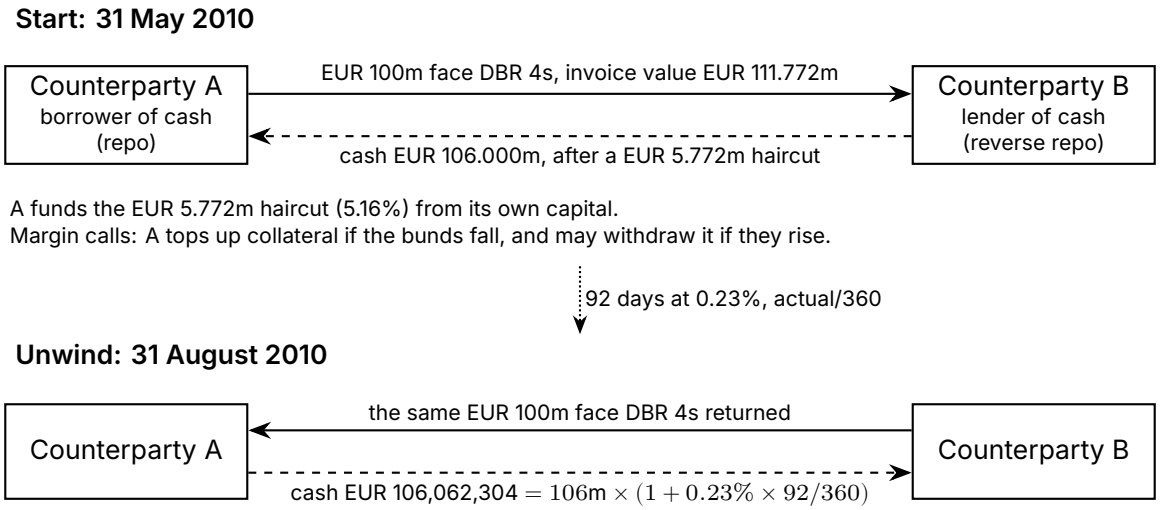


Figure 1. The two legs of GARP's repo example with the haircut it describes. Solid arrows carry securities, dashed arrows carry cash; the dotted arrow is the term. The repurchase price is fixed at the start; without the haircut it would be EUR 111,837,697.10 (eq. 14.1).

2. The closed-book template

$$\text{Repurchase price} = \left[\text{Face} \times \frac{\text{Clean price} + \text{Accrued}}{100} \times (1 - h)\right] \times \left(1 + r_{\text{repo}} \times \frac{\text{days}}{360}\right)$$

The bracket is the cash lent at the start; the haircut part, $\text{Face} \times \frac{\text{full price}}{100} \times h$, is the borrower's own capital. Check the stem for all three inputs before computing: price (not face), accrued interest, haircut.

Example 1 (Schweser, no haircut or accrued). On 1 May, A borrows USD 11 million for 31 days by selling ABC bonds with a face value of USD 10 million and a market value of USD 11 million, at 0.30%:

$$11{,}000{,}000 \times \left(1 + 0.003 \times \frac{31}{360}\right) = \text{USD } 11{,}002{,}841.67$$

Using the face value gives USD 10,002,583.33; a 365-day year gives USD 11,002,802.74.

Example 2 (all three inputs). A dealer repos out USD 20 million face of a 3% semi-annual note at a clean price of 101.25, 60 days into a 182-day coupon period, with a 2% haircut, for 45 days at 4.80%.

$$\begin{aligned}
\text{Accrued} &= \tfrac{3}{2} \times \tfrac{60}{182} = 0.4945, \qquad \text{full price} = 101.25 + 0.4945 = 101.7445 \\
\text{Invoice value} &= 20{,}000{,}000 \times 1.017445 = \text{USD } 20{,}348{,}901 \\
\text{Cash lent} &= 20{,}348{,}901 \times 0.98 = \text{USD } 19{,}941{,}923 \qquad (\text{own capital USD } 406{,}978) \\
\text{Repurchase price} &= 19{,}941{,}923 \times \left(1 + 0.048 \times \tfrac{45}{360}\right) = 19{,}941{,}923 \times 1.006 = \text{USD } 20{,}061{,}575
\end{aligned}$$

$\Rightarrow$ The dealer controls USD 20.35 million of notes with USD 0.41 million of its own money and pays USD 119,652 of interest for 45 days.

3. Terms, rolls and the rate

- **Overnight, open, term.** Most repo is overnight; an open repo is a one-day repo that renews daily until cancelled; term repo runs out to several months. Term repo usually includes **rights of substitution**: a borrower of cash that needs to sell a bond being financed may replace it with other bonds of comparable value and quality.

- **Rolling.** A dealer that has not sold the bonds by the repo's expiry rolls the repo. With the same counterparty this looks like a new trade. With a new counterparty C, A repays B the amount due on the old contract and repos the bonds to C at their *current* price, so the roll leaves a cash surplus or deficit (offset by the P&L on any hedge).
- **Customer financing** is done back to back: the dealer lends cash to the customer against its bonds and repos the same bonds out to another lender; each leg's haircut depends on that counterparty's creditworthiness and negotiating power.
- **Which rate.** A funding trade on a bond that is not trading special is done at the GC rate; the lower special rate belongs to a bond that cash lenders want to borrow (LTR-13 e).

Formula card: LTR-13 a

Full (invoice) price	clean price + accrued interest; accrued $= \frac{c}{2} \times \frac{\text{days since last coupon}}{\text{days in period}}$ (or the stem's convention)
Cash lent at the start	$\text{Face} \times \frac{\text{full price}}{100} \times (1 - h)$
Borrower's own capital	$\text{Face} \times \frac{\text{full price}}{100} \times h$
Repo interest	cash lent $\times\, r \times \frac{\text{days}}{360}$ (actual days, actual/360)
Repurchase price	cash lent $\times \left(1 + r \times \frac{\text{days}}{360}\right)$, fixed at the start
Margin call	restore the collateral to its agreed value (GARP: post the fall in value)
Rate to use	GC for a bond not trading special; special rate only for the bond in demand

Exam traps

- **Face instead of price.** Interest accrues on the cash lent, which is based on the invoice (market) value, not the face amount.
- **Dropped accrued interest.** The collateral is valued at its full price; the clean price understates both the cash lent and the repurchase price.
- **Dropped haircut.** The lender advances less than the collateral value; forgetting the haircut gives zero own capital and a repurchase price on the full invoice value.
- **Day count.** Actual/360, not 365; scale the annual rate by the term (a full year's interest is the planted error); count actual days between the dates (1 January to 1 July 2015 is 181 days).
- **Intermediate result.** The interest alone, or the cash lent, is not the repurchase price.
- **Names.** "Repo" is the cash borrower's side, "reverse repo" the cash lender's. The repurchase price is fixed at the start, not set at the market price on the unwind date.
- **GC versus special.** A non-special bond is financed at GC; the special rate is a wrong-input twin.

Practice questions: LTR-13 a

Question LTR-13 a.1

Bank Q441 • AnalystPrep Q.5378

A risk manager compares repurchase agreements (repos) and reverse repos. Which statement is correct?

A. In a repo, the seller of securities agrees to buy them back later at a slightly lower price.

B. In a reverse repo, the buyer of securities agrees to sell them back later at a slightly lower price.

C. Repos are typically used to raise cash, while reverse repos are typically used to invest cash.

D. Repos are initiated by the buyer of securities, while reverse repos are initiated by the seller of securities.

Question LTR-13 a.2

Bank Q430 • AnalystPrep Q.2240 (rebuilt)

Bolny Bank enters into a reverse repurchase agreement with Kovit Bank. In the first leg of the transaction, Bolny lends cash to Kovit and receives US government bonds. In the second leg, Bolny will:

A. Sell the bonds back to Kovit at the market price prevailing on the settlement date and keep the cash.

B. Sell the bonds back to Kovit at the price fixed at inception (cash lent plus agreed interest), thereby recovering its cash with interest.

C. Keep the bonds permanently if Kovit has not asked for them back by the settlement date.

D. Pay Kovit additional cash and receive the bonds in return.

Question LTR-13 a.3

Bank Q424 • GARP book Ch.13 Q (GARP-LR-17), rebuilt with options

Deep Valley Bank borrows USD 125 million overnight through a repurchase agreement (RP) collateralized by Treasury bills. The current RP rate is 2.50 percent (annualized, actual/360 day count). How much will the bank pay in interest on this borrowing?

A. USD 8,561.64

B. USD 8,680.56

C. USD 260,416.67

D. USD 3,125,000.00

Question LTR-13 a.4

Bank Q442 • BT P2.T7.513.1

At initiation of a repurchase agreement (repo), Counterparty A sells a security to Counterparty B for settlement on June 1, 2015, at an invoice price of USD 180.0 million. Counterparty A agrees to repurchase the security three months later, for settlement on September 1, 2015, at a purchase price equal to the invoice price plus interest at a repo rate of 0.90%. Using the actual/360 convention, which is nearest to the repurchase price?

A. USD 414,000

B. USD 180,000,000

C. USD 180,414,000

D. USD 181,620,000

Question LTR-13 a.5

Bank Q443 • Schweser Module Quiz 78.1 Q1 (rebuilt)

Pasquini Investments, a private brokerage, needs 30-day financing of USD 25 million. It will post AAA-rated government bonds with a face value of USD 27 million and a market value of USD 25 million as collateral, and was quoted a rate of 0.5%. Assuming no haircut and the actual/360 convention, which of the following reflects the contract type from Pasquini's perspective and the repurchase price it must pay?

A. Repo; USD 25,010,417

B. Reverse repo; USD 25,010,417

C. Repo; USD 25,125,000

D. Repo; USD 27,011,250

Question LTR-13 a.6

Bank Q426 • GARP PE1 Q80 • Mock 3 Q1

A bank buys a bond on its coupon payment date. Three months later, to generate immediate liquidity, the bank repos the bond. Assume a 30/360-style simple accrual and simple repo interest on a half-year term.

Item	Value
Notional (USD)	100,000
Coupon (semi-annual)	5%
Current bond price (clean, per 100)	98
Repo haircut	5%
Repo interest rate	3%

If the repo contract expires 6 months from now, what is the bank's expected cash outflow at the end of the repo transaction?

A. USD 94,497

B. USD 95,702

C. USD 97,630

D. USD 100,739

Question LTR-13 a.7 Workbook D6.1 • S1 • Map #43

A treasury analyst at a Canadian hedge fund is arranging the financing of a new long position. The fund buys CAD 50 million face of a 4% semi-annual Government of Canada bond and finances it with a 91-day term repo from a dealer bank, delivering the bond as collateral:

- Clean price: 97.50 per 100 face; 76 days have elapsed in the current 182-day coupon period
- Haircut: 4% of the invoice (full) price
- The bond is an off-the-run issue that is not trading special; the dealer's 91-day general collateral repo rate is 3.20%
- The on-the-run issue of similar maturity trades special at a 91-day repo rate of 1.40%
- Repo interest accrues on an actual/360 basis

Which of the following is closest to the cash the fund must supply from its own capital at initiation and to the repurchase price it must pay at the end of the repo?

A. Own capital CAD 0.000 million; repurchase price CAD 49.565 million

B. Own capital CAD 1.950 million; repurchase price CAD 47.179 million

C. Own capital CAD 1.967 million; repurchase price CAD 47.368 million

D. Own capital CAD 1.967 million; repurchase price CAD 47.583 million

Solutions • LTR-13 a

Solution LTR-13 a.1

Answer: C

Bank Q441 • AnalystPrep Q.5378

Step 1 – Name the sides

In a repo the seller of the security receives cash now and repurchases later at a higher price: it raises cash.

Step 2 – The same trade from the other side

The buyer of the security pays cash now and sells the security back later at the higher price: for it the trade is a reverse repo, an investment of cash.

$\Rightarrow$ Repos raise cash; reverse repos invest cash.

Interpretation

One transaction, two names. GARP notes the terminology is "particularly confusing" because the borrower of a *security* does a reverse (it lends cash), so always say whether cash or securities are being borrowed or lent.

Why the other options are wrong

A is incorrect. The repurchase price is higher, not lower: the difference is the interest on the cash.

B is incorrect. The reverse-repo party sells back at the higher repurchase price, which is how it earns interest.

D is incorrect. Swaps the sides. The names describe sides, not initiators: the repo is the seller's (cash borrower's) side, and a specials trade, for one, is initiated by the cash lender.

LO: LTR-13 a: Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction. • **Reference:** Tuckman and Serrat, Fixed Income Securities 3e, Ch.12, 14.1 Repurchase Agreements: Structure and Uses

Solution LTR-13 a.2

Answer: B

Bank Q430 • AnalystPrep Q.2240

Step 1 – Roles

Kovit sold the bonds and borrowed cash: it did the repo. Bolny lent cash and holds the bonds as collateral: it did the reverse.

Step 2 – The second leg

Bolny returns the bonds and receives the repurchase price agreed at inception, the cash lent plus interest at the repo rate.

$\Rightarrow$ Bolny recovers its cash with interest, whatever the bonds are worth on that date.

Interpretation

The fixed repurchase price is what makes a repo a loan rather than two outright trades: the cash lender is exposed to the bonds' value only if the borrower fails to repay (LTR-13 c).

Why the other options are wrong

A is incorrect. Pricing the second leg at the market price would make it an ordinary sale and purchase, with price risk on both sides.

C is incorrect. The repurchase commitment is binding; the bonds are collateral and go back on the agreed date.

D is incorrect. Reverses the cash flows: Bolny receives cash in the second leg.

LO: LTR-13 a: Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo trans-

action. • **Reference:** Tuckman and Serrat, Ch.12, 14.1

Solution LTR-13 a.3 Answer: B

Bank Q424 • GARP book Ch.13 Q

Step 1 – Interest on an actual/360 basis

$$\text{Interest} = \text{principal} \times r \times \frac{\text{days}}{360} = 125{,}000{,}000 \times 0.025 \times \frac{1}{360} = \text{USD } 8{,}680.56$$

$\Rightarrow$ The bank repays USD 125,008,680.56 the next morning.

Interpretation

Overnight repo against Treasury bills is one of the cheapest ways for a bank to fund itself, which is why the question sits in GARP's chapter on non-deposit liabilities (LTR-12). The original question had no options; the distractors below are the usual scaling errors.

Why the other options are wrong

A is incorrect. Used a 365-day year: $3{,}125{,}000/365 = 8{,}561.64$.
C is incorrect. Charged one month instead of one day: $3{,}125{,}000/12 = 260{,}416.67$.
D is incorrect. A full year of interest with no scaling: $125\text{m} \times 2.5\% = 3{,}125{,}000$.

LO: LTR-13 a: Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction. • **Reference:** Rose and Hudgins, Bank Management & Financial Services 9e, Ch.13 (GARP Ch.13, Managing Nondeposit Liabilities, end-of-chapter problem 5); Tuckman and Serrat, Ch.12, eq. (14.1)

Solution LTR-13 a.4 Answer: C

Bank Q442 • BT P2.T7.513.1

Step 1 – Count the actual days

1 June to 1 July is 30 days, to 1 August 31, to 1 September 31: 92 days.

Step 2 – Interest

$$180{,}000{,}000 \times 0.0090 \times \frac{92}{360} = \text{USD } 414{,}000$$

Step 3 – Repurchase price

$$180{,}000{,}000 + 414{,}000 = \text{USD } 180{,}414{,}000$$

$\Rightarrow$ A pays back the invoice price plus 92 days of interest.

Interpretation

The structure is GARP's own example (31 May to 31 August, 92 days); only the size and rate differ.

Why the other options are wrong

A is incorrect. The interest alone: the invoice price was not added back.
B is incorrect. The invoice price with no interest.
D is incorrect. A full year of interest, $180\text{m} \times 0.90\% = 1.62\text{m}$, with no 92/360 scaling: 181,620,000.

LO: LTR-13 a: Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction. • **Reference:** Tuckman and Serrat, Ch.12, 14.1, eq. (14.1)

Solution LTR-13 a.5 — Answer: A

Bank Q443 • Schweser Module Quiz 78.1 Q1

Step 1 – Contract type

Pasquini sells the bonds now and buys them back later to raise cash: from its side this is a repo (the lender does the reverse).

Step 2 – The base

The cash raised is the USD 25 million market value; the USD 27 million face is irrelevant.

Step 3 – Repurchase price

$$25{,}000{,}000 \times \left(1 + 0.005 \times \frac{30}{360}\right) = 25{,}000{,}000 + 10{,}416.67 = \text{USD } 25{,}010{,}417$$

$\Rightarrow$ Pasquini pays USD 10,417 for 30 days of secured funding.

Interpretation

Two traps in one stem: the name of the side and the base for interest. With a haircut the cash, and so the repurchase price, would be lower than USD 25 million.

Why the other options are wrong

B is incorrect. Right number, wrong side: the reverse repo is the cash lender's view.
C is incorrect. Applied the full annual rate with no 30/360 scaling: $25\text{m} \times 1.005 = 25{,}125{,}000$.
D is incorrect. Accrued interest on the face value: $27\text{m} \times (1 + 0.005 \times 30/360) = 27{,}011{,}250$.

LO: LTR-13 a: Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction. • **Reference:** Tuckman and Serrat, Ch.12, 14.1

Solution LTR-13 a.6 — Answer: B

Bank Q426 • GARP PE1 Q80

Step 1 – Accrued interest

The semi-annual coupon on USD 100,000 is USD 2,500; three months into the six-month period, accrued is $2{,}500 \times 0.5 = \text{USD } 1{,}250$.

Step 2 – Full (invoice) value

$$98{,}000 + 1{,}250 = \text{USD } 99{,}250$$

Step 3 – Cash received after the haircut

$$99{,}250 \times (1 - 0.05) = \text{USD } 94{,}287.50$$

$\Rightarrow$ The bank funds the USD 4,962.50 haircut itself.

Step 4 – Repayment at the unwind

$$94{,}287.50 \times (1 + 0.03 \times 0.5) = \text{USD } 95{,}701.81 \approx 95{,}702$$

Interpretation

This is the closed-book template the map reports as recurring across GARP practice papers. Each distractor drops one input, so check for price, accrued interest and haircut before computing.

Why the other options are wrong

A is incorrect. Left out accrued interest: $98{,}000 \times 0.95 \times 1.015 = 94{,}496.50$.
C is incorrect. Valued the bond at par plus accrued instead of its price: $101{,}250 \times 0.95 \times 1.015 = 97{,}630.31$.

D is incorrect. Ignored the haircut: $99{,}250 \times 1.015 = 100{,}738.75$.

LO: LTR-13 a: Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction. • **Reference:** Tuckman and Serrat, Ch.12, 14.1, eq. (14.1) and haircuts

Solution LTR-13 a.7 — Answer: D

Workbook D6.1 • S1

Step 1 – Accrued interest and full price

$$\text{accrued} = \frac{4}{2} \times \frac{76}{182} = 0.8352, \qquad \text{full price} = 97.50 + 0.8352 = 98.3352$$

$\Rightarrow$ The repo lender values the collateral at the invoice price the buyer actually pays, coupon accrual included.

Step 2 – Invoice value of the collateral

$$50{,}000{,}000 \times 0.983352 = \text{CAD } 49{,}167{,}582$$

Step 3 – Cash lent and the fund's own capital

$$\text{Cash}_0 = 49{,}167{,}582 \times (1 - 0.04) = \text{CAD } 47{,}200{,}879$$

$$\text{own capital} = 49{,}167{,}582 - 47{,}200{,}879 = \text{CAD } 1{,}966{,}703$$

$\Rightarrow$ The dealer lends 96% of the full value; the fund pays the 4% haircut, CAD 1.967 million, from its own capital.

Step 4 – Choose the repo rate

This is a funding trade on a bond that is not trading special, so it is done at the GC rate of 3.20%. The 1.40% special rate applies only to the on-the-run bond that cash lenders want to borrow.

Step 5 – Repurchase price

$$\text{Cash}_T = 47{,}200{,}879 \times \left(1 + 0.032 \times \frac{91}{360}\right) = 47{,}200{,}879 \times 1.0080889 = \text{CAD } 47{,}582{,}682$$

$\Rightarrow$ The fund pays CAD 381,803 of repo interest over 91 days to get its bonds back.

Interpretation

The fund controls CAD 49.17 million of bonds with CAD 1.97 million of its own money (leverage of 25 times). That leverage is only as durable as the dealer's willingness to roll the repo at maturity, which is the vulnerability examined in Question LTR-13 c.6 (workbook D6.2).

Why the other options are wrong

A is incorrect. Dropped the haircut: the dealer is assumed to lend the full invoice value, so own capital is zero and the repurchase price is $49{,}167{,}582 \times 1.0080889 = 49{,}565{,}294$.

B is incorrect. Dropped the accrued interest: invoice $= 50\text{m} \times 0.975 = 48{,}750{,}000$, own capital $= 0.04 \times 48{,}750{,}000 = 1{,}950{,}000$, repurchase $= 46{,}800{,}000 \times 1.0080889 = 47{,}178{,}560$.

C is incorrect. Wrong-input twin: used the on-the-run special rate instead of GC, $47{,}200{,}879 \times (1 + 0.014 \times 91/360) = 47{,}367{,}918$. Own capital is unaffected, so the error shows only in the second number.

LO: LTR-13 a: Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction. LTR-13 e: Compare the use of general and special collateral in repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.1 and 14.3, eq. (14.1)

LTR-13 • Repurchase Agreements and Financing • Vardeez: VERY IMPORTANT

LTR-13 c

P3 CONCEPT

Explain how counterparty risk and liquidity risk can arise through the use of repo transactions.

Workbook: D6 Repurchase Agreements • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D6, reported in 2 of 16 recall sources (Map #43): "repo calculation: which party is more vulnerable if one bank is not allowing the repo to roll over?" The candidate's own answer was the hedge fund.
Vardeez: no pointer of its own; his note ranks the two risks (counterparty risk lower because of the collateral, liquidity risk the bigger concern).

Reported exam shapes

- **S2** Which party is more vulnerable when a bank will not roll the repo over.

What the questions look like: statement questions on which risk a repo creates for whom, which mitigant addresses which risk (collateral, haircuts, margin calls, shorter terms, better collateral), and who is exposed when financing is withdrawn. No calculations.

1. Counterparty risk

Recap: the cash lender holds the borrower's securities as collateral, usually with a haircut and margin calls (LTR-13 e, Section 1; LTR-13 a, Section 1).

- **Definition.** "The counterparty risk of lending money through a repo is that the borrower defaults and the value of the collateral turns out to be insufficient to cover the loan amount."
- **Why it is small.** "Holding collateral makes the lender less vulnerable to the creditworthiness of a counterparty": on default, the lender sells the collateral "to recover any amounts owed". Repos are short and collateralised, so counterparty risk is generally the smaller concern.
- **What is left.** The borrower may default *at the same time* as the collateral falls in value, so the sale might not cover the loan; hence haircuts. With high-grade government collateral GARP calls this a **right-way risk**: a dealer default would probably trigger a flight to quality that *raises* the value of government bonds.
- **Operational reality.** In the 2007 to 2009 crisis many repo lenders found they were not well positioned, in expertise or operations, to take possession of and liquidate collateral.
- **The borrower of cash is exposed too.** It has delivered securities worth more than the cash it received (the haircut). If the lender fails, or will not return the collateral, the borrower can lose up to that excess. Lehman's estate accused JPMorgan of taking billions more collateral than it needed (LTR-13 d).

2. Liquidity risk

Two different liquidity risks arise, one on each side of the trade.

- **Collateral liquidity risk (the cash lender).** The collateral may lose value during the term or become hard to sell; in turbulent markets values can fall sharply and liquidity can dry up. Even a lender untroubled by the borrower's credit is exposed. Managed by **haircuts, margin calls, shorter**

repo terms and higher-quality collateral. This is the risk Schweser and Vardeez rank as the bigger concern.

- **Funding liquidity risk (the cash borrower).** Repo is cheap and relatively liquid, but its short maturities put it on the less stable side of the funding spectrum (Figure 1), although more stable than short-term unsecured borrowing because collateral "should prevent repo lenders from bolting too quickly". If lenders lose confidence, the financing "can disappear as fast as the repos mature, which is mostly overnight". The borrower can no longer hold inventory or finance customers, may cease to be an acceptable counterparty, and must sell positions, likely at fire-sale prices. "While significant business losses rather than financing are the usual cause of a financial institution's difficulties, the loss of financing is often the killing blow"; the same holds for all leveraged investors, "like part of the hedge fund world".
- **Tension over term.** Borrowers want longer repo terms (time to refinance, raise capital or sell businesses); lenders want shorter ones (less exposure to defaults). Repo rates allocate terms, but "the financial system as a whole cannot both extend the maturities of secured financing and contract the maturities of secured lending."
- **How it played out.** Before the crisis, borrowers used repo to finance lower-quality collateral, such as weaker corporate bonds and mortgage-backed securities, at low rates and haircuts. In the crisis the worst hit could not meet margin calls, could not post collateral against higher haircuts, and could not replace lost financing: collateral liquidations, losses, capital depletion and failure. Bear Stearns' repo lenders declined to roll "even when the loans were supported by high-quality collateral such as agency securities" (LTR-13 d).

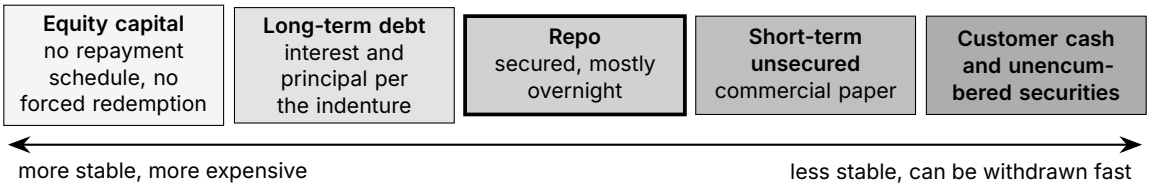


Figure 1. The funding stability spectrum in GARP Section 14.2, grey shading darkening toward the unstable end. Repo (thick border) is cheaper than equity or long-term debt and more stable than unsecured short-term borrowing, but it can vanish as fast as the repos mature. GARP places customer cash and unencumbered securities, which "can be withdrawn without notice at any time", at the extreme.

Risk	Who bears it	How it arises	Mitigants
Counterparty (credit) risk	Cash lender	Borrower fails to repay while the collateral is worth less than the loan	Collateral; haircut; right-way government collateral
Counterparty risk on the collateral	Cash borrower	Lender fails or will not return collateral worth more than the cash (the haircut)	Moderate haircuts; sound counterparties
Collateral liquidity risk	Cash lender	Collateral value falls or it cannot be sold during the term	Haircuts, margin calls, shorter terms, higher-quality collateral
Funding liquidity (rollover) risk	Cash borrower	Lenders refuse to roll, raise haircuts or demand better collateral	Longer terms, stable funding mix, unencumbered liquid assets

Exam traps

- **Ranking.** For the cash lender, collateral makes counterparty risk the smaller concern; the value and liquidity of the collateral is the bigger one.
- **Mitigant mapping.** Collateral addresses counterparty risk; haircuts, margin calls, short terms and quality address collateral liquidity risk. Options that swap them are wrong.
- **Role misassignment.** A refusal to roll hurts the *borrower* of cash, who must refinance or sell; the lender just gets its cash back and is over-collateralised meanwhile.
- **Absolute claims.** Collateral does not make repo funding stable or riskless; a 5% haircut covers only a 5% fall.
- **Both sides.** The borrower of cash also has counterparty exposure, to the excess collateral it has delivered.
- **Margin calls work both ways.** Extra collateral in falling markets; withdrawals in rising ones.

Practice questions: LTR-13 c

Question LTR-13 c.1

Bank Q450 • AnalystPrep Q.2297 (rebuilt)

In a repurchase agreement, what is the purpose of a collateral "haircut"?

A. It discounts the collateral value relative to its risk, so the cash advanced is less than the collateral's market value, protecting the lender against a decline in collateral value.

B. It is the fee the borrower pays the lender, deducted from the repo rate.

C. It is the additional collateral the lender must return when prices rise.

D. It increases the borrower's cash proceeds when collateral is riskier.

Question LTR-13 c.2

Bank Q456 • Schweser Module Quiz 78.1 Q2 (rebuilt)

In repo transactions, posting collateral and requiring collateral haircuts are important risk mitigants. Which of the following correctly matches each mitigant with the primary risk it addresses?

A. Posting collateral and haircuts both mitigate counterparty risk only.

B. Posting collateral mitigates liquidity risk; haircuts mitigate counterparty risk.

C. Posting collateral mitigates counterparty risk; haircuts mitigate liquidity risk (the risk of a decline in collateral value).

D. Posting collateral and haircuts both mitigate liquidity risk only.

Question LTR-13 c.3 Bank Q455 • Mock 2 Q58

A bank proposes using very short-term repo transactions to earn additional investment income. The collateral is provided by the borrower and the haircut is 5%. To which of the following risks is the bank exposed?

A. Counterparty risk only.

B. Liquidity risk only.

C. Both counterparty risk and liquidity risk.

D. Neither counterparty risk nor liquidity risk.

Question LTR-13 c.4 Bank Q451 • AnalystPrep Q.1870 (rebuilt)

A repo is a short-term loan secured by collateral. The cash lender's main residual exposure is that the collateral loses value during the term of the repo. Which of the following statements about this risk and its mitigation is most accurate?

A. If the collateral value falls, the borrower may be called to post additional collateral (a margin call), or the repo may be repriced so that the loan amount is reduced.

B. Once a haircut has been applied the collateral no longer poses any risk, so margin calls are unnecessary.

C. If the collateral value rises the lender suffers a loss, whereas if it falls the lender is fully protected.

D. The risk is eliminated because repos are legally long-term loans, which removes the effect of short-term price moves.

Question LTR-13 c.5 Bank Q454 • BT notes P2.T6.406.2

Consider a classic repurchase agreement in which Bank A sells securities to Counterparty B and receives cash, and later repurchases the securities. Bank A is the repo seller (borrows cash) and Counterparty B is the repo buyer (lends cash). Each of the following statements about this repo is true EXCEPT:

A. A higher haircut (overcollateralization) reduces the counterparty risk faced by the lender, Counterparty B.

B. The securities sold by Bank A become collateral owned by Counterparty B, who can liquidate them to recover the cash lent if Bank A defaults.

C. Bank A, as the cash borrower, incurs no counterparty risk; only Counterparty B does.

D. Counterparty B is exposed to credit risk, market risk, liquidity risk of the collateral, operational risk and legal risk.

Question LTR-13 c.6 Workbook D6.2 • S2 • Map #43

A hedge fund has financed a leveraged long position in BBB-rated corporate bonds for several months through overnight repo with Bank K, rolling the repo each day at a 5% haircut. After market rumours about the fund's trading losses, Bank K tells the fund that it will not roll the repo when it matures tomorrow morning. A risk consultant reviewing the episode is asked which party is more vulnerable to the non-renewal. Which assessment is correct?

A. The hedge fund is more vulnerable: as the borrower of cash it must replace the financing or sell the bonds at short notice, possibly at fire-sale prices, while Bank K's loan is secured by collateral worth more than the cash it lent.

B. Bank K is more vulnerable: as the lender of cash it must find a buyer for the bonds at short notice, possibly at fire-sale prices, while the fund's cash is protected by the haircut it received at the start of the repo.

C. Neither party is materially vulnerable: because the repo is fully collateralised with a haircut, the non-renewal simply unwinds the trade, and the fund can always refinance the same bonds with another repo lender.

D. The hedge fund is more vulnerable, but only if the bonds have fallen in value since the repo began, because otherwise returning the cash and taking back the bonds leaves the fund with no loss and no funding need.

Solutions • LTR-13 c

Solution LTR-13 c.1

Answer: A

Bank Q450 • AnalystPrep Q.2297

Step 1 – What the lender is exposed to

GARP: "a lender of cash faces the risk that a borrower defaults at the same time those securities decline in value. In that eventuality, selling the collateral might not fully cover the loss of the loan amount."

Step 2 – The haircut as a cushion

"Repo agreements often provide haircuts through which investors require borrowers to deliver securities worth more than the amount of the loan": EUR 106 million against EUR 111.772 million in GARP's example.

$\Rightarrow$ The haircut protects the lender against a fall in collateral value.

Interpretation

Riskier or less liquid collateral gets a larger haircut, and the borrower funds the haircut from its own capital (LTR-13 a). A haircut is set at the start; margin calls keep the cushion in place as prices move.

Why the other options are wrong

B is incorrect. Confuses the haircut with interest; interest is the repo rate on the cash lent.

C is incorrect. Describes the margin mechanism in a rising market, and it is the borrower who withdraws excess collateral then.

D is incorrect. Polarity flip: riskier collateral gets a larger haircut and so *less* cash per unit of collateral.

LO: LTR-13 c: Explain how counterparty risk and liquidity risk can arise through the use of repo transactions. •
Reference: Tuckman and Serrat, Fixed Income Securities 3e, Ch.12, 14.1 Repos and Cash Management

Solution LTR-13 c.2

Answer: C

Bank Q456 • Schweser Module Quiz 78.1 Q2

Step 1 – Counterparty risk

The risk that the borrower does not repay. Posted collateral lets the lender sell it to recover amounts owed, so the lender depends less on the borrower's creditworthiness.

Step 2 – Liquidity risk

The risk that the collateral falls in value or becomes illiquid during the term. The haircut is a buffer against that fall, helped by margin calls, shorter terms and better collateral.

$\Rightarrow$ Collateral for counterparty risk; haircuts for liquidity risk.

Interpretation

The layers work together: collateral turns a default into a collateral sale, and the haircut makes it likely the sale still covers the loan after prices have moved against the lender.

Why the other options are wrong

A is incorrect. Haircuts address the collateral's value, not only default.

B is incorrect. Reverses the mapping.

D is incorrect. Collateral itself is the primary counterparty risk mitigant.

LO: LTR-13 c: Explain how counterparty risk and liquidity risk can arise through the use of repo transactions. •
Reference: Tuckman and Serrat, Ch.12, 14.1 Repos and Cash Management

Solution LTR-13 c.3 — Answer: C

Bank Q455 • Mock 2 Q58

Step 1 – The bank's side

It lends cash and holds the borrower's collateral: it is the cash lender.

Step 2 – Counterparty risk

The borrower may fail to repay. The collateral reduces the loss but does not remove the exposure: recovery depends on selling the collateral.

Step 3 – Liquidity risk

The collateral can fall by more than the 5% haircut, or be hard to sell when needed.
$\Rightarrow$ Both risks are present.

Interpretation

Very short terms reduce both risks (less time for prices to move, frequent chances to stop lending) but eliminate neither.

Why the other options are wrong

A is incorrect. Ignores the risk that the collateral falls by more than 5% or becomes illiquid.
B is incorrect. Ignores the borrower's default, which still matters if the collateral proves insufficient.
D is incorrect. Treats a secured loan as riskless; the haircut is only a partial buffer.

LO: LTR-13 c: Explain how counterparty risk and liquidity risk can arise through the use of repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.1

Solution LTR-13 c.4 — Answer: A

Bank Q451 • AnalystPrep Q.1870

Step 1 – The exposure

A fall in collateral value during the term erodes the lender's cover.

Step 2 – The remedy

Repos are subject to margin calls: "the borrower of cash supplies extra collateral in declining markets but may withdraw collateral in advancing markets." In GARP's example, if the bunds fall from EUR 111.772 million to EUR 110 million, the borrower posts EUR 1.772 million, so the lender still holds EUR 111.772 million against EUR 106 million.
$\Rightarrow$ Margin calls, or repricing to a smaller loan, restore the lender's cover.

Interpretation

The haircut sets the cushion; margin calls keep it intact as prices move. Together they manage the collateral liquidity risk.

Why the other options are wrong

B is incorrect. A haircut is a fixed buffer: a larger fall uses it up, so margin calls are still needed.
C is incorrect. Reversed: a rise in collateral value does not hurt the lender, and a fall is not fully covered.
D is incorrect. Repos are short-term; the term does not remove price risk.

LO: LTR-13 c: Explain how counterparty risk and liquidity risk can arise through the use of repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.1 Repos and Cash Management

Solution LTR-13 c.5 — Answer: C

Bank Q454 • BT notes P2.T6.406.2

Step 1 – Check the true statements

A: a bigger haircut gives the lender more collateral per unit of cash. B: on default the lender sells the collateral to recover the cash. D: the lender faces credit risk and also market, collateral liquidity, operational and legal risks.

Step 2 – The false one

Bank A has delivered securities worth more than the cash it received. If Counterparty B defaults or fails to return them, A loses up to the excess.
$\Rightarrow$ Both sides of a repo carry counterparty risk; C is the exception.

Interpretation

The borrower's exposure grows with the haircut and with any extra collateral posted under margin calls. Lehman's estate argued that JPMorgan, its tri-party clearing agent, took billions more collateral than it needed in its final days (LTR-13 d).

Why the other options are wrong

A is incorrect. True: overcollateralisation protects the lender.
B is incorrect. True: liquidating collateral on default is the lender's main protection.
D is incorrect. True: the lender's risks go beyond credit risk.

LO: LTR-13 c: Explain how counterparty risk and liquidity risk can arise through the use of repo transactions. •
Reference: Tuckman and Serrat, Ch.12, 14.1; Case Study: JPMorgan Chase's Repo Exposure to Lehman Brothers

Solution LTR-13 c.6 — Answer: A

Workbook D6.2 • S2

Step 1 – Name the roles

The fund sold the bonds to Bank K with an agreement to buy them back: it is the borrower of cash (it did the repo). Bank K is the lender of cash and holds the bonds as collateral, worth about 105% of its loan $(1/0.95)$.

Step 2 – What non-renewal does to each side

Bank K simply receives its cash back plus interest; until then its loan is over-collateralised and it can sell the collateral "to recover any amounts owed" if the fund defaults. The fund must hand over cash it does not have: Tuckman and Serrat note that a borrower's repo financing "can disappear as fast as the repos mature, which is mostly overnight", after which it "would have to sell inventory" in sales that "would likely turn into fire sales", and that the argument "applies to all leveraged investors, like part of the hedge fund world."
$\Rightarrow$ The exposure created by a refusal to roll is funding liquidity risk, and it sits with the borrower of cash.

Interpretation

This is the Bear Stearns pattern in miniature: repo lenders "declined to roll over or renew repo loans, even when the loans were supported by high-quality collateral". "The loss of financing is often the killing blow."

Why the other options are wrong

B is incorrect. Role misassignment. The roles are reversed: the lender of cash gets its cash back and is protected by the haircut it charges; it is the borrower of cash who must raise funds or sell into a fire sale.
C is incorrect. Absolute claim. Collateral makes repo more stable than unsecured funding but does not guarantee refinancing; in the crisis borrowers "were unable to replace lost financing

arrangements when lower-quality collateral found fewer and fewer takers".
D is incorrect. Scope/condition error. The fund's problem exists even at unchanged prices: it must still find the full cash to repay the loan, and only then does it get the bonds back. A price fall adds a loss but is not the condition for vulnerability.

LO: LTR-13 c: Explain how counterparty risk and liquidity risk can arise through the use of repo transactions. • **Reference:** Tuckman and Serrat, Ch.12, 14.1 Repos and Cash Management, and 14.2 Repo, Liquidity Management, and the Financial Crisis

LTR-16 • Covered Interest Parity Lost: Understanding the Cross-Currency Basis • Vardeez: IMPORTANT

LTR-16 a

P3 CONCEPT

Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps.

Workbook: No workbook section • Vardeez focus

WHAT IS TESTED HERE

Why P3. The reading is Tier D workbook topic D8 (Map #46, 2 of 16 recall sources) and Vardeez rates it IMPORTANT: "Short Reading and anything can be tested here - so know it well". The map carries no workbook section for this LO; workbook D8.2 tags it (option C is the FX swap versus cross-currency swap sibling swap) and is worked as Question LTR-16 b.6.
Vardeez: no LO-level pointer. His notes single out two discriminators: tenor, and the rate at which principal is re-exchanged (forward rate for the FX swap, initial spot rate for the cross-currency swap). This block carries the shared background of the reading (CIP, the two swaps, the basis); LTR-16 b and c point back here.

Reported exam shapes: none listed. The questions below follow the LO wording.

What the questions look like: feature-matching statements ("which correctly differentiates", "each is true EXCEPT"), the definition of the basis, and one hedging application. No calculations under this CONCEPT LO; the numerical basis questions sit in LTR-16 c.

1. Covered interest parity

Borio, McCauley, McGuire and Sushko (BIS Quarterly Review, 2016; GARP Ch.17) call CIP "the closest thing to a physical law in international finance". It is a no-arbitrage condition: interest rates on two otherwise identical assets in two currencies must be equal once the currency risk is hedged.

$$\boxed{\frac{F}{S} = \frac{1+r}{1+r^*}} \quad \Longleftrightarrow \quad F - S = S\left(\frac{1+r}{1+r^*} - 1\right)$$

S and F are the spot and forward rates in **US dollars per unit of foreign currency**; r is the dollar rate and r^* the foreign currency rate. The second form says the forward points $(F-S)$ equal the spot rate times the (compounded) rate differential. The intuition is two ways to invest dollars: directly at r, or by selling dollars spot for the foreign currency, investing it at r^* and selling the proceeds forward at F. Both lock in a dollar return, so they must pay the same. CIP is read off FX swaps and cross-currency swaps.

2. FX swap

One party **borrows one currency from, and simultaneously lends another currency to**, a second party. The amounts are exchanged at the **spot rate** S and repaid at the **pre-agreed forward rate** F at maturity. There are no interim payments: the implicit interest is set by $F - S$, and the contract is quoted in **forward points** $(F - S)$. FX swaps are mostly short-dated (78% of turnover under seven days, LTR-15 a). They let a bank fund a foreign currency asset with domestic currency: the bank pays the dollar interest through the forward points.

Example. A euro area bank needs USD 100 million for three months. $S = 1.2500$ USD per EUR; three-month rates are 4.00% a year for dollars (1.00% for the quarter) and 1.00% for euros (0.25%). Under CIP $F = 1.25 \times 1.01/1.0025 = 1.259352$.

- Start: the bank delivers EUR 80 million and receives USD 100 million (at S).

- Maturity: the bank pays $80 \times 1.259352 =$ USD 100.748 million and receives EUR 80 million back (at F).

$\Rightarrow$ The 0.748 million of forward points is the dollar interest (USD 1.000 million) less the euro interest the bank gave up ($0.2 \times 1.259352 = 0.252$ million in dollars). When CIP holds, the swap-implied dollar rate $\frac{F}{S}(1 + r^*) - 1$ equals the cash rate, here exactly 1.00% for the quarter.

3. Cross-currency swap

A **longer-term** instrument, typically **above one year**, in which the two parties also simultaneously borrow and lend equivalent amounts in two currencies. Three differences from the FX swap:

1. at maturity the amounts are exchanged back at the **initial spot rate** S, not at a forward rate;
2. during the life the counterparties **periodically exchange interest payments**;
3. in a **cross-currency basis swap** the reference rates are the respective Libor rates **plus the basis** b.

The re-exchange can use the old spot rate only *because* interest is paid along the way: the interest legs do the work that the forward points do in an FX swap. Same bank, three years: it receives USD 100 million and delivers EUR 80 million at 1.25; each quarter it pays dollar Libor (plus any basis) on USD 100 million and receives euro Libor on EUR 80 million; at maturity it repays USD 100 million and receives EUR 80 million at 1.25, whatever the spot rate is then (Figure 1).

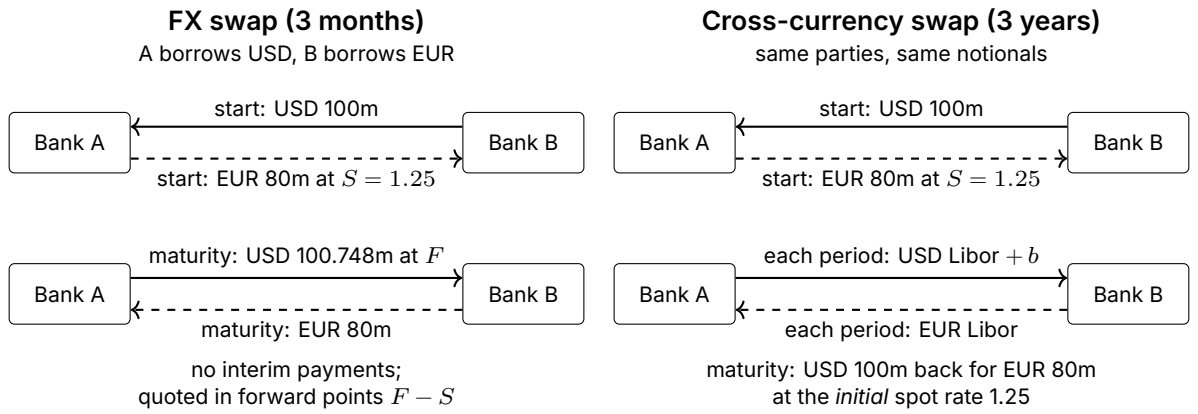


Figure 1. Cash flows of the two swaps for a bank (A) borrowing dollars against euros. Solid arrows: dollar flows; dashed arrows: euro flows. The FX swap has two exchanges, the second at the forward rate. The cross-currency swap adds periodic interest legs and re-exchanges principal at the initial spot rate; when CIP fails, the dollar borrower pays the basis b on top of dollar Libor.

4. The cross-currency basis

The basis is "the amount by which the interest paid to borrow one currency by swapping it against another differs from the cost of directly borrowing this currency in the cash market". A non-zero basis is a CIP violation. Since 2007 the dollar has typically commanded a premium in FX swaps: forward points are **too wide**, $F - S > S\left(\frac{1+r}{1+r^*} - 1\right)$, so a party lending dollars sells the foreign currency forward at a higher dollar price than the rate differential warrants, and a party borrowing dollars through the swap pays more than the cash rate. For one period, b is the adjustment to the dollar rate that restores parity:

$$F - S = S\,\frac{1 + r + b}{1 + r^*} - S \qquad \Longleftrightarrow \qquad b = \frac{F}{S}(1 + r^*) - (1 + r)$$

- **Who pays.** The party **borrowing dollars pays the basis on top of the dollar rate**; the counterparty borrowing the other currency in effect gets an equal discount on it.
- **Zero basis.** If CIP holds throughout, $b = 0$ and the cross-currency basis swap is a plain floating-for-floating currency swap.
- **Sign convention.** Box 17.1 attaches b to the *dollar* leg, so a dollar premium is a *positive* b (GARP practice questions and Schweser use this). The reading's text and graphs, like market quotes, attach the basis to the *non-dollar* leg, so the same premium is a *negative* EUR or JPY basis. Same economics: dollars are dearer through swaps than in the cash market. The Australian dollar basis has been the other way round.

	FX swap	Cross-currency swap
Mechanics	Borrow one currency, lend another	Same, for equivalent amounts
Start	Exchange at spot S	Exchange at spot S
Interim payments	None	Periodic interest (Libor $+ b$ in a basis swap)
Maturity exchange	Pre-agreed forward rate F	Initial spot rate S
Tenor	Short, mostly days to months	Longer, typically above one year
Quoted as	Forward points $F - S$	The basis b on a Libor leg
CIP violation shows as	Forward points too wide	A non-zero basis
Typical use	Short-term funding of a foreign currency asset	Longer-horizon funding and hedging (bond issuers)

Exam traps

- **Forward versus initial spot.** FX swap unwinds at the pre-agreed forward F; cross-currency swap at the initial spot S. Swapping them reads well and is wrong.
- **Interim interest.** Only the cross-currency swap pays interest during its life; the FX swap's "interest" is in the forward points.
- **Tenor.** Cross-currency swaps are the longer instrument (typically above one year), not the reverse.
- **CIP can fail in both.** It shows as too-wide forward points in FX swaps and as a non-zero basis in cross-currency swaps.
- **Who pays b.** The dollar borrower pays it on top of the dollar rate; the other side pays less on the foreign currency.
- **Zero basis is normal, not void.** $b = 0$ makes the instrument an ordinary floating-for-floating currency swap.
- **Basis is a rate.** It is an interest rate adjustment, not the forward points and not the rate differential.
- **Sign.** Positive b on the dollar leg (Box 17.1) and negative EUR or JPY basis (market quote) describe the same dollar premium.

Practice questions: LTR-16 a

Question LTR-16 a.1

Bank Q523 • AnalystPrep Q.4191

FX swaps and cross-currency swaps are derivative instruments used to hedge foreign currency exposures. Which of the following correctly identifies a similarity between the two?

A. Both instruments involve an exchange of currencies.

B. Both instruments involve the exchange of principal and interest payments in different currencies.

C. Both instruments are used primarily to manage long-term interest rate risk.

D. Both instruments are commonly used by central banks to manage their currency reserves.

Question LTR-16 a.2

Bank Q520 • AnalystPrep Q.4185 (rebuilt)

Company A wants to convert USD 60 million of floating-rate debt into a GBP loan. On the trade date, A exchanges USD 60 million with Company B for GBP 48 million in a 10-year cross-currency swap. At maturity, A and B swap the principal amounts back. What exchange rate (USD per GBP) applies to the final exchange?

A. 0.80, the initial spot rate expressed as GBP per USD

B. The 10-year forward rate implied by covered interest parity

C. The spot rate prevailing at maturity

D. 1.25, the initial spot rate

Question LTR-16 a.3

Bank Q527 • Schweser Module Quiz 81.1 (rebuilt, key corrected)

Which of the following statements correctly differentiates a foreign exchange (FX) swap from a cross-currency basis swap?

A. Both contracts re-exchange the principal at maturity at the original spot rate, and both pay periodic interest during the life of the contract.

B. An FX swap exchanges periodic interest payments based on each currency's interest rate plus a basis, whereas a cross-currency basis swap pays no interim cash flows.

C. An FX swap re-exchanges the currencies at a pre-agreed forward rate and is quoted in forward points $(F - S)$, whereas a cross-currency basis swap re-exchanges at the original spot rate and exchanges interest payments based on each currency's rate plus the basis during its life.

D. A cross-currency basis swap is typically a short-term contract of under one year, whereas an FX swap is typically a longer-term contract of more than one year.

Question LTR-16 a.4

Bank Q525 • Schweser Module Quiz 81.1 Q1

The cross-currency swap basis is the:

A. interest rate differential in a cross-currency swap.

B. amount by which the interest rate of one currency must be adjusted in a cross-currency swap so that covered interest parity (CIP) holds.

C. price to the long position in a cross-currency swap.

D. difference between the forward and spot exchange rates in a cross-currency swap.

Question LTR-16 a.5

Bank Q518 • AP Mock 3 Q25 (rebuilt)

Walter runs a US business with dollar revenues and has a long-dated mortgage on a French property with fixed payments in euros. The dollar is weakening against the euro, so each payment costs more dollars. Which action best protects him over the full life of the mortgage?

A. Enter a cross-currency swap in which he receives euro payments and pays dollar payments, with the final principal exchange at today's spot rate.

B. Enter a one-month FX swap once, swapping dollars for euros at spot and back at the forward rate.

C. Enter a cross-currency swap in which he receives dollar payments and pays euro payments.

D. Wait until the FX market stabilises and then convert dollars to euros at the prevailing spot rate each period.

Question LTR-16 a.6

Bank Q524 • BT P2.T8.20.17.2

In regard to the differences between a foreign exchange (FX) swap and a cross-currency swap, each of the following is true EXCEPT which statement is false?

A. An FX swap is typically a shorter-term instrument, while a cross-currency swap is a longer-term instrument (often above one year).

B. Covered interest rate parity (CIRP) cannot fail in the case of an FX swap, while CIRP cannot even be tested on a cross-currency swap because the final exchange uses the initial spot rate.

C. At maturity, the FX swap has the counterparties repay the borrowed amounts at the pre-agreed forward rate, but the cross-currency swap exchanges the borrowed amounts at the initial spot rate.

D. The FX swap typically has only two cash flow legs (the initial simultaneous lending and the repayment at maturity), but the cross-currency swap exchanges several interim interest payments.

Solutions • LTR-16 a

Solution LTR-16 a.1 — Answer: A

Bank Q523 • AnalystPrep Q.4191

Step 1 – What both swaps do at the start

In both, the parties exchange currencies at the current spot rate: each borrows one currency and lends the other.

Step 2 – Where they part

They differ in the maturity exchange (forward rate versus initial spot rate), in tenor (short versus typically above one year) and in interim interest (none versus periodic).

$\Rightarrow$ The shared feature is the exchange of currencies.

Interpretation

Read "similarity" questions against the comparison table: only the first two rows (mechanics and the start) are common to both.

Why the other options are wrong

B is incorrect. Only cross-currency swaps exchange interest; an FX swap has just the two principal exchanges.

C is incorrect. Both address currency funding and exposure, not primarily long-term interest rate risk; FX swaps are mostly short-term.

D is incorrect. Reserve management is not a defining feature of either instrument in the reading.

LO: LTR-16 a: Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 a.2 — Answer: D

Bank Q520 • AnalystPrep Q.4185

Step 1 – Initial spot rate

$60/48 = 1.25$ USD per GBP.

Step 2 – Re-exchange rule

In a cross-currency swap the borrowed amounts are exchanged back at the initial spot rate.

$\Rightarrow$ USD 60 million for GBP 48 million at 1.25, whatever happens to the market over 10 years.

Interpretation

The fixed re-exchange works because the interest legs carry the rate differential over the life of the swap. In an FX swap there are no interest legs, so the differential has to sit in the forward rate.

Why the other options are wrong

A is incorrect. 0.80 is the same rate quoted the other way round (GBP per USD); the question asks for USD per GBP.

B is incorrect. The forward rate is the re-exchange rate of an FX swap, the sibling instrument.

C is incorrect. The future spot rate is unknown at inception; the swap fixes the rate in advance.

LO: LTR-16 a: Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 a.3 — Answer: C

Bank Q527 • Schweser Module Quiz 81.1

Step 1 – FX swap

Exchange at spot S today and back at the pre-agreed forward F; quoted in forward points $(F-S)$; no interim interest.

Step 2 – Cross-currency basis swap

Exchange at spot today; periodic interest on each currency's reference rate plus the basis b; principal re-exchanged at the original spot rate; typically longer than one year.
$\Rightarrow$ Only C states both sets of features correctly.

Interpretation

The bank rebuilt this question and corrected its key; C is the textbook pairing of Box 17.1.

Why the other options are wrong

A is incorrect. Gives the FX swap the cross-currency swap's features: an FX swap unwinds at F and pays no interim interest.
B is incorrect. Reverses the two contracts: the interest-plus-basis payments belong to the cross-currency swap.
D is incorrect. Reverses the tenors.

LO: LTR-16 a: Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps. •
Reference: Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 a.4 — Answer: B

Bank Q525 • Schweser Module Quiz 81.1 Q1

Step 1 – Where the basis sits

In a cross-currency basis swap the reference rates are the two Libor rates plus the basis b.

Step 2 – What it measures

If the forward points are wider than CIP warrants, b is the amount by which the rate on one leg must be adjusted so that parity with FX swap pricing holds: $F - S = S(1+r+b)/(1+r^*) - S$.
$\Rightarrow$ The basis is the rate adjustment that restores CIP; $b = 0$ when CIP holds.

Interpretation

Box 17.1 puts b on the dollar leg (positive when the dollar is at a premium); market quotes put it on the non-dollar leg (negative). Both are "the amount by which the interest rate of one currency must be adjusted".

Why the other options are wrong

A is incorrect. The rate differential $r - r^*$ is part of CIP itself, not the deviation from it.
C is incorrect. The basis is a spread on a reference rate, not a price paid by the long side.
D is incorrect. $F - S$ is the forward points of an FX swap; the basis is the interest rate equivalent of the deviation from parity.

LO: LTR-16 a: Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps. •
Reference: Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 a.5 — Answer: A

Bank Q518 • AP Mock 3 Q25

Step 1 – The exposure

Walter owes a long stream of euro payments and earns dollars: he is short euros over many years.

Step 2 – The instrument

The cross-currency swap is the longer-term instrument: it exchanges interest during its life and re-exchanges principal at the initial spot rate, which fixes the dollar cost of the euro stream.

Step 3 – The direction

To hedge he must receive euros (to pay the mortgage) and pay dollars (from his revenues).
$\Rightarrow$ A.

Interpretation

Tenor decides the instrument and the exposure decides the direction. An FX swap would have to be rolled every month at new forward points, leaving him exposed to the euro over the life of the mortgage.

Why the other options are wrong

B is incorrect. A one-month FX swap covers one month and then unwinds at the forward rate; the rest of the mortgage stays unhedged.
C is incorrect. Receiving dollars and paying euros adds to his short-euro exposure instead of offsetting it.
D is incorrect. Waiting leaves the exposure open, and a weakening dollar keeps raising the cost.

LO: LTR-16 a: Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps. •
Reference: Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 a.6 **Answer: B**

Bank Q524 • BT P2.T8.20.17.2

Step 1 – Check A, C and D

A: true, the cross-currency swap is longer-term. C: true, forward rate versus initial spot rate. D: true, only the cross-currency swap has interim interest payments.

Step 2 – Test B

CIP can fail through both instruments: in FX swaps as forward points wider than the rate differential warrants, in cross-currency swaps as a non-zero basis b.
$\Rightarrow$ B is the false statement.

Interpretation

Box 17.1 states that "the relationship between F and S is read off market transactions in FX instruments, notably FX swaps and cross-currency swaps". The whole reading is about CIP failing in exactly these instruments since 2007.

Why the other options are wrong

A is incorrect. True, so not the exception.
C is incorrect. True, so not the exception.
D is incorrect. True, so not the exception.

LO: LTR-16 a: Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps. •
Reference: Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

LTR-16 • Covered Interest Parity Lost: Understanding the Cross-Currency Basis • Vardeez: IMPORTANT

LTR-16 b

P3 CONCEPT

Identify key factors that affect the cross-currency swap basis.

Workbook: D8 US Dollar Shortage and the Cross-Currency Basis • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D8 (Map #46, 2 of 16 recall sources) in a reading Vardeez rates IMPORTANT ("Short Reading and anything can be tested here").
Vardeez: no LO-level pointer. His notes give two factors, market liquidity and risk premia, with two named measures to keep apart: counterparty credit risk is read from Libor-OIS spreads, sovereign credit risk from sovereign CDS spreads.

Reported exam shapes

- **S2** Covered interest parity violations and the cross-currency basis (workbook D8.2, which tags LTR-16 a, b and c and is placed here).

What the questions look like: statement questions matching a factor to its mechanism or its measure, and the bridge to LTR-16 c: why even small risk premia matter when hedging demand is large. The bank has three questions for this LO; GARP PE1 Q35 is re-filed here from LTR-16 c and two new questions cover Box 17.1's scaling result and the regression evidence. The numerical basis question filed here in the bank is moved to LTR-16 c (BOTH).

1. Why the basis should be zero, and what stops it

Cash market rates and the spot rate can be taken as given (those markets are far larger than FX derivatives), so it is shifts in demand for FX swaps and cross-currency swaps that push the **forward rate** F away from parity. In an ideal world any deviation would trigger arbitrage at once, because CIP arbitrage is treated as riskless: an FX swap has no open currency position, credit, counterparty, market and liquidity risks are assumed negligible, and arbitrageurs have unimpaired access to cash and derivatives markets. Each factor below breaks one of those assumptions. (Definitions of the swaps and of b: LTR-16 a, Sections 2 to 4.)

2. The factors

Market liquidity. In a crisis, liquidity in the spot and forward FX markets can evaporate, so the bid-ask spread becomes non-trivial. An arbitrageur buys at the ask and sells at the bid; with $1 + r^* \approx 1$ the deviation that survives is

$$b = \frac{F^b - S^a}{S^a} - (r - r^*),$$

with S^a the spot ask and F^b the forward bid. Wider spreads raise the cost of executing the swaps and can remove the arbitrage profit.

Credit risk in the underlying instruments (risk premia).

- Arbitrage by global banks borrowing and lending in the **Libor (interbank) markets**: a rise in **counterparty credit risk**, measured by **Libor-OIS spreads**, opens the basis.
- Arbitrage by banks or asset managers using **government bonds** in the two currencies: differences in **sovereign credit risk**, measured by **sovereign CDS spreads**, open it.
- In general, with risk-free r and r^* and a risk premium rp on the underlying investment over the life of the swap, $b = \frac{F-S}{S} - (r - r^*) = rp$.

Hedging demand, which scales the premium. Even if risk premia are low, CIP deviations can arise when the demand to hedge one currency is large, because small premia are multiplied by the large balance sheet exposures needed to meet hedgers' demand. Sushko et al. show the deviation can be proportional to hedging demand times the per-dollar balance sheet cost of FX derivatives exposures:

$$b = \frac{F - S}{S} - (r - r^*) \;\propto\; rp \times \text{FX hedging demand}$$

The price set in the market is the forward leg F: arbitrageurs pass their balance sheet costs on as **wider forward points**. Markets clear, so arbitrageurs' aggregate position equals hedgers' net position. In non-crisis times rp is whatever cost participants assign to deploying balance sheet: absolute credit limits, and the funding cost of capital held against current and potential future derivatives exposures (banks); the cost of placing bonds in different currencies (top-rated supranationals, which arbitrage the long-term basis by issuing in dollars and swapping); the price and availability of repo funding (hedge funds). Why these costs and that demand rose after the crisis is LTR-16 c.

What the regressions show (JPY/USD, December 2007 to December 2015). The standard model of the three-month basis uses counterparty risk (Libor-OIS), funding liquidity (US minus Japan repo spreads) and market liquidity (FX bid-ask), all significant. Adding hedging demand improves it: the **interaction** of Libor-OIS with hedging demand matters (1% higher Libor-OIS with 1% higher demand: about **45 basis points** wider), and Libor-OIS alone is no longer significant. For the **two-year** basis, hedging demand has a **direct** effect: 1% more demand, about **1 basis point** wider. Long-term swaps are the more direct barometer of hedged cross-currency flows; short-term FX swaps are more sensitive to risk premia and bank funding strains, especially in crises (Figure 2).

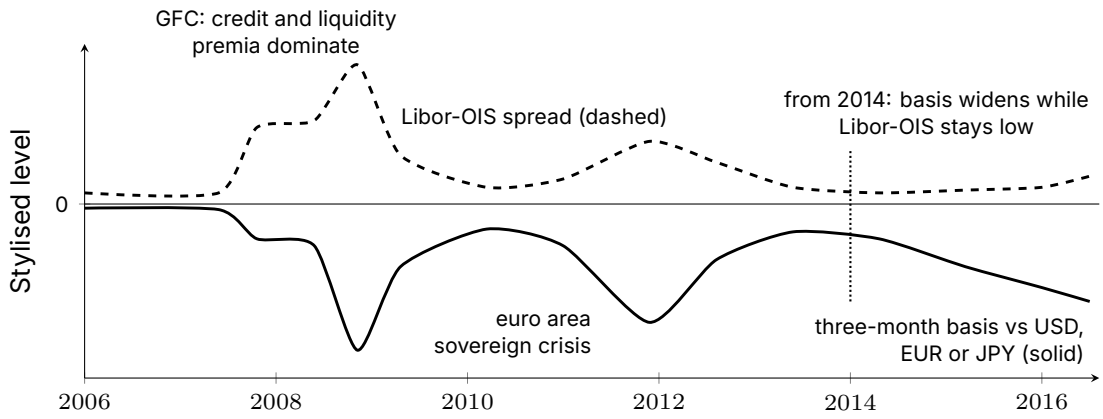


Figure 2. Stylised shapes only, after GARP Graph 1; not data, and no common scale. The basis is drawn as the market quotes it on the non-dollar leg (negative when dollars are dearer through swaps). In 2008 and 2011 to 2012 the basis moved with Libor-OIS (counterparty risk). After 2014 it widened with Libor-OIS low: the hedging-demand and balance sheet story of LTR-16 c (dotted line marks 2014).

Correction to the source notes. Vardeez (following Schweser) places the counterparty credit risk "in FX markets". In GARP it is the counterparty risk of the *interbank* (*Libor*) *markets* in which banks borrow and lend the two currencies to run the arbitrage, which is why Libor-OIS is the measure. The FX swap itself carries no open currency position.

Factor	Measure	How it moves the basis
Market liquidity	FX bid-ask spread (spot ask, forward bid)	Raises execution cost; arbitrage no longer pays
Counterparty credit risk	Libor-OIS spread	Risk premium when arbitrage runs through interbank borrowing and lending
Sovereign credit risk	Sovereign CDS spread	Risk premium when arbitrage runs through government bonds
Funding liquidity	Repo spread differential	Costlier collateralised funding for arbitrageurs
Hedging demand $\times$ balance sheet cost	Hedging volumes; per-dollar cost rp	Small rp times large demand gives a wide, persistent basis

Exam traps

- **Measure swap.** Libor-OIS for counterparty (interbank) risk; sovereign CDS for sovereign risk in bond-based arbitrage; bid-ask for market liquidity.
- **Wider spreads, wider basis.** Higher bid-ask spreads make arbitrage *costlier*, so the basis persists; they do not make the deviation more profitable.
- **"Small" is enough.** Even small risk premia produce a wide basis when hedging demand is large: the effect is $rp\times$ demand.
- **Price set on the forward leg.** Arbitrageurs pass costs on through F (wider forward points), not through the spot rate or cash rates.
- **Tenor.** Three-month basis: interaction of Libor-OIS and demand (about 45 bp). Two-year basis: demand directly (about 1 bp per 1%).
- **Hedging demand mattered before 2014 too.** The regressions find it at work throughout, not only after 2014.

Practice questions: LTR-16 b

Question LTR-16 b.1

Bank Q529 • written for the bank, GARP style

A treasury analyst at a bank observes that during a sudden market disruption the bid-ask spreads on spot and forward euro/dollar transactions widened sharply, and the cross-currency basis moved away from zero. The analyst wants to explain why arbitrageurs did not close the basis. Which of the following statements is correct?

A. Wider bid-ask spreads affect the basis only through a higher Libor-OIS spread, which measures counterparty credit risk.

B. Wider bid-ask spreads make the CIP deviation more profitable to exploit, so the basis should close more quickly.

C. Wider bid-ask spreads raise the transaction cost of executing the arbitrage, which can eliminate the profit from exploiting the CIP deviation.

D. Wider bid-ask spreads matter only if sovereign CDS spreads on the underlying government bonds have also widened.

Question LTR-16 b.2 Bank Q531 • written for the bank, GARP style

A portfolio manager at an asset management firm runs CIP arbitrage by borrowing and lending government bonds in two currencies and hedging the currency exposure with FX swaps. A risk analyst is asked which indicator best captures the risk premium factor that could cause the manager's arbitrage to leave a non-zero cross-currency basis. Which of the following is most appropriate?

A. Sovereign CDS spreads, because differences in sovereign credit risk between the two bond markets can create CIP deviations.

B. Libor-OIS spreads, because counterparty credit risk in interbank markets drives deviations in all CIP arbitrage.

C. The bid-ask spread on spot transactions, because it measures the credit risk premium of the bonds.

D. The VIX, because equity volatility is the only factor that determines the risk premium of government bonds.

Question LTR-16 b.3 Bank Q532 • GARP PE1 Q35 (re-filed from LTR-16 c)

The CIO of a UK-based hedge fund has asked the fund's portfolio managers to identify potential trading opportunities for the fund. After reviewing several suggestions, the CIO decides that trading FX swaps may offer lucrative returns. The CIO asks a portfolio manager to identify the roles that the hedge fund and other market participants can play in FX swap markets as well as factors that may affect the pricing of the cross-currency FX swap basis. Which of the following statements would be correct for the manager to make?

A. A persistent and positive FX swap basis is likely to result from non-financial firms issuing debt denominated in their home currency in offshore capital markets rather than in their home country's debt market.

B. A persistent and positive FX swap basis may exist because of the decision of institutional investors to reduce their hedges of foreign currency investments.

C. A persistent and positive FX swap basis is likely to result from international banks concentrating their lending activities in their home markets and reducing loan exposures in offshore markets.

D. A positive FX swap basis may exist because arbitrageurs pass on their balance sheet borrowing costs through the forward FX swap price.

Question LTR-16 b.4 New • built from GARP Ch.17, Box 17.1

A rates strategist at a dealer notes that the per-dollar cost banks attach to using their balance sheets for CIP arbitrage is small, yet the dollar/yen basis is wide. According to the framework in Box 17.1, which of the following statements is correct?

A. Small per-dollar balance sheet costs cannot produce a wide basis, so the basis must reflect counterparty risk alone.

B. The deviation can be proportional to the per-dollar balance sheet cost multiplied by FX hedging demand, so even small costs produce a wide basis when hedging demand for one currency is large.

C. Arbitrageurs pass their balance sheet costs on through the spot exchange rate, which moves away from the level implied by the interest rate differential.

D. When hedging demand is large, arbitrageurs' aggregate position is small, so the basis is set by liquidity in the cash money markets rather than in FX swaps.

Question LTR-16 b.5 New • built from GARP Ch.17, Table 17.1

A quantitative analyst at a central bank replicates the regression evidence on the JPY/USD basis from December 2007 to December 2015. Which finding is consistent with the reading?

A. For the three-month basis, the Libor-OIS spread on its own remains the dominant driver once hedging demand is added, and hedging demand has no effect at any tenor.

B. The two-year basis is driven mainly by FX bid-ask spreads, while the three-month basis is the more direct barometer of hedged cross-currency flows.

C. For the three-month basis, the Libor-OIS spread interacted with hedging demand is significant (1% more of each, about 45 basis points wider), while for the two-year basis hedging demand has a direct effect of about 1 basis point per 1% of demand.

D. Hedging demand began to matter only after 2014; before 2014 the basis was explained entirely by counterparty risk.

Question LTR-16 b.6 Workbook D8.2 • S2 • Map #46

The treasurer of a Japanese life insurer that hedges its US dollar bond portfolio with three-month FX swaps is presenting to the investment committee. The committee asks why the cost of hedging has stayed high since 2014, even though concerns about bank credit quality have faded and wholesale dollar funding has recovered. Which statement is correct for the treasurer to make?

A. A persistently negative USD/JPY basis means that borrowing dollars through the FX swap market is cheaper than borrowing them directly in the dollar cash market, which has lowered the insurer's hedging cost.

B. The basis has persisted because heavy demand for dollar hedges from Japanese banks and insurers has met tighter limits to arbitrage, as balance sheet capacity has become costly for the banks that would close it.

C. Unlike the insurer's FX swaps, a cross-currency swap exchanges no interest during its life and re-exchanges the principal at a pre-agreed forward rate rather than at the initial spot rate.

D. Japanese banks, which fund their dollar assets through FX swaps, act mainly as arbitrageurs in the yen/dollar market, so their growing reliance on swaps has narrowed the basis since 2014.

Solutions • LTR-16 b

Solution LTR-16 b.1

Answer: C

Bank Q529 • written for the bank

Step 1 – Name the factor

Wider spot and forward bid-ask spreads are the market liquidity factor.

Step 2 – Its mechanism

The arbitrageur buys at the ask and sells at the bid, so the measured deviation must exceed the round-trip cost: $b = (F^b - S^a)/S^a - (r - r^*)$. If the cost exceeds the deviation, the arbitrage does not pay and the basis stays open.

$\Rightarrow$ C.

Interpretation

Liquidity and credit are separate factors with separate measures. In 2008 both moved together, which is why the crisis basis was so wide.

Why the other options are wrong

A is incorrect. Libor-OIS measures counterparty credit risk, the risk premium factor, not the liquidity channel.

B is incorrect. Reads the wider spread as a larger mispricing; it is a larger cost, which reduces the profit.

D is incorrect. Sovereign CDS spreads belong to the risk premium factor in bond-based arbitrage; liquidity works on its own through transaction costs.

LO: LTR-16 b: Identify key factors that affect the cross-currency swap basis. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 b.2

Answer: A

Bank Q531 • written for the bank

Step 1 – Identify the arbitrage

The manager runs CIP arbitrage through government bonds in the two currencies.

Step 2 – Match the measure

Box 17.1: with government bonds, deviations can come from differences in sovereign credit risk, typically measured by sovereign CDS spreads.

$\Rightarrow$ A.

Interpretation

The measure follows the instrument. Arbitrage through interbank (Libor) borrowing and lending is exposed to counterparty risk, measured by Libor-OIS; arbitrage through government bonds is exposed to sovereign risk, measured by sovereign CDS.

Why the other options are wrong

B is incorrect. Libor-OIS is the measure for interbank-based arbitrage; it is not universal.

C is incorrect. The bid-ask spread measures market liquidity (transaction cost), not a credit risk premium.

D is incorrect. The reading names counterparty and sovereign credit risk as the risk premium drivers; the VIX is not the measure, and "only" is an absolute claim.

LO: LTR-16 b: Identify key factors that affect the cross-currency swap basis. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 b.3 — Answer: D

Bank Q532 • GARP PE1 Q35

Step 1 – How the basis is priced

The price actually set in FX derivatives is the forward leg F. CIP arbitrageurs taking the other side of hedging demand pass their balance sheet costs on as wider forward points $(F - S)$.

Step 2 – Read the sign

In GARP's convention (Box 17.1) a dollar premium is a positive basis on the dollar leg, so "a positive FX swap basis" is the post-crisis case.
$\Rightarrow$ D: balance sheet costs passed through the forward price.

Interpretation

The three distractors invert the three sources of hedging demand of LTR-16 c (banks' currency mismatches, institutional investors' hedges, non-financial firms issuing *foreign* currency debt and swapping back). Each describes *less* hedging demand, which would narrow the basis.

Why the other options are wrong

A is incorrect. Hedging demand arises when firms issue in a *foreign* currency and swap back (US firms issuing in euros); home-currency issuance creates none.
B is incorrect. Reversed: institutional investors maintaining or raising their hedges add pressure; reducing hedges eases it.
C is incorrect. Banks' currency mismatches (foreign lending funded in another currency) create the demand; concentrating lending at home reduces it.

LO: LTR-16 b: Identify key factors that affect the cross-currency swap basis. LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1 and Section 17.1

Solution LTR-16 b.4 — Answer: B

New • built from GARP Ch.17, Box 17.1

Step 1 – The scaling result

Box 17.1: even if risk premia are low, CIP deviations can arise if the demand to hedge one currency is large, because small premia have big effects "when scaled by the large size of the balance sheet exposures needed to meet the hedgers' demand". Sushko et al.: $b \propto rp \times$ FX hedging demand.

Step 2 – Where the cost shows up

Arbitrageurs pass the per-dollar cost rp on as wider forward points.
$\Rightarrow$ Small rp times large demand gives a wide basis: B.

Interpretation

This is the bridge from LTR-16 b to c: the factors set the per-dollar cost, the post-crisis hedging demand sets the scale.

Why the other options are wrong

A is incorrect. Ignores the scaling by hedging demand; the reading's point is that small costs can produce a wide basis.
C is incorrect. The spot rate and cash rates are taken as given; the price set is the forward leg F.
D is incorrect. Markets clear, so arbitrageurs' aggregate position equals hedgers' net position: large demand means large arbitrage positions, and the basis is set in FX swaps.

LO: LTR-16 b: Identify key factors that affect the cross-currency swap basis. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 b.5 Answer: C

New • built from GARP Ch.17, Table 17.1

Step 1 – Three-month basis

The standard specification (Libor-OIS, repo spread difference, FX bid-ask) is significant. Adding hedging demand and its interaction with Libor-OIS improves it: the interaction coefficient implies about 45 basis points wider for 1% more of each, and Libor-OIS alone is no longer significant.

Step 2 – Two-year basis

Hedging demand has a direct (linear) effect: a coefficient of about -0.91, roughly 1 basis point wider per 1% more demand; the interaction is not significant.
$\Rightarrow$ C.

Interpretation

Long-term swaps are the more direct barometer of hedged cross-currency flows; short-term FX swaps are more sensitive to risk premia and funding strains, particularly in crises.

Why the other options are wrong

A is incorrect. Once demand is added, Libor-OIS on its own loses significance, and demand matters at both tenors.
B is incorrect. Reverses the tenors: the long-term basis tracks hedged flows; FX bid-ask enters the three-month specification, not the two-year one.
D is incorrect. The evidence shows hedging demand "played a role even before the widening of the basis from 2014".

LO: LTR-16 b: Identify key factors that affect the cross-currency swap basis. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Section 17.3, Table 17.1

Solution LTR-16 b.6 Answer: B

Workbook D8.2 • S2

Step 1 – Read the sign

Since 2007 the basis for lending dollars against the yen has been negative: dollars obtained by swapping yen cost more than dollars borrowed in the cash market. The insurer pays that premium in its forward points.

Step 2 – Why it opens

Hedging demand that is "largely insensitive to the size of the basis" presses on it: Japanese banks fund dollar lending with yen deposits via swaps, and life insurers hedge 60 to 70% of their US dollar bonds. Aggregate Japanese hedging need rose from USD 0.9 trillion (2009) to over USD 1.2 trillion (2015), and the basis tracked it.

Step 3 – Why it does not close

Arbitrage now "incurs a cost per unit of balance sheet": CVA and PFE capital charges, leverage ratios, quarter-end reporting, costlier dollar repo relative to yen repo, and internal limits on Japanese official sector exposures. "Balance sheet space is rented, not free."
$\Rightarrow$ With bank credit concerns gone, the persistence is explained by hedging demand meeting limits to arbitrage, not by counterparty risk.

Why the other options are wrong

A is incorrect. Polarity flip. A negative basis makes swapped dollars more expensive than cash dollars, which raises the insurer's hedging cost.
C is incorrect. Sibling swap. These are FX swap features reversed: a cross-currency swap does exchange interest periodically (Libor plus the basis) and re-exchanges principal at the initial spot

rate; it is the FX swap that settles at the forward rate with no interim interest.
D is incorrect. Role misassignment. Japanese banks are hedgers who add to dollar demand; their reliance on swaps "has reduced their capacity to serve as counterparties to non-bank hedgers" and to arbitrage, widening, not narrowing, the basis.

LO: LTR-16 a: Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps. LTR-16 b: Identify key factors that affect the cross-currency swap basis. LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko, "Covered Interest Parity Lost", BIS Quarterly Review (2016), Box 17.1, sections 17.1 and 17.3

LTR-16 • Covered Interest Parity Lost: Understanding the Cross-Currency Basis • Vardeez: IMPORTANT

LTR-16 c

P3 BOTH

Assess the causes of covered interest rate parity violations after the financial crisis of 2008.

Workbook: D8 US Dollar Shortage and the Cross-Currency Basis • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D8 (Map #46, 2 of 16 recall sources) in a reading Vardeez rates IMPORTANT. The TYPE is BOTH by the user's decision (29 October): the numerical CIP and basis questions in the bank are kept here.
Vardeez: no LO-level pointer. His notes split the LO into two halves, why the basis opens (hedging demand that is insensitive to the basis) and why it does not close (a cost per unit of balance sheet).

Reported exam shapes

- **S2** Covered interest parity violations and the cross-currency basis (D8.2 is worked under LTR-16 b as Question LTR-16 b.6; Questions LTR-16 c.7 and c.8 here).

What the questions look like: two kinds. Statement questions on the causes (GARP PE2 Q5 asks for "the most appropriate explanation" of a persistent basis). Numerical questions: a CIP forward, an arbitrage profit, the basis from S, F, r, r^*, and the cost of funding through a basis swap. Two new questions cover the three hedging-demand sources, the sign of the basis across countries and Box 17.2's issuance arithmetic, which no bank question tests.

1. The puzzle

The first explanation of the post-2007 basis was strain in interbank markets: counterparty risk and constrained access to wholesale dollar funding inhibited arbitrage during the GFC and again in the euro area sovereign crisis (LTR-16 b). The puzzle is that the violations **persisted, and widened from 2014, for short and long tenors**, after banks had strengthened their balance sheets, credit concerns had faded and wholesale funding had recovered. Banks also drew very little on central bank swap lines in this period. The authors' answer has two halves: **hedging demand opens the basis** and **limits to arbitrage stop it closing**.

2. Why the basis opens: demand for currency hedges

The authors focus on three sources of hedging demand that are **rather insensitive to the size of the basis**, and so exert sustained pressure on it even when it is non-zero:

1. **Banks** (structural): banks run currency mismatches given their deposit base and close the remaining gaps with FX swaps. Japanese banks fund dollar lending with yen deposits through swaps; their dollar funding gap grew from about USD 0.6 to 0.9 trillion.
2. **Institutional investors** (insurance companies and pension funds) hedge foreign currency investments strategically. Unconventional monetary policy compressed term and credit spreads and pushed them abroad for yield and duration; their hedge ratios are quite insensitive to hedging costs and move slowly. Japanese life insurers hedge 60 to 70% of their US dollar bonds.
3. **Non-financial firms** borrow where credit spreads are narrow and swap back. With ECB purchases compressing euro spreads, US firms issued "reverse yankee" euro bonds and swapped the proceeds into dollars, lending euros and borrowing dollars through swaps (Box 17.2).

Across currencies the sign of the basis follows the net hedging position against the dollar. Where banks *compete* with institutional investors for swapped dollars (Japan; the euro area once reverse yankee issuance is added), the basis is negative in market quotes. Where banks *offset* investors, because they fund a large domestic mortgage book by borrowing abroad and swapping into local currency (Australia; Sweden against the euro), the basis is positive. Banks are usually the swing factor. **Over time**, Japanese aggregate hedging need rose from USD 0.9 trillion (2009) to over USD 1.2 trillion (2015), and after a clear break in the GFC the basis tracked it closely. ECB and BoJ easing announcements widened the basis; the "disappointing" ECB announcement of 3 December 2015 narrowed it.

3. Why the basis does not close: limits to arbitrage

"**Balance sheet space is rented, not free.**" As a result of tighter management of risks and related balance sheet constraints, arbitrage now incurs a **cost per unit of balance sheet**, passed on into FX swap prices as a premium. Arbitrage is costly and risky: it enlarges the balance sheet, takes credit risk on both the borrowing and the investing leg, and may bring mark-to-market and liquidity risk (collateral transfers). These risks always existed but were not fully priced before the GFC, when dealers raised leverage to dangerous levels; since then shareholders, creditors and prudential authorities have hard-wired them. The evidence:

- **Regulation:** CVA charges price counterparty risk in derivatives; potential future exposure add-ons in Basel III and US leverage ratios require capital against derivatives exposures.
- **Quarter-end spikes** in the basis and repo rates since 2014, with more weight on quarter-end reporting and ratios.
- **Shift to repo funding:** secured funding needs collateral; BoJ bond purchases made yen collateral scarce while the Fed stopped buying, so dollar repo became dearer relative to yen repo and the basis widened with the VIX in normal ranges.
- **Internal limits:** foreign banks' claims on the Japanese official sector rose from about USD 200 billion (Q4 2014) to USD 323 billion (end-2015), by then 10% to 19% of their equity; the basis widened after S&P's 2015 downgrade of Japan.
- **Related markets:** supranationals' "issuance arbitrage" (issue in dollars, swap into euros) slowed when the dollar swap rate fell below the Treasury yield in Q3 2015.
- **Dual roles:** Japanese banks' own reliance on swaps reduced their capacity to arbitrage or to supply hedges.

"**CIP deviations look to be here to stay**" while hedging demand is high and imbalanced across currencies (Figure 3).

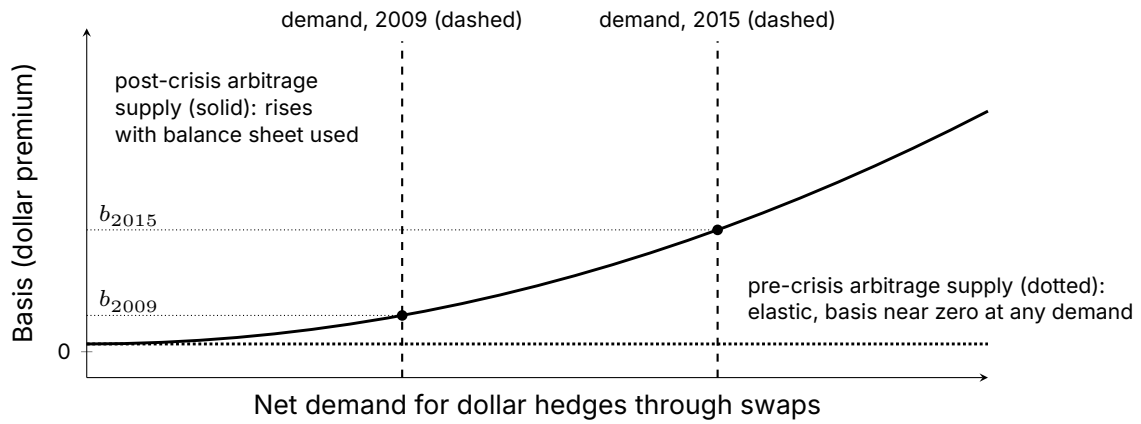


Figure 3. The two halves of the explanation (schematic). Hedging demand is insensitive to the basis (vertical dashed lines). Before the GFC, arbitrage supplied any quantity at a near-zero basis (dotted line), so demand did not matter. After it, each extra dollar of arbitrage uses costly balance sheet (solid curve), so the basis is set where demand meets that curve and widens as demand grows.

Correction to the source notes. Vardeez (following Schweser) describes the second source of hedging demand as "insurance companies and banks". GARP names institutional investors "such as insurance companies and pension funds"; banks are the first source. (Japan's pension funds, led by the largely unhedged GPIF, have low hedge ratios and "ought not to affect the basis"; its life insurers are the hedgers.)

4. The calculations

Example. A funding analyst sees $S = 1.2000$ USD per EUR, a one-year forward $F = 1.2270$, a one-year dollar rate $r = 2.50\%$ and a euro rate $r^* = 0.50\%$.

- CIP forward: $1.2 \times 1.025/1.005 = 1.223881$. The market forward points (0.0270) are wider than parity (0.023881).
- Basis, exact one-period form: $b = \frac{F}{S}(1+r^*) - (1+r) = 1.0225 \times 1.005 - 1.025 = 0.0026$, about **26 basis points**.
- Basis, the approximation with $1+r^* \approx 1$: $b = \frac{F-S}{S} - (r - r^*) = 2.25\% - 2.00\% =$ **25 basis points**.
- Swap-implied dollar rate: $\frac{F}{S}(1+r^*) - 1 = 2.76\%$, against 2.50% in the cash market. A euro area insurer hedging dollar bonds pays the premium; an arbitrageur lending dollars through the swap would earn it, but only if its balance sheet cost is below about 26 basis points a year.

Quote convention. With the foreign currency as the quote currency (yen per dollar), CIP inverts: $F = S\,(1 + r_{\text{JPY}})/(1 + r_{\text{USD}})$. Always put the rate of the *quote* currency on top.

Formula card: LTR-16 c

CIP (S, F in USD per FC)	$F/S = (1+r)/(1+r^*)$
Forward points	$F - S = S\left[(1+r)/(1+r^*) - 1\right]$
FC per USD quotes	$F = S\,(1+r_{\text{FC}})/(1+r_{\text{USD}})$ (rate of the quote currency on top)
Basis, one period	$F - S = S(1+r+b)/(1+r^*) - S \quad \Rightarrow \quad b = \frac{F}{S}(1+r^*) - (1+r)$
Basis, approximation	$b \approx \frac{F-S}{S} - (r - r^*) \quad$ (with $1 + r^* \approx 1$)
Swap-implied USD rate	$\frac{F}{S}(1+r^*) - 1$; exceeds r when the dollar borrower pays b
Basis swap, USD borrower	net cost $= (r_{\text{USD}} + b) - r_{\text{FC}}$
Arbitrage profit	covered return $\frac{F}{S}(1+r^*)$ against the funding cost $1+r$
Factors (LTR-16 b)	$b = \frac{F^b - S^a}{S^a} - (r - r^*); \quad b = rp; \quad b \propto rp\times$ hedging demand
Issuance arbitrage (Box 17.2)	saving $=$ spread in USD $-$ (spread in FC $+ b$); break-even b equals the spread gap

Exam traps

- **Post-2014 cause.** Credit risk explains the GFC and euro-crisis episodes; the persistence after 2014, with Libor-OIS normal, is hedging demand meeting balance sheet constraints.
- **Demand insensitive to price.** The three hedgers do not back off when the basis widens; that is why the pressure is sustained.
- **Demand rose, not fell.** A distractor saying hedging demand declined, or that the basis appears only in crises, contradicts the reading.
- **Cost per unit of balance sheet.** Arbitrage is riskless in payoff but not free; it does not "eliminate" counterparty risk, and regulation *tightened*, not loosened, balance sheet capacity.
- **Japanese banks are hedgers.** Their reliance on swaps reduced arbitrage capacity and widened the basis.
- **Sign across countries.** Banks competing with investors for dollars: negative quoted basis (Japan, euro area). Banks offsetting investors: positive (Australia).
- **Calculation slips.** Inverting the CIP ratio for the quote convention; subtracting instead of compounding rates; forgetting to subtract the rate differential from the forward premium; reading a positive b as cheaper dollars.

Practice questions: LTR-16 c

Question LTR-16 c.1

Bank Q541 • BT Mock Q14 (rebuilt)

The spot USDJPY rate is 110.00 (JPY per USD; USD is the base currency). The one-year USD interest rate is 2.50% and the one-year JPY interest rate is 0.450%, both with annual compounding. If covered interest parity holds exactly, what is the one-year swap rate, defined as $F - S$?

A. -2.26 JPY

B. -2.20 JPY

C. Zero

D. $+2.24$ JPY

Question LTR-16 c.2

Bank Q534 • AnalystPrep Q.4188 (rebuilt)

The one-year USD interest rate is 5% and the one-year GBP interest rate is 8%. The GBP/USD spot rate is 1.2500 (USD per GBP) and the one-year forward rate is 1.2400. An arbitrageur borrows USD 2,000 for one year, converts it to GBP at spot, buys a one-year GBP deposit, and sells the GBP proceeds forward. What is the riskless USD profit at maturity?

A. USD 42.72

B. USD 60.00

C. USD 160.00

D. USD 2,142.72

Question LTR-16 c.3

Bank Q530 • written for the bank, GARP style (re-filed from LTR-16 b)

A funding desk analyst at a US bank collects the following one-year data: spot rate $S = 1.2000$ USD per EUR, forward rate $F = 1.2240$ USD per EUR, USD interest rate $r = 2.00\%$, and EUR interest rate $r^* = 0.50\%$. Using the relationship $b = \frac{F-S}{S} - (r - r^*)$, the analyst computes the deviation from covered interest parity. Which of the following is correct?

A. The deviation is $+0.50\%$, so a party borrowing US dollars through an FX swap effectively pays a lower rate than in the cash market.

B. The deviation is $+0.50\%$, so a party borrowing US dollars through an FX swap effectively pays a higher rate than in the cash market.

C. The deviation is $+1.50\%$, so CIP holds because the forward premium equals the interest rate differential.

D. The deviation is $+2.00\%$, so a party borrowing US dollars through an FX swap effectively pays a higher rate than in the cash market.

Question LTR-16 c.4 Bank Q528 • AnalystPrep Q.4184 (rebuilt; re-filed from LTR-16 a)

A bank (Counterparty A) enters a cross-currency basis swap in which it borrows USD and lends an equivalent amount of EUR. USD Libor is 1.0% and EUR Libor is 0.5%. A dollar shortage means the USD borrower must pay USD Libor plus a basis premium of 40 bps, while it receives EUR Libor on the EUR leg. What is the net annual cost of the swap to Counterparty A?

A. 0.1%

B. 0.9%

C. 1.4%

D. 1.9%

Question LTR-16 c.5 New • built from GARP Ch.17, Box 17.2

The treasurer of a BBB-rated US telecoms firm needs five-year US dollar funding. The firm's bonds yield 100 basis points over the swap rate in US dollars and 50 basis points over the swap rate in euros. If it issues in euros, it will swap the proceeds into dollars with a cross-currency basis swap in which, as the dollar borrower, it pays a basis of 30 basis points on top of the dollar rate. The firm's analyst also notes that the five-year euro swap rate is 70 basis points below the dollar swap rate. Relative to issuing in dollars, the firm's saving from issuing in euros and swapping is closest to:

A. 20 basis points

B. 50 basis points

C. 80 basis points

D. 90 basis points

Question LTR-16 c.6 New • built from GARP Ch.17, Sections 17.1 and 17.2

An FX strategist compares the three-year cross-currency basis against the US dollar across jurisdictions, quoted as in the reading's graphs on the non-dollar currency (negative when dollars are dearer through swaps than in the cash market). In Japan, banks fund dollar lending with yen deposits through FX swaps and life insurers hedge most of their US dollar bonds. In Australia, banks fund a large Australian dollar mortgage book partly by borrowing abroad and swapping the proceeds into Australian dollars, while Australian institutional investors hedge their foreign assets. Which statement is consistent with the reading?

A. Both bases are negative, because after the crisis the US dollar commands the same premium against every currency.

B. The JPY basis is positive, because Japanese banks act mainly as arbitrageurs that supply dollars in the swap market.

C. The JPY basis is negative because Japanese banks and insurers both borrow dollars through swaps; the AUD basis is positive because Australian banks lend dollars in the swap market and offset investors' hedging demand.

D. The AUD basis is negative, because Australian banks add to institutional investors' demand for dollars through swaps.

Question LTR-16 c.7 Bank Q542 • BT P2.T8.20.17.3

Since the global financial crisis (GFC), covered interest rate parity (CIP) violations have persisted. Two explanations are offered:

I. The basis opens up because banks, institutional investors and non-financial firms demand currency hedges (for example, banks closing balance sheet currency mismatches with FX swaps).

II. Arbitrageurs now price the costs and hidden risks of arbitrage, for example the cost of enlarging the balance sheet, which limits arbitrage.

Which statement(s) explain the post-GFC violations of CIP?

A. Neither I nor II

B. Only I

C. Only II

D. Both I and II

Question LTR-16 c.8 Bank Q543 • GARP PE2 Q5

A portfolio manager at a US-based hedge fund notes a positive cross-currency basis between two currencies that has persisted since the global financial crisis (GFC) of 2007 to 2009. What is the most appropriate explanation for this persistence?

A. The costs of credit value adjustments have increased, as arbitrage positions typically eliminate counterparty risks.

B. Regulatory changes have permitted an increase in US banks' speculative proprietary trading activities.

C. The addition of a liquidity risk cost to swap pricing is no longer required given the decline in the overall level of interest rates.

D. The costs for arbitrageurs to finance their positions are increasingly reflected in the basis.

Solutions • LTR-16 c

Solution LTR-16 c.1

Answer: B

Bank Q541 • BT Mock Q14

Step 1 – Put the quote currency's rate on top

The rate is JPY per USD, so JPY is the quote currency: $F = S\,(1 + r_{\text{JPY}})/(1 + r_{\text{USD}})$.

Step 2 – Forward

$$F = 110 \times \frac{1.0045}{1.025} = 107.80$$

Step 3 – Swap points

$$F - S = 107.80 - 110.00 = -2.20 \text{ JPY}$$

$\Rightarrow$ The dollar, the higher-yielding currency, trades at a forward discount.

Interpretation

The forward discount offsets the 2.05% rate advantage of the dollar: a yen investor who converts to dollars earns more interest but buys back fewer yen forward. In GARP's USD-per-FC notation this is the same CIP with the ratio flipped.

Why the other options are wrong

A is incorrect. Linear approximation $110 \times (0.0045 - 0.025) = -2.255$, without dividing by $1 + r_{\text{USD}}$.

C is incorrect. Zero holds only if the two rates are equal.

D is incorrect. Inverted the ratio: $110 \times 1.025/1.0045 = 112.24$, a forward premium for the high-yield currency.

LO: LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 c.2

Answer: A

Bank Q534 • AnalystPrep Q.4188

Step 1 – Convert and invest

$$2{,}000/1.25 = \text{GBP } 1{,}600, \qquad 1{,}600 \times 1.08 = \text{GBP } 1{,}728$$

Step 2 – Convert back at the forward

$$1{,}728 \times 1.24 = \text{USD } 2{,}142.72$$

Step 3 – Repay the loan

$$2{,}000 \times 1.05 = \text{USD } 2{,}100, \qquad \text{profit} = 2{,}142.72 - 2{,}100 = \text{USD } 42.72$$

$\Rightarrow$ The covered sterling investment beats the dollar funding cost: CIP is violated.

Interpretation

The CIP forward is $1.25 \times 1.05/1.08 = 1.2153$; the quoted 1.2400 is above it, so the covered return $\frac{F}{S}(1 + r^*) = 1.0714$ exceeds 1.05. In the reading's world this profit exists on paper and still may not be taken, because the trade uses balance sheet that now has a price (Section 3).

Why the other options are wrong

B is incorrect. Simple rate differential $2{,}000 \times (8\% - 5\%) = 60$, ignoring the forward discount on sterling.

C is incorrect. The 8% sterling interest on USD 2,000, ignoring both the loan cost and the conversions.

D is incorrect. Gross proceeds; the USD 2,100 loan repayment was not deducted.

LO: LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1 and introduction

Solution LTR-16 c.3

Answer: B

Bank Q530 • written for the bank

Step 1 – Forward premium

$$\frac{F - S}{S} = \frac{0.0240}{1.2000} = 2.00\%$$

Step 2 – Rate differential

$$r - r^* = 2.00\% - 0.50\% = 1.50\%$$

Step 3 – Basis

$$b = 2.00\% - 1.50\% = +0.50\%$$

$\Rightarrow$ Forward points are wider than CIP warrants, so the dollar borrower through the swap pays more than the cash rate.

Interpretation

The exact one-period form gives $\frac{F}{S}(1 + r^*) - (1 + r) = 1.02 \times 1.005 - 1.02 = 0.51\%$; the approximation in the stem drops the small cross term. Who receives the 50 basis points? The dollar lender, which is the arbitrageur's side, and only if its balance sheet cost is lower.

Why the other options are wrong

A is incorrect. Right number, wrong direction: a positive b means swapped dollars are dearer.

C is incorrect. Reports the rate differential as the deviation; CIP would hold only if the forward premium equalled it, and it is 2.00%.

D is incorrect. Reports the forward premium and forgets to subtract the rate differential.

LO: LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 c.4

Answer: B

Bank Q528 • AnalystPrep Q.4184

Step 1 – Dollar leg, paid

A borrows dollars, so it pays dollar Libor plus the basis: $1.0\% + 0.40\% = 1.40\%$.

Step 2 – Euro leg, received

A lends euros, so it receives euro Libor: 0.50%.

Step 3 – Net

$$1.40\% - 0.50\% = 0.90\%$$

$\Rightarrow$ The dollar borrower carries the basis on top of the cash-market differential of 0.50%.

Interpretation

Without a basis the net cost would be the 0.50% differential, exactly what CIP prices. The extra 40 basis points is the CIP violation, paid by the side that needs the scarce currency.

Why the other options are wrong

A is incorrect. Subtracted the basis as if received: $1.0\% - 0.4\% - 0.5\% = 0.1\%$.
C is incorrect. Ignored the euro leg: dollar Libor plus basis only.
D is incorrect. Added euro Libor instead of receiving it: $1.4\% + 0.5\%$.

LO: LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.1

Solution LTR-16 c.5 — Answer: A

New • built from GARP Ch.17, Box 17.2

Step 1 – Cost of issuing in dollars

Dollar swap rate + 100 basis points.

Step 2 – Cost of issuing in euros and swapping

The swap turns the euro coupon (euro swap rate + 50) into a dollar cost: dollar swap rate + 50, plus the 30 basis point basis the dollar borrower pays, so dollar swap rate + 80.

Step 3 – Saving

$$100 - (50 + 30) = 20 \text{ basis points}$$

$\Rightarrow$ The basis eats 30 of the 50 basis point spread advantage.

Interpretation

Box 17.2: if CIP held the firm would save the full 50 basis points and would issue all its debt in euros. The firm's hedging demand widens the basis, which "tends to reconcile the different spreads in the two markets"; issuance pays until b reaches the 50 basis point spread gap. The euro swap rate's 70 basis point discount is a red herring: once the proceeds are swapped, the rate level difference is priced out by the swap.

Why the other options are wrong

B is incorrect. Ignored the basis: the 50 basis point saving that applies only if CIP holds.
C is incorrect. Treated the basis as received: $100 - (50 - 30) = 80$.
D is incorrect. Counted the lower euro swap rate as a saving: $70 + 100 - 50 - 30 = 90$.

LO: LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, Box 17.2 Reverse Yankee Issuance

Solution LTR-16 c.6 — Answer: C

New • built from GARP Ch.17, Sections 17.1 and 17.2

Step 1 – Japan

Banks use yen deposits to fund dollar lending through swaps, and life insurers hedge dollar bonds: both borrow dollars through swaps, so they compete for swapped dollars. Dollar borrowing via swaps is in higher demand and dearer than in the cash market: a negative quoted basis.

Step 2 – Australia

Banks with a mortgage book larger than their deposit base borrow abroad and swap into Aus-

tralian dollars, so in the swap market they lend dollars. That offsets investors' hedging demand, and the basis is positive.
⇒ The sign follows the net hedging position against the dollar, with banks as the swing factor.

Interpretation

Sweden looks like Japan against the dollar but like Australia against the euro, because the euro swap market is the marginal funding source for Swedish banks. Pick the right currency pair before reading the sign.

Why the other options are wrong

A is incorrect. Absolute claim: the reading shows a positive Australian dollar basis.
B is incorrect. Role misassignment: Japanese banks are hedgers adding to dollar demand; their reliance on swaps reduced their arbitrage capacity.
D is incorrect. Reverses Australian banks' role: they offset investors, which is why Australia is the exception.

LO: LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, 17.1 Demand for Currency Hedges; 17.2 The Currency Basis in the Cross Section

Solution LTR-16 c.7 — Answer: D

Bank Q542 • BT P2.T8.20.17.3

Step 1 – Statement I

True: hedging demand from banks, institutional investors and non-financial firms, insensitive to the size of the basis, is why the basis opens up.

Step 2 – Statement II

True: arbitrage enlarges the balance sheet and brings credit, funding and liquidity risks that are now priced, so limits to arbitrage stop the basis closing.
⇒ Both: the reading's two-part explanation.

Interpretation

Each half alone is incomplete: demand without limits to arbitrage would be met by arbitrageurs at a zero basis; limits without demand pressure would leave nothing to arbitrage (Figure 3).

Why the other options are wrong

A is incorrect. Denies both causes.
B is incorrect. Omits limits to arbitrage, which explain why the basis persists.
C is incorrect. Omits hedging demand, which explains why the basis opens.

LO: LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, 17.1 A Framework

Solution LTR-16 c.8 — Answer: D

Bank Q543 • GARP PE2 Q5

Step 1 – What arbitrage requires

Closing the basis means enlarging the balance sheet, with credit risk on both legs and mark-to-market and liquidity risk.

Step 2 – What changed after the crisis

Those risks are now actively managed and charged for (CVA, PFE add-ons, leverage ratios, internal limits): a cost per unit of balance sheet, passed into FX swap prices.
⇒ The basis persists because arbitrageurs' financing and balance sheet costs are reflected in it.

Interpretation

"Positive" is GARP's Box 17.1 sign for a dollar premium. The workbook's D8.2 (LTR-16 b.6) asks the same question from a hedger's side.

Why the other options are wrong

A is incorrect. CVA charges did rise, but arbitrage positions carry counterparty risk on both legs; they do not eliminate it.

B is incorrect. Regulation tightened banks' balance sheet capacity; it did not permit more proprietary trading.

C is incorrect. Liquidity and funding risks are priced more carefully post-crisis; the level of interest rates does not remove them.

LO: LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. • **Reference:** Borio, McCauley, McGuire and Sushko (2016), GARP Ch.17, 17.1 Limits to Arbitrage